Biotechnology

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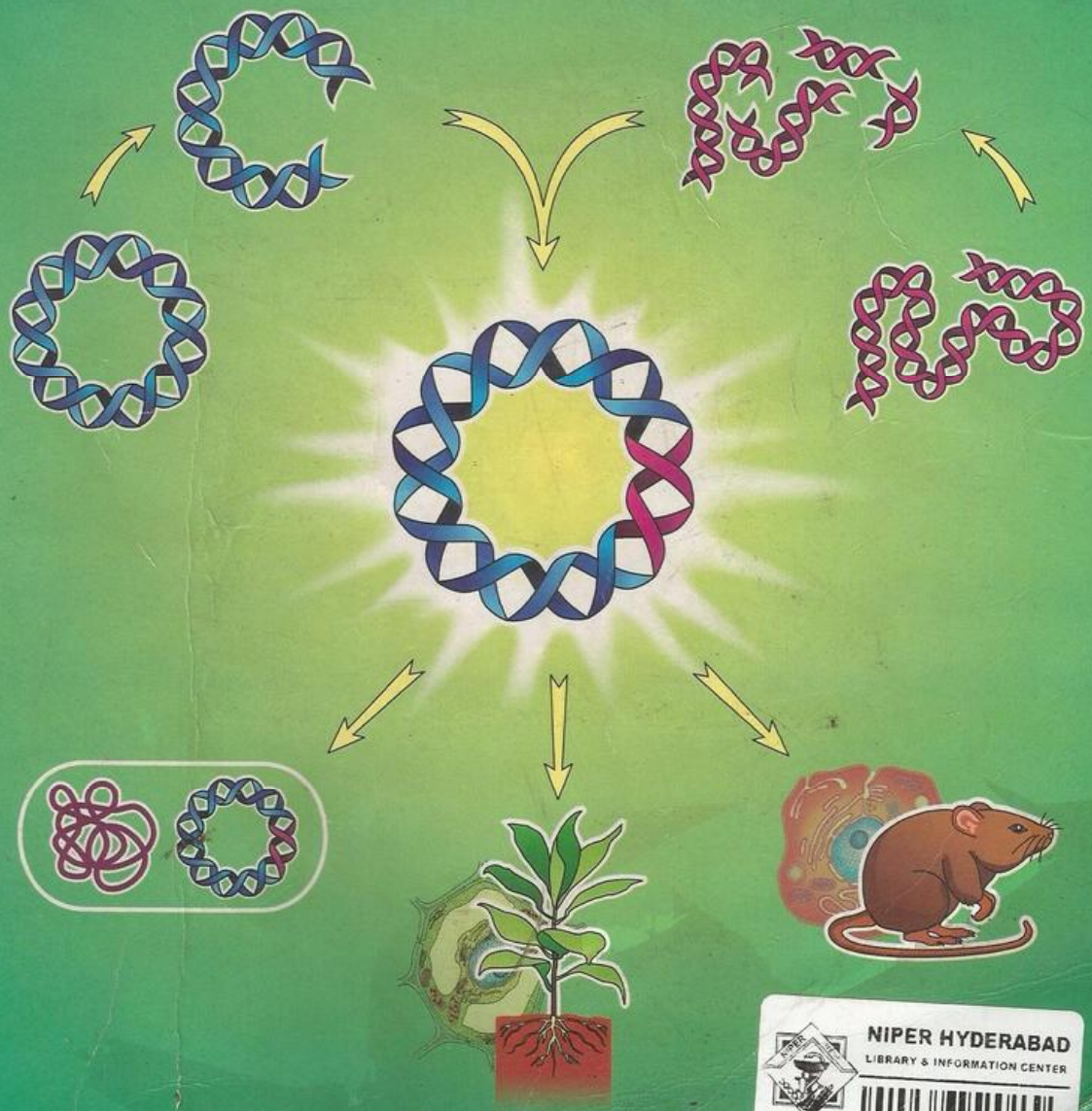

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Contents in Brief

SECTION I

Introduction to Biotechnology

1 The Scope of Biotechnology 3

SECTION II

Introduction to Molecular Biology

2 DNA and RNA-Composition and Structure 11
3 DNA-Replication, Recombination, and Repair 23
4 Transcription and Translation 38
5 Regulation of Gene Expression 59

SECTION III

Genetic Engineering/ Recombinant DNA Technology

6 Introduction to Genetic Engineering 75
7 Basic Techniques in Genetic Engineering 92
8 Polymerase Chain Reaction (DNA Amplification) 112
9 Gene Libraries 120
10 Site-directed Mutagenesis and Protein Engineering 129
11 Manipulation of Gene Expression in Host Cells 137
12 Human Genome Project 148

SECTION IV

Medical/Pharmaceutical Biotechnology (Biotechnology in Healthcare)

13 Gene Therapy 157
14 DNA in Disease Diagnosis and Medical Forensics 173
15 Pharmaceutical Products of DNA Technology 189
16 Recombinant Vaccines 199
17 Monoclonal Antibodies (Hybridoma Technology) 213
18 Assisted Reproductive Technology 227

SECTION V

Microbial/Industrial Biotechnology

19 Bioprocess/Fermentation Technology 239
20 Downstream Processing 270
21 Enzyme Technology 281
22 Biotransformation 306
23 Microbial Production of Organic Solvents 311

24 Microbial Production of Organic Acids 318
25 Microbial Production of Antibiotics 329
26 Microbial Production of Amino Acids 344
27 Microbial Production of Vitamins 355
28 Microbial Production of Foods and Beverages 362
29 Single-Cell Protein, and Mushrooms 373
30 Polysaccharides, Polyhydroxyalkanoates, and Lipids 382
31 Biomass and Bioenergy 393
32 Microbial Mining and Metal Biotechnology 400

SECTION VI

Animal Cell/Animal Biotechnology

33 Animal Cell Culture — Fundamentals, Facilities and Applications 407
34 Culture Media for Animal Cells 415
35 Cultured Cells-Biology and Characterization 423
36 Primary Culture and Cell Lines 437
37 Animal Cell Culture—General Considerations and Scale-up 450
38 Cell Viability and Cytotoxicity 459
39 Cell Transformation and Cell Cloning 463
40 Organ and Histotypic Cultures, and Tissue Engineering 468
41 Transgenic Animals 480

SECTION VII

Plant/Agricultural Biotechnology

42 Plant Tissue Culture — General 497
43 Plant Tissue Culture Media 517
44 Protoplast Culture and Somatic Hybridization 523
45 Production of Haploid Plants 538
46 Somaclonal Variations 546
47 Clonal Propagation (Micropropagation) 552
48 Germplasm Conservation and Cryopreservation 565
49 Genetic Engineering of Plants — Methodology 572
50 Applications of Plant Transformation/ Transgenic Plants 596
51 Transgenic Plants as Bioreactors 622
52 Growth Promoting Bacteria in Plants 639
53 Molecular Marker-Aided Plant Breeding 648

SECTION VIII

Environmental Biotechnology

54 Environmental Pollution — General 657
55 Air Pollution and Control 669
56 Water Pollution and Sewage 679
57 Sewage/Waste Water Treatment 686
58 Sludge and Solid Wastes-Treatment and Disposal 708
59 Biodegradation and Bioremediation 718
60 Global Environmental Problems 730

SECTION IX

Biotechnology and Society

61 Biotechnology — Society, Risks, Ethics, Patenting 739

SECTION X

Basics to Learn Biotechnology

62 Cell — The Basic Unit of Life 749
63 Microorganisms 754
64 Bioorganic and Biophysical Chemistry, and Biochemistry Tools 758
65 Biomolecules 771
66 Enzymology 786
67 Metabolisms 796
68 Immunology 816
69 Genetics 822
70 Bioinformatics 827

Appendices

Glossary 833
Abbreviations 852
Index 855

Contents in Detail

SECTION I

Introduction to Biotechnology

1 The Scope of Biotechnology 3

Old and new biotechnology 3
Definition(s) of biotechnology 3
History of biotechnology 4
Biotechnology — a multidisciplinary growing tree 4
Commercialization of biotechnology 7
Public perception of biotechnology 7
The future of biotechnology 7
Organization of the book 7

SECTION II

Introduction to Molecular Biology

2 DNA and RNA — Composition and Structure 11

Nucleotides 11
Structure of nucleotides 11
Structure of DNA 14
DNA double helix 14
Other types of DNA structure 17
The size of DNA molecule-units of length 17
Denaturation of DNA strands 18
Organization of DNA in the cell 19
Structure of RNA 19
Types of RNA 21
Catalytic RNAs-ribozymes 22

3 DNA-Replication, Recombination and Repair 23

Replication of DNA 23
Replication in prokaryotes 24
Replication in eukaryotes 27
Process of replication in eukaryotes 27
Inhibitors of DNA replication 29
Cell cycle and DNA replication 29
Telomeres and telomerase 30
Telomere in senescence and cancer 31
Recombination 31
Homologous recombination 31
Non-homologous recombination 32
Damage and repair of DNA 33
Types of DNA damages 34
Mutations 34
Repair of DNA 35
Defects in DNA repair and cancer 37

4 Transcription and Translation 38

Genome 38
Transcriptome 38
Proteome 38

Transcription 39
Transcription in prokaryotes 39
Transcription in eukaryotes 42
Post-transcriptional modifications 43
Cellular RNA contents 45
Reverse transcription 46
Translation 46
Genetic code 46
Protein biosynthesis 48
Requirement of the components 49
Activation of amino acids 50
Protein synthesis proper 50
Initiation of translation 51
Elongation of translation 53
Termination of translation 53
Inhibitors of protein synthesis 53
Chaperones and protein folding 55
Post-translational modifications of proteins 56
Protein targeting 57
Mitochondrial DNA, transcription and translation 58

5 Regulation of Gene Expression 59
Gene regulation — general 59
The operon concept 60
Lactose (lac) operon 60
Tryptophan operon 62
Gene expression in eukaryotes 62
Chromatin structure and gene expression 63
Enhancers and tissue-specific gene expression 65
Combination of DNA elements and proteins in gene expression 65
Motifs in proteins and gene expression 65
Gene regulation in eukaryotes 67
Methods to study gene expression/regulation 68
Gene analysis by T-DNA and transposon tagging 70
Methods to study protein-protein interactions 70
Phage display 70
Yeast two-hybrid system 72

SECTION III

Genetic Engineering/Recombinant DNA Technology

6 Introduction to Genetic Engineering 75
Brief history of recombinant DNA technology 75
Molecular tools of genetic engineering 76
Restriction endonucleases-DNA cutting enzymes 76
DNA ligases-DNA joining enzymes 78
Alkaline phosphatase 79
DNA modifying enzymes 79
Host cells — the factories of cloning 81
Prokaryotic hosts 82
Eukaryotic hosts 82
Vectors — the cloning vehicles 82
Plasmids 83
Bacteriophages 85
Cosmids 86
Artificial chromosome vectors 88
Shuttle vectors 89
Choice of a vector 89
Methods of gene transfer 89
Transformation 89
Conjugation 89
Electroporation 89
Liposome-mediated gene transfer 90
Transduction 90
Direct transfer of DNA 91
Gene cloning strategies 91
Cloning from genomic DNA or mRNA? 91
Genetic engineering of plants 92
Genetic engineering guidelines 92
Asilomar recommendations 92
NIH guidelines 92
The future of genetic engineering 93

7 Basic Techniques in Genetic Engineering 95
Agarose gel electrophoresis 95
Pulsed-field gel electrophoresis (PFGE) 97

Contour-clamped homogeneous electrical-field electrophoresis (CHEF) 93
Polyacrylamide gel electrophoresis (PAGE) 93
Other uses of gel electrophoresis 94
Isolation and purification of nucleic acids 94
Purification of cellular DNA 94
Purification of plasmid DNA 95
Purification of mRNA 96
Isolation of chromosomes 96
Fluorescence-activated cell sorting 96
Collection of chromosomes with identical sizes 97
Nucleic acid blotting techniques 97
Southern blotting 99
Northern blotting 99
Dot-blotting 100
Western blotting 100
Autoradiography 100
Colony and plaque blotting 100
DNA sequencing 101
Maxam and Gilbert technique 101
Dideoxynucleotide method 102
Bacteriophage M_{13} as a cloning and DNA sequencing vector 103
DNA sequencing by primer walking 104
Chromosome walking in DNA sequencing 105
Automated DNA sequencing 106
Alternative methods of DNA sequencing 107
Pyrosequencing 107
DNA chips (microarrays) 108
Chemical synthesis of DNA 109
The phosphoramidite method 109
Synthesis of genes 111

8 Polymerase Chain Reaction (DNA Amplification) 112
Technique of PCR 112
Sources of DNA polymerase 114
Key factors for optimal PCR 114
Variations of PCR 114
Nested PCR 115
Inverse PCR 115
Anchored PCR 115
Reverse transcription PCR 115
Asymmetric PCR 115
Real-time quantitative PCR 116
Random amplified polymorphic DNA (RAPD) 116
Amplified fragment length polymorphism (AFLP) 117
Rapid amplification of cDNA ends (RACE) 118
Applications of PCR 118
PCR in clinical diagnosis 118
PCR in DNA sequencing 119
PCR in gene manipulation and expression studies 119
PCR in comparative studies of genomes 119
PCR in forensic medicine 119
PCR in comparison with gene cloning 119

9 Gene Libraries 120
Creating a gene library 120
PCR as an alternative to genomic library construction 123
Complementary DNA libraries 123
Screening strategies 125
Screening by DNA hybridization 125
DNA probes 126
Screening by colony hybridization 126
Screening by PCR 128
Screening by immunological assay 128
Screening by protein function 128

10 Site-directed Mutagenesis and Protein Engineering 129
Strategies to improve *in vitro* activities of enzymes 129
Oligonucleotide—directed mutagenesis 130
Casettee mutagenesis 131
PCR-amplified oligonucleotide-directed mutagenesis 132
Protein engineering 133
Increasing the stability and biological activity of proteins 133
Addition of disulfide bonds 133
Changing asparagine to other amino acids 133
Reducing the free sulfhydryl groups 133
Single amino acid changes 134

Improving kinetic properties of enzymes 135
Protein engineering by use of gene families 136
Protein engineering through chemical modifications 136
Protein engineering — an ever expanding field 136

11 Manipulation of Gene Expression in Host Cells 137
Selection of host cells for gene expression 137
Manipulation of gene expression in prokaryotes 138
Manipulation of gene expression in eukaryotes 139
Saccharomyces cerevisiae — the yeast in expressing cloned genes 140
Other yeast expression systems 141
Insect cell expression systems 141
Mammalian cell expression vectors 143
Gene expression to produce proteins 143
Collection and purification of recombinant proteins 144
Purification of recombinant proteins 146

12 Human Genome Project 148
The birth and activity of human genome project 148
Mapping of the human genome 149
Approaches for genome sequencing 149
Human genome sequence-results summarised 149
Genomes of some other organisms sequenced 153
Benefits/applications of human genome sequencing 153
Ethics and human genome 154

SECTION IV

Medical/Pharmaceutical Biotechnology (Biotechnology in Healthcare)

13 Gene Therapy 157
Approaches for gene therapy 157
***Ex vivo* gene therapy**
Vectors in gene therapy
Viruses
Human artificial chromosome
Bone marrow cells
Selected examples of *ex vivo* gene therapy
Therapy for adenosine deaminase deficiency
Therapy for familial hypercholesterolemia
Therapy for Lesch-Nyhan syndrome
Therapy for hemophilia
Ex vivo gene therapy with non-autologous cells
***In vivo* gene therapy**
Gene delivery by viruses
Gene delivery by non-viral systems
Gene therapy strategies for cancer
Gene therapy for AIDS
Antigene and antisense therapy
Antisense therapy for cancer
Antisense therapy for AIDS
Antisense oligonucleotides as therapeutic agents
Chimeric oligonucleotides in gene correction
Aptamers as therapeutic agents
Ribozymes as therapeutic agents
The future of gene therapy

14 DNA in Disease Diagnosis and Medical Forensics 173
Methods of DNA assay
Nucleic acid hybridization
DNA probes
The DNA chip-microarray of gene probes
DNA in the diagnosis of infectious diseases
Tuberculosis
Malaria
Chaga's disease
Acquired immunodeficiency syndrome (AIDS)

Human papilloma virus 178
Lyme disease 178
Periodontal disease 178
DNA probes for other diseases 178
DNA in the diagnosis of genetic diseases 179
Cystic fibrosis 179
Sickle-cell anemia 179
Duchenne's muscular dystrophy 179
Huntington's disease 180
Fragile X syndrome 180
Other triple repeat diseases 180
Alzheimer's disease 181
Amyotrophic lateral sclerosis 181
Cancers 181
Diabetes 182
Obesity 183
DNA analysis for other human diseases 183
Gene banks — a novel concept 184
DNA analysis for environmental monitoring 184
Water quality testing 184
DNA fingerprinting or DNA profiling 184
DNA markers in disease diagnosis and fingerprinting 185
Restriction fragment length polymorphisms (RFLPs) 185
Variable number tandem repeats (VNTRs) 186
Microsatellites (simple tandem repeats) 188
Single nucleotide polymorphisms (SNPs) 188
Current technology of DNA fingerprinting 188

15 Pharmaceutical Products of DNA Technology 189
Human protein replacements 189
Insulin 189
Human growth hormone 192
Clotting factor VIII 194
Therapeutic agents for human diseases 194
Tissue plasminogen activator 194
Interferons 196
Erythropoietin 198
Deoxyribonuclease I (DNase I) 198
Alginate lyase 198
Second generation therapeutic proteins (muteins) 198
Vaccines 198

16 Recombinant Vaccines 199
Traditional vaccines 199
Recombinant vaccines — general 200
Subunit vaccines 201
Hepatitis B 201
Foot and mouth disease 202
Herpes simplex virus 203
Tuberculosis 203
Meningitis 203
AIDS (Acquired immunodeficiency syndrome) 204
DNA vaccines (genetic immunization) 205
DNA vaccine and immunity 205
RNA vaccines 207
Plants as edible subunit vaccines 207
Attenuated recombinant vaccines 208
Cholera 208
Salmonella species 210
Leishmania species 210
Vector recombinant vaccines 211
Vaccines against viruses — vaccinia virus 211
Delivery of antigens by bacteria 212

17 Monoclonal Antibodies 213
Principles for creation of hybridoma cells 213
Production of monoclonal antibodies 214
Large scale production of MAbs 216
Human monoclonal antibodies 216
Genetic engineering strategies for the production of human-mouse MAbs 217
Production of MAbs in *E. coli* 218
Second generation monoclonal antibodies 218
Advantages of monoclonal antibodies 219
Limitations of monoclonal antibodies 219

Applications of monoclonal antibodies 219
Diagnostic applications 219
MAbs in biochemical analysis 219
MAbs in diagnostic imaging 220
Therapeutic applications 222
MAbs as direct therapeutic agents 222
MAbs as targeting agents in therapy 223
Protein purification 225
Miscellaneous applications 226
Catalytic MAbs (abzymes) 226
Antibody fingerprinting 226

18 Assisted Reproductive Technology 227
Manipulation of reproduction in animals 227
Artificial insemination 227
Embryo transfer 228
In vitro fertilization 230
Embryo cloning 231
Manipulations of reproduction in humans 231
Causes of infertility and application of ART 231
Intrauterine insemination (IUI) 232
In vitro fertilization and embryo transfer (IVF and ET) 232
Methodology of IVF 232
Success rates of IVF 233
The world's picture of test tube babies 233
Gamete intrafallopian transfer (GIFT) 233
Zygote intrafallopian transfer (ZIFT) 234
Intravaginal culture (IVC) 234
Cytoplasmic transfer (CT) 234
Micromanipulation 234
Cryopreservation 235
Assisted hatching (AH) 235
Preimplantation of genetic diagnosis (PED) 236
The negative aspects of ART 236

SECTION V

Microbial/Industrial Biotechnology

19 Bioprocess/Fermentation Technology 239
Bioreactors/Fermenters 239
Types of bioreactors 239
A conventional bioreactor — common features 244
Operation of a conventional bioreactor 245
Sterilization 245
Inoculation and sampling 245
Aeration 245
Control systems 246
Cleaning 246
Solid substrate (solid state) fermentation 247
Media (substrates) for industrial fermentation 248
Substrates used as carbon sources 249
Substrates used as nitrogen sources 249
Sources of growth factors 250
Sterilization of culture media and gases 250
Sterilization of culture media 250
Sterilization of air 251
Isolation of microorganisms 252
Preservation of microorganisms 254
Microbial metabolic products — low molecular weight compounds 254
Primary metabolites 254
Secondary metabolites 255
Bioconversions 257
Microbial metabolic products — high molecular weight compounds 257
Genetic improvement (development) of strains 257
Methods of strain development 258
Mutation 258
Genetic recombination 259
Principles of microbial growth and culture system 259
Batch culture or batch fermentation 259
Fed batch culture or fed-batch fermentation 261
Semi-continuous culture or semi-continuous fermentation 261
Continuous culture or continuous fermentation 262
Growth kinetics of microorganisms 263
Classification of fermentation processes 264
The fermentation process 265
Measurement and control of bioprocess parameters 267
Scale-up 268
Food technology 268
Food preservation 269

20 Downstream Processing 270

Stages in downstream processing 270
Solid-liquid separation 270
Flotation 271
Flocculation 271
Filtration 271
Centrifugation 272
Release of intracellular products 273
Cell disruption 274
Concentration 276
Evaporation 276
Liquid-liquid extraction 276
Membrane filtration 277
Precipitation 277
Adsorption 278
Purification by chromatography 278
Formulation 279
Integration of different processes 280

21 Enzyme Technology 281

Applications of enzymes 281
Commercial production of enzymes 281
The technology of enzyme production — general considerations 283
Regulation of microbial enzyme production — general considerations 286
Genetic engineering for microbial enzyme production 286
Immobilization of enzymes and cells 288
Methods of immobilization 288
Choice of immobilization technique 290
Stabilization of soluble enzymes 290
Immobilization of cells 293
Effect of immobilization on enzyme properties 293
Immobilized enzyme reactors 294
Applications of immobilized enzymes and cells 295
Manufacture of commercial products 295
Immobilized enzymes and cells — analytical applications 296
Biosensors 297
Types of biosensors 298
Electrochemical biosensors 299
Thermometric biosensors 300
Optical biosensors 301
Piezoelectric biosensors 302
Whole cell biosensors 302
Immunobiosensors 303
Applications of biosensors 304
Immobilized enzymes and cells — therapeutic applications 304

22 Biotransformation 306

Types of biotransformation reactions 306
Sources of biocatalysts and techniques for biotransformation 306
Product recovery in biotransformation 307
Examples of biotransformation 308
Biotransformation to produce commercial products 308
Biotransformation of steroids 308
Biotransformation of antibiotics 308
Biotransformation of arachidonic acid to prostaglandins 310
Biotransformation for the production of ascorbic acid 310
Biotransformation of glycerol to dihydroxyacetone 310
Biotransformation for the production of indigo 310

23 Microbial Production of Organic Solvents 311

Alcohol 311
Production of ethanol by fermentation 312
Acetone and butanol 315
Glycerol 316

24 Microbial Production of Organic Acids 318

Citric acid 318
Factors in the regulation of citric acid production 320
Production process for citric acid 321
Surface processes 321
Submerged processes 322
Production of citric acid from alkanes 322
Recovery of citric acid 322
Gluconic acid 324

Lactic acid 325
Acetic acid 326
L-Ascorbic acid 327
Itaconic acid 328

25 Microbial Production of Antibiotics 329
Antibiotics — general 329
Applications of antibiotics 330
Production of antibiotics — a major pharmaceutical industry 330
Penicillins 331
Production process of penicillin 332
Production of 6-amino penicillanic acid 334
Cephalosporins 334
Production process of cephalosporin 334
Production of 7-aminocephalosporanic acid 335
New β-lactam technology for production of 7-ACA 336
Aminoglycosides 336
Production process of streptomycin 337
Tetracyclines 338
Production process of chlortetracycline 339
Production of tetracycline-different processes 339
Macrolides 340
Production process of erythromycin 340
Aromatic antibiotics 341
Chloramphenicol 341
Griseofulvin 341
Nucleoside antibiotics 342
Genetic manipulations of *Streptomyces* 342
Good antibiotic manufacturing practices 343

26 Microbial Production of Amino Acids 344
Amino acid production — general considerations 344
Commercial applications of amino acids 344
Methods for production of amino acids 345
Strain development for amino acid production 346
L-Glutamic acid 346
L-Lysine 350
L-Threonine 352
L-Phenylalanine 353
L-Tryptophan 353
L-Aspartic acid 354

27 Microbial Production of Vitamins 355
Vitamin B_{12} 355
Commercial production of vitamin B_{12} 356
Riboflavin 357
Commercial production of riboflavin 358
β-Carotene 359
Commercial production of β-carotene 359
Gibberellins-plant growth stimulants 361
Microbial production of gibberellins 361

28 Microbial Production of Foods and Beverages 362
Fermented Foods 362
Cheese 362
Yoghurt 365
Sauekraut 365
Bread 365
Baker's yeast 365
Other vegetarian foods 366
Sweeteners 367
Thaumatin — a sweet protein 367
Monellin — a candidate for sugar substitute 367
Flavour enhancers 367
Alcoholic beverages 367
Alcoholic beverage production — general aspects 368
Beer 369
Wines 370
Other microbially-derived food products 371
Enzymes in food and beverage industries 371
Enzymes in the preparation of fruit juices 371
Diagnostics in food biotechnology 372
Food biotechnology — public acceptance 372

29 Single-Cell Protein, and Mushrooms 373
Microorganisms and substrates used for production of SCP 374

Production of SCP from high energy sources 375
Production of SCP from wastes 378
Production of SCP from wood 379
Production of SCP from CO_2 379
Production of SCP from sewage 379
Genetically engineered artificial protein as animal feed 380
Mushrooms 380
Production of edible mushrooms 381

30 Polysaccharides, Polyhydroxyalkanoates and Lipids 382
Polysaccharides 382
General features of microbial polysaccharides 383
Xanthan 383
Dextran 385
Alginate 385
Scleroglucan 385
Gellan 386
Pollulan 386
Curdlan 386
Polyhydroxyalkanoates 386
PHA-chemistry and properties 386
Polyhydroxybutyrate (PHB) 387
Biosynthesis of other PHA 389
Biopol — a biodegradable plastic 390
Genetic engineering for PHA production 390
Microbial lipids 391
Production of single-cell oils 391
Microbial rubber 391
Microbial adhesive biopolymer 392
Microbial production of indigo 392

31 Biomass and Bioenergy 393
Biomass 393
Sources and utilization of biomass 394
Production of alcohol from biomass 395
Production of biogas from biomass 395
Process of biogas production (biogas plant) 397
Hydrogen — a new biofuel 398
Production of biohydrogen 398
Energy-rich crops 399
The future of biomass 399

32 Microbial Mining and Metal Biotechnology 400
Bioleaching 400
Commercial process of bioleaching 401
Bioleaching of copper 402
Bioleaching of uranium 402
Bioleaching of other metals 403
Advantages of bioleaching 403
Biosorption 403
Microbial recovery of petroleum 404

SECTION VI

Animal Cell/Animal Biotechnology

33 Animal Cell Culture — Fundamentals, Facilities and Applications 407
Facilities for animal cell culture 407
Minimal requirements for cell culture 408
Infrastructure 408
Equipment 408
Culture vessels 408
Use of non-adhesive substrates in tissue culture 409
Contamination, aseptic conditions, and sterilization 409
Aseptic conditions 409
Sterilization 410
Advantages and limitations of tissue culture 411
Applications of animal cell cultures 411
Medical/pharmaceutical products of animal cell cultures 412
Genetic engineering of animal cells and their applications 413
Risks in a tissue culture laboratory and safety 413
Biohazards 414

34 Culture Media for Animal Cells 415
Natural media 415

Artificial media 415
Physicochemical properties of culture media 415
Balanced salt solutions 417
Complete culture media 417
Serum 419
Selection of medium and serum 420
Supplementation of the medium with tissue extracts 420
Serum-free media 421
Advantages and disadvantages of serum-free media 421
Development of serum-free media 422
Commonly used serum-free media 422

35 Cultured Cells—Biology and Characterization 423
Characteristics of cultured cells 423
Cell adhension 423
Cell proliferation 424
Cell differentiation 424
Metabolism of cultured cells 425
Initiation of cell culture 425
Evolution and development of cell lines 427
Characterization of cultured cells 428
Morphology of cells 428
Species of origin of cells 429
Identification of tissue of origin 429
Transformed cells 430
Identification of specific cell lines 430
Measurement of growth parameters of cultured cells 432
Growth cycle of cultured cells 432
Plating efficiency of cultured cells 433
Cell synchronization 433
Cell separation by physical means 433
Cell separation by chemical blockade 434
Some highlights of cell synchronization 435
Senescence and apoptosis 435
Cellular senescence 435
Measurement of senescence 435
Apoptosis 435
Measurement of apoptosis 436

36 Primary Culture and Cell Lines 437
Primary cell culture 437
Techniques for primary culture 437
Mechanical disaggregation 437
Enzymatic disaggregation 438
Primary explant technique 441
Separation of viable and non-viable cells 441
Medical ethics and safety measures in culture techniques 441
Cell lines 442
Finite cell lines 442
Continuous cell lines 443
Nomenclature of cell lines 443
Selection of cell lines 443
Maintenance of cell cultures 444
Subculture 445
Monolayer cultures 445
Suspension cultures 446
Stem cell cultures 447
Embryonic stem (ES) cells 447
Epithelial cells 448
Maintenance of stem cells in culture 448
Characterization of stem cells 448
Applications of cultured stem cells 449

37 Animal Cell Culture — General Considerations and Scale-up 450
Cell culture-general considerations 450
Cell quantitation 450
Equipment and medium 451
pH and buffer systems 451
Oxygen 451
Growth kinetics 451
Types of culture processes 452
Other practical considerations 452
Scale-up 453
Scale-up in suspension 453
Factors in scaling up 453
Stirrer culture 453
Continuous flow culture 454
Air-lift fermenter culture 454
Other systems for suspension culture 455
Scale-up in monolayer 455
Roller bottle culture 456
Multisurface culture 456
Multiarray disks, spirals and tubes 456

Microcarrier culture 456
Perfused monolayer culture 457
Monitoring of cell growth in scale-up 458
Monitoring of suspension cultures 458
Monitoring of monolayer cultures 458

38 Cell Viability and Cytotoxicity 459
Limitations of *in vitro* cytotoxicity studies 459
Assays for cell viability and cytotoxicity 460
Cytotoxicity and viability assays 460
Survival assays 461
Metabolic assays 462
Transformation assays 462
Inflammation assays 462

39 Cell Transformation and Cell Cloning 463
Transformation of cells 463
Genetic instability 463
Immortalization 463
Aberrant growth control 464
Tumorigenicity 465
Cell cloning 465
Dilution cloning 465
Stimulation of plating efficiency 466
Suspension cloning 467
Isolation of clones 467

40 Organ and Histotypic Cultures, and Tissue Engineering 468
Organotypic models 468
Organ cultures 468
Techniques for organ culture 469
Histotypic cultures 470
Three dimensional cultures 471
Organotypic cultures 472
Tissue engineering 472
Cell source and culture 473
Culture of cells 473
Cell orientation 474
Cell support materials 474
Design and engineering of tissues 475
Tissue modelling 477
Embryonic stem cell engineering 477
ES cell cultures to produce differentiated cells 477
Human embryonic stem cell research 477

41 Transgenic Animals 480
Importance of transgenic animals — general 480
Transgenic mice 481
Retroviral vector method 481
Microinjection method 481
Embryonic stem cell method 482
Gene knockout 484
Yeast artificial chromosome in transgenesis 485
Transgenic mice and their applications 485
The human mouse 485
The Alzheimer's mouse 485
The oncomouse 486
The prostate mouse 486
The knockout mouse 486
Transgenic mice for human diseases 487
Transgenesis in large animals 487
Transgenic cattle 488
Transgenic sheep and goats 488
Transgenic pigs 488
Transgenic chickens 489
Transgenic fish 490
Position effects 490
Animal bioreactors 490
Dolly — the transgenic clone 490
Cloning of pet animals 491
Transgenic animals produced by cloning fetal cells 491
Transgenic animals in xenotransplantation 492
Pigs in xenotransplantation? 492
Transgenic organisms to interrupt disease cycles 493
Transgenic snails 493
Transgenic mosquitoes 493
Transgenic tsetse flies 493
Transgenic bollworms 493
Transgenic medflies 493

SECTION VII

Plant/Agricultural Biotechnology

42 Plant Tissue Culture — General 497
Benefits of plant tissue culture 497
Basic structure and growth of a plant 497
Conventional plant breeding and plant tissue culture 498
Terms used in tissue culture 498
Brief history of plant tissue culture 499
Types of culture 499
Basic technique of plant tissue culture 499
Callus culture 500
Cell culture 502
Isolation of single cells 502
Suspension cultures — growth and subculture 502
Types of suspension cultures 503
Synchronization of suspension cultures 504
Measurement of growth of cultures 505
Measurement of viability of cultured cells 505
Culture of isolated single cells (single cell clones) 505
Applications of plant tissue cultures 506
Secondary metabolites in plant culture 507
Applications of secondary metabolites 508
Production of secondary metabolites 509
Selection of cell lines for high yield of secondary metabolites 509
Large scale (mass) cultivation of plant cells 511
Medium composition and effect of nutrients 512
Elicitor-induced production of secondary metabolites 514
Effect of environmental factors 515
Biotransformation using plant cell cultures 516
Secondary metabolite release and analysis 516

43 Plant Tissue Culture Media 517
Major types of media 517
Constituents of media 519
Preparation of media 522
Selection of a suitable medium 522

44 Protoplast Culture and Somatic Hybridization 523
Importance of protoplasts and their cultures 523
Isolation of protoplasts 513
Mechanical method 524
Enzymatic method 524
Culture of protoplasts 526
Culture media 526
Culture methods 527
Regeneration of protoplasts 527
Sub-protoplasts 528
Somatic hybridization 528
Fusion of protoplasts 529
Selection of hybrid cells 531
Identification of hybrid (cells) plants 533
Chromosome number in somatic hybrids 534
Cybrids 535
Applications of somatic hybridization 536
Limitations of somatic hybridization 537

45 Production of Haploid Plants 538
In vivo and *in vitro* approaches 538
Androgenesis 539
Anther culture 539
Pollen (microspore) culture 539
Development of androgenic haploids 540
Factors affecting androgenesis 540
Gynogenesis 542
Identification of haploids 543
Morphological approach 543
Genetic approach 543
Diploidization of haploid plants (production of homozygous plants) 543
Colchicine treatment 543
Endomitosis 544
Applications of haploids 544
Crops developed through anther culture 545
Limitations of haploids 545

46 Somaclonal Variations 546
Basis of somaclonal variations 546

Isolation of somaclonal variants 547
Without *in vitro* selection 547
With *in vitro* selection 547
Factors affecting production of somaclonal variants 549
Applications of somaclonal variations 549
Limitations of somaclonal variations 550
Gametoclonal variations 551

47 Clonal Propagation (Micropropagation) **552**
Technique of micropropagation 552
Multiplication by axillary buds and apical shoots 553
Multiplication by adventitious shoots 556
Factors affecting micropropagation 556
Organogenesis 556
Somatic embryogenesis 557
Applications of micropropagation 559
Disadvantages of micropropagation 559
Production of disease-free plants 560
Methods to eliminate viruses in plants 560
Elimination of pathogens other than viruses 562
Merits and demerits of disease-free plant production 562
Embryo culture 562
Mature embryo culture 562
Embryo rescue 562
Nutritional requirements of embryo culture 563
Applications of embryo culture 564

48 Germplasm Conservation and Cryopreservation **565**
In-situ conservation 565
Ex-situ conservation 565
Cryopreservation 566
Techniques of cryopreservation 567
Cold storage 569
Low-pressure and low-oxygen storage 570
Applications of germplasm storage 571
Limitations of germplasm storage 571

49 Genetic Engineering of Plants-Methodology **572**
Gene transfer methods 573
Vector-mediated gene transfer 573
Agrobacterium-mediated gene transfer 573
Crown gall disease and Ti plasmid 574
Hairy root disease of A. rhizogenes — *R_1 plasmids* 577
Ti plasmid-derived vector systems 577
Plant transformation technique using *Agrobacterium* 580
Virus-mediated gene transfer (plant viruses as vectors) 581
Caulimoviruses as vectors 581
Geminiviruses as vectors 582
RNA plant viruses as vectors 582
Limitations of viral vectors in gene transfer 583
Direct or vectorless DNA transfer 583
Physical gene transfer methods 583
Electroporation 583
Particle bombardment (biolistics) 584
Microinjection 586
Liposome-mediated transformation 586
Silicon carbide fibre-mediated transformation 587
Chemical gene transfer methods 587
Polyethylene glycol-mediated transformation 587
DEAE dextran-mediated transfer 588
Calcium phosphate co-precipitation-mediated transfer 588
DNA imbibition by cells/tissues 588
Marker genes for plant transformation 588
Selectable marker genes 588
Antibiotic resistance genes 588
Antimetabolite marker genes 589
Herbicide resistance genes 589
Production of marker-free transgenic plants 590
Clean gene technology 590
Reporter genes 590
Promoters and terminators 592
Transgene stability, expression and gene silencing 593
Scaffold attachment regions and gene stability 593
Introns and gene expression 593
Gene silencing 593
Chloroplast transformation 594
Chloroplast engineering 594

50 Applications of Plant Transformation/ Transgenic Plants 596

Resistance to biotic stresses 596
Insect (pest) resistance 597
Resistance genes from microorganisms 597
Resistance genes from higher plants 600
Proteinase (protease) inhibitors 601
Lectins 601
Resistance genes from animals 601
Insect resistance through copy nature strategy 601
Virus resistance 602
Fungal and bacterial diseases 604
Pathogenesis-related (PR) proteins 605
Ribosome-inactivating proteins (RIPs) 605
Phytoalexins 606
Nematode resistance 606
Resistance to abiotic stresses 607
Herbicide resistance 607
Glyphosate resistance 607
Phosphinothricin resistance 609
Sulfonylureas and imidazolinones resistance 610
Environmental impact of herbicide-resistant crops 610
Tolerance to water deficit stresses 610
Resistance against ice-nucleating bacteria 611
Improvement of crop yield and quality 612
Green revolution 612
Genetic engineering for extended shelf-life of fruits 612
Biochemical changes during tomato ripening 613
Genetic manipulation of fruit ripening 614
Longer shelf-life of fruits and vegetables 615
Genetic engineering for prevention of discoloration 616
Genetic engineering for flower pigmentation 616
Genetic engineering for male sterility 617
Transgenic plants with improved nutrition 617
Amino acids of seed storage proteins 617
Genetic engineering for improving palatability of foods 618
Golden rice — the provitamin A enriched rice 619
Genetic engineering to increase vitamins and minerals 620
Commercial transgenic crop plants 620
Transgenic plants as bioreactors 621

51 Transgenic Plants as Bioreactors 622

Metabolic engineering of carbohydrates 622
Starch 622
Cyclodextrins 623
Fructans 623
Trehalose 624
Metabolic engineering of lipids 624
Production of shorter chain fatty acids 624
Production of longer chain fatty acids 625
Production of fatty acids with modified degree of saturation 625
Production of rare fatty acids 626
Biodegradable plastics (bioplastics) 626
Genetically engineered plants as protein factories 628
Approaches for protein production in plants 628
Oleosin partition technology 631
Production of industrial enzymes in plants 631
Production of lysosomal enzymes in plants 633
Production of antibodies (plantibodies) in plants 634
Production of vaccines in plants 636
Production therapeutic proteins in plants 637

52 Growth-Promoting Bacteria in Plants 639

Biological nitrogen fixation 639
Nitrogen cycle 639
Nitrogen fixing bacteria 639
Mechanism of nitrogen fixation 641
Genetic manipulations for nitrogen fixation 641
Genetic engineering of nitrogenase gene 642
Genetic engineering of hydrogenase gene 644
Genetic engineering of nodulation genes 644
Biocontrol of phytopathogens 644
Siderophores 644
Antibiotics 645
Enzymes 645

Biofertilizers 645
Symbiotic nitrogen fixers 646
Asymbiotic nitrogen fixers 646
Phosphate solubilizing bacteria 646
Organic fertilizers 647
Benefits of biofertilizers 647
Limitations of biofertilizers 647

53 Molecular Marker-Aided Plant Breeding 648
Molecular markers 648
Markers based on DNA hybridization 649
Markers based on PCR amplification 650
Molecular marker assisted selection 652
Molecular breeding 652
Quantitative trait loci 652
Arid and semi-arid plant biotechnology 653
Greenhouse and greenhome technology 653

SECTION VIII

Environmental Biotechnology

54 Environmental Pollution — General 657
Environment-basic concepts 657
Atmosphere 657
Hydrosphere 657
Lithosphere 658
Biosphere 658
Environmental pollution-sources and nature 658
Sources of pollution 658
Nature of pollutants 658
Pollution monitoring/measurement 659
Biotechnological methods for measurement of pollution 659
Criteria for biomonitoring of pollution 659
Bioassays in environmental monitoring 660
Cell biology in environmental monitoring 661
Molecular biology in environmental monitoring 662
Biosensors in environmental monitoring 663
Biotechnological methods for management of pollution 663
Atmospheric CO_2 reduction 664
Photosynthesis to reduce atmospheric CO_2 664
Biological calcification to reduce atmospheric CO_2 664
Sewage treatment by bacteria and algae 664
Eutrophication and phosphorus pollution 665
Biological removal of phosphorus 666
Management of metal pollution 666
Bioscavengers of metals 666
Mechanism of metal scavenging 666
Immobilized cells in the management of pollution 667

55 Air Pollution and Control 669
Sources of air pollutants 669
Classification of air pollutants 670
Effects of air pollution 671
Biomonitoring of air pollution 672
Air pollution control 673
Control devices for particulate pollutants 673
Control devices for gaseous pollutants 675
Control devices for volatile organic compounds (VOCs) 678

56 Water Pollution and Sewage 679
Dissolved oxygen 679
Water pollution 680
Nature of water pollutants 680
Organic pollutants 680
Inorganic pollutants 680
Microbiological pollutants 680
Radioactive pollutants 681
Waste water and sewage 682
Composition of sewage 682
Types of sewage 682
Measurement of water pollution 683
Measurement of organic matter of sewage 683
Detection of pathogenic organisms of sewage 684
Laboratory methods for detection of coliform organisms 684
Techniques to distinguish fecal from non-fecal bacteria 685

57 Sewage/Waste Water Treatment 686
Classification of treatment processes 686

Preliminary treatment 687
Primary treatment 688
Secondary or biological treatment 689
Aerobic suspended-growth treatment processes 690
Activated sludge process 690
Aerated lagoons 691
Sequencing batch reactor 691
Aerobic digestion 692
Aerobic attached-growth treatment processes 692
Trickling filters 692
Roughing filters 693
Rotating biological contactors (RBC) 693
Packed-bed reactors (fluidized bed reactors) 693
Anaerobic suspended-growth treatment processes 694
Anaerobic digestion 695
Anaerobic contact process 696
Anaerobic attached-growth treatment processes 696
Anaerobic filter process 696
Expanded-bed process 696
Pond treatment processes 696
Aerobic ponds 696
Anaerobic ponds 697
Facultative ponds 697
Tertiary treatment 698
Solids removal 699
Biological nitrogen removal 700
Biological phosphorus removal 701
Disinfection 703
Sewage/waste water treatment — a summary 703
Waste water treatment of some industries 703
Waste water treatment for dairies 703
Waste water treatment for distillery 704
Waste water treatment for tannery 704
Waste water treatment for sugar industry 705
Treatment of antibiotics in waste water 705
Water recycling 705
Small sewage/waste water treatment systems 706
Oxidation ditch 706
Septic tanks 706
Imhoff tanks 707

58 Sludge and Solid Wastes-Treatment and Disposal 708
Sources and characteristics of sludge 708
Preliminary operations 708
Sludge thickening (concentration) 709
Sludge stabilization 710
Composting 711
Mechanism of composting 712
Methods of composting 712
Vermicomposting 714
Conditioning of sludge 714
Disinfection of sludge 714
Dewatering 714
Heat drying 715
Thermal reduction of sludge 715
Ultimate disposal of sludge 715
Landfilling 715
Lagooning 716
Septage and septage disposal 716
Treatment and disposal of solid wastes 716
Separation and composting plants 717

59 Biodegradation and Bioremediation 718
Pseudomonas — the predominant microorganism for bioremediation 718
Recalcitrant xenobiotics 720
Types of bioremediation 720
Metabolic effects of microorganisms on xenobiotics 721
Types of reactions in bioremediation 721
Biodegradation of hydrocarbons 722
Biodegradation of pesticides and herbicides 722
Biodegradation of some other important compounds 724
Genetic engineering for more efficient bioremediation 724
Genetically engineered microorganisms (GEMs) in bioremediation 726
Bioremediation of contaminated soils and waste lands 727
Techniques of soil bioremediation 727
Bioremediation of ground water 728

60 Global Environmental Problems 730
Green house effect and global warming 730
Green house gases 730
Measures to control green house effect 732
The problem of ozone 732
Depletion of ozone 733
Ozone hole 733
Effects of ozone depletion 733
Measures to control ozone depletion 734
Acid rain 734
Development of acid rain 734
Effects of acid rain 734
Measures to control acid rain 735
Environmental sustainability and biotechnology 735

SECTION IX

Biotechnology and Society

61 Biotechnology — Society, Risks, Ethics, Patenting 739
Benefits of biotechnology 739
ELSI of biotechnology 739
Recombinant therapeutic products for human healthcare 740
Genetic modifications and food consumption 740
Recombinant foods and religious beliefs 742
Are GM foods safe? 742
Release of genetically engineered organisms 742
Applications of human genetic rDNA research 743
Human embryonic stem cell research 744
Cloning humans? 744
Patenting biotechnology inventions 744
Biotechnology products 744
Biotechnology processes 744
What is a patent? 744
Intellectual property rights 744
The process of patenting 745
Plant breeders' rights 746
Patenting and biotechnology research 746
Biotechnology and the developing countries 746

SECTION X

Basics to Learn Biotechnology

62 Cell — The Basic Unit Life 749
Prokaryotic and eukaryotic cells 749
Eukaryotic cell 750
Animal cell 750
Plant cell 752

63 Microorganisms 754
Bacteria 754
Characteristics of bacteria 754
Importance of bacteria 755
Control of pathogenic bacteria 756
Fungi 756
Characteristics of fungi 756
Importance of fungi 756
Control of fungi 757
Viruses 757
Characteristics of viruses 757
Biological status of viruses 757
Importance of viruses 757

64 Bioorganic and Biophysical Chemistry, and Biochemistry Tools 758
Introduction to bioorganic chemistry 758
Most common organic compounds found in living system 758
Common functional groups in biochemistry 758
Common ring structures in biochemistry 758
Isomerism 759
Overview of biophysical chemistry 762
Acids and bases 762
Buffers 762
Solutions 762
Colloidal state 763
Diffusion 763
Osmosis 764
Viscosity 764
Surface tension 765
Isotopes 765

Tools of Biochemistry 766
Chromatography 766
Electrophoresis 767
Photometry — colorimer and spectrophotometer 767
Ultracentrifugation 768
Radioimmunoassay 769
Enzyme-linked immunosorbant assay 770

65 Biomolecules 771
Chemical molecules of life 771
Carbohydrates 772
Classification of carbohydrates 772
Monosaccharides 773
Glycosides 773
Derivatives of monosaccharides 773
Disaccharides 774
Polysaccharides 774
Homopolysaccharides 774
Heteropolysaccharides 775
Lipids 775
Classification of lipids 775
Fatty acids 776
Triacylglycerols 776
Phospholipids 776
Lipoproteins 777
Steroids 777
Proteins and amino acids 777
Amino acids 778
Structure of proteins 781
Primary structure of protein 781
Secondary structure of protein 782
Tertiary structure of protein 783
Quaternary structure of protein 783
Classification of proteins 783
Isoprenoids and pigments 784
Isoprenoids 785
Pigments 785

66 Enzymology 786
Nomenclature and classification 786
Chemical nature of enzymes 787
Factors affecting enzyme activity 787
Active site 789
Enzyme inhibition 790
Enzyme specificity 791
Coenzymes 792
Mechanism of enzyme action 793
Diagnostic importance of enzymes 795
Isoenzymes 795

67 Metabolisms 796
Introduction to metabolism 796
Metabolism of carbohydrates 798
Glycolysis 798
Conversion of acetyl CoA to pyruvate 798
Citric acid cycle 799
Gluconeogenesis 800
Glycogen metabolism 800
Glycogenesis 800
Glycogenolysis 800
Hexose monophosphate shunt 800
Glyoxylate cycle 803
Photosynthesis 803
Metabolism of lipids 804
Fatty acid oxidation 804
Biosynthesis of fatty acids 806
Metabolism of cholesterol 806
Cholesterol biosynthesis 806
Degradation of cholesterol 806
Metabolism of Amino acids 806
Metabolism of amino acids — general aspects 808
Transamination 809
Deamination 809
Urea cycle 809
Metabolism of individual amino acids 809
Glycine 809
Phenylalanine and tyrosine 809
Tryptophan 811
Sulfur amino acids 811
Glutamate and glutamine 811
Fate of carbon skeleton of amino acids 811
Integration of metabolism 812
Energy demand and supply 812
Integration of major metabolic pathways and energy metabolism 813
Regulation of metabolic pathways 815

68 Immunology 816
Innate immunity 816
The immune system 816
Organization of immune system 817
Cells of the immune system 817
Immunoglobulins 817
Structure of immunoglobulins 817
Classes of immunoglobulins 818
Synthesis of immunoglobulins 819
Major histocompatibility complex 819
The immune response 820
Cytokines 820
Immunity in health and disease 821
Autoimmune diseases 821
Organ transplantation 821
Cancers 821
AIDS 821

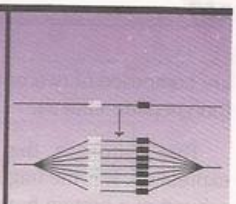
69 Genetics 822
Brief history and development of genetics 822
Basic principles of heredity in humans 823
Patterns of inheritance 823
Genetic diseases in humans 824
Eugenics 825

70 Bioinformatics 827
Broad coverage of bioinformatics 827
Components of bioinformatics 828
Bioinformatics and the internet 828
Biological databases 828
Applications of bioinformatics 830

Appendices 831
Glossary 833
Abbreviations 852

Index 855

Regulation of Gene Expression



DNA, the chemical vehicle of heredity, is composed of ***functional units***, namely ***genes***. The term ***genome*** refers to the total genetic information contained in a cell. The bacterium *Escherichia coli* contains about 4,400 genes present on a single chromosome. The ***genome of humans*** is more complex, with 23 pairs of (diploid) chromosomes containing 6 billion (6×10^9) base pairs of DNA, with an estimated ***30,000–40,000 genes***. At any given time, only a fraction of the genome is expressed.

The living cells possess a remarkable property to adapt to changes in the environment by regulating the gene expression. For instance, insulin is synthesized by specialized cells of pancreas and not by cells of other organs (say kidney, liver), although the nuclei of all the cells of the body contain the insulin genes. Molecular regulatory mechanisms facilitate the expression of insulin gene in pancreas, while preventing its expression in other cells.

GENE REGULATION—GENERAL

The ***regulation of the expression of genes*** is absolutely ***essential for the growth, development, differentiation*** and the very existence of an organism. There are two types of gene regulation-positive and negative.

1. **Positive regulation :** The gene regulation is said to be positive when its expression is increased by a regulatory element (positive regulator).

2. **Negative regulation :** A decrease in the gene expression due to the presence of a regulatory element (negative regulator) is referred to as negative regulation.

It may be noted here that double negative effect on gene regulation results in a positive phenomenon.

Constitutive and inducible genes

The genes are generally considered under two categories.

1. **Constitutive genes :** The products (proteins) of these genes are required all the time in a cell. Therefore, the constitutive genes (or housekeeping genes) are expressed at more or less constant rate in almost all the cells and, further, they are not subjected to regulation e.g. the enzymes of citric acid cycle.

2. **Inducible genes :** The concentration of the proteins synthesized by inducible genes is regulated by various molecular signals. An inducer increases the expression of these genes while a repressor decreases, e.g. tryptophan pyrrolase of liver is induced by tryptophan.

One cistron-one subunit concept

The chemical product of a gene expression is a protein which may be an enzyme. It was originally believed that each gene codes for a specific enzyme, leading to the popular concept, one gene-one enzyme. This however, is not necessarily valid due to the fact that several enzymes (or proteins)

are composed of two or more nonidentical subunits (polypeptide chains).

The cistron is the smallest unit of genetic expression. It is the fragment of DNA coding for the subunit of a protein molecule. The original concept of ***one gene-one enzyme is replaced by one cistron-one subunit.***

Models for the study of gene expression

Elucidation of the regulation of gene expression in prokaryotes has largely helped to understand the principles of the flow of information from genes to mRNA to synthesize specific proteins. Some important features of prokaryotic gene expression are described first. This is followed by a brief account of eukaryotic gene expression.

THE OPERON CONCEPT

The operon is the coordinated unit of genetic expression in bacteria. The concept of operon was introduced by Jacob and Monod in 1961 (Nobel Prize 1965), based on their observations on the regulation of lactose metabolism in *E. coli*. This is popularly known as ***lac operon***.

LACTOSE (LAC) OPERON

Structure of lac operon

The lac operon (***Fig. 5.1***) consists of a regulatory gene (I; I for inhibition), operator gene (O) and three structural genes (Z, Y, A). Besides these genes, there is a promoter site (P), next to the operator gene, where the enzyme RNA polymerase binds. The structural genes Z, Y and A respectively, code for the enzymes β-galactosidase, galactoside permease and galactoside acetylase. β-Galactosidase hydrolyses lactose (β-galactoside) to galactose and glucose while permease is responsible for the transport of lactose into the cell. The function of acetylase (coded by A gene) remains a mystery.

The structural genes Z, Y and A transcribe into a single large mRNA with 3 independent translation units for the synthesis of 3 distinct enzymes. An mRNA coding for more than one protein is known as ***polycistronic mRNA***. Prokaryotic organisms contain a large number of polycistronic mRNAs.

Repression of lac operon

The regulatory gene (I) is constitutive. It is expressed at a constant rate leading to the synthesis of lac repressor. Lac repressor is a tetrameric (4 subunits) regulatory protein (total mol. wt. 150,000) which specifically binds to the operator gene (O). This prevents the binding of the enzyme RNA polymerase to the promoter site (P), thereby blocking the transcription of structural genes (Z, Y and A). This is what happens in the absence of lactose in *E. coli*. The ***repressor molecule acts as a negative regulator of gene expression***.

Derepression of lac operon

In the presence of lactose (inducer) in the medium, a small amount of it can enter the *E. coli* cells. The repressor molecules have a high affinity for lactose. The lactose molecules bind and induce a conformational change in the repressor. The result is that the repressor gets inactivated and, therefore, cannot bind to the operator gene (O). The RNA polymerase attaches to the DNA at the promoter site and transcription proceeds, leading to the formation of polycistronic mRNA (for genes Z, Y and A) and, finally, the 3 enzymes. Thus, lactose induces the synthesis of the three enzymes β-galactosidase, galactoside permease and galactoside acetylase. ***Lactose acts by inactivating the repressor molecules***, hence this process is known as derepression of lac operon.

Gratuitous inducers : There are certain structural analogs of lactose which can induce the lac operon but are not the substrates for the enzyme β-galactosidase. Such substances are known as gratuitous inducers. Isopropylthiogalactoside (IPTG) is a gratuitous inducer, extensively used for the study of lac operon.

The catabolite gene activator protein : The cells of *E. coli* utilize glucose in preference to lactose; when both of them are present in the medium. After the depletion of glucose in the medium, utilization of lactose starts. This indicates that glucose somehow interferes with the induction of lac operon. This is explained as follows.

The attachment of RNA polymerase to the promoter site requires the presence of a ***catabolite gene activator protein (CAP)*** bound to cyclic AMP (***Fig. 5.2***). The presence of glucose lowers the intracellular concentration of cAMP by inactivating

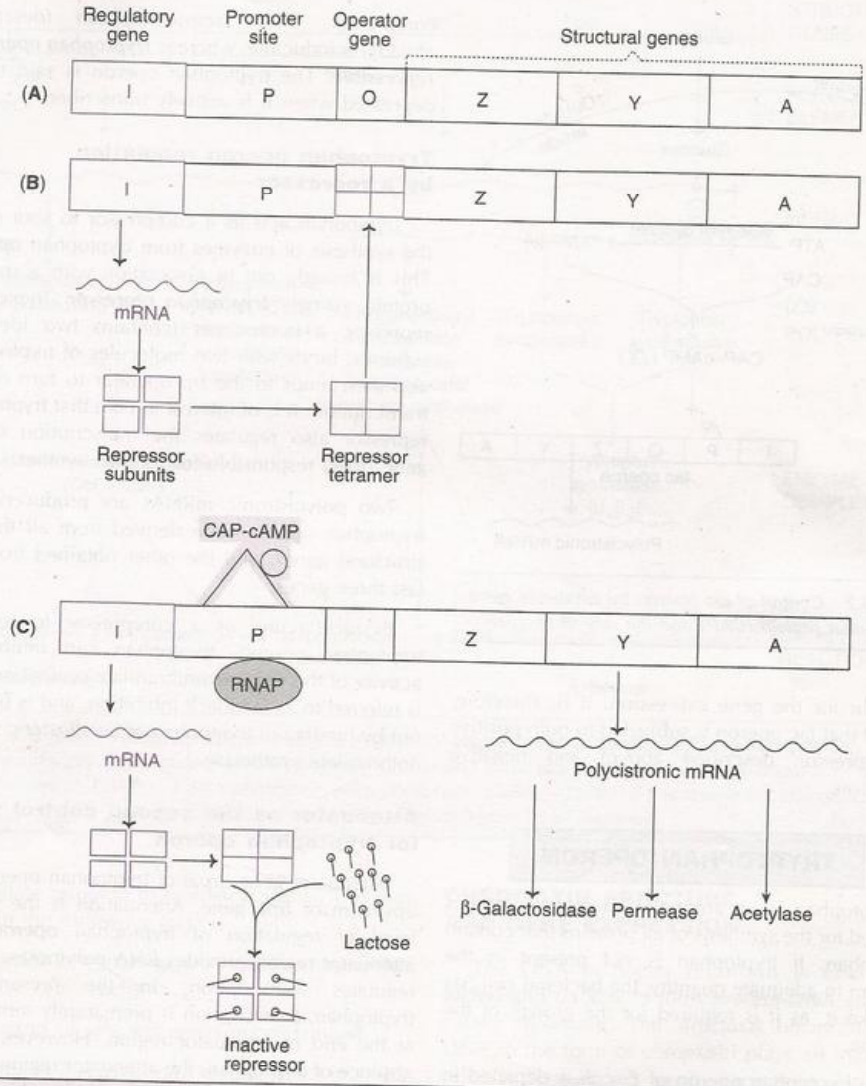


Fig. 5.1 : Model of lactose operon in E.coli *(A) Structure of lac operon (B) Repression of lac operon (C) Derepression of lac operon. (CAP—cAMP—catabolite gene activator protein bound to cAMP; RNAP—RNA polymerase).*

the enzyme adenylyl cyclase responsible for the synthesis of cAMP. Due to the diminished levels of cAMP, the formation of CAP-cAMP is low. Therefore, the binding of RNA polymerase to DNA (due to the absence of CAP-cAMP) and the transcription is almost negligible in the presence of glucose. Thus, glucose interferes with the expression of lac operon by depleting cAMP levels. Addition of exogenous cAMP is found to initiate the transcription of many inducible operons, including lac operon.

It is now clear that the presence of CAP-cAMP is essential for the transcription of structural genes of lac operon. Thus, CAP-cAMP acts as a positive

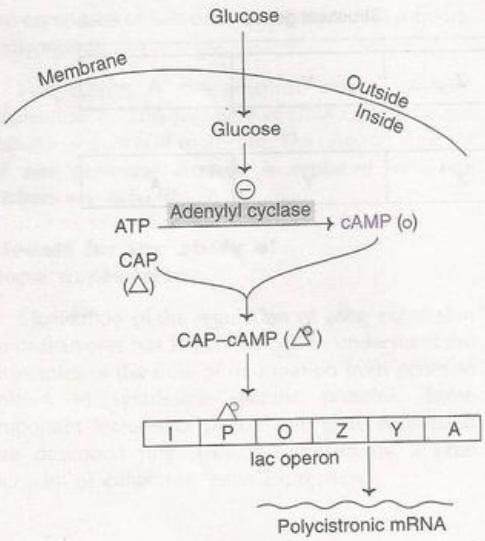


Fig. 5.2 : Control of lac operon by catabolite gene activator protein (CAP) and the role of glucose.

regulator for the gene expression. It is, therefore, evident that lac operon is subjected to both positive (by repressor, described above) and negative regulation.

TRYPTOPHAN OPERON

Tryptophan is an aromatic amino acid, and is required for the synthesis of all proteins that contain tryptophan. If tryptophan is not present in the medium in adequate quantity, the bacterial cell has to make it, as it is required for the growth of the bacteria.

The tryptophan operon of *E. coli* is depicted in ***Fig. 5.3***. This operon contains five structural genes (*trpE, trpD, trpC, trpB, trpA*), and the regulatory elements—primary promoter (*trpP*), operator (*trpO*), attenuator (*trpa*), secondary internal promoter ($TrpP_2$), and terminator (*trpt*).

The five structural genes of tryptophan operon code for three enzymes (two enzymes contain two different subunits) required for the synthesis of tryptophan from chlorismate.

The tryptophan repressor is always turned on, unless it is repressed by a specific molecule called corepressor. Thus lactose operon (described already) is inducible, whereas ***tryptophan operon is repressible***. The tryptophan operon is said to be depressed when it is actively transcribed.

Tryptophan operon regulation by a repressor

Tryptophan acts as a corepressor to shut down the synthesis of enzymes from tryptophan operon. This is brought out in association with a specific protein, namely ***tryptophan repressor***. Tryptophan repressor, a homodimer (contains two identical subunits) binds with two molecules of tryptophan, and then binds to the *trp* operator to turn off the transcription. It is of interest to note that tryptophan repressor also regulates the transcription of the gene (*trpR*) responsible for its own synthesis.

Two polycistronic mRNAs are produced from tryptophan operon—one derived from all the five structural genes, and the other obtained from the last three genes.

Besides acting as a corepressor to regulate tryptophan operon, tryptophan can inhibit the activity of the enzyme anthranilate synthetase. This is referred to as feedback inhibition, and is brought out by binding of tryptophan at an allosteric site on anthranilate synthetase.

Attenuator as the second control site for tryptophan operon

Attenuator gene (*trpa*) of tryptophan operon lies upstream of *trpE* gene. Attenuation is the second level of regulation of tryptophan operon. The attenuator region provides RNA polymerase which regulates transcription. In the presence of tryptophan, transcription is prematurely terminated at the end of attenuator region. However, in the absence of tryptophan, the attenuator region has no effect on transcription. Therefore, the polycistronic mRNA of the five structural genes can be synthesized.

GENE EXPRESSION IN EUKARYOTES

Each cell of the higher organism contains the entire genome. As in prokaryotes, gene expression in eukaryotes is regulated to provide the

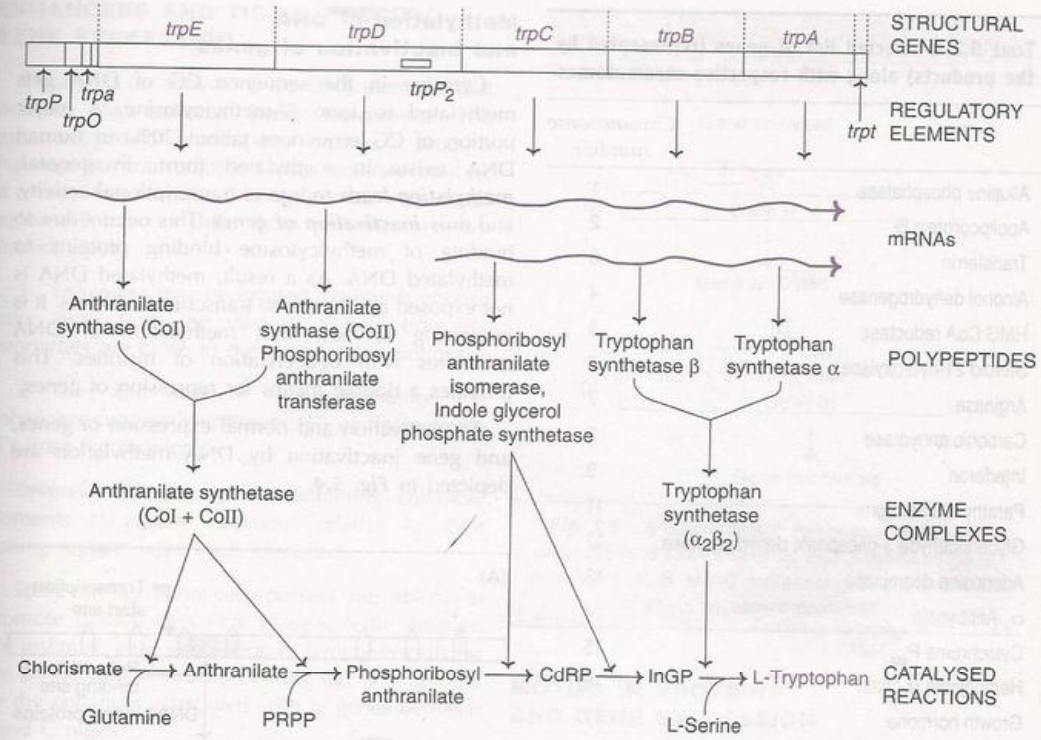


Fig. 5.3 : *Tryptophan operon in* E.coli *[regulatory elements are promoter (trpP), operator (trpO), attenuator (trpa), secondary internal promoter ($trpP_2$) and terminator (trpt); CoI, CoII — Component I and component II; PRPP — 5-Phosphoribosyl 1-pyrophosphate; CdRP — Carboxyl-phenylamino 1 deoxyribulose 5-phosphate; InGP — Indole 3-glycerol phosphate].*

appropriate response to biological needs. This may occur in the following ways

- Expression of certain genes (housekeeping genes) in most of the cells.
- Activation of selected genes upon demand.
- Permanent inactivation of several genes in all but a few types.

In case of prokaryotic cells, most of the DNA is organized into genes which can be transcribed. In contrast, in mammals, ***very little of the total DNA is organized into genes*** and their associated regulatory sequences. The function of the bulk of the extra DNA is not known.

Eukaryotic gene expression and its regulation are highly complex. Some of the important aspects are briefly described.

CHROMATIN SRUCTURE AND GENE EXPRESSION

The DNA in higher organisms is extensively folded and packed to form ***protein-DNA complex*** called chromatin. The structural organization of DNA in the form of chromatin plays an important role in eukaryotic gene expression. In fact, chromatin structure provides an additional level of control of gene expression.

A selected list of genes (represented by the products) along with the respective chromosomes on which they are located is given in ***Table 5.1***.

In general, the genes that are transcribed within a particular cell are less condensed and more open in structure. This is in contrast to genes that are not transcribed which form highly condensed chromatin.

TABLE 5.1 A selected list of genes (represented by the products) along with respective chromosomes

Genes	*Chromosome number*
Alkaline phosphatase	1
Apolipoprotein B	2
Transferrin	3
Alcohol dehydrogenase	4
HMG CoA reductase	5
Steroid 21-hydroxylase	6
Arginase	7
Carbonic anhydrase	8
Interferon	9
Parathyroid hormone	11
Glyceraldehyde 3-phosphate dehydrogenase	12
Adenosine deaminase	13
α_1-Antitrypsin	14
Cytochrome P_{450}	15
Hemoglobin α-chain	16
Growth hormone	17
Prealbumin	18
Creatine phosphokinase (M chain)	19
Adenosine deaminase	20
Superoxide dismutase	21
Immunoglobulin (λ chain)	22
Glucose 6-phosphate dehydrogenase	X
Steroid sulfatase	Y

Histone acetylation and deacetylation

Eukaryotic DNA segments are wrapped around histone proteins to form nucleosome. Acetylation or deacetylation of histones is an ***important factor in determining the gene expression***. In general, acetylation of histones leads to activation of gene expression while deacetylation reverses the effect.

Acetylation predominantly occurs on the lysine residues in the amino terminal ends of histones. This modification in histones reduces the positive charges of terminal ends (tails), and decreases their binding affinity to negatively charged DNA. Consequently, nucleosome structure is disrupted to allow transcription.

Methylation of DNA and inactivation of genes

Cytosine in the sequence CG of DNA gets methylated to form 5'-methylcytosine. A major portion of CG sequences (about 20%) in human DNA exists in methylated form. In general, ***methylation leads to*** loss of transcriptional activity, and thus ***inactivation of genes***. This occurs due to binding of methylcytosine binding proteins to methylated DNA. As a result, methylated DNA is not exposed and bound to transcription factors. It is interesting to note that methylation of DNA correlates with deacetylation of histones. This provides a double means for repression of genes.

The activation and normal expression of genes, and gene inactivation by DNA methylation are depicted in ***Fig. 5.4***.

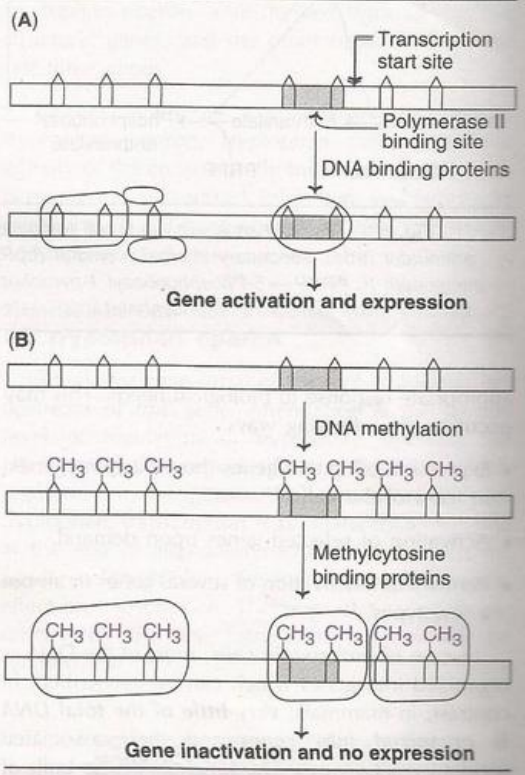


Fig. 5.4 : *Methylation of DNA and inactivation of genes (**A**) Gene activation in the absence of DNA methylation (**B**) Gene inactivation due to methylation (⬡ represent CG sequences).*

ENHANCERS AND TISSUE-SPECIFIC GENE EXPRESSION

Enhancers (or activators) are DNA elements that facilitate or ***enhance gene expression***. The enhancers provide binding sites for specific proteins that regulate transcription. They facilitate binding of the transcription complex to promoter regions. Enhancers differ from promoters in two distinct ways

1. Enhancers may be located thousands of base pairs away from the start of transcription site (promoters are close to the site of transcription).

2. They can work in either orientation i.e. enhancers can work upstream (5′) or downstream (3′) from the promoter.

Several eukaryotic genes containing enhancer elements at various locations relative to their coding regions have been identified.

Some of the enhancers possess the ability to promote transcription in a tissue-specific manner. For instance, gene expression in lymphoid cells for the production immunoglobulins (Ig) is promoted by the enhancer associated with Ig genes between J and C regions.

Transgenic animals are frequently used for the study of tissue-specific expression. The available evidence from various studies indicates that the ***tissue-specific gene expression is largely mediated through the involvement of enhancers.***

COMBINATION OF DNA ELEMENTS AND PROTEINS IN GENE EXPRESSION

Gene expression in mammals is a complicated process with several environmental stimuli on a single gene. The ultimate response of the gene which may be positive or negative is brought out by the association of DNA elements and proteins.

In the illustration given in the ***Fig. 5.5***, gene I is activated by a combination of activators 1, 2 and 3. Gene II is more effectively activated by the combined action of 1, 3 and 4. Activator 4 is not in direct contact with DNA, but it forms a bridge between activators 1 and 3, and activates gene II. As regards gene III, it gets inactivated by a combination of 1, 5 and 3. In this case, protein 5 interferes with the binding of protein 2 with the DNA and inactivates the gene.

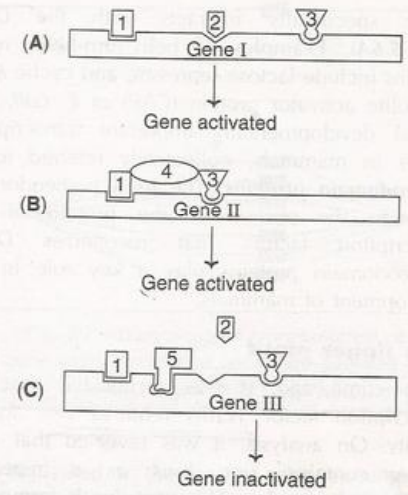


Fig. 5.5 : A diagrammatic representation of the association of DNA elements and proteins in gene regulation. **A**, **B** *and* **C** *represent genes I, II and III (1...5 represent proteins).*

MOTIFS IN PROTEINS AND GENE EXPRESSION

A motif literally means a dominant element. Certain motifs in proteins mediate the binding of regulatory proteins (transcription factors) to DNA. The specific control of transcription occurs by the binding of regulatory proteins with high affinity to the correct regions of DNA.

A great majority of specific ***protein-DNA interactions*** are brought out by four unique motifs.

- Helix-turn-helix (HTH)
- Zinc finger
- Leucine zipper
- Helix-loop-helix (HLH).

The above listed ***amino acid motifs*** bind with high affinity to the specific site and low affinity to other parts of DNA. The motif-DNA interactions are maintained by hydrogen bonds and van der Waals forces.

Helix-turn-helix motif

The helix-turn-helix (***HTS***) motif is about 20 amino acids which represents a small part of a large protein. HTS is the domain part of the protein

which specifically interacts with the DNA (***Fig. 5.6A***). Examples of helix-turn-helix motif proteins include lactose repressor, and cyclic AMP catabolite activator protein (CAP) of *E. coli*, and several developmentally important transcription factors in mammals, collectively referred to as ***homeodomain proteins***. The term homeodomain refers to the portion of the protein of the transcription factors that recognizes DNA. Homeodomain proteins play a key role in the development of mammals.

Zinc finger motif

Sometime ago, it was recognized that the transcription factor TFIIIA requires zinc for its activity. On analysis, it was revealed that each TFIIIA contains zinc ions as a repeating coordinated complex. This complex is formed by the closely spaced amino acids cysteine and cysteine, followed by a histidine—histidine pair. In some instances, His-His is replaced by a second Cys-Cys pair (***Fig. 5.6B***).

The zinc fingers bind to the major groove of DNA, and lie on the face of the DNA. This binding makes a contact with 5 bp of DNA. The ***steroid hormone receptor transcription factors*** use zinc finger motifs to bind to DNA.

The occurrence of a mutation resulting in a single amino acid change of zinc finger may lead to resistance to the action of certain hormones on gene expression. A mutated zinc finger resistant to the action of calcitriol (active form of vitamin D) has been identified. This may ultimately result in rickets (vitamin D deficiency).

Leucine zipper motif

The ***b***asic regions of leucine ***zip***per (bZIP) proteins are rich is the amino acid leucine. There occurs a periodic repeat of leucine residues at every seventh position. This type of repeat structure allows two identical monomers or heterodimers to ***zip together*** and form a dimeric complex. This protein-protein complex associates and interacts with DNA (***Fig. 5.6C***). Good examples of leucine zipper proteins are the enhancer binding proteins (EBP)—fos and jun.

Helix-loop-helix motif

Two amphipathic (literally means a feeling of closeness) α-helical segments of proteins can form helix-loop-helix motif and bind to DNA. The

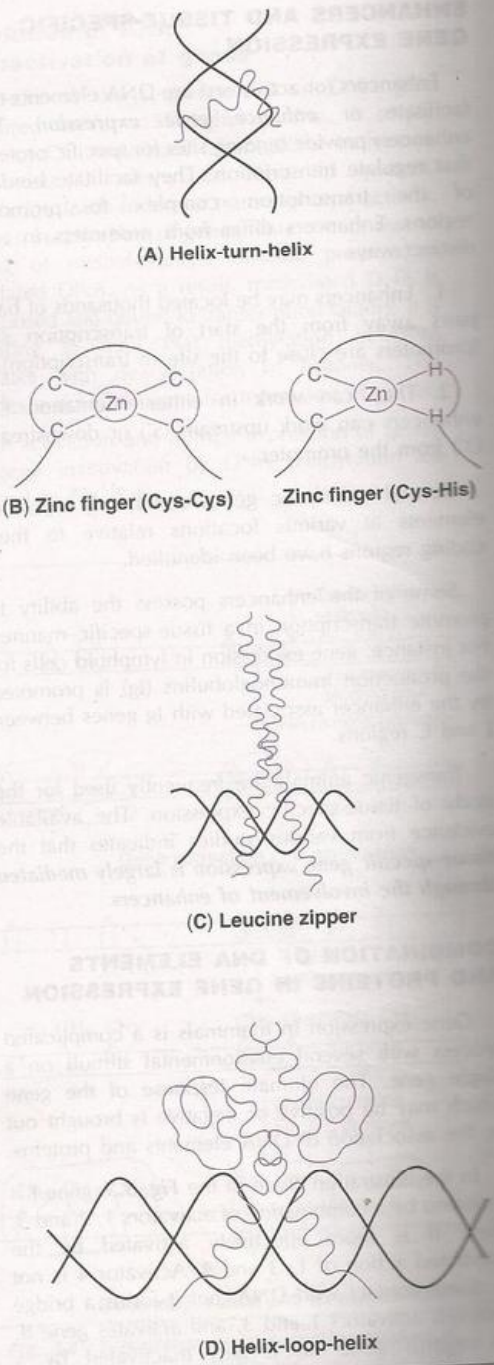


Fig. 5.6 : Diagrammatic representation of common motifs in proteins interacting with DNA

dimeric form of the protein actually binds to DNA (***Fig. 5.6D***).

GENE REGULATION IN EUKARYOTES

The important features of eukaryotic gene expression along with the regulatory aspects are described in the preceeding pages. Besides transcription, eukaryotic cells also employ variety of other mechanisms to regulate gene expression. The most important ones are listed below, and briefly described next.

1. Gene amplification
2. Gene rearrangement
3. Processing of RNA
4. Alternate mRNA splicing
5. Transport of mRNA from nucleus to cytoplasm
6. Degradation of mRNA.

Gene amplification

In this mechanism, the expression of a gene is increased several fold. This is commonly observed during the developmental stages of eukaryotic organisms. For instance, in fruit fly (*Drosophila*), the amplification of genes coding for egg shell proteins is observed during the course of oogenesis. The amplification of the gene (DNA) can be observed under electron microscope (***Fig. 5.7***).

The occurrence of gene amplification has also been reported in humans. Methotrexate is an anticancer drug which inhibits the enzyme dihydrofolate reductase. The malignant cells develop drug resistance to long term administration of methotrexate by amplifying the genes coding for dihydrofolate reductase.

Gene rearrangment

The body possesses an enormous capacity to synthesize a wide range of antibodies. It is estimated that the human body can produce about 10 billion (10^{10}) antibodies in response to antigen stimulations. The molecular mechanism of this antibody diversity was not understood for long. It is now explained on the basis of gene rearrangement or ***transposition of genes*** or ***somatic recombination of DNA***.

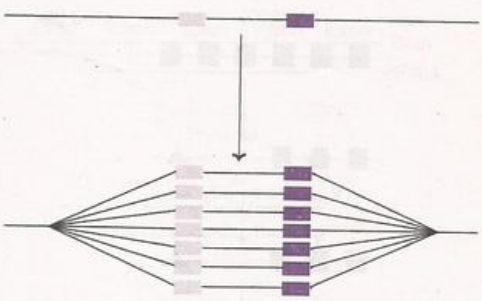


Fig. 5.7 : *Diagrammatic representation of gene amplification (the genes are depicted in colour shade and colour).*

The structure of a typical immunoglobulin molecule consists of two light (L) and two heavy (H) chains. Each one of these chains (L or H) contains an N-terminal variable (V) and C-terminal constant (C) regions. The V regions of immunoglobulins are responsible for the recognition of antigens. The phenomenon of gene rearrangement can be understood from the mechanism of the synthesis of light chains of immunoglobulins (***Fig. 5.8***).

Each light chain can be synthesized by three distinct DNA segments, namely the variable (V_L), the joining (J_L) and the constant (C_L). The mammalian genome contains about 500 V_L segments, 6 J_L segments and 20 C_L segments. During the course of differentiation of B-lymphocytes, one V_L segment (out of the 500) is brought closer to J_L and C_L segments. This occurs on the same chromosome. For the sake of illustration, 100^{th} V_L, 3^{rd} J_L and 10^{th} C_L segments are rearranged in ***Fig. 5.8***. The rearranged DNA (with V_L, J_L and C_L fragments) is then transcribed to produce a single mRNA for the synthesis of a specific light chain of the antibody. By innumerable combinations of V_L, J_L and C_L segments, the body's immune system can generate millions of antigen specific immunoglobulin molecules.

The formation of heavy (H) chains of immunoglobulins also occurs by rearrangement of 4 distinct genes—variable (V_H), diversity (D), joining (J_H) and constant (C_H).

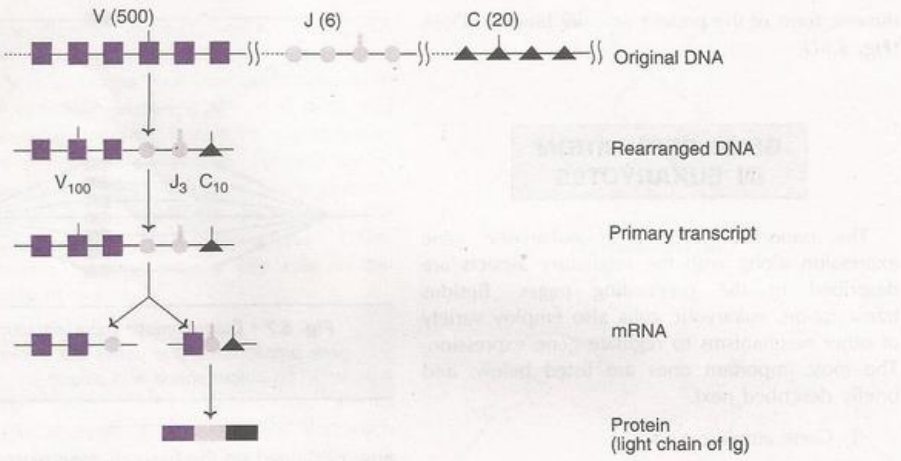


Fig. 5.8 : Diagrammatic representation of gene rearrangement for the synthesis of light chain of immunoglobulin.

Processing of RNA

The RNA synthesized in transcription undergoes modifications resulting in a functional RNA. The changes include intron-exon splicing, polyadenylation etc. (Chapter 4).

Alternate mRNA splicing

Eukaryotic cells are capable of carrying out alternate mRNA processing to control gene expression. Different mRNAs can be produced by alternate splicing which code for different proteins (for more details, Refer Chapter 4).

Degradation of mRNA

The expression of genes is indirectly influenced by the stability of mRNA. Certain hormones regulate the synthesis and degradation of some mRNAs. For instance, estradiol prolongs the half-life of vitellogenin mRNA from a few hours to about 200 hours.

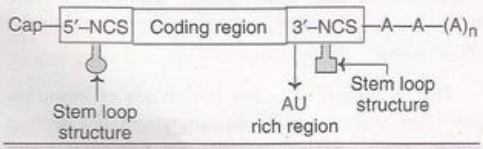


Fig. 5.9 : A diagrammatic representation of a typical *eukaryotic mRNA. (NCS-Non-coding sequences)*

It appears that the ends of mRNA molecules determine the stability of mRNA. A typical eukaryotic mRNA has 5′-non-coding sequences (5′-NCS), a coding region and a 3′-NCS. All the mRNAs are capped at the 5′ end, and most of them have a polyadenylate sequence at the 3′ end (***Fig. 5.9***). The 5′ cap and poly (A) tail protect the mRNA against the attack by exonuclease. Further, stem-loop structures in NCS regions, and AU rich regions in the 3′ NCS also provide stability to mRNA.

METHODS TO STUDY GENE EXPRESSION/REGULATION

Gene expression or gene regulation is usually studied at the transcriptional level, i.e. production of mRNA from the gene. The methods to elucidate gene expression are designed to provide information on one or more of the following

- Sequence of the gene
- Size of the transcript (mRNA)
- Starting and finishing points of genes to produce the transcript.
- Number and position of introns on the genes.
- The activity of the promoter.

Some of the important and general methods employed to study the gene regulation are briefly described.

Southern blot

Southern blot is a novel technique to detect a known fragment of DNA in the DNA preparation of an organism. This technique is particularly useful to detect the presence of a foreign DNA in the genetically modified organisms or to identify the presence and copy number of genes in an organism's genome. The details on this technique are given elsewhere (Chapter 7).

Northern blot

Northern blot specifically detects the size and sequence of the mRNA. The total mRNA is extracted from a cell or tissue suspension, and separated by agarose gel electrophoresis and then detected by hybridization (Refer Chapter 7).

Nuclease SI mapping

Nuclease SI is an enzyme that can specifically degrade single-stranded nucleic acids. Nuclease SI mapping is used *to* ***determine the number of introns*** present in a gene (***Fig. 5.10***).

The mature mRNA is hybridized with its corresponding gene (i.e. genomic DNA). The portion of the intron on the gene which is not transcribed is looped out. This looped-out intron can be specifically digested by nuclease SI, which degrades the single-stranded DNA. The number and presence of introns can be identified by analysing the fragmented DNAs.

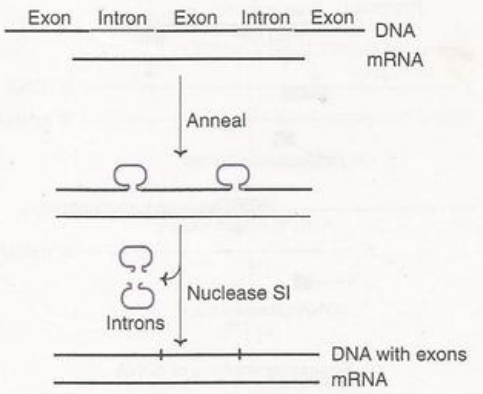


Fig. 5.10 : A diagrammatic representation of mapping of introns by nuclease SI digestion.

Nuclease protection assay

In nuclease protection assay, the test transcript (mRNA) is hybridized with excess quantities of *in vitro* synthesized and radioactively labeled DNA molecules (usually obtained from cloned genes). The annealed hybrids which are labeled, are subjected to digestion by nuclease SI which degrades single-stranded nucleic acids. The nuclease-treated and untreated hybridized molecules are separated by agar gel electrophoresis and identified (***Fig. 5.11***).

Nuclease protection assay is a variant of nuclease SI mapping and provides information as regards the presence of introns, transcriptional termini and the test transcript proper.

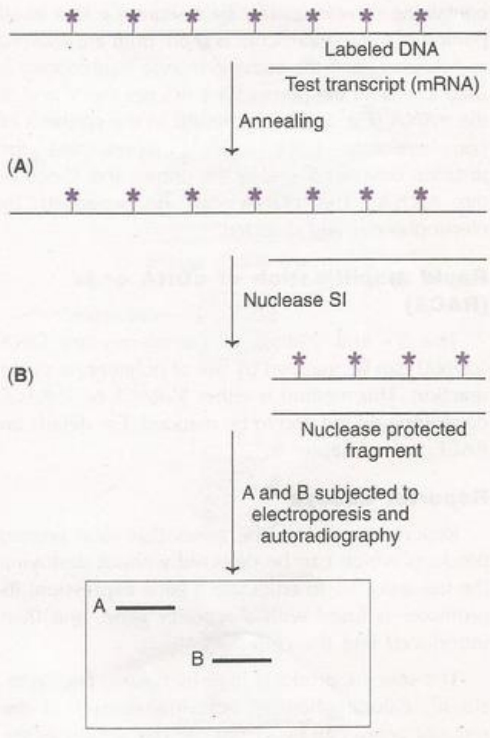


*Fig. 5.11 : A diagrammatic representation of nuclease protection assay (**A**) Hybridized DNA not treated with nuclease SI (**B**) Nuclease SI treated hybridized DNA.*

Primer extension

Primer extension method is a reliable technique to determine the 5′ end of the transcripts. For this purpose, a synthetic 5′-labeled oligonucleotide primer

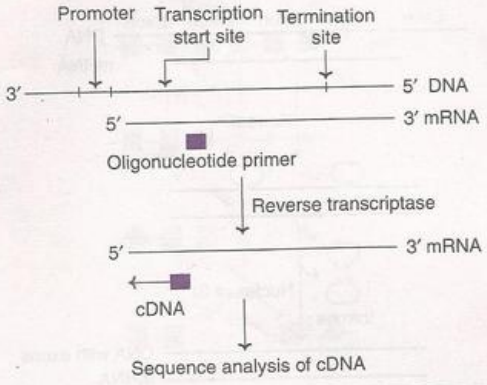


Fig. 5.12 : A diagrammatic representation of primer extension technique to detect the transcription start site.

containing complementary base sequence to a small portion of the test transcript is used. Both are allowed to hybridize, and the enzyme reverse transcriptase is used to extend the primer till it reaches the 5′ end of the mRNA (***Fig. 5.12***). This results in the synthesis of complementary DNA (cDNA) representing the distance between 5′-end of the primer and 5′-end of the mRNA. The cDNA can be separated by electrophoresis and detected.

Rapid amplification of cDNA ends (RACE)

The 5′- and 3′-ends of complementary DNA (cDNA) can be mapped by use of polymerase chain reaction. This method is either 5′-RACE or 3′-RACE depending on the end to be mapped. For details on RACE, refer Chapter 8.

Reporter assays

Reporter genes are the genes that form protein products which can be detected without destoying the tissues/cells. To elucidate a gene expression, its promoter is fused with a reporter gene, and then introduced into the cells.

The specific products (e.g. luciferase, β-galactosidase, chloramphenicol acetyltransferase) of the reporter genes can be identified. The activity of the reporter gene reflects the activity of the promoter gene, and consequently the gene expression.

Reporter assays are very useful for the study of gene expression *in vivo* in the tissues/cells.

GENE ANALYSIS BY T-DNA AND TRANSPOSON TAGGING

Gene tagging broadly involves the ***insertion of a recognizable DNA fragment within a gene*** so that the function of a gene is disrupted, and the gene is identified by virtue of the inserted DNA fragment.

T-DNA (transferred DNA) is the part of tumor-inducing plasmid (Ti plasmid) DNA found in the soil bacterium *Agrobacterium tumefaciens* (For details, Refer Chapter 49) Transposons or transposable elements are mobile genetic elements (DNA pieces) that can move from one place to another in a DNA molecule (Refer Chapter 3). Transposons or T-DNA can be used in gene tagging and gene analysis.

The transposon tagging of a gene is depicted in ***Fig. 5.13***. When a transposon in a plasmid is introduced into a cell, it gets incorporated into DNA, and the gene gets disrupted. Consequently, transposon insertion produces a mutant (A^-). This

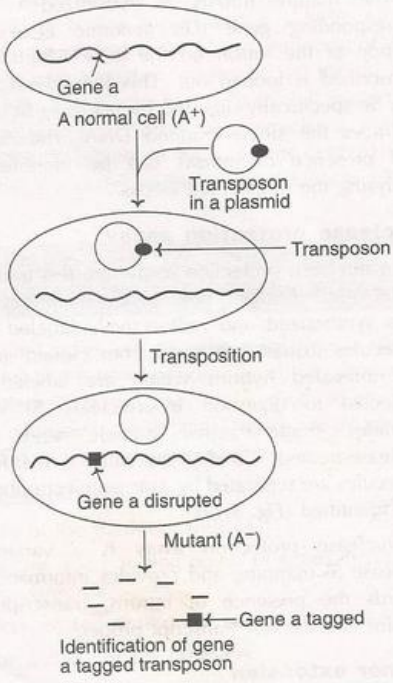


Fig. 5.13 : Transposon tagging of a gene in gene analysis.

mutant can be identified by its phenotype and a gene library. Further, the mutant can be screened for the presence of transposon. ***By identifying the location of insertion of transposon, the location of the specific gene can be identified.***

METHODS TO STUDY PROTEIN–PROTEIN INTERACTIONS

The operation of the genome can be evaluated by the study of proteome. Thus, by studying the functions of proteins, it is possible to understand how the genome operates and how a dysfunctional genome activity can result in disease states such as cancer.

Proteomics broadly involves the methodology for characterizing the protein content of the cell. This can be done by protein electrophoresis, mass spectrometry etc.

Identification of protein-protein interaction is a recent approach to study proteome. The ***protein interaction maps*** can be constructed to understand the relation between the proteome and cellular biochemistry. ***Phage display*** and ***yeast two-hybrid system are commonly used to study protein-protein interactions***.

PHAGE DISPLAY

Phage display is a novel technique to evaluate genome activity with particular reference to identify proteins that interact with one another. It basically involves insertion of a foreign DNA into phage genome, and its expression as fusion product with a phage coat protein (***Fig. 5.14A***). This is followed by screening of test protein by phage display library (***Fig. 5.14B***). The technique is briefly described below.

A special type of cloning vector such as a bacteriophage or filamentous bacteriophage (e.g. M13) are used for phage display. A fragment of DNA coding for the test protein is inserted into the vector DNA (adjacent to phage coat protein gene). After transformation of *E. coli*, this recombinant gene (fused frame of DNA) results in the synthesis of ***hybrid protein***. The new protein is made up of the test protein fused with the phage coat protein. The phage particles produced in the transformed *E. coli* display the test protein in their coats.

The test protein interaction can be identified by using a phage display library. For this purpose, the test protein is immobilized within a well of a

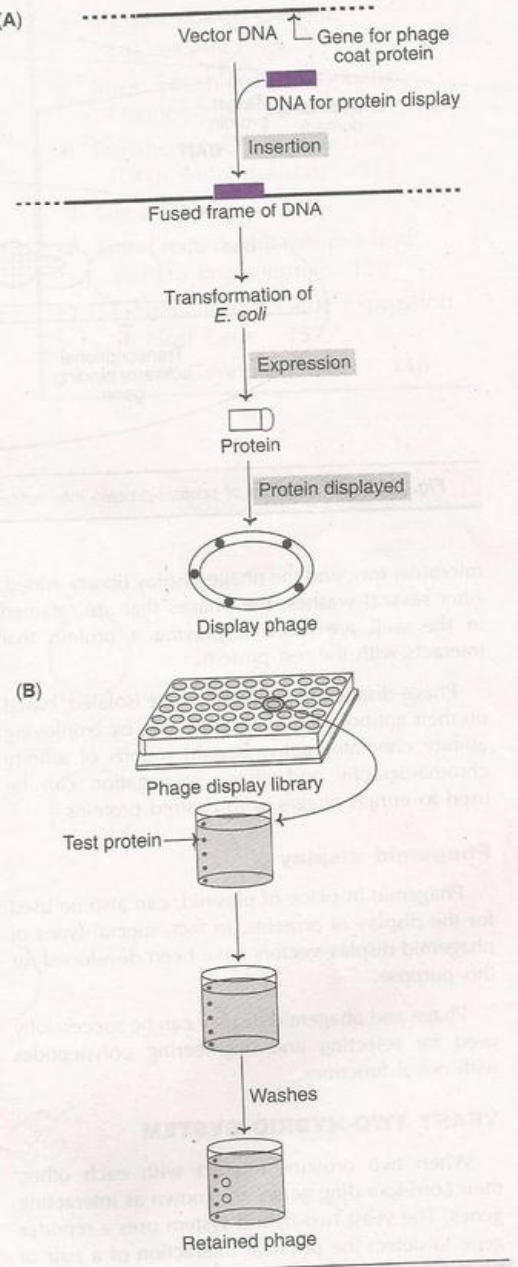


Fig. 5.14 : *Elucidation of protein–protein interaction by phage display (**A**) Production of fusion protein displayed on phage (**B**) Screening of test protein by phage display library.*

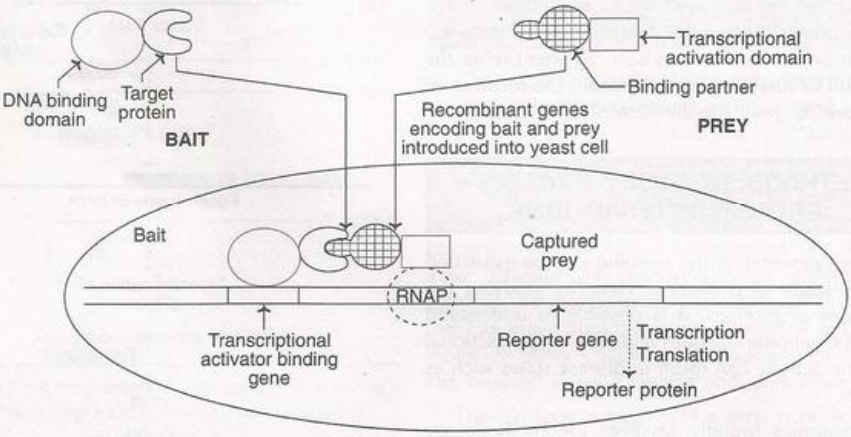


Fig. 5.15 : Elucidation of protein-protein interaction by yeast two-hybrid system (RNAP-RNA polymerase)

microtiter tray, and the phage display library added. After several washes, the phages that are retained in the well are those displaying a protein that interacts with the test protein.

Phage-displaying peptides can be isolated, based on their antibody-binding properties, by employing affinity chomatography. Several rounds of affinity chromatography and phage propagation can be used to enrich phages with desired proteins.

Phagemid display

Phagemid in place of plasmid, can also be used for the display of proteins. In fact, special types of phagemid display vectors have been developed for this purpose.

Phage and phagemid display can be successfully used for selecting and engineering polypeptides with novel functions.

YEAST TWO-HYBRID SYSTEM

When two proteins interact with each other, their corresponding genes are known as interacting genes. The yeast two-hybrid system uses a reporter gene to detect the physical interaction of a pair of proteins inside a yeast nucleus.

The two-hybrid method is based on the observation that most of the transcriptional proteins (i.e. the proteins involved in promoting transcription of a gene) contain two distinct domains — DNA binding domain and transcriptional activation domain. When these two domains are phyically separated, the protein loses its activity. However, the same protein can be reactivated when the domains are brought together. These proteins can bind to DNA and activate transcription.

The target protein is fused to a DNA-binding domain to form a ***bait***. When this target protein binds to another specifically designed protein namely the ***prey*** in the nucleus, they interact, which in turn switches on the expression of the reporter gene (***Fig. 5.15***). The reporter genes can be detected by growing the yeast on a selective medium.

It is possible to generate the bait and prey fusion proteins by standard recombinant DNA techniques. A single baid protein is frequently used to fish out interacting partners among the collection of prey proteins. A large number of prey proteins can be produced by ligating DNA encoding the activation domain of a transcriptional activator to a misture of DNA-fragments from a cDNA library.

Yeast three-hybrid system

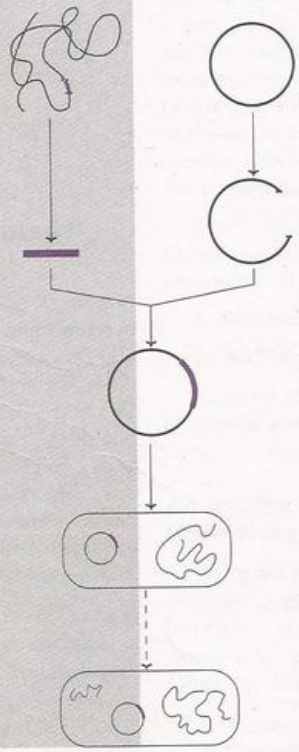
The interactions between protein and RNA molecules can be investigated by using a technique known as yeast three-hybrid system.

SECTION III

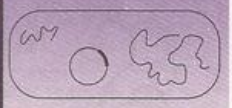
GENETIC ENGINEERING AND RECOMBINANT DNA TECHNOLOGY

CONTENTS

6 Introduction to Genetic Engineering 75
7 Basic Techniques in Genetic Engineering 92
8 Polymerase Chain Reaction (DNA Amplification) 112
9 Gene Libraries 120
10 Site-directed Mutagenesis and Protein Engineering 129
11 Manipulation of Gene Expression in Host Cells 137
12 Human Genome Project 148



Introduction to Genetic Engineering



Genetic engineering primarily involves the ***manipulation of genetic material (DNA)*** to achieve the desired goal in a pre-determined way. Some ***other terms*** are also in common use to describe genetic engineering.

- ***Gene manipulation***
- ***Recombinant DNA (rDNA) technology***
- ***Gene cloning (molecular cloning)***
- ***Genetic modifications***
- ***New genetics.***

BRIEF HISTORY OF RECOMBINANT DNA TECHNOLOGY

The present day DNA technology has its roots in the experiments performed by ***Boyer and Cohen in 1973***. In their experiments, they successfully recombined two plasmids (pSC 101 and pSC 102) and cloned the new plasmid in *E.coli*. Plasmid pSC 101 possesses a gene resistant to antibiotic tetracycline while plasmid pSC 102 contains a gene resistant to another antibiotic kanamycin. The newly developed recombined plasmid when incorporated into the bacteria exhibited resistance to both the antibiotics-tetracycline and kanamycin.

The second set of experiments of Boyer and Cohen were more organized. A gene encoding a protein (required to form rRNA) was isolated from the cells of African clawed frog *Xenophs laevis*, by use of a restriction endonuclease enzyme (*ECoRI*). The same enzyme was used to cut open plasmid pSC 101 DNA. Frog DNA fragments and plasmid DNA fragments were mixed, and pairing occurred between the complementary base pairs. By the addition of the enzyme DNA ligase, a recombined plasmid DNA was developed. These new plasmids, when introduced into *E.coli*, and grown on a nutrient medium resulted in the production of an extra protein (i.e. the frog protein). Thus, the genes of a frog could be successfully transplanted, and expressed in *E.coli*. This made the real beginning of modern rDNA technology and laid foundations for the present day molecular biotechnology.

Some biotechnologists who admire Boyer-Cohen experiments divide the subject into two chronological categories.

1. ***BBC***-biotechnology ***B***efore ***B***oyer and ***C***ohen.
2. ***ABC***-biotechnology ***A***fter ***B***oyer and ***C***ohen.

More information on the historical developments of genetic engineering and biotechnology is given under the scope of biotechnology (Chapter 1).

An outline of recombinant DNA technology

There are many diverse and complex techniques involved in gene manipulation. However, the basic principles of recombinant DNA technology are reasonably simple, and broadly involve the following stages (***Fig. 6.1***).

1. Generation of DNA fragments and selection of the desired piece of DNA (e.g. a human gene).

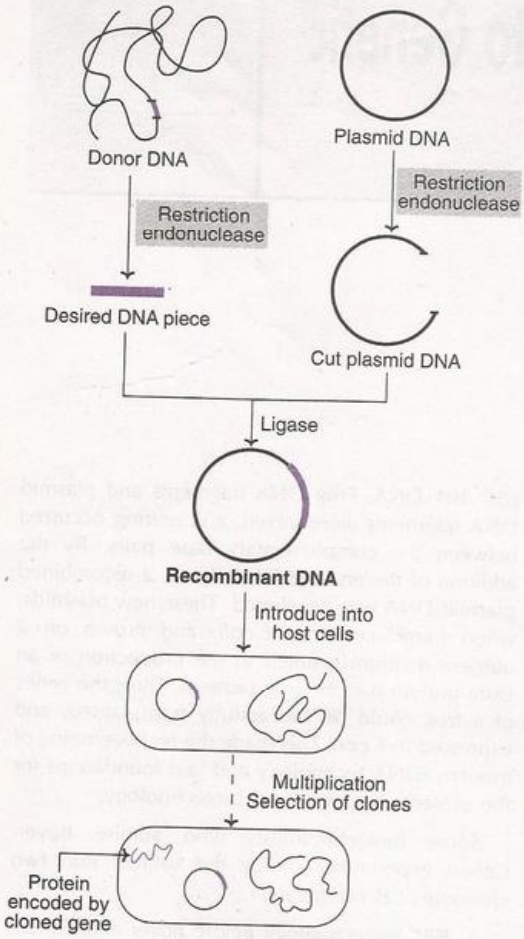


Fig. 6.1 : *The basic principle of recombinant DNA technology.*

2. Insertion of the selected DNA into a cloning vector (e.g. a plasmid) to create a ***recombinant DNA*** or ***chimeric DNA*** (Chimera is a monster in Greek mythology that has a lion's head, a goat's body and a serpent's tail. This may be comparable to Narasimha in Indian mythology).

3. Introduction of the recombinant vectors into host cells (e.g. bacteria).

4. Multiplication and selection of clones containing the recombinant molecules.

5. Expression of the gene to produce the desired product.

Recombinant DNA technology with special reference to the following aspects is described in this chapter.

1. Molecular tools of genetic engineering.
2. Host cells-the factories of cloning.
3. Vectors-the cloning vehicles.
4. Methods of gene transfer.
5. Gene cloning strategies.
6. Genetic engineering guidelines.
7. The future of genetic engineering.

MOLECULAR TOOLS OF GENETIC ENGINEERING

An engineer is a person who designs, constructs (e.g. bridges, canals, railways) and manipulates according to a set plan. The term genetic engineer may be appropriate for an individual who is involved in genetic manipulations. The ***genetic engineer's toolkit*** or molecular tools namely the enzymes most commonly used in recombinant DNA experiments are briefly described.

RESTRICTION ENDONUCLEASES—DNA CUTTING ENZYMES

Restriction endonucleases are one of the most important groups of enzymes for the manipulation of DNA. These are the bacterial enzymes that can cut/split DNA (from any source) at specific sites. They were first discovered in *E.coli* restricting the replication of bacteriophages, by cutting the viral DNA (The host *E.coli* DNA is protected from cleavage by addition of methyl groups). Thus, the enzymes that restrict the viral replication are known as ***restriction enzymes*** or restriction endonucleases.

Hundreds of restriction endonucleases have been isolated from bacteria, and some of them are commercially available. The progress and growth of biotechnology is unimaginable without the availability of restriction enzymes.

Nomenclature

Restriction endonucleases are named by a standard procedure, with particular reference to the bacteria from which they are isolated. The first letter (in italics) of the enzymes indicates the genus

name, followed by the first two letters (also in italics) of the species, then comes the strain of the organism and finally a Roman numeral indicating the order of discovery. A couple of examples are given below.

EcoRI is from *Escherichia* (***E***) *coli* (***co***), strain Ry13 (***R***) and first endonuclease (***I***) to be discovered. ***HindIII*** is from ***Haemophilus*** (***H***) influenzae (***in***), strain Rd (***d***) and the third endonucleases (***III***) to be discovered.

Types of endonucleases

At least 4 different types of restriction endonucleases are known-type 1 (e.g *Ecok12*), type II (e.g. *EcoRI*), type III (e.g. *EcoPI*) and type IIs. Their characteristic features are given in ***Table 6.1***. Among these, type II restriction endonucleases are most commonly used in gene cloning.

Table 6.1 Characteristics of different types of restriction endonucleases

Type	*Salient features*
I	A single enzyme with 3 subunits for recognition, cleavage and methylation. It can cleave up to 1000 bp from recognition site.
II	Two different enzymes either to cleave or modify the recognition sequence. Cleavage site is the same or close to recognition site.
III	A single enzyme with 2 subunits for recognition and cleavage. Cleavage site is 24–26 bp from recognition site.
IIs	Two different enzymes, cleavage site is up to 20 bp from recognition site.

Table 6.2 Some restriction enzymes with sources, recognition sequences and the products formed

Enzyme (source)	*Recognition sequence*	*Products*
EcoRI (*Escherichia coli*)	5′……G✂A–A–T–T–C……3′ 3′……C–T–T–A–A✂G……5′	A–A–T–T–C…… G…… ……G C–T–T–A–A
BamHI (*Bacillus amylolique faciens*)	5′……G✂G–A–T–C–C……3′ 3′……C–C–T–A–G✂G……5′	G–A–T–C–C…… G…… ……G ……C–C–T–A–G
HaeIII (*Haemophilus aegyptius*)	5′……G–G✂C–C……3′ 3′……C–C✂G–G……5′	*C–C…… G–G…… ……*G–G ……C–C
HindIII (*Haemophilus influenzae*)	5′……A✂A–G–C–T–T……3′ 3′……T–T–C–G–A✂A……5′	A–G–C–T–T…… A…… ……A ……T–T–C–G–A
NotI (*Nocardia otitidis*)	5′……G–C✂G–G–C–C–G–C……3′ 3′……C–G–C–C–G–G✂C–G……5′	G–G–C–C–G–C…… C–G…… ……G–C ……C–G–C–C–G–G……

(***Note :*** *Scissors indicate the sites of cleavage.* * *The products are with blunt ends while for the rest, the products are with sticky ends).*

Recognition sequences

Recognition sequence is the ***site where the DNA is cut*** by a restriction endonuclease. Restriction endonucleases can specifically recognize DNA with a particular sequence of 4-8 nucleotides and cleave. Each recognition sequence has two fold rotational symmetry i.e. the same nucleotide sequence occurs on both strands of DNA which run in opposite direction (***Table 6.2***). Such sequences are referred to as palindromes, since they read similar in both directions (forwards and backwards).

Cleavage patterns

Majority of restriction endonucleases (particularly type II) cut DNA at defined sites within recognition sequence. A selected list of enzymes, recognition sequences, and their products formed is given in ***Table 6.2.***

The cut DNA fragments by restriction endonucleases may have mostly ***sticky ends*** (cohesive ends) or ***blunt ends***, as given in Table 6.2. DNA fragments with sticky ends are particularly useful for recombinant DNA experiments. This is because the single-stranded sticky DNA ends can easily pair with any other DNA fragment having complementary sticky ends.

DNA LIGASES — DNA JOINING ENZYMES

The cut DNA fragments are covalently joined together by DNA ligases. These enzymes were originally isolated from viruses. They also occur in *E.coli* and eukaryotic cells. DNA ligases actively participated in cellular DNA repair process.

The action of DNA ligases is absolutely required to permanently hold DNA pieces. This is so since the hydrogen bonds formed between the complementary bases (of DNA strands) are not strong enough to hold the strands together. DNA ligase joins (seals) the DNA fragments by forming a phosphodiester bond between the phosphate group of 5′-carbon of one deoxyribose with the hydroxyl group 3′-carbon of another deoxyribose (***Fig. 6.2***).

Phage T4 DNA ligase requires ATP as a cofactor while *E.coli* DNA ligase is dependent on NAD^+. In each case, the cofactor (ATP or NAD^+) is split to form an enzyme—AMP complex that brings about the formation of phosphodiester bond. The action of DNA ligase is the ultimate step in the formation of a recombinant DNA molecule.

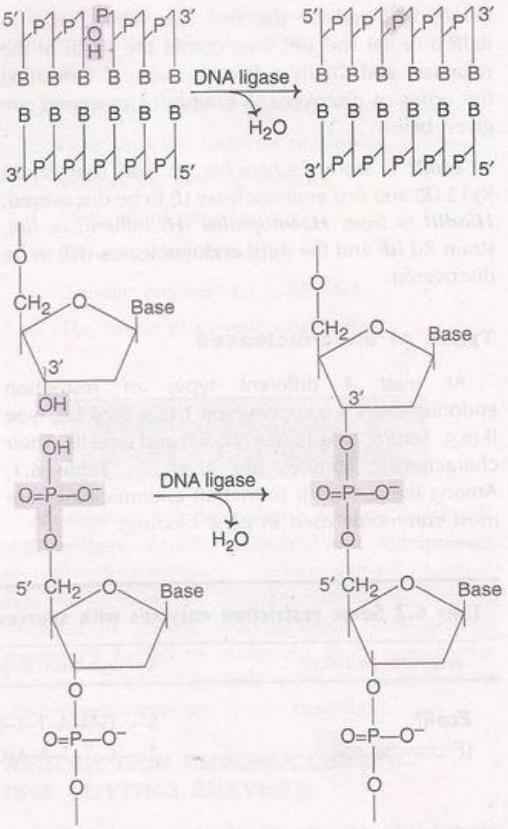


Fig. 6.2 : Action of DNA ligase in the formation of phosphodiester bond (B-base).

Homopolymer tailing

The complementary DNA strands can be joined together by ***annealing.*** This principle is utilized in homopolymer tailing. The technique involves the addition of oligo (dA) to 3′-ends of some DNA molecules and the addition of oligo (dT) to 3′-ends of other molecules. The homopolymer extensions (by adding 10-40 residues) can be synthesized by using terminal deoxynucleotidyltransferase (of calf thymus). Homopolymer tailing, achieved by annealing is illustrated in ***Fig. 6.3.***

Linkers and adaptors

Linkers and adaptors are ***chemically synthesized, short, double-stranded DNA molecules***. Linkers

possess restriction enzyme cleavage sites. They can be ligated to blunt ends of any DNA molecule and cut with specific restriction enzymes to produce DNA fragments with sticky ends (***Fig. 6.4***).

Adaptors contain preformed sticky or cohesive ends. They are useful to be ligated to DNA fragments with blunt ends. The DNA fragments held to linkers or adaptors are finally ligated to vector DNA molecules (***Fig 6.4***).

ALKALINE PHOSPHATASE

Alkaline phosphatase is an enzyme involved in the removal of phosphate groups. This enzyme is useful to prevent the unwanted ligation of DNA molecules which is a frequent problem encountered in cloning experiments.

When the linear vector plasmid DNA is treated with alkaline phosphatase, the 5'-terminal phosphate is removed (***Fig 6.5***). This prevents both recircularization and plasmid DNA dimer formation. It is now possible to insert the foreign DNA through the participation of DNA ligase.

DNA MODIFYING ENZYMES

Some authors prefer to use the broad term DNA modifying enzymes to all the enzymes involved in recombinant DNA technology. These enzymes represent the ***cutting*** and ***joining*** functions in DNA manipulation. They are broadly categorized as nucleases, polymerases and enzymes modifying ends of DNA molecules, and briefly described below, and illustrated in ***Fig. 6.6***.

Nucleases

Nucleases are the enzymes that break the phosphodiester bonds (that hold nucleotides together) of DNA. ***Endonucleases*** act on the internal phosphodiester bonds while ***exonucleases*** degrade DNA from the terminal ends (***Fig 6.6A***). Restriction endonucleases, described already, are good examples of endonucleases. Some other examples of endo- and exonucleases are listed.

Endonucleases

- Nuclease S_1 specifically acts on single-stranded DNA or RNA molecules.
- Deoxyribonuclease I (DNase I) cuts either single or double-stranded DNA molecules at random sites.

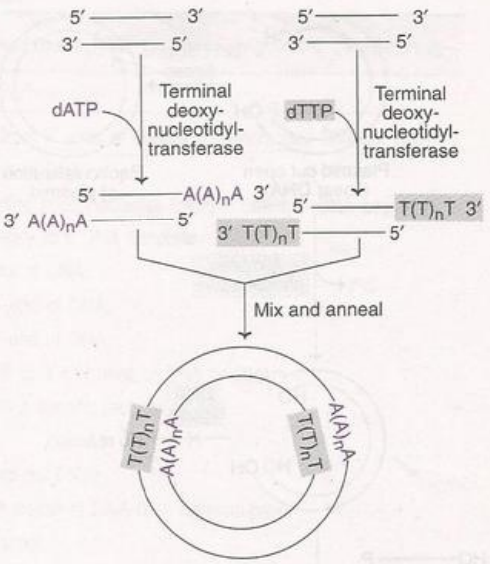


Fig. 6.3 : Homopolymer tailing.

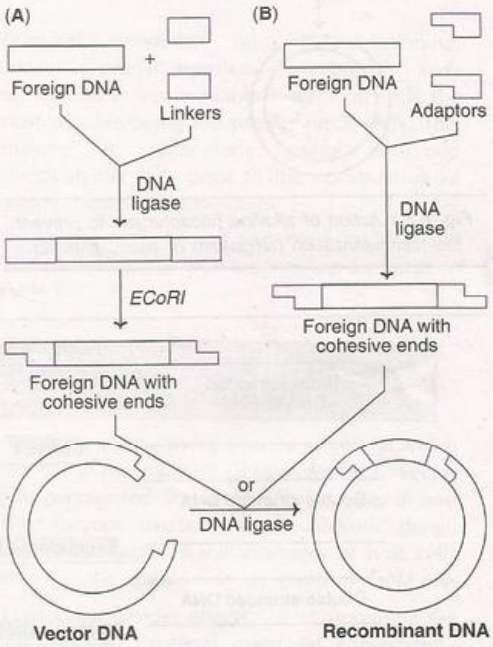


*Fig. 6.4 : Use of linkers (**A**) and adaptors (**B**) in the formation of recombinant DNA (Note that the linkers have blunt ends while adaptors have sticky ends).*

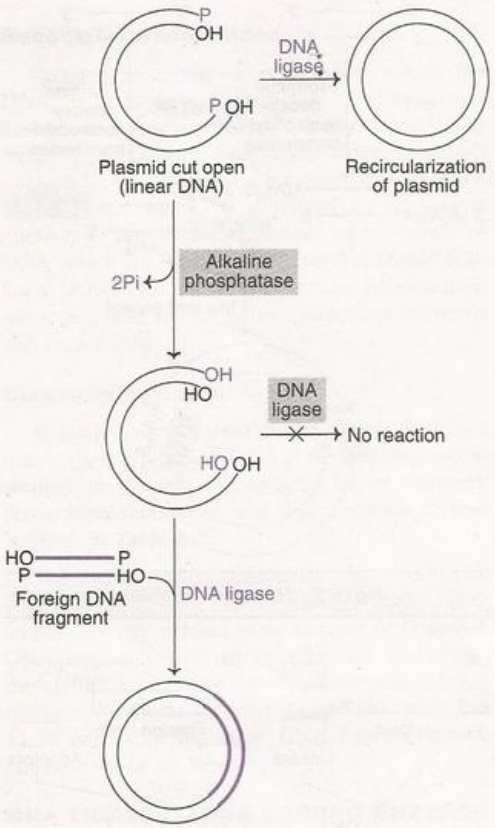


Fig. 6.5 : Action of alkaline phosphatase to prevent the recircularization (religation) of vector plasmid.

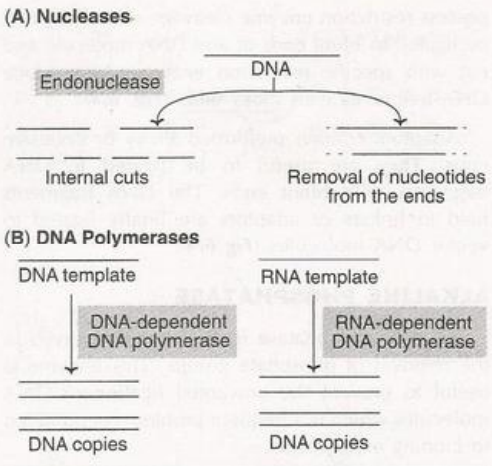


*Fig. 6.6 : An outline of the activities of nucleases (**A**), and DNA polymerases (**B**).*

Exonucleases

- Exonuclease III cuts DNA and generates molecules with protruding 5′-ends.
- Nuclease Bal 31 is a fast acting 3′-exonuclease. Its action is usually coupled with slow acting endonucleases.

A diagrammatic representation of the action of endo-and exonucleases is given in ***Fig. 6.7***.

Besides the DNA cutting enzymes, there are RNA specific nucleases, which are referred to as ***ribonucleases*** (***RNases***).

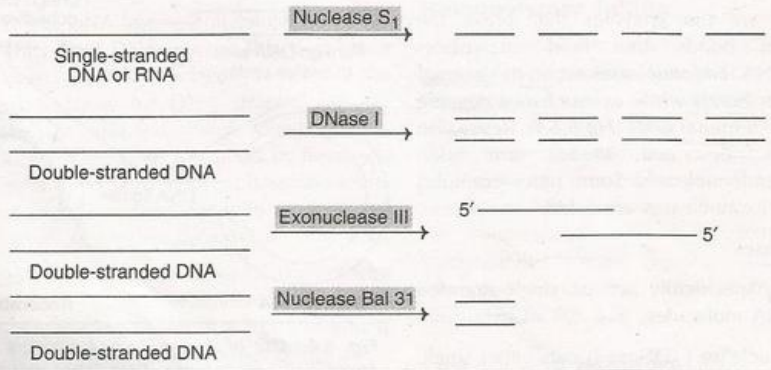


Fig. 6.7 : Mode of action of selected endo– and exonucleases (DNase I–Deoxyribonuclease I).

TABLE 6.3 The most commonly used enzymes in recombinant DNA technology/genetic engineering

Enzyme	*Use/reaction*
Alkaline phosphatase	Removes phosphate groups from 5′-ends of double/single-stranded DNA (or RNA).
Bal 31 nuclease	For the progressive shortening of DNA.
DNA ligase	Joins DNA molecules by forming phosphodiester linkages between DNA segments.
DNA polymerase I	Synthesizes DNA complementary to a DNA template.
DNase I	Produces single-stranded nicks in DNA.
Exonuclease III	Removes nucleotides from 3′-end of DNA.
λ exonuclease	Removes nucleotides from 5′-end of DNA.
Polynucleotide kinase	Transfers phosphate from ATP to 5′-OH ends of DNA or RNA.
Restriction enzymes	Cut double-stranded DNA with a specific recognition site.
Reverse transcriptase	Synthesizes DNA from RNA.
RNase A	Cleaves and digests RNA (and not DNA).
RNase H	Cleaves and digests the RNA strand of RNA-DNA heteroduplex.
Taq DNA polymerase	Used in polymerase chain reaction
SI nuclease	Degrades single-stranded DNA and RNA.
Terminal transferase	Adds nucleotides to the 3′-ends of DNA or RNA. Useful in homopolymer tailing.

Polymerases

The group of enzymes that catalyse the synthesis of nucleic acid molecules are collectively referred to as polymerases. It is customary to use the name of the nucleic acid template on which the polymerase acts (***Fig. 6.6B***). The three important polymerases are given below.

- ***DNA-dependent DNA polymerase*** that copies DNA from DNA.
- ***RNA-dependent DNA polymerase*** (reverse transcriptase) that synthesizes DNA from RNA.
- ***DNA-dependent RNA polymerase*** that produces RNA from DNA.

For more details on the functions of these enzymes, refer Chapter 4.

Enzymes modifying the ends of DNA

There are certain enzymes that act on the terminal ends of DNA and modify these molecules. The important ones are listed.

- ***Alkaline phosphatase*** that removes the terminal phosphate group (see ***Fig. 6.5***).
- ***Polynucleotide kinase*** involved in the addition of phosphate groups.
- ***Terminal transferase*** (also called terminal deoxynucleotidyl transferase) repeatedly adds nucleotides to any available 3′-terminal ends, the most suitable being the protruding 3′-ends. This enzyme is particularly useful to add homopolymer tails prior to the construction of recombinant DNA molecules.
- The most commonly used enzymes in recombinant DNA technology/genetic engineering are listed in ***Table 6.3***.

HOST CELLS — THE FACTORIES OF CLONING

The hosts are the ***living systems or cells*** in which the carrier of recombinant DNA molecule or ***vector can be propagated***. There are different types of host cells-prokaryotic (bacteria) and eukaryotic (fungi, animals and plants). Some examples of host cells used in genetic engineering are given in ***Table 6.4***.

Host cells, besides effectively incorporating the vector's genetic material, must be conveniently cultivated in the laboratory to collect the products. In general, ***microorganisms*** are ***preferred*** as host cells, since they ***multiply faster*** compared to cells of higher organism (plants or animals).

PROKARYOTIC HOSTS

Escherichia coli

The bacterium, *Escherichia coli* was the first organism used in the DNA technology experiments and continues to be ***the host of choice*** by many workers. Undoubtedly, *E.coli*, the simplest Gram negative bacterium (a common bacterium of human and animal intestine), has played a key role in the development of present day biotechnology. Under suitable environment, *E.coli* can double in number every 20 minutes. Thus, as the bacteria multiply, their plasmids (along with foreign DNA) also multiply to produce millions of copies, referred to as colony or in short ***clone***. The term clone is broadly used to a mass of cells, organisms or genes that are produced by multiplication of a single cell, organism or gene.

Limitations of *E. coli* : There are certain limitations in using *E.coli* as a host. These include-causation of diarrhea by some strains, formation of endotoxins that are toxic, and a low export ability of proteins from the cell. Another ***major drawback*** is that *E.coli* (or even other prokaryotic organisms) ***cannot perform post-translational modifications***.

Bacillus subtilis

Bacillus subtilis is a rod shaped non-pathogenic bacterium. It has been used as a host in industry for the production of enzymes, antibiotics, insecticides etc. Some workers consider *B.subtilis* as an ***alternative to E.coli***.

EUKARYOTIC HOSTS

Eukaryotic organisms are preferred to produce human proteins since these hosts with complex structure (with distinct organelles) are more suitable to synthesize complex proteins. The ***most commonly used*** eukaryotic organism is the ***yeast, Saccharomyces cerevisiae***. It is a non-pathogenic organism routinely used in brewing and baking industry. Certain fungi have also been used in gene cloning experiments.

Mammalian cells

Despite the practical difficulties to work with and high cost factor, mammalian cells (such as mouse cells) are also employed as hosts. The advantage is that certain complex proteins which cannot be synthesized by bacteria can be produced by mammalian cells e.g. tissue plasminogen activator. This is mainly because the mammalian cells ***possess the machinery to modify the protein to its final form*** (post-translational modifications).

Table 6.4 Some examples of host cells used in genetic engineering

Group	*Examples*
Prokaryotic	
Bacteria	*Escherichia coli*
	Bacillus subtilis
	Streptomyces sp
Eurkaryotic	
Fungi	*Saccharomyces cerevisiae*
	Aspergillus nidulans
Animals	Insect cells
	Oocytes
	Mammalian cells
	Whole organisms
Plants	Protoplasts
	Intact cells
	Whole plants

It may be noted here that the gene manipulation experiments in higher animals and plants are usually carried out to alter the genetic make up of the organism to create transgenic animals (Chapter 41) and transgenic plants (Chapter 49), rather than to isolate genes for producing specific proteins.

VECTORS — THE CLONING VEHICLES

Vectors are the DNA molecules, which can ***carry a foreign DNA fragment to be cloned***. They are self-replicating in an appropriate host cell. The most important vectors are plasmids, bacteriophages, cosmids and phasmids.

Characteristics of an ideal vector

An ideal vector should be small in size, with a single restriction endonuclease site, an origin of replication and 1-2 genetic markers (to identify recipient cells carrying vectors). Naturally occurring plasmids rarely possess all these characteristics.

PLASMIDS

Plasmids are extrachromosomal, double-stranded, circular, self-replicating DNA molecules. Almost all the bacteria have plasmids containing a low copy number (1-4 per cell) or a high copy number (10-100 per cell). The size of the plasmids varies from 1 to 500 kb. Usually, plasmids contribute to about 0.5 to 5.0% of the total DNA of bacteria (**Note :** A few bacteria contain linear plasmids e.g. *Streptomyces* sp, *Borella burgdorferi*).

Types of plasmids

There are many ways of grouping plasmids. They are categorized as ***conjugative*** if they carry a set of transfer genes (*tra* genes) that facilitates bacterial conjugation, and ***non-conjugative***, if they do not possess such genes.

Another classification is based on the copy number. ***Stringent plasmids*** are present in a limited number (1-2 per cell) while ***relaxed plasmids*** occur in large number in each cell.

F-plasmids possess genes for their own transfer from one cell to another, while ***R-plasmids*** carry genes resistance to antibiotics.

In general, the conjugative plasmids are large, show stringent control of DNA replication, and are present in low numbers. On the other hand, non-conjugative plasmids are small, show relaxed control of DNA replication, and are present in high numbers.

Nomenclature of plasmids

It is a common practice to designate ***p***lasmid by a lower case ***p***, followed by the first letter(s) of researcher(s) names and the numerical number given by the workers. Thus, ***pBR322*** is a ***p***lasmid discovered by ***B***olivar and ***R***odriguez who designated it as ***322.*** Some plasmids are given names of the places where they are discovered e.g. ***pUC*** is ***p***lasmid from ***U***niversity of ***C***alifornia.

pBR322 — the most common plasmid vector

pBR322 of *E.coli* is the most popular and widely used plasmid vector, and is appropriately regarded as the parent or grand parent of several other vectors.

pBR322 has a DNA sequence of 4,361 bp. It carries genes resistance for ampicillin (Amp^r) and tetracycline (Tel^r) that serve as markers for the identification of clones carrying plasmids. The plasmid has unique recognition sites for the action of restriction endonucleases such as *EcoRI, HindIII, BamHI, SalI* and *PstII* (***Fig. 6.8***).

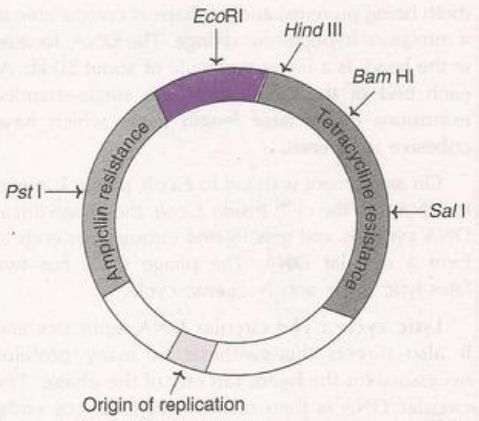


Fig. 6.8 : *Genetic map of plasmid cloning vector pBR322.*

Other plasmid cloning vectors

The other plasmids employed as cloning vectors include pUC19 (2,686 bp, with ampicillin resistance gene), and derivatives of pBR322–pBR325, pBR328 and pBR329.

BACTERIOPHAGES

Bacteriophages or simply ***phages*** are the ***viruses*** that ***replicate within the bacteria***. In case of certain phages, their DNA gets incorporated into the bacterial chromosome and remains there permanently. Phage vectors can accept short fragments of foreign DNA into their genomes. The advantage with phages is that they ***can take up larger DNA segments than plasmids***. Hence phage vectors are preferred for working with genomes of human cells.

Bacteriophage λ

Bacteriophage lambda (or simply ***phage λ***), a virus of *E.coli*, has been most thoroughly studied and developed as a vector. In order to understand how bacteriophage functions as a vector, it is desirable to know its structure and life cycle (***Fig. 6.9***). Phage λ consists of a head and a tail

(both being proteins) and its shape is comparable to a miniature hypodermic syringe. The DNA, located in the head, is a linear molecule of about 50 kb. At each end of the DNA, there are single-stranded extensions of 12 base length each, which have cohesive (cos) ends.

On attachment with tail to *E.coli*, phage λ injects its DNA into the cell. Inside *E.coli*, the phage linear DNA cyclizes and gets ligated through cos ends to form a circular DNA. The phage DNA has two fates-lytic cycle and lysogenic cycle.

Lytic cycle : The circular DNA replicates and it also directs the synthesis of many proteins necessary for the head, tail etc, of the phage. The circular DNA is then cleaved (to form cos ends) and packed into the head of the phage. About 100 phage particles are produced within 20 minutes after the entry of phage into *E.coli*. The host cell is then subjected to lysis and the phages are released. Each progeny phage particle can infect a bacterial cell, and produce several hundreds of phages. It is estimated that by repeating the lytic cycle four times, a single phage can cause the death of more than one billion bacterial cells. ***If a foreign DNA is spliced into phage DNA, without causing harm to phage genes, the phage will reproduce (replicate the foreign DNA) when it infects bacterial cell.*** This has been exploited in phage vector employed cloning techniques.

Lysogenic cycle : In this case, the phage DNA (instead of independently replicating) becomes integrated into the *E.coli* chromosome and replicates along with the host genome. No phage particles are synthesized in this pathway.

Use of phage λ as a vector

Only about 50% of phage λ DNA is necessary for its multiplication and other functions. Thus, as much as 50% (i.e.up to 25kb) of the phage DNA can be replaced by a donor DNA for use in cloning experiments. However, several restriction sites are present on phage λ which is not by itself a suitable vector. The λ-based phage vectors are modifications of the natural phage with much reduced number of restriction sites. Some of them are discussed hereunder.

Insertion vectors : They have just one unique cleavage site, which can be cleaved, and a foreign DNA ligated in. It is essential that sufficient DNA (about 25%) has to be deleted from the vector to make space for the foreign DNA (about 18kb).

Replacement vectors : These vectors have a pair of restriction sites to remove the non-essential DNA (***stuffer DNA***) that will be replaced by a foreign DNA. Replacement vectors can accommodate up to 24kb, and propagate them.

Many phage vector derivations (insertion/ replacement) have been produced by researchers for use in recombinant DNA technology.

The main advantage of using phage vectors is that the foreign DNA can be packed into the phage (*in vitro* packaging), the latter in turn can be injected into the host cell very effectively (**Note :** No transformation is required).

Phage M_{13} vectors

Phage M_{13} (bacteriophage M_{13}) is a single-stranded DNA phage of *E.coli*. Inside the host cell, M_{13} synthesizes the complementary strand to form a double-stranded DNA (replicative form DNA; ***RF DNA***). For use as a vector, RF DNA is isolated and a foreign DNA can be inserted on it. This is then returned to the host cell as a plasmid. Single-stranded DNAs are recovered from the phage particles. Phage M_{13} is useful for sequencing DNA through Sanger's method (Refer Chapter 7).

COSMIDS

Cosmids are the vectors possessing the characteristics of both plasmid and bacteriophage λ. ***Cosmids*** can be constructed by adding a fragment of phage λ DNA including ***cos*** site, to plas***mids***. A foreign DNA (about 40 kb) can be inserted into cosmid DNA. The recombinant DNA so formed can be packed as phages and injected into *E.coli* (***Fig. 6.10***). Once inside the host cell, cosmids behave just like plasmids and replicate. The advantage with cosmids is that they can carry larger fragments of foreign DNA compared to plasmids.

Phasid vectors

Phasids are the combination of plasmid and phage and can function as either one (i.e as plasmid or phage). Phasids possess functional origins of replication of both plasmid and phage λ, and therefore can be propagated (as plasmid or phage) in appropriate *E.coli*. The vectors phasids may be used in many ways in cloning experiments.

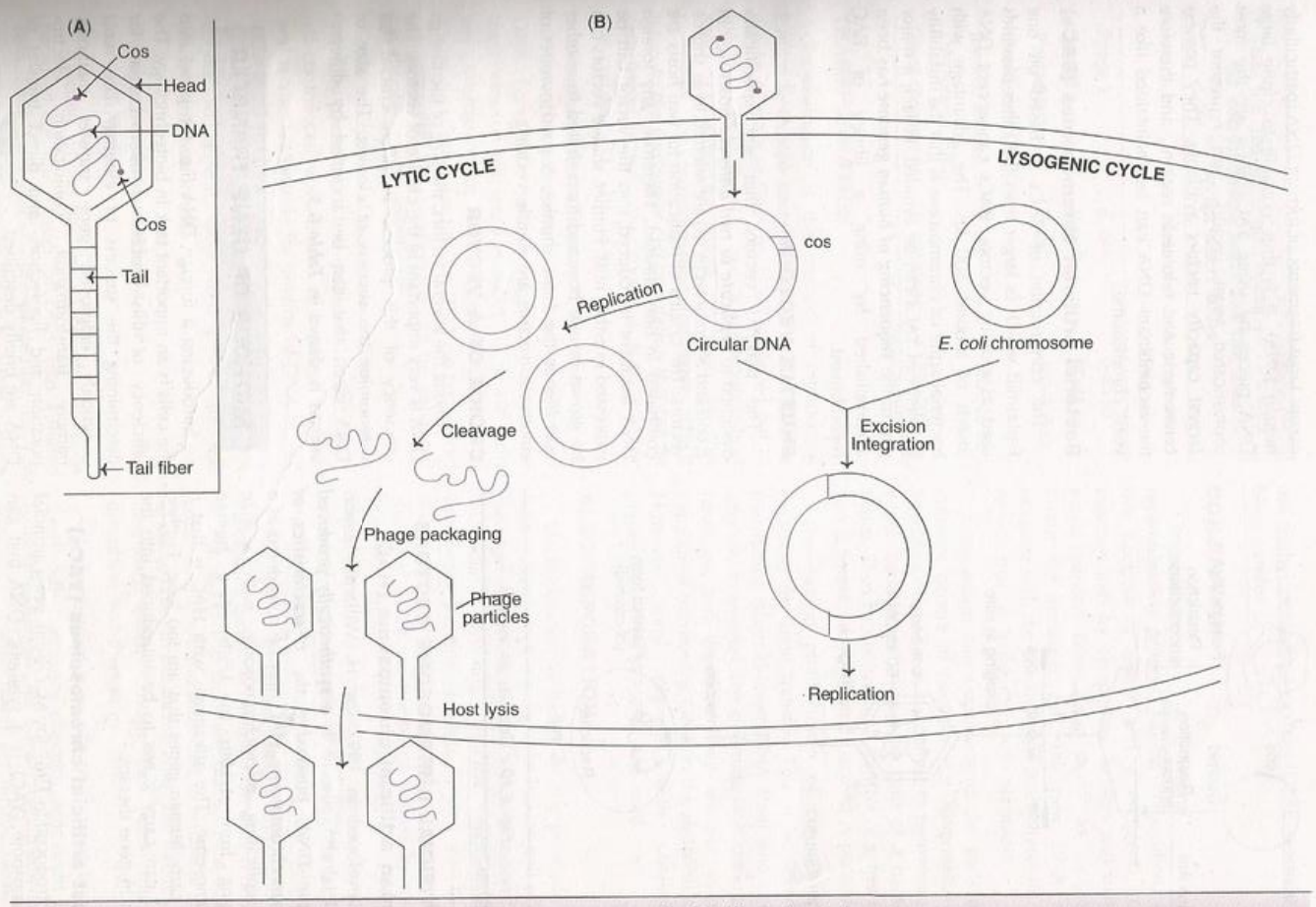


Fig. 6.9 : (A) Bacteriophage λ (B) Life cycle of phage λ.

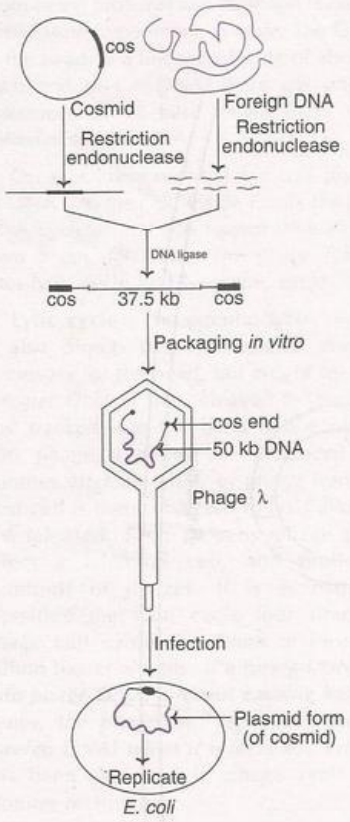


Fig. 6.10 : Cosmids as vectors.

ARTIFICIAL CHROMOSOME VECTORS

Human artificial chromosome (HAC)

Developed in 1997 (by H. Willard); human artificial chromosome is a ***synthetically produced vector DNA, possessing the characteristics of human chromosome***. HAC may be considered as a self-replicating microchromosome with a size ranging from 1/10th to 1/5th of a human chromosome. The advantage with HAC is that it can carry human genes that are too long. Further, HAC can carry genes to be introduced into the cells in gene therapy.

Yeast artificial chromosomes (YACs)

Introduced in 1987 (by M. Olson), yeast artificial chromosome (YAC) is a synthetic DNA that can accept large fragments of foreign DNA (particularly human DNA). It is thus possible to clone large DNA pieces by using YAC. YACs are the most sophisticated yeast vectors, and represent the ***largest capacity vectors*** available. They possess centromeric and telomeric regions, and therefore the recombinant DNA can be maintained like a yeast chromosome.

Bacterial artificial chromosomes (BACs)

The construction of BACs is based on one F-plasmid which is larger than the other plasmids used as cloning vectors. BACs can accept DNA inserts of around 300 kb. The advantage with bacterial artificial chromosome is that the instability problems of YACs can be avoided. In fact, a major part of the ***sequencing of human genome*** has been accomplished by using a library of BAC recombinant.

SHUTTLE VECTORS

The plasmid vectors that are specifically designed to ***replicate in two different hosts*** (say in *E.coli* and *Streptomyces* sp) are referred to as shuttle vectors. The origins of replication for two hosts are combined in one plasmid. Therefore, any foreign DNA fragment introduced into the vector can be expressed in either host. Further, shuttle vectors can be grown in one host and then shifted to another host (hence the name shuttle). A good number of eukaryotic vectors are shuttle vectors.

CHOICE OF A VECTOR

Among the several factors, the size of the foreign DNA is very important in the choice of vectors. The efficiency of this process in often crucial for determining the success of cloning. The size of DNA insert that can be accepted by different vectors is shown in ***Table 6.5***.

METHODS OF GENE TRANSFER

Introducing a foreign DNA (i.e. the gene) into the cells is an important task in biotechnology. The efficiency of this process is often crucial for determining the success of cloning. The most commonly employed gene transfer methods, namely transformation, conjugation, electroporation and lipofection, and direct transfer of DNA are briefly described.

TABLE 6.5 The different cloning vectors with the corresponding hosts and the sizes of foreign insert DNAs

Vector	*Host*	*Foreign insert DNA size*
Phage λ	*E. coli*	5–25 kb
Cosmid λ	*E. coli*	35–45 kb
Plasmid artifical chromosome (PAC)	*E. coli*	100–300 kb
Bacterial artificial chromosome (BAC)	*E. coli*	100–300 kb
Yeast chromosome	*S. cerevisiae*	200–2000 kb

TRANSFORMATION

Transformation is the method of introducing foreign DNA into bacterial cells (e.g. *E.coli*). The uptake of plasmid DNA by *E.coli* is carried out in ice-cold $CaCl_2$ (0-5°C), and a subsequent heat shock (37–45°C for about 90 sec). By this technique, the ***transformation frequency***, which refers to ***the fraction of cell population that can be transferred***, is reasonably good e.g. approximately one cell for 1000 (10^{-3}) cells.

Transformation efficiency : It refers to the number of transformants per microgram of added DNA. For *E.coli*, transformation by plasmid, the transformation efficiency is about 10^7 to 10^8 cells per microgram of intact plasmid DNA. The bacterial cells that can take up DNA are considered as competent. The competence can be enhanced by altering growth conditions.

The mechanism of the transformation process is not fully understood. It is believed that the $CaCl_2$ affects the cell wall, breaks at localized regions, and is also responsible for binding of DNA to cell surface. A brief heat shock (i.e. the sudden increase in temperature from 5°C to 40°C) stimulates DNA uptake. In general, large-sized DNAs are less efficient in transforming.

Other chemical methods for transformation

Calcium phosphate (in place of $CaCl_2$) is preferred for the transfer of DNA into cultured cells. Sometimes, calcium phosphate may result in precipitate and toxicity to the cells. Some workers use diethyl aminoethyl dextran (DEAE -dextran) for DNA transfer.

CONJUCTION

Conjugation is ***a natural microbial recombination process***. During conjugation, two live bacteria (a donor and a recipient) come together, join by cytoplasmic bridges and transfer single-stranded DNA (from donor to recipient). Inside the recipient cell, the new DNA may integrate with the chromosome (rather rare) or may remain free (as is the case with plasmids).

Conjugation can occur among the cells from different genera of bacteria (e.g *Salmonella* and *Shigella* cells). This is in contrast to transformation which takes place among the cells of a bacterial genus. Thus by conjugation, transfer of genes from two different and unrelated bacteria is possible.

The natural phenomenon of conjugation is exploited for gene transfer. This is achieved by transferring plasmid-insert DNA from one cell to another. In general, the plasmids lack conjugative functions and therefore, they are not as such capable of transferring DNA to the recipient cells. However, some plasmids with conjugative properties can be prepared and used.

ELECTROPORATION

Electroporation is based on the principle that high voltage electric pulses can induce cell plasma membranes to fuse. Thus, electroporation is a technique involving ***electric field-mediated membrane permeabilization.*** Electric shocks can also induce cellular uptake of exogenous DNA (believed to be via the pores formed by electric pulses) from the suspending solution. Electroporation is a simple and rapid technique for introducing genes into the cells from various organisms (microorganisms, plants and animals).

The basic technique of electroporation for transferring genes into mammalian cells is depicted in ***Fig. 6.11***. The cells are placed in a solution containing DNA and subjected to electrical shocks to cause holes in the membranes. The foreign DNA fragments enter through the holes into the cytoplasm and then to nucleus.

Electroporation is an effective way to transform *E.coli* cells containing plasmids with insert DNAs longer than 100 kb. The transformation efficiency is around 10^9 transformants per microgram of DNA

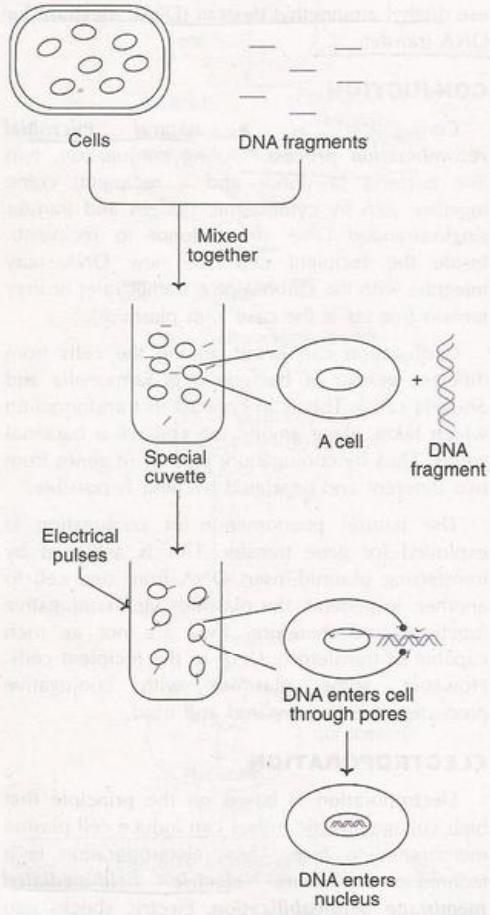


*Fig. 6.11 : Gene transfer by electroporation. (**Note** : Magnification depicted on right side)*

for small plasmids (about 3kb) and about 10^6 for large plasmids (about 130 kb).

LIPOSOME-MEDIATED GENE TRANSFER

Liposomes are circular lipid molecules, which have an aqueous interior that can carry nucleic acids. Several techniques have been developed to encapsulate DNA in liposomes. The liposome-mediated gene transfer, referred to as ***lipofection***, is depicted in ***Fig. 6.12***.

On treatment of DNA fragment with liposomes, the DNA pieces get encapsulated inside liposomes. These liposomes can adher to cell membranes and fuse with them to transfer DNA fragments. Thus, the DNA enters the cell and then to the nucleus. The positively charged liposomes very efficiently complex with DNA, bind to cells and transfer DNA rapidly.

Lipofection is a very efficient technique and is used for the transfer of genes to bacterial, animal and plant cells.

The different methods of gene transfer are briefly described above. More details are given in the respective chapters applying these techniques to achieve gene transfer.

TRANSDUCTION

Sometimes, the foreign DNA can be packed inside animal viruses. These viruses can naturally

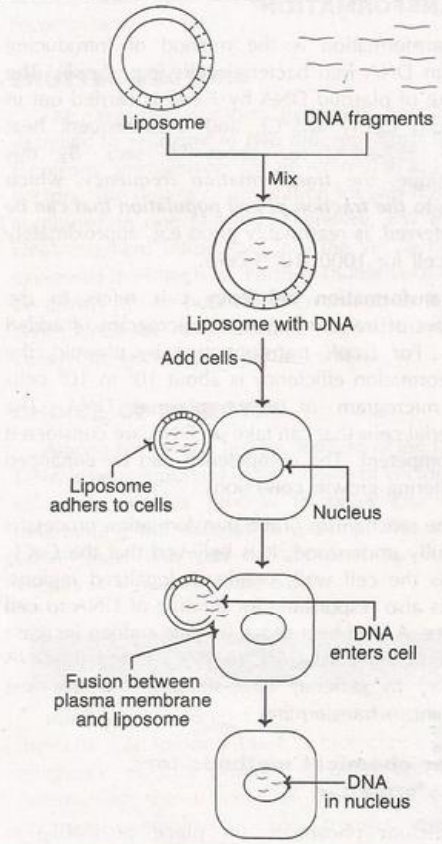


*Fig. 6.12 : Liposome-mediated gene transfer (**Note** : For clarity, the native cell DNA is not shown).*

infect the cells and introduce the DNA into host cells. The transfer of DNA by this approach is referred to as transduction.

DIRECT TRANSFER OF DNA

It is possible to directly transfer the DNA into the cell nucleus. Microinjection and particle bombardment are the two techniques commonly used for this purpose.

Microinjection

DNA transfer by microinjection is generally used for the cultured cells. This technique is also useful to introduce DNA into large cells such as oocytes, eggs and the cells of early embryos.

Some more details on gene transfer methods including particle bombardment are given in Chapter 49.

The term ***transfection is used for the transfer DNA into eukaryotic cells,*** by various physical or chemical means.

GENE CLONING STRATEGIES

A clone refers to ***a group of*** organisms, cells, ***molecules*** or other objects, ***arising from a single individual.*** Clone and colony are almost synonymous.

Gene cloning strategies in relation to recombinant DNA technology broadly involve the following aspects (***Fig. 6.13***).

- Generation of desired DNA fragments.
- Insertion of these fragments into a cloning vector.
- Introduction of the vectors into host cells.
- Selection or screening of the recipient cells for the recombinant DNA molecules.

It is obvious that for a good understanding (or account) of gene cloning strategies, this chapter has to be learnt in detail. In addition, the reader must invariably refer gene libraries (Chapter 9) for ***cloning of genes and screening strategies,*** and the ***polymerase chain reaction*** (Chapter 8) for *in vitro* generation of large quantities of DNA. Further, gene cloning also involves the expression of genes which is described under ***manipulation of gene expression in host cells*** (Chapter 11).

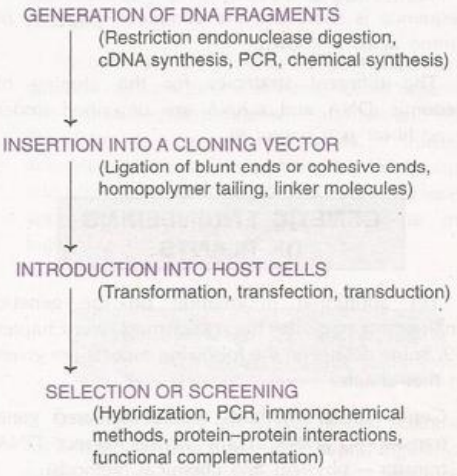


Fig. 6.13 : An overview of cloning strategies in recombinant DNA technology.

CLONING FROM GENOMIC DNA OR MRNA?

DNA represents the complete genetic material of an organism which is referred to as genome. Theoretically speaking, cloning from genomic DNA is supposed to be ideal. But the DNA contains non-coding sequences (introns), control regions and repetitive sequences. This complicates the cloning strategies, hence DNA as a source material is not preferred, by many workers. However, if the objective of cloning is to elucidate the control of gene expression, then genomic DNA has to be invariably used in cloning.

The use of mRNA in cloning is preferred for the following reasons.

- mRNA represents the actual genetic information being expressed.
- Selection and isolation mRNA is easy.
- As introns are removed during processing, mRNA reflects the coding sequence of the gene.
- The synthesis of recombinant protein is easy with mRNA cloning.

Besides the direct use of genomic DNA or mRNA, it is possible to synthesize DNA in the laboratory (Chapter 7), and use it in cloning

experiments. This approach is useful if the gene sequence is short and the complete sequence of amino acids is known.

The different strategies for the cloning of genomic DNA and mRNA are described under gene libraries (Chapter 9).

GENETIC ENGINEERING OF PLANTS

For additional information on the genetic engineering of plants, the reader must refer Chapter 49. Some details on the following aspects are given in that chapter

- Gene transfer methods (vector-mediated gene transfer — T DNA, plant viruses; direct DNA transfer — physical and chemical methods).
- Marker genes for plant transformation.
- Promoters and terminators.
- Transgene stability, expression and gene silencing.
- Chloroplast transformation.

GENETIC ENGINEERING GUIDELINES

With the success of Boyer-Cohen experiments (in 1973), it was realised that recombinant DNA technology could be used to create organisms with novel genes. This created worldwide commotion (among scientists, public and government officials) about the safety, ethics and unforeseen consequences of genetic manipulations. Some of the phrases quoted in media in those days are given.

- Manipulation of life.
- Playing God.
- Man made evolution.
- The most threatening scientific research.

It was feared that some new organisms, created inadvertently or deliberately for warfare, would cause epidemics and environmental catastrophes. Due to the fears of the dangerous consequences, a cautious approach on recombinant DNA experiments was suggested.

In 1974, a group of ten scientists led by Paul Berg wrote a letter that simultaneously appeared in the prestigeous journals-Nature, Science and Proceedings of the National Academy of Sciences. The dangers of DNA technology were printed out in that letter (highlights given below) :

"Recent advances in techniques for isolation and rejoining of segments of DNA now permit construction of biologically active recombinant DNA molecules in vitro. Although such experiments are likely to facilitate the solution of important theoretical and practical biological problems, they would also result in creation of novel types of DNA elements whose biological properties cannot be completely predicted. There is a serious concern that some of these DNA molecules could prove biologically hazardous".

The letter also appealed to molecular biologists worldwide for a moratorium on many kinds of recombinant DNA research, particularly those involving pathogenic organisms.

ASILOMAR RECOMMENDATIONS

In February 1975, a group of 139 scientists from 17 countries held a conference at Asilomar, a conference center in California, USA. They assured the uneasy public that the microorganisms used in DNA experiments were specifically bred and could not survive outside the laboratory. These scientists ***formulated guidelines and recommendations for conducting experiments in genetic engineering.***

NIH GUIDELINES

National Institute of Health (NIH), USA constituted the ***Recombinant DNA Advisory Committee (RAC)*** which issued a set of stringent guidelines to conduct research on DNA. RAC was in fact overseeing the research projects involving gene splicing and recombinant DNA.

Some of the important original NIH recommendations on recombinant DNA research relate to the following aspects.

- Physical (laboratory) containment levels for conducting expermiments.
- Biological containment-the host into which foreign DNA is inserted should not proliferate outside the laboratory or transfer its DNA into other organisms.

- For research on pathogenic organisms, elaborate, controlled and self-contained rooms were recommended.
- For research on less dangerous organisms, units equipped with high quality filter systems should be used.
- No deliberate release of any organism containing recombinant DNA into the environment.

It may be noted here that although the NIH guidelines did not have the legal status, most institutions, companies and scientists voluntarily complied.

Relaxation of NIH guidelines

It was in 1980, the original NIH guidelines were considerably relaxed by NIH-RAC, based on the experience and experimental data obtained from the NIH-sponsored studies on recombinant DNA research. It was almost agreed that the original apprehensions on recombinant DNA research were unfounded.

It is a fact that the genetic engineering research flourished and progressed rapidly after relaxation of NIH guidelines. It may however be noted that NIH-RAC continues to be a watchdog over the DNA technology experiments.

Pharmaceutical products of recombinant DNA

As the recombinant DNA technology progressed, many pharmaceutical compounds of human health care are being produced through genetic manipulations. Most countries consider that the existing regulations for approval of pharmaceuticals of commercial use are adequate to ensure safety since the process by which the product manufactured is irrelevant. Thus, the recombinant DNA product (protein, vaccine, drug) is evaluated for its safety and efficacy like any other pharmaceutical product.

Genetically engineered organisms (GEOs)

Recombinant DNA research has resulted in the creation of many genetically engineered organisms. These include microorganisms, animals and plants. The latter two respectively result in transgenic animals and transgenic plants. The safety aspects and other related matters of GEOs are discussed at appropriate places in text, and some major highlights are given in Chapter 61.

THE FUTURE OF GENETIC ENGINEERING

DNA technology has largely helped scientists to understand the structure, function and regulation of genes. The development of new/modern biotechnology is primarily based on the success of DNA technology. Thus, the ***present biotechnology (more appropriately molecular biotechnology) has its main roots in molecular biology.***

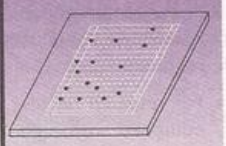
Biotechnology is an interdisciplinary approach for applications to human health, agriculture, industry and environment. The major objective of biotechnology is to solve problems associated with human health, food production, energy production and environmental control.

The major contributions of genetic engineering through the new discipline biotechnology are given in this book.

It is an accepted fact that the recombinant DNA technology has entered the main stream of human life and has become one of the most significant applications of scientific research. ***Biotechnology is regarded as more an art than a science.***

After the successful sequencing of human genome, many breakthroughs in biotechnology are expected in future.

Basic Techniques in Genetic Engineering



There are several techniques used in recombinant DNA technology or gene manipulation. The most frequently used methods are listed.

- Agarose gel electrophoresis.
- Isolation and purification of nucleic acids.
- Isolation of chromosomes.
- Nucleic acid blotting techniques.
- DNA sequencing.
- Alternate methods of DNA sequencing.
- Chemical synthesis of DNA.
- Methods of gene transfer (Chapters 6 and 49).
- Polymerase chain reaction (Chapter 8).
- Production of monoclonal antibodies (Chapter 17).
- Construction of gene library (Chapter 9).
- Radiolabeling of nucleic acids (Chapter 9).

Some details on the first seven techniques are described in this chapter while other technique are discussed elsewhere (as referred in the corresponding chapters). It may however, be noted that some specific applications of these techniques are frequently referred to as when needed in the corresponding chapters.

AGAROSE GEL ELECTROPHORESIS

Electrophoresis refers to the ***movement of charged molecules in an electric field***. The negatively charged molecules move towards the positive electrode while the positively charged molecules migrate towards the negative electrode.

Gel electrophoresis is a routinely used analytical technique for the separation/purification of specific DNA fragments. The gel is composed of either polyacrylamide or agarose. Polyacrylamide gel electrophoresis (PAGE) is used for the separation of smaller DNA fragments while agarose electrophoresis is convenient for the separation of DNA fragments ranging in size from 100 base pairs to 20 kb pairs. Gel electrophoresis can also be used for the separation of RNA molecules.

Agarose is a ***polysaccharide*** derived from seaweeds. It forms a solid gel when dissolved in aqueous solution at concentrations between 0.5 and 2.0% (w/v). It may be noted here that the agarose used for electrophoresis is more purified form of agar when compared to that used for culture purposes.

A diagrammatic view of the agarose gel electrophoresis unit is shown in ***Fig. 7.1***.

The DNA samples are placed in the wells of the gel surface and the power supply is switched on. As the DNA is negatively charged, DNA fragments

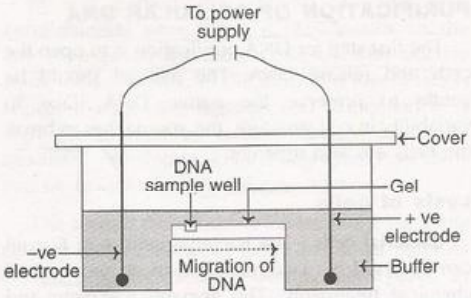


Fig. 7.1 : A diagrammatic representation of agarose gel electrophoresis system.

move through the gel towards the positive electrode. The rate of migration of DNA is dependent on the size and shape. In general, smaller linear fragments move faster than the larger ones. Hence, gel electrophoresis can be conveniently used for the separation of a mixture of DNA fragments, based on their size.

Agarose forms gels with pore sizes ranging from 100 to 300 nm in diameter. The actual pore size depends on the concentration of the agarose. The size of the pores determines the range of DNA fragments that can be separated on electrophoresis. For instance, a 0.3% agarose is used for the separation of DNA fragments between 5 and 50 kb, while a 5% agarose can separate 100–500 bp molecules.

The migration of DNA fragments during the course of electrophoresis can be monitored by using dyes with known migration rates. These dyes are added to the DNA samples before loading. The bands of the DNA can be detected by soaking the gel in ***ethidium bromide*** solution. When activated by ultraviolet radiation, DNA base pairs in association with ethidium bromide emit orange fluorescence. And in this way the DNA fragments separated in agarose electrophoresis can be identified (***Fig. 7.2***).

PULSED-FIELD GEL ELECTROPHORESIS (PFGE)

The large sized DNA molecules (~ 10 Mb) can be separated in agarose gels by using PFGE. This is made possible by periodically altering the direction of DNA migration by changing the orientation of electric field with respect to the gel. As the electric field orientation changes, the DNA molecules align themselves and migrate in new directions.

Limitation of PFGE : The major limitation of pulsed-field gel electrophoresis is that the samples do not migrate in straight lines. This makes the further analysis of DNA rather difficult.

CONTOUR-CLAMPED HOMOGENEOUS ELECTRICAL-FIELD ELECTROPHORESIS (CHEF)

CHEF is an improved method to apply alternating electrical fields to separate DNA molecules in electrophoresis. In this approach, the reorientation angle of electrical field is fixed at 120°C (or even at 90°) and electrophoresis carried out. By using CHEF, large DNA fragments (200–300 kb) can be routinely separated in a matter of hours.

POLYACRYLAMIDE GEL ELECTROPHORESIS (PAGE)

Polyacrylamide gel is composed of chains of acrylamide monomers cross-linked with methylenebisacrylamide units. The pore size of the gel is dependent on the total concentration of the monomers and the cross links.

Polyacrylamide gel electrophoresis (PAGE) is used ***for the separation of single-stranded DNA molecuels that differ in length by just one nucleotide***. Agarose gels cannot be used for this purpose. This is because polyacrylamide gels have

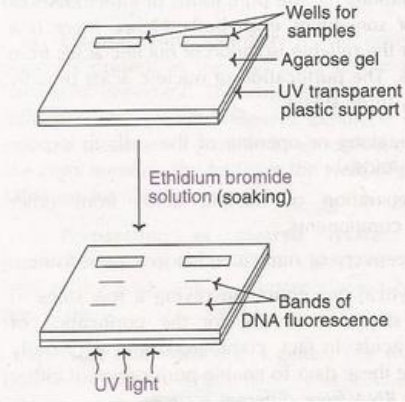


Fig. 7.2 : Identification of DNA bands separated by agarose gel electrophoresis.

smaller pore sizes than agarose gels and allow precise separation of DNA molecules from 10–1500 bp.

Polyacrylamide gel electrophoresis is a wonderful technique for the fine separation of DNA molecules that differ from each other even by 1-3 nucleotides. PAGE is used in DNA sequencing, and for the identification of amplified products of DNA by polymerase chain reaction (PCR).

OTHER USES OF GEL ELECTROPHORESIS

Besides the separation of DNA fragments, gel electrophoresis is also useful for understanding the molecular configuration of DNA molecules, and protein-nucleic acid interactions. The latter is based on the principle that binding of a protein to DNA usually results in the reduction of electrophoretic mobility.

The electrophoretic technique for protein-nucleic acid interaction involves the addition of a desired protein to double-stranded DNA fragments, separation of this complex and naked DNA by electrophoresis. The separated patterns can be visualized to understand the protein-nucleic acid interactions.

ISOLATION AND PURIFICATION OF NUCLEIC ACIDS

Almost all the experiments dealing with gene manipulations require pure forms of either DNA or RNA, or sometimes even both. Hence there is a need for the reliable isolation of nucleic acids from the cells. The purification of nucleic acids broadly involves three stages.

1. Breaking or opening of the cells to expose nucleic acids.
2. Separation of nucleic acids from other cellular components.
3. Recovery of nucleic acids in a pure form.

Analytical procedures involving a few steps to several steps are in use for the purification of nucleic acids. In fact, commercial kits are readily available these days to enable purification of either DNA or RNA from different sources.

The basic principles and procedures for nucleic acid purification are briefly described.

PURIFICATION OF CELLULAR DNA

The first step for DNA purification is to open the cells and release DNA. The method should be gentle to preserve the native DNA. Due to variability in cell structure, the approaches to break the cells are also different.

Lysis of cells

Bacterial cells : The bacterial cells (e.g. *E. coli*) can be lysed by a combination of enzymatic and chemical treatments. The enzyme lysozyme and the chemical ethylenediamine tetraacetate (EDTA) are used for this purpose. This is followed by the addition of a detergent such as sodium dodecyl sulfate (SDS).

Animal cells : Animal cells, particularly cultured animal cells, can be easily opened by direct treatment of cells with detergents (SDS).

Plant cells : Plant cells with strong cell walls require harsh treatment to break open. The cells are frozen and then ground in a morter and pestle. This is an effective way of breaking the cellulose walls.

Methods to purify DNA

There are two different approaches to purify DNA from the cellular extracts.

1. **Purification of DNA by removing cellular components :** This involves the degradation or complete removal of all the cellular components other than DNA. This approach is suitable if the cells do not contain large quantities of lipids and carbohydrates.

The cellular extract is centrifuged at a low speed to remove the debris (e.g. pieces of cell wall) that forms a pellet at the bottom of the tube. The supernatant is collected and treated with phenol to precipitate proteins at the interface between the organic and aqueous layers. The aqueous layer, containing the dissolved nucleic acids, is collected and treated with the enzyme ribonuclease (RNase). The RNA is degraded while the DNA remains intact. This DNA can be precipitated by adding ethanol and isolated after centrifugation, and suspended in an appropriate buffer.

2. **Direct purification of DNA :** In this approach, the DNA itself is selectively removed from the cellular extract and isolated. There are two ways for direct purification of DNA.

In one method, the addition of a detergent ***cetyltrimethyl ammonium*** (CTAB) results in the formation of an insoluble complex with nucleic acids. This complex, in the form of a precipitate is collected after centrifugation and suspended in a high-salt solution to release nucleic acids. By treatment with RNase, RNA is degraded. Pure DNA can be isolated by ethanol precipitation.

The second technique, is based on the principle of tight binding between DNA and silica particles in the presence of a denaturing agent such as guanidinium thiocyanate. The isolation of DNA can be achieved by the direct addition of silica particles and guanidinium thiocyanate to the cellular extract, followed by centrifugation. Alternately, a column chromatography containing silica can be used, and through this the extract and guanidinium thiocyanate are passed. The DNA binds to the silica particles in the column which can be recovered.

PURIFICATION OF PLASMID DNA

Pure forms of plasmid DNA are often required in genetic engineering experiments. The prerequisite for a successful isolation of plasmid DNA is the appropriate and gentle lysis of the host cells, so that the plasmid DNA is released without too much contamination of chromosomal DNA. The high molecular weight host cell chromosomal DNA can be removed along with the cell debris by high-speed centrifugation. This results in the formation of a ***cleared lysate*** from which pure plasmid DNA can be isolated. Of the several methods in use to isolate plasmid DNA, two are briefly described.

Isopycnic centrifugation method

The cleared lysate (referred above) is treated with a solution of cesium chloride (CsCl) containing ethidium bromide (EtBr). ***EtBr can bind with linear DNA molecules*** (chromosomal DNA fragments and open plasmid DNA) and not with circular plasmid DNA. The density of DNA-EtBr complex is much lower than the plasmid DNA. The circular plasmid DNA can be separated by isopycnic centrifugation in a CsCl-EtBr gradient (***Fig. 7.3***).

Birnboim and Doly method

The technique developed by Birnboim and Doly (1979) is widely used for the purification of plasmid DNA. This method is based on the principle that

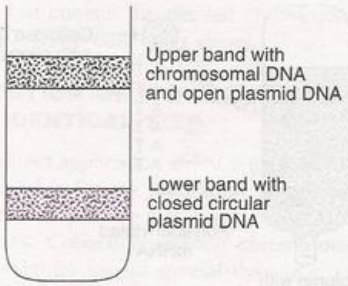


Fig. 7.3 : Purification of plasmid DNA by isopycnic centrifugation in CsCl-EtBr gradient.

there is a narrow range of ***pH (12.0–12.5) at which denaturation of linear DNA and not closed circular DNA occurs.***

When the cleared lysate is subjected to carefully monitored alkaline pH (12.0 to 12.5), the plasmid DNA molecules remain in solution while all other DNA molecules get denatured and precipitate. This precipitate can be removed by centrifugation. The plasmid DNA in the supernatant can be concentrated by ethanol precipitation. Sometimes, further purification of plasmid DNA may be necessary which can be achieved by gel filtration.

Factors affecting plasmid DNA purification

Several factors influence the purification of plasmid DNA. The most important ones are briefly described.

1. **Plasmid copy number :** The number of plasmids present in the host cells at the time of harvest is important. The copy number in turn is influenced by growth medium, genotype of host cell and the stage of growth. In general, the ***higher the copy number, the better is the yield of plasmid DNA on purification.***

2. **Preparation of cleared lysate :** The methodology adopted and the care taken for the lysis of the host cells to prepare cleared lysate is important.

3. **Host cells with *end A* gene :** The wild-type host cells posses *endA* gene that encodes the enzyme endonuclease I. The functions of this enzyme are not clearly known. It is however, observed that cell strains bearing *endA*

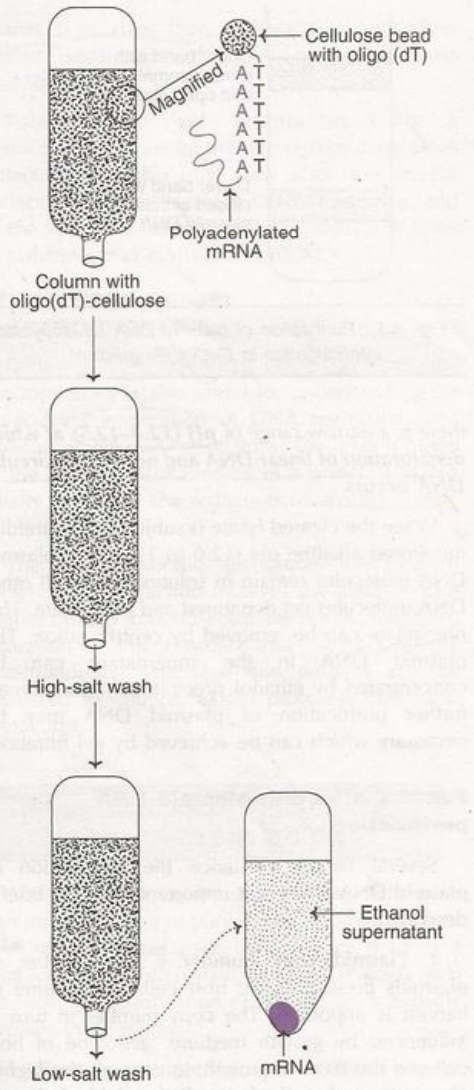


Fig. 7.4 : Purification of mRNA by affinity chromatography with oligo(dT)-cellulose.

mutations improve the stability and yield of plasmid DNA.

PURIFICATION OF mRNA

Among the RNAs, mRNA is frequently required in a pure form for genetic experiments.

After the cells are disrupted on lysis by different techniques (See p. 99), the cellular extract is deproteinised by treatment with phenol or phenol/chloroform mixtures. On centrifugation, the nucleic acids get concentrated in the upper aqueous phase which may then be precipitated by using isopropanol or ethanol.

The purification of mRNA can be achieved by affinity chromatography using oligo (dT)-cellulose (***Fig. 7.4***). This is based on the principle that oligo (dT)-cellulose can specifically bind to the poly (A) tails of eukaryotic mRNA. Thus, by this approach, it is possible to isolate mRNA from DNA, rRNA and tRNA.

As the nucleic acid solution is passed through an affinity chromotographic column, the oligo(dT) binds to poly(A) tails of mRNA. By washing the column with high-salt buffer, DNA, rRNA and tRNA can be eluted, while the mRNA is tightly bound. This mRNA can be then eluted by washing with low-salt buffer. The mRNA is precipitated with ethanol and collected by centrifugation (***Fig. 7.4***).

ISOLATION OF CHROMOSOMES

Separation of large chromosomes of eukaryotes is not possible by conventional electrophoresis. The individual chromosomes of eukaryotes can be separated by ***fluorescence-activated cell sorting (FACS)***, also known as ***flow cytometry*** or ***flow karyotyping***.

FLUORESCENCE-ACTIVATED CELL SORTING

To carry out FACS, the dividing cells (with condensed chromosomes) are carefully broken open, and a mixture of intact chromosomes is prepared. These chromosomes are then stained with a fluorescent dye. The quantity of the dye that binds to a chromosome depends on its size. Thus, larger chromosomes (with more DNA) bind more dye and fluoresce more brightly than the smaller ones.

The dye-mixed chromosomes are diluted and passed through a fine aperture that results in the formation of a stream of droplets. Each droplet contains a single chromosome. The fluorescence of the chromosomes is detected by a laser. When the fluorescence indicates that the chromosome illuminated by the laser is the one desired,

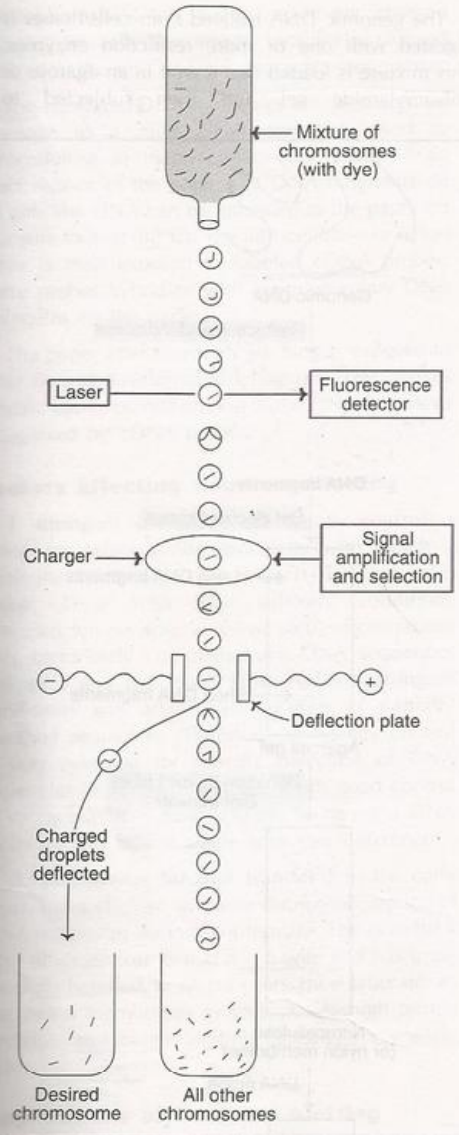


Fig. 7.5 : Separation of chromosomes by fluorescence-activated cell sorting (FACS).

electrical charge is specifically applied to these droplets (and no others) which get charged. This results in the deflection of the droplets with the desired chromosome which can be separated from the rest and collected (***Fig. 7.5***). Uncharged droplets that do not contain the desired chromosome pass through a waste collection vessel.

COLLECTION OF CHROMOSOMES WITH IDENTICAL SIZE

The direct application of FACS (described above) is not suitable for the separation of chromosomes with identical sizes. e.g. chromosomes 21 and 22 in humans. Collection of such chromosomes can be achieved by use of special dyes (e.g. Hoechst 33258 and chromomycin A_3) which bind to AT-rich DNA or GC-rich DNA. The rest of the procedure is the same that has been described.

It is convenient to separate two or more chromosomes with identical sizes by different dyes. This is possible since no two chromosomes are likely to contain identical GC/AT contents.

NUCLEIC ACID BLOTTING TECHNIQUES

Blotting techniques are very widely used analytical tools for the specific ***identification of desired DNA or RNA fragments*** from thousands of molecules. Blotting refers to the ***process of immobilization of sample nucleic acids or solid support*** (nitrocellulose or nylon membranes). The blotted nucleic acids are then used as targets in the hybridization experiments for their specific detection. An outline of the nucleic acid blotting technique is depicted in ***Fig. 7.6.***

Types of blotting techniques

The most comonly used blotting techniques are listed below

- Southern blotting (for DNA)
- Northern blotting (for RNA)
- Dot blotting (DNA/RNA)
- Colony and plaque hybridization (cells from colony).

The Southern blotting is named after the scientist Ed Southern (1975) who developed it. The other names Northern blotting and Western blotting are laboratory jargons which are now accepted. ***Western blotting involves the transfer of protein blots and their identification by using specific antibodies.***

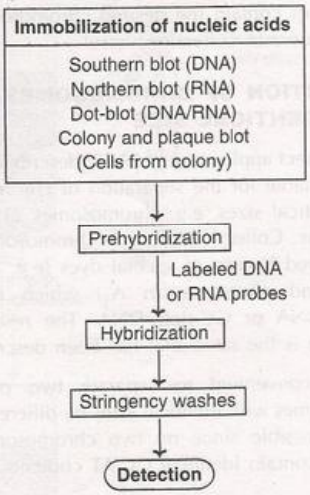


Fig. 7.6 : An outline of the nucleic acid blotting techniques.

A diagrammatic representation of a typical blotting apparatus is depicted in ***Fig. 7.7***.

SOUTHERN BLOTTING

Southern blotting technique is the first nucleic acid blotting procedure developed in 1975 by Southern. It is depicted in ***Fig. 7.8***, and briefly described.

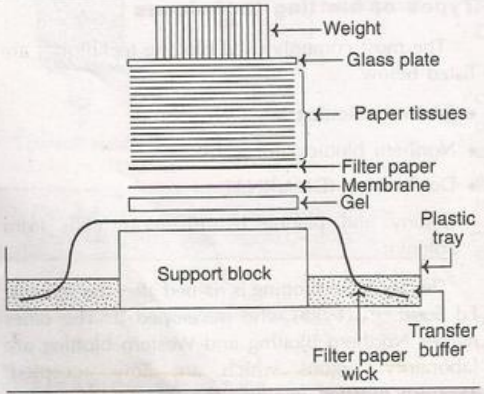


Fig. 7.7 : Diagrammatic representation of a typical blotting apparatus.

The genomic DNA isolated from cells/tissues is digested with one or more restriction enzymes. This mixture is loaded into a well in an agarose or polyacrylamide gel and then subjected to

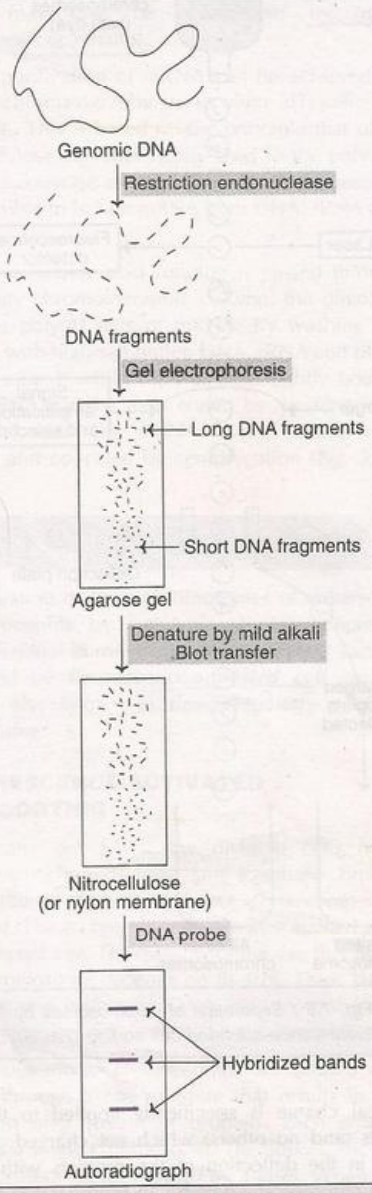


Fig. 7.8 : Southern blotting technique.

electrophoresis. DNA, being negatively charged migrates towards the anode (positively charged electrode); the smaller DNA fragments move faster.

The separated DNA molecules are denatured by exposure to a mild alkali and transferred to nitrocellulose or nylon paper. This results in an exact replica of the pattern of DNA fragments on the gel. The DNA can be annealed to the paper on exposure to heat (80°C). The nitrocellulose or nylon paper is then exposed to labeled cDNA probes. These probes hybridize with complementary DNA molecules on the paper.

The paper after thorough washing is exposed to X-ray film to develop autoradiograph. This reveals specific bands corresponding to the DNA fragments recognized by cDNA probe.

Factors affecting Southern blotting

1. **Stringent conditions (stringency control) :** Stringency refers to the specificity with which a particular DNA target sequence is detected by a probe. Thus, with high stringent conditions (elevated temperature and low salt concentration) only completely complementary DNA sequences will bind and hybridize. However, low stringent conditions will allow hybridization of partially matched sequences. Therefore, stringency control is very essential for specific detection of DNA molecules in Southern blotting. With good control of stringency, it is now possible to detect a DNA molecule just with a single base pair difference.

2. **Membranes for blot transfer :** In the early years, nitrocellulose was used for immobilization of DNA molecules during blot transfer. The drawback with nitrocellulose is that it is fragile and has to be carefully handled. In recent years, most laboratories use nylon membranes as they posses high tensile strength, and better binding capacity for nucleic acids.

Applications of Southern blotting

Southern blotting technique is extremely specific and sensitive, although it is a simple technique. Some of the applications are listed.

- It is an invaluable method in gene analysis.
- Important for the confirmation of DNA cloning results.
- Useful for mapping restriction sites around a single copy gene sequence.

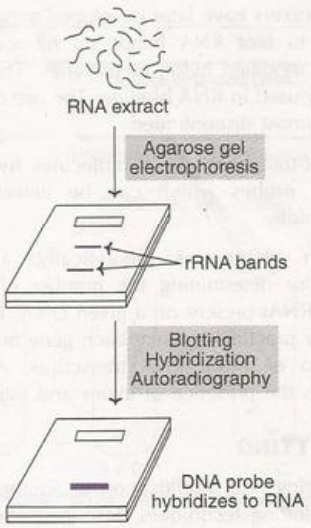


Fig. 7.9 : An outline of Northern blotting.

- Forensically applied to detect minute quantities of DNA (to identify parenthood, thieves, rapists etc.).
- Highly useful for the determination of restriction fragment length polymorphism (RFLP) associated with pathological conditions.
- DNA pieces from one species (e.g. human) can be used to detect DNA molecules from related species (e.g. chimpanze, cow). This technique is referred to as ***Zoo-blotting***.

NORTHERN BLOTTING

Northern blotting is the technique for the ***specific identification of RNA molecules***. The procedure adopted is almost similar to that described for Southern blotting and is depicted in ***Fig. 7.9***. RNA molecules are subjected to electrophoresis, followed by blot transfer, hybridization and autoradiography.

RNA molecules do not easily bind to nitrocellulose paper or nylon membranes. Blot-transfer of RNA molecules is carried out by using a chemically reactive paper prepared by diazotization of aminobenzyloxymethyl to create diazobenzyloxymethyl (DBM) paper. The RNA can covalently bind to DBM paper.

Some workers have later developed appropriate conditions to blot RNA bands on nitrocellulose paper, and modified nylon membranes. These are now widely used in RNA blotting. The use of DBM papers is almost discontinued.

The blot-transferred RNA molecules hybridize with DNA probes which can be detected by autoradiography.

Northern blotting is theoretically, a good technique for determining the number of genes (through mRNA) present on a given DNA. But this is not really practicable since each gene may give rise to two or more RNA transcripts. Another drawback is the presence of exons and introns.

DOT-BLOTTING

Dot-blotting is a modification of Southern and Northern blotting techniques described above. In this approach, the ***nucleic acids (DNA or RNA) are directly spotted onto the filters***, and not subjected to electrophoresis. The hybridization procedure is the same as in original blotting techniques.

Dot-blotting technique is particularly useful in obtaining quantitative data for the evaluation of gene expression.

WESTERN BLOTTING

Western blotting involves the ***identification of proteins***. It is to very useful to understand the nucleic acid functions, particularly during the course of gene manipulations.

The technique of Western blotting involves the transfer of electrophoresed protein bands from polyacrylamide gel to nylon or nitrocellulose membrane. These proteins can be detected by specific protein-ligand interactions. Antibodies or lectins are commonly used for this purpose.

AUTORADIOGRAPHY

Autoradiography is the process of ***localization and recording of a radiolabel*** within a solid specimen, with the production of an ***image in a photographic emulsion***. These emulsions are composed of silver halide crystals suspended in gelatin.

When a β-particle or a γ-ray from a radiolabel passes through the emulsions, silver ions are converted to metallic silver atoms. This results in the development of a visible image which can be easily detected.

Direct autoradiography

Direct autoradiography is ideally suited for the detection of weak to medium strength β-emitting radionuclides (^{3}H, ^{14}C, ^{35}S). In this technique, the sample is placed in direct contact with the film. The radioactive emissions result in the development of black areas.

Indirect autoradiography

For the detection of highly energetic β-particles (e.g. ^{32}P, ^{125}I), direct autoradiography is not suitable. This is because these emissions pass through and beyond the film, and a major part of the energy gets wasted.

Indirect autoradiography is useful for the detection of highly energetic β-particles. In this technique, the β-particle energy is first converted to light by a scintillator, which then emits photons on exposure to photographic emulsion.

Applications of autoradiography

As already described, autoradiography is closely associated with blotting techniques for the detection of DNA, RNA and proteins.

COLONY AND PLAQUE BLOTTING

Colony and plaque blotting is a process of hybridization for the specific ***identification and purification of colonies*** (e.g. bacterial clones). This technique is depicted in ***Fig. 7.10***, and briefly described hereunder.

The desired bacteria are grown as colonies on an agar plate. When a nitrocellulose filter paper is overlaid on the agar plate, the colonies get transferred. They are permanently fixed to the paper by heat. On treatment with alkali (NaOH), the cells lyse and the DNA gets denatured. When these DNA prints are exposed to specific probes (radiolabel), hybridization occurs. The hybrid complex can be localized and detected by autoradiography.

A similar strategy (described above for bacteria) can be used for the identification of phages and fragments of phage libraries.

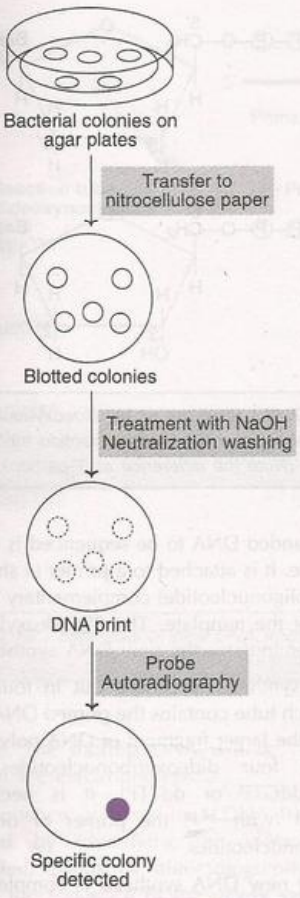


Fig. 7.10 : Colony and plaque blotting technique.

DNA SEQUENCING

Determination of nucleotide sequence in a DNA molecule is the basic and fundamental requirement in biotechnology. DNA sequencing is important to understand the functions of genes, and basis of inherited disorders. Further, DNA cloning and gene manipulation invariably require knowledge of accurate nucleotide sequence.

MAXAM AND GILBERT TECHNIQUE

The first DNA sequencing technique, ***using chemical reagents***, was developed by Maxam and Gilbert (1977). This method is briefly described below (**Fig. 7.11**).

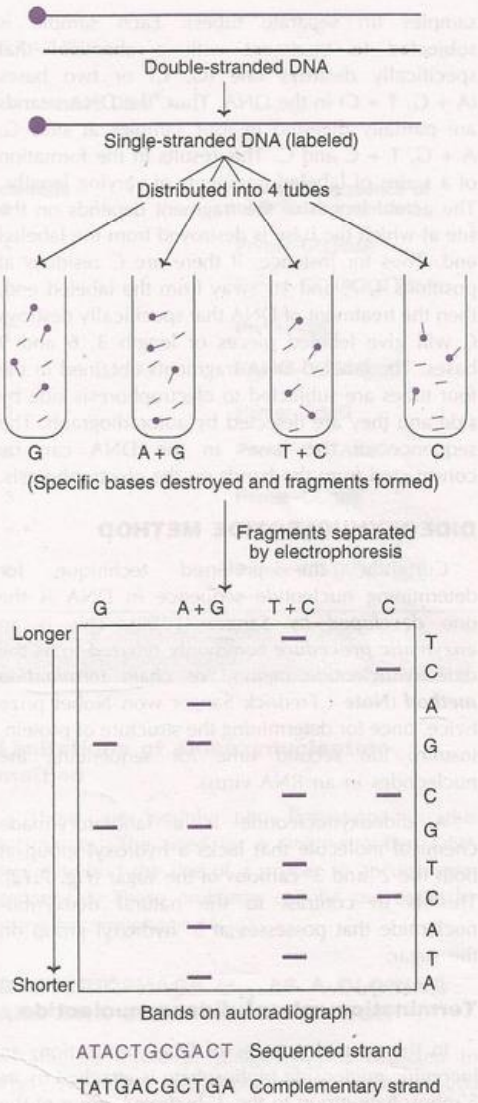


Fig. 7.11 : Maxam and Gilbert method for DNA sequencing.

A strand of source DNA is labeled at one end with ^{32}P. The two strands of DNA are then separated. The labeled DNA is distributed into four

samples (in separate tubes). Each sample is subjected to treatment with a chemical that specifically destroys one (G, C) or two bases (A + G, T + C) in the DNA. Thus, the DNA strands are partially digested in four samples at sites G, A + G, T + C and C. This results in the formation of a series of labeled fragments of varying lengths. The actual length of the fragment depends on the site at which the base is destroyed from the labeled end. Thus for instance, if there are C residues at positions 4, 7, and 10 away from the labeled end, then the treatment of DNA that specifically destroys C will give lebeled pieces of length 3, 6 and 9 bases. The labeled DNA fragments obtained in the four tubes are subjected to electrophoresis side by side and they are detected by autoradiograph. The sequence of the bases in the DNA can be constructed from the bands on the electrophoresis.

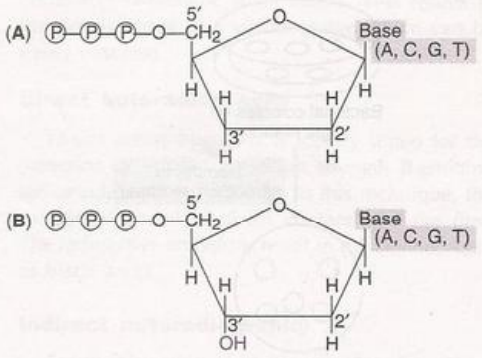


Fig. 7.12 : Structure of (A) dideoxynucleotide triphosphate and (B) deoxynucleotide triphosphate (Note the difference at 3′-carbon).

DIDEOXYNUCLEOTIDE METHOD

Currently, the preferred technique for determining nucleotide sequence in DNA is the one ***developed by Sanger*** (1980). This is an ***enzymatic procedure*** commonly referred to as the dideoxynucleotide method or ***chain termination method*** (**Note :** Fredrick Sanger won Nobel prize twice, once for determining the structure of protein, insulin; the second time for sequencing the nucleotides in an RNA virus).

A dideoxynucleotide is a laboratory-made chemical molecule that lacks a hydroxyl group at both the 2′ and 3′ carbons of the sugar (***Fig. 7.12***). This is in contrast to the natural deoxyribonucleotide that possesses at 3′ hydroxyl group on the sugar.

Termination role of dideoxynucleotide

In the normal process of DNA replication, an incoming nucleoside triphosphate is attached by its 5′-phosphate group to the 3′-hydroxyl group of the last nucleotide of the growing chain (Refer Chapter 3) when a dideoxynucleotide is incorporated to the growing chain, no further replication occurs. This is because dideoxynucleotide, lacking a 3′-hydroxyl group, cannot form a phosphodiester bond and thus the DNA synthesis terminates.

Sequencing method

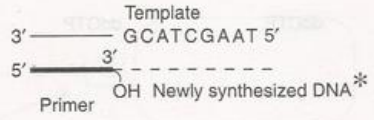
The process of sequencing DNA by dideoxynucleotide method is briefly described. A single-stranded DNA to be sequenced is chosen as a template. It is attached to a primer (a short length of DNA oligonucleotide) complementary to a small section of the template. The 3′-hydroxyl group of the primer initiates the new DNA synthesis.

DNA synthesis is carried out in four reaction tubes. Each tube contains the primed DNA, ***Klenow subunit*** (the larger fragment of DNA polymerase of *E. coli*), four dideoxyribonucleotides (ddATP, ddCTP, ddGTP or ddTTP). It is necessary to radiolabel (with ^{32}P) the primer or one of the deoxyribonucleotides.

As the new DNA synthesis is completed, each one of the tubes contains fragments of DNA of varying length bound to primer. Let us consider the first reaction tube with dideoxyadenosine (ddATP). In this tube, DNA synthesis terminates whenever the growing chain incorporated ddA (complementary to dT on the template strand). Therefore, this tube will contain a series of different length DNA fragments, each ending with ddA. In a smilar fashion, for the other 3 reaction tubes, DNA synthesis stops as the respective dideoxynucleotides are incorporated.

The synthesis of new DNA fragments in the four tubes is depicted in ***Fig. 7.13***.

The DNA pieces are denatured to yield free strands with radiolabel. The samples from each tube are separated by polyacrylamide gel electrophoresis. This separation technique resolves DNA pieces, different in size even by a single



Reaction tube with dideoxynucleotide	Primer with nucleotide extended	Primer with sequence of nucleotides extended
ddATP	Primer + 4	Primer–CGTddA
	Primer + 9	Primer–CGTAGCTTddA
ddCTP	Primer + 1	Primer–ddC
	Primer + 6	Primer–CGTAGddC
ddGTP	Primer + 2	Primer–C**ddG**
	Primer + 5	Primer–CGTA**ddG**
ddTTP	Primer + 3	Primer–CGddT
	Primer + 7	Primer–CGTAGCddT
	Primer + 8	Primer–CGTAGCTddT

***Fig. 7.13** : Synthesis of new DNA fragments in the presence of dideoxynucleotides (*the size of the new DNA is variable, depending on the chain termination).*

nucleotide. The shortest DNA will be the fastest moving on the electrophoresis.

The sequence of bases in a DNA fragment is determined by identifying the electrophoretic (radiolabeled) bands by autoradiography. In the ***Fig. 7.14***, the sequence of the newly synthesized DNA fragment that is complementary to the original DNA piece is shown. It is conventional to read the bands from bottom to top in 5′ to 3′ direction. By noting the order of the bands first C, second G, third T and so on, the sequence of the DNA can be determined accurately. As many as 350 base sequences of a DNA fragment can be clearly identified by using autoradiographs.

Modifications of dideoxynucleotide method

Replacement of ^{32}P-radiolabel by ^{33}P or ^{35}S improves the sharpness of autoradiographic images. DNA polymerase of the thermophilic bacterium, *Thermus aquaticus* (in place of Klenow fragment of *E. coli* DNA polymerase I) or a modified form of phage T7 DNA polymerase (sequenase) improves the technique.

Limitations of dideoxynucleotide method

There are mainly two limitations in this techniques. The need for a single-stranded DNA template and the use of a primer to an unknown sequences. These problems can be overcome by using bacteriophage M_{13}.

BACTERIOPHAGE M_{13} AS A CLONING AND DNA SEQUENCING VECTOR

The life cycle of phage M_{13} is depicted in ***Fig. 7.15***. This virus contains a single-stranded circular DNA. When it infects *E. coli*, the double-stranded DNA is synthesized which is a replicative form (RF). RF DNA replicates until 50–100 copies are produced. Now the mode of replication is changed to synthesize single-stranded DNA (ssDNA) molecules. Each of the ssDNA of phage M_{13} is coated with a protein. The phage particles then freely pass through the membrane to be released out. Thus, the *E. coli* cells infected with M_{13} can continuously secrete infective particles, without undergoing lysis.

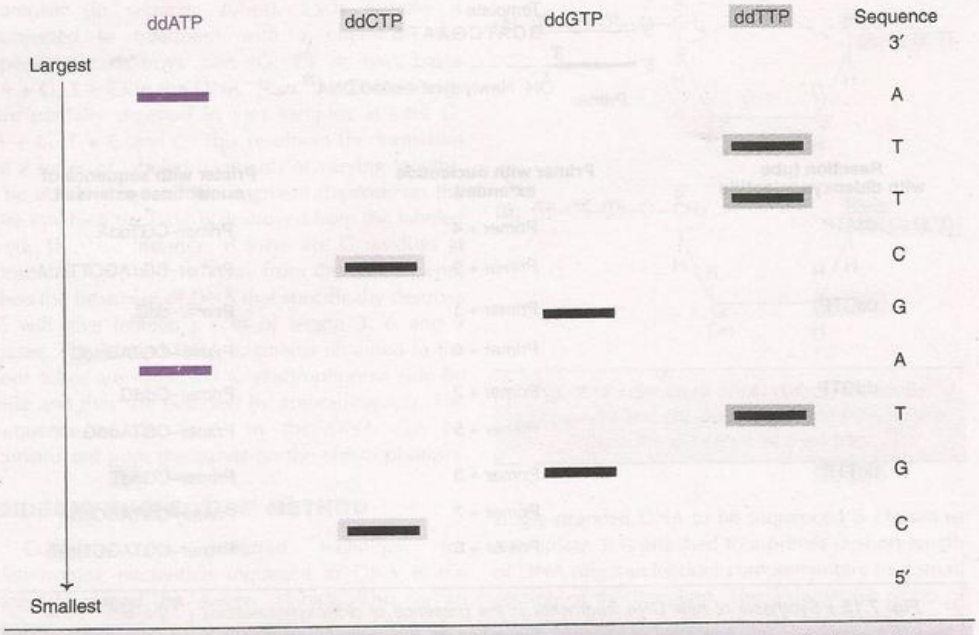


Fig. 7.14 : Sequence of the newly synthesized DNA fragment (complementary to original strand).

Bacteriophage M_{13} is employed as a cloning and DNA sequence determining vector. This is possible since the double-stranded replicative form can be isolated and used as a plasmid, while the single-stranded phage DNA can be employed as a template for DNA sequencing. Usually, a target DNA with less than 500 bp can be cloned and sequenced in phage M_{13}.

This double-stranded RF forms of phage M_{13} are isolated and the DNA to be cloned is inserted. This phage which is comparable to a plasmid is returned to *E. coli*. (The uptake of phages with inserts can be screened by using *lac Z* marker). As the phage undergoes its life cycle (***Fig. 7.15***), single-stranded DNA molecules of the inserted DNA are recovered. Sequence determination of the single-stranded DNA can be done by dideoxynucleotide method (See p. 102). The primer used is complementary to a part of phage M_{13} DNA which is close to the point of insertion. Therefore, a single primer can be used for different DNA molecules inserted.

For determining the sequence of a larger DNA (≤ 2,000 bp), it is necessary to clone different pieces of the DNA. From the sequences of these fragments (frequently overlapping), the final sequence can be built.

DNA SEQUENCING BY PRIMER WALKING

Primer walking or ***primer extension technique*** is employed for sequence determination of long pieces of DNA (≥ 5,000 bp). In this method, plasmid cloned DNA containing target DNA is annealed with a primer (synthetic oligonucleotide). The primer binds with the DNA close to the inserted DNA. Now the target DNA (250–350 nucleotides) is sequenced, most frequently by ***dideoxynucleotide method*** (described already).

From the base sequence of a fragment of the target DNA (determined in the first round), the second primer is chosen. It is an oligonucleotide to bind to a region approximately 300 nucleotides away from first primer binding. The next fragment of target DNA can be sequenced by another 250–300 bases (second round). This sequence will be useful to choose the third primer and determine the next 250–350 bases (third round). In this fashion, the primer walking is continued until the

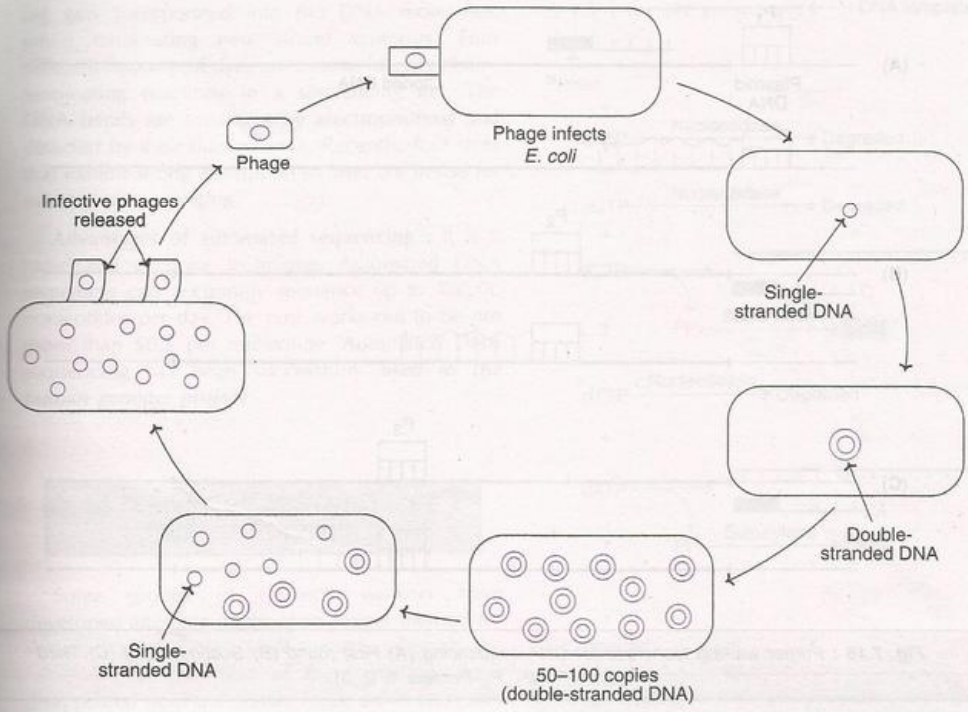


Fig. 7.15 : The life cycle of bacteriophage M_{13}

complete target DNA is sequenced. Three rounds of primer walking technique for DNA sequencing are depicted in ***Fig. 7.16***.

CHROMOSOME WALKING IN DNA SEQUENCING

Chromosome walking is a DNA base sequencing method in which the chromosome is analysed by extending one tip to reach the other. The technique basically involves systematically ***moving along the chromosome from a known location to an unknown location***, and thus ***determining the DNA sequence***. The method adopted in chromosome walking is depicted in ***Fig. 7.17***, and briefly described below.

A DNA fragment with approximately 40,000 base pairs can be sequenced. To start with a marker is used to identify the appropriate gene probe (from the DNA probe library). This gene probe must have a section of the base pair identical to the base pairs of the marker. This initial probe will hybridize only with the clones containing fragment A. This fragment A can be isolated, cloned and used as a probe to detect fragment B. This procedure of cloning and probing with fragments is repeated again and again until fragment D hybridizes with fragment E.

As is evident from the above description, chromosome walking uses probes derived from the ends of overlapping clones to facilitate a walk along with the DNA sequence. In the process of this walk, the sequence of the desired target gene can be identified.

Chromosome jumping

An improved strategy of chromosome walking is chromosome jumping. Many regions of DNA that are difficult to be cloned by walking can be jumped. The procedure involves the circularization of large genomic fragments generated by the

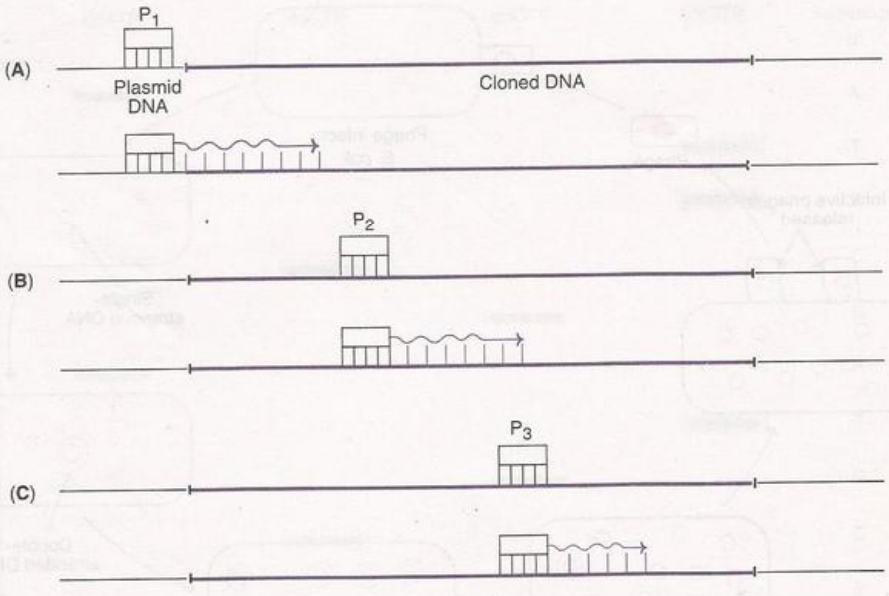


*Fig. 7.16 : Primer walking technique for DNA sequencing (**A**) First round (**B**) Second round (**C**) Third round (P_1, P_2, P_3–Primers 1, 2, 3).*

digestion of endonucleases. This is followed by cloning of the region lowering the closure of the fragment. By this approach, the ***DNA sequences located at far off places can be brought together and cloned***. A chromosome jumping can be constructed in this manner and used for long distance chromosome walks.

Applications of chromosome walking and jumping

The gene responsible for the disease ***cystic fibrosis (CF)*** has been ***identified*** by chromosome walking and jumping. The gene of protein responsible for CF is called cystic fibrosis transmembrane conductance regulator (CFTR) and is located on the long arm of chromosome 7.

AUTOMATED DNA SEQUENCING

DNA sequencing in the recent years is carried out by an ***automated DNA sequencer***. In this technique, flourescent tags are attached to chain-terminating nucleotides (dideoxynucleotides). This

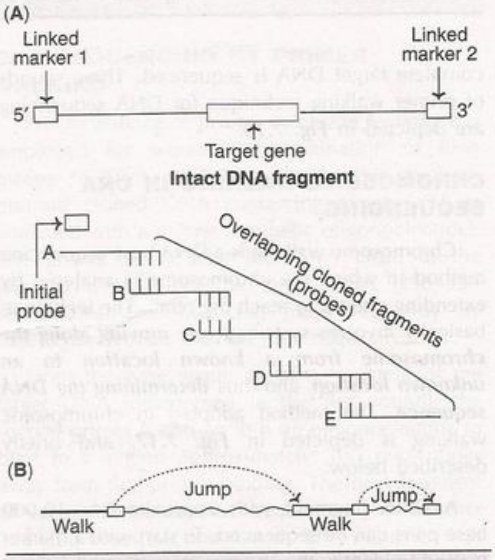


*Fig. 7.17 : (**A**) Chromosome walking (**B**) Chromosome jumping.*

tag gets incorporated into the DNA molecules, while terminating new strand synthesis. Four different fluorescent dyes are used to identify chain-terminating reactions in a sequencing gel. The DNA bands are separated by electrophoresis and detected by their fluorescence. Recently, four dyes that exhibit strong absorption in laser are in use for automated sequencing.

Advantages of automated sequencing : It is a rapid and accurate technique. Automated DNA sequencer can accurately sequence up to 100,00 nucleotides per day. The cost works out to be not more than $0.2 per nucleotide. Automated DNA sequencing has been successfully ***used in the human genome project***.

ALTERNATIVE METHODS OF DNA SEQUENCING

Some groups of research workers have developed alternate methods to Sanger method for sequencing of DNA. Unfortunately, despite the initial excitement, most of these methods have disappeared from the scene. There are at least two methods with some promise for DNA sequencing- ***pyrosequencing***, and ***gene chips*** (microarrays).

PYROSEQUENCING

Pyrosequencing is a DNA sequencing method that is based on the principle of ***determining which one of the four bases (A, G, C, T) is incorporated at each step while a DNA template is copied***. In this technique, the template copies in a straight forward manner whithout added dideoxy-nucleotides (ddNTPs). As the new strand is synthesized, the order in which the deoxy-nucleotides (dNTPs) are incorporated is detected, and the sequence can be read as the reaction proceeds (***Fig. 7.18***). The identification of base addition becomes possible since the addition of a nucleotide is accompanied by the release of a molecule of pyrophosphate. This can be detected by chemiluminescence technique.

In the procedure of pyrosequencing, each dNTP is added individually (not all four together), along with a nucleotidase enzyme. This enzyme degrades the dNTP if it is not incorporated into the newly synthesized DNA strand. Once the appropriate

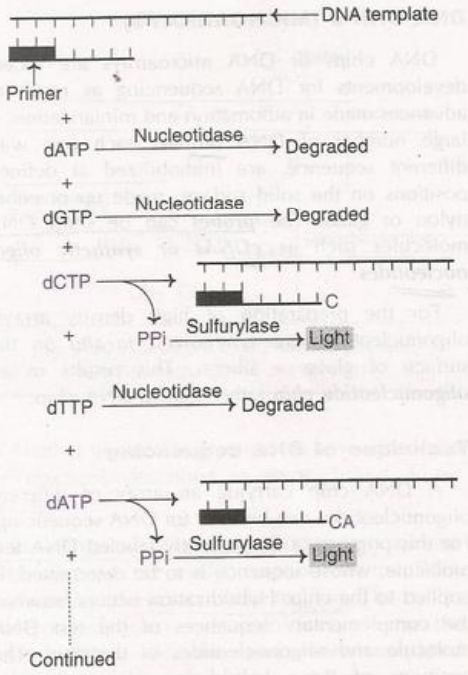


Fig. 7.18 : The technique of pyrosequencing.

nucleotide is incorporated into the new DNA strand, a molecule of pyrophosphate is released. This can be converted by the enzyme sulfurylase into a flash of light (chemiluminescence). By the addition of each dNTP separately, one after another, the order of the nucleotides added to the growing DNA strand can be followed, and the sequence can be identified.

Detection of the molecule ***pyro***phosphate is the basis of DNA ***sequencing***, hence the name ***pyrosequencing***. Since pyrosequencing does not require electrophoresis or any other DNA fragment separation technique, it is more rapid than chain termination sequencing.

The major ***limitation*** of pyrosequencing is that it is suitable to detect the ***sequence of about 200 nucleotides***. This is much less than the Sanger method. Many improvements are being made in pyrosequencing so that much longer DNA molecules can be sequenced. In fact, an automated system for pyrosequencing is also available now.

DNA CHIPS (MICROARRAYS)

DNA chips or DNA microarrays are recent developments for DNA sequencing as result of advances made in automation and miniarization. A large number of DNA probes, each one with different sequence, are immobilized at defined positions on the solid surface, made up of either nylon or glass. The ***probes*** can be short DNA molecules such as ***cDNAs or synthetic oligonucleotides***.

For the preparation of high density arrays, oligonucleotides are synthesized *in situ* on the surface of glass or silicon. This results in an ***oligonucleotide chip*** rather than a DNA chip.

Technique of DNA sequencing

A DNA chip carrying an array of different oligonucleotides can be used for DNA sequencing. For this purpose, a fluorescently labeled DNA test molecule, whose sequence is to be determined, is applied to the chip. Hybridization occurs between the complementary sequences of the test DNA molecule and oligonucleotides of the chip. The positions of these hybridizing oligonucleotides can be determined by confocal microscopy. Each hybridizing oligonucleotide represents an 8-nucleotide sequence that is present in the DNA probe. The sequence of the test DNA molecule can be deduced from the overlaps between the sequences of the hybridizing oligonucleotides (***Fig. 7.19***).

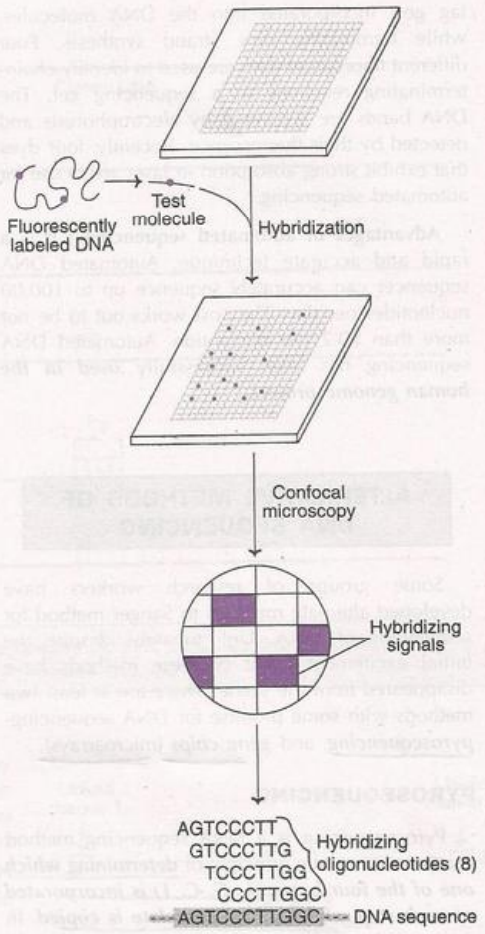


Fig. 7.19 : Microarray (or chip) technology in DNA sequencing.

Applications of DNA chips

There have been many successes with this relatively new technology of DNA chips. Some of them are listed.

- Identification of genes responsible for the development of nervous systems.
- Detection of genes responsible for inflammatory diseases.
- Construction of microarrays for every gene in the genome of *E. coli*, and almost all the genes of the yeast *Saccharomyces cerevisiae*.
- Expression of several genes in prokaryotes has been identified.
- Detection and screening of single nucleotide polymorphisms (SNPs).
- Rapid detection of microorganisms for environmental monitoring.

The future of DNA chips

The major limitation of DNA chips at present is the unavailability of complete genome arrays for higher eukaryotes, including humans. It is expected that within the next few years such DNA chips will be available. This will help the biotechnologists to capture the functional snapshots of the genome in action for higher organisms.

CHEMICAL SYNTHESIS OF DNA

Advances in the laboratory techniques have made it possible to chemically synthesize DNA in a short period. Thus, oligonucleotides of about 100 bases can be produced in about 10 hours. Laboratory synthesis of DNA (recently by use of ***DNA synthesizers or gene machines***) with specific sequence of nucleotides, rapidly and inexpensively, has significantly contributed to cloning. Chemical synthesis of DNA is based on the ability to protect the reactive – 5′ and – 3′ ends by blocking (protecting) them.

THE PHOSPHORAMIDITE METHOD

The phosphoramidite method is the technique of choice, currently in use, for the synthesis of DNA. The synthesis is carried out on a solid phase system i.e. by attaching the growing DNA strand to a solid support, in a reaction vessel. The procedure involves the following stages.

1. **Nucleoside attachment to solid support :** The initial nucleoside (base + sugar) is attached at 3′ end to an inert solid support, usually to glass beads with uniform pores, referred to as controlled pore glass (CPG) beads.

2. **Preparation of phosphoramidites :** For each one of the four bases (A, G, C and T), phosphoramidites can be synthesized. This is achieved by attaching dimethoxytrityl (DMT) group to the 5′-hydroxyl group of deoxyribose and a diisopropylamine group to the 3′-phosphite group of the nucleoside. A methyl residue protects the 3′-phosphite group. The general structure of phosphoramidite is depicted in ***Fig. 7.20***.

3. **Coupling :** A nucleotide precursor (i.e. a phosphoramidite) with its 3′-phosphorus couples with 5′-hydroxyl of the initial nucleoside bound to a CPG bead.

4. **Oxidation and deprotection :** The unstable trivalent phosphite is oxidized to pentavalent stable phosphate. This is followed by the removal of DMT that protects (deprotection) 5-hydroxyl group. For convenience, oxidation and deprotection are considered together in ***Fig. 7.21*** also, they are actually two independent reactions.

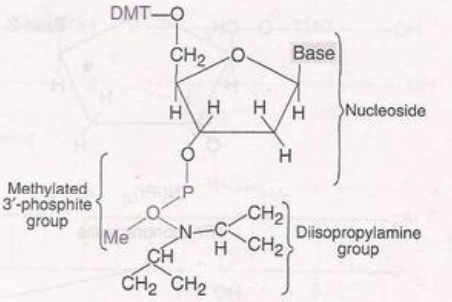


Fig. 7.20 : Structure of a phosphoramidite. (DMT-Dimethoxytrityl; Me-Methyl)

Another nucleotide precursor is now added and the reactions described in steps 3 and 4 are repeated.

Coupling, oxidation and deprotection are the three steps in the cyclic reaction for the DNA synthesis by phosphoramidite method. The cycles are repeated again and again to synthesise the desired DNA. At the end, the completed oligonucleotide is removed from the glass support and the groups protecting the bases and phosphates are cleaved.

Applications of synthesized oligonucleotides

Chemically synthesized oligonucleotides have a number of uses in biotechnology.

1. The linkers and adaptors are the oligonucleotides commonly employed in the preparation of recombinant DNA (Chapter 6).

2. *Chemically synthesized DNAs with known* sequences can be used for the synthesis of proteins/ polypeptides.

3. DNA probes, particularly the single-stranded oligonucleotide probes, can be synthesized in the laboratory. This is based on the codon sequence of mRNA which in turn is dependent on the amino acid sequence.

4. Single-stranded oligonucleotides are used as primers in polymerase chain reaction (PCR).

5. For *in vitro* mutagenesis, single-stranded oligonucleotides are employed.

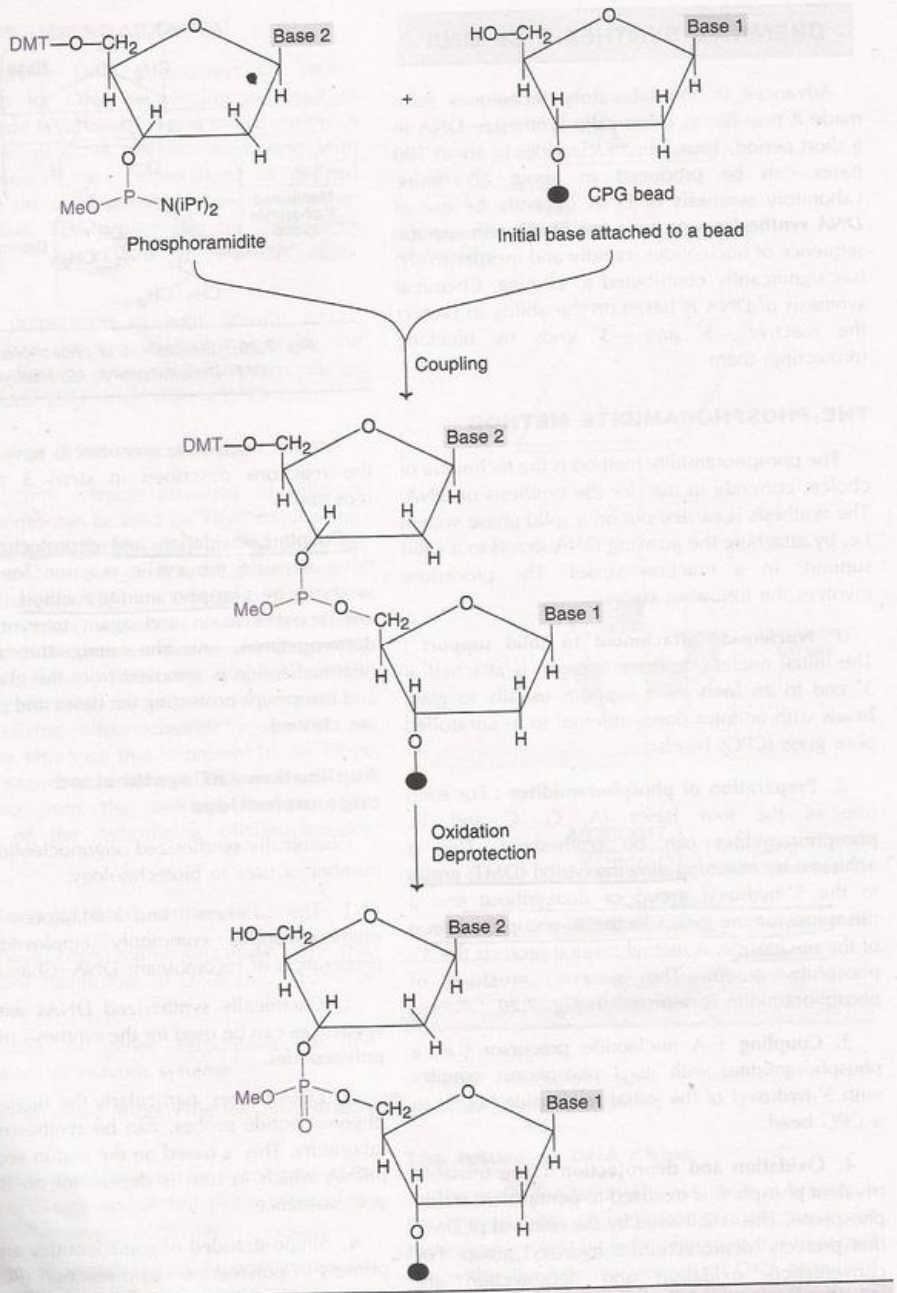


Fig. 7.21 : Chemical synthesis of DNA by phosphoramidite method (CPG-Controlled pore glass; DMT-Dimethoxytrityl; Me-Methyl; $N(iPr)_2$-Diisopropylamine).

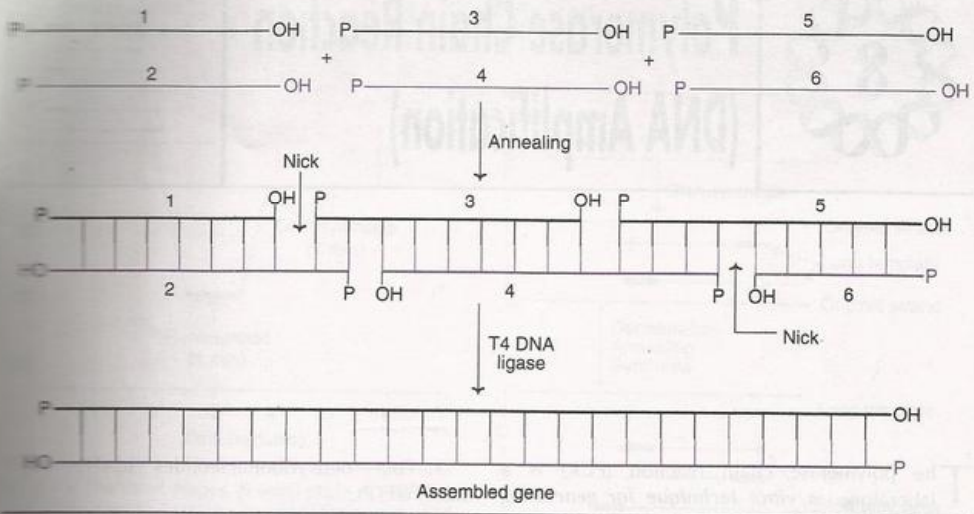


Fig. 7.22 : Chemical synthesis of a gene.

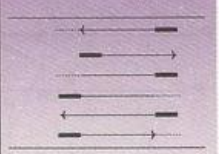
SYNTHESIS OF GENES

Gene is a double-stranded DNA; two single-stranded complementary oligonucleotides can be synthesized, separately, and on annealing they form double strands. This can be conveniently achieved for smaller genes (60–90 bp). However, the synthesis of longer genes (> 300 bp) is associated with some practical difficulties. This is mainly due to the fact that the coupling efficiency for the chemical synthesis of DNA is never 100%. To overcome this problem, small DNA fragments are synthesized and assembled (***Fig. 7.22***). Sealing of the nicks is achieved by use of T4 DNA ligase.

Another way of synthesizing genes is to produce overlapping oligonucleotides which on annealing contain large gaps. These gaps can be filled by enzymatic synthesis of DNA by DNA polymerase—while synthesizing genes in the laboratory, utmost care should be taken to see that the nucleotides are in the correct sequence (this can be checked by DNA sequence technique).

An Indian born and American settled scientist Har Gobind Khorana and his associates were the first to synthesize an entire tRNA gene for yeast alanyl tRNA in 1972.

Polymerase Chain Reaction (DNA Amplification)



The polymerase chain reaction (PCR) is a laboratory (*in vitro*) ***technique for generating large quantities of a specified DNA***. Obviously, PCR is a ***cell-free amplification technique*** for synthesizing multiple identical copies (billions) of ***any DNA*** of interest. Developed in 1984 by Karry Mullis (Nobel laurate, 1993), PCR is now considered as a basic tool for the molecular biologist. As is a photocopier a basic requirement in an office, so is the PCR machine in a molecular biology laboratory!

Principle of PCR

The double-stranded DNA of interest is denatured to separate into two individual strands. Each strand is then allowed to hybridize with a primer (renaturation). The primer-template duplex is used for DNA synthesis (the enzyme-DNA polymerase). These three steps—***denaturation, renaturation*** and ***synthesis*** are repeated again and again to generate multiple forms of target DNA.

TECHNIQUE OF PCR

The essential requirements for PCR are listed below

1. A target DNA (100–35,000 bp in length).

2. Two primers (synthetic oligonucleotides of 17–30 nucleotides length) that are complementary to regions flanking the target DNA.

3. Four deoxyribonucleotides (dATP, dCTP, dGTP, dTTP).

4. A DNA polymerase that can withstand at a temperature upto 95° C (i.e., thermostable).

The reaction mixture contains the target DNA, two primers (in excess), a thermostable DNA polymerase (isolated from the bacterium *Thermus aquaticus* (i.e., *Taq* DNA polymerase) and four deoxyribonucleoties. The actual technique of PCR involves repeated cycles for amplification of target DNA. Each cycle has three stages.

1. **Denaturation :** On raising the temperature to about 95° C for about one minute, the DNA gets denatured and the two strands separate.

2. **Renaturation or annealing :** As the temperature of the mixutre is slowly cooled to about 55° C, the primers base pair with the complementary regions flanking target DNA strands. This process is called renaturation or annealing. High concentration of primer ensures annealing between each DNA strand and the primer rather than the two strands of DNA.

3. **Synthesis :** The initiation of DNA synthesis occurs at 3'-hydroxyl end of each primer. The primers are extended by joining the bases complementary to DNA strands. The synthetic process in PCR is quite comparable to the DNA replication of the leading strand (Refer Chapter 3). However, the temperature has to be kept optimal as required by the enzyme DNA polymerase. For *Taq*

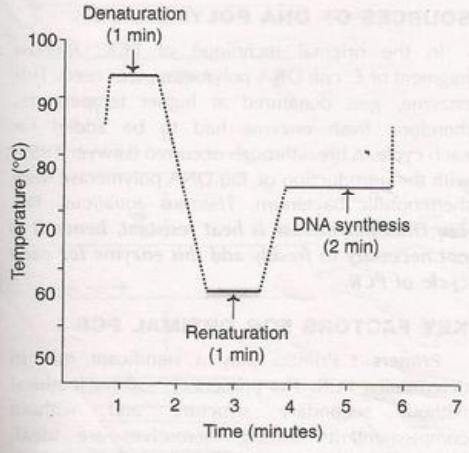


***Fig. 8.1** : The three stages in each cycle of PCR in relation to temperature and time (Each cycle takes approximately 3-5 minutes).*

DNA polymerase, the optimum temperature is around 75° C (for *E. coli* DNA polymerase, it is around 37° C). The reaction can be stopped by raising the temperature (to about 95° C).

The 3 stages of PCR in relation to temperature and time are depicted in ***Fig. 8.1***. Each cycle of PCR takes about 3-5 minutes. In the normal practice, the PCR is carried out in an automated machine.

As is evident from the ***Fig. 8.2*** (cycle I), the new DNA strand joined to each primer is beyond the sequence that is complementary to the second primer. These new strands are referred to as ***long templates*** and they will be used in the second cycle.

For the second cycle of PCR, the DNA strands (original + newly synthesized long template) are denatured, annealed with primers and subjected to DNA synthesis. At the end of second round, long templates, and short templates (DNA strands with primer sequence at one end, and sequence complementary to the other end primer) are formed.

In the third cycle of PCR, the original DNA strands along with long and short templates are the starting materials. The technique of denaturation, renaturation and synthesis are repeated. This procedure is repeated again and again for each cycle. It is estimated that at the end of 32nd cycle of PCR, about a million-fold target DNA is synthesized (***Table 8.1***). The short templates possessing precisely the target DNA as double-stranded molecules accumulate.

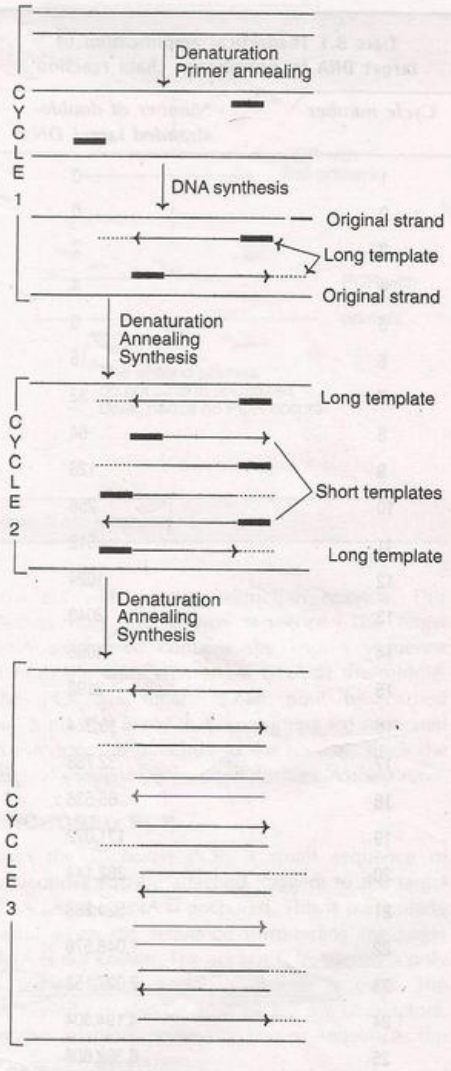


***Fig. 8.2** : The polymerase chain reaction (PCR) representing the initial three cycles (▬ indicate primers).*

TABLE 8.1 Theoretical amplification of target DNA by polymerase chain reaction

Cycle number	*Number of double-stranded target DNA*
1	0
2	0
3	2
4	4
5	8
6	16
7	32
8	64
9	128
10	256
11	512
12	1024
13	2048
14	4096
15	8192
16	16,384
17	32,768
18	65,536
19	131,072
20	262,144
21	524,288
22	1,048,576
23	2,097,152
24	4,194,304
25	8,388,608
26	16,777,216
27	33,544,432
28	67,108,864
29	134,217,728
30	268,435,456
31	536,870,912
32	1,073,741,824

SOURCES OF DNA POLYMERASE

In the original technique of PCR, Klenow fragment of *E. coli* DNA polymerase was used. This enzyme, gets denatured at higher temperature, therefore, fresh enzyme had to be added for each cycle. A breakthrough occurred (Lawyer 1989) with the introduction of *Taq* DNA polymerase from thermophilic bacterium, *Thermus aquaticus*. The ***Taq DNA polymerase is heat resistant, hence it is not necessary to freshly add this enzyme for each cycle of PCR.***

KEY FACTORS FOR OPTIMAL PCR

Primers : Primers play a significant role in determining PCR. The primers (17–30 nucleotides) without secondary structure and without complementarity among themselves are ideal. The complementary primers can hybridize to form ***primer dimer*** and get amplified in PCR. This prevents the multiplication of target DNA.

DNA polymerase : As already described, *Taq* DNA polymerase is preferred as it can withstand high temperature. In the ***hot-start protocol***, DNA polymerase is added after the heat denaturation step of the first cycle. This avoids the extension of the mismatched primers that usually occur at low temperature.

Taq polymerase lacks proof reading exonuclease (3'-5') activity which might contribute to errors in the products of PCR. Some other thermostable DNA polymerases with proof-reading activity have been identified e.g., *Tma* DNA polymerase from *Thermotoga maritama; Pfu* DNA polymerase from *Pyrococcus furiosus*.

Target DNA : In general, the shorter the sequence of target DNA, the better is the efficiency of PCR. However, in recent years, amplification of DNA fragments upto 10 kb has been reported. The sequence of target DNA is also important in PCR. Thus, GC-rich regions of DNA strand hinder PCR.

Promoters and inhibitors : Addition of proteins such as bovine serum albumin (BSA) enhances PCR by protecting the enzyme DNA polymerase. Humic acids, frequently found in archeological samples of target DNA inhibit PCR.

VARIATIONS OF PCR

The basic technique of the PCR has been described. Being a versatile technique, PCR is

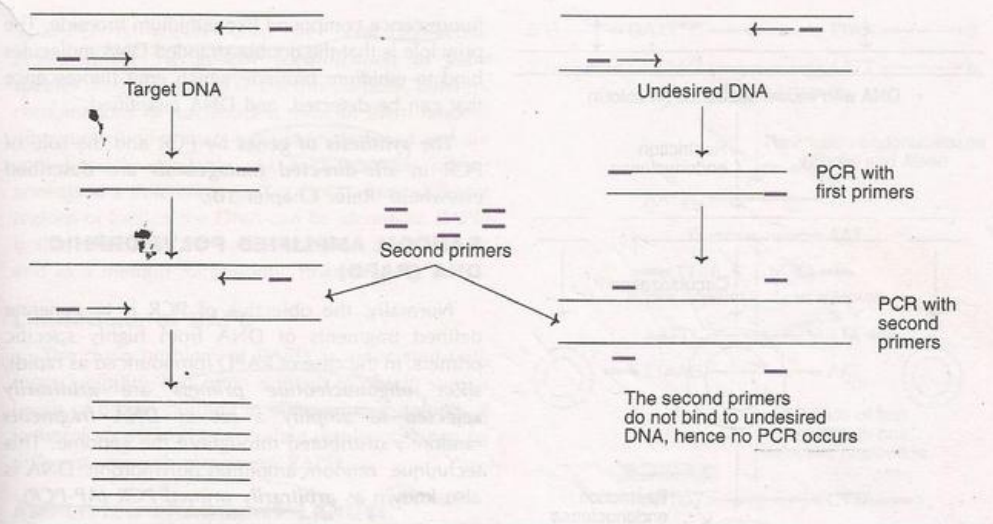


Fig. 8.3 : Nested polymerase chain reaction.

modified as per the specific demands of the situation. Thus, there are many variations in the original PCR, some of them are discussed, hereunder.

NESTED PCR

Sequence similarities between the target DNA and related DNA are very frequently seen. As a result of this, the primers may bind to both the DNAs and therefore even the undesired DNA also gets amplified in PCR. Use of ***nested primers*** increases the specificity of PCR, and selectively amplifies target DNA. Nested PCR is illustrated in ***Fig. 8.3***. In the first cycle of PCR, the products are both from target DNA and undesired DNA. A second set of internal primers is now used. They will selectively bind to target DNA and amplification proceeds.

INVERSE PCR

In the inverse PCR, amplification of DNA of the unknown sequences is carried out from the known sequence (***Fig. 8.4***). The target DNA is cleaved with a restriction endonuclease which does not cut the known sequence but cuts the unknown sequence on either side. The DNA fragments so formed are inverted and get circularized (DNA ligase is employed as a sealing agent). The circle containing the known sequences is now cut with another restriction enzyme. This cleaves only the known sequence. The target DNA so formed contains the known sequence at both the ends with target DNA at the middle. The PCR amplification can now be carried out. It may be noted that the primers are generated in the opposite direction to the normal, since the original sequence is inverted during circularization.

ANCHORED PCR

In the anchored PCR, a small sequence of nucleotides can be attached (tagged) to the target DNA i.e., the DNA is anchored. This is particularly useful when the sequence surrounding the target DNA is not known. The anchor is frequently a poly G tail to which a poly C primer is used. The anchoring can also be done by the use of adaptors. As the adaptors possess a known sequence, the primer can be chosen.

REVERSE TRANSCRIPTION PCR

The PCR technique can also be employed for the amplification of RNA molecules in which case it is referred to as reverse transcription — PCR (***RT-PCR***). For this purpose, the RNA molecule (mRNA) must be first converted to complementary DNA (cDNA) by the enzyme ***reverse transcriptase***. The cDNA then serves as the template for PCR. Different primers can be employed for the synthesis

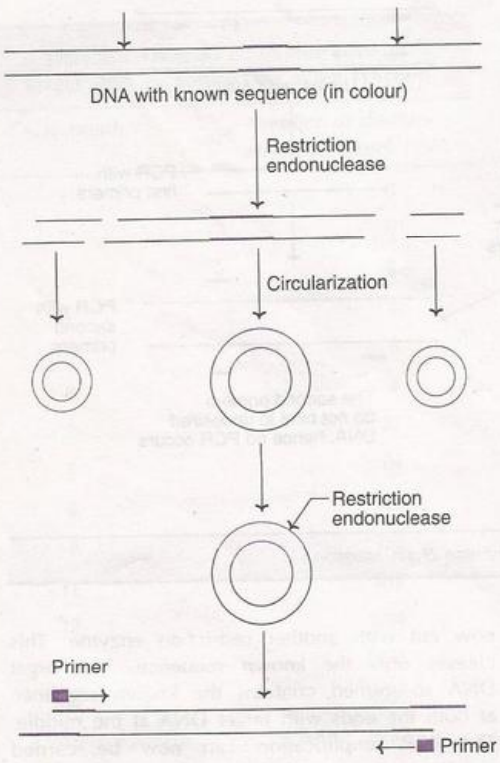


Fig. 8.4 : Inverse PCR.

of first strand of cDNA. These include the use of random primers, oligo dT primer and a sequence specific primer (***Fig. 8.5***).

ASYMMETRIC PCR

PCR technique can also be used for the synthesis of single-stranded DNA molecules, particularly useful for DNA sequencing (Chapter 7). In the asymmetric PCR, two primers in a ratio of 100 : 1 are used. After 20–25 cycles of PCR, one primer is exhausted. The result is that in the next 5–10 PCR cycles, only single-stranded DNAs are generated.

REAL-TIME QUANTITATIVE PCR

The quantification of PCR products in different cycles is not as simple as projected by thereotical considerations (***Table 8.1***). In practice, large variations occur. The most commonly used technique for measuring the quantity of PCR is by employing a fluorescence compound like eithidium bromide. The principle is that the double-stranded DNA molecules bind to ethidium bromide which emit fluorescence that can be detected, and DNA quantified.

The ***synthesis of genes*** by PCR and the role of PCR in ***site-directed mutagenesis*** are described elsewhere (Refer Chapter 10).

RANDOM AMPLIFIED POLYMORPHIC DNA (RAPD)

Normally, the objective of PCR is to generate defined fragments of DNA from highly specific primers. In the case of RAPD (pronounced as rapid), short ***oligonucleotide primers*** are ***arbitrarily selected to amplify*** a set of ***DNA fragments*** randomly distributed throughout the genome. This technique, random amplified polymorphic DNA is also known as ***arbitrarily primed PCR (AP-PCR)***.

The procedure of RAPD is comparable to the general technique of PCR. This method basically involves the use of a single primer at low stringency. A single short oligonucleotide (usually a 9–10 base primer) binds to many sites in the genome and the DNA fragments are amplified from them. The stringency of primer binding can be increased after a few PCR cycles. This allows the amplification of best mismatches. RAPD can be carefully designed so that it finally yields genome-specific band patterns that are useful for comparative analysis. This is possible since genomic DNA from two different individuals often produces different amplified patterns by RAPD. Thus, a particular DNA fragment may be generated for one individual and not for the other, and this represents DNA polymorphism which can be used as a genetic marker.

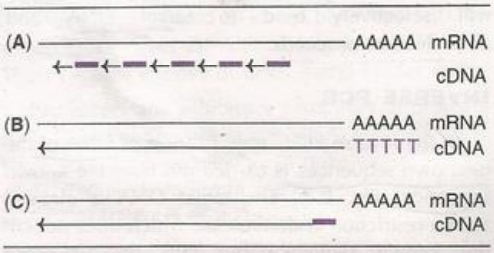


Fig. 8.5 : *Synthesis of first strand of cDNA in reverse transcription-PCR with different primers (**A**) Random primers (**B**) oligo dT primer (**C**) Sequence specific primer (**Note :** The primers are shown in colour).*

RAPD is widely used by plant molecular biologists for the ***genetic identification of plant species*** (Refer Chapter 53). For this purpose, different combinations of nucleotides, most of them random oligonucleotide primers have been designed and are commercially available. As each random primer anneals to a different region of DNA, many different regions of loci on the DNA can be identified. RAPD is thus useful for the construction of genetic maps and as a method for genomic fingerprinting.

Limitations of RAPD

The main problem of RAPD is associated with reproducibility. It is often difficult to obtain similar levels of primer binding in different experiments. It is therefore difficult to correlate results obtained by different research groups on RAPD.

AMPLIFIED FRAGMENT LENGTH POLYMORPHISM (AFLP)

AFLP is a very sensive method for detecting polymorphism in the genome. It is based on the principle of restriction fragment length polymorphism (RFLP, Refer Chapter 14) and RAPD. AFLP may be appropriately regarded as a diagnostic fingerprinting technique that detects genomic restriction fragments.

In the AFLP, PCR amplification rather than Southern blotting (mostly used in RFLP) is used for the detection of restriction fragments. It may be noted that AFLP is employed to detect the presence or absence of restriction fragments, and not the lengths of these fragments. This is the major difference between AFLP and RFLP. AFLP is very ***widely used in plant genetics***. It has not proved useful in the mapping of animal genomes, since this technique is mainly based on the presence of high rates of substitutional variations which are not found in animals. On the other hand, substitutional variations resulting in RFLPs are more common in plants.

The basic principle of AFLP involves the amplification of subsets of RFLPs using PCR (***Fig. 8.6***).

A genomic DNA is isolated and digested simultaneously with two different restriction endonucleases — *EcoRI* with a 6 base pair recognition site and *MseI* with a 4 base pair recognition site. These two enzymes can cleave the DNA and result in small fragments (< 1 kb) which can be amplified by PCR. For this purpose the DNA

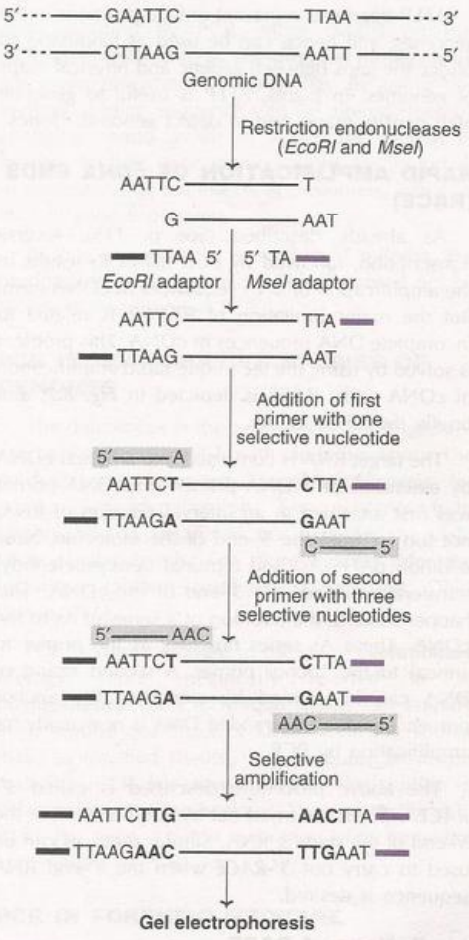


Fig. 8.6 : The technique of amplified fragment length of polymorphism (AFLP).

fragments are ligated with *EcoRI* and *MseI* adaptors. These common adaptor sequences (flanking genomic sequences) serve as primer binding sites on the restriction fragments. The DNA fragments can be amplified with AFLP primers each having only one selective nucleotide. These PCR products are diluted and used as templates for the selective amplification employing two new AFLP primers that have 2 or 3 selective nucleotides.

After the selective amplification by PCR, the DNA products are separated on a gel. The resultant DNA fingerprint is identified by autoradiography.

AFLP fragments represent unique positions in the genomes, and hence can be used as landmarks to bridge the gaps between genetic and physical maps of genomes. In plants, AFLP is useful to generate high density maps, and to detect genomic clones.

RAPID AMPLIFICATION OF cDNA ENDS (RACE)

As already described (See p. 115), reverse transcription, followed by PCR (RT-PCR) results in the amplification of RNA sequences in cDNA form. But the major limitation of RT-PCR is related to incomplete DNA sequences in cDNA. This problem is solved by using the technique rapid amplification of cDNA ends. RACE is depicted in ***Fig. 8.7***, and briefly described below.

The target RNA is converted into a partial cDNA by extension of a DNA primer. This DNA primer was first annealed at an interval position of RNA, not too far from the 5′-end of the molecule. Now addition dATP (As) and terminal deoxynucleotidyl transferase extends the 3′-end of the cDNA. This happens due to the addition of a series of As to the cDNA. These As series now act as the primer to anneal to the anchor primer. A second strand of DNA can be formed by extending the anchor primer. The double-stranded DNA is now ready for amplification by PCR.

The above procedure described is called ***5′-RACE***, since it is carried out by amplification of the 5′-end of the starting RNA. Similar protocol can be used to carry out ***3′-RACE*** when the 3′-end RNA sequence is desired.

Limitations of RACE

Since a specific primer is used, the specificity of amplification of RACE may not be very high. Another disadvantage is that the reverse transcriptase may not fully reach the 5′-ends of RNA, and this limits the utility of RACE. In recent years, some modifications have been done to improve RACE.

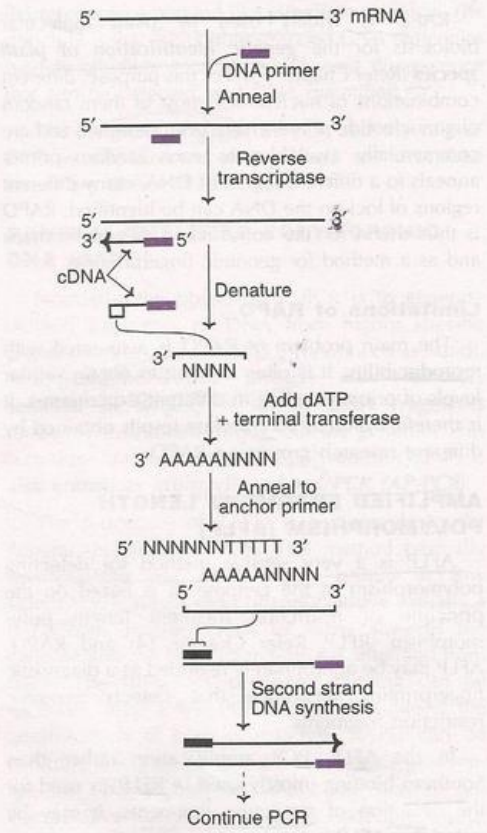


Fig. 8.7 : Rapid amplification of cDNA ends (RACE).

APPLICATIONS OF PCR

The advent of PCR had, and continues to have tremendous impact on molecular biology. The applications of PCR are too many to be listed here. Some of them are selectively and very briefly described. Other applications of PCR are discussed at appropriate places.

PCR IN CLINICAL DIAGNOSIS

The specificity and sensitivity of PCR is highly useful for the diagnosis of various diseases in humans. These include diagnosis of inherited disorders (genetic diseases), viral diseases, bacterial diseases etc.

The occurrence of genetic diseases frequently identified by restriction fragment length polymorphism (RFLP) can be employed only when there is a mutation resulting in a detectable change in the length of restriction fragment. Many genetic diseases occur without the involvement of RFLP.

For all such disorders, PCR technique is a real boon, as it provides direct information of DNA. This is done by amplification of DNA of the relevant region, followed by the direct analysis of PCR products.

Prenatal diagnosis of inherited diseases : PCR is employed in the prenatal diagnosis of inherited diseases by using chorionic villus samples or cells from amniocentesis. Thus, diseases like sickle-cell anemia, β-thalassemia and phenylketonuria can be detected by PCR in these samples.

Diagnosis of retroviral infections : PCR from cDNA is a valuable tool for diagnosis and monitoring of retroviral infections, e.g., HIV infection.

Diagnosis of bacterial infections : PCR is used for the detection of bacterial infection e.g., tuberculosis by *Mycobacterium tuberculosis*.

Diagnosis of cancers : Several virally-induced cancers (e.g., cervical cancer caused by human papilloma virus) can be detected by PCR. Further, some cancers which occur due to chromosomal translocation (chromosome 14 and 18 in follicular lymphoma) involving known genes are identified by PCR.

PCR in sex determination of embryos : Sex of human and live stock embryos fertilized *in vitro*, can be determined by PCR, by using primers and DNA probes specific for sex chromosomes. Further, this technique is also useful to detect sex — linked disorders in fertilized embryos.

PCR IN DNA SEQUENCING

As the PCR technique is much simpler and quicker to amplify the DNA, it is conveniently used for sequencing. For this purpose, single-strands of DNA are required. In asymmetric PCR (See p. 116), preferential amplification of a single-strand is carried out. In another method, ***strand removal*** can be achieved by digesting one strand (usually done by exonuclease by its action on 5'-phosphorylated strand).

PCR IN GENE MANIPULATION AND EXPRESSION STUDIES

The advantage with PCR is that the primers need not have complementary sequences for the target DNA. Therefore, the sequence of nucleotides in a piece of the gene (target DNA) can be manipulated and amplified by PCR. More details on this technique, referred to as ***site-directed mutagenesis***, are given elsewhere (Chapter 10). By using this method, coding sequence can be altered (thereby changing amino acids) to synthesize protein of interest. Further, gene manipulations are important in understanding the effects of promoters, initiators etc., in gene expression.

PCR is important in the study of mRNAs, the products of gene expression. This is carried out by reverse transcription — PCR.

PCR IN COMPARATIVE STUDIES OF GENOMES

The differences in the genomes of two organisms can be measured by PCR with random primers. The products are separated by electrophoresis for comparative identification. Two genomes from closely related organisms are expected to yield more similar bands. For more details, refer the technique random amplified polymorphic DNA (See p. 116).

PCR is very important in the study ***evolutionary biology***, more specifically referred to as ***phylogenetics***. As a technique which can amplify even minute quantities of DNA from any source (hair, mummified tissues, bone, or any fossilized material), PCR has revolutionized the studies in palaeontology and archaelogy. The movie 'Jurassic Park', has created public awareness of the potential applications of PCR!

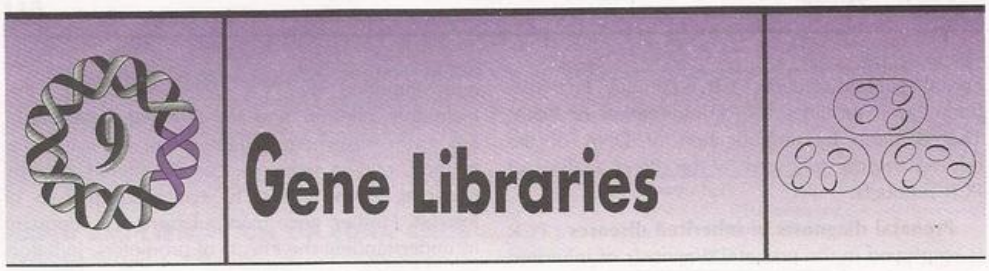
PCR IN FORENSIC MEDICINE

A single molecule of DNA from any source (blood strains, hair, semen etc.) of an individual is adequate for amplification by PCR. Thus, PCR is very important for identification of criminals.

The reader may refer DNA finger printing technique described elsewhere (Chapter 14).

PCR IN COMPARISON WITH GENE CLONING

PCR has several advantages over the traditional gene cloning techniques (Chapter 6). These include better efficiency, minute quantities of starting material (DNA), cost-effectiveness, minimal technical skill, time factor etc. In due course of time, PCR may take over most of the applications of gene cloning.



9 Gene Libraries

The ***collection of DNA fragments*** (specifically genes) from a particular species represents gene libraries. The creation or construction of gene libraries (broadly ***genomic libraries***) is accomplished by isolating the complete genome (entire DNA from a cell) which is cut into fragments, and cloned in suitable vectors. Then the specific clone carrying the desired (target) DNA can be identified, isolated and characterized. In this manner, a library of genes or clones (appropriately considered as ***gene bank***) for an the entire genome of a species can be constructed. The sizes of genomes in different species are variable (***Table 9.1***). A complete gene library for each organism contains all the genomic DNA.

Biotechnologists are particularly interested in the isolation of genes (and therefore creation of gene libraries) which encode for proteins. There is a distinct difference in the genes of prokaryotic and eukaryotic cells. In prokaryotic organisms, the structural genes coding for proteins are continuous. However, in case of eukaryotes, the coding regions (exons) of structural genes are separated by non-coding regions (introns). For this reason, the construction of gene libraries for eukaryotes is more complicated.

TABLE 9.1 Genome sizes of some organisms

Organism	*Genome size* (kb)*
Escherichia coli	4.0×10^3
Saccharomyces cerevisiae	1.35×10^4
Tobacco	1.6×10^6
Wheat	5.9×10^6
Drosophila melanogaster	1.8×10^5
Mouse	3.3×10^6
Human	3.2×10^6

* *Haploid where appropriate*

CREATING A GENE LIBRARY

The DNA from the source organism is digested by restriction endonuclease (e.g., *EcoRI*), to result in fragments. It is desirable to create conditions so that partial digestion and not complete digestion occurs. By this way, all possible DNA fragments of variable size can be produced. The partial digestion of a DNA with a restriction endonuclease is depicted in ***Fig. 9.1***. The cleavages occur at different sites to result in DNA fragments of varying lengths, some of them may be large while others are small. In practice, a combination of restriction enzymes are used to digest source DNA to release a large number of DNA fragments. The desired fragments can be isolated and cloned.

Some workers use the term shotgun experiment (or shotgun approach) for ceation of random clones; (without necessarily identifying all of them) from a genomic DNA. In ***shotgun approach, the DNA is subjected to random cleavage by restriction endonucleases***.

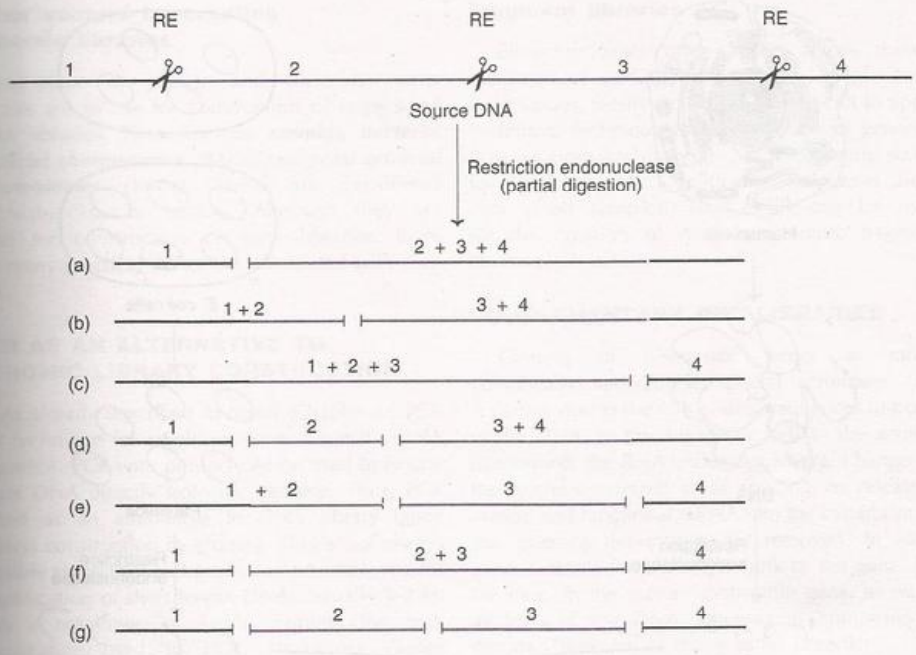


Fig. 9.1 : *Partial digestion of a DNA by the enzyme restriction endonuclease (RE) at three sites [**Note :** the coloured fragment represents the desired gene and it is one (shown in f) among several fragments formed].*

Maniatis technique for creating gene library

In the technique developed by Maniatis *et al* (1978), two restriction endonucleases are used to cut the target DNA. Partial digestion allows the formation of majority of DNA fragments with a length of 10–30 kb. The fragments are frequently overlapping and they can be fractionated by gel electrophoresis. The ***isolated fragments*** (approximately 20 kb in size) ***are inserted into λ phage vector and cloned.***

Establishing a gene library for humans

The human cellular DNA (the entire genome) may be subjected to digestion by restriction endonucleases (e.g., *EcoRI*). The fragments formed on an average are of about 4 kb size. (i.e., 4000 nitrogenes bases). Each human chromosome, containing approximately 100,000 kb can be cut into about 25,000 DNA fragments. As the humans have 23 different chromosomes (24 in man), there are a total of 575,000 fragments of 4 kb length formed. Among these 575,000 DNA fragments is the DNA or gene of interest (say insulin gene).

Now is the selection of a vector and cloning process. *E.coli*, a harmless bacterium to humans is most commonly used. The ***plasmids from E. coli*** are isolated. They are digested by the same restriction enzyme as was used for cutting human genome to form open plasmids. The human chromosomal DNA fragments and open plasmids are joined to produce recombined plasmids. These plasmids contain different DNA fragments of humans. The recombined plasmids are inserted into *E. coli* and the cells multiply (***Fig. 9.2***). The *E. coli* cells possess all the human DNA in fragments. It must, however be remembered that each ***E. coli cell contains different DNA fragments.*** All the *E. coli* cells put together collectively represent ***genomic library*** (containing about 575,000 DNA fragments).

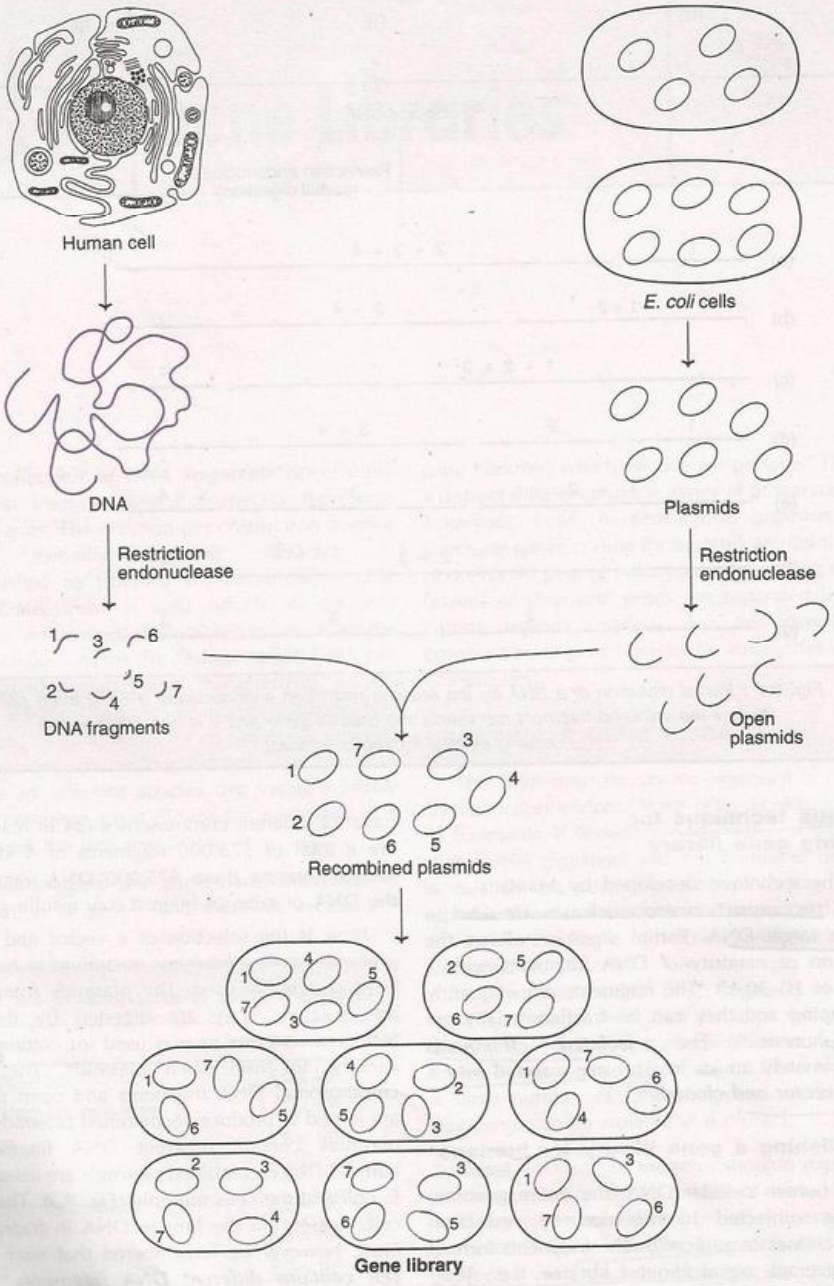


*Fig. 9.2 : Creation of a genomic library for humans. (**Note :** Double-stranded DNA is represented by single lines or circles for clarity; human DNA fragments are coloured).*

Other vectors for creating genomic libraries

In place of phages and plasmids, other vectors are in use for construction of large sized DNA libraries. These include ***cosmids, bacterial artificial chromosomes (BACs)*** and ***yeast artificial chromosomes (YACs)***. These are considered as high capacity vectors. Although they are ideal for construction of gene libraries, there are many practical difficulties associated with their use.

PCR AS AN ALTERNATIVE TO GENOMIC LIBRARY CONSTRUCTION

As already described in detail (Chapter 8), PCR is a technique for amplification of a specific DNA sequence. PCR with primers can be used to isolate target DNA directly from the genome. Thus, PCR serves as an alternative to DNA library (gene library) construction by cloning. This is not always possible since PCR technique can be employed for amplification of short length DNAs (usually 1-2 kb with a maximum of 5 kb). Further, the high temperature used in PCR, sometimes causes damage to bases and generates nicks in DNA strands.

Long PCR

With some modifications in the PCR, it is now possible to amplify DNA fragments up to a length of 22 kb from the human genomic DNA. This is achieved by using a combination of two DNA polymerase enzymes, besides lowering the reaction temperature. One of the DNA polymerases has proof-reading activity to remove the mismatched bases. Some commercial companies in fact provide enzyme cocktails ideally suited for long PCR e.g., TaqPlus Long PCR system marketed by ***Stratagene***. It contains *Taq* polymerase and the thermostable proof-reading enzyme *Pfee* polymerase.

Long PCR has been applied for the structured analysis of human genes and genomes of HIV.

Long PCR is unlikely to replace the construction of genomic libraries. This is because creation of DNA libraries is permanent while long PCR is temporary. Further, PCR can be employed for amplifying selected DNA fragments (of interest) from genomic libraries.

Fragment libraries

Biotechnologists often come across minute quantities of starting materials e.g., single cells, fixed tissues, fossils etc. It is quite difficult to apply traditional technique for construction of genomic libraries from such samples. PCR is ideally suited for isolation and amplification of genes from very small samples. Thus, PCR can be used for the creation of random genomic fragment libraries.

COMPLEMENTARY DNA LIBRARIES

Cloning of eukaryotic genes is rather complicated and requires special techniques. This is mainly due to the non-coding sequences (introns) in the DNA. In the eukaryotic cell as the gene is transcribed, the RNA undergoes several changes in the nucleus (referred to as splicing) to release a mature and functional mRNA into the cytoplasm. In this manner, the introns are removed. In some genes, introns form a major bulk of the gene. For instance, in the human dystrophin gene, as much as 99% of the DNA sequence is composed of introns. There are as many as 79 introns!

The prokaryotes, particularly the bacteria, do not possess the ability to remove the introns. Hence the functional mRNA is not correctly formed in a prokaryotic cell for an eukaryotic gene. Thus, cloning of eukaryotic genes becomes a difficult task.

Synthesis of complementary DNA

Complementary DNA (cDNA) is a double-stranded complement of an mRNA. cDNA can be synthesized from mRNA by ***reverse transcription***. An eukaryotic functional mRNA which does not have introns possesses a G cap at 5′ and a poly (A) tail at the 3′ end (approximately 200 adenine residues).

The requisite mRNA is isolated and purified (particularly from cells which are rich in the specific mRNA e.g., pancreatic cells for insulin mRNA). An oligo-dT primer is added to bind to the short segment of poly A tail region (by annealing). This primer provides 3′-hydroxyl group for the synthesis of a DNA strand. With the addition of the enzyme reverse transcriptase and four deoxynucleotides (dATP, dTTP, dGTP and dCTP), DNA synthesis proceeds. For the bases ***A, G, C*** and

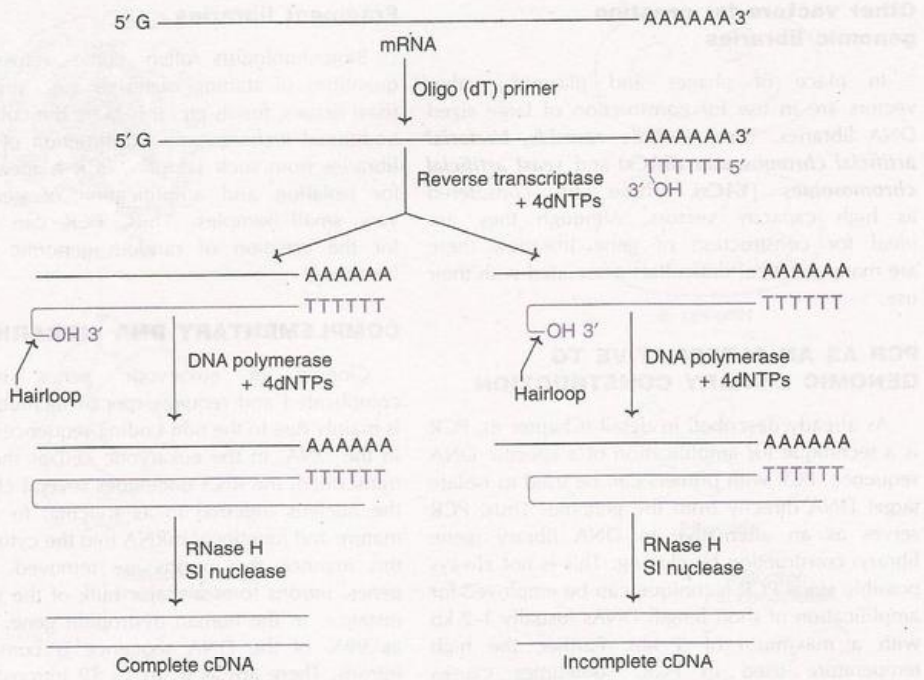


Fig. 9.3 : Synthesis of complementary DNA (cDNA).

U in the template (mRNA), the ***corresponding complementary bases*** in DNA respectively are ***T, C, G*** and ***A***. The newly synthesized first DNA strand has a tendency to fold back on to itself for a few nucleotides to form a hairpin loop (***Fig. 9.3***).

The loop of the first DNA strand serves as the template for the synthesis of second DNA strand. By the addition of *E. coli* DNA polymerase (Klenow fragment), the second DNA strand synthesis occurs starting from the end of the hairpin loop. On treatment with the enzyme RNase H, mRNA molecules are degraded. The enzyme SI nuclease cleaves the hairpin loops and degrades single-stranded DNA extensions. The final products are complementary DNA copies of original mRNA, some of them are complete while others are incomplete.

Limitation of the technique : The main disadvantage with the hairpin method is the loss of a small sequence at the 5′ end of cDNA due to cleavage by SI nuclease.

Improved method for cDNA synthesis

To overcome the limitation described above, some improvements have been made in cDNA synthesis. One such improved technique is shown in ***Fig. 9.4***.

As the first strand of cDNA is synthesized, it is ***tailed with cytidine residues*** with the help of the enzyme terminal transferase. The mRNA strand is hydrolysed with alkali, and the full length cDNA is recovered. A synthetic oligo-dG primer is then annealed to oligo-dC. This in turn enables the synthesis of the second strand of cDNA. By this improved technique, a full length of cDNA corresponding to mRNA (in turn the gene) is obtained. But the efficiency of this method is comparatively lower.

Construction of cDNA libraries

The complementary DNA molecules can be cloned in cloning vector (e.g., plasmid), for creating cDNA libraries. The cDNA insertion into the vector should have correct orientation. This is achieved by

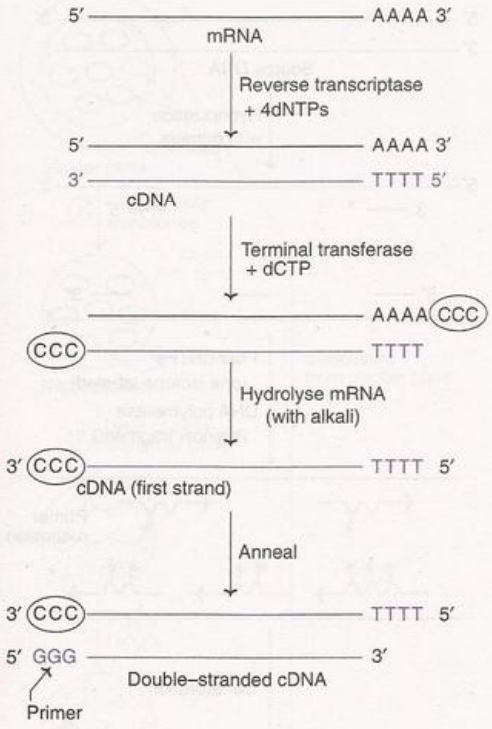


Fig. 9.4 : *Improved method for complementary DNA synthesis.*

the addition of a synthetic linker to the double-stranded cDNA.

In a technique developed by Okayama and Berg (1982), the mRNA is first linked to the plasmid cloning vector and then cDNA synthesis is carried out.

RT-PCR as an alternative to cDNA cloning

Reverse transcription followed by PCR (RT-PCR) can amplify the mRNA to give cDNA (For details Refer Chapter 8).

RT-PCR is very rapid hence cDNA molecules can be obtained in a short period. Further, even the long length mRNAs can be conveniently used in RT-PCR.

There are some disadvantages also in RT-PCR. The DNA polymerase used in RT-PCR is error-prone, and even a very minute contamination of mRNA (with other mRNAs) will give false results.

SCREENING STRATEGIES

Once a DNA library or a cDNA library is created, the clones (i.e., the cell lines) must be screened for identification of specific clones. The screening techniques are mostly based on the sequence of the clone or the structure/function of its product.

SCREENING BY DNA HYBRIDIZATION

The target sequence in a DNA can be determined with a ***DNA probe*** (***Fig. 9.5***). To start with, the double-stranded DNA of interest is converted into single strands by heat or alkali (denaturation). The two DNA strands are kept apart by binding to solid matrix such as nitrocellulose or nylon membrane. Now, the single strands of DNA probe (100–1,000 bp) labeled with radioisotope are added. Hybridization (i.e., base pairing) occurs between the complementary nucleotide sequences

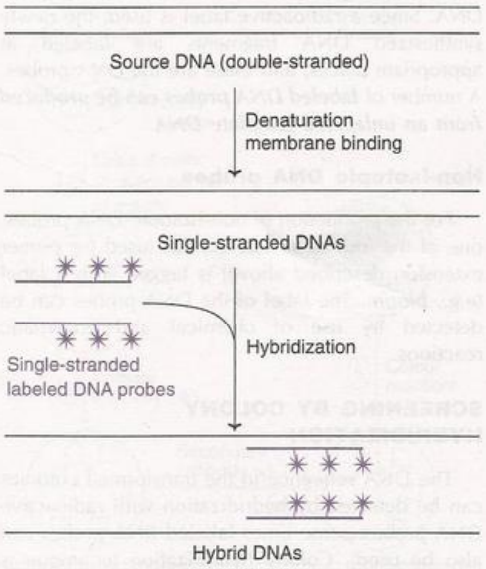


Fig. 9.5 : *Screening by DNA hybridization (✳ indicates radioisotope label in the DNA probe)*

of the target DNA and the probe. For a stable base pairing, at least 80% of the bases in the two strands (target DNA and the probe) should be matching. The hybridized DNA can be detected by ***autoradiography***.

DNA PROBES

The DNA probes used for screening purpose can be synthesized in many ways.

Random primer method

Radioisotope labeled DNA primers can be produced by this technique (***Fig. 9.6***). The double-stranded DNA containing the sequence needed to serve as a probe is denatured. ***A mixture of synthetic oligonucleotides, with all possible combinations of bases*** (*A, G, C* and *T*), with a length of 6 nucleotides each ***serve as primers***. Some of these primers with complementary sequences will hybridize with the template DNA. This occurrence is entirely by chance and the probability is reasonably good.

By the addition of four deoxyribonucleotides (one of them is radiolabeled) and in the presence of the enzyme DNA polymerase of *E. coli* (Klenow fragment), the primers are extended on the template DNA. Since a radioactive label is used, the newly synthesized DNA fragments are labeled at appropriate places, and these are the DNA probes. A number of ***labeled DNA probes can be produced from an unlabeled template DNA***.

Non-isotopic DNA probes

For the production of non-isotopic DNA probes, one of the four deoxynucleotides (used for primer extension described above) is tagged with a label (e.g., ***biotin***). The label of the DNA probes can be detected by use of chemical and enzymatic reactions.

SCREENING BY COLONY HYBRIDIZATION

The DNA sequence in the transformed colonies can be detected by hydridization with radioactive DNA probes (some times labeled RNA probes can also be used). Colony hybridization technique is also referred to as ***replica plating*** by some authors. The technique depicted in ***Fig. 9.7*** is briefly described.

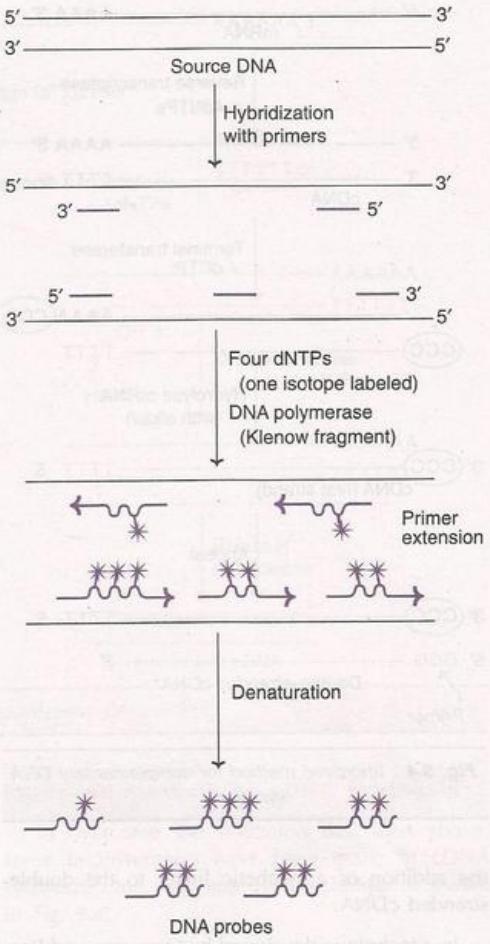


Fig. 9.6 : *Synthesis of radioisotope labeled DNA probes (**Note :** Non-isotopic DNA probes can be prepared by tagging with chemical labels e.g. biotin)*

The transformed cells are grown as colonies on a master plate. Samples of each colony are transferred to a solid matrix such as nitrocellulose or nylon membrane. The transfer is carefully carried out to retain the pattern of the colonies on the master plate. Thus, the nitrocellulose paper contains a photocopy pattern of the master plate colonies. The colony cells are lysed and deproteinized. The DNA is denatured and irreversibly bound to matrix. Now a radiolabeled DNA probe is added which hybridizes with the

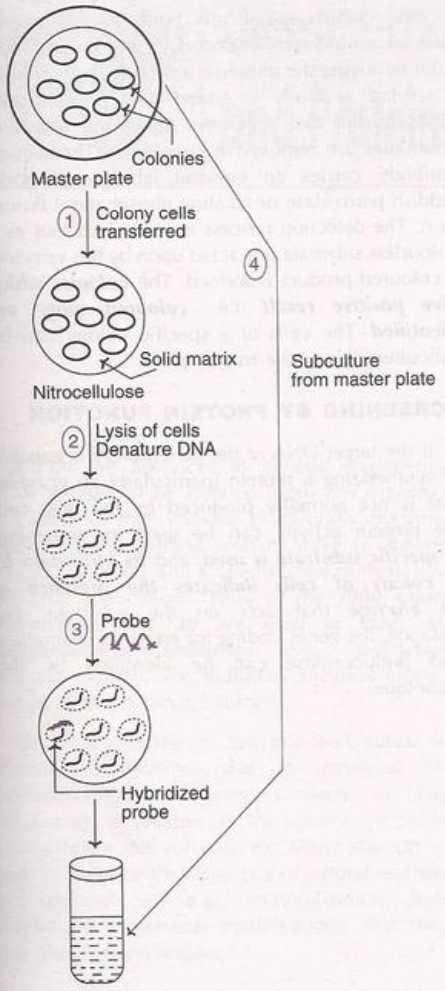


*Fig. 9.7 : Screening by colony hybridization (**Note** : Step 4 is carried out for the colony identified in step 3).*

complementary target DNA. The non-hybridized probe molecules are washed away. The colony with hybridized probe can be identified on autoradiograph. The cells of this colony (from the master plate) can be isolated and cultured.

Many a times multiple colonies are detected on hybridization by a DNA probe. This is due to overlapping sequences. To identify which colony has the complete sequence of the target gene, data observed from the restriction endonuclease analysis will be helpful.

Modifications of colony hybridization technique

Several improvements in the colony hybridization technique, described above, have been made in recent years. In the ***plaque lift technique***, nitrocellulose paper is directly applied on the upper surface of master agar plate making a direct contact. By this way, plaques can be lifted and several identical DNA prints can be made from a single plate. This technique increases reliability. More recently, screening of DNA libraries is carried out by automated techniques.

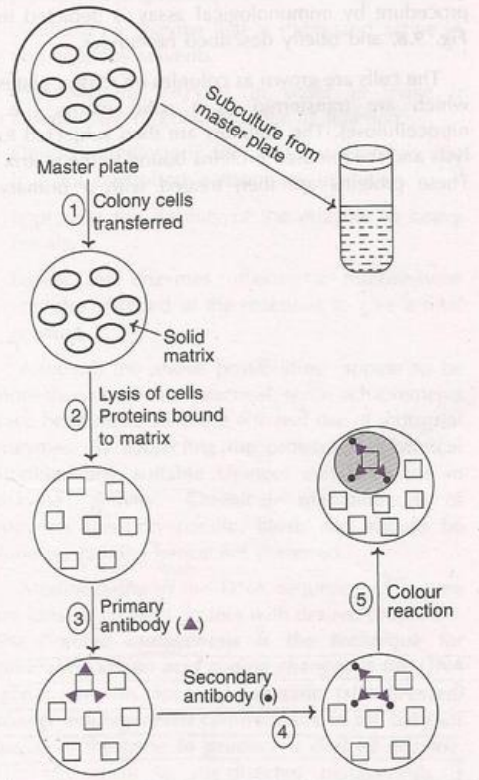


Fig. 9.8 : Immunological assay for screening a gene library.

SCREENING BY PCR

Polymerase chain reaction (PCR) is as good as hybridization technique for screening DNA libraries. But adequate information (on the franking sequences of target DNA) must be available to prepare primers for this method. The colonies are maintained in multiwell plates, each well is screened by PCR and the positive wells are identified.

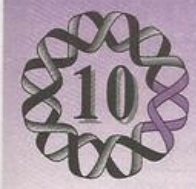
SCREENING BY IMMUNOLOGICAL ASSAY

Immunological techniques can be used for the detection of a protein or a polypeptide, synthesized by a gene (through transcription followed by translation). The procedure adopted for immunological assay and hybridization technique (described already) are quite comparable. Screening procedure by immunological assay is depicted in ***Fig. 9.8***, and briefly described hereunder.

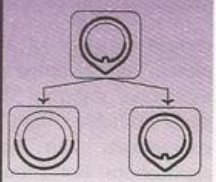
The cells are grown as colonies on master plates which are transferred to a solid matrix (i.e., nitrocellulose). The colonies are then subjected to lysis and the released proteins bound to the matrix. These proteins are then treated with a primary antibody which specifically binds to the protein (acts as an antigen), encoded by the target DNA. After removing the unbound antibody by washings, a second antibody is added which specifically binds to the first antibody. Again the unbound antibodies are removed by washings. The second antibody carries an enzyme label (e.g., horse raddish peroxidase or alkaline phosphatase) bound to it. The detection process is so devised that as a colourless substrate it is acted upon by this enzyme, a coloured product is formed. The ***colonies which give positive result*** (i.e., ***coloured spots***) ***are identified***. The cells of a specific colony can be subcultured from the master plate.

SCREENING BY PROTEIN FUNCTION

If the target DNA of the gene library is capable of synthesizing a protein (particularly an enzyme) that is not normally produced by the host cell, the protein activity can be used for screening. A ***specific substrate is used***, and ***its utilization by a colony of cells indicates the presence of an enzyme*** that acts on the substrate. For instance, the genes coding for enzymes α-amylase, and β-glucosidase can be identified by this technique.



Site-directed Mutagenesis and Protein Engineering



With the advances in genetic engineering, genes can be isolated (from any organism) and used for the synthesis of naturally occurring proteins. Some of these proteins serve as enzymes (i.e., biocatalysts), and a selected few of them have industrial applications. Thus, ***of the 2000 natural enzymes, around 20 are used in food, and pharmaceutical industries*** e.g., α-amylase, cellulase, papain. The industrial enzymes enjoy a special status in biotechnology.

The natural enzymes are not well suited for industrial applications due to unnatural and unphysiological environments such as high temperature, unsuitable pH, the presence of certain chemicals (organic solvents) etc. Many attempts are made to improve the efficiency of natural enzymes for industrial use e.g., immobilization (Refer Chapter 21), chemical modifications. But these have very limited impact.

STRATEGIES TO IMPROVE *IN VITRO* ACTIVITIES OF ENZYMES

Several possibilities can be thought of to improve the *in vitro* activities of the enzymes. Some of them are listed below.

- Increasing substrate affinity to enzyme.
- Making the enzyme thermal tolerant (active at high temperature) and/or pH stable.
- Enhancing the substrate specificity by modifying the substrate binding site of the enzyme.
- Designing the enzyme to make it resistant to proteolytic degradation.
- Synthesizing enzyme that is stable and active in non-aqueous solvents.
- Changing the enzyme in order to make it independent of cofactor(s) for its function.
- Eliminating allosteric sites in the enzyme so that it is not controlled by feedback inhibition.
- Improving the stability of the enzyme to heavy metals.
- Fusing the enzymes (making a multienzyme complex) needed in the reactions to give a final product.

Although the above possibilities appear to be more theoretical than practical, some achievements have been made for more efficient use of industrial enzymes. By subjecting the proteins to chemical modifications, suitable changes can be made in enzyme activity. Chemical manipulation of enzymes are non-specific, harsh and should be done repeatedly, hence not preferred.

Modifications in the DNA sequence of a gene are ideal to create a protein with desired properties. ***Site-directed mutagenesis is the technique for generating amino acid coding changes in the DNA (gene)***. By this approach ***specific (site-directed) change (mutagenesis)*** can be made in the base (or bases) of the gene ***to produce a desired enzyme***. The net result in site-directed mutagenesis is incorporation of a desired amino acid (of one's

choice) in place of a specific amino acid in a protein or a polypeptide. By employing this technique, enzymes that are more efficient and more suitable than the naturally occurring counterparts can be created for industrial applications. But it must be remembered that site-directed mutagenesis is a trial and error method that may or may not result in a better protein. Further, the detailed information on the structure and functions of a protein are desirable to undertake site-directed mutagenesis. It is customary to give emphasis for the creation of enzymes by this technique because of their commercial value through industrial use. The fact is that any naturally occurring protein with improved functions can be synthesized by site-directed mutagenesis.

There are many methods employing site-directed mutagenesis. A selected few of them are briefly described.

OLIGONUCLEOTIDE-DIRECTED MUTAGENESIS

In the technique of oligonucleotide-directed mutagenesis, ***the primer is a chemically synthesized oligonucleotide*** (7–20 nucleotides long). It is complementary to a position of a gene around the site to be mutated. But it contains mismatch for the base to be mutated.

The basic technique is depicted in ***Fig. 10.1***. The starting material is a single-stranded DNA (to be mutated) carried in an M_{13} phage vector. On mixing this DNA with primer, the oligonucleotide hybridizes with the complementary sequences, except at the point of mismatched nucleotide. Hybridization (despite a single base mismatch) is possible by mixing at low temperature with excess of primer, and in the presence of high salt concentration.

By the addition of 4-deoxyribonucleoside triphosphates and DNA polymerase (usually Klenow fragment of *E.coli* DNA polymerase) replication occurs. Thus the oligonucleotide primer is extended to form a complementary strand of the DNA. The ends of the newly synthesized DNA are sealed by the enzyme DNA ligase. The double- stranded DNA (i.e., M_{13} phage molecule) containing the mismatched nucleotides is

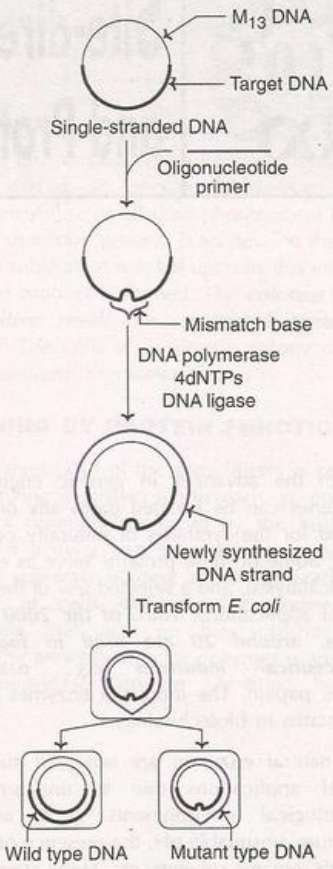


Fig. 10.1 : *Oligonucleotide-directed mutagenesis (dNTPs–Deoxyribonucleoside triphosphates)*

introduced into *E. coli* by transformation. The infected *E. coli* cells produce M_{13} virus particles containing either the original wild type sequence or the mutant sequence. The virus particles lyse the cells and form plaques.

Theoretically, it is expected that half of the phage M_{13} particles should carry wild type sequence while the other half mutant sequence (since the DNA replicates semiconservatively). But in actual practice, due to technical reasons, only ***1-5% of the viruses with mutated sequences are recovered.***

The double-stranded DNAs of M_{13} are isolated. The mutated genes are cut with restriction enzymes,

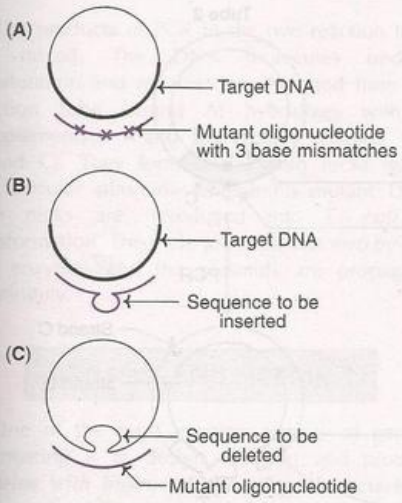


Fig. 10.2 : Variations in oligonucleotide-directed mutagenesis (A) Multiple point mutagenesis (B) Insertion mutagenesis (C) Deletion mutagenesis.

ligated to a plasmid vector of *E. coli*. The altered protein is produced in the *E. coli* which can be isolated and purified.

Oligonucleotide-directed mutagenesis by using plasmid DNA (instead of M_{13}) is also in use. This technique reduces the number of steps since mutagenesis and cloning the target DNA are in the same vector.

Variations in oligonucleotide-directed mutagenesis

There are some variations in use in the oligonucleotide-directed mutagenesis, as the situation demands.

1. **Multiple point mutagenesis :** Oligonucleotide-directed mutagenesis can be used to create DNAs with multiple point mutations with the requisite number of base mismatches (***Fig. 10.2A***).

2. **Insertion mutagenesis :** In this case, the mutant oligonucleotide carries a sequence to be inserted (sandwiched between two sites with complementary sequences). This can bind with the target DNA on either side (***Fig. 10.2B***).

3. **Deletion mutagenesis :** The mutant oligonucleotide binds to two separate sites on either side of the target DNA. This enables a small position of target DNA to be deleted (***Fig. 10.2C***).

CASETTEE MUTAGENESIS

In casettee mutagenesis, a synthetic double-stranded oligonucleotide (a small DNA fragment i.e., casettee) containing the requisite/desired mutant sequence is used. It replaces the corresponding sequence in the wild type DNA. ***Casettee mutagenesis is possible if the fragment of the gene to be mutated lies between two restriction enzyme cleavage sites***. This intervening sequence can be cut and replaced by the synthetic oligonucleotide (with mutation).

The outline of the casettee mutagenesis technique is depicted in ***Fig. 10.3***. The plasmid DNA is cut with restriction enzymes (such as *EcoR1* and *Hind111*). The synthetic oligonucleotide containing mutant sequence is also cleaved and ligated to plasmid DNA. The recombinant plasmids that multiply in *E. coli* are all mutants.

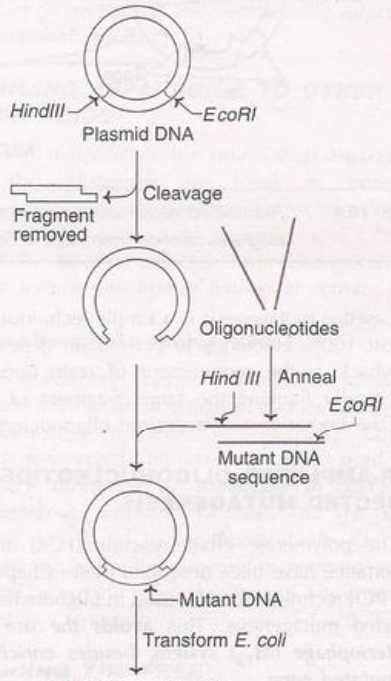


Fig. 10.3 : Casettee mutagenesis.

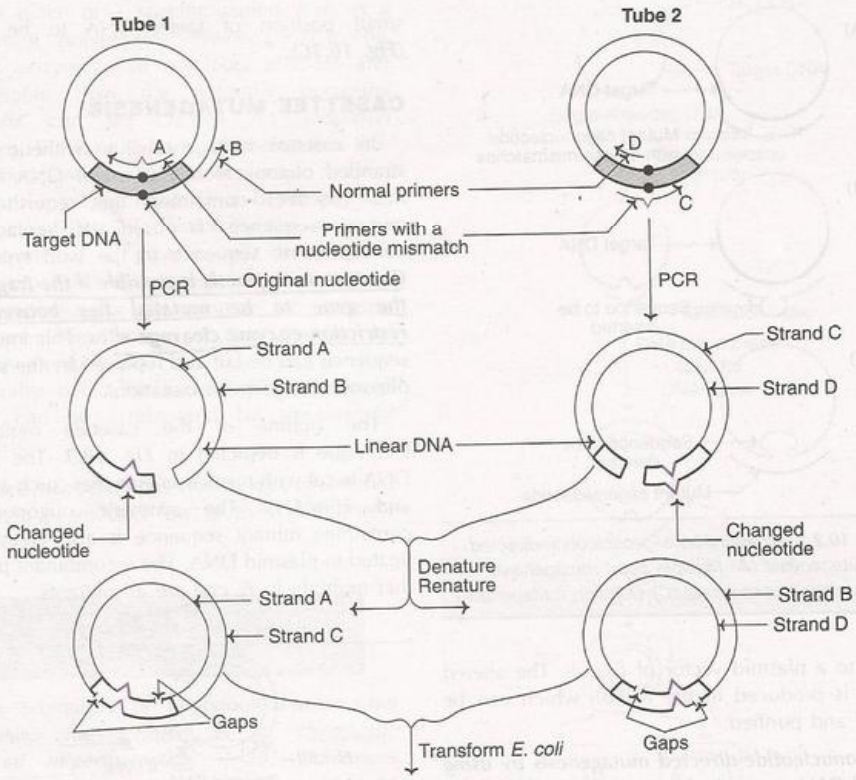


Fig. 10.4 : PCR-amplified oligonucleotide-directed mutagenesis (⟶-Normal primers; ⟿-Primers with a single nucleotide mismatch; ●-Original nucleotides; ≈-Changed nucleotides).

Casettee mutagenesis is a simple technique with almost 100% efficiency to get mutant genes. The drawback is the requirement of restriction sites specifically flanking the target fragment of DNA and the limitation to get desired oligonucleotides.

PCR-AMPLIFIED OLIGONUCLEOTIDE-DIRECTED MUTAGENESIS

The polymerase chain reaction (PCR) and its importance have been described (Refer Chapter 8). The PCR technique can be used in oligonucleotide-directed mutagenesis. This ***avoids the use of a bacteriophage (M_{13}) system, besides enrichment of mutated gene.***

The PCR-based mutagenesis technique commonly employed is depicted in ***Fig. 10.4***. First the target DNA (gene) is cloned on to a plasmid vector and distributed into two reaction tubes. To each tube are added two primers (oligonucleotides synthesized by using PCR). One primer (A in tube 1 and C in tube 2) is complementary to a region in one strand of the cloned gene except for one nucleotide mismatch (i.e., the one targeted for a change). The other primer (B in tube 1 and D in tube 2) is fully complementary to a sequence in the other strand, within or adjacent to the cloned gene. The placement of primers for hybridization (with the DNA strands) in each tube is done in opposite direction. Now, the PCR technique is carried out for amplification of the DNA molecules. These DNAs are actually linear. In ***Fig. 10.4***, they are shown as discontinuous circles for understanding subsequent steps.

The products of PCR in the two reaction tubes are mixed. The DNA molecules undergo denaturation and renaturation. A strand from one reaction tube (strand A) hybridizes with its complementary strand from other reaction tube (strand C). They form circles with nicks (gaps). The circular plasmids (containing mutant DNA) with nicks are introduced into *E. coli* by transformation. The nicks are sealed *in vivo* by host cell enzymes and the plasmids are propagated indefinitely.

PROTEIN ENGINEERING

One of the most exciting aspects of genetic engineering is to design, develop and produce ***proteins with improved operating characteristics*** (increased stability, improved kinetics etc.), besides sometimes ***creating even novel proteins***. The techniques such as site-directed mutagenesis and gene cloning are utilized for this pupose.

INCREASING THE STABILITY AND BIOLOGICAL ACTIVITY OF PROTEINS

By increasing the half-lives or thermostability of enzymes/proteins, their industrial applications or therapeutic uses can be more appropriately met. Some of the approaches for producing proteins with enhanced stability are described below.

ADDITION OF DISULFIDE BONDS

Significant ***increase in the thermostability of enzymes*** is observed ***by adding disulfide bonds***. However, the additional disulfide bonds should not interfere with the normal enzyme function.

In general, the new protein with added disulfide bonds does not readily unfold at high temperatures, and further it is resistant to denaturation at non-physiological conditions (high pH, presence of organic solvents). These characteristics are particularly important for industrial applications of certain enzymes.

T4 Lysozyme

This an enzyme of bacteriophage T4. Good success has been achieved in introducing disulfide bonds in T4 lysozyme. This was done by changing two, four or six amino acids (in close proximity) to cysteine residues to respectively form one, two or three disulfide bonds.

In the wild type (native) T4 lysozyme, there are two cysteine residues at positions 54 and 97 which however are not held together by a disulfide bond. By oligonucleotide-directed mutagenesis, cysteine residues created disulfide bonds between positions (numbered form N-terminal end) 3 and 97, 9 and 164, and 21 and 142 of the enzyme. Introduction of disulfide bonds increases the folded structure and thermostability of the enzyme. Thus, ***T4 lysozyme with three disulfide bonds is very stable with good biological activity***. (**Note :** T4 lysozyme has no industrial or therapeutic applications. It is discussed here due to the success achieved in protein engineering).

Xylanase

This is an enzyme used in the industry for manufacture of paper from wood pulp. Xylanase has to be catalytically active at high temperature. Introduction of ***disulfide bonds*** (one, two or three) makes it thermostable, and substantially ***improves its functional efficiency***.

CHANGING ASPARAGINE TO OTHER AMINO ACIDS

At high temperature, the amino acids ***asparagine*** and also ***glutamine*** are likely to ***undergo deamidation*** (releasing ammonia) to respectively form aspartic acid and glutamic acid. These alterations are often associated with changes in the protein folding and ***loss of biological activity***.

Triosephosphate isomerase

This is a dimeric enzyme with identical subunits, each one having two asparagine residues which are thermosensitive (undergo deamidation). Oligonucleotide-directed mutagenesis was used to ***introduce threonine or isoleucine in place of asparagine*** at positions 14 and 78. The new enzyme was found to be ***thermostable***. On the other hand, when the asparagine residues were replaced by aspartic acid, the enzyme was unstable even at low temperature, with reduced activity.

REDUCING THE FREE SULFHYDRYL GROUPS

Sometimes, the presence of free sulfhydryl groups (contributed by cysteine residues) in more

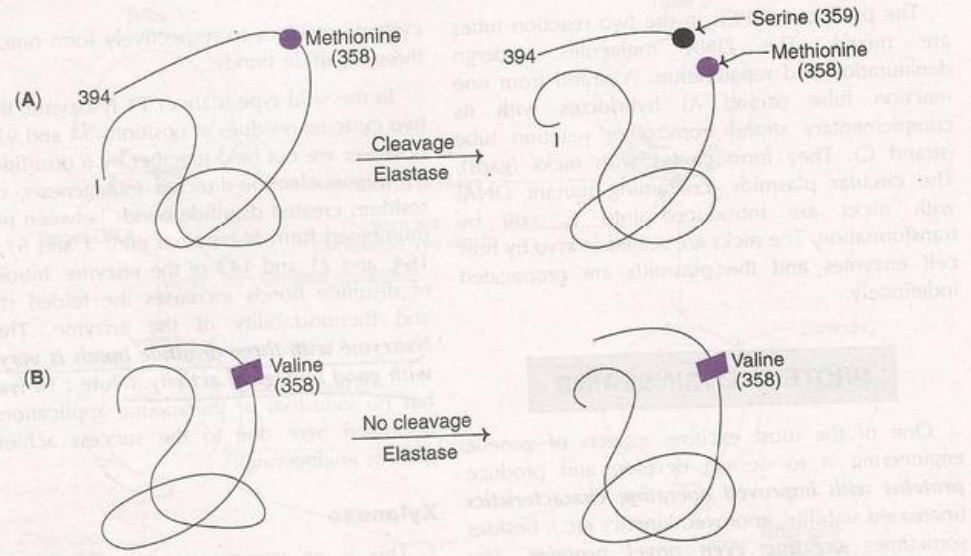


Fig. 10.5 : (A) The cleavage of α_1–antitrypsin by binding to elastase (B) No cleavage occurs when methionine (358) is replaced by valine.

numbers may be responsible for the low activity of the protein. In such a case, the protein or ***enzyme stability and its activity can be increased by reducing the number of sulfhydryl groups.***

Human β-interferon

The antiviral activity of human β-interferon (IFN β) produced in *E. coli* by genetic engineering was found to be only 10% of the original glycosylated form. Further, IFN β was found to exist as dimers and oligomers which are almost inactive. In fact, the cysteine residues were involved in intermolecular disulfide bonding resulting in the formation of dimers and oligomers. This happens in *E. coli* cells and not in human cells. This problem was successfully overcome by replacing cysteine residues by serine. It may be noted that the structures of these two amino acids are similar except that serine has oxygen in place of sulfur in cysteine. Consequently, ***introduction of serine in place of cysteine reduces free sulfhydryl groups.***

By the above process, IFN β with increased stability and good biological activity can be produced. Increased stability is in fact required for storage and therapeutic use of proteins such as interferons.

SINGLE AMINO ACID CHANGES

Some of the recombinant proteins can be improved in their stability and biology activity by a second generation variants. These have been frequently achieved by a single amino acid change.

α_1-Antitrypsin

α_1-Antitrypsin inhibits the action of neutrophil elastase (elastase is an enzyme that damages the lung tissues, often resulting in emphysema i.e. abnormal distension of lungs by air). α_1-Antitrypsin binds to elastase and prevents its action. In this process, α_1-antitrypsin gets cleaved between serine (359 residue) and methionine (358 residue). The free methionine is oxidized to methionine sulfoxide, making α_1-antitrypsin a poor inhibitor of elastase.

By ***replacing methionine at 358 position by valine, an oxidative-resistant variant of α_1-antitrypsin has been created (Fig. 10.5).*** This new enzyme is particularly important in treating patients with genetic deficiency of α_1-antitrypsin.

Insulin

In the neutral solution, therapeutic insulin is present mostly as zinc—containing hexamer. By introducing single amino acid substitutions, insulins

were found to be in monomeric state with good stability and biological activity.

Tissue plasminogen activator (tPA)

Tissue plasminogen activator is therapeutically used to lyse the blood clots that cause myocardial infarction. Due to its shorter half-life (around 5 minutes), tPA has to be repeatedly administered. By ***replacing asparagine residue*** (at position 120) ***with glutamine, the half-life of tPA can be substantially increased***. This is due to the fact that glutamine is less glycosylated than asparagine and this makes a difference in the half-life of tPA.

Hirudin

Hirudin is a protein secreted by leech salivary gland, and is a strong thrombin inhibitor (i.e., acts as an anticoagulant). By ***replacing asparagine*** (at 47 position) ***with lysine, the potency of hirudin can be increased*** several-fold.

Dihydrofolate reductase

The enzyme dihydrofolate reductase (DHFR) catalyses the conversion of 7, 8-dihydrofolate to 5, 6, 7, 8-tetrahydrofolate. The latter coenzyme is closely involved in one carbon metabolism, which ultimately results in the synthesis of nucleic acids and amino acids. The inhibition of DHFR (conventionally by folate analogues such as methotrexate) will restrict the growth of tumor cells. Therefore the enzyme DHFR has some therapeutic applications.

By employing site-directed mutagenesis, ***replacement of glycine*** (at position 95) ***by alanine*** **was found to produce *DHFR* which is *completely inactive***.

T4 lysozyme

Replacement of glycine by any other amino acid in the protein structure, in general, decreases the stability. On the other hand, proline residues increase protein stability. In T4 lysozyme, substitution of glycine (at position 77) by alanine, and alanine (at position 82) by proline are found to increase the enzyme stability.

IMPROVING KINETIC PROPERTIES OF ENZYMES

It is possible to improve the functional activities of enzymes, by improving their kinetic properties (K_m, specificity etc.) through oligonucleotide-directed mutagenesis. This is particularly required for enzymes with industrial and therapeutic applications.

Subtilisin

Subtilisin is a major industrial enzyme secreted by Gram-positive bacteria (*Bacillus* species). It is a serine protease very widely ***used as an enzyme detergent*** (cleaning agent in laundries). However, the large scale industrial use of native subtilisin is restricted due to the oxidative inactivation of this enzyme. This occurs in a manner analogous to that already discussed for α_1-antitrypsin. The reason being that ***methionine*** (at position 222) lying close to the active site ***gets oxidized, making the enzyme inactive***. Logically, ***replacement of methionine*** (at position 222) ***by other amino acids will solve the problem***. This has been done, and in fact, all the 19 other amino acids have been tried as substitutes for methionine, with varying success.

Subtilisin is an enzyme that has been extensively exploited for genetic manipulations over the past 15 years. The result is that ***about 50% of the native amino acids of this enzyme have been changed by in vitro mutagenesis***. And almost every property of subtilisin has been altered. These include its stability, substrate specificity, thermal and alkaline inactivation.

Modifying metal requirement of subtilisin : The enzyme subtilisin binds to calcium and gets stabilized. But in many industries where subtilisin is used, metal-chelating agents are also employed. They bind to calcium and the enzyme becomes inactive. This problem can be solved by oligo-nucleotide-directed mutagenesis. The nucleotide sequence in the DNA encoding for the amino acids (residues 75 to 83) that bind to calcium is removed. Then, several other amino acids are modified in subtilisin by trial and error method. In fact, success has been achieved in creating calcium-free subtilisin with good stability and activity.

Asparaginase

This is an enzyme used ***in the control of leukemia*** (uncontrolled growth of white blood cells). Intravenous administration of asparaginase cleaves asparagine to aspartate (the reduced availability of asparagine restricts cell prolifiration).

Surprisingly, the asparaginases from different sources exhibited differences in their effectiveness to control leukemia. This is due to the variations in the kinetic properties. Thus, the ***asparaginase with a low K_m value*** (i.e., with high affinity for asparagine hence more breakdown) has to be selected ***for therapeutic use in the control of leukemia***.

Tyrosyl t-RNA synthetase

The enzyme tyrosyl t-RNA synthetase from *Bacillus stearothermophilus* has been modified with regard to substrate binding i.e., K_m value. This enzyme catalyses the two reactions given below to finally give tyrosine t-RNA.

$$\text{Tyrosine} + \text{ATP} \longrightarrow \text{Tyrosyladenylate} + \text{PPi}$$

$$\text{Tyrosyladenylate} + \text{t-RNA} \longrightarrow \text{Tyrosine t-RNA} + \text{AMP}$$

Replacement of threonine (at position 51), ***by*** either ***alanine or proline***, variants of tyrosyl t-RNA synthetase have been produced. Alanine variant has a two fold binding affinity (low K_m) for ATP. And for proline variant ATP binds about 100-fold more tightly (very low K_m). (**Note :** The proline in general distorts the helical structure and reduces the enzyme affinity to the substrate. The one mentioned here is surprisingly, an exception).

Restriction endonucleases

The protein engineering studies can be successfully employed for modifying enzyme specificities. This is what has been achieved in oligonucleotide-directed mutagenesis with respect to restriction endonucleases.

So far, about 2500 restriction endonucleases are known. However, since most of these enzymes recognize the same sequence on the DNA for their action, there are only about 200 different recognition sites. Thus there is an overlap in the recognition sites of several restriction enzymes. As such, there are two types of restriction endonucleases—the ***frequent cutters*** which recognize a sequence of 4-6 bp and ***rare cutters*** recognizing a sequence above 8 bp. The rare cutters are more useful for producing large DNA fragments.

By ***using protein engineering technique***, the existing ***restriction endonucleases have been suitably modified to produce rare cutters***.

PROTEIN ENGINEERING BY USE OF GENE FAMILIES

The recent development in protein engineering is ***DNA shuffling***, also known as ***molecular breeding*** (developed by Nesse in 2000). This method can be applied to a protein if it belongs to a known protein family. The technique primarily involves isolation of genes from each species, and then creation of hybrids in different combinations. While applying this approach to subtilisin, genes from 26 species were mixed to create a library. Among these, 4 enzymes were found to have improved properties.

PROTEIN ENGINEERING THROUGH CHEMICAL MODIFICATIONS

Although with limited success, chemical modifications of proteins have been attempted to increase the stability of proteins to high temperature and organic solvents. The amino acid lysine residues can be cross-linked by use of chemical linkers.

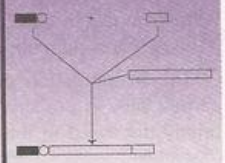
The most extensively used ***protein cross linker*** is ***glutaraldehyde***. It stabilizes the proteins in solutions. By using glutaraldehyde, certain proteins (hemoglobin, insulin, phosphofructokinase, lactate dehydrogenase) have been stabilized.

PROTEIN ENGINEERING—AN EVER EXPANDING FIELD

Selected examples of protein engineering are described above only to highlight the scope of protein engineering. The techniques of enzyme engineering are rapidly expanding and several newer developments are in the offing to develop proteins / enzymes for industrial and therapeutic purposes.

The question before biotechnologists specialized in protein engineering is not 'what modifications are possible', but 'how do I achieve the modifications I wish to make'.

Manipulation of Gene Expression in Host Cells



The preparation of recombinant DNAs, their insertion into vectors, and the cloning methods have been described (Chapter 6). The expression of cloned genes in the host organisms is dealt with in this chapter.

SELECTION OF HOST CELLS FOR GENE EXPRESSION

The nature of a host cell or an organism is as important as the nature of a vector. The most important requirements of a good host include its suitable cultivation in the laboratory, besides incorporating the vector's genetic material. Several prokaryotes and eukaryotes are employed as hosts to express foreign genes (Refer Chapter 6).

Prokaryotic hosts

The bacterium ***Escherichia coli*** was the first organism to be used in recombinant DNA technology experiments, and continues to be ***a host of choice for commercial production of proteins***. The extensive use of *E. coli* is mainly due to its high rate of reproduction (the cells double in number, every twenty minutes), besides good knowledge on its biochemistry, physiology and molecular biology. There are a few disadvantages also in using *E. coli* as a host. These include a relatively poor export system for proteins and the production of endotoxins which are often difficult to remove (from other useful products).

The other host bacterium in use is *Bacillus subtilis*. This organism is widely employed for commercial production of antibiotics, industrial enzymes, insecticides etc.

Eukaryotic hosts

It is often desirable to use eukaryotic organisms, with a well defined nucleus and celluar organelles, as hosts. The ***main advantage*** with eukaryotes is that they bring about several ***post-translational modifications*** to make viable and functional proteins. Further, use of eukaryotic hosts is not associated with the generation and interference of toxins which is the case with some prokaryotes. Thus, eukaryotic gene expression systems are preferred for the production of proteins and therapeutic agents that are useful for humans and animals.

The yeast ***Saccharomyces cerevisiae is widely used*** as a host for the expression of cloned eukaryotic genes. The other yeasts in use include *Kluveromyces lactis, Schizosaccharomyces pombe* and *Picha pastoris*.

The ***insect cells*** which are ***infected by baculovirus*** are in use as hosts in recent years. The baculovirus system can carry and express hundreds of genes in insect cells. Another advantage is that the safety factor since baculoviruses do not infect humans, other vertebrates or plants.

Mammalian cells such as ***mouse cells*** can be used as hosts to produce complex proteins with optimal biological functions. But the limitations

with mammalian cells are that the techniques are tedious, often difficult, and also expensive.

The manipulation of gene expression, as it is carried out, in prokaryotic and eukaryotic cells is briefly discussed in the following pages.

MANIPULATION OF GENE EXPRESSION IN PROKARYOTES

The ***prime objective of gene cloning*** is to finally result in the ***large scale production of proteins*** for a variety of purposes (industrial, commercial, human health and welfare). This is achieved by the maximal expression of cloned genes through manipulations. The following are the important features of gene expression that can be considered for manipulation.

- The presence of regulatable promoters.
- The number of copies of cloned genes.
- The location of the cloned genes whether inserted into a plasmid or integrated into host genome.
- The translation efficiency of the host.
- The cellular location of the foreign protein and its stability in the host cell.

Some of the strategies that are employed for the manipulation of gene expression in *E. coli* are discussed hereunder.

Regulatable promoters

The presence of a strong regulatable promoter sequence is essential for an effective expression of a cloned gene. This is achieved by employing the promoters of *E. coli lac* (lactose) operon or *trp* (tryptophan) operon. These promoters have strong affinity for RNA polymerase, and consequently the dounstream region (of cloned gene) is transcribed. The promoters thus provide a switch for turning on or turning off the transcription of a cloned gene.

Fusion proteins

The ***combination of a foreign protein*** (encoded by a cloned gene) ***with the host protein*** is referred to as a fusion protein.

In general, the foreign proteins synthesized are rapidly degraded. This can be reduced by covalently linking a stable host protein to the foreign proteins (i.e., fusion proteins). The fusion proteins in fact protect the proteolytic degradation of cloned gene product. The synthesis of fusion proteins is achieved by ligating the coding sequences of two genes (cloned gene and host gene). However, it is absolutely essential to ensure that cloned gene contains the correct sequence for the synthesis of the target protein.

Cleavage of fusion proteins : The fusion proteins, as such, interfere with the biological activity of the target protein. Therefore, these proteins should be cleaved to release the specific desired functional proteins.

Uses of fusion proteins : The purification of recombinant proteins is much easier in the form of fusion proteins. Fusion proteins are also useful for generating antibodies against target proteins.

Tandem gene arrays

In general, increase in the number of plasmids (containing cloned gene) proportionately increases the production of recombinant protein. This has a drawback. As the plasmid number increases, the genes coding for antibiotic resistance also increase. The overall effect is that the regular metabolic activites of the host cell are disturbed for the synthesis of plasmid proteins. Consequently, the yield of cloned gene product is not optimum.

An alternative approach is to clone ***multiple copies of the target gene on a single plasmid*** (instead of a single gene on a plasmid). In this manner, tandem arrays of a gene can be created. However, each sequence of the genes should be in correct orientation for transcription and translation.

Efficiency of translation

The quantity of the cloned gene product produced depends on the efficiency of translation. In general, the binding ability of mRNA with the ribosomal RNA, at translational initiation signal called ***ribosome binding site determines translation***. Thus, the efficiency of translation is better if the binding of mRNA to rRNA is stronger. The actual binding between mRNA and rRNA occurs by complementary base pairing of a sequence of 6-8 nucleotides.

To achieve maximum translation, the *E. coli* expression vectors are designed to possess a strong ribsome binding site.

Stability of proteins

The half-lives of recombinant proteins are highly variable, ranging from minutes to hours. The stability of proteins can be increased by adding amino acids at the N-terminal end of the proteins. Thus, by attaching methionine, serine and alanine to the N-terminal end, the half-life of β-galactosidase can be increased from 2 minutes to 20 hours! Frequently, a single ***amino acid addition at N-terminal end stabilizes the protein.*** The yield of recombinant DNA proteins can be enhanced by increasing half-lives.

Secretion of proteins

The stability of a protein and its secretion are interrelated. An amino acid sequence (signal peptide) may be attached to a protein to facilitate its secretion through cell membrane. Recombinant proteins secreted into the growth medium can be easily purified.

Integration of cloned DNA into the host chromosome

The use of plasmids for transcription and translation of cloned DNA imposes a metabolic load (discussed later) on the host. In addition, there is often a chance of losing plasmids during cell multiplication. These problems can be overcome by integrating the cloned DNA directly into host chromosomal DNA. Once the ***cloned DNA becomes a part of genome, it can be maintained for several generations.***

Cloned DNA integration into the host DNA is possible only when there is a complementary sequence of about 50 nucleotides between them. The exchange of DNAs occurs by a recombination process (***Fig. 11.1***). The cloned DNA lies in the middle of plasmid DNA. On physical contact with chromosomal DNA, base pairing occurs between plasmid DNA (x and y) and chromosomal DNA (x′ and y′). And the cloned DNA is transferred to host chromosomal DNA by a physical exchange i.e., recombination.

Metabolic load

The presence of ***cloned DNA alters the metabolism*** and cellular functions ***of the host organism***. Such metabolic changes are collectively referred to as metabolic load, ***metabolic drain*** or ***metabolic burden***. There are several causes for the metabolic load. These include increased utilization of energy for replication and maintenance of plasmids, overproduction of proteins (also drains amino acids, tRNAs), and interference of foreign proteins on the host cell function.

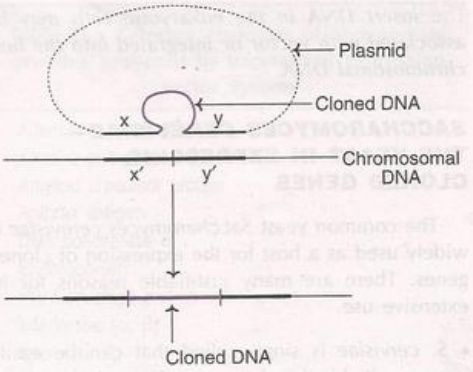


Fig. 11.1 : Integration of a cloned DNA into chromosomal DNA.

MANIPULATION OF GENE EXPRESSION IN EUKARYOTES

Expression of cloned genes in eukaryotes has certain ***advantages***. The most important being the ***ability of eukaryotic organisms to bring about post-translational modifications***—glycosylation, phosphorylation, correct disulfide bond formation, proteolytic cleavage etc. Eukaryotic expression systems produce stable and biologically active proteins. This is in contrast to the prokaryotic expression of cloned genes.

In general, the eukaryotic expression of cloned genes is quite comparable to that occurs in the prokaryotes. However, from the technical perspective, it is more difficult to conduct experiments with eukaryotic cells. Many a times, vectors with two distinct origins of replication are used. They serve as shuttle vectors and function in prokaryotic as well as eukaryotic hosts.

The insertion of a foreign DNA into bacterial and yeast cells is referred to as ***transformation*** (Refer Chapter 6). The term ***transfection*** is used for the introduction of a foreign DNA into animal cells.

The ***insert DNA in the eukaryotic cells may be associated with vector or integrated into the host chromosomal DNA.***

SACCHAROMYCES CEREVISIAE— THE YEAST IN EXPRESSING CLONED GENES

The common yeast *Saccharomyces cerevisiae* is widely used as a host for the expression of cloned genes. There are many justifiable reasons for its extensive use.

- *S. cervisiae* is single-celled that can be easily grown. Its biochemistry, genetics and physiology are quite known.
- It has a naturally occurring plasmid and strong promoters for efficient expression.
- *S. cerevisiae* can bring about many post-translational changes in proteins.
- The secreted recombinant proteins can be easily isolated, since very few host proteins are secreted.
- The U.S. Food and Drug Administration has certified *S. cerevisiae* as a ***generally recognized as a safe*** (***GRAS***) organism.

As such, *S. cerevisiae* has been in use for several decades in baking and brewing industries. Biotechnologists work quite comfortably with this yeast to ***produce a large number of recombinant proteins***. These include ***insulin, α_1-antitrypsin, hepatitis B virus surface antigen***, platelet derived growth factor, fibroblast growth factor and HIV-I antigens. These products are in use as diagnostic agents, vaccines, and therapeutic agents.

Vectors for *S. cerevisiae*

There are three types of vectors for *S. cerevisiae* :

1. Episomal or plasmid vectors.
2. Integrating vectors.
3. Yeast artificial chromosomes (YACs).

1. **Plasmid vectors :** Among the vectors, plasmids with single cloned genes are widely used. Manipulation with growth conditions increase the vector stability and expression efficiency. Use of tandem gene arrays has not met with success, since they are unstable.

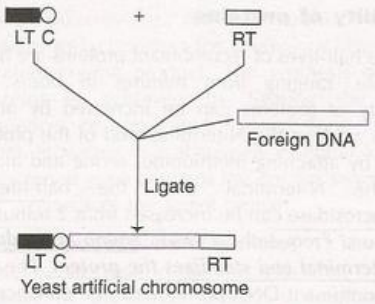


Fig. 11.2 : *Construction of yeast artificial chromosome (LT-Left telomere; C-Centromere; RT-Right telomere).*

2. **Integrating vectors :** They are basically the integration of cloned genes with chromosomal DNA. These are not frequently used, since the protein production is low.

3. **Yeast artificial chromosome (YAC) :** Introduced in 1987, YAC is a fragment of yeast DNA that will accept a foreign DNA of about 250–500 kb in length. In fact, the yeast DNA is only about 1% of the total DNA which however, is very important, since it contains three essential genes required for replication. These are the genes for ***telomere*** (that protects DNA from nuclease degradation and thus maintains stability), ***centromere*** (forms spindles during cell division) and the ***origin of replication*** (where DNA polymerase initiates replication). YAC behaves just like a chromosome and replicates.

The construction of the yeast artificial chromosome is depicted in ***Fig. 11.2***. Two opposite ends of a yeast chromosome namely the left telomere and right telomere are chosen. The left telomere is then attached to a centromere. A large segment of the foreign DNA is added and all the three are ligated. Unlike the plasmid vectors, the stability of YAC increases as the size of insert DNA increases.

YACs have not been used for commercial production of recombinant proteins. However, they have been employed successfully for physical mapping of genomic DNAs, particularly in human genome project, (For details, Refer Chapter 12).

Post-translational modifications by *S. cerevisiae*

The heterologous proteins synthesized by *S. cerevisiae* undergo post-translational changes while they are being exported into the extracellular environment. To facilitate protein secretion, a single (leader) peptide is attached to the protein. This peptide is removed by the yeast endoprotease.

OTHER YEAST EXPRESSION SYSTEMS

Despite the very successful use of *S. cerevisiae* for generating recombinant proteins, there are certain limitations. These include a very low or a limited yield, difficulty in secretion of some proteins and hyperglycosylation. Attempts are being made to explore the utility of other yeasts for the production of hepatitis B virus surface antigen (HBsAg) and bovine lysozyme. The yeast, ***Hansenula polymorpha***, is employed for the ***synthesis of α- and β-globin chains of human hemoglobin***.

INSECT CELL EXPRESSION SYSTEMS

Cultured insect cells are in use for expressing cloned DNAs. *Baculoviruses* exclusively infect insect cells. The DNA of these viruses encode for several products and their productivity in cells is very high to the extent of more than 10,000 times compared to mammalian cells. Besides carrying a large number of foreign genes, the baculoviruses can effectively express and process the products formed. Another advantage with these viruses is that they ***cannot infect humans***, other vertebrates or plants. Thus, ***baculoviruses are safe vectors***.

Polyhedrin gene of baculovirus

The polyhedrin gene is responsible for the synthesis of a matrix protein-polyhedrin. This protein is synthesized in large quantities by baculovirus during the infection cycle. Polyhedrin protects the virus from being inactivated by environmental agents. The promoter for polyhedrin gene is very strong. However, the life-cycle of baculovirus does not depend on the presence of this gene. Polyhedrin gene can be replaced by a cloned gene, and the genetically engineered baculovirus can infect the cultured insect cells. The cloned gene expresses, and large quantities of recombinant proteins are produced. Because of a close similarity in the post-translational modifications between insects and mammals, biologically active proteins can be produced by this approach. And in fact, by using baculovirus as an expression vector system, a good number of mammalian and viral proteins have been synthesized (***Table 11.1***).

Table 11.1 Selected examples of recombinant proteins produced by baculovirus expression vector system

Adenosine deaminase
Alkaline phosphatase
Amyloid precursor protein
Anthrax antigen
DNA polymerase α
Erythropoietin
HIV-I envelope protein
Interferons (α, β)
Interleukin-2
Malaria proteins
Pancreatic lipase
Polio virus proteins
Rabies virus proteins
Rhodopsin
Simian rotavirus capsid antigen
Tissue plasminogen activator

Baculovirus expression vector system

The most commonly used baculovirus is ***Autographa californica multiple nuclear polyhedrosis virus (AcMNPV)***. It can grow on the insect cell lines (e.g., derived from fall army worm) and produce high levels of polyhedrin or a recombinant protein.

The organization of a baculovirus (AcMNPV) transfer vector is shown in ***Fig. 11.3A***. It consists of an *E. coli*-based plasmid vector along with the DNA of baculovirus. This in turn has AcMNPV DNA, a polyhedrin promoter region, cloning site for insert DNA and polyhedrin termination region.

When the insect culture cells, transfected with AcMNPV are mixed with transfer vector carrying a cloned gene, a double crossover occurs. The result is that the cloned gene with polyhedrin promoter and termination sequences gets integrated into

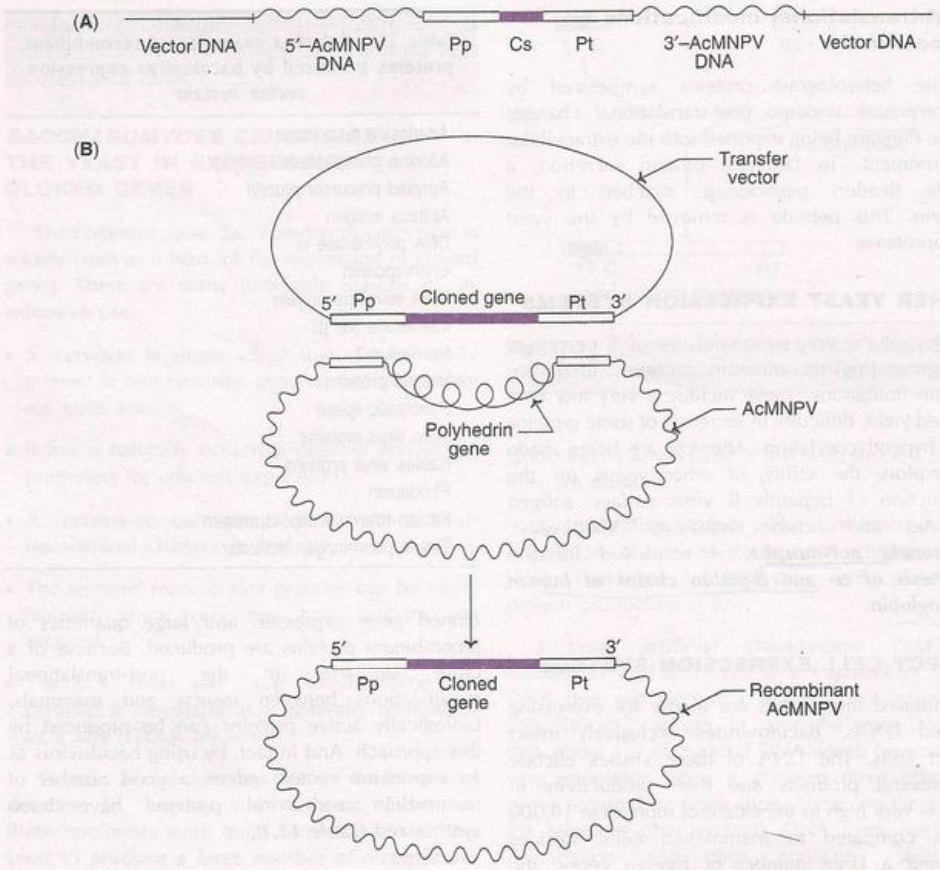


*Fig. 11.3 : Baculovirus expression vector system (**A**) Organization of baculovirus transfer vector (**B**) Replacement of the polyhedrin gene of baculovirus with a cloned gene from a transfer vector (AcMNPV–*Autographa californica *multiple nuclear polyhedrosis virus; Pp–Polyhydrin gene promoter; Cs–Cloning site; Pt–Polyhedrin gene termination; **Note :** The coding region of polyhedrin gene not shown in A).*

AcMNPV DNA (***Fig. 11.3B***). In this process, polyhedrin gene is lost. The recombinant baculovirus containing cloned gene is isolated.

The host insect culture cells, on infection with recombinant baculovirus, produce heterologous proteins. A large number and a wide variety of recombinant proteins (around 500) have been synthesized in the laboratory. A majority of them (>95%) have the requisite post-translational modifications. A selected list of recombinant proteins is given in ***Table 11.1***.

Modifications in the production of recombinant baculovirus

The original method of creating recombinant baculovirus has undergone several changes. Incorporation of a unique Bsu 361 restriction endonuclease site on the polyhedrin gene increases the yield of recombinant baculovirus production to about 30% from the normal 1%.

Bacmid : This is shuttle vector for *E. coli* and insect cell baculovirus. Construction of a recombinant bacmid is a novel approach to carry

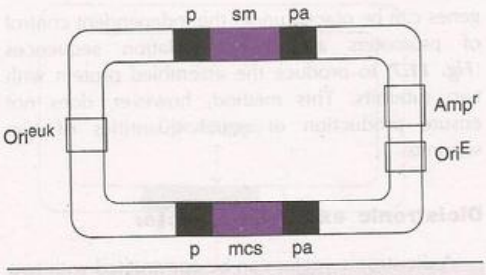


***Fig. 11.4** : A diagrammatic representation of mammalian expression vector (p–Promoter sequence; Pa–polyadenylation sequence; mcs–Multiple cloning site; sm-Selectable marker gene; Ori^{euk}–Origin of eukaryotic replication; Ori^E–Origin of* E. coli *replication; Amp^r–Ampicilin resistant marker gene).*

out all the genetic manipulations including the expression of baculovirus vector in *E. coli*.

Use of yeast cells : The genetic manipulations of AcMNPV genome can be done in yeast cells with yeast-insect shuttle vector. Then the recombinant baculovirus is introduced into insect cells.

MAMMALIAN CELL EXPRESSION VECTORS

Mammalian expression vectors are ***useful for the production*** of specific and authentic ***recombinant proteins*** (for use as therapeutic agents). In addition, they are also helpful for studying the function and regulation of mammalian genes. In general, the mammalian expression vectors are quite comparable to other eukaryotic expression vectors. However, large-scale production of recombinant proteins with engineered mammalian cells is costly.

A diagrammatic representation of mammalian vector is shown in ***Fig. 11.4***. It contains a eukaryotic origin of replication from an animal virus such as Simian virus 40 (SV40) and a prokaryotic origin of replication (from *E. coli*). The mammalian vector has a multiple cloning site and a selectable marker gene. Both of them are under the control of eukaryotic promoter and polyadenylation sequences. These sequences are obtained from either animal viruses (SV40, herpes simplex virus) or mammalian genes (growth hormone, metallothionein). The promoter sequences facilitate the transcription of cloned genes (at the multiple cloning site) and the selectable marker genes. On the other hand, the polyadenylation sequences terminate the transcription. Ampicillin resistant marker gene can be used for selecting the transformed *E. coli* cells.

Markers for mammalian expression vectors

There are several markers in use for the selection of transformed mammalian cells. The bacterial gene (Neo^r) that encodes for neomycin phosphotransferase is frequently used. The other markers are the genes that encode for the enzyme dihydrofolate reductase (DHFR), and glutamine synthetase (GS).

GENE EXPRESSION TO PRODUCE PROTEINS

Several proteins have been produced by mammalian expression vectors. The yield of protein synthesis in some cases, is increased by inserting an intron between the promoter and the cloned gene. However, the reason for this is not clear.

Coordinated expression

By coordinating the expression of a cloned gene and a selected marker, the production of recombinant protein can be increased.

The coordinated expression of DHFR (marker) and a cloned gene is illustrated in ***Fig. 11.5***. The DHFR gene is inserted close to a cloned gene and both of them are under the control of a single promoter and termination (polyadenylation) sequences. The DHFR gene is flanked by intron-removing sequence. As the genes are expressed, DHFR is synthesized by primary transcript while the recombinant protein is formed from spliced mRNA.

Expression of two cloned genes

Some proteins are composed of two or more subunits. For example, hemoglobin is a tetramer with two copies of each subunit i.e., $\alpha_2\beta_2$. Each subunit can be separately synthesized by the corresponding cloned gene, and they are then mixed to form the multimeric protein *in vitro*. This method is not very satisfactory since the *in vitro* assembly of multimeric proteins is not efficient. Techniques have been developed in recent years to produce two different proteins (i.e., subunits of a multimeric protein) in the same cell simultaneously.

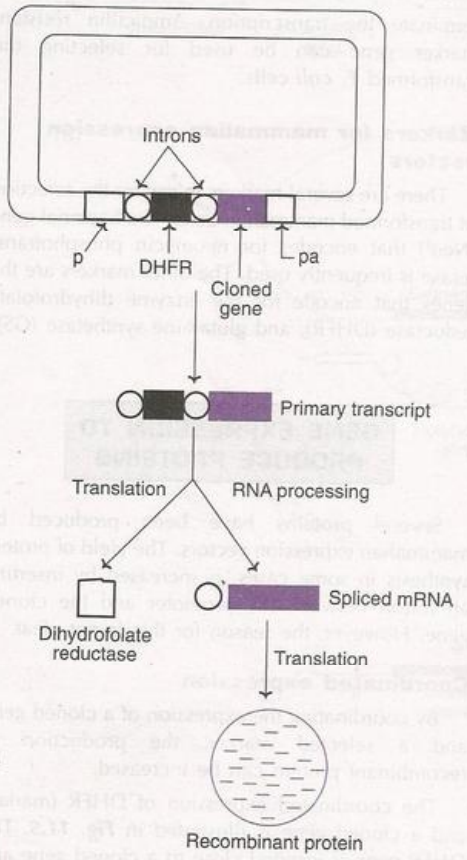


Fig. 11.5 : Coordinated expression of DHFR and a cloned gene (DHFR–Dihydrofolate reductase; p–Promoter sequence; pa–Polyadenylation sequence).

Two-vector expression system

Two expression vectors, each carrying a cloned gene for each subunit, are co-transfected into host cells (***Fig. 11.6***). These genes encode for the corresponding protein subunits. They then assemble to form a functional protein. The two-vector expression system has certain limitations — loss of one vector, overexpression of cloned gene in one of the vectors. This causes an imbalance in the formation and assembly of subunits.

Two-gene expression vector

A single vector carrying two cloned genes is used to overcome the above problem. The two genes can be placed under the independent control of promoters and polyadenylation sequences (***Fig. 11.7***) to produce the assembled protein with two subunits. This method, however, does not ensure production of equal quantities of two subunits.

Dicistronic expression vector

A dicistronic vector can be constructed with two cloned genes joined together to a small sequence of DNA that contains an ***internal ribosomal entry site*** (***IRES***). IRESs, found in mammalian viruses, allow a simultaneous transcription and translation to produce the assembled protein (***Fig. 11.8***). The transcription of gene-IRES-gene is under the control of a singe promoter and termination signals. By using dicistronic expression vector, ***the synthesis of equal amounts of the protein subunits can be achieved.***

COLLECTION AND PURIFICATION OF RECOMBINANT PROTEINS

As the recombinant proteins are produced by the cloned genes, they start accumulating. The next task is to collect and purify the specific gene product i.e., the requisite protein. This is not an easy job since many a times the recombinant protein is foreign to the host cell which possesses an enzyme machinery to degrade the outside proteins. Thus, human insulin produced in the bacterial host cells can be degraded by the proteases. This problem can be solved by using bacterial strains (e.g., *E. coli*) deficient in proteases. But there is a disadvantage since the proteases are defensive enzymes and hence the new strains lacking these enzymes are susceptible for easy destruction.

In an alternative approach, the ***recombinant proteins are fused with the native host proteins*** (Refer fusion proteins p. 138). The fusion proteins are ***resistant to protease activity***. Some times, foreign proteins accumulate as aggregates in the host organism, minimizing the protease degradation. The problem with protein aggregates is that the biological activity may be lost while extracting the protein from the aggregates.

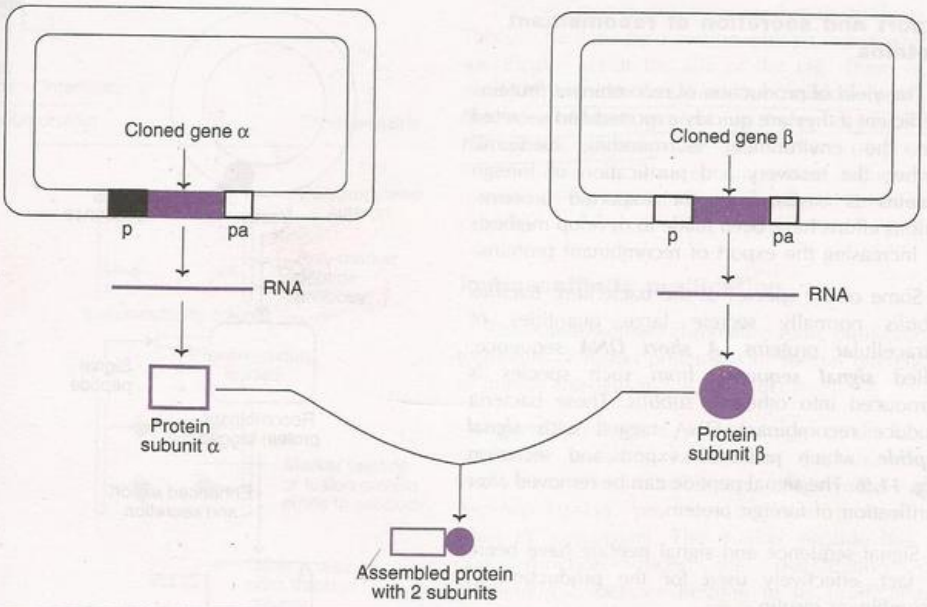


Fig. 11.6 : Two–vector expression system for producing a dimeric protein (p–Promoter sequence; pa–Polyadenylation sequence)

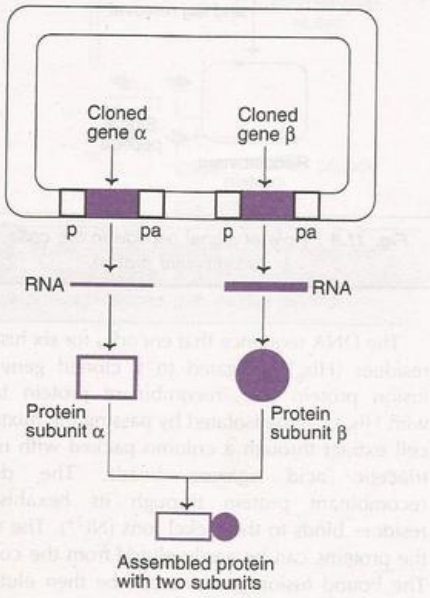


Fig. 11.7 : Two–gene expression vector for producing a dimeric protein (p–Promoter sequence; pa–Polyadenylation sequence).

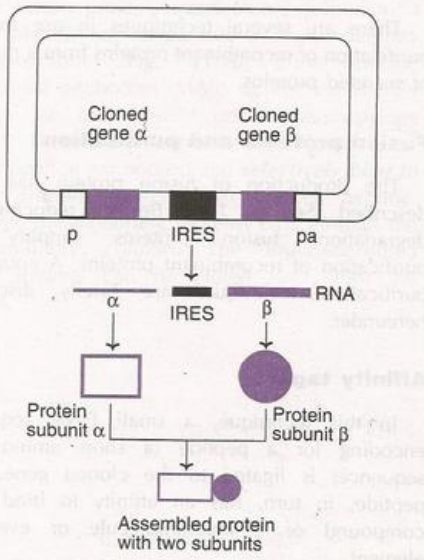


Fig. 11.8 : Dicistronic expression vector for producing a dimeric protein (p–Promoter sequence; pa–Polyadenylation sequence; IRES–Internal ribosomal entry site).

Export and secretion of recombinant proteins

The yield of production of recombinant proteins is efficient if they are quickly exported and secreted into the environment (surrounding medium). Further, the recovery and purification of foreign proteins is easier from the exported proteins. Serious efforts have been made to develop methods for increasing the export of recombinant proteins.

Some of the species of the bacterium, *Bacillus subtilis* normally secrete large quantities of extracellular proteins. ***A short DNA*** sequence, called ***signal sequence*** from such species is introduced into other *B. subtilis*. These bacteria produce recombinant DNA tagged with ***signal peptide***, which promotes export and secretion (***Fig. 11.9***). The signal peptide can be removed after purification of foreign protein.

Signal sequence and signal peptide have been, in fact, effectively used for the production of recombinant insulin.

PURIFICATION OF RECOMBINANT PROTEINS

There are several techniques in use for the purification of recombinant proteins from a mixture of secreted proteins.

Fusion proteins and purification

The production of fusion protein has been described (See p. 138). Besides reducing the degradation, fusion proteins simplify the purification of recombinant proteins. A couple of purification techniques are briefly discussed hereunder.

Affinity tagging

In this technique, a small DNA sequence encoding for a peptide (a short amino acid sequence) is ligated to the cloned gene. This peptide, in turn, has an affinity to bind to a compound or a macromolecule or even an element.

The tagged amino acid sequence, which forms a part of the recombinant protein acts as an ***affinity tag*** or ***identification tag***. The use of ***hexahistidine tag*** is briefly described in the next column.

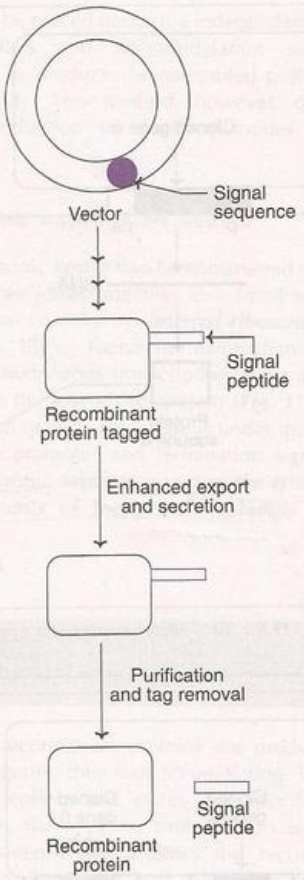


Fig. 11.9 : Role of signal peptide in the collection of recombinant protein.

The DNA sequence that encodes for six histidine residues (His_6) is ligated to a cloned gene. The fusion protein (i.e., recombinant protein tagged with His_6) can be isolated by passing the mixture of cell extract through a column packed with nickel-triacetic acid agarose beads. The desired recombinant protein through its hexahistidine residues binds to the nickel ions (Ni^{2+}). The rest of the proteins can be easily eluted from the column. The bound fusion proteins can be then eluted by lowering the pH of the buffer solution. Alternately, the fusion proteins can be selectively removed by binding of heaxahistidine tag to a competitor compound such as imidazole.

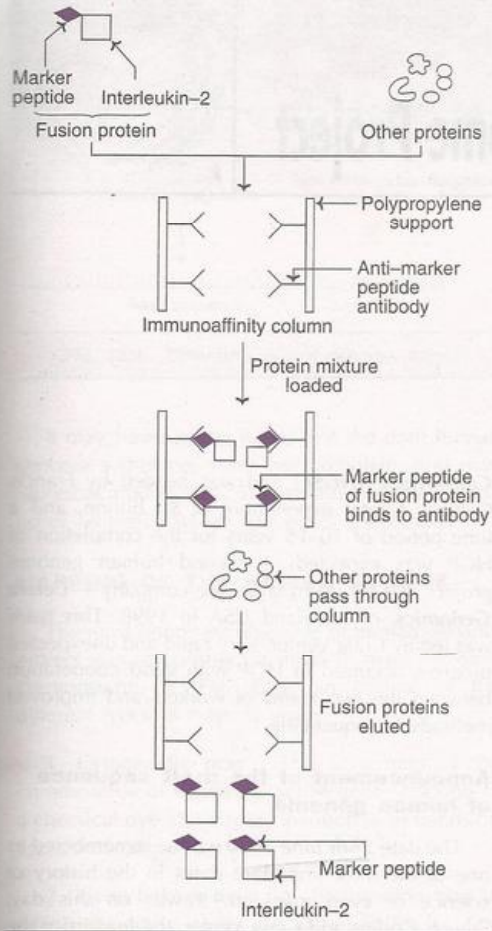


***Fig. 11.10** : Immunoaffinity chromatographic purification of a fusion protein (Interleukin-tagged with marker peptide).*

The next step is to remove the hexahistidine tag. This can be done by digestion with proteases which specifically act at the site of the tag. There is a need ***to remove hexahistidine residues only if the recombinant protein is being used as a therapeutic agent.*** For all other purposes, hexahistidine tag is acceptable, since it does not interfere in the normal structure and function of the proteins.

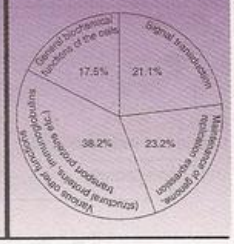
Immunoaffinity purification

The immunoaffinity chromatographic purification technique of fusion proteins with special reference to human interleukin-2 is described.

Interleukin-2 gene joined to a small DNA sequence encoding a marker peptide (that synthesizes 8 amino acids, Asp-Tyr-Lys-Asp-Asp-Asp-Asp-Lys) produces a fusion protein in yeast (*S. cerevisiae*). The marker peptide has a dual function-reduces the degradation of interleukin-2, besides helping in its purification (**Note :** The eight-amino acid marker peptide is commercially available under the brand name ***Flag peptide***).

The fusion protein, interleukin-2 joined to a marker peptide can be purified by immunoaffinity chromatography (***Fig. 11.10***). The specific monoclonal antibodies (MAb) against the marker peptide, are immobilized on a polypropylene support. These ***MAb serve as the ligand*** (anti-marker peptide antibodies) and ***selectively bind to fusion proteins*** tagged with marker peptide. However, the remaining proteins pass through the immunoaffinity column. The immunopurified fusion proteins can be eluted later from the column.

Human Genome Project



The most important features of a DNA molecule are the nucleotide sequences, and the identification of genes and their activities. Since 1920, scientists have been working to determine the sequences of pieces of DNA. This was further extended for the complete sequence determination of genome of certain lower organisms e.g. plasmid pBR 322 in 1979. The mitochondrial genome was sequenced in 1981.

THE BIRTH AND ACTIVITY OF HUMAN GENOME PROJECT

The human genome project (***HGP***) was conceived in 1984, and officially begun in earnest in October 1990. The primary objective of HGP was to determine the nucleotide sequence of the entire human nuclear genome. In addition, HGP was also entrusted to elucidate the genomes of several other model organisms e.g. *Escherichia coli*, *Saccharomyces cerevisiae* (yeast), *Caenorhabditis elegans* (roundworm), *Mus musculus* (mouse). James Watson (who elucidated DNA structure) was the first Director of HGP.

In 1997, United States established the ***National Human Genome Research Institute*** (***NHGRI***). The HGP was an international venture involving research groups from six countries — USA, UK, France, Germany, Japan and China, and several individual laboratories and a large number of scientists and technicians from various disciplines. This collaborative venture was named as ***International Human Genome Sequencing Consortium*** (***IHGSC***) and was headed by Francis Collins. A total expenditure of $3 billion, and a time period of 10–15 years for the completion of HGP was expected. A second human genome project was set up by a private company — ***Celera Genomics***, of Maryland USA in 1998. This team was led by Craig Venter. Very rapid and unexpected progress occurred in HGP with good cooperation between the two teams of workers and improved methods in sequencing.

Announcement of the draft sequence of human genome

The date ***26th June 2000*** will be remembered as one of the most important dates in the history of science or even mankind. It was on this day, Francis Collins and Craig Venter, the leaders of the two human genome projects, in the presence of the President of U.S., jointly announced the working drafts of human genome sequence. The detailed results of the teams were later published in February 2001 in scientific journals Nature (IHGSC) and Science (Celera Genomics).

The human genome project results attracted worldwide attention. This achievement was hailed with many descriptions in the media.

- The mystery of life unravelled.
- The library of life.
- The periodic table of life.
- The Holy grail of human genetics.

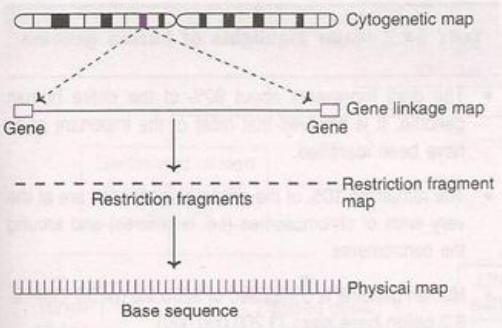


Fig. 12.1 : Different types of genome maps.

It may however, be noted that the draft human genome sequences were not complete, and may represent around 90%. The remaining 10% is made up of sequence where few genes are located.

MAPPING OF THE HUMAN GENOME

The most important objective of human genome project was to construct a series of maps for each chromosome. In ***Fig. 12.1***, an outline of the different types of maps is given.

1. **Cytogenetic map :** This is a map of the chromosome in which the active genes respond to a chemical dye and display themselves as bands on the chromosome.

2. **Gene linkage map :** A chromosome map in which the active genes are identified by locating closely associated marker genes. The most commonly used DNA markers are ***restriction fragment length polymorphism (RFLP), variable number tandems repeats (VNTRs) and short tandem repeats (STRs)***. VNTRs are also called as ***minisatellites*** while STRs are ***microsatellites***.

3. **Restriction fragment map :** This consists of the random DNA fragments that have been sequenced.

4. **Physical map :** This is the ultimate map of the chromosome with highest resolution base sequence. The methods for DNA sequencing are given in Chapter 7. Physical map depicts the location of the active genes and the number of bases between the active genes.

APPROACHES FOR GENOME SEQUENCING

A list of different methods used for mapping of human genomes is given in ***Table 12.1***. These techniques are also useful for the detection of normal and disease genes in humans.

For elucidating human genome, different approaches were used by the two HGP groups. ***IHGSC*** predominantly employed map first and sequence later approach. The principal method was ***heirarchical shotgun sequencing***. This technique involves fragmentation of the genome into small fragments (100–200 kb), inserting them into vectors (mostly bacterial artificial chromosomes, BACs) and cloning. The cloned fragments could be sequenced.

Celera Genomics used ***whole genome shotgun approach***. This bypasses the mapping step and saves time. Further, Celera group was lucky to have ***high-throughput sequenators*** and ***powerful computer programmes*** that helped for the early completion of human genome sequence.

Whose genome was sequenced?

One of the intriguing questions of human genome project is whose genome is being sequenced and how will it relate to the 6 billion or so population with variations in world? There is no simple answer to this question. However, looking from the positive side, it does not matter whose genome is sequenced, since the phenotypic differences between individuals are due to variations in just 0.1% of the total genome sequences. Therefore many individual genomes can be used as source material for sequencing.

Much of the human genome work was performed on the material supplied by the Centre for Human Polymorphism in Paris, France. This institute had collected cell lines from sixty different French families, each spanning three generations. The material supplied from Paris was used for human genome sequencing.

HUMAN GENOME SEQUENCE-RESULTS SUMMARISED

The information on the human genome projects is too vast, and only some highlights can be given (***Table 12.2***). Some of them are briefly described.

Table 12.1 A list of principal methods used for mapping of genomes (and also normal and disease genes in humans)

Method	*Comments*
DNA sequencing	Physical map of DNA can be identified with highest resolution.
Use of probes	To identify RFLPs, STS and SNPs.
Radiation hybrid mapping	Fragment genome into large pieces and locate markers and genes. Requires somatic cell hybrids.
Fluorescence *in situ* hybridization (FISH)	To localize a gene on chromosome.
Sequence tagged site (STS) mapping	Applicable to any part of DNA sequence if some sequence information is available.
Expressed sequence tag (EST) mapping	A variant of STS mapping; expressed genes are actually mapped and located.
Pulsed-field gel electrophoresis (PFGE)	For the separation and isolation of large DNA fragments.
Cloning in vectors (plasmids, phages, cosmids, YACs, BACs)	To isolate DNA fragments of variable lengths.
Polymerase chain reaction (PCR)	To amplify gene fragments
Chromosome walking	Useful for cloning of overlapping DNA fragments (restricted to about 200 kb).
Chromosome jumping	DNA can be cut into large fragments and circularized for use in chromosome walking.
Detection of cytogenetic abnormalities	Certain genetic diseases can be identified by cloning the affected genes e.g. Duchenne muscular dystrophy.
Databases	Existing databases facilitate gene identification by comparison of DNA and protein sequences.

(RFLP–Restriction fragment length polymorphism; STS–Sequence tagged site; SNP–Single nucleotide polymorphism; YAC–Yeast artificial chromosome; BAC–Bacterial artificial chromosome)

Table 54.2 Major highlights of human genome

- The draft represents about 90% of the entire human genome. It is believed that most of the important parts have been identified.
- The remaining 10% of the genome sequences are at the very ends of chromosomes (i.e. telomeres) and around the centromeres.
- Human genome is composed of 3200 Mb (or 3.2 Gb) i.e. 3.2 billion base pairs (3,200,000,000).
- Approximately 1.1 to 1.5% of the genome codes for proteins.
- Approximately 24% of the total genome is composed of introns that split the coding regions (exons), and appear as repeating sequences with no specific functions.
- The number of protein coding genes is in the range of 30,000–40,000.
- An average gene consists of 3000 bases, the sizes however vary greatly. Dystrophin gene is the larget known human gene with 2.4 million bases.
- Chromosome 1 (the larget human chromosome) contains the highest number of genes (2968), while the Y chromosome has the lowest. Chromosomes also differ in their GC content and number of transposable elements.
- Genes and DNA sequences associated with many diseases such as breast cancer, muscle diseases, deafness and blindness have been identified.
- About 100 coding regions appear to have been copied and moved by RNA–based transposition (retro-transposons).
- Repeated sequences constitute about 50% of the human genome.
- A vast majority of the genome (~ 97%) has no known functions.
- Between the humans, the DNA differs only by 0.2% or one in 500 bases.
- More than 3 million single nucleotide polymorphisms (SNPs) have been identified.
- Human DNA is about 98% identical to that of chimpanzees.
- About 200 genes are close to that found in bacteria.

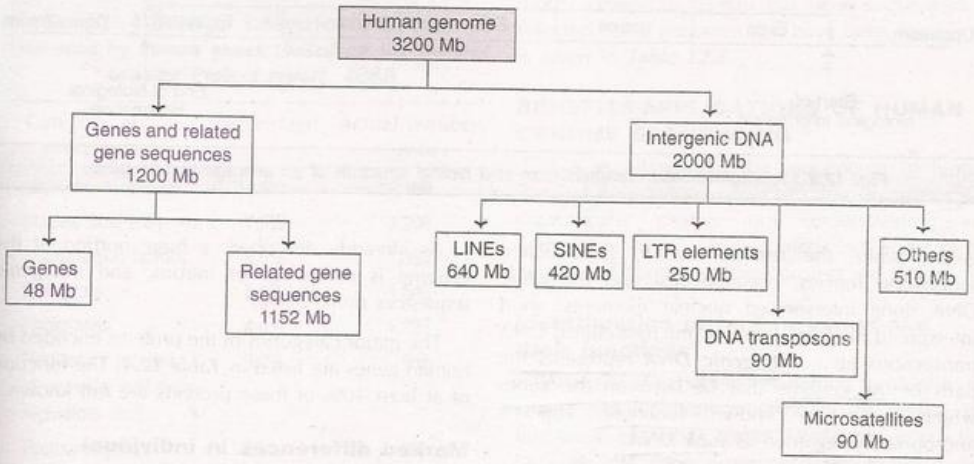


Fig. 12.2 : An overview of the organization of human genome (LINEs-Long interspersed nuclear elements; SINEs–Short interspersed nuclear elements; LTR-Long terminal repeats).

TABLE 12.3 Some interesting analogs/sidelights about human genome

- The base sequence in human genome would fill about 200 telephone books of 1000 pages each.
- If the genome is recited at the rate of one base per second for 24 hours a day, it would take a century to recite the book of life.
- If a typist types at the rate of 60 words per minute (i.e. 360 letters) for 8 hours a day, he/she would take around 50 years to type human genome.
- If the DNA sequence is typed in lines 10 cm containing 60 nucleotide bases and printed the human genome sequence (from a single cell) would stretch a distance of 5000 km.
- If the DNA in the entire human body is put end to end, it would reach to the sun and back over 600 times (**Note :** The human body contains 100 trillion cells; the length of DNA in a cell is 6 feet; the distance between the sun and earth is 93 million miles).
- The total expenditure for human genome project was $3 billion. The magnitude of this huge amount has to be appreciated. If one starts counting at a non–stop rate of a dollar per second, it would take about 90 years to complete.

Most of the genome sequence is identified

About 90% of the human genome has been sequenced. It is composed of 3.2 billion base pairs (3200 Mb or 3.2 Gb). If written in the format of a telephone book, the base sequence of human genome would fill about 200 telephone books of 1000 pages each. Some other interesting analogs/ sidelights of genome are given in ***Table 12.3***.

Individual differences in genomes : It has to be remembered that every individual, except identical twins, have their own versions of genome sequences. The differences between individuals are largely due to ***single nucleotide polymorphisms (SNPs)***. SNPs represent positions in the genome where some individuals have one nucleotide (i.e. an A), and others have a different nucleotide (i.e. a G). The frequency of occurrence of SNPs is estimated to be one per 1000 base pairs. About 3 million SNPs are believed to be present and at least half of them have been identified.

Organization of human genome

An outline of the organization of the human genome is given in ***Fig. 12.2***. Of the 3200 Mb, only a small fraction (48 Mb) represents the actual

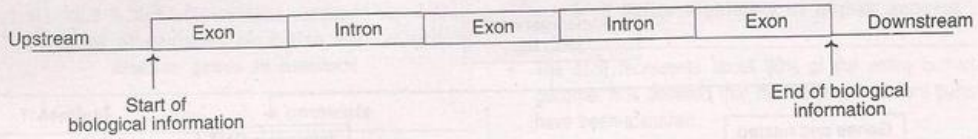


Fig. 12.3 : A diagrammatic representation of a typical structure of an average human gene.

genes, while the rest is due to gene-related sequences (introns, pseudogenes) and intergenic DNA (long interspersed nuclear elements, short *interspread nuclear elements, microsatellites,* DNA transposons etc.). ***Intergenic DNA*** represents the parts of the genome that lie between the genes which have no known function. This is appropriately regarded as ***junk DNA***.

Genes present in human genome

The two genome projects differ in their estimates of the total number of genes in humans. Their figures are in the range of 30,000–40,000 genes. The main reason for this variation is that it is rather difficult to specifically recognize the DNA sequences which are genes and which are not.

Before the results of the HGP were announced, the best guess of human genes was in the range of 80,000–100,000. This estimate was based on the fact that the number of proteins in human cells are 80,000–100,000, and thus so many genes expected. The fact that the number of genes is much lower than the proteins suggests that the ***RNA editing*** (RNA processing) is widespread, so that a single mRNA may code for more than one protein.

A diagrammatic representation of a typical structure of an average human gene is given in ***Fig. 12.3***. It has exons and introns.

A broad categorization of human gene catalog in the form of a pie chart is depicted in ***Fig. 12.4***. About 17.5% of the genes participate in the general biochemical functions of the cells, 23% in the maintenance of genome, 21% in signal transduction while the remaining 38% are involved in the production of structural proteins, transport proteins, immunoglobins etc.

Human genes encoding proteins

It is now clear that only 1.1-1.5% of the human genome codes for proteins. Thus, this figure 1.1-1.5% represents exons of genome.

As alreaddy described, a huge portion of the genome is composed of introns, and intergenic sequences (junk DNA).

The major categories of the proteins encoded by human genes are listed in ***Table 12.4***. The function of at least 40% of these proteins are not known.

Marked differences in individual chromosomes

The landscape of human chromosomes varies widely. This includes many features such as gene number per megabase, GC content, density of SNPs and number of transposable elements. For instance, chromosome 19 has the richest gene content (23 genes per megabase) while chromosome 13 and Y chromosome have the least gene content (5 genes per megabase).

Other interesting / important features of human genome

For more interesting features of human genome, refer ***Table 12.3***.

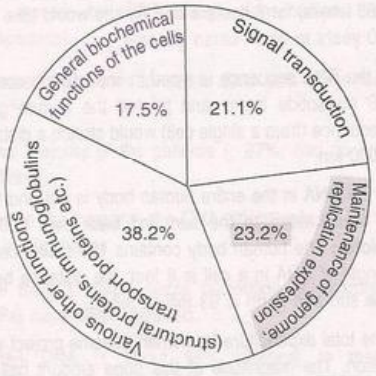


Fig. 12.4 : A pie chart showing a broad categorization of the human gene catalog (About 13000 genes whose functions are not known are not included).

TABLE 12.4 Different categories of proteins encoded by human genes (*based on the Human Genome Project report, 2001*)

Category of proteins	*Percentage*	*Actual number of genes*
Unknown functions	41.0%	12,809
Nucleic acid enzymes	7.5%	2,308
Transcription factors	6.0%	1,850
Receptors	5.0%	1,543
Hydrolases	4.0%	1,227
Regulatory proteins (G-proteins, cell cycle regulators etc.)	3.2%	988
Protooncogenes	2.9%	902
Structural proteins of cytoskeleton	2.8%	876
Kinases	2.8%	868

*(**Note** : This table is based on the rough draft of human genome reported by Celera Genomics. The percentages are derived from a total of 26,383 genes)*

- It is surprising to note that the number of genes found in humans is only twice that present in the roundworm (19,099) and thrice that of fruit fly (13,001).
- Around 200 genes appear to have been derived from bacteria by lateral transfer. Surprisingly, none of these genes are present in non-vertebrate eukaryotes.
- The proteins encoded by human genes are more complex than that of invertebrates.
- The flood of the data of human genome projects will be highly useful for bioinformatics and biotechnology.

GENOMES OF SOME OTHER ORGANISMS SEQUENCED

Sequencing of genomes is not confined to humans. For obvious reasons and significance, human genome sequencing attracted worldwide attention.

In fact, the first genome sequence of the bacteriophage QX174 was determined in 1977. Yeast was the first eukaryotic organism to be sequenced (1986). Recently, the mouse, an animal model closest to human has been sequenced. A selected list of genomes that have been sequenced is given in ***Table 12.5***.

BENEFITS/APPLICATIONS OF HUMAN GENOME SEQUENCING

It is expected that the sequencing of human genome, and the genomes of other organisms will dramatically change our understanding and perceptions of biology and medicine. Some of the benefits of human genome project are given.

Identification of human genes and their functions

Analysis of genomes has helped to identify the genes, and functions of some of the genes. The functions of other genes and the interaction between the gene products needs to be further elucidated.

TABLE 12.5 A selected list of genomes that have been sequenced

Name of the species	*Genome size (Mb/kb)*	*Comments (year)*
Bacteriophage QX174	5.38 kb	First genome sequenced (1977).
Plasmid pBR 322	4.3 kb	First plasmid sequenced (1979).
Yeast chromosome III	315 kb	First chromosome sequenced (1992).
Haemophilus influenzae	1.8 Mb	First genome of cellular organism to be sequenced (1995).
Saccharomyces cerevisiae	12 MB	First eukaryotic organism to be sequenced (1996).
Arabidopsis thaliana	125 MB	First plant genome to be sequenced (2000).
Homo sapiens (human)	3200 MB	First mammalian genome to be sequenced (2001).
Oryza sativa (rice)	430 MB	First crop plant genome to be sequenced (2002).
Mus musculis (mouse)	3300 MB	Animal model closest to human (2003).

Understanding of polygenic disorders

The biochemistry and genetics of many single-gene disorders have been elucidated e.g. sickle-cell anemia, cystic fibrosis, retinoblastoma. A majority of the common diseases in humans, however, are polygenic in nature e.g. cancer, hypertension, diabetes. At present, we have very little knowledge about the causes of these diseases. The information on the genome sequence will certainly help to unravel the mysteries surrounding polygenic diseases.

Improvements in gene therapy

At present, human gene therapy is in its infancy for various reasons. Genome sequence knowledge will certainly help for more effective treatment of genetic diseases by gene therapy.

Improved diagnosis of diseases

In the near future, probes for many genetic diseases will be available for specific identification and appropriate treatment.

Development of pharmacogenomics

The drugs may be tailored to treat the individual patients. This will become possible considering the variations in enzymes and other proteins involved in drug action, and the metabolism of the individuals.

Genetic basis of psychiatric disorders

By studying the genes involved in behavioural patterns, the causation of psychiatric diseases can be understood. This will help for the better treatment of these disorders.

Understanding of complex social trait

With the genome sequence now in hand, the complex social traits can be better understood. For instance, recently genes controlling speech have been identified.

Knowledge on mutations

Many events leading to the mutations can be uncovered with the knowledge of genome.

Better understanding of developmental biology

By determining the biology of human genome and its regulatory control, it will be possible to understand how humans develop from a fertilized eggs to adults.

Comparative genomics

Genomes from many organisms have been sequenced, and the number will increase in the coming years. The information on the genomes of different species will throw light on the major stages in evolution.

Development of biotechnology

The data on the human genome sequence will spur the development of biotechnology in various spheres.

ETHICS AND HUMAN GENOME

The research on human genomes will make very sensitive data available that will affect the personal and private lives of individuals. For instance, once it is known that a person carries genes for an incurable disease, what would be the strategy of an insurance company? How will the society treat him/her? There is a possibility that ***individuals with substandard genome sequences may be discriminated***. Human genome results may also promote racial discrimination categorizing the people with ***good*** and ***bad genome sequences***. Considering the gravity of ethics related to a human genome, about 3% of the HGP budget was earmarked for ethical research.

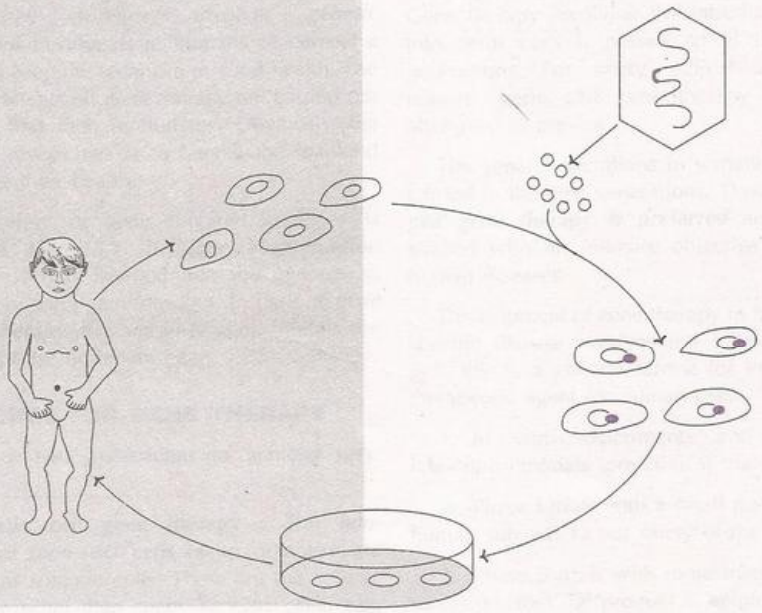
In the 1990s, there was a move by some scientists to patent the genes they discovered. This created an uproar in the public and scientific community. Fortunately, the idea of patenting genes (of human genome sequencing) was dropped. The fear still exists that genetic information will be used for commercial purposes.

SECTION IV

MEDICAL/ PHARMACEUTICAL BIOTECHNOLOGY (BIOTECHNOLOGY IN HEALTHCARE)

CONTENTS

13 Gene Therapy 157
14 DNA in Disease Diagnosis and Medical Forensics 173
15 Pharmaceutical Products of DNA Technology 189
16 Recombinant Vaccines 199
17 Monoclonal Antibodies (Hybridoma Technology) 213
18 Assisted Reproductive Technology 227



Gene Therapy

Advances in biochemistry and molecular biology have helped to understand the genetic basis of inherited diseases. It was a dream of the researchers to replace the defective genes with good ones, and cure the genetic disorders.

Gene therapy is the process of inserting genes into cells to treat diseases. The newly introduced genes will encode proteins and correct the deficiencies that occur in genetic diseases. Thus, gene therapy primarily involves ***genetic manipulations*** in animals or humans ***to correct a disease,*** and keep the organism in good health. The initial experiments on gene therapy are carried out in animals, and then in humans. Obviously, the goal of the researchers is to benefit the mankind and improve their health.

An overview of gene therapy strategies is depicted in ***Fig. 13.1***. In ***gene augmentation therapy***, a DNA is inserted into the genome to replace the missing gene product. In case of ***gene inhibition therapy***, the antisense gene inhibits the expression of the dominant gene.

APPROACHES FOR GENE THERAPY

There are two approaches to achieve gene therapy.

1. **Somatic cell gene therapy :** The ***non-reproductive (non-sex) cells*** of an organism are referred to as somatic cells. These are the cells of an organism other than sperm or eggs cells, e.g., bone marrow cells, blood cells, skin cells, intestinal cells. At present, all the research on gene therapy is directed to correct the genetic defects in somatic cells. In essence, somatic cell gene therapy involves the ***insertion of a fully*** functional and ***expressible gene into a target somatic cell*** to correct a genetic disease permanently.

2. **Germ cell gene therapy :** The ***reproductive (sex) cells*** of an organism constitute germ cell line. Gene therapy involving the introduction of DNA into germ cells is passed on to the successive generations. For safety, ethical and technical reasons, germ cell gene therapy is not being attempted at present.

The genetic alterations in somatic cells are not carried to the next generations. Therefore, ***somatic cell gene therapy is preferred*** and extensively studied with an ultimate objective of correcting human diseases.

Development of gene therapy in humans for any specific disease involves the following steps. In fact, this is a general format for introducing any therapeutic agent for human use.

1. *In vitro* experiments and research on laboratory animals (pre-clinical trials).

2. Phase I trials with a small number (5–10) of human subjects to test safety of the product.

3. Phase II trials with more human subjects to assess whether the product is helpful.

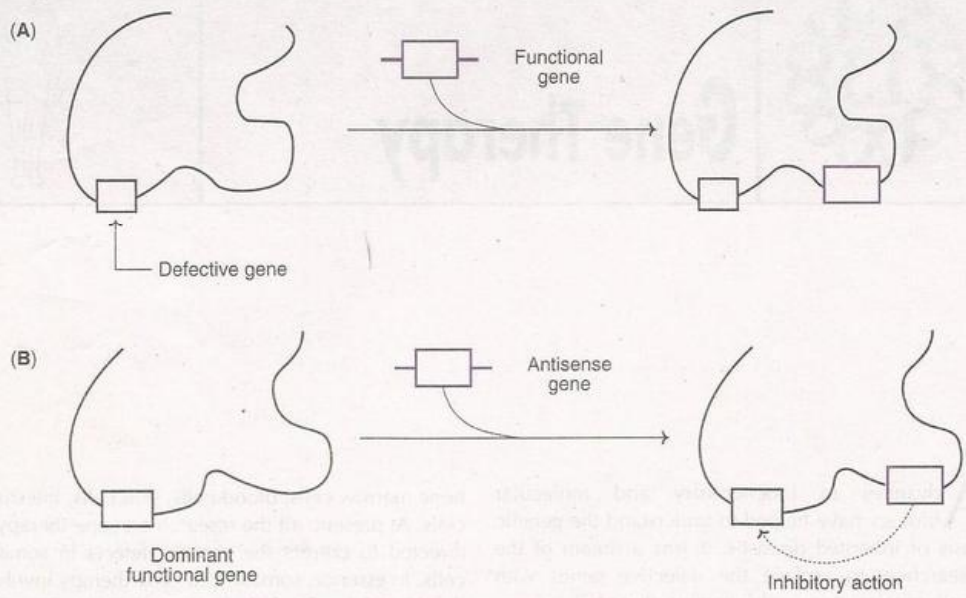


Fig. 13.1 : Overview of two major gene therapy strategies (A) Gene augmentation therapy (B) Gene inhibition therapy.

4. Phase III trials in large human samples for a final and comprehensive analysis of the safety and efficacy of the product.

As such, gene therapy involves a great risk. There are several regulatory agencies whose permission must be sought before undertaking any work related to gene therapy. ***Recombinant DNA Advisory Committee (RAC)*** is the supervisory body of the National Institute of Health, U.S.A., that clears proposals on experiments involving gene therapy. A large number of genetic disorders and other diseases are currently at various stages of gene therapy trials. A selected list of some important ones is given in ***Table 13.1***.

There are two types of gene therapies.

I. ***Ex vivo* gene therapy :** This involves the ***transfer of genes in cultured cells*** (e.g., bone marrow cells) which are then reintroduced into the patient.

II. ***In vivo* gene therapy :** The ***direct delivery of genes into the cells*** of a particular tissue is referred to as *in vivo* gene therapy.

EX VIVO GENE THERAPY

The *ex vivo* gene therapy can be applied to only selected tissues (e.g., bone marrow) whose cells can be cultured in the laboratory.

The technique of *ex vivo* gene therapy involves the following steps (***Fig. 13.2***).

1. Isolate cells with genetic defect from a patient.

2. Grow the cells in culture.

3. Introduce the therapeutic gene to correct gene defect.

4. Select the genetically corrected cells (stable transformants) and grow.

5. Transplant the modified cells to the patient.

The procedure basically involves the ***use of the patient's own cells for culture and genetic correction, and then their return back to the patient***. This technique is therefore, not associated with adverse immunological responses after transplanting the cells. *Ex vivo* gene therapy is

Table 13.1 Human gene therapy trials

Disease	*Gene therapy*
Severe combined immunodeficiency (SCID)	Adenosine deaminase (ADA).
Cystic fibrosis	Cystic fibrosis transmembrane regulator (CFTR).
Familial hypercholesterolemia	Low density lipoprotein (LDL) receptor.
Emphysema	α_1-Antitrypsin
Hemophilia B	Factor IX
Thalassemia	α- or β-Globin
Sickle-cell anemia	β-Globin
Lesch-Nyhan syndrome	Hypoxanthine-guanine phosphoribosyltransferase (HGPRT).
Gaucher's disease	Glucocerebrosidase
Peripheral artery disease	Vascular endothelial growth factor (VEGF)
Fanconi anemia	Fanconi anemia C
Melanoma	Tumor necrosis factor (TNF)
Melanoma, renal cancer	Interleukin-2 (IL-2)
Glioblastoma (brain tumor), AIDS, ovarian cancer	Thymidine kinase (herpes simplex virus)
Head and neck cancer	p^{53}
Breast cancer	Multidrug resistance I
AIDS	*rev* and *env*
Colorectal cancer, melanoma, renal cancer	Histocompatability locus antigen-B_7 (HLA-B_7)
Duchenne muscular dystrophy	Dystrophin
Short stature*	Growth hormone
Diabetes*	Glucose transporter-2, (GLUT-2), glucokinase
Phenylketonuria*	Phenylalanine hydroxylase
Citrullinemia*	Arginosuccinate synthetase

**Mostly confined to animal experiments*

efficient only, if the therapeutic gene (remedial gene) is stably incorporated and continuously expressed. This can be achieved by use of vectors.

VECTORS IN GENE THERAPY

The ***carrier particles or molecules used to deliver genes to somatic cells*** are referred to as vectors. The important vectors employed in *ex vivo* gene therapy are listed below and briefly described next.

- Viruses
- Human artificial chromosome
- Bone marrow cells.

VIRUSES

The vectors frequently used in gene therapy are viruses, particularly ***retroviruses***. RNA is the genetic material in retroviruses. As the retrovirus enters the host cell, it synthesizes DNA from RNA (by reverse transcription). The so formed viral DNA (referred to as provirus) gets incorporated into the DNA of the host cell. The proviruses are normally harmless. However, there is a tremendous risk, since some of the retroviruses can convert normal cells into cancerous ones. Therefore, it is absolutely essential to ensure that such a thing does not happen.

Making retroviruses harmless

Researchers employ certain biochemical methods to convert harmful retroviruses to harmless ones, before using them as vectors. For instance, by artificially ***removing a gene that encodes for the viral envelope, the retrovirus*** can be crippled and ***made harmless***. This is because, without the

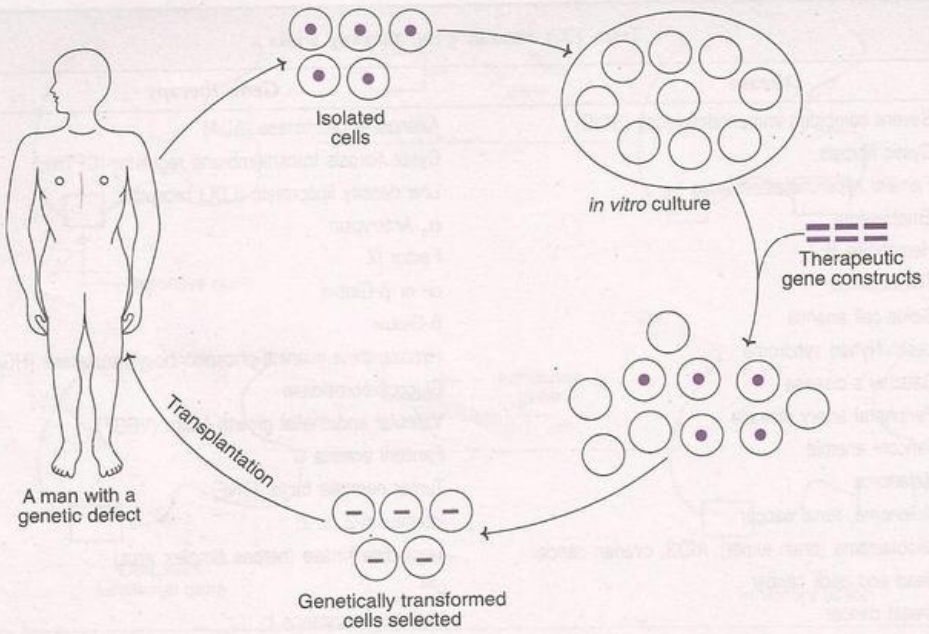


Fig. 13.2 : The procedure for ex vivo *gene therapy.*

envelope, retrovirus cannot enter the host cell. The production of a large number (billions) of viral particles can be achieved, starting from a single envelope defective retrovirus (***Fig. 13.3***). This is made possible by using helper viruses which contain normal gene for envelope formation. Along with the helper virus, the vector (with defective envelope gene) can enter the host cell and both of them multiply. By repeated multiplication in host cells, billions of vector and helper viruses are produced. The vector viruses can be separated from the helper viruses and purified. Isolation of vector viruses, totally free from helper viruses, is absolutely essential. Contamination of helper viruses is a big threat to the health of the patients undergoing gene therapy.

Retroviruses in gene therapy

The genetic map of a typical retrovirus is depicted in ***Fig. 13.4A***. In general, the retrovirus particle has RNA as a genome organized into six regions. It has a 5′-long terminal repeat (5′-LTR), a non-coding sequence required for packaging RNA designated as psi (Ψ), a gene ***gag*** coding for structural protein, a gene ***pol*** that codes for reverse transcriptase, a gene ***env*** coding for envelope protein and a 3-LTR sequence.

For use of a retrovirus as a vector, the structural genes ***gag*** and ***pol*** are deleted (besides ***env*** gene, as described above). These genes are actually adjacent to Ψ region. In addition, a promoter gene is also included (***Fig. 13.4B***). This vector design allows the synthesis of cloned genes. A ***retroviral vector can carry a therapeutic DNA of maximum size of 8 kb***.

A retroviral vector DNA can be used to transform the cells. However, the efficiency of delivery and integration of therapeutic DNA are very low. In recent years, techniques have been developed to deliver the vector RNA to host cells at a high frequency. For this purposes, packaged retroviral RNA particles are used. This technique allows a high efficiency of integration of pharmaceutical DNA into host genome.

Several ***modified viral vectors*** have been developed in recent years for gene therapy. These include ***oncoretrovirus***, ***adenovirus***, ***adeno-***

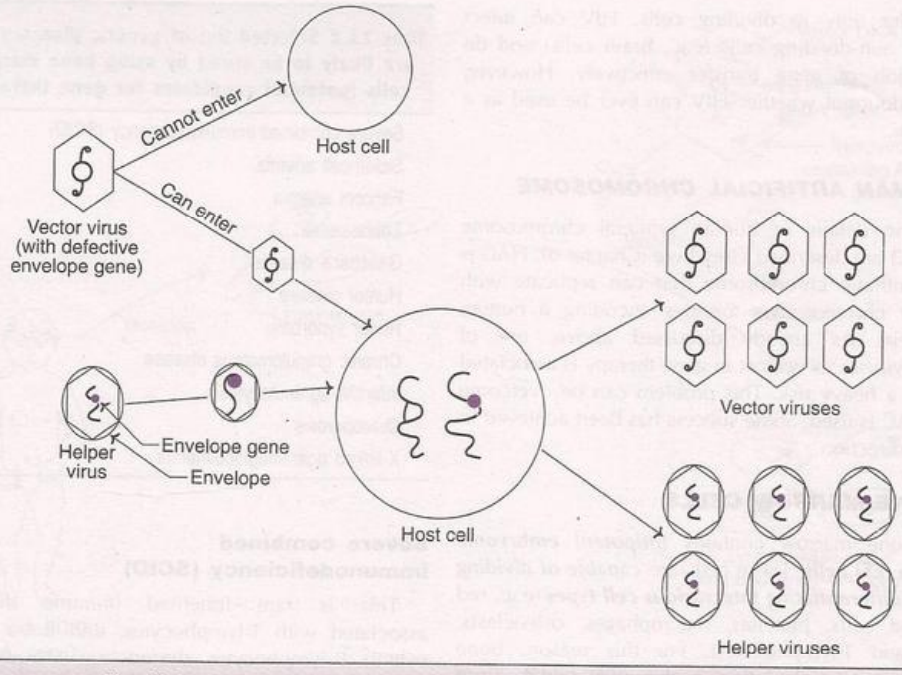


Fig. 13.3 : Large scale production of vector viruses by using helper viruses.

associated virus, ***herpes virus*** and a number of hybrid vectors combining the good characters of the parental vectors.

Murine leukemia viruses in gene therapy

This is a retrovirus that causes a type of leukemia in mice. It can react with human cells as well as the mouse cells, due to a similarity in the surface receptor protein. Murine leukemia virus (MLV) is frequently used in gene transfer.

AIDS virus in gene therapy?

It is suggested that the human immunodeficiency virus (HIV) can be used as a vector in gene transfer. But this is ***bound to create public uproar***. Some workers have been successful in creating a harmless HIV (crippled HIV) by removing all the genes related to reproduction. At the same time, the essential genes required for gene transfer are retained. There is a distinct advantage with HIV when compared with MLV. MLV is capable of bringing out gene

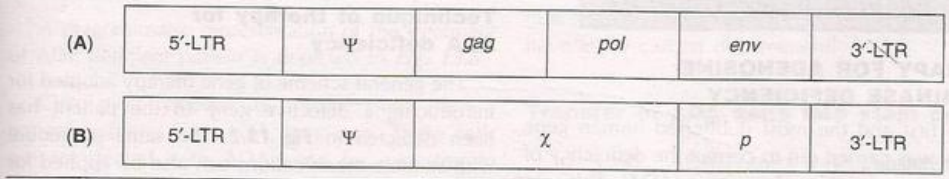


Fig. 13.4 : A retrovirus used in gene therapy. (A) General map of a typical retrovirus (B) Gene map of a modified retrovirus for use in gene therapy (LTR-Long terminal repeat; Ψ-Packaging signal sequence; gag-*Coding sequence for structural protein;* pol-*Coding sequence for reverse transcriptase;* env-*Envelope protein coding sequence; χ-Therapeutic gene; p–Promoter gene).*

transfer only in dividing cells. HIV can infect even non-dividing cells (e.g., brain cells) and do the job of gene transfer effectively. However, it is doubtful whether HIV can ever be used as a vector.

HUMAN ARTIFICIAL CHROMOSOME

The details of human artificial chromosome (HAC) are described elsewhere (Chapter 6). HAC is a synthetic chromosome that can replicate with other chromosomes, besides encoding a human protein. As already discussed above, use of retroviruses as vectors in gene therapy is associated with a heavy risk. This problem can be overcome if HAC is used. Some success has been achieved in this direction.

BONE MARROW CELLS

Bone marrow contains ***totipotent embryonic stem (ES) cells***. These cells are ***capable of dividing and differentiating into various cell types*** (e.g., red blood cells, platelets, macrophages, osteoclasts, B- and T-lymphocytes). For this reason, bone marrow transplantation is the most widely used technique for several genetic diseases. And there is every reason to believe that the genetic disorders that respond to bone marrow transplantation are likely to respond to ***ex vivo*** gene therapy also (***Table 13.2***). For instance, if there is a gene mutation that interferes with the function of erythrocytes (e.g., sickle-cell anemia), bone marrow transplantation is done. Bone marrow cells are the potential candidates for gene therapy of sickle-cell anemia. However, this is not as simple as theoretically stated.

TABLE 13.2 Selected list of genetic diseases that are likely to be cured by using bone marrow cells (potential candidates for gene therapy)

Severe combined immunodeficiency (SCID)
Sickle-cell anemia
Fanconi anemia
Thalassemia
Gaucher's disease
Hunter disease
Hurler syndrome
Chronic granulomatous disease
Infantile agranulocytosis
Osteoporosis
X-linked agammaglobulinemia

SELECTED EXAMPLES OF *EX VIVO* GENE THERAPY

THERAPY FOR ADENOSINE DEAMINASE DEFICIENCY

The first and the most publicised human gene therapy was carried out to correct the deficiency of the enzyme adenosine deaminase (ADA). This was done on September 14, 1990 by a team of workers led by Blaese and Anderson at the National Institute of Health, USA (The girl's name is Ashanti, 4 years old then).

Severe combined immunodeficiency (SCID)

This is rare inherited immune disorder associated with T-lymphocytes, and (to a lesser extent) B-lymphocytes dysfunction. About 50% of SCID patients have a defect in the gene (located on chromosome 20, and has 32,000 base pairs and 12 exons) that encodes for adenosine deaminase. In the deficiency of ADA, deoxyadenosine and its metabolites (primarily deoxyadenosine 5'-triphosphate) accumulate and destroy T-lymphocytes. T-Lymphocytes are essential for body's immunity. Besides participating directly in body's defense, they promote the function of B-lymphocytes to produce antibodies. Thus, the ***patients of SCID*** (lacking ADA) ***suffer from infectious diseases and die at an young age***. Previously, the children suffering from SCID were treated with conjugated bovine ADA, or by bone marrow transplantation.

Technique of therapy for ADA deficiency

The general scheme of gene therapy adopted for introducing a defective gene in the patient has been depicted in ***Fig 13.2***. The same procedure with suitable modifications can also be applied for other gene therapies.

A plasmid vector bearing a proviral DNA is selected. A part of the proviral DNA is replaced by the ADA gene and a gene (G 418) coding for

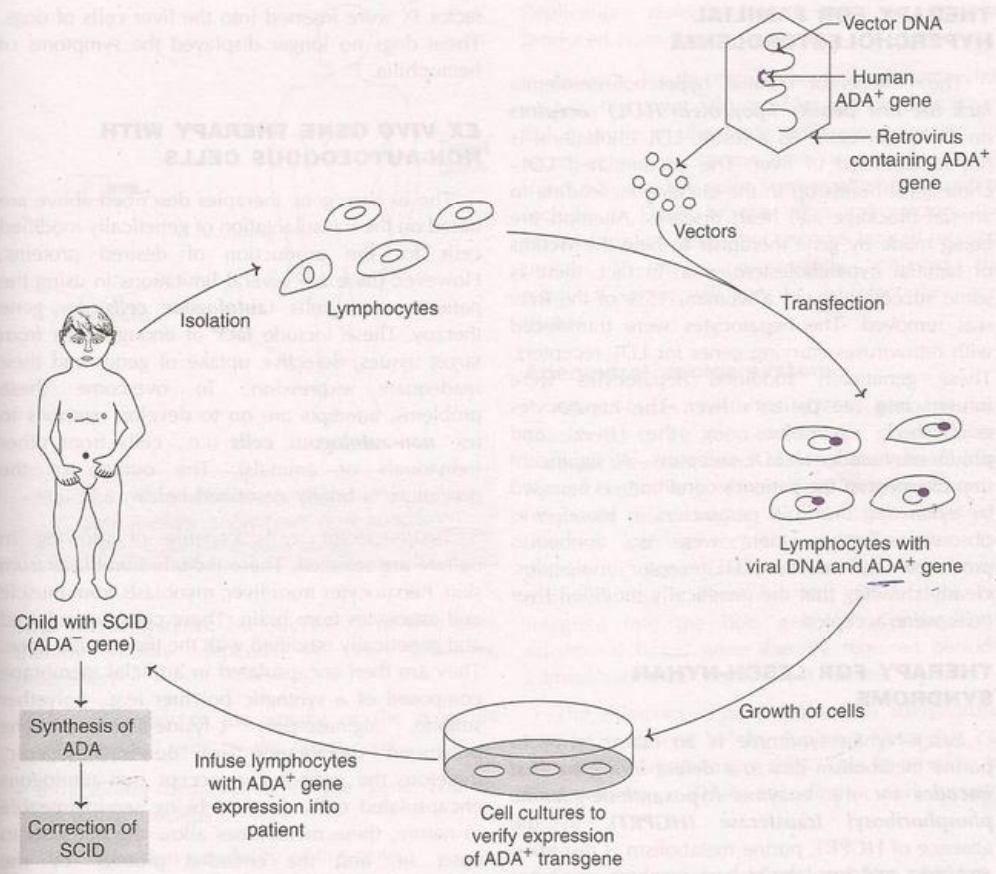


Fig. 13.5 : *Treatment of adenosine deaminase (ADA) deficient patient by somatic* ex vivo *gene therapy (SCID-Severe combined immunodeficiency disease).*

antibiotic resistance, and then cloned. The antibiotic resistance gene will help to select the desired clones with ADA gene.

A diagrammatic representation of the treatment of ADP deficient patient is depicted in ***Fig. 13.5***.

Circulating lymphocytes are removed from a patient suffering from ADA deficiency. These cells are transfected with ADA gene by exposing to billions of retroviruses carrying the said gene. The genetically-modified lymphocytes are grown in cultures to confirm the expression of ADA gene and returned to the patient. These lymphocytes persist in the circulation and synthesize ADA. Consequently, the ability of the patient to produce antibodies is increased. However, there is a limitation. The lymphocytes have a short life span (just live for a few months), hence the transfusions have to be carried out frequently.

Transfer of ADA gene into stem cells

In 1995, ADA gene was transfered into the stem cells, obtained from the umbilical cord blood, at the time of baby's delivery. Four days after birth, the infant received the modified cells back. By this way, a permanent population of ADA gene producing cells was established.

THERAPY FOR FAMILIAL HYPERCHOLESTEROLEMIA

The patients of familial hypercholesterolemia ***lack the low density lipoprotein (LDL) receptors*** on their liver cells. As a result, LDL cholesterol is not metabolised in liver. The accumulated LDL-cholesterol builds up in the circulation, leading to arterial blockage and heart diseases. Attempts are being made by gene therapists to help the victims of familial hypercholesterolemia. In fact, there is some success also. In a woman, 15% of the liver was removed. The hepatocytes were transduced with retroviruses carrying genes for LDL receptors. These genetically modified hepatocytes were infused into the patient's liver. The hepatocytes established themselves in the liver and produced functional LDL-receptors. A significant improvement in the patient's condition, as assessed by estimating the lipid parameters in blood, was observed. Further, there were no antibodies produced against the LDL-receptor molecules, clearly showing that the genetically modified liver cells were accepted.

THERAPY FOR LESCH-NYHAN SYNDROME

Lesch-Nyhan syndrome is an inborn error in purine metabolism due to a ***defect in a gene that encodes*** for the enzyme ***hypoxanthine-guanine phosphoribosyl transferase (HGPRT)***. In the absence of HGPRT, purine metabolism is disturbed and uric acid level builds up, resulting in severe gout and kidney damage. The victims of Lesch-Nyhan syndrome exhibit symptoms of mental retardation, besides an urge to bite lips and fingers, causing self-mutilation.

By using retroviral vector system, HGPRT producing genes were successfully inserted into cultured human bone marrow cells. The major problem in humans is the involvement of brain. Experiments conducted in animals are encouraging. However, it is doubtful whether good success can be achieved by gene therapy for Lesch-Nyhan syndrome in humans, in the near future.

THERAPY FOR HEMOPHILIA

Hemophilia is a genetic disease due ***lack of a gene that encodes for clotting factor IX***. It is characterized by excessive bleeding. By using a retroviral vector system, genes for the synthesis of factor IX were inserted into the liver cells of dogs. These dogs no longer displayed the symptoms of hemophilia.

EX VIVO GENE THERAPY WITH NON-AUTOLOGOUS CELLS

The *ex vivo* gene therapies described above are based on the transplantation of genetically modified cells for the production of desired proteins. However, there are several limitations in using the patient's own cells (***autologous cells***) for gene therapy. These include lack of enough cells from target tissues, defective uptake of genes and their inadequate expression. To overcome these problems, attempts are on to develop methods to use ***non-autologous cells*** (i.e., cells from other individuals or animals). The outline of the procedure is briefly described below.

Tissue-specific cells capable of growing in culture are selected. These include fibroblasts from skin, hepatocytes from liver, myoblasts from muscle and astrocytes from brain. These cells are cultured and genetically modified with the therapeutic gene. They are then encapsulated in artificial membrane composed of a synthetic polymer (e.g., polyether sulfone, alginase-poly L-lysine-alginate). The polymeric membranes are non-immunogenic, therefore the patient can accept non-autologous encapsulated cells. Further, being semipermeable in nature, these membranes allow the nutrients to enter in, and the encoded protein (by the therapeutic gene) to pass out.

Experiments conducted in animals have shown some encouraging results for using non-autologous cells in gene therapy. The encapsulated cells were found to proliferate and produce the required protein. However, the success has been very limited in human trials.

IN VIVO GENE THERAPY

The ***direct delivery of the therapeutic gene*** (DNA) ***into the target cells*** of a particular tissue of a patient constitutes *in vivo* gene therapy (***Fig. 13.6***). Many tissues are the potential candidates for this approach. These include liver, muscle, skin, spleen, lung, brain and blood cells. Gene delivery can be carried out by ***viral*** or ***non-***

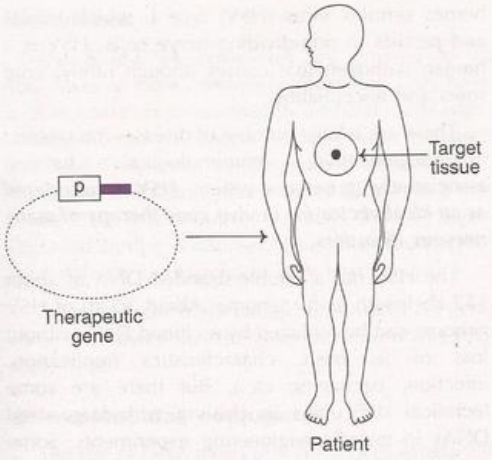


Fig. 13.6 : Diagrammatic representation of in vivo *gene therapy. (p-Promoter gene specific for therapeutic gene)*

viral vector systems. The success of ***in vivo*** gene therapy mostly depends on the following parameters

- The efficiency of the uptake of the remedial (therapeutic) gene by the target cells.
- Intracellular degradation of the gene and its uptake by nucleus.
- The expression capability of the gene.

In vivo gene therapy with special reference to gene delivery systems (viral, non-viral) with suitable examples is described.

GENE DELIVERY BY VIRUSES

Many viral vector systems have been developed for gene delivery. These include retroviruses, adenoviruses, adeno-associated viruses and herpes simplex virus.

Retrovirus vector system

Replication defective retrovirus vectors that are harmless are being used (See p. 159). A ***plasmid in association with a retrovirus, a therapeutic gene and a promoter is referred to as plasmovirus***. The plasmovirus is capable of carrying a DNA (therapeutic gene) of size less than 3.4 kb. Replication defective virus particles can be produced from the plasmovirus.

As such, for the delivery of genes by retroviral vectors, the target cells must be in a dividing stage. But majority of the body cells are quiescent. In recent years, viral vectors have been engineered to infect non-dividing cells. Further, attempts are on to include a DNA in the retroviral vectors (by engineering *env* gene) that encodes for cell receptor protein. If this is successfully achieved, the retroviral vector will specifically infect the target tissues.

Adenoviral vector system

Adenoviruses (with a ***DNA genome***) are considered to be good vectors for gene delivery because they ***can infect most of the non-dividing human cells***. A common cold adenovirus is a frequently used vector. As the target cells are infected with a recombinant adenovirus, the therapeutic gene (DNA) enters the nucleus and expresses itself. However, this DNA does not integrate into the host genome. Consequently, adenoviral based gene therapy required periodic administration of recombinant viruses.

The efficiency of gene delivery by adenoviruses can be enhanced by developing a virus that can specifically infect target cells. This is possible by incorporating a DNA encoding a cell receptor protein.

Adeno-associated virus vector system

Adeno-associated virus is a human virus that can integrate into chromosome 19. It is a single-stranded, non-pathogenic small DNA virus (4.7 kb). As the adeno-associated virus enters the host cell, the DNA becomes double-stranded, gets integrated into chromosome and expresses.

Adeno-associated viruses can serve as good vectors for the delivery of therapeutic genes. Recombinant viruses are created by using two plasmids and an adenovirus (i.e., helper virus) by a special technique. Some attempts were made to use therapeutic genes for the treatment of the human diseases-hemophilia (for production of blood clotting factor IX) and cystic fibrosis (for synthesis of cystic fibrosis transmembrane regulator protein) by employing adeno-associated viruses.

Therapy for cystic fibrosis

Cystic fibrosis (CF) is one of the most common (frequence 1 : 2,500) and fatal genetic diseases. It is characterized by the accumulation of sticky, dehydrated mucus in the respiratory tract and lungs. Patients of CF are highly susceptible to bacterial infections in their lungs and most of them die before reaching the age of thirty.

Cystic fibrosis can be traced in European folklore, the following statement used to be said "Woe to that child which when kissed on the forehead tastes salty. He is be witched and soon must die".

Biochemical basis : In the normal persons the chloride ions of the cells are pushed out through the participation of a protein called ***cystic fibrosis transmembrane regulator (CFTR)***. In the patients of cystic fibrosis, the CFTR protein is not produced due to a gene defect. Consequently, the chloride ions concentrate within the cells which draw water from the surroundings. As a result, the respiratory tract and the lungs become dehydrated with sicky mucus, an ideal environment for bacterial infections.

Gene therapy : As the defective gene for cystic fibrosis was identified in 1989, researchers immediately started working on gene therapy for this disease. Adenoviral vector systems have been used, although the success has been limited. The major drawback is that the benefits are short-lived, since the adenoviruses do not integrate themselves into host cells. Multiple administration of recombinant adenovirus caused immunological responses that destroyed the cells.

By using adeno-associated virus vector system, ***some encouraging results were reported in the gene therapy of CF***. In the phase I clinical trials with CF patients, the vector persisted for about 70 days and some improvement was observed in the patients.

Some researchers are trying to insert CF gene into the developing fetal cells (in experimental animals such as mice) to produce CFTR protein. But ***a major breakthrough is yet to come***.

Herpes simplex virus vector system

The retroviruses and adenoviruses (discussed above) employed in *in vivo* gene therapy are engineered to infect specific target cells. There are some viruses which have a natural tendency to infect a particular type of cells. The best example is herpes simplex virus (HSV) type I, which infects and persists in non-dividing nerve cells. HSV is a human pathogen that causes (though rarely) cold sores and encephalitis.

These are a large number of diseases (metabolic, neurodegenerative, immunological, tumors) associated with nervous system. ***HSV is considered as an ideal vector for in vivo gene therapy of many nervous disorders***.

The HSV has a double-stranded DNA of about 152 kb length as its genome. About 30 kb of HSV genome can be replaced by a cloned DNA without loss of its basic characteristics (replication, infection, packaging etc.). But there are some technical difficulties in dealing with large-sized DNAs in genetic engineering experiments. Some modified HSV vectors with reduced genomic sizes have been developed.

Most of the work on the gene therapy, related to the use of HSV as a vector, is being conducted in experimental animals. And the results are quite encouraging. HSV vectors could deliver therapeutic genes to the brain and other parts of nervous system. These genes are well expressed and maintained for long periods. More research, however, is needed before going for human trials.

If successful, HSV may help to treat many neurodegenerative syndromes such as Parkinson's disease and Alzheimer's disease by gene therapy.

GENE DELIVERY BY NON-VIRAL SYSTEMS

There are certain limitations in using viral vectors in gene therapy. In addition to the prohibitive cost of maintaining the viruses, the viral proteins often induce inflammatory responses in the host. Therefore, there is a continuous search by researchers to find alternatives to viral vector systems.

Pure DNA constructs

The ***direct introduction of pure DNA constructs into the target tissue*** is quite simple. However, the efficiency of DNA uptake by the cells and its expression are rather low. Consequently, large quantities of DNA have to be injected periodically. The therapeutic genes produce the proteins in the target cells which enter the circulation and often get degraded.

Lipoplexes

The ***lipid-DNA complexes*** are referred to as lipoplexes or more commonly ***liposomes***. They have a DNA construct surrounded by artificial lipid layers. A large number of lipoplexes have been prepared and used. They are non-toxic and non-immunogenic. The major limitation with the use of lipoplexes is that as the DNA is taken up by the cells, most of it gets degraded by the lysosomes. Thus, the efficiency of gene delivery by lipoplex is very low.

Some clinical trials using liposome-CFTR gene complex showed that the gene expression was very short-lived.

DNA-molecular conjugates

The use of DNA-molecular conjugates avoids the lysosomal breakdown of DNA. Another advantage of using conjugates is that large-sized therapeutic DNAs (> 10 kb) can be delivered to the target tissues. The most commonly used synthetic conjugate is ***poly-L-lysine***, bound to a specific target cell receptor. The therapeutic DNA is then made to combine with the conjugate to form a complex (***Fig. 13.7***). This DNA molecular conjugate binds to specific cell receptor on the target cells. It is engulfed by the cell membrane to form an endosome which protects the DNA from being degraded. The DNA released from the endosome enters the nucleus where the therapeutic gene is expressed.

Human artificial chromosome

Human artificial chromosome (***HAC***) which can carry a large DNA (one or more therapeutic genes with regulatory elements) is ***a good and ideal vector***. Studies conducted in cell cultures using HAC are encouraging. But the major ***problem is the delivery*** of the large-sized chromosome ***into the target cells***. Researchers are working to produce cells containing genetically engineered HAC. There exists a possibility of encapsulating and implanting these cells in the target tissue. But a long way to go!

Efficiency of gene delivery by non-viral vectors

Although the efforts are continuously on to find suitable non-viral vectors for gene delivery, the success has been very limited. This is mainly due to the following two reasons.

1. The efficiency of transfection is very low.

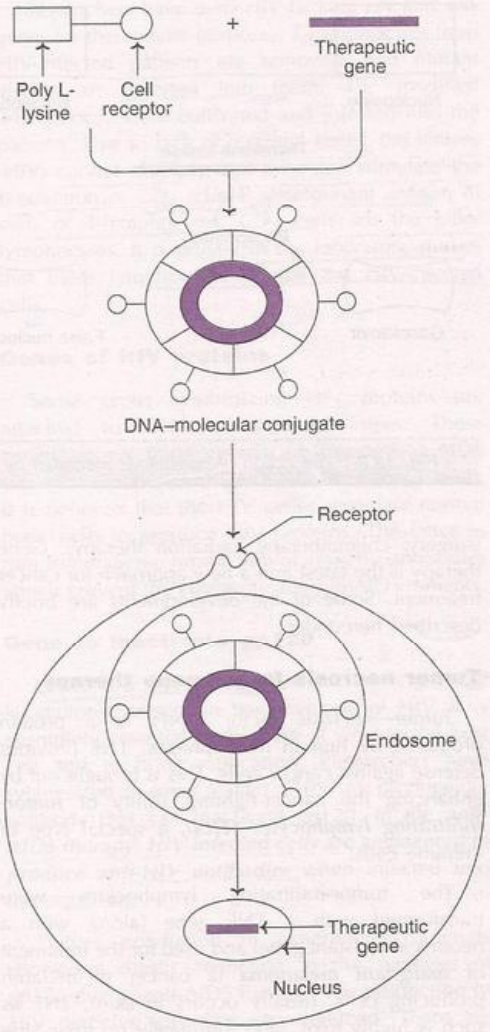


Fig. 13.7 : DNA-molecular conjugate in the delivery of therapeutic gene.

2. The expression of the therapeutic gene is for a very short period, consequently there is no effective treatment of the disease.

GENE THERAPY STRATEGIES FOR CANCER

Cancer is the leading cause of death throughout the world, despite the intensive treatment strategies

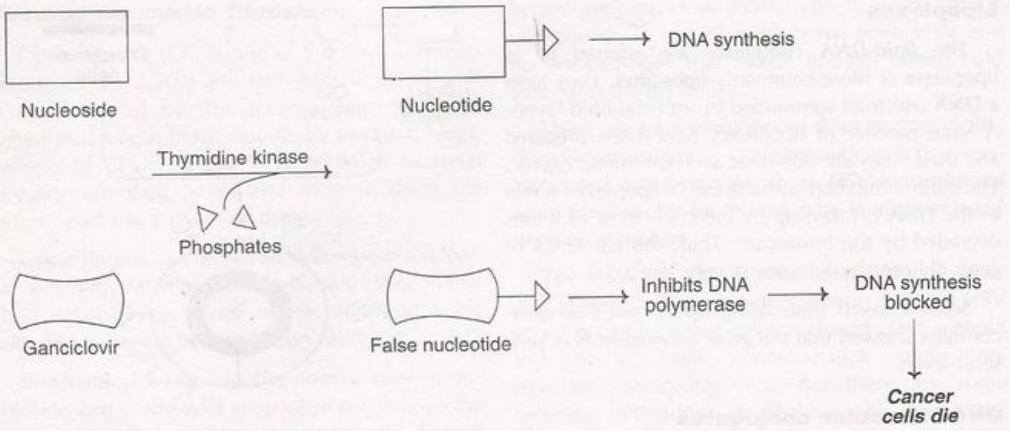


Fig. 13.8 : The action of ganciclovir mediated by thymidine kinase to inhibit the growth of cancer cells.

(surgery, chemotherapy, radiation therapy). Gene therapy is the latest and a new approach for cancer treatment. Some of the developments are briefly described hereunder.

Tumor necrosis factor gene therapy

Tumor necrosis factor (TNF) is a protein produced by human macrophages. TNF provides defense against cancer cells. This is brought out by enhancing the cancer-fighting ability of ***tumor-infiltrating lymphocytes (TILs)***, a special type of immune cells.

The tumor-infiltrating lymphocytes were transformed with a TNF gene (along with a neomycin resistant gene) and used for the treatment of ***malignant melanoma*** (a cancer of melanin producing cells, usually occurs in skin). TNF as such is highly toxic, and fortunately no toxic side effects were detected in the melanoma patients injected with genetically altered TILs with TNF gene. Some improvement in the cancer patients was observed.

Suicide gene therapy

The gene encoding the enzyme ***thymidine kinase*** is often referred to as suicide gene, and is used for the treatment of certain cancers.

Thymidine kinase (TK) phosphorylates nucleosides to form nucleotides which are used for the synthesis of DNA during cell division. The drug ***ganciclovir*** (***GCV***) bears a close structural resemblance to certain nucleosides (thymidine). By mistake, TK phosphorylates ganciclovir to form triphosphate-GCV, a false and unsuitable nucleotide for DNA synthesis. ***Triphosphate-GCV inhibits DNA polymerase (Fig. 13.8)***. The result is that the elongation of the DNA molecule abruptly stops at a point containing the false nucleotide (of ganciclovir). Further, the triphospate-GCV can enter and kill the neighbouring cancer cells, a phenomenon referred to as ***bystander effect***. The ultimate result is that the cancer cells cannot multiply, and therefore die. Thus, the drug ganciclovir can be used to kill the cancer cells.

Ganciclovir is frequently referred to as a ***prodrug*** and this type of approach is called ***prodrug activation gene therapy***. Ganciclovir has been used for treatment of brain tumors (e.g., glioblastoma, a cancer of glial cells in brain), although with a limited success.

In the suicide gene therapy, the ***vector used*** is ***herpes simplex virus (HSV)*** with a gene for thymidine kinase (TK) inserted in its genome. Normal brain cells do not divide while the brain tumor cells go on dividing unchecked. Thus, there is a continuous DNA replication in tumor cells. By using GCV-HSVTK suicide gene therapy, some reduction in proliferating tumor cells was reported. Several new strategies are being developed to increase the delivery of HSVTK gene to all the cells throughout a tumor.

Two-gene cancer therapy

For treatment of certain cancers, two gene systems are put together and used. For instance, ***TK suicide gene*** (i.e., GCV-HSVTK) is clubed with ***interleukin-2 gene*** (i.e. a gene promoting immunotherapy). Interleukin-2 produced mobilizes immune response. It is believed that certain proteins are released from the tumor cells on their death. These proteins, in association with immune cells, reach the tumor and initiate immunological reactions directed against the cancer cells.

Two-gene therapies have been carried out in experimental animals with colon cancer and liver cancer, and the results are encouraging.

Gene replacement therapy

A gene named ***p*^{53}** codes for a ***p***rotein with a molecular weight of ***53*** kilodaltons (hence p^{53}). p^{53} is considered to be a ***tumor-suppressor gene***, since the protein it encodes binds with DNA and inhibits replication. The tumor cells of several tissues (breast, brain, lung, skin, bladder, colon, bone) were found to have altered genes of p^{53} (mutated p^{53}), synthesizing different proteins from the original. These altered proteins cannot inhibit DNA replication. It is believed that the damaged p^{53} gene may be a causative factor in tumor development.

Some workers have tried to replace the damaged p^{53} gene by a normal gene by employing adenovirus vector systems (See p. 165). There are some encouraging results in the patients with liver cancer.

The antisense therapy for cancer is discussed elsewhere (See p. 170).

GENE THERAPY FOR AIDS

AIDS is a global disease with an alarming increase in the incidence every year. It is invariably fatal, since there is no cure. Attempts are being made to relieve the effects of AIDS by gene therapy. Some of the approaches are discussed hereunder.

rev and *env* genes

A mutant strain of human immunodeficiency virus (HIV), lacking ***rev*** and ***env*** genes has been developed. The regulatory and envelope proteins of HIV are respectively produced by ***rev*** and ***env*** genes. Due to lack of these genes, the virus cannot replicate.

Researchers have used HIV lacking ***rev*** and ***env*** genes for therapeutic purposes. T-Lymphocytes from HIV-infected patients are removed, and mutant viruses are inserted into them. The modified T-lymphocytes are cultivated and injected into the patients. Due to lack of essential genes, the viruses (HIV) cannot multiply, but they can stimulate the production of CD_8 (cluster determinant antigen 8) cells of T-lymphocytes. CD_8 cells are the killer lymphocytes. It is proved in the laboratory studies that these lymphocytes destroy the HIV-infected cells.

Genes of HIV proteins

Some genes synthesizing HIV proteins are attached to DNA of mouse viruses. These genetically-modified viruses are injected to AIDS patients with clinical manifestations of the disease. It is believed that the HIV genes stimulate normal body cells to produce HIV proteins. The latter in turn stimulate the production of anti-HIV antibodies which prevent the HIV replication in AIDS patients.

Gene to inactivate *gp120*

gp120 is a ***g***lyco***p***rotein (molecular weight ***120*** kilodaltons) present in the envelope of HIV. It is absolutely essential for binding of virus to the host cell and to bring replication. Researchers have synthesized a gene (called F105) to produce an antibody that can inactivate gp120. In the ***anti-AIDS therapy, HIV-infected cells are engineered to produce anti-HIV antibodies*** when injected into the organism.

Studies conducted in experimental animals showed a drastic reduction in the synthesis of gp120 due to anti-AIDS therapy. The production of HIV particles was also very reduced. There are some attempts to prevent AIDS by antisense therapy (See p. 171).

ANTIGENE AND ANTISENSE THERAPY

In general, gene therapy is carried out by introducing a therapeutic gene to produce the defective or the lacking protein. But there are certain disorders (cancer, viral and parasitic infections, inflammatory diseases) which result in an overproduction of certain normal proteins. It is

possible to treat these diseases by ***blocking transcription*** using a single-stranded nucleotide sequence (antigene oligonucleotide) that hybridizes with the specific gene, and this is called ***antigene therapy***. ***Antisense therapy*** refers to the ***inhibition of translation*** by using a single-stranded nucleotide (antisense oligonucleotide). Further, it is also possible to inhibit both transcription and translation by blocking (with oligonucleotides) the transcription factor responsbile for the specific gene expression.

Nucleic acid therapy refers to the ***use of DNA or RNA molecules for therapeutic purposes***, as stated above. The naturally occurring sequences of DNA and RNA (with suitable modifications) or the synthetic ones can be employed in nucleic acid therapy. Theoretically, there is a vast potential for use of nucleic acids as therapeutic agents. But most of the work that is being carried out relates to the use of RNA in antisense therapy. Some of these are described below (**Note** : Some authors use antisense therapy in a broad sense to reflect antigene therapy as well as antisense therapy, discussed in the previous paragraph).

ANTISENSE THERAPY FOR CANCER

Oncogenes are the genes responsible for the causation of cancer. The dominantly acting oncogenes can be targeted in antisense technology by using antisense transgenes or oligonucleotides. Antisense oligonucleotides are used for the treatment of myeloid leukemia in as early as 1991.

Antisense RNA molecules are more frequently used in cancer therapy. This approach is effective only if the antisense oligonucleotide (antisense mRNA) specifically binds to the target mRNA and blocks protein biosynthesis (translation). This can be achieved in two ways, as illustrated in ***Fig. 13.9***.

The antisense cDNA can be cloned and transfected into cells. Antisense mRNA is synthesized by transcription. This can readily bind with the specific mRNA and block translation (***Fig. 13.9A***). The mRNA is actually formed by a gene containing exons and introns through transcription, followed by processing.

The other way to block translation is to directly introduce antisense RNA into the cells. This hybridizes with target mRNA and blocks translation (***Fig. 13.9B***).

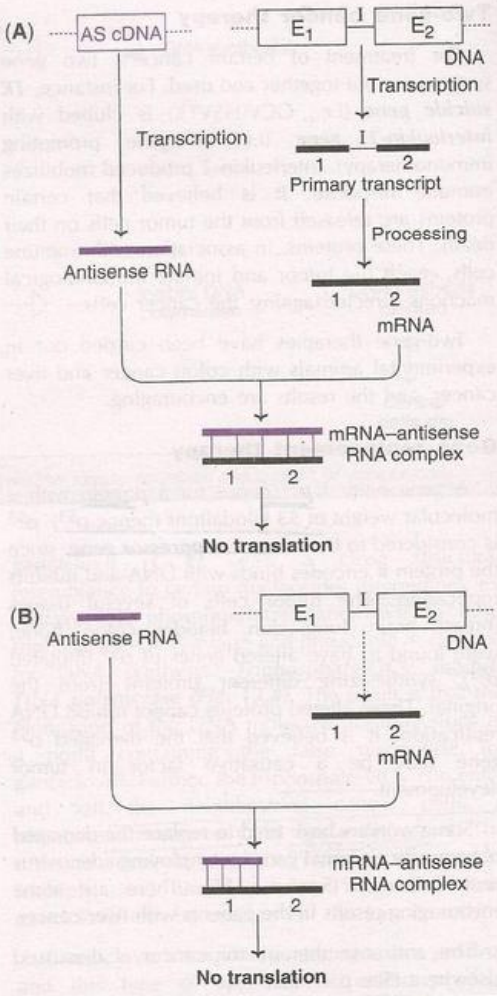


***Fig. 13.9** : Inhibition of translation by antisense RNA (**A**) The cloned AS cDNA introduced into cells to produce antisense RNA (**B**) Antisense RNA directly introduced into cells. (AS cDNA = Antisense complementary DNA; E_1, E_2-Exons in a gene; I-Intron)*

The antisense mRNA therapy was tried for the treatment of a brain tumor namely malignant glioma and the cancer of prostate gland. In case malignant glioma, the protein insulin-like growth factor I (IGF-I) is overproduced, while in prostate cancer, insulin-like growth factor I receptor (IGF-IR) protein is more synthesized. For both these cancers, the respective antisense cDNAs can be used to synthesize, antisense mRNA molecules. These in

turn, are used to block translation, as briefly described above and illustrated in ***Fig. 13.9***.

ANTISENSE THERAPY FOR AIDS

Attempts are also being made to prevent HIV infection through antisense therapy. The basic principle is briefly described (***Fig. 13.10***).

The target cells for HIV infection are genetically engineered to contain a gene that can express a complementary copy of HIV genome. This gene produces antisense RNA. When the cells containing antisense RNA are infected by HIV, it binds to viral RNA forming a double-stranded RNA-RNA hybrid molecules. This double-stranded molecule cannot be used by the enzyme reverse transcriptase. Consequently, a DNA copy of the HIV genome cannot be made and incorporated into the genome.

ANTISENSE OLIGONUCLEOTIDES AS THERAPEUTIC AGENTS

It is found that oligodeoxynucleotides (with about 15-20 units) can hybridize with specific mRNA and block translation with an ultimate effect of reducing the specific protein. Oligonucleotides can hybridize with different types of RNA transcripts-mRNAs, intron-exons, double-stranded RNAs. The major limitation of using naturally occurring oligonucleotides is that they are degraded by intracellular nucleases.

Certain modifications in the nucleotides (bases or sugars, phosphate) without affecting their hybridization ability have been prepared. However, these modified oligonucleotides are resistant to degradation by nucleases. The most widely used antisense oligonucleotide has a sulfur group in place of the free oxygen in phosphodiester bond. The modified bond is referred to as ***phosphorothioate linkage***. These phosphorothioate antisense oligonucleotides are resistant to degradation by nucleases, besides being water soluble. Some other modifications of oligo-nucleotides are with phosphoramidite linkage and 2-0-methylribose (***Fig. 13.11***). The modified antisense oligonucleotides are in fact used as therapeutic agents in the clinical trials for the treatment of certain cancers, bowel disease, AIDS, malaria and viral infections.

The free oligonucleotides do not readily enter the cells. Therefore, they are encapsulated in liposomes for efficient delivery to inhibit the mRNA translation.

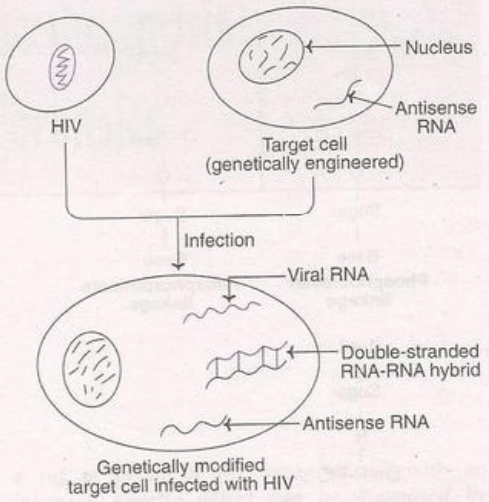


Fig. 13.10 : Antisense therapy to prevent AIDS (HIV-Human immunodeficiency virus)

Possible inhibition of smooth muscle cell proliferation : Proliferation of smooth muscle cells (along with secretion of extracellular matrix) has been implicated in hypertension, failure of coronary bypass grafts, hypertension and atherosclerosis. There exists a possibility of controlling smooth muscle cell proliferation by using antisense therapy.

CHIMERIC OLIGONUCLEOTIDES IN GENE CORRECTION

Many genetic diseases are due to single base pair mutations. A strategy has been devised to correct such defects by using a ***chimeric oligonucleotides*** composed of RNA-DNA oligonucleotide with 68 units. This chimeric oligonucleotide has hairpin caps and methylated ribose sugars (at 2nd carbon). The hairpin caps protect the molecule from the degradation by exonucleases while methylated ribose prevents digestion by RNase.

The correction of a single base pair mutation by using a chimeric oligonucleotide is depicted in ***Fig. 13.12***. The RNA-DNA molecules (chimeric oligonucleotides) bind more readily with duplex DNA and bring about the correction in the mutated base pair.

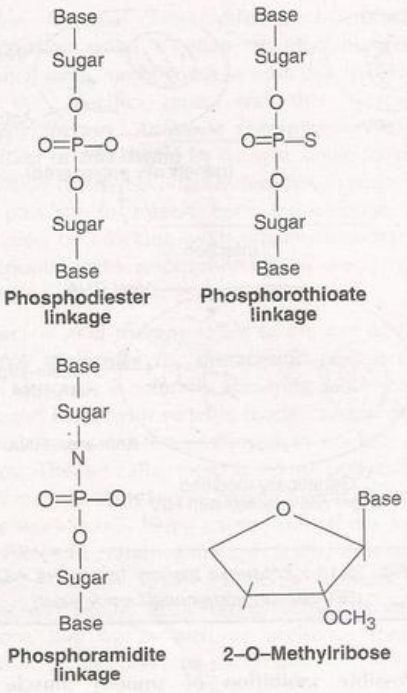


Fig. 13.11 : Certain modifications in antisense oligonucleotides.

APTAMERS AS THERAPEUTIC AGENTS

A special type of ***oligonucleotides*** that can specifically ***bind to target proteins*** and not to nucleic acids are referred to as aptamers. Aptamers are useful as therapeutic agents. For instance, an aptamer that can bind to thrombin inhibits blood clotting. Such oligonucleotides can be used in surgical procedures.

RIBOZYMES AS THERAPEUTIC AGENTS

RNA molecules that can serve as ***enzymes*** (biocatalysts) are referred to as ribozymes. The naturally occurring ribozymes are to be modified so that they can specifically hybridize with mRNA sequences and block protein biosynthesis (translation). There is a limitation however, in directly inserting ribozymes into target cells, as they are easily degraded. This can be overcome by binding ribozyme with an oligodeoxynucleotide. There is a possibility of treating certain cancers and viral diseases by genetically engineered ribozymes.

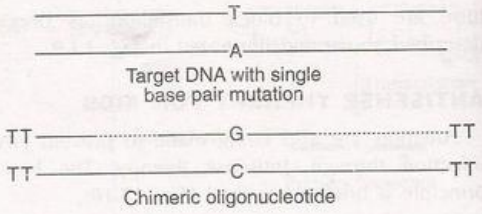


Fig. 13.12 : Chimeric oligonucleotide in gene correction (The dotted lines represent ribonucleotides while the continuous lines correspond to deoxyribonucleotides; the coloured base pair is the correct one to replace the mutated base pair)

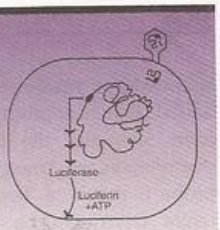
THE FUTURE OF GENE THERAPY

Theoretically, gene therapy is ***the permanent solution for genetic diseases***. But it is not as simple as it appears since gene therapy has several inbuilt complexicities. Gene therapy broadly involves isolation of a specific gene, making its copies, inserting them into target tissue cells to make the desired protein. The story does not end here. It is absolutely essential to ensure that the gene is harmless to the patient and it is appropriately expressed (too much or too little will be no good). Another concern in gene therapy is the body's immune system which reacts to the foreign proteins produced by the new genes.

The public, in general, have an ***exaggerated expectations*** on gene therapy. The researchers, at least for the present, are ***unable to satisfy them***. As per the records, by 1999 about 1000 Americans had undergone clinical trails involving various gene therapies. Unfortunately, the gene therapists are unable to categorically claim that gene therapy has permanently cured any one of these patients! Some people in the media (leading news papers and magazines) have openly questioned whether it is worth to continue research on gene therapy!!

It may be true that as of now, gene therapy due to several limitations, has not progressed the way it should, despite intensive research. But ***a break-through may come anytime***, and of course, this is only possible with persistent research. And a day may come (it might take some years) when almost every disease will have a gene therapy, as one of the treatment modalities. And gene therapy will revolutionize the practice of medicine!

14 DNA in Disease Diagnosis and Medical Forensics



Diagnosis of diseases due to pathogens (bacteria, viruses, fungi etc.) or due to inherent genetic defects is necessary for appropriate treatment. Traditional diagnostic methods for parasite infections include microscopic examination, *in vitro* culture, and detection of antibodies in serum. And for the genetic diseases, the procedures such as estimation of metabolites (blood/urine) and enzyme assays are employed. These laboratory techniques are indirect and not always specific. Scientists are in continuous search for specific, sensitive and simple diagnostic techniques for identification of diseases.

DNA, being the genetic material of the living organisms, contains the information that contributes to various characteristic features of the specific organism. Thus, ***the presence of a disease-causing pathogen can be detected by identifing a gene or a set of genes of the organism***. Likewise an ***inherited genetic defect can be diagnosed by identifying the alterations in the gene***. In the modern laboratory diagnostics, DNA analysis is a very useful and sensitive tool.

The basic principles underlying the DNA diagnostic systems, and their use in the diagnosis of certain pathogenic and genetic diseases are described. Besides these, the various approaches for DNA fingerprinting (or DNA profiling) are also discussed.

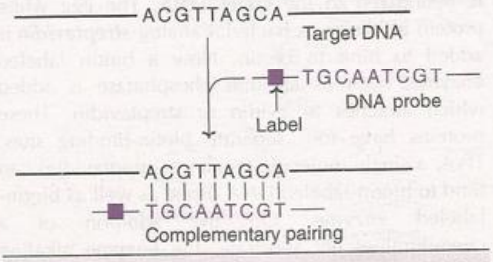


***Fig. 14.1** : Hybridization of target DNA with DNA probe (with radioactive isotope label).*

METHODS OF DNA ASSAY

The specific ***identification of the DNA sequence*** is absolutely ***essential in the laboratory diagnostics***. This can be achieved by employing the following principles/tools.

NUCLEIC ACID HYBRIDIZATION

Hybridization of nucleic acids (particularly DNA) is the basis for reliable DNA analysis. Hybridization is based on the principle that a single-stranded DNA molecule recognizes and specifically binds to a complementary DNA strand amid a mixture of other DNA strands. This is comparable to a specific key and lock relationship. The general procedure adopted for nucleic acid hybridization is as follows (***Fig. 14.1***).

The single-stranded target DNA is bound to a membrane support. Now the DNA probe (single-

stranded and labeled with a detector substance) is added. Under appropriate conditions (temperature, ionic strength), the DNA probe pairs with the complementary target DNA. The unbound DNA probe is removed. Sequence of nucleotides in the target DNA can be identified from the known sequence of DNA probe.

There are two types of DNA hybridization-radioactive and non-radioactive respectively using DNA probes labeled with isotopes and non-isotopes (discussed below) as detectors.

DNA PROBES

A DNA probe or a ***gene probe*** is a synthetic, ***single-stranded DNA molecule that can recognize and specifically bind to a target DNA*** (by complementary base pairing) in a mixture of biomolecules. DNA probes are either long (> 100 nucleotides) or short (< 50 nucleotides), and may bind to the total or a small portion of the target DNA. There is a wide variation in the size of DNA probes used (may range from 10 bases to 10,000 bases). The most important requirement is their specific and stable binding with target DNAs.

Methods employed to obtain DNA probes

A great majority of DNA probes are chemically synthesized in the laboratory. There are, however, many other ways of obtaining them-isolation of selected regions of genes, cloning of intact genes, producing from mRNAs.

Isolation of selected regions of genes : The DNA from an organism (say a pathogen) can be cut by using restriction endonucleases. These DNA fragments are cloned in vectors and the DNA probes can be selected by screening.

Synthesis of DNA probes from mRNA : The mRNA molecules specific to a particular DNA sequence (encoding a protein) are isolated. By using the enzyme reverse transcriptase, complementary DNA (cDNA) molecules are synthesized. This ***cDNA*** can be used ***as a probe*** to detect the target DNA.

Mechanism of action of DNA probes

The basic principle of DNA probes is based on the ***denaturation and renaturation (hybridization) of DNA***. When a double-stranded DNA molecule is subjected to physical (temperature > 95°C or pH < 10.5) or chemical (addition of urea or formaldehyde) changes, the hydrogen bonds break and the complementary strands get separated. This process is called denaturation. Under suitable conditions (i.e., temperature, pH, salt concentration), the two separated single DNA strands can reassemble to form the original double-stranded DNA, and this phenomenon is referred to as renaturation or hybridization.

Radioactive detection system

The ***DNA probe*** is usually ***tagged with a radioactive isotope*** (commonly phosphorus-32). The target DNA is purified and denatured, and mixed with DNA probe. The isotope labeled DNA molecules specifically hybridizes with the target DNA (***Fig. 14.1***). The non-hybridized probe DNA is washed away. The presence of radioactivity in the hybridized DNA can be detected by autoradiography. This reveals the presence of any bound (hybridized) probe molecules and thus the complementary DNA sequences in the target DNA.

Non-radioactive detection system

The disadvantage with the use of radioactive label is that the isotopes have short half-lives and involve risks in handling, besides requiring special laboratory equipment. So, non-radioactive detection systems (e.g., biotinylation) have also been developed. ***Biotin-labeled (biotinylated) nucleotides are incorporated into the DNA probe.*** The detection system is based on the enzymatic conversion of a chromogenic (colour producing) or chemiluminescent (light emitting) substrates.

The procedure commonly adopted for chemiluminescent detection of target DNA is depicted in ***Fig. 14.2***. A biotin labeled DNA probe is hybridized to the target DNA. The egg white protein ***avidin*** or its bacterial analog ***streptavidin*** is added to bind to biotin. Now a biotin labeled enzyme, such as alkaline phosphatase is added which attaches to avidin or streptavidin. These proteins have four separate biotin-binding sites. Thus, a single molecule (avidin or streptavidin) can bind to biotin-labeled DNA probe as well as biotin-labeled enzyme. On the addition of a chemiluminescent substrate, the enzyme alkaline phosphatase acts and converts it to a light emitting product which can be measured. (**Note :** When the enzyme peroxidase is employed in place of alkaline phosphatase, chromogenic substrates will be used, and the colour of the product formed is measured).

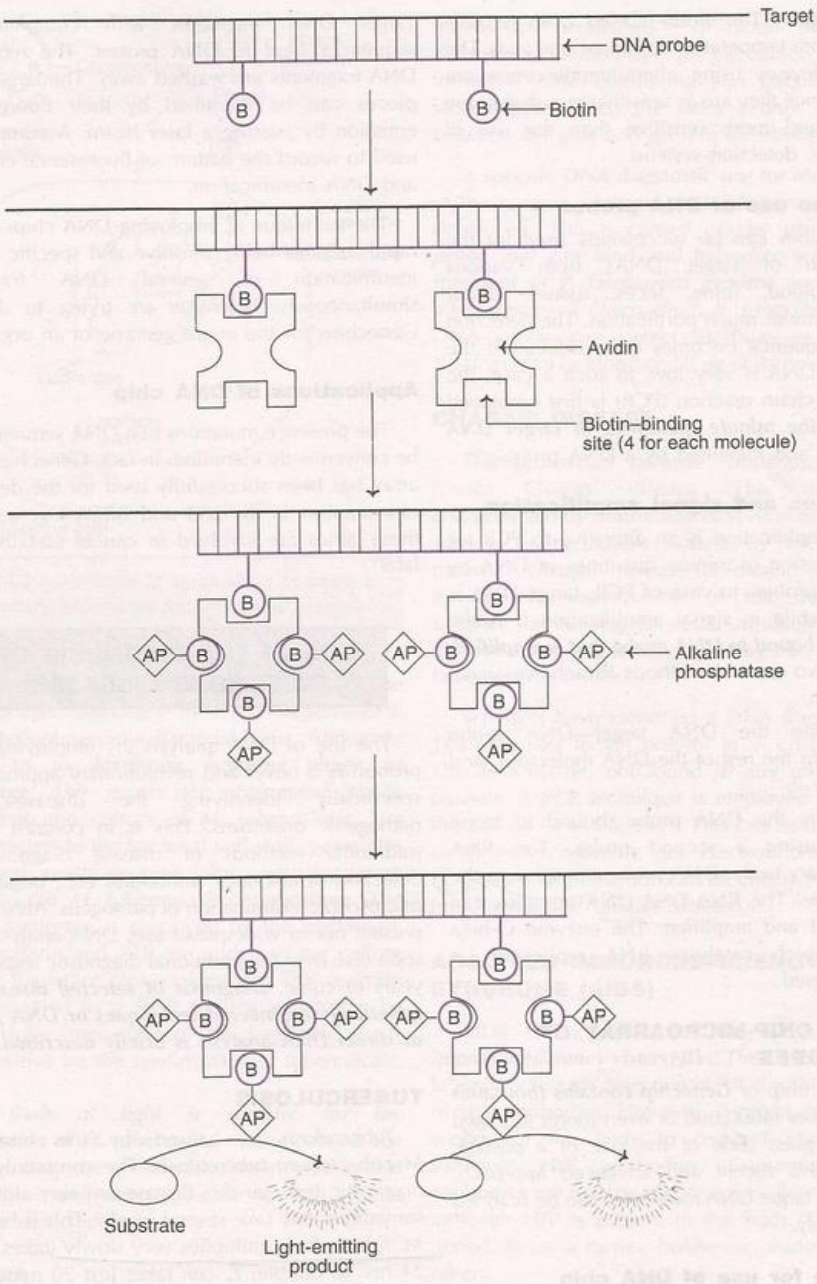


Fig. 14.2 : *Non-radioactive detection system of target DNA by using biotin-labeled DNA probe*
*(**Note** : Chemiluminescent detection is illustrated; for chromogenic detection, the enzyme peroxidase and a chromogenic substrate are used)*

Advantages : The biotin-labeled DNA is quite stable at room temperature for about one year. The detection devises using chemiluminescence are preferred, since they are as sensitive as radioisotope detection, and more sensitive than the use of chromogenic detection systems.

PCR in the use of DNA probes

DNA probes can be successfully used for the identification of target DNAs from various samples — blood, urine, feces, tissues, throat washings without much purification. The detection of target sequence becomes quite difficult if the quantity of DNA is very low. In such a case, the polymerase chain reaction (PCR) is first employed to ***amplify the minute quantities of target DNA*** (Chapter 8), and identified by a DNA probe.

DNA probes and signal amplification

Signal amplification is an alterative to PCR for the identification of minute quantities of DNA by using DNA probes. In case of PCR, target DNA is amplified, while in signal amplification it is the ***target DNA bound to DNA probe that is amplified.*** There are two general methods to achieve signal amplification.

1. Separate the DNA target—DNA probe complex from the rest of the DNA molecules, and then amplify it.

2. Amplify the DNA probe (bound to target DNA) by using a second probe. The RNA complementary to the DNA probe can serve as the second probe. The RNA-DNA-DNA complex can be separated and amplified. The enzyme O-beta replicase which catalyses RNA replication is commonly used.

THE DNA CHIP-MICROARRAY OF GENE PROBES

The DNA chip or ***Genechip contains thousands of DNA probes*** (4000,000 or even more) arranged on a small glass slide of the size of a postage stamp. By this recent and advanced approach, thousands of target DNA molecules can be scanned simultaneously.

Technique for use of DNA chip

The unknown DNA molecules are cut into fragments by restriction endonucleases. Fluorescent markers are attached to these DNA fragments. They are allowed to react to the probes of the DNA chip. Target DNA fragments with complementary sequences bind to DNA probes. The remaining DNA fragments are washed away. The target DNA pieces can be identified by their fluorescence emission by passing a laser beam. A computer is used to record the pattern of fluoresence emission and DNA identification.

The technique of employing DNA chips is very rapid, besides being sensitive and specific for the identification of several DNA fragments simultaneously. Scientists are trying to develop Genechips for the entire genome of an organism.

Applications of DNA chip

The presence mutations in a DNA sequence can be conveniently identified. In fact, Genechip probe array has been successfully used for the detection of mutations in the p53 and BRCA I genes. Both these genes are involved in cancer (details given later).

DNA IN THE DIAGNOSIS OF INFECTIOUS DISEASES

The use of DNA analysis (by employing DNA probes) is a novel and revolutionary approach for specifically identifying the disease-causing pathogenic organisms. This is in contrast to the traditional methods of disease diagnosis by detection of enzymes, antibodies etc., besides the microscopic examination of pathogens. Although at present not in widespread use, DNA analysis may soon take over the traditional diagnostic tests in the years to come. ***Diagnosis of selected diseases by genetically engineered techniques or DNA probes or direct DNA analysis is briefly described.***

TUBERCULOSIS

Tuberculosis is caused by the bacterium *Mycobacterium tuberculosis.* The commonly used diagnostic tests for this disease are very slow and sometimes may take several weeks. This is because *M. tuberculosis* multiplies very slowly (takes about 24 hrs. to double; *E. coli* takes just 20 minutes to double).

A novel diagnostic test for tuberculosis was developed by genetic engineering, and is illustrated in ***Fig. 14.3.*** A gene from ***firefly***, encoding the

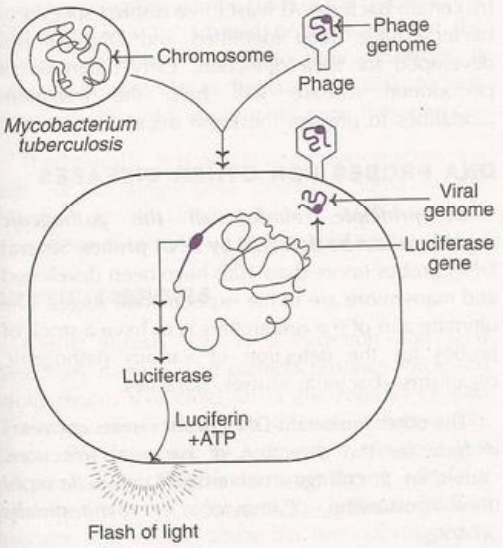


Fig. 14.3 : Diagnosis of tuberculosis by using a genetically engineered bacteriophage (phage).

enzyme ***luciferase*** is introduced into the bacteriophage specific for *M. tuberculosis*. The bacteriophage is a bacterial virus, frequently referred to as ***luciferase reporter phage or mycophage***. The genetically engineered phage is added to the culture of *M. tuberculosis*. The phage attaches to the bacterial cell wall, penetrates inside, and inserts its gene (along with luciferase gene) into the *M. tuberculosis* chromosome. The enzyme luciferase is produced by the bacterium. When luciferin and ATP are added to the culture medium, luciferase cleaves luciferin. This reaction is accompanied by a flash of light which can be detected by a luminometer. This diagnostic test is quite sensitive for the confirmation of tuberculosis.

The ***flash of light is specific for the identification of M. tuberculosis*** in the culture. For other bacteria, the genetically engineered phage cannot attach and enter in, hence no flash of light would be detected.

MALARIA

Malaria, mainly caused by *Plasmodium falciparum* and *P. vivax*, affects about one-third of the world's population. The commonly used laboratory tests for the diagnosis of malaria include microscopic examination of blood smears, and detection of antibodies in the circulation. While the former is time consuming and frequently gives false-negative tests, the latter cannot distinguish between the past and present infections.

A ***specific DNA diagnostic test*** for identification of the current infection of *P. falciparum* has been developed. This is carried out by ***using a DNA probe that can bind and hybridize with a DNA fragment of P. falciparum genome*** and not with other species of *Plasmodium*. It is reported that this DNA probe can detect as little as 1 ng of *P. falciparum* in blood or 10 pg of its purified DNA.

CHAGAS' DISEASE

The protozoan parasite *Trypanosoma cruzi* causes Chagas' disease. This disease is characterized by destruction of several tissues (liver, spleen, brain, lymph nodes) by the invading parasite. Chagas' disease is diagnosed by the microscopic examination of the fresh blood samples. Immunological tests, although available, are not commonly used, since they frequently give false-positive results.

Scientists have identified a DNA fragment with 188-base pair length present in *T. cruzi* genome. This is however, not found in any other related parasite. A PCR technique is employed to amplify the 188 bp DNA fragment. This can be detected by using polyacrylamide gel electrophoresis. Thus, PCR-based amplification can be effectively used for the diagnosis of Chagas' disease.

ACQUIRED IMMUNODEFICIENCY SYNDROME (AIDS)

AIDS is caused by the virus, ***human immunodeficiency virus*** (***HIV***). The commonly used laboratory test for detection of AIDS is the detection of HIV antibodies. However, it might take several weeks for the body to respond and produce sufficient HIV antibodies. Consequently, the antibodies test may be negative (i.e., false-negative), although HIV is present in the body. During this period, being a carrier, he/she can transmit HIV to others.

DNA probes, with radioisotope label, for HIV DNA are now available. By ***using PCR and DNA probes, AIDS can be specifically diagnosed in the laboratory***. During the course of infection cycle,

HIV exists as a segment of DNA integrated into the T-lymphocytes of the host. The T-lymphocytes of a suspected AIDS patient are isolated and disrupted to release DNA. The so obtained DNA is amplified by PCR, and to this DNA probes are added. If the HIV DNA is present, it hybridizes with the complementary sequence of the labeled DNA probe which can be detected by its radioactivity. The advantage of DNA probe is that it can detect the virus when there are no detectable antibodies in the circulation.

HIV diagnosis in the newborn

Detection of antibodies is of no use in the newborn to ascertain whether AIDS has been transmitted from the mother. This is because the antibodies might have come from the mother but not from the virus. This problem can be solved by using DNA probes to detect HIV DNA in the newborn.

HUMAN PAPILLOMA VIRUS

Human papilloma virus (HPV) causes ***genital warts***. HPV is also associated with the cervical cancer in women. The ***DNA probe*** (trade name ***Virapap*** detection kit) that specifically detects HPV has been developed. The tissue samples obtained from woman's cervix are used. HPV DNA, when present hybridizes with DNA probe by complementary base pairing, and this is the positive test.

LYME DISEASE

Lyme disease is caused by the bacterium, *Borrelia burgdorferi*. This disease is characterized by fever, skin rash, arthritis and neurological manifestations.

The diagnosis of Lyme disease is rather difficult, since it is not possible to see *B. burgdorferi* under microscope and the antibody detection tests are not very reliable.

Some workers have used PCR to amplify the DNA of *B. burgdorferi*. By using appropriate DNA probes, the bacterium causing Lyme disease can be specifically detected.

PERIODONTAL DISEASE

Periodontal disease is characterized by the degenerative infection of gums that may ultimately lead to tooth decay and loss. This disease is caused by certain bacteria. At least three distinct species of bacteria have been identified and DNA probes developed for their detection. Early diagnosis of periodontal disease will help the treatment modalities to prevent the tooth decay.

DNA PROBES FOR OTHER DISEASES

In principle, almost all the pathogenic organisms can be detected by DNA probes. Several DNA probes (more than 100) have been developed and many more are in the experimental stages. The ultimate aim of the researchers is to have a stock of probes for the detection of various pathogenic organisms—bacteria, viruses, parasites.

The other important DNA probes in recent years include for the detection of bacterial infections caused by *E. coli* (gastroenteritis) *Salmonella typhi* (food poisoning), *Campylobacter hyoitestinalis* (gastritis).

Diagnosis of tropical diseases

Malaria, filariasis, tuberculosis, leprosy, schistosomiasis, leishmaniasis and trypanosomiasis are the tropical diseases affecting millions of people throughout the world. As already described (See p. 177), for the diagnosis of malaria caused by *P. falciparum*, a DNA probe has been developed. A novel diagnostic test, by genetic manipulations, has been devised for the diagnosis of tuberculosis (See p. 176). Scientists are continuously working to develop better diagnostic techniques for other tropical diseases.

DNA IN THE DIAGNOSIS OF GENETIC DISEASES

Traditional laboratory tests for the diagnosis of genetic diseases are mostly based on the estimation of metabolites and/or enzymes. This is usually done after the onset of symptoms.

The laboratory tests based on DNA analysis can specifically diagnose the inherited diseases at the genetic level. DNA-based tests are useful to discover, well in advance, whether the individuals or their offsprings are at risk for any genetic disease. Further, such tests can also be employed for the prenatal diagnosis of hereditary disorders, besides identifying the carriers of genetic diseases.

By knowing the genetic basis of the diseases, the individuals can be advised on how to limit the transmission of the disease to their offsprings. It may also be possible, in due course of time, to treat genetic diseases by appropriate gene therapies.

Theoretically, it is possible to develop screening tests for all single-gene diseases. Some of the important genetic diseases for which DNA analysis is used for diagnosis are briefly described.

CYSTIC FIBROSIS

Cystic fibrosis (CF) is a common and fatal hereditary disease. The patients produce thick and sticky mucus that clogs lungs and respiratory tract (For some more details on CF, refer Chapter 13). Cystic fibrosis is due to a defect in *cftr* gene that encodes ***cystic fibrosis transmembrane regulator protein***. *cftr* gene is located on chromosome 7 in humans, and a DNA probe has been developed to identify this gene.

The genetic disease cystic fibrosis is inherited by a recessive pattern, i.e., the disease develops when two recessive genes are present. It is now possible to detect CF genes in duplicate in the fetal cells obtained from samples of amniotic fluid. As the test can be done months before birth, it is possible to know whether the offspring will be a victim of CF. One group of researchers have reported that CF gene can be detected in the eight-celled embryo obtained through *in vitro* fertilization.

SICKLE-CELL ANEMIA

Sickle-cell anemia is a genetic disease characterized by the irregular sickle (crescent like) shape of the erythrocytes. Biochemically, this disease results in severe anemia and progressive damage to major organs in the body (heart, brain, lungs, joints).

Sickle-cell anemia occurs due to a single amino acid change in the β-chain of hemoglobin. Specifically, the amino acid ***glutamate at the 6th position of β-chain is replaced by valine***. At the molecular level, sickle-cell anemia is due to a single-nucleotide change (***A → T***) in the β-globin gene of coding (or antisense) strand. In the normal β-globin gene the DNA sequence is CCTGAGGAG, while in sickle-cell anemia, the sequence is CCTGTGGAG. This single-base mutation can be detected by using restriction enzyme *MstII* to cut

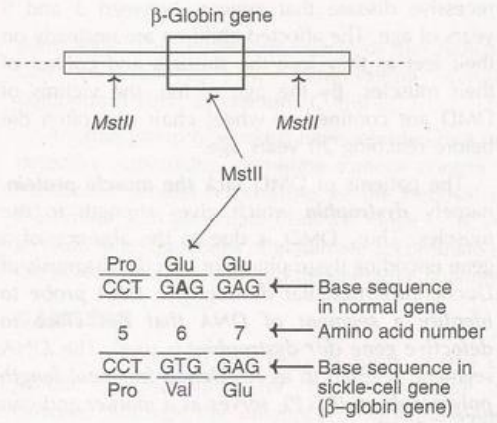


Fig. 14.4 : *Single-base change resulting in sickle-cell anemia. (**Note :** A small and relevant DNA fragment of β-globin gene magnified and shown with encoded amino acids).*

DNA fragments in and around β-globin gene, followed by the electrophoretic pattern of the DNA fragments formed. The change in the base from A to T in the β-globin gene destroys the recognition site (CCTGAGG) for *MstII* (***Fig. 14.4***). Consequently, the DNA fragments formed from a sickle-cell anemia patient for β-globin gene differ from that of a normal person. Thus, sickle-cell anemia can be detected by digesting mutant and normal β-globin gene by restriction enzyme and performing a ***hybridization with a cloned β-globin DNA probe***.

Single-nucleotide polymorphisms

The ***single base changes*** that occur in some of the genetic diseases (e.g., sickle-cell anemia) are collectively referred to as single-nucleotide polymorphisms (***SNPs***, pronounced ***snips***). It is estimated that the frequency of SNPs is about one in every 1000 bases. Sometimes SNPs are associated with amino acid change in the protein that is encoded. A point mutation in α_1-antitrypsin gene is also a good example of SNPs, besides sickle-cell anemia.

DUCHENNE'S MUSCULAR DYSTROPHY

Duchenne's muscular dystrophy (***DMD***) is a genetic abnormality characterized by progressive wasting of leg and pelvic muscles. It is a sex-linked

recessive disease that appears between 3 and 5 years of age. The affected children are unsteady on their feet as they lose the strength and control of their muscles. By the age of ten, the victims of DMD are confined to wheel chair and often die before reaching 20 years age.

The patients of DMD ***lack the muscle protein***, namely ***dystrophin*** which gives strength to the muscles. Thus, DMD is due to the absence of a gene encoding dystrophin. For specific diagnosis of Duchenne's muscular dystrophy, a ***DNA probe to identify a segment of DNA that lies close to defective gene (for dystrophin)*** is used. This DNA segment, referred to as ***restriction fragment length polymorphism (RFLP), serves as a marker*** and can detect DMD with 95% certainity.

In the DNA diagnostic test using RFLP for DMD, DNA samples must be obtained from as many blood relatives (parents, grand-parents, uncles, aunts etc.) as possible. The RFLP patterns, constructed for the entire family are thoroughly checked for the affected and unaffected relatives. This is required since there is a wide variation in RFLPs from family to family. Thus, there is no single identifying test for the diagnosis of genetic diseases based on RFLPs analysis.

HUNTINGTON'S DISEASE

Huntington's disease is a genetic disease (caused by a dominant gene) characterized by progressive deterioration of the nervous system, particularly the destruction of brain cells. The victims of this disease (usually above 50 years of age) exhibit thrashing (jerky) movements and then insanity [older name was Huntington's chorea; *chorea (Greek) means to dance*]. Huntington's disease is invariably fatal.

The molecular basis of Huntington's disease has been identified. The gene responsible for this disease lies on chromosome number 4, and is characterized by ***excessive repetition of the base triplet CAG***. The victims of Huntington's disease have CAG triplet repeated 42–66 times, against the normal 11–34 times. The ***triplet CAG encodes*** for the amino acid ***glutamine***. It is believed that the ***abnormal protein*** (with very high content of glutamine) ***causes the death of cells*** in the basal ganglia (the part of the brain responsible for motor function).

Huntington's disease can be detected by the analysis of RFLPs in blood related individuals. The clinical manifestations of this disease are observed after middle age, and by then the person might have already passed on the defective gene to his/her offsprings.

FRAGILE X SYNDROME

Fragile X syndrome, as the name indicates, is due to a genetic defect in X chromosome (a sex chromosome) and affects both males and females. The victims of this disease are characterized by mental retardation.

Researchers have found that sufferers of fragile X syndrome have the three nucleotide bases (***CGG***) repeated again and again. It is believed that these ***trinucleotide repeats block the transcription process resulting in a protein deficiency***. This protein is involved in the normal function of the nerve cells, and its deficiency results in mental retardation.

A ***DNA probe*** has been developed for the detection of fragile X syndrome in the laboratory.

OTHER TRIPLE REPEAT DISEASES

Excessive ***repetition*** of ***triplet bases*** in DNA are now known to result in several diseases which are collectively referred to as triple repeat diseases. Besides Huntington's disease and fragile X syndrome (discussed above), some more triple repeats are given below.

Friedreich's ataxia

The trinucleotide GAA repeats 200 to 900 times on chromosome 9 in Friedreich's ataxia. This disease is associated with degradation of spinal cord. ***Spinocerebellar ataxia*** is another triplet disease, characterized by neuromuscular disorder, and is due to trinucleotide repeats of CAG by 40 to 80 times on chromosome 6.

There are a few triple repeat diseases in which the repeats tend to increase with each generation and the diseases become more severe. This also results in the onset of clinical manifestations at early ages. ***Kennedy's disease***, also called ***spinobulbar muscular atropy*** (***CAG repeat***) and ***myotonic dystrophy*** (***CTG***) are good examples.

Are triple repeat diseases confined to humans?

Triple repeat diseases have so far not been detected in any other organisms (bacteria, fruit flies, other mammals) except in humans. More

studies however, may be needed to confirm this. The occurrence of triple repeat diseases indicates that the structure of DNA may be rather unstable and dynamic. This is in contrast to what molecular biologists have been thinking all along.

ALZHEIMER'S DISEASE

Alzheimer's disease is characterized by loss of memory and impaired intellectual function (dementia). The victims of this disease cannot properly attend to their basic needs, besides being unable to speak and walk.

The patients of Alzheimer's disease were found to have a specific protein, namely ***amyloid*** in the plaques (or clumps) of dead nerve fibers in their brains. A group of researchers have identified a specific ***gene on chromosome 21*** that is believed to be ***responsible for familial Alzheimer's disease***.

A ***DNA probe*** has been developed to locate the genetic marker for Alzheimer's disease. The present belief is that many environmental factors, and a virus may also be responsible for the development of this disease. It may be possible that in the individuals with genetic predisposition, the outside factors may be stimulatory for the onset of the disease.

AMYOTROPHIC LATERAL SCLEROSIS

Amyotrophic lateral sclerosis (***ALS***) is characterized by degenerative changes in the motor neurons of brain and spinal cord. A gene to explain the inherited pattern of ALS was discovered.

The gene, known as ***sodI***, encoding for the enzyme ***superoxide dismutase*** is located on chromosome 21. This gene was found to be defective in families suffering from amyotrophic lateral sclerosis. In fact, certain point mutations in the *sodI* resulting in single amino acid changes in superoxide dismutase have been identified.

Superoxide dismutase is a key enzyme in eliminating the highly toxic free radicals that damage the cells (free radicals have been implicated in aging and several disease e.g. cancer, cataract, Parkinson's disease, Alzheimer's disease). On the basis of the function of superoxide dismutase, it is presumed that ALS occurs as a result of free radical accumulation due to a defective enzyme (as a consequence of mutated gene *sodI*). The deleterious effects of free radicals can be reduced by administering certain compounds such as vitamins C and E.

Another group of workers have reported that the defective superoxide dismutase cannot control a transporter protein responsible for the removal of the amino acid glutamate from the nerve cells. As a result, large quantities of glutamate accumulate in the nervous tissue leading to degenerative changes.

CANCERS

It is now agreed that there is some degree of genetic predisposition for the occurrence of cancers, although the influence of environmental factors cannot be underestimated. In fact, cancer susceptible genes have been identified in some families e.g., genes for melanoma susceptibility in humans are located on chromosomes 1 and 9.

p^{53} gene

The gene p^{53} encodes for a *p*rotein with a molecular weight ***53*** kilodaltons (hence the name). It is believed that the protein produced by this gene helps DNA repair and suppresses cancer development. Certain damages that occur in DNA may lead to unlimited replication and uncontrolled multiplication of cells. In such a situation, the protein encoded by p^{53} gene binds to DNA and blocks replication. Further, it facilitates the faulty DNA to get repaired. The result is that the cancerous cells are not allowed to establish and multiply. Thus, ***p^{53} is a cancer-suppressor gene*** and acts as a guardian of cellular DNA.

Any ***mutation in the gene p^{53}*** is likely to alter its tumor suppressor function that ***lead to cancer development***. And in fact, the altered forms of p^{53} recovered from the various tumor cells (breast, bone, brain, colon, bladder, skin, lung) confirm the protective function of p^{53} gene against cancers.

It is believed that the environmental factors may cause mutations in p^{53} gene which may ultimately lead to cancer. Some of the mutations of p^{53} gene may be inherited, which probably explains the occurrence of certain cancers in some families.

Genes of breast cancer

Two ***genes***, namely ***BRCAI*** and ***BRCAII***, implicated in certain ***hereditary forms of breast cancer*** in women, have been identified. It is estimated that about 80% of inherited breast cancers are due to mutations in either one of these

two genes — BRCAI or BRCAII. In addition, there is a high risk for ovarian cancer due to mutations in BRCAI.

It is suggested that the normal genes BRCAI and BRCAII encode proteins (with 1863 and 3418 amino acids respectively) that function in a manner comparable to gene p^{53} protein (as described above). As such, BRCAI and BRCAII are DNA-repair and tumor-suppressor genes. Some researchers believe that these two proteins act as gene regulators.

Diagnostic tests for the analysis of the genes BRCAI and BRCAII were developed. Unfortunately, their utility is very limited, since there could be hundreds of variations in the base sequence of these genes.

Genes of colon cancer

The occurrence of colon cancer appears to be genetically linked since it runs in some families. Some researchers have identified a gene linked with ***hereditary nonpolyposis colon cancer*** or ***HNPCC*** (some times called ***Lynch syndrome***). This gene encoded a protein that acts as a guardian and brings about DNA repair whenever there is a damage to it. However, as and when there is a mutation to this protective gene, an altered protein is produced which cannot undo the damage done to DNA. This leads to HNPCC. It is estimated that the occurrence of this altered gene is one in every 200 people in general population.

Microsatellite marker genes

Microsatellites refer to the short repetitive sequences of DNA that can be employed as markers for the identification of certain genes. For colon cancer, microsatellite marker genes have been identified on chromosome 2 in humans. There is a lot of variability in the sequence of microsatellites (as is the case with RFLPs described on p. 185).

Early detection of the risk for colon cancer by DNA analysis is a boon for the would be victims of this disease. The suspected individuals can be periodically monitored for the signs and treated appropriately. Unlike many other cancers, the chances of cure for colon cancer are reasonably good.

Gene of retinoblastoma

Retinoblastoma is a rare cancer of the eye. If detected early, it can be cured by radiation therapy and laser surgery or else the eyeball has to be removed.

Scientists have identified a missing or a defective (mutated) gene on chromosome number 13, being responsible for retinoblastoma. The normal gene when present on chromosome 13 is anticancer and does not allow retinoblastoma to develop.

DIABETES

Diabetes mellitus is a clinical condition characterized by ***increased blood glucose level (hyperglycemia)*** due to insufficient or inefficient (incompetent) insulin. In other words, individuals with diabetes cannot utilize glucose properly in their body.

A rare form of ***type II diabetes*** (i.e., non-insulin dependent diabetes mellitus, NIDDM) is ***maturity onset diabetes of the young (MODY)***. MODY, occurring in adolescents and teenagers, is found to have a genetic basis. A ***gene, synthesizing the enzyme glucokinase***, located on chromosome 7, is found to be ***defective in MODY patients***.

Glucokinase is a key enzyme in glucose metabolism. Besides its involvement in the metabolism, glucokinase in the pancreatic cells serves as a detector for glucose concentration in the blood. This detection stimulates β-cells of the pancreas to secrete insulin. A gene modification that results in a defective or an altered glucokinase hampers pancreatic insulin secretion. Later work has shown that glucokinase gene is defective in the common form of type II diabetes.

DNA probes for type II diabetes

The glucokinase genes from normal and type II diabetes patients were cloned and scanned with DNA probes. It was found that ***a single base mutation of the gene led to a defective glucokinase production that is largely responsible for MODY***, and also a majority of individuals with type II diabetes. Later, some workers reported a possibility of at least a dozen mutations (though less important than the one discussed) in glucokinase gene for type II diabetes.

Genes responsible for type I diabetes

Type I diabetes or insulin-dependent diabetes mellitus (IDDM) mainly occurs in childhood, particularly between 12–15 years of age. IDDM is characterized by almost total deficiency of insulin. Researchers have identified at least 18 different chromosome regions linked with type I diabetes. These DNA sequences are located on chromosomes 6, 11 and 18.

OBESITY

Obesity is an abnormal increase in the body weight due to fat deposition. Men and women are considered obese if their weight due to fat, respectively exceeds more than 20% and 25% of the body weight. Obesity increases the risk of high blood pressure, diabetes, atherosclerosis and other life-threatening conditions.

Although many believed that obesity could be genetically inherited, the molecular basis was not known for long. It was in 1994, a group of workers identified a mutated gene that caused obesity in mice. Later, a similar gene was found in humans also.

The gene designated ***ob*** (for ***ob***ese) is located on chromosome 6 in mouse. The DNA of ob gene contains 650 kb and encodes a protein with 167 amino acids in adipose tissue. This protein is responsible to keep the weight of the animals under control. The genetically ***obese mice have mutated ob gene*** and therefore the weight-control protein is not produced. It is believed that this protein functions like a hormone, acts on the hypothalamus, and controls the site of hunger and energy metabolism (these two factors are intimately linked with obesity).

With the discovery of ***ob*** gene, the treatment for inherited obesity may soon become a reality. In fact, one multinational biotechnology company has started producing ***ob protein*** that can be used for weight reduction in experimental mice.

Besides the ***ob*** gene, a few other genes (***fat*** gene, ***tub*** gene) that might be associated with obesity have also been discovered.

DNA ANALYSIS FOR OTHER HUMAN DISEASES

There is a continuous search for the identification of more and more genes that are responsible for human diseases. Such an approach will ultimately help in the specific diagnosis of these diseases before their actual occurrence. In addition to human diseases described above, some more are given below.

Deafness

The deafness, inherited in some families, has genetic basis. A team of workers have identified a gene on chromosome 5, encoding a protein that facilitates the assembly of actin (protein) molecules in the cochlea of inner ear. The association of actin is very essential for the detection of sound waves by the ear. A mutation of the gene on chromosome 5 results in a defective protein synthesis and non-assembly of actin molecules which cause deafness. Some other genes, besides the one described here, have also been found to be associated with deafness.

Glaucoma

Glaucoma is a disease of the eye that may often lead to blindness. It occurs as a result of damage to the optic nerve due to pressure that builds up in the eye. A gene responsible for the hereditary glaucoma in teenagers has been detected on chromosome 1. Another group of researchers have found a gene on chromosome 3 which is linked with the adult-onset glaucoma.

Baldness

There is an inherited form of baldness, called ***alopecia universalis***. This is found to be associated with a gene located on chromosome 12.

Parkinson's disease

Parkinson's disease is a common disorder in many elderly people, with about 1% of the population above 60 years being affected. It is characterized by muscular regidity, tremors, expressionless face, lethargy, involuntary movements etc. In the victims of Parkinson's disease, there is ***degeneration of brain cells, besides a low concentration of dopamine*** (a neuro-transmitter). Researchers have identified that a gene-encoded protein namely ***α-synuclein*** plays a significant role in the development of Parkinson's disease. An altered form of α-synuclein (due to a mutation in the gene) accumulates in the brain as ***Lewy bodies***. This is responsible for nerve cells degeneration and their death in the Parkinson's disease.

Hemochromatosis

Hemochromatosis is an iron-overload disease in which iron is directly deposited in the tissues (liver, spleen, heart, pancreas and skin). An abnormal gene on chromosome 6 is linked with hemochromatosis. The amino acid ***tyrosine***, in the normal protein encoded by this gene is ***replaced by cysteine***. This abnormal protein is responsible for ***excessive iron absorption*** from the intestine which accumulates in the various tissues leading to their damage and malfunction.

Menke's disease

Menke's disease, a copper deficiency disorder, is characterized by decreased copper in plasma, depigmentation of hair, degeneration of nerve cells and mental retardation. A gene located on X-chromosome, encoding a transport protein, is linked with Menke's disease. A defect in the gene, consequently in the protein, impairs copper absorption from the intestine.

GENE BANKS — A NOVEL CONCEPT

As the search continues by scientists for the identification of more and more genes responsible for various diseases, the enlightened public (particularly in the developed countries), is very keen to enjoy the fruits of this research outcome. As of now, DNA probes are available for the detection a limited number of diseases. Researchers continue to develop DNA probes for a large number of genetically predisposed disorders.

Gene banks are the centres for the storage of individual's DNAs for future use to diagnose diseases. For this purpose, the DNA isolated from a person's cells (usually white blood cells) is stored. As and when a DNA probe for the detection of a specific disease is available, the stored DNA can be used for the diagnosis or risk assessment of the said genetic disease.

In fact, some institutions have established gene banks. They store the DNA samples of the interested customers at a fee (one firm was charging $ 200) for a specified period (say around 20–25 years). For the risk assessment of any disease, it is advisable to have the DNAs from close relatives of at least 2-3 generations.

DNA ANALYSIS FOR ENVIRONMENTAL MONITORING

Environmental pollution (particularly water and foods) by pathogenic microorganisms and viruses is a common occurrence. In the traditional approach, the pathogens are identified by cultivating them in the laboratory. In recent years, techniques have been developed for their detection by DNA analysis.

WATER QUALITY TESTING

The quality of potable water can be tested by detecting the ***indicator bacteria*** such as ***Escherichia coli***. For the DNA analysis, water is filtered and the bacteria trapped on the filter are broken to release DNA. By use of DNA probes to detect specific genes, the presence of *E. coli* can be specifically detected. The DNA analysis technique is very sensitive and is capable of identifying a single *E. coli* cell in 100 ml of water. It is also possible to detect other pathogenic organisms such as *Salmonella*, *Vibrio* and *Shigella* in due course of time.

DNA analysis can also be employed for the specific detection of ***viruses and protozoal pathogens.***

DNA FINGERPRINTING OR DNA PROFILING

DNA fingerprinting is the present day ***genetic detective*** in the practice ***of modern medical forensics***. The underlying principles of DNA fingerprinting are briefly described.

The structure of each person's genome is unique. The only exception being monozygotic identical twins (twins developed from a single fertilized ovum). The unique nature of genome structure provides a good opportunity for the specific identification of an individual.

It may be remembered here that in the traditional fingerprint technique, the individual is identified by preparing an ink impression of the skin folds at the tip of the person's finger. This is based on the fact that the nature of these skin folds

is genetically determined, and thus the fingerprint is unique for an individual. In contrast, the ***DNA fingerprint is an analysis of the nitrogenous base sequence in the DNA of an individual.***

History and terminology

The original DNA fingerprinting technique was developed by Alec Jaffreys in 1985. Although the DNA fingerprinting is commonly used, a more general term ***DNA profiling*** is preferred. This is due to the fact that a wide range of tests can be carried out by DNA sequencing with improved technology.

Applications of DNA fingerprinting

The amount of DNA required for DNA fingerprint is remarkably small. The ***minute quantities of DNA*** from blood strains, body fluids, hair fiber or skin fragments ***are enough. Polymerase chain reaction is used to amplify*** this DNA for use in fingerprinting. DNA profiling has wide range of applications—most of them related to medical forensics. Some important ones are listed below.

- Identification of criminals, rapists, thieves etc.
- Settlement of paternity disputes.
- Use in immigration test cases and disputes.

In general, the fingerprinting technique is carried out by collecting the DNA from a suspect (or a person in a paternity or immigration dispute) and matching it with that of a reference sample (from the victim of a crime, or a close relative in a civil case).

DNA MARKERS IN DISEASE DIAGNOSIS AND FINGERPRINTING

The ***DNA markers are highly useful for genetic mapping of genomes.*** There are four types of DNA sequences which can be used as markers.

1. Restriction fragment length polymorphisms (***RFLFs***, pronounced as rif-lips).

2. Minisatellites or variable number tandem repeats (***VNTRs***, pronounced as vinters).

3. Microsatellites or simple tandem repeats (***STRs***).

4. Single nucleotide polymorphisms (***SNPs***, pronounced as snips).

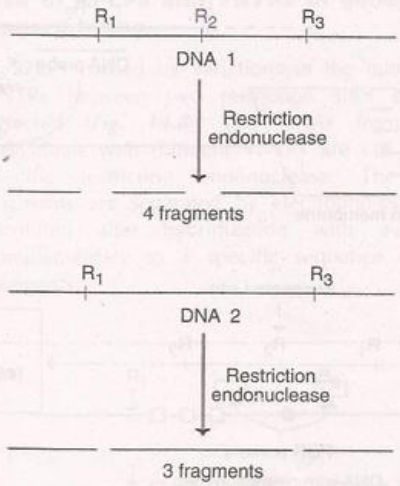


Fig. 14.5 : *An outline of the restriction fragment length polymorphism (RFLP) (R_1, R_2, R_3 represent the sites for the action of restriction endonucleases).*

The general aspects of the above DNA markers are described along with their utility in disease diagnosis and DNA fingerprinting.

RESTRICTION FRAGMENT LENGTH POLYMORPHISMS (RFLPs)

A RFLP represents a stretch of DNA that serves as a marker for mapping a specified gene. RFLPs are located randomly throughout a person's chromosomes and have no apparent function.

A DNA molecule can be cut into different fragments by a group of enzymes called restriction endonucleases (For details, Refer Chapter 6). These fragments are called ***polymorphisms*** (literally means ***many forms***).

An outline of RFLP is depicted in ***Fig. 14.5***. The DNA molecule 1 has three restriction sites (R_1, R_2, R_3), and when cleaved by restriction endonucleases forms 4 fragments. Let us now consider DNA 2 with an inherited mutation (or a genetic change) that has altered some base pairs. As a result, the site (R_2) for the recognition by restriction endonuclease is lost. This DNA molecule 2 when cut by restriction endonuclease forms only 3 fragments (instead of 4 in DNA 1).

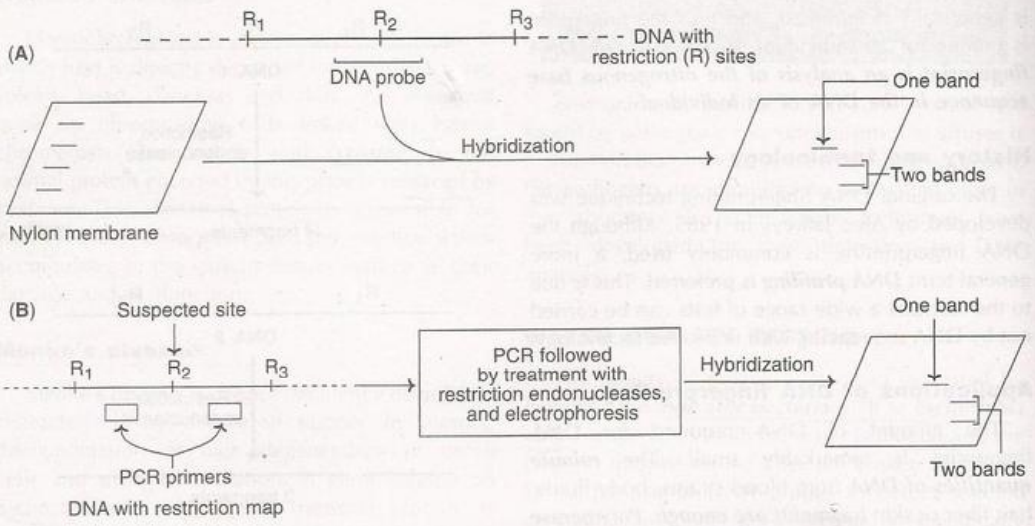


Fig. 14.6 : Two common methods used for scoring restriction fragment length polymorphism (RFLP) (A) RFLP by Southern hybridization (B) RFLP by polymerase chain reaction (PCR).

As is evident from the above description, a stretch of DNA exists in ***fragments*** of various ***lengths*** (***polymorphisms***), derived by the action of ***restriction*** enzymes, hence the name restriction fragment length polymorphisms.

RFLPs in the diagnosis of diseases

If the RFLP lies within or even close to the locus of a gene that causes a particular disease, it is possible to trace the defective gene by the analysis of RFLP in DNA. The person's cellular DNA is isolated and treated with restriction enzymes. The DNA fragments so obtained are separated by electrophoresis. The RFLP patterns of the disease suspected individuals can be compared with that of normal people (preferably with the relatives in the same family). By this approach, it is possible to determine whether the individual has the marker RFLP and the disease gene. With 95% certainity, RFLPs can ***detect single gene-based diseases***.

Methods of RFLP scoring : Two methods are in common use for the detection of RFLPs (***Fig. 14.5***).

1. **Southern hybridization :** The DNA is digested with appropriate restriction enzyme, and separated by agarose gel electrophoresis. The so obtained DNA fragments are transferred to a nylon membrane. A DNA probe that spans the suspected restriction site is now added, and the hybridized bands are detected by autoradiograph. If the restriction site is absent, then only a single restriction fragment is detected. If the site is present, then two fragments are detected (***Fig. 14.6A***).

2. **Polymerase chain reaction :** RFLPs can also be scored by PCR. For this purpose, PCR primers that can anneal on either side of the suspected restriction site are used. After amplification by PCR, the DNA molecules are treated with restriction enzyme and then analysed by agarose gel electrophoresis. If the restriction site is absent only one band is seen while two bands are found if the site is found (***Fig. 14.6B***).

Applications of RFLPs : The approach by RFLP is very powerful and has helped many genes to be mapped on the chromosomes. e.g. sickle-cell anemia (chromosome 11), cystic fibrosis (chromosome 7), Huntington's desease (chromosome 4), retinoblastoma (chromosome 13), Alzheimer's disease (chromosome 21).

VARIABLE NUMBER TANDEM REPEATS (VNTRs)

VNTRs, also known as minisatellites, like RFLPs, are DNA fragments of different length. The main

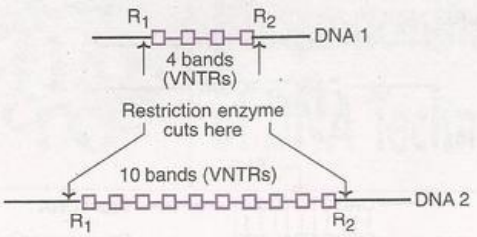


Fig. 14.7 : A diagrammatic representation of variable number tandem repeats (VNTRs). Each band (or copy) represents a repeating sequence in the DNA (e.g. 100 base pairs each). R_1 and R_2 indicate the sites cut by a restriction enzyme.

difference is that RFLPs develop from random mutations at the site of restriction enzyme activity while VNTRs are formed due to different number of base sequences between two points of a DNA molecule. In general, VNTRs are made up of tandem repeats of short base sequences (10–100 base pairs). The number of elements in a given region may vary, hence they are known as variable number tandem repeats.

An individual's genome has many different VNTRs and RFLPs which are unique to the individual. The ***pattern of VNTRs and RFLPs forms the basis of DNA fingerprinting*** or DNA profiling.

In the ***Fig. 14.7***, two different DNA molecules with different number of copies (bands) of VNTRs are shown. When these molecules are subjected to restriction endonuclease action (at two sites R_1 and R_2), the VNTR sequences are released, and they can be detected due to variability in repeat sequence copies. These can be used in mapping of genomes, besides their utility in DNA fingerprinting.

VNTRs are useful for the detection of certain genetic diseases associated with alterations in the degree of repetition of microsatellites e.g. Huntington's chorea is a disorder which is found when the VNTRs exceed 40 repeat units.

Limitations of VNTRs : The major drawback of VNTRs is that they are not evenly distributed throughout the genome. VNTRs tend to be localized in the telomeric regions at the ends of the chromosomes.

Use of RFLPs and VNTRs in genetic fingerprinting

RFLPs caused by variations in the number of VNTRs between two restriction sites can be detected (***Fig. 14.8***). The DNAs from three individuals with different VNTRs are cut by the specific restriction endonuclease. The DNA fragments are separated by electrophoresis, and identified after hybridization with a probe complementary to a specific sequence on the fragments.

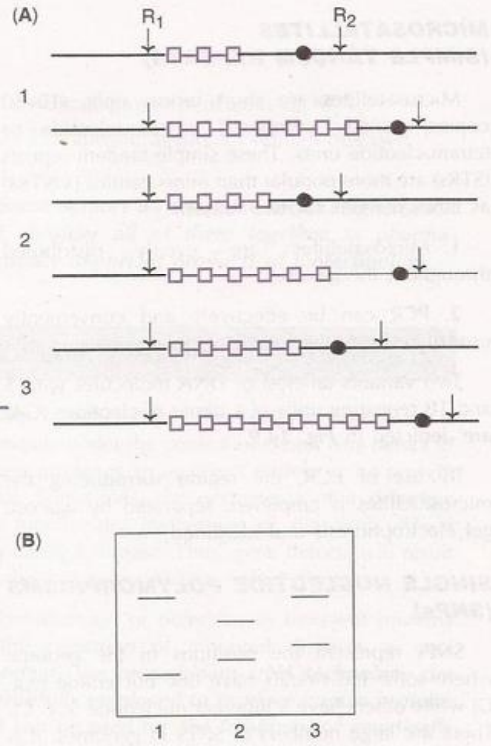


*Fig. 14.8 : Use of restriction fragment length polymorphisms (RFLPs) caused by variable number tandem repeats (VNTRs) in genetic fingerprinting (**A**) An illustration of DNA structure from three individuals (**B**) Hybridized pattern of DNA fragment with a probe complementary to the sequence shown in black circles (1, 2 and 3 represent the individuals; R_1 and R_2 indicate restriction sites; coloured squares are the number of VNTRs)*

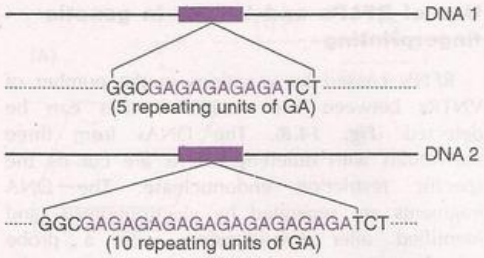


Fig. 14.9 : Two alleles of DNA molecules representing 5 and 10 dimer repeating units.

MICROSATELLITES (SIMPLE TANDEM REPEATS)

Microsatellites are short repeat units (10–30 copies) usually composed of dinucleotide or tetranucleotide units. These simple tandem repeats (STRs) are more popular than minisatellites (VNTRs) as DNA markers for two reasons.

1. Microsatellites are evenly distributed throughout the genome.

2. PCR can be effectively and conveniently used to identify the length of polymorphism.

Two variants (alleles) of DNA molecules with 5 and 10 repeating units of a dimer nucleotides (GA) are depicted in ***Fig. 14.9***.

By use of PCR, the region surrounding the microsatellites is amplified, separated by agarose gel electrophoresis and identified.

SINGLE NUCLEOTIDE POLYMORPHISMS (SNPs)

SNPs represent the positions in the genome where some individuals have one nucleotide (e.g. G) while others have a different nucleotide (e.g. C). There are large numbers of SNPs in genomes. It is estimated that the human genome contains at least 3 million SNPs. Some of these SNPs may give rise to RFLPs.

SNPs are highly useful as DNA markers since there is no need for gel electrophoresis and this saves a lot of time and labour. The detection of SNPs is based on the oligonucleotide hybridization analysis (***Fig. 14.10***).

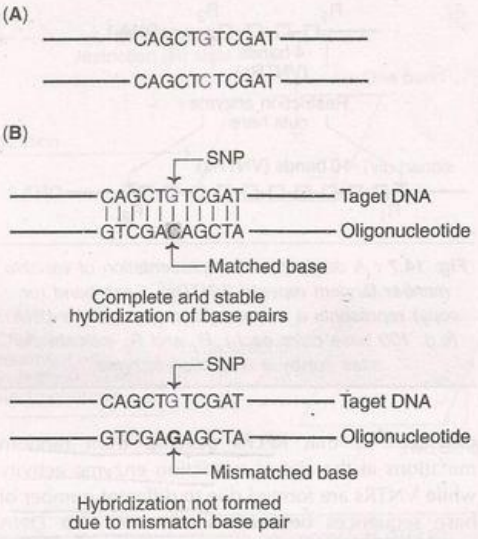


Fig. 14.10 : (A) An illustration of single nucleotide polymorphism (SNP) (B) Oligonucleotide hybridization to detect SNP.

An oligonucleotide is a short single-stranded DNA molecule synthesized in the laboratory with a length not usually exceeding 50 nucleotides. Under appropriate conditions, this nucleotide sequence will hybridize with a target DNA strand if both have completely base paired structure. Even a single mismatch in base pair will not allow the hybridization to occur.

DNA chip technology is most commonly used to screen SNPs hybridization with oligonucleotide (Refer Chapter 7).

CURRENT TECHNOLOGY OF DNA FINGERPRINTING

In the forensic analysis of DNA, the original techniques based on RFLPs and VNTRs are now largely replaced by microsatellites (short tandem repeats). The basic principle involves the amplification of microsatellites by polymerase chain reaction followed by their detection.

It is now possible to generate a DNA profile by automated DNA detection system (comparable to the DNA sequencing equipment).

15 Pharmaceutical Products of DNA Technology

A few decades ago, it was realised that certain proteins could be used as pharmaceutical agents for the treatment of human diseases. e.g. insulin for diabetes mellitus, interferon for viral diseases. However, the availability of such therapeutic/pharmaceutical products was very limited due to costly and cumbersome procedures involved in their isolation/production. Further, their use in humans was associated with several complications. For instance, administration of pig insulin to diabetic patients results in the development of antibodies.

The advent of recombinant DNA technology heralded a new chapter for the production of a wide range of therapeutic agents in sufficient quantities for human use. The commercial exploitation of ***recombinant DNA (rDNA) technology*** began in late 1970s by a few biotechnological companies to produce proteins. There are at least 400 different proteins being produced (by DNA technology) which may serve as therapeutic agents for humans. A selected list of some important human proteins produced by recombinant DNA technology potential for the treatment of human disorders is given in ***Table 15.1***. As of now, only a selected few of them (around 30) have been approved for human use, and the most important among these are given in ***Table 15.2***.

The pharmaceutical products of recombinant DNA technology are broadly divided into the following three categories and briefly discussed with important examples.

1. Human protein replacements.
2. Therapeutic agents for human diseases.
3. Vaccines.

Some authors do not make such categorization and consider all of them together as pharmaceutically important products of biotechnology.

HUMAN PROTEIN REPLACEMENTS

The synthesis of the cellular proteins is ultimately under the control of genes. Any defect in a gene produces an incorrect protein or no protein at all. Sometimes, the occurrence of a defective (i.e. functionally ineffective) or deficient protein may cause a disease. Thus, gene defects will result in inherited or genetically linked diseases.

Identification of defective or deficient proteins in the causation of inherited diseases is very important. The ***recombinant DNA technology can be fruitfully employed to produce human proteins that can be used for the treatment of genetically linked diseases***. This is referred to as human protein replacement strategy in biotechnology.

INSULIN

The hormone insulin is produced by the β-cells of islets of Langerhans of pancreas. Human insulin contains 51 amino acids, arranged in two polypeptide chains. The chain A has 21 amino

Table 15.1 A selected list of human proteins produced by recombinant DNA technology for treatment of human disorders

Disorder	*Recombinant protein(s)*
Anemia	Hemoglobin, erythropoietin
Asthma	Interleukin-I receptor
Atherosclerosis	Platelet-derived growth factor
Delivery	Relaxin
Blood clots	Tissue plasminogen activator, urokinase
Burns	Epidermal growth factor
Cancer	Interferons, tumor necrosis factor, colony stimulating factors, interleukins, lymphotoxin, macrophage-activating factor
Diabetes	Insulin, insulin-like growth factor
Emphysema	α_1-Antitrypsin
Female infertility	Chorionic gonadotropin
Free radical damage (minimizing)	Superoxide dismutase
Growth defects	Growth hormone, growth hormone-releasing factor, somatomedin-C
Heart attacks	Prourokinase
Hemophilia A	Factor VIII
Hemophilia B	Factor IX
Hepatitis B	Hepatitis B vaccine
Hypoalbuminemia	Serum albumin
Immune disorders	Interleukins, β-cell growth factors
Kidney disorders	Erythropoietin
Low Gehrig's disease (amytrophic lateral sclerosis)	Brain-derived neurotropic factor
Multiple sclerosis	Interferons (α, β, γ)
Nerve damage	Nerve growth factor
Osteomalacia	Calcitonin
Pain	Endorphins and enkephalins
Rheumatic disease	Adrenocorticotropic hormone
Ulcers	Urogastrone
Viral infections	Interferons (α, β, γ)

acids while B has 30 amino acids. Both are held together by disulfide bonds.

Diabetes mellitus

Diabetes mellitus affects about 2–3% of the general population. It is a genetically linked disease characterized by ***increased blood glucose concentration*** (***hyperglycemia***). The occurrence of diabetes is due to insufficient or inefficient (incompetent) insulin. Insulin facilitates the cellular uptake and utilization of glucose for the release of energy. In the absence of insulin, glucose accumulates in the blood stream at higher concentration, usually when the blood glucose concentration exceeds about 180 mg/dl, glucose is excreted into urine. The patients of diabetes are weak and tired since the production of energy (i.e. ATP) is very much depressed.

Table 15.2 A selected list of recombinant proteins that have been approved for human use by U.S. Food and Drug Administration

Recombinant protein	*Disorder(s) treated*
Coagulation factor VIII	Hemophilia A
Coagulation factor IX	Christmas disease
DNase I	Cystic fibrosis
Erythropoietin	Anemia, kidney disease
Glucocerebrosidase	Gaucher's disease
Growth hormone	Growth defects in children
Granulocyte colony stimulating factor	Cancer
Hepatitis B surface antigen (vaccine)	Hepatitis B
Insulin	Diabetes mellitus
Interferon α	Leukemia, kaposi sarcoma genital warts, hepatitis B
Interferon β	Multiple sclerosis
Interferon γ	Chronic granulomatous disease
Interleukin-2	Renal cell carcinoma
Interleukin-10	Thrombocytopenia
Somatotropin	Growth defects
Tissue plasminogen activator	Acute myocardial infarction, pulmonary embolism

The more serious complications of uncontrolled diabetes include kidney damage (nephropathy), eye damage (retinopathy), nerve diseases (neuropathy) and circulatory diseases (atherosclerosis, stroke). In fact, diabetes is the third leading cause of death (after heart disease and cancer) in many developed countries.

In the early years, insulin isolated and purified from the pancreases of pigs and cows was used for the treatment of diabetics. There is a slight difference (by one to three amino acids) in the structure of animal insulin compared to human insulin. This resulted in allergy in some of the diabetics when animal insulin was administered. Another problem with animal insulin is that large number of animals have to be sacrificed for extracting insulin from their pancreases. For instance, about 70 pigs (giving about 5 kg pancreatic tissue) have to be killed to get insulin for treating a single diabetic patient just for one year!

Production of recombinant insulin

Attemps to produce insulin by recombinant DNA technology started in late 1970s. The ***basic technique consisted of inserting human insulin gene and the promoter gene of lac operon on to the plasmids of E. coli***. By this method human insulin was produced. It was in July 1980, seventeen human volunteers were, for the first time, administered recombinant insulin for treatment of diabetes at Guy's Hospital, London. And in fact, insulin was the first ever pharmaceutical product of recombinant DNA technology administered to humans. Recombinant insulin worked well, and this gave hope to scientists that DNA technology could be successfully employed to produce substances of medical and commercial importance. An ***approval***, by the concerned authorities, for using recombinant insulin for the treatment of diabetes mellitus was ***given in 1982***. And in 1986, Eli Lilly company received approval ***to market human insulin*** under the trade name ***Humulin***.

Technique for recombinant insulin production : The orginal technique (described briefly above) of insulin synthesis in *E. coli* has undergone several changes, for improving the yield. e.g. addition of signal peptide, synthesis of A and B chains separately etc.

The procedure employed for the synthesis of two insulin chains A and B is illustrated in

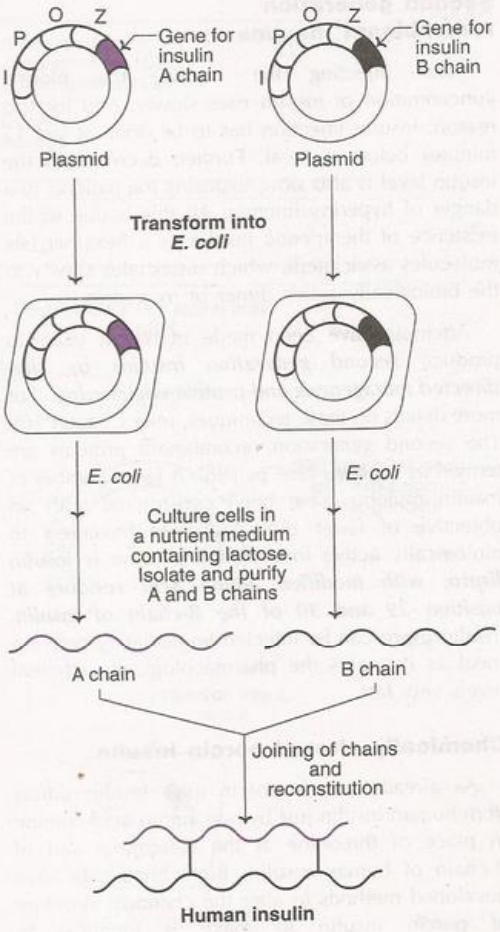


Fig. 15.1 : *The production of recombinant insulin in* E. coli *(I-Inducer gene, P-Promoter gene, O-Operator gene, Z-β-Galactosidase gene; all these genes are of lac operon system).*

Fig. 15.1. The genes for insulin A chain and B chain are separately inserted to the plasmids of two different *E. coli* cultures. The *lac* operon system (consisting of inducer gene, promoter gene, operator gene and structural gene Z for β-galactosidase) is used for expression of both the genes. The presence of lactose in the culture medium induces the synthesis of insulin A and B chains in separate cultures. The so formed insulin chains can be isolated, purified and joined together to give a full-pledged human insulin.

Second generation recombinant insulins

After injecting the insulin, the plasma concentration of insulin rises slowly. And for this reason, insulin injection has to be done at last 15 minutes before a meal. Further, decrease in the insulin level is also slow, exposing the patients to a danger of hyperinsulinemia. All this is due to the existence of therapeutic insulin as a hexamer (six molecules associated), which dissociates slowly to the biologically active dimer or monomer.

Attempts have been made in recent years to produce ***second generation insulins by site-directed mutagenesis and protein engineering***. (For more details on these techniques, refer Chapter 10). The second generation recombinant proteins are termed as ***muteins*** (See p. 198). A large number of insulin muteins have been constructed with an objective of faster dissociation of hexamers to biologically active forms. Among these is ***insulin lispro, with modified amino acid residues at position 29 and 30 of the B-chain of insulin***. Insulin lispro can be injected immediately before a meal as it attains the pharmacologically efficient levels very fast.

Chemically altered porcin insulin

As already stated, porcin (pig) insulin differs from human insulin just by one amino acid-alanine in place of threonine at the C-terminal and of B-chain of human insulin. Biotechnologists have developed methods to alter the chemical structure of porcin insulin to make it identical to human insulin. And this chemically modified porcin insulin can also be employed for the treatment diabetes mellitus.

HUMAN GROWTH HORMONE

Growth hormone is produced by the pituitary gland. It regulates the growth and development. Growth hormone stimulates overall body growth by increasing the cellular uptake of amino acids, and protein synthesis, and promoting the use of fat as body fuel.

Insufficient human growth hormone (hGH) in young children results in retarded growth, clinically referred to as ***pituitary dwarfism***. The child usually is less than four feet in height, and has chubby face and abundant fat around the waist.

Traditional treatment for dwarfism : The children of pituitary dwarfism were treated with regular ***injections of growth hormone extracted from the brains of deceased humans***. It may be noted that only human growth hormone is effective for treatment of dwarfism. (This is in contrast to diabetes where animal insulins are employed). At least eight pituitary glands from cadavers must be extracted to get hGH adequate for treating a dwarf child just for one year! And such treatment has to be continued for 8–10 years!! Further, administration hGH isolated from human brains exposes the children to a great risk of transmitting the cadaver brain diseases (through virus or viral-like agents) e.g. Creuzfeldt-Jacob (CJ) syndrome characterized by convulsions, wasting of muscle etc.

Production of recombinant hGH

Biotechnologists can now produce hGH by genetic engineering. The technique adopted is quite comparable with that of insulin production. The procedure essentially consists of inserting hGH gene into *E. coli* plasmid, culturing the cells and isolation of the hGH from the extracellular medium.

Limitation in hGH production : The hGH is a protein comprised of 191 amino acids. During the course of its natural synthesis in the body, hGH is tagged with a single peptide (with 26 amino acids). The signal peptide is removed during secretion to release the active hGH for biological functions. The entire process of hGH synthesis goes on in an orderly fashion in the body. However, ***signal peptide interrupts hGH production by recombinant technology***. The complementary DNA (cDNA) synthesized from the mRNA encoding hGH is inserted into the plasmid. The plasmid containing *E. coli* when cultured, produces full length hGH along with signal peptide. But *E. coli* cannot remove the signal peptide. Further, it is also quite difficult to get rid of signal peptide by various other means. Theoretically, cDNA encoding signal peptide can be cut to solve these problems. Unfortunately, there is no restriction endonuclease to do this job, hence this is not possible.

A novel approach for hGH production : Biotechnologists have resolved the problem of signal peptide interruption by a novel approach (***Fig. 15.2***). The base sequence in cDNA encoding signal peptide (26 amino acids) plus the neighbouring 24 amino acids (i.e. a total 50 amino

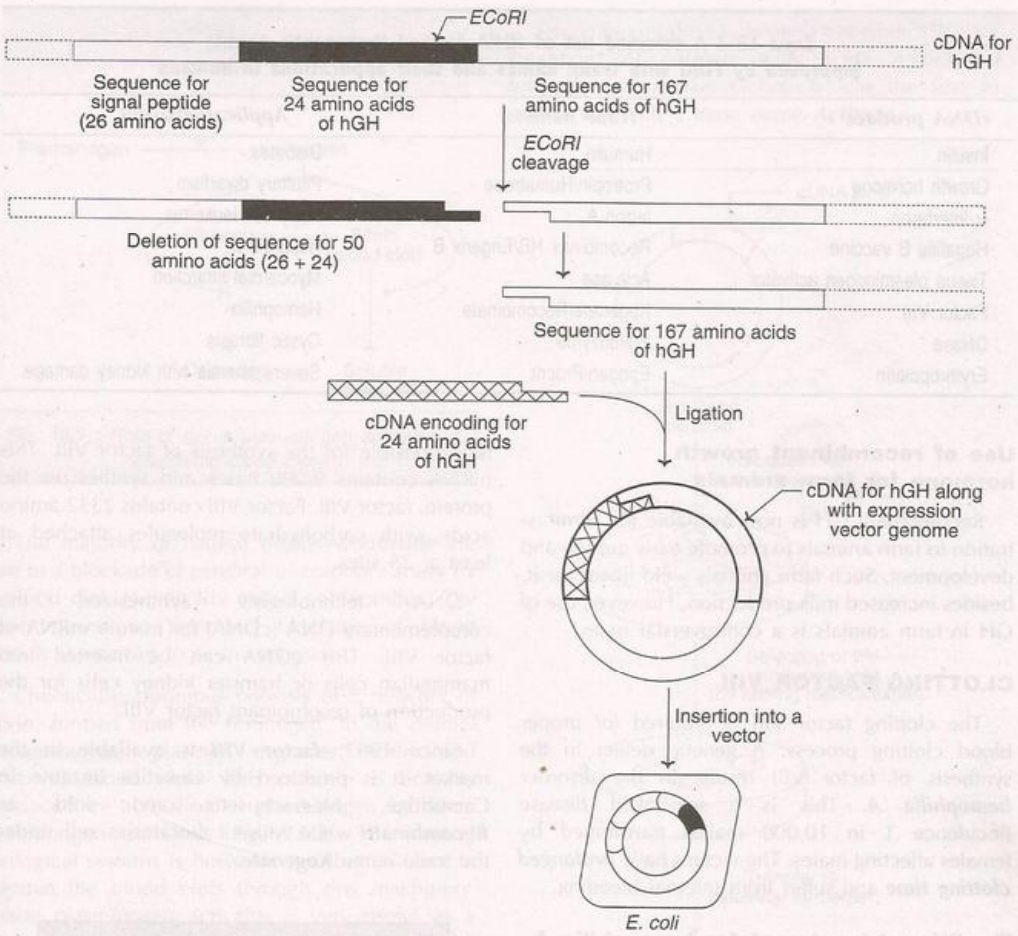


Fig. 15.2 : *The production of recombinant human growth hormone (cDNA–Complementary DNA, hGH–Human growth hormone)*

acids) is cut by restriction endonuclease *ECoRI*. Now a gene (cDNA) for 24 amino acid sequence of hGH (that has been deleted) is freshly synthesized and ligated to the remaining hGH cDNA. The so constitued cDNA, attached to a vector, is inserted into a bacterium such as *E. coli* for culture and production of hGH. In this manner, the biologically functional hGH can be produced by DNA technology.

Recombinant hGH was approved for human use in 1985. It is marketed as ***Protropin*** by Genetech company and ***Humatrope*** by Eri Lilly company.

Controversy over the use of hGH

Recombinant hGH can be administered to children of very short stature. It has to be given daily for many years with an annual cost of about $ 20,000. Some workers have reported substantial increase in the height of growth retarded children. One group of workers observed that the normal growth pattern in children was not restored on hGH administration, although there was an initial spurt. Another big question raised by the opponents of hGH therapy is whether it is necessary to consider short stature as a disorder at all for treatment!

Table 15.3 A selected list of rDNA-derived therapeutic agents (approved by FDA) with trade names and their applications in humans

rDNA product	*Trade name(s)*	*Applications/uses*
Insulin	Humulin	Diabetes
Growth hormone	Protropin/Humatrope	Pituitary dwarfism
α-Interferon	Intron A	Hairy cell leukemia
Hepatitis B vaccine	Recombinax HB/Engerix B	Hepatitis B
Tissue plasminogen activator	Activase	Myocardial infarction
Factor VIII	Kogenate/Recombinate	Hemophilia
DNase	Pulmozyme	Cystic fibrosis
Erythropoietin	Epogen/Procrit	Severe anemia with kidney damage

Use of recombinant growth hormone for farm animals

Recombinant GH is now available for administration to farm animals to promote early growth and development. Such farm animals yield linear meat, besides increased milk production. However, use of GH in farm animals is a controversial issue.

CLOTTING FACTOR VIII

The clotting factor VIII is required for proper blood clotting process. A genetic defect in the synthesis of factor VIII results in the disorder ***hemophilia A***. This is a sex-linked disease (incidence 1 in 10,000 males) transmitted by females affecting males. The victims have ***prolonged clotting time*** and suffer from internal bleeding.

Traditional treatment for hemophilia A

Clotting factor VIII was isolated from the whole blood and administered to the patients of hemophilia A. This approach requires large quantities of blood. Another problem is the risk of transmission of certain diseases like AIDS to the hemophiliacs.

Production of recombinant factor VIII

The gene for the formation of factor VIII is located on X chromosome. It is a complex gene of 186 kb (i.e., 186,000 base pairs) in size, organized into 26 exons of varying length. In between the exons, many introns are present. The introns vary in their size, starting from 200 base pairs to as high as 32,000 base pairs.

Biotechnologists were able to isolate mature mRNA (containing only exons and no introns) that is responsible for the synthesis of factor VIII. This mRNA contains 9,000 bases and synthesizes the protein, factor VIII. Factor VIII contains 2332 amino acids, with carbohydrate molecules attached at least at 25 sites.

DNA technologists synthesized the complementary DNA (cDNA) for mature mRNA of factor VIII. This cDNA can be inserted into mammalian cells or hamster kidney cells for the production of recombinant factor VIII.

Since 1992, ***factor VIII*** is available in the market. It is produced by Genetics Institute in Cambridge, Massachusetts and sold as ***Recombinate*** while Miles Laboratories sell under the trade name ***Kogenate***.

THERAPEUTIC AGENTS FOR HUMAN DISEASES

Biotechnology is very useful for the production of several therapeutic products for treating human diseases. A selected list of rDNA-derived therapeutic agents along with trade names and their uses in humans are given in ***Table 15.3***. Some of these are described above (under human protein replacements) while the remaining are discussed below.

TISSUE PLASMINOGEN ACTIVATOR

Tissue plasminogen activator (tPA) is a naturally occurring protease enzyme that helps ***to dissolve blood clots***. tPA is a boon for **patients suffering *from thrombosis***.

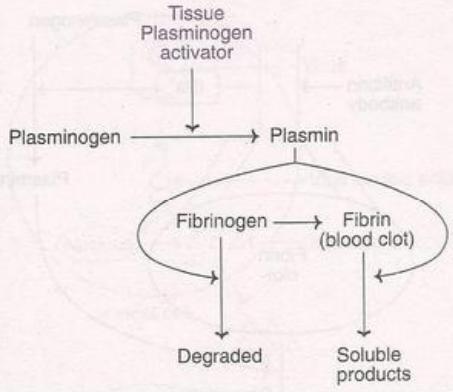


Fig. 15.3 : Role of tissue plasminogen activator in dissolving blood clots.

The majority of natural deaths worldwide are due to a blockade of cerebral or coronary artery by a blood clot, technically called as thrombus. The phenomenon of thrombus blockage of blood vessels is referred to as thrombosis.

Chemically, thrombus consists of a network of fibrin, formed from the fibrinogen. In the normal circumstances, plasmin degrades fibrin and dissolves blood clots. This plasmin is actually produced by activation of plasminogen by tissue plasminogen activator (***Fig. 15.3***). The natural biological systems is however, not that efficient to remove the blood clots through this machinery. Tissue plasminogen activator is very useful as a therapeutic agent in dissolving blood clots (thrombi) by activating plasminogen. By removing the arterial, thrombi, the possible damage caused by them on heart and brain could be reduced.

Production of recombinant tPA

DNA technologists synthesized the complementary DNA (cDNA) molecule for tissue plasminogen activator. This cDNA was then attached to a synthetic plasmid and introduced into mammalian cells (***Fig. 15.4***). They were cultured and tPA-producing cells were selected by using methotrexate to the medium. tPA-producing cells were transferred to an industrial tank (fermenter). tPA, secreted into the culture medium, is isolated for therapeutic purpose. It may be noted here that ***tPA was the first pharmaceutical product to be produced by mammalian cell culture.***

Recombinant tPA has been in use since 1987 for ***treatment of patients with acute myocardial infarction or stroke.*** Genetech was the first to market tPA with a trade name ***Activase.***

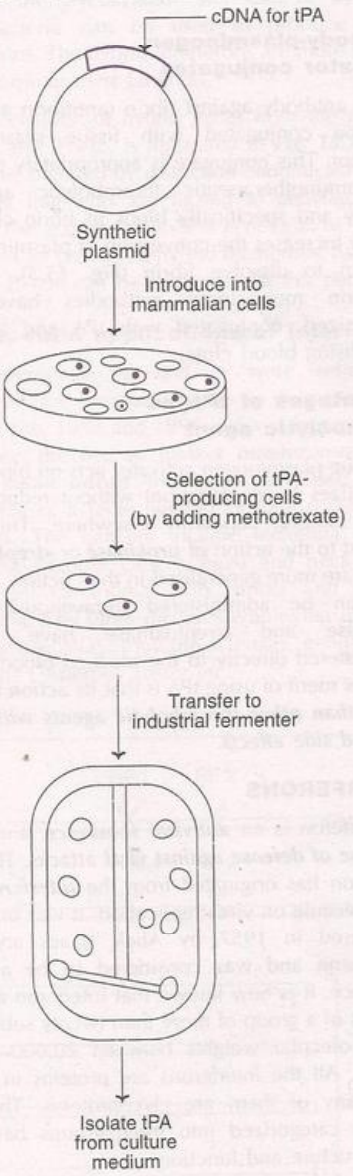


Fig. 15.4 : The production of recombinant tissue plasminogen activator (tPA).

Alteplase and Reteplase : These are the second generation recombinant tPAs. They have ***increased in vivo half-lives*** and are functionally more efficient. The general aspects of second generation recombinant proteins are given elsewhere (See p. 198).

Antibody-plasminogen activator conjugates

An antibody against fibrin (antifibrin antibody) can be conjugated with tissue plasminogen activator. This conjugate is appropriately regarded as immunotherapeutic thrombolytic agent. It quickly and specifically binds to fibrin clots and locally increases the conversion of plasminogen to plasmin to dissolve fibrin (***Fig. 15.5***). In fact, antifibrin monoclonal antibodies have been synthesized, conjugated with tPA and tried for solubilising blood clots.

Advantages of tPA as thrombolytic agent

Tissue plasminogen activator acts on blood clots (solubilizes by degradation) without reducing the blood clotting capability elsewhere. This is in contrast to the action of ***urokinase*** or ***streptokinase*** which are more generalised in their action. Further, tPA can be administered intravenously while urokinase and streptokinase have to be administered directly to the blocked blood vessel. Another merit of using tPA is that its ***action*** is much ***faster than other thrombolytic agents with much reduced side effects***.

INTERFERONS

Interferon is an ***antiviral substance***, and is the ***first line of defense against viral attacks***. The term interferon has originated from the ***interference*** of this molecule on virus replication. It was originally discovered in 1957 by Alick Isaacs and Jean Lindemann and was considered to be a single substance. It is now known that interferon actually consists of a group of more than twenty substances with molecular weights between 20,000–30,000 daltons. All the interferons are proteins in nature and many of them are glycoproteins. They are broadly categorized into three groups based on their structure and function

Interferon-α (IFN-α)
Interferon-β (IFN-β)
Interferon-γ (IFN-γ)

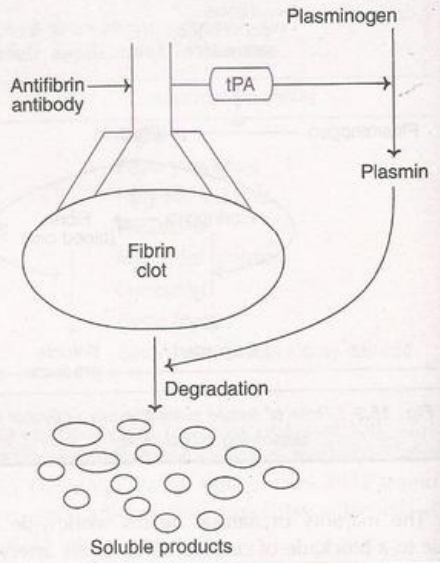


Fig. 15.5 : Action of antifibrin antibody—tPA conjugate in dissolving blood clot (tPA–tissue plasminogen activator).

Mechanism of action of interferons

Interferons are produced by mammalian cells when infected by viruses. As the virus releases its nucleic acid into cellular cytoplasm, it stimulates the host DNA to produce interferons. These interferons, secreted by the cells, bind to the adjacent cells. Here, they stimulate the cellular DNA to produce a series of antiviral enzymes. The so formed proteins inhibit viral replication and protect the cells (***Fig. 15.6***). It is believed that the protective (enzymes) proteins bind to mRNA of viruses and block their protein synthesis. The action of interferons appears to be species specific. Thus, human interferons operate in humans. Other animal (dog, mouse) interferons are ineffective in man.

Isolation of interferons in the early years

Blood was the only source of interferons earlier. The procedure was very tedious and the quantity of interferons isolated was very little. Thus, as much as 50,000 litres of human blood was required to get just 100 mg of interferons! Therefore, it was very difficult to conduct research or use interferons for therapeutic purposes.

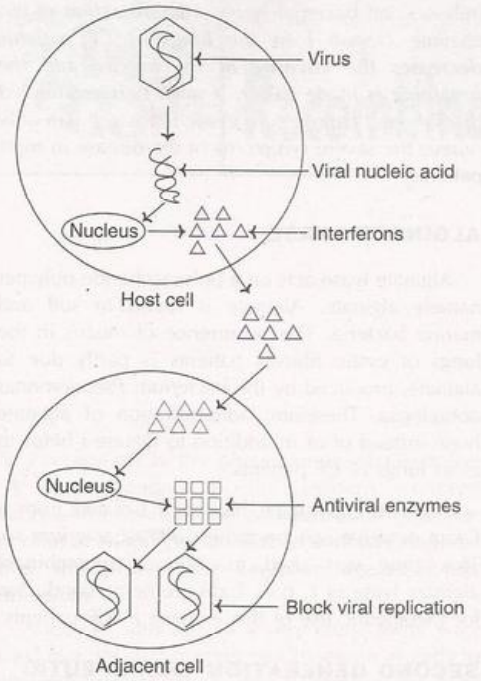


Fig. 15.6 : The mechanism of action of interferons.

Production of recombinant interferons

The complementary DNA (cDNA) was synthesized from the mRNA of a specific interferon. This is inserted to a vector (say plasmid) which is introduced into *E. coli* or other cells. The interferon can be isolated from the culture medium. This is the basic mechanism of producing recombinant interferons.

The production of interferons is relatively less in bacterial hosts, although *E. coli* was the first to be used. This is mainly because most interferons are glycoproteins in nature and bacteria do not possess the machinery for glycosylation of proteins.

Production interferons by yeasts : The yeast *Saccharomyces cerevisiae* is more suitable for the production of recombinant interferons. This is mainly because the yeast possess the mechanism to carry out glycosylation of proteins, similar to that occurs in mammalian cells. The DNA sequence coding for specific human interferon can be attached to the yeast alcohol dehydrogenase gene in a plasmid and introduced into 4 yeast cells. The yield of interferons is several fold higher compared to *E. coli*.

Production of hybrid interferons : Several attempts have been made to produce hybrid interferons. This is advantageous since ***different interferons with different antiviral activities can be combined to produce a more efficient interferon***. Further, the glycosylation step can be bypassed, and bacteria can be used to produce hybrid interferons. The hybrid interferons are more reactive in performing their function.

The creation of hybrid genes from the genes of IFN-α_2 and IFN-α_3 is illustrated in ***Fig. 15.7***. These genes are digested by restriction endonucleases. The resulting fragments are ligated to generate hybrid genes. The appropriate hybrid genes can be selected and used for producing hybrid interferons. As already stated, *E. coli* can be employed for this purpose.

Therapeutic applications of interferons

Interferons-α, -β and -γ were respectively approved for therapeutic use in humans in the years 1986, 1993 and 1990. A Swiss biotechnology firm was the first to market ***interferon-α*** with a ***trade name Intron***. Interferons are used for the treatment of a large number of viral diseases and cancers. The cancers include leukemia, kaposis sarcoma, bladder cancer, head and neck cancer, renal cell carcinoma, skin cancer and multiple myeloma. The other diseases employing interferon therapy are AIDS, multiple sclerosis, genital warts, hepatitis C, herpes zoster etc.

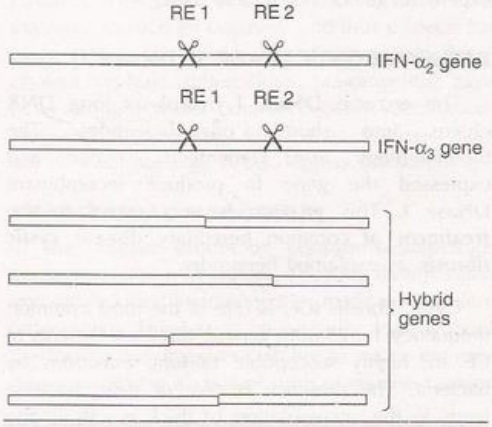


Fig. 15.7 : Creation of hybrid interferon genes (RE-Restriction endonuclease, IFN-Interferon).

Interferons are also employed in the treatment of common colds and influenza. For this purpose, interferons can be used as nasal sprays.

The basic mechanism of action of interferons against viruses has already been described. Interferons are found to cause the death of cancerous cells. This is brought out by stimulating the action of ***natural killer (NK) cells***, a specialised form of lymphocytes that can destroy cancerous cells.

Despite the widespread therapeutic applications of interferons, they are not within the reach of a large number of common people due to the cost factor (the cost of production being very high).

ERYTHROPOIETIN

Erythropoietin is a hormone synthesized by the kidneys. It stimulates the stem cells of bone marrow to produce mature erythrocytes. Biotechnologists were successful in producing recombinant erythropoietin. An approval for its therapeutic use in humans was obtained in the year 1989. Amgen Inc. first ***marked erythropoietin*** with a trade name ***Epogen***. It is useful in treating the patients with severe anemia that accompanies kidney disease. Another firm Ortho-Biotech company produced ***Procrit***, a genetically engineered erythropoietin in 1997. Procrit acts like the natural hormone and stimulates the production of erythrocytes. It is used in anemic patients undergoing non-cardiac, non-vascular surgery. Procrit administration before surgery serves as an alternative to blood transfusion. However, therapeutic use of procrit is quite expensive, hence not widely used.

DEOXYRIBONUCLEASE I (DNase I)

The enzyme DNase I hydrolysis long DNA chains into shorter oligonucleotides. The biotechnology firm Genentech isolated and expressed the gene to produce recombinant DNase I. This enzyme is very useful in the ***treatment of*** common hereditary disease ***cystic fibrosis***, as explained hereunder.

Cystic fibrosis (CF) is one of the most common (frequency 1 : 25,000) genetic diseases. Patients of CF are highly susceptible to lung infections by bacteria. The presence of live or dead bacteria leads to the accumulation of thick mucus in the lungs making the breathing very difficult. The major constituent of this mucus is the bacterial DNA (released on bacterial lysis). ***Administration of*** the enzyme ***DNase I to the lungs of CF patients decreases the viscosity of the mucus, and the breathing is made easier***. It must be remembered that DNase I cannot cure cystic fibrosis. It can only relieve the severe symptoms of the disease in most patients.

ALGINATE LYASE

Alginate lyase acts on a polysaccharide polymer namely alginate. Alginate is found in soil and marine bacteria. The occurrence of mucus in the lungs of cystic fibrosis patients is partly due to alginate, produced by the bacterium *Pseudomonas aeruginosa*. Therefore, administration of alginate lyase instead of or in addition to DNase I helps to clear lungs of CF patients.

Alginate lyase gene has been isolated from a Gram-negative soil bacterium, *Flavobacterium* sp. This gene was used to produce recombinant alginate lyase in *E. coli*. Trails are being conducted for therapeutic use of this enzyme in CF patients.

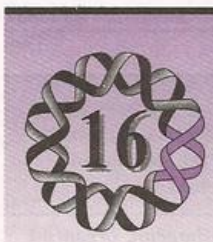
SECOND GENERATION THERAPEUTIC PROTEINS (MUTEINS)

By employing ***site-directed mutagenesis***, the amino acid sequence of a recombinant protein can be suitably modified as desired, by a technique referred to as ***protein engineering*** (For details Refer Chapter 10). The mutated proteins are collectively referred to as ***muteins***. Protein engineering is a rational approach to modify a protein with regard to its stability, solubility, specificity, substrate affinity, pharmacokinetics etc.

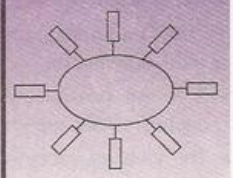
The muteins obtained by protein engineering technique are considered as ***Second generation of therapeutic proteins***. Selected examples of such proteins (e.g. ***insulin lispro***, ***Alteplase***) are already described (See p. 192 and p. 196).

VACCINES

Vaccines are another important group of pharmaceutical products of recombinant DNA technology. They are separately dealt with in Chapter 16. However, the reader must treat and learn vaccines also as a part of this chapter.



Recombinant Vaccines



Vaccination is the phenomenon of preventive immunization. In the modern concept, vaccination involves the ***administration*** (injection or oral) ***of an antigen to elicit an antibody response that will protect the organism against future infections.***

Vaccines, although in a very crude form, were used in a peculiar manner by Chinese in as early as eleventh century. They used to remove the dried scabs of smallpox patients and ground them. The so prepared powder was sprayed in the noses of healthy people. It was observed that these people were less susceptible to smallpox when there was an outbreak of epidemic.

Smallpox is a virulent disease with a high death rate. Even the survivors become victims of permanent disfigurement, blindness and mental retardation. Edward Jenner, an English Physician, observed that the farmers and milkmaids working with cows developed a mild form of smallpox called cowpox or vaccinia. And the cowpox infection could protect these people from the infection of smallpox. It was in 1796, Jenner experimentally tested this observation. He inoculated an 8–year old boy with exudate from a cowpox lesion, and he repeated it twice with a gap of some weeks. He then inoculated more human volunteers. Jenner noted that smallpox did not develop in these volunteers. This was the first discovery of the principle of vaccination.

It took nearly hundred years for the scientists to clearly understand the basis of smallpox immunity. We now know that cowpox viruses (vaccinia) inoculated by Jenner, stimulate the body's immune system to produce antibodies which neutralise the cowpox as well as smallpox viruses. The present day vaccines which are more refined work in a similar fashion.

Vaccines are mainly of three types.

1. ***Dead bacteria or inactivated viruses.***

2. ***Live non-virulent or weakened (attenuated) bacteria/or viruses.***

3. ***Viral fragments or bacterial molecules (subunit vaccines).***

A vaccine triggers the body's immune system to produce antibodies against a specific disease—causing organism (virus, bacterium or other parasite). This provides surveillance against future exposure to such an organism and thus protects the body. Many communicable diseases (smallpox, cholera, typhoid, tuberculosis, poliomyelitis) have been brought under control through vaccination. However, as on today, for several diseases, there are no vaccines e.g., AIDS, leprosy, filariasis.

TRADITIONAL VACCINES

The disease producing organisms (infectious agents) are grown in culture. They are then purified and either killed (inactivated) or made non-virulent (attenuated). This has to be carefully done without the loss of the organism's ability to evoke immune response against a virulent form of disease-causing organism.

The traditional production of vaccines has several drawbacks.

- It is not possible to develop vaccines for the organisms not grown in culture.
- The yield of vaccines is very low.
- Cell cultures are costly to maintain.
- There is a danger of non-virulent organisms getting converted to virulent ones. Vaccinations by such organisms may cause the disease itself.
- It is not possible to prevent all the diseases by use of traditional vaccines e.g., AIDS.

Purified antigen vaccines

Some improvements in traditional vaccines have been made by isolating the antigens from the pathogenic organisms. The antigens of bacterial cell walls (e.g., *Streptococcus pneumoniae* causing pneumonia), and the endotoxins are good examples of purified vaccines. The endotoxins that do not possess toxicity but retain immunogenecity are referred to as ***toxoids*** e.g. toxods of tetanus, diphtheria etc.

RECOMBINANT VACCINES—GENERAL

Recombinant DNA technology in recent years, has become a boon to produce ***new generation vaccines***. By this approach, some of the limitations (listed above) of traditional vaccine production could be overcome. In addition, several new strategies, involving gene manipulation are being tried to create novel recombinant vaccines.

The list of diseases for which recombinant vaccines are developed or being developed is given in ***Table 16.1***. It may be stated here that due to very stringent regulatory requirements to use in humans, the new generation vaccines are first tried in animals, and it may ***take some more years before most of them are approved for use in humans***.

Types of recombinant vaccines

The recombinant vaccines may be broadly categorized into three groups :

1. **Subunit recombinant vaccines :** These are the components of the pathogenic organisms. Subunit vaccines include proteins, peptides and DNA.

2. **Attenuated recombinant vaccines :** These are the genetically modified pathogenic organisms (bacteria or viruses) that are made non-pathogenic and used as vaccines.

TABLE 16.1 A selected list of diseases along with the pathogenic organisms for which recombinant vaccines are developed or being developed

Disease	*Pathogenic organism*
Viral diseases	
Accute infantile gastroenteritis	Rotavirus
Acute respiratory diseases	Influenza A and B viruses
AIDS	Human immunodeficiency virus
Chicken pox	Viricella–zoster virus
Encephalitis	Japanese encephalitis virus
Genital ulcers	Herpes simplex virus type–2
Hemorrhagic fever	Dengue virus
Liver damage	Hepatitis A virus
Liver damage	Hepatitis B virus
Upper and lower respiratory tract lesions	Yellow fever virus
Bacterial diseases	
Cholera	*Vibrio cholerae*
Diarrhea	*E. coli*
Dysentery	*Shigella* strain
Gonorrhea	*Neissria gonorroheae*
Leprosy	*Mycobacterium leprae*
Meningitis	*Neisseria meningitidis*
Pneumonia	*Streptococcus pneumoniae*
Rheumatic fever	*Streptococcus* group A
Tetanus	*Clostridium tetani*
Tuberculosis	*Mycobacterium tuberculosis*
Typhoid	*Salmonella typhi*
Urogenital tract infection	*Streptococcus* group B
Parasitic diseases	
Filariasis	*Wuchereria bancrofti*
Malaria	*Plasmodium* sp
River blindness	*Onchocerca volvulus*
Schistosomiasis	*Schistosoma mansoni*
Sleeping sickness	*Trypanosoma* sp

3. **Vector recombinant vaccines :** These are the genetically modified viral vectors that can be used as vaccines against certain pathogens.

Some of the developments made in the production of recombinant vaccines against certain diseases are briefly described.

SUBUNIT VACCINES

As already stated, subunit recombinant vaccines are the ***components (proteins, peptides, DNAs) of the pathogenic organisms***. The advantages of these vaccines include their purity in preparation, stability and safe use. The disadvantages are — high cost factor and possible alteration in native conformation. Scientists carefully evaluate the pros and cons of subunit vaccines for each disease, and proceed on the considered merits.

HEPATITIS B

Hepatitis B is a widespread disease in man. It primarily affects liver causing ***chronic hepatitis, cirrhosis and liver cancer***. Hepatitis B virus is a 42 nm particle, called ***Dane particle***. It consists of a core containing a viral genome (DNA) surrounded by a phospholipid envelope carrying surface antigens (***Fig. 16.1A***). Infection with hepatitis B virus produced Dane particles and 22 nm sized particles. The latter contain surface antigens which are more immunogenic. It is however, very difficult to grow hepatitis B virus in mammalian cell culture and produce surface antigens.

The gene encoding for ***hepatitis B surface antigen (HBsAg)*** has been identified. ***Recombinant hepatitis B vaccine as a subunit vaccine, is produced by cloning HbsAg gene in yeast cells***. *Saccharomyces cerevisiae,* a harmless baking and brewing yeast, is used for this purpose (***Fig. 16.1B***). The gene for HBsAg is inserted (pMA 56) which is linked to the alcohol dehydrogenase promoter. These plasmids are then transferred and cultured. The cells grown in tryptophan, free medium are selected and cloned. The yeast cells are cultured. The HBsAg gene is expressed to produce 2nm sized particles similar to those found in patients infected with hepatitis B. (These particles are immunoreactive with anti-HBsAg antibodies). The subunit HBsAg as 22 nm particles can be isolated and used to immunize individuals against hepatitis B.

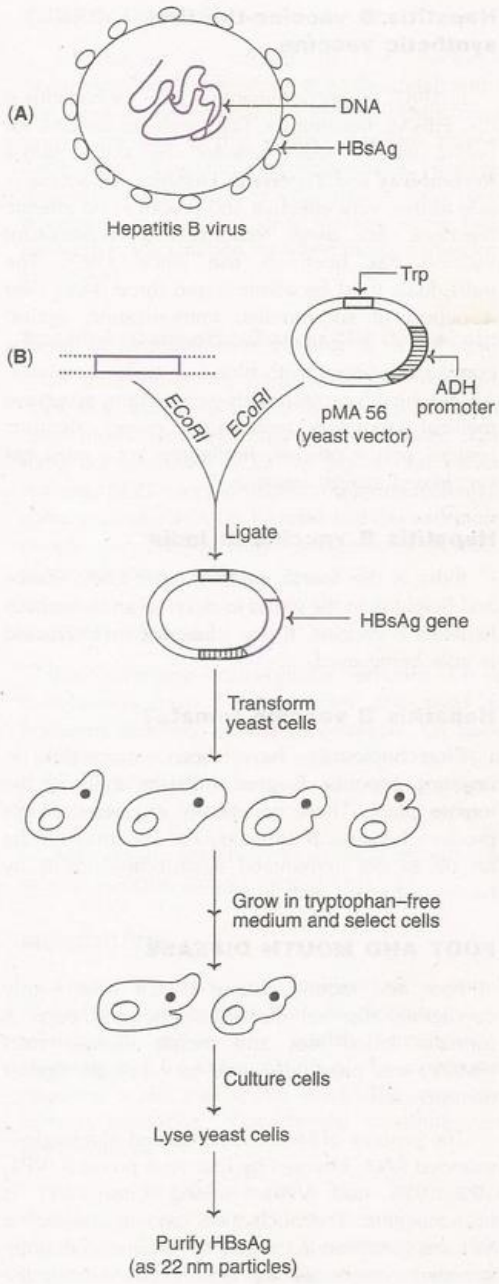


Fig. 16.1 : (A) Hepatitis B virus–Dane particle (42 nm particle); (B) Production of hepatitis B surface antigen (HBsAg) in yeast cells (Trp–Tryptophan, ADH–Alcohol dehydrogenase).

Hepatitis B vaccine-the first synthetic vaccine

In 1987, the recombinant vaccine for hepatitis B (i.e. HBsAg) became the first synthetic vaccine for public use. It was marketed by trade names ***Recombivax*** and ***Engerix-B***. Hepatitis B vaccine is safe to use, very effective and produces no allergic reactions. For these reasons, this recombinant vaccine has been in use since 1987. The individuals must be administered three doses over a period of six months. Immunization against hepatitis B is strongly recommended to anyone coming in contact with blood or body secretions. All the health professionals—physicians, surgeons, medical laboratory technicians, nurses, dentists, besides police officers, firefighters etc., must get vaccinated against hepatitis B.

Hepatitis B vaccine in India

India is the ***fourth country*** (after USA, France and Belgium) in the world to develop an indigenous hepatitis B vaccine. It was launched in 1997, and is now being used.

Hepatitis B vaccine tomato?

Biotechnologists have been successful in inserting hepatitis B gene into the cells of the tomato plant. These genetically enginered plants produce hepatitis B antigens. The day may not be far off to get immunized against hepatitis B by having a tomato with lunch!

FOOT AND MOUTH DISEASE

Foot and mouth disease (FMD) is a highly contagious disease affecting cattle and pigs. A formalin killed foot and mouth disease virus (FMDV) was previously used to vaccinate against this disease.

The genome of FMDV is composed of a single-stranded RNA, covered by four viral proteins (VP1, VP2, VP3 and VP4). Among these, VPI is immunogenic. The nucleotide sequence encoding VPI was identified in the FMDV genome. A double-stranded complementary DNA (cDNA) from the single-stranded viral RNA (genome) was synthesized. This cDNA was then digested with restriction enzymes and the fragments were cloned by using plasmid pBR322 in *E. coli*. The recombinant vaccine for FMDV in the form of viral protein 1 was used to vaccinate animals. However, VPI vaccination was found to be less effective than that of the whole virus in protecting FMD. Further, studies are being pursued to improve the efficiency of subunit vaccine.

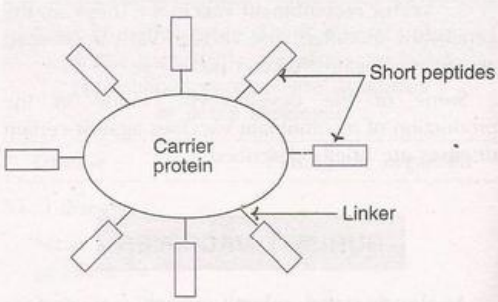


Fig. 16.2 : A general structure of a synthetic peptide vaccine.

The concept of peptide vaccines

Theoretically, it is expected that only ***small portions of a*** given ***protein*** (i.e., domains) are ***immunogenic*** and bind to antibodies. Logically, it is possible to use short peptides that are immunogenic as vaccines. These are referred to as peptide vaccines.

Peptide vaccines for foot and mouth disease

Some details on the FMD are described above. The domains of viral protein I (VPI) of FMDV were chemically synthesized. From the C-terminal end of VPI, amino acids 141 to 160, 151 to 160 and 200 to 213, and from N-terminal end, amino acids 9 to 24, 17 to 32 and 25 to 41 were synthesized. Each one of these ***short peptides*** (***domains***) was ***bound to the surface of a carrier protein*** (***Fig. 16.2***) and ***used as a vaccine***. Among the peptides used, the one corresponding to amino acids 141 to 160 was found to be effective in immunizing guinea pigs against FMD. In addition, when two peptides were joined together (amino acids 141 to 158, and amino acids 200 to 213), they served as more efficient recombinant vaccines.

The success so far to use recombinant peptides as vaccines has been very limited. This is mainly because a short peptide usually is not enough to be

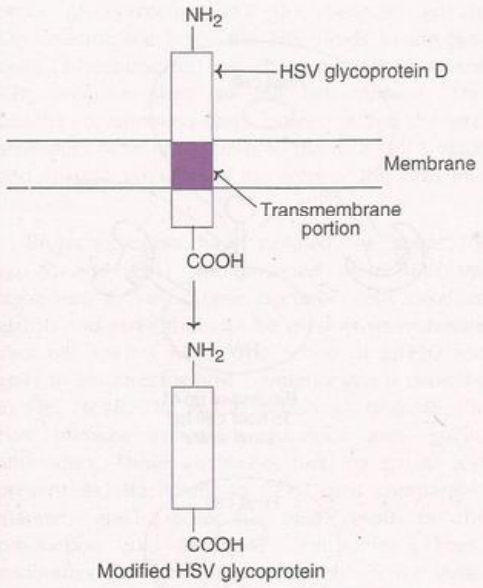


Fig. 16.3 : Modification of HSV glycoprotein D by deleting transmembrane portion. (HSV–Herpes simplex virus)

sufficienty immunogenic, since it may not have the same conformation as that of the original viral particle. However, scientists continue their search for specific, inexpensive and safe synthetic peptide vaccines for various diseases.

HERPES SIMPLEX VIRUS

Herpes simplex virus (HSV) is an oncogenic (cancer-causing) virus. In addition, it also causes sexually transmitted diseases, encephalitis and severe eye infections. Attempts have been made to produce subunit vaccines against HSV.

An envelope glycoprotein D (gD) of HSV that can elicit antibody production has been identified. This is a membrane bound protein, and difficult to isolate and purify. The ***glycoprotein D was modified by deleting the transmembrane portion of the protein*** (***Fig. 16.3***) and the gene was modified. This gene for gD was cloned in a mammalian vector and expressed in Chinese hamster ovary (CHO). The advantage here is that the protein can get glycosylated (unlike in *E. coli* system). In the experimental trials, the modified form of gD was found to be effective against HSV.

TUBERCULOSIS

Tuberculosis is caused by the bacterium *Mycobacterium tuberculosis*. It is often fatal, and as per some estimates nearly 3 million deaths occur every year due to this highly infectious disease. Antibiotics are used to treat tuberculosis. However, drug-resistant *M. tuberculosis* strains have been developed making the drug therapy some times ineffective. Vaccination for tuberculosis is therefore, advocated.

Bacillus Calmette-Guerin (BCG) vaccine

In some countries, particularly the developing ones, BCG vaccine is widely used to protect against tuberculosis. However, countries like United States have not approved BCG vaccination for various reasons. BCG vaccine itself causes tuberculosis in some individuals (AIDS victims) and the vaccinated people respond positively for laboratory diagnosis of tuberculosis.

Subunit vaccines

The secretory (extracellular proteins of *M. tuberculosis* have been purified and used for immunoprotection against tuberculosis. Of about 100 such proteins, six were found to be useful (either individually or in combination) to immunize guinea pigs.

Attempts are underway to develop recombinant subunit vaccine against tuberculosis.

MENINGITIS

Group B strain of meningococci, namely *Neisseria meningitidis* causes meningitis in adolescents and young adults. Meningitis is characterized by inflammation of the membranes covering brain and spinal cord. The symptoms include headache, photophobia, irritability, and neck stiffness.

Pizza *et al* (2000) made a novel approach to develop a vaccine against meningitis. They identified 350 proteins (potential protective antigens) and the entire sequence of genome coding for these proteins in *N. meningitidis*. All the 350 candidate antigens were expressed in *E. coli*, purified and used to immunize mice. A good bactericidal antibody response was observed in these mice.

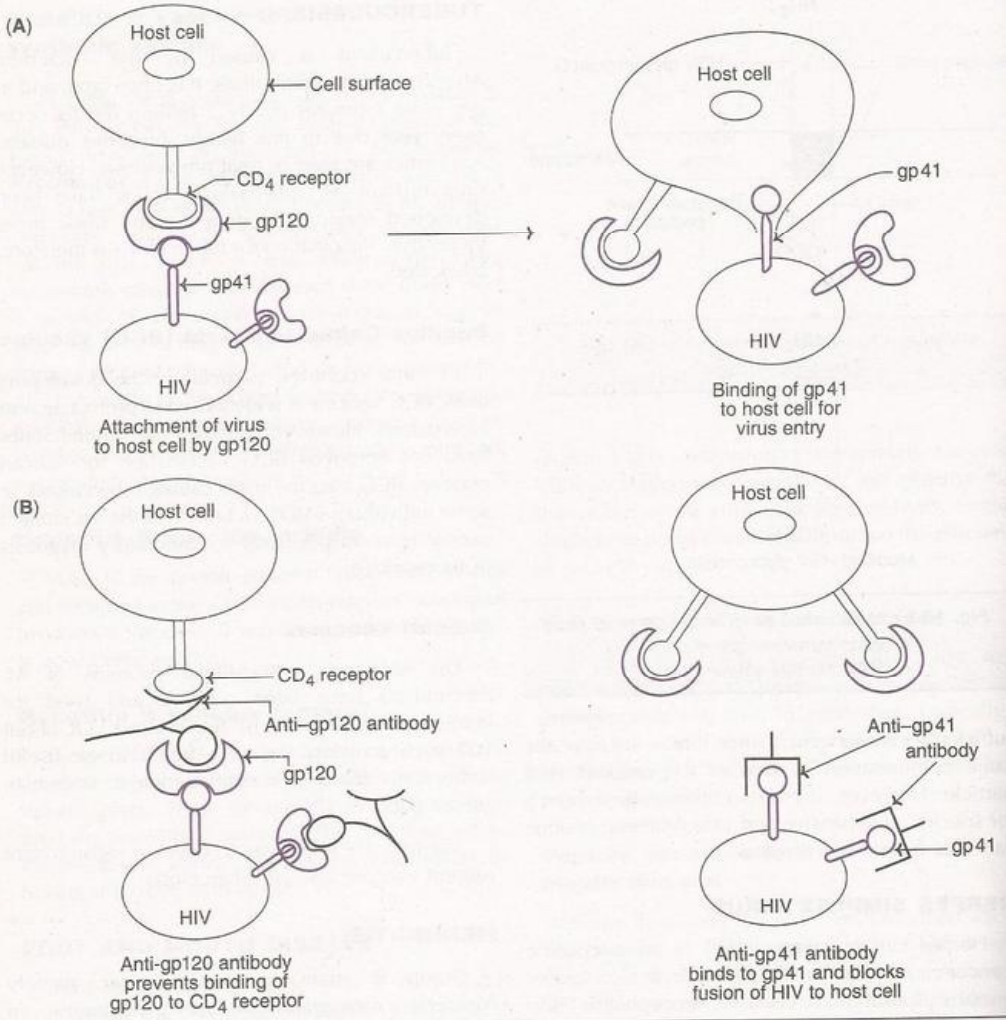


*Fig. 16.4 : Subunit recombinant vaccines against AIDS. (**A**) The functions of gp120 and gp41 of HIV; (**B**) The action of anti–gp120 and anti–gp41 antibodies to prevent the entry of HIV into host cell.*

AIDS (ACQUIRED IMMUNODEFICIENCY SYNDROME)

*AIDS is a retroviral disease caused by **human immunodeficiency virus** (**HIV**).* This disease is characterized by immunosuppression, neoplasma and neurological manifestations. AIDS is invariably fatal, since as of now there is no cure. Development of a vaccine against AIDS is a top priority by DNA technologists worldover. In fact, vaccines are being continuously developed and field tested, although there has been no success so far.

Subunit vaccines

The development of two subunit vaccines, specifically the glycoproteins of HIV envelope is described here. The functions of gp120 and gp41 of HIV are illustrated in ***Fig. 16.4A***. The glycoprotein gp120 projects out of the HIV envelope while the

other glycoprotein gp41 lies beneath gp120. On entering the body, the HIV binds to the host cells (T-lymphocytes) by attaching gp120 to the CD_4 receptor sites on the cell surface. This attachment uncovers gp41 molecules and the viral envelope. Now gp41 binds to the host cell surface and opens a passage for the entry of the virus into the cell.

Biotechnologists have isolated the genes for gp120 and gp41 and inserted them into the bacterium *E. coli*. These bacterial cells produce gp120 and gp41 that can be used as recombinant vaccines against AIDS. The action of gp120 and gp41 in immunizing host T-lymphocytes is depicted in ***Fig. 16.4B***. The gp120 molecules stimulate the host immune system to produce anti- gp120 antibodies. These antibodies bind to gp120 and prevent its attachment to CD_4. In a comparable manner, gp41 molecules also result in the production of anti-gp41 antibodies. These antibodies also bind to gp41 and block the virus-host cell union. The net result of ***using gp120 and gp41 vaccines is that the entry of HIV into the host cells is prevented.***

Vaccine against AIDS— not yet a reality

The description of vaccine development against AIDS (given above), which appears attractive is not so simple. The most important limitations are that the ***HIV has high frequency of mutations***. Therefore the ***vaccines developed cannot bind to the new virus*** (i.e., mutated one). In addition, gp120 and gp41 are very poor stimulators of immune system. Despite these limitations, scientists have not lost hope, and continue their research to develop vaccines against AIDS.

DNA VACCINES (GENETIC IMMUNIZATION)

Genetic immunization by using DNA vaccines is a novel approach that came into being in 1990. The ***immune response of the body is stimulated by a DNA molecule***. A DNA vaccine consists of a gene encoding an antigenic protein, inserted onto a plasmid, and then incorported into the cells in a target animal. The plasmid carrying DNA vaccine normally contains a promoter site, cloning site for the DNA vaccine gene, origin of replication, a selectable marker sequence (e.g. a gene for ampicillin resistance) and a terminator sequence (a poly—A tail).

DNA vaccine—plasmids can be administered to the animals by one of the following delivery methods.

- Nasal spray
- Intramuscular injection
- Intravenous injection
- Intradermal injection
- Gene gun or biolistic delivery (involves pressure delivery of DNA-coated gold beads).

DNA VACCINE AND IMMUNITY

An illustration of a DNA vaccine and the mechanism of its action in developing immunity is given in ***Fig. 16.5***. The plasmid vaccine carrying the DNA (gene) for antigenic protein enters the nucleus of the inoculated target cell of the host. This DNA produces RNA, and in turn the specific antigenic protein. The antigen can act directly for developing humoral immunity or as fragments in association with major histocompatability class (MHC) molecules for developing cellular immunity.

Humoral immunity

As the antigen molecules bind to B-lymphocytes, they trigger the production of antibodies which can destroy the pathogens. Some of the B-lymphocytes become memory cells that can protect the host against future infections.

Cellular immunity

The protein fragments of the antigen bound to MHC molecules can activate the cytotoxic T-lymphocytes. They are capable of destroying the infected pathogenic cells. Some of the activated T-lymphocytes become memory cells which can kill the future infecting pathogens.

Complementary DNA vaccines

For genetic immunization, complementary DNA (cDNA) vaccines can also be used. Some workers have successfully used cDNA as vaccines e.g. immunization of mice against influenza.

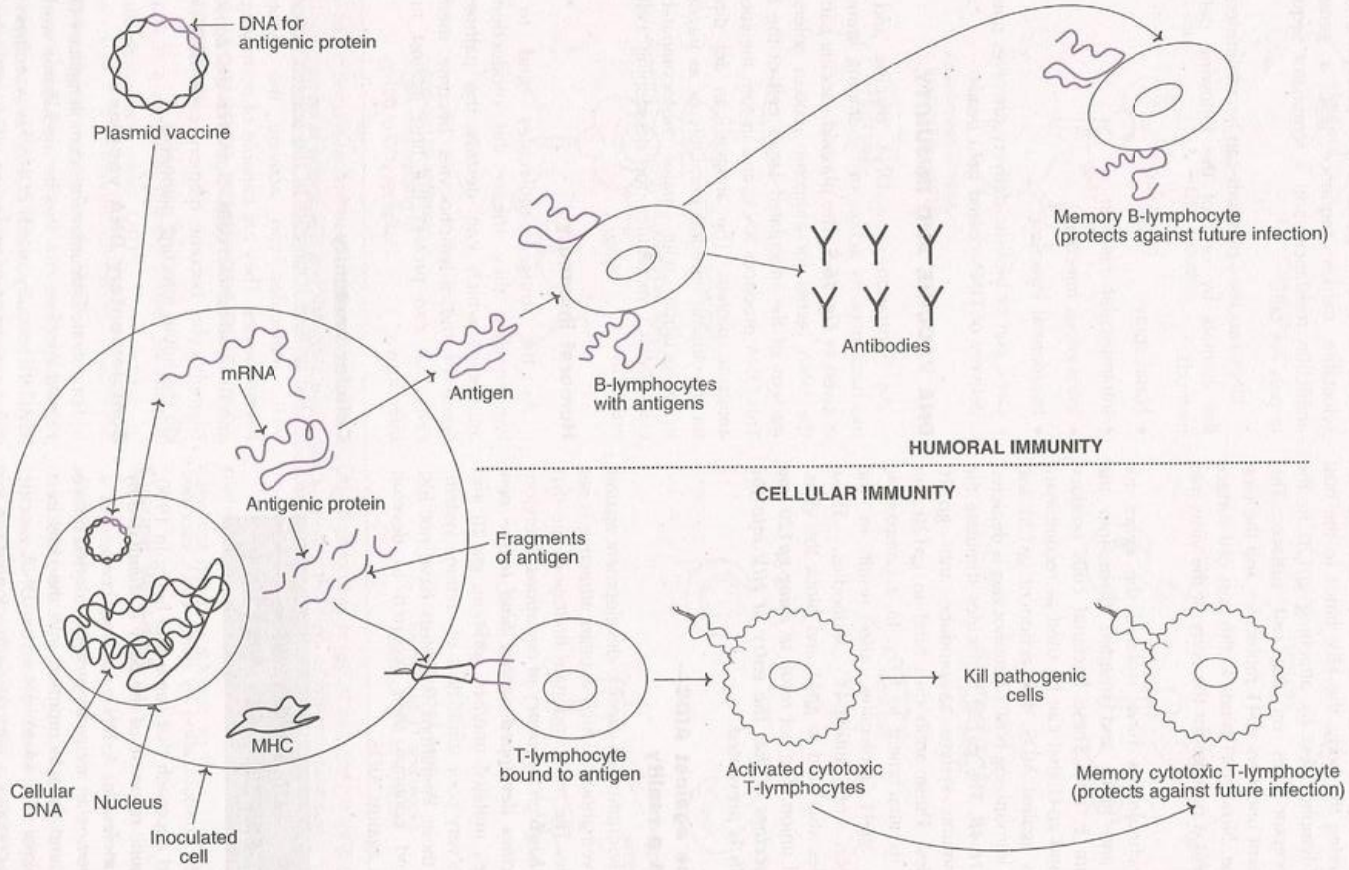


Fig. 16.5 : DNA vaccine and mechanism of its action in developing immunity (MHC–Major histocompatability complex molecule)

DNA vaccines for production of antigens and antibodies

A novel approach for the production of antigens as well as antibodies by DNA vaccine was developed in 1997. In the experiments conducted in mice, researchers ***injected plasmids containing the genes for malarial parasite and also the genes for the antibodies against malarial parasite.*** The B-lymphocytes of these mice performed a double duty, and produced antigens for, and antibodies against malarial parasite. The antigens stimulate to produce more and more antibodies. The antibodies so produced react with malarial parasite. The generation of antigens and antibodies by using a DNA vaccine is a recent development in immunology, and is referred to as ***antigenic antibody approach of DNA vaccine.***

Screening of pathogenic genome for selecting DNA vaccines

The ultimate goal of scientists is to choose the right DNA fragment from the pathogen to serve as a vaccine for the strongest immune response against the invading pathogen. For this purpose, the ***pathogen's DNA can be broken into fragments and a large number of vaccine DNA—plasmids can be prepared.*** The immune response for each one of the DNA vaccines can be studied by injecting the pathogen. By screening the DNA fragments of the pathogenic genome, it is possible to choose one or few DNA vaccines that can offer maximal immune protection.

Advantages of DNA vaccines

There are several advantages of using DNA vaccines in immunization.

1. The tedious and costly procedures of purifying antigens or creating recombinant vaccines are not necessary.

2. DNA vaccines are very specific in producing the target proteins (antigens or antibodies). Thus, they trigger immune response only against the specific pathogen.

3. In general, DNA vaccines elicit much higher immune response compared to other kinds of vaccines.

4. DNA vaccines are more stable for temperature variations (low or high) than the conventional vaccines. Thus, the storage and transport problems associated with vaccines are minimal.

5. The delivery methods to the host are simpler for DNA vaccines.

Disadvantages of DNA vaccines

1. The fate of the DNA vaccine in the host cells is not yet clear. There is a possibility of this DNA getting integrated into the host genome and this may interrupt the normal functions.

2. There also exists a danger of cancer due to DNA vaccines.

3. The post-translational modification of the gene (DNA vaccine) product in host cells may not be the same as that found in the native antigen.

Present status of DNA vaccines

Since 1990, several groups of workers world-over have been trying to develop DNA vaccines against various diseases in experimental animals. Genetic immunization has been done against a number of pathogenic organisms. These include influenza A virus, rabies virus, hepatitis B virus, bovine herpes virus, HIV type I, and *Plasmodium* species (malarial parasite).

It must be noted that ***DNA vaccines have not been tried in humans for*** obvious reasons. The most important being the ***unknown risks*** of these foreign DNAs in human subjects.

RNA VACCINES

Several workers are trying to use RNA molecules as vaccines. These RNAs can readily synthesize the antigenic proteins and offer immunity. But unfortunately, ***RNAs are less stable than DNAs.*** This poses a big problem for RNA vaccine manufacture and distribution. Therefore, the ***progress*** in the development of RNA vaccines has been ***rather slow*** compared to DNA vaccines.

PLANTS AS EDIBLE SUBUNIT VACCINES

Plants serve as a cheap and safe production systems for subunit vaccines. The edible vaccines can be easily ingested by eating plants. This eliminates the processing and purification procedures that are otherwise needed. ***Transgenic plants*** (tomato, potato) have been ***developed for***

Table 16.2 A selected list of plant edible subunit vaccines

Antigen	*Host plant*
Rabies glycoprotein	Tomato
Foot and mouth virus (VPI)	*Arabidopsis*
Herpes virus B surface antigen	Tobacco
Cholera toxin B subunit	Potato
Human cytomegalovirus glycoprotein B	Tobacco

expressing antigens derived from animal viruses (rabies virus, herpes virus). A selected list of recombinant vaccines against animal viruses produced in plants is given in ***Table 16.2***.

Edible vaccine production and use

The production of vaccine potatoes is illustrated in ***Fig. 16.6***. The bacterium, *Agrobacterium tumefaciens* is commonly used to deliver the DNA (genetic material) for bacterial or viral antigens. A plasmid carrying the antigen gene and an antibiotic resistance gene are incorporated into the bacterial cells (*A. tumefaciens*). The cut pieces of potato leaves are exposed to an antibiotic which can kill the cells that lack the new genes. The surviving cells (i.e., gene altered ones) can multiply and form a callus (clump of cells). This callus is allowed to sprout shoots and roots, which are grown in soil to form plants. In about three weeks, the plants bear potatoes with antigen vaccines.

The ***first clinical trials*** in humans, using a plant-derived vaccine were conducted in 1997. This involved the ingestion ***of transgenic potatoes with a toxin of E. coli causing diarrhea***. Some success was reported. For some more details on edible vaccines, the reader must refer Chapter 51.

ATTENUATED RECOMBINANT VACCINES

In the early years of vaccine research, attenuated strains of some pathogenic organisms were prepared by prolonged cultivation — weeks, months or even years. Although the reasons are not known, the infectious organism would lose its ability to cause disease but retains its capability to act as an immunizing agent. This type of approach is almost outdated now.

It is now possible to ***genetically engineer the organisms (bacteria or viruses) and use them as live vaccines***, and such vaccines are referred to as attenuated recombinant vaccines. The genetic manipulations for the production of these vaccines are broadly of two types

1. Deletion or modification of virulence genes of pathogenic organisms.

2. Genetic manipulation of non-pathogenic organisms to carry and express antigen determinants from pathogenic organisms.

The advantage with attenuated vaccines is that the native conformation of the immunogenic determinants is preserved, hence the immune response is substantially high. This is in contrast to purified antigens which often elicit poor immunological response.

Some of the important attenuated vaccines developed by genetic manipulations are briefly described.

CHOLERA

Cholera is an intestinal disease characterized by diarrhea, dehydration, abdominal pain and fever. It is caused by the bacterium, *Vibrio cholerae*. This pathogenic organism is transmitted by drinking water contaminated with fecal matter. Cholera epidemics are frequently seen in developing countries where the water purification and sewage disposal systems are not well developed.

On entering the small intestine, *V. cholerae* colonizes and starts producing large amounts of a ***toxic protein***, a ***hexameric enterotoxin***. This enterotoxin stimulates the cells lining intestinal walls to release sodium, bicarbonate and other ions. Water accompanies these ions leading to severe diarrhea, dehydration, and even death.

The currently used cholera vaccine is composed of phenol-killed *V. cholerae*. The immuno-protection, lasting for 3–6 months is just moderate. Attempts are being made to develop better vaccines.

The DNA technologists have identified the gene encoding enterotoxin (toxic protein). Enterotoxin, an hexamer, consists of one A subunit and five identical B subunits. The A subunit has two functional domains-the A_1 peptide which possesses

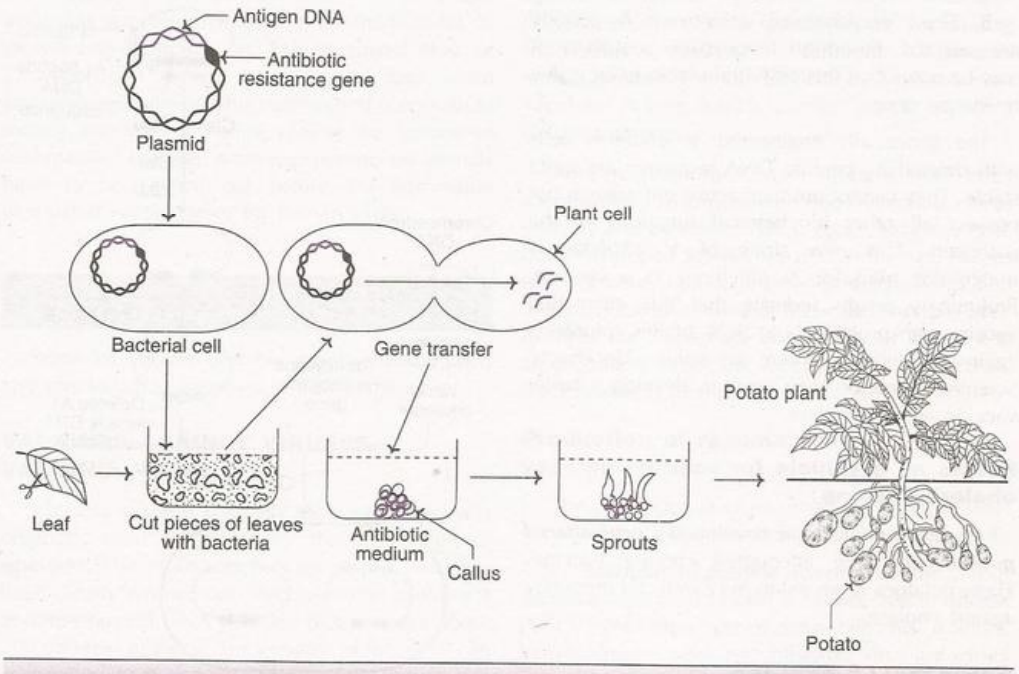


Fig. 16.6 : An illustration for edible vaccine production.

the toxic activity and A_2 peptide that joins A subunit to B subunits. By genetic engineering, it was possible to delete the DNA sequence encoding A_1 peptide and create a new strain of *V. cholerae*. This strain is non-pathogenic, since it cannot produce enterotoxin. The ***genetically engineered V. cholerae is a good candidate to serve as an attenuated vaccine.***

Creating a new strain of *V. cholerae*

The development of a new strain of *Vibrio cholerae* that can effectively serve as an attenuated recombinant vaccine is depicted in ***Fig. 16.7***, and briefly described below.

1. A tetracycline resistance gene was inserted into the A_1 peptide sequence of *V. cholerae* chromosome. This destroys the DNA sequence encoding for A_1 peptide, besides making the strain resistant to tetracycline. Unfortunately, the tetracycline resistant gene is easily lost and the enterotoxin activity is restored. Because of this, the new strain of *V. cholerae* as such cannot be used as a vaccine.

2. The DNA sequence of A_1 peptide is incorporated into a plasmid, cloned and digested with restriction enzymes (*ClaI and XbaI*). In this manner, the A_1 peptide coding sequence is deleted (the DNA encoding for 183 of the 194 amino acids of the A_1 peptide is actually removed). By using T_4 DNA ligase, the plasmid is recircularized. This plasmid contains a small portion of A_1 peptide coding sequence.

3. The plasmid, containing the deleted A_1 peptide sequence is transferred by conjugation into the *V. cholerae* strain carrying a tetracycline resistance gene.

4. Recombination can occur between the plasmid (containing a small portion of peptide A_1 coding sequence) and the chromosome of *V. cholerae* (carrying tetracycline resistance gene). The result of this double crossover is the formation of ***V. cholerae containing a chromosomal DNA lacking A_1 peptide DNA sequence***. As the bacterium, *V. cholerae* multiplies, the plasmids are lost in the next few generations.

5. The *V. cholerae* cells defective in A_1 peptide are selected, based on tetracycline sensitivity. It may be noted that this new strain lacks tetracycline resistance gene.

The genetically engineered *V. cholerae* cells with deleted A_1 peptide DNA sequence are quite stable. They cannot produce active enterotoxin but possess all other biochemical functions of the pathogen. This new strain of *V. cholerae* is undergoing trials for its efficiency as a vaccine. Preliminary results indicate that this attenuated vaccine can protect about 90% of the volunteers against cholera. But there are some side effects. Scientists continue their work to develop a better vaccine against cholera.

Potato as a vehicle for cholera vaccine

A group workers have developed a ***gene altered potato*** containing attenuated cholera vaccine. These potatoes when fed to mice induced immunity against cholera.

SALMONELLA SPECIES

The different strains of Salmonella genus are responsible for causing ***typhoid, enteric fever, food poisoning and infant death***. Immunoprotection against *Salmonella* pathogens is really required.

Some workers have been successful in deleting ***aro genes*** and ***pur genes*** in *Salmonella*. *Aro* genes encode for the enzymes responsible for the biosynthesis of aromatic compounds, while *pur* genes encode for enzymes of purine metabolism. The new strains of *Salmonella* can be grown *in vitro* on a complete medium. The ***doubly deleted strains*** have a very restricted growth *in vivo*, while they ***can stimulate immunological response***. The genetically engineered attenuated vaccines of *Salmonella* have been shown to be effective as oral vaccines in experimental animals (mice, cattle, sheep, chickens). Some workers claim that the new strain of *Salmonella* offers immunoprotection in humans also.

LEISHMANIA SPECIES

Leishmania species are flagellated protozoan parasites and are responsible for the disease ***leishmaniasis***. This disease is characterized by ***cutaneous***, ***visceral*** and ***mucosal leisons***. Leishmaniasis is transmitted by sand flies.

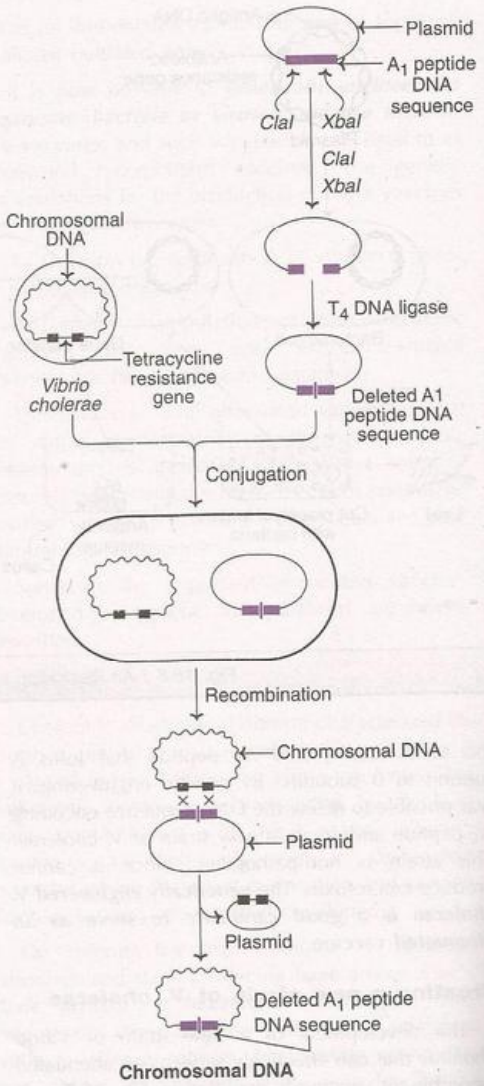


Fig. 16.7 : *Development of a new strain of* V. cholerae *as an attenuated recombinant vaccine (*ClaI *and* XbaI *are restriction endonucleases).*

An ***attenuated strain of leishmania*** has been created and successfully ***used in mice to offer immunoprotection*** against leishmaniasis. In *Leishmania major*, the genes encoding dihydrofolate reductase-thymidylate synthase can be replaced by the genes encoding resistance to antibiotics G–418

and hygromycin. This new strain of *L. major* invariably requires thymidine in the medium for its growth and multiplication. The attenuated strain of *L. major* can survive only a few days when administered to mice. This short period is enough to induce immunity in mice against the lesions of leishmania. However, more experiments on animals have to be carried out before the leishmania attenuated vaccine goes for human trials.

VECTOR RECOMBINANT VACCINES

Some of vectors can be genetically modified and employed as vaccines against pathogens.

VACCINES AGAINST VIRUSES— VACCINIA VIRUS

Vaccinia viruse is basically the vaccine that was originally used by Jenner for the eradication of ***smallpox***. The molecular biology of this virus has been clearly worked out. Vaccinia virus contains a double-stranded DNA (187 kb) that encodes about 200 different proteins. The genome of this virus can accommodate stretches of foreign DNA which can be expressed along with the viral genes.

The vaccinia virus can replicate in the host cell cytoplasm (of the infected cells) rather than the nucleus. This is possible since the vaccinia virus possesses the machinery for DNA replication, transcription-DNA polymerase, RNA polymerase etc. The foreign genes inserted into the vaccinia virus can also be expressed along with the viral genome. Thus, the foreign DNA is under the control of the virus, and is expressed independently from the host cell genome.

The vaccinia viruses are generally harmless, relatively easy to cultivate and stable for years after lyophilization (freeze-drying). All these features make the vaccinia virues strong candidates for vector vaccine. The cloned foreign genes (from a pathogenic organism) can be inserted into vaccinia virus genome for encoding antigens which in turn produces antibodies against the specific disease-causing agent. The advantage with ***vector vaccine***, is that it ***stimulates B-lymphocytes*** (to produce antibodies) ***and T-lymphocytes*** (to kill virus infected cells). This is in contrast to a subunit vaccine which can stimulate only B-lymphocytes. Thus, vaccinia virus can provide a high level of immunoprotection against pathogenic organisms. Another advantage of vaccinia virus is the ***possibility of vaccinating individuals against different diseases simultaneously***. This can be done by a recombinant vaccinia viruses which carries genes encoding different antigens.

Antigen genes for certain diseases have been successfully incorporated into vaccinia virus genome and expressed. Thus, vector vaccines have been developed against hepatitis, influenza, herpes simplex virus, rabies, angular stomatitis virus and malaria. However, none of ***these vaccines has been licensed for human use due to fear of safety***. It is argued that recombinant vaccinia virus might create life threatening complications in humans.

Production of recombinant vaccinia viruses

The development of recombinant vaccinia virus is carried out by a two-step procedure (***Fig. 16.8***).

1. **Assembly of plasmid insertion vector :** Fresh vaccinia (cow pox) viruses are processed to release their DNAs. Now genes from hepatitis B virus, herpes simplex virus and influenza virus are added one after another and inserted into vaccinia virus genome. These DNA clusters are cloned in *E. coli* for increasing their number and to produce plasmid insertion vectors. The plasmid contains the foreign subunit genes, the natural vaccinia genes, including the promoter genes. The recombinant plasmids are isolated and purified and serve as plasmid insertion vectors.

2. **Production of recombinant vaccinia viruses :** The animal cells are infected with plasmid insertion vectors and normal vaccinia viruses. As the viral replication occurs, the plasmids are taken up to produce recombinant vaccinia viruses. The plasmid insertion vector incorporates its genes into vaccinia virus genome at a place that encodes for the enzyme thymidine kinase (TK). Thus the recombinant viruses have lost their ability to produce TK. There are two advantages of loss of TK gene. One is that it is easy to select recombined vaccinia viruses that lack TK gene and the second is that these viruses are less infectious than the normal viruses.

The recombinant vaccinia viruses, released from the cultured animal cells, can be successfully used as vaccines. These live viral vaccines have some advantages over the killed or subunit vaccines.

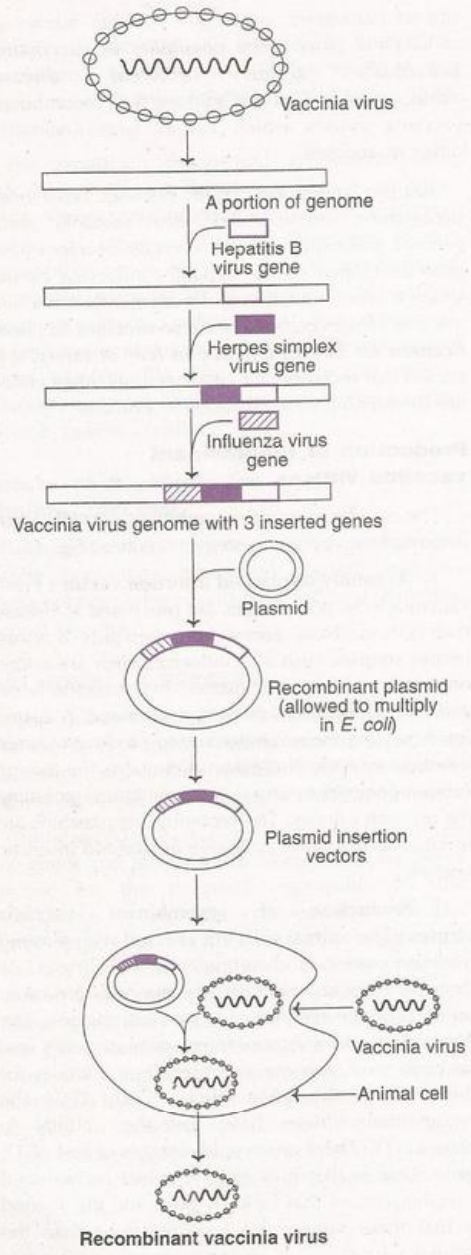


Fig. 16.8 : Production of recombinant vaccinia virus.

Advantages

1. Authenticated antigens that closely resemble natural antigens can be produced.

2. The virus can replicate in the host cells. This enables the amplification of the antigens for their action on B-lymphocytes and T-lymphocytes.

3. There is a possibility of vaccinating several diseases with one recombinant vaccinia virus.

Disadvantages

1. The most important limitation is the yet unknown risks of using these vaccines in humans.

2. There may be serious complications of using recombinant viral vaccines in immunosuppressed individuals such as AIDS patients.

Other viral recombinant vaccines

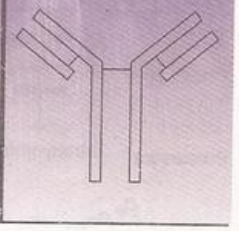
Most of the work on the development of live viral vaccines has been carried out on vaccinia virus. Other viruses such as ***adenovirus, poliovirus and varicella-zoster virus are also being tried*** as recombinant vaccines. Scientists are attracted to develop a recombinant poliovirus as it can be orally administered. It might take many more years for the recombinant viral vaccines to become a reality for human use.

DELIVERY OF ANTIGENS BY BACTERIA

It is known that the antigens located on the surface of a bacterial cell are more immunogenic than the antigens in the cytoplasm. Based on this observation, scientists have developed strategies to coat the surfaces of non-pathogenic organisms with antigens of pathogenic bacteria.

Flagellin is a protein present in the fragella (thread like filaments) of *Salmonella*. A synthetic oligonucleotide encoding the epitope of cholera toxin B subunit was inserted into *Salmonella* flagellin gene. This epitope was in fact found on the flagellum surface. These flagella-engineered bacteria, when administered to mice, raised antibodies against the cholera toxin B subunit peptide. It may be possible in future to incorporate multiple epitopes (2 or 3) into the flagellin gene to create multivalent bacterial vaccines.

17 Monoclonal Antibodies (Hybridoma Technology)



Antibodies or ***immunoglobulins*** are protein molecules produced by a specialized group of cells called ***B-lymphocytes*** (plasma cells) in mammals. The structures, characteristics and various other aspects of immunoglobulins (Igs) are described elsewhere (Refer Chapter 68). Antibodies are a part of the defense system to protect the body against the invading foreign substances namely ***antigens***. Each antigen has ***specific antigen determinants (epitopes)*** located on it. The antibodies have ***complementary determining regions (CDRs)*** which are mainly responsible for the antibody specificity.

In response to an antigen (with several different epitopes), B-lymphocytes gear up and produce many different antibodies. This type of antibodies which can react with the same antigen are designated as ***polyclonal antibodies***. The polyclonal antibody production is variable and is dependent on factors such as epitopes, response to immunity etc. Due to lack of specificity and heterogenic nature, there are several limitations on the utility of polyclonal antibodies for therapeutic and diagnostic purposes.

Monoclonal antibody (MAb) is a single type of antibody that is directed against a specific antigenic determinant (epitope). It was a dream of scientists to produce MAbs for different antigens. In the early years, animals were immunized against a specific antigen, B-lymphocytes were isolated and cultured *in vitro* for producing MAbs. This approach was not successful since culturing normal B-lymphocytes is difficult, and the synthesis of MAb was short-lived and very limited.

It is interesting that immortal monoclonal antibody producing cells do exist in nature. They are found in the patients suffering from a disease called ***multiple myeloma*** (a cancer of B-lymphocytes). It was in 1975. George Kohler and Cesar Milstein (Nobel Prize, 1984) achieved large scale production of MAbs. They could successfully ***hybridize*** antibody—producing ***B-lymphocytes with myeloma cells in vitro*** and create a ***hybridoma***. The result is that the artificially immortalized B-lymphocytes can multiply indefinitely *in vitro* and produce MAbs. The hybridoma cells possess the growth and multiplying properties of myeloma cells but secrete antibody of B-lymphocytes. The ***production of monoclonal antibodies by the hybrid cells is referred to as hybridoma technology***.

PRINCIPLE FOR CREATION OF HYBRIDOMA CELLS

The myeloma cells used in hybridoma technology must not be capable of synthesizing their own antibodies. The selection of hybridoma cells is based on inhibiting the nucleotide (consequently the DNA) synthesizing machinery. The mammalian cells can synthesize nucleotides by two pathways—*de novo* synthesis and salvage pathway (***Fig. 17.1***).

The ***de novo synthesis*** of nucleotides requires tetrahydrofolate which is formed from dihydro-

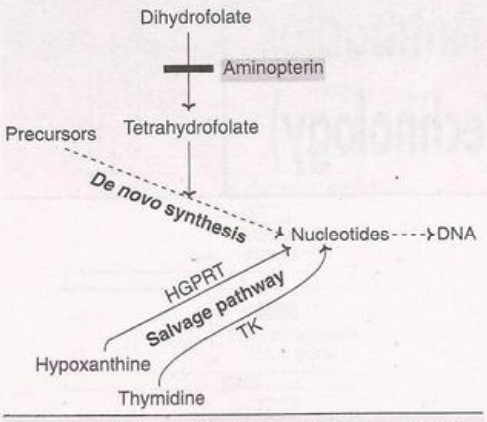


Fig. 17.1 : Pathways for the synthesis of nucleotides (HGPRT–Hypoxanthine guanine phosphoribosyl transferase; TK–Thymidine kinase)

folate. The formation of tetrahydrofolate (and therefore nucleotides) can be ***blocked by the inhibitor aminopterin***.

The ***salvage pathway*** involves the direct conversion of purines and pyrimidines into the corresponding nucleotides. Hypoxanthine guanine phosphoribosyl transferase (HGPRT) is a key enzyme in the salvage pathway of purines. It converts hypoxanthine and guanine respectively to inosine monophosphate and guanosine monophosphate. Thymidine kinase (TK), involved in the salvage pathway of pyrimidines converts thymidine to thymidine monophosphate (TMP). Any mutation in either one of the enzymes (HGPRT or TK) blocks the salvage pathway.

When cells deficient (mutated cells) in HGPRT are grown in a medium containing ***h***ypoxanthine ***a***minopterin and ***t***hymidine (***HAT*** medium), they cannot survive due to inhibition of *de novo* synthesis of purine nucleotides (**Note :** Salvage pathway is not operative due to lack of HGPRT). Thus, cells lacking HGPRT, grown in HAT medium die.

The hybridoma cells possess the ability of myeloma cells to grow *in vitro* with a functional HGPRT gene obtained from lymphocytes (with which myeloma cells are fused). Thus, only the hybridoma cells can proliferate in HAT medium, and this procedure is successfully used for their selection.

PRODUCTION OF MONOCLONAL ANTIBODIES

The establishment of hybridomas and production of MAbs involves the following steps (***Fig. 17.2***).

1. Immunization
2. Cell fusion
3. Selection of hybridomas
4. Screening the products
5. Cloning and propagation
6. Characterization and storage.

1. **Immunization :** The very first step in hybridoma technology is to immunize an animal (usually a mouse), with appropriate antigen. The antigen, along with an adjuvant like ***Freund's complete or incomplete adjuvant*** is injected subcutaneously (adjuvants are non-specific potentiators of specific immune responses). The injections at multiple sites are repeated several times. This enables increased stimulation of B-lymphocytes which are responding to the antigen. Three days prior to killing of the animal, a final dose of antigen is intravenously administered. The immune-stimulated cells for synthesis of antibodies have grown maximally by this approach. The concentration of the desired antibodies is assayed in the serum of the animal at frequent intervals during the course of immunization.

When the serum concentration of the antibodies is optimal, the animal is sacrificed. The spleen is aseptically removed and disrupted by mechanical or enzymatic methods to release the cells. The lymphocytes of the spleen are separated from the rest of the cells by density gradient centrifugation.

2. **Cell fusion :** The thoroughly washed lymphocytes are mixed with HGPRT defective myeloma cells. The mixture of cells is exposed to polyethylene glycol (PEG) for a short period (a few minutes), since it is toxic. PEG is removed by washing and the cells are kept in a fresh medium. These cells are composed of a mixture of hybridomas (fused cells), free myeloma cells and free lymphocytes.

3. **Selection of hybridomas :** When the cells are cultured in HAT medium (the principle described above), only the hybridoma cells grow, while the rest will slowly disappear. This happens in 7–10 days of culture.

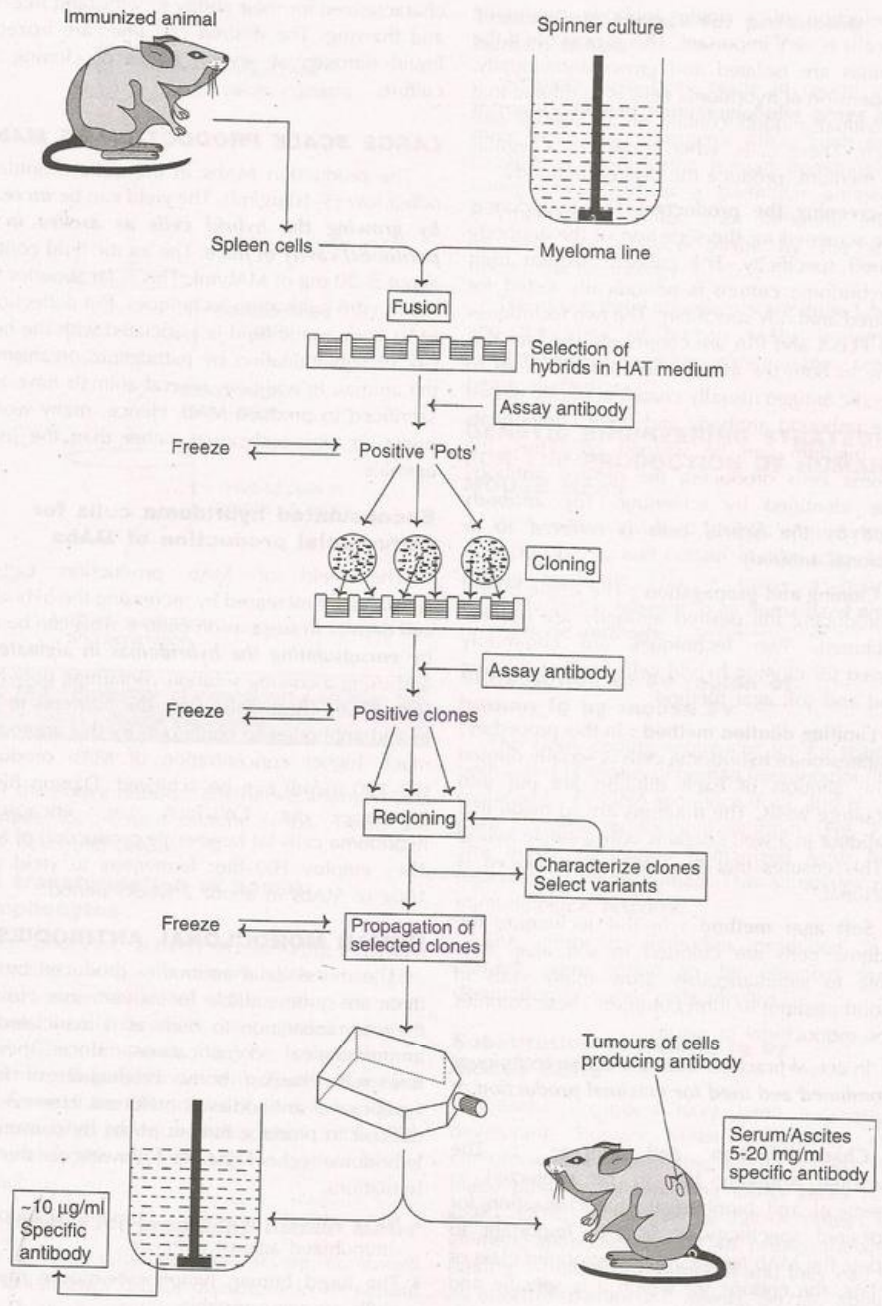


Fig. 17.2 : Basic protocol for the derivation of monoclonal antibodies from hybrid myelomas.

Selection of a single antibody producing hybrid cells is very important. This is possible if the hybridomas are isolated and grown individually. The suspension of hybridoma cells is so diluted that the individual aliquots contain on an average one cell each. These cells, when grown in a regular culture medium, produce the desired antibody.

4. **Screening the products :** The hybridomas must be screened for the secretion of the antibody of desired specificity. The culture medium from each hybridoma culture is periodically tested for the desired antibody specificity. The two techniques namely ELISA and RIA are commonly used for this purpose. In both the assays, the antibody binds to the specific antigen (usually coated to plastic plates) and the unbound antibody and other components of the medium can be washed off. Thus, the hybridoma cells producing the desired antibody can be identified by screening. ***The antibody secreted by the hybrid cells is referred to as monoclonal antibody.***

5. **Cloning and propagation :** The single hybrid cells producing the desired antibody are isolated and cloned. Two techniques are commonly employed for cloning hybrid cells-limiting dilution method and soft agar method.

Limiting dilution method : In this procedure, the suspension of hybridoma cells is serially diluted and the aliquots of each dilution are put into microculture wells. The dilutions are so made that each aliquot in a well contains only a single hybrid cell. This ensures that the antibody produced is monoclonal.

Soft agar method : In this technique, the hybridoma cells are cultured in soft agar. It is possible to simultaneously grow many cells in semisolid medium to form colonies. These colonies will be monoclonal in nature.

In actual practice, ***both the above techniques are combined and used for maximal production of MAbs.***

6. **Characterization and storage :** The monoclonal antibody has to be subjected to biochemical and biophysical characterization for the desired specificity. It is also important to elucidate the MAb for the immunoglobulin class or sub-class, the epitope for which it is specific and the number of binding sites it possesses.

The stability of the cell lines and the MAbs are important. The cells (and MAbs) must be characterized for their ability to withstand freezing, and thawing. The desired cell lines are frozen in liquid nitrogen at several stages of cloning and culture.

LARGE SCALE PRODUCTION OF MAbs

The production MAbs in the culture bottles is rather low (5–10 µg/ml). The yield can be ***increased by growing the hybrid cells as ascites in the peritoneal cavity of mice.*** The ascitic fluid contains about 5–20 mg of MAb/ml. This is far superior than the *in vitro* cultivation techniques. But collection of MAb from ascitic fluid is associated with the heavy risk of contamination by pathogenic organisms of the animal. In addition, several animals have to be sacrificed to produce MAb. Hence, many workers prefer *in vitro* techniques rather than the use of animals.

Encapsulated hybridoma cells for commercial production of MAbs

The yield of MAb production can be substantially increased by increasing the hybridoma cell density in suspension culture. This can be done by ***encapsulating the hybridomas in alginate gels*** and using a coating solution containing poly-lysine (***Fig. 17.3***). These gels allow the nutrients to enter in and antibodies to come out. By this approach, a much higher concentration of MAb production (10–100 µg/ml) can be achieved. Damon Biotech company and Cell-Tech use encapsulated hybridoma cells for large-scale production of MAbs. They employ 100–liter fermenters to yield about 100g of MAbs in about 2 weeks period.

HUMAN MONOCLONAL ANTIBODIES

The monoclonal antibodies produced by using mice are quite suitable for *in vitro* use. However, their administration to humans is associated with immunological complications, since they are foreign to human body. Production of human monoclonal antibodies is preferred. However, it is difficult to produce human MAbs by conventional hybridoma technology. The following are the major limitations.

- For ethical reasons, humans cannot be immunized against antigens.
- The fused human lymphocyte-mouse myeloma cells are very unstable.
- There are no suitable myeloma cells in humans that can replace mouse myeloma cells.

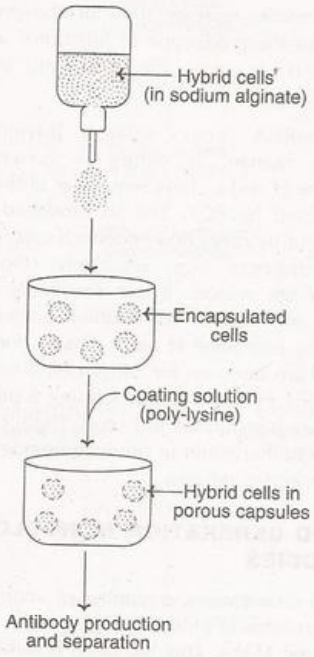


Fig. 17.3 : Production of monoclonal antibodies by microencapsulation.

For the above reasons, alternative arrangements are made to produce human MAbs. These are briefly described below.

Viral transformation of human B-lymphocytes

B-Lymphocytes, actively synthesizing antibody, are treated with fluorescent-labeled antigen. The fluorescent-activated cells are separated. However, B-cells on their own, cannot grow in culture. This limitation can be overcome by transforming B-lymphocytes with ***Epstein-Bar virus*** (EBV). Some of the EBV-transformed cells can grow in culture and produce monoclonal antibodies. Unfortunately, the yield of MAb is very low by this approach.

SCID mouse for producing human MAbs

The mouse suffering from severe combined immunodeficiency (SCID) disease lacks its natural immunological system. Such mouse can be challenged with appropriate antigens to produce human MAbs (For details, Refer Chapter 17).

Transgenic mouse for producing human MAbs

Attempts have been made in recent years to ***introduce human immunoglobulin genes into the mice*** to develop transgenic mice. Such mice are capable of synthesizing human immunoglobulins when immunized to a particular antigen. The B-lymphocytes isolated from transgenic mice can be used to produce MAbs by the standard hybridoma technology.

The above three approaches are quite laborious, and the yield of human MAbs is very low. Consequently, researchers continue their search for better alternatives.

GENETIC ENGINEERING STRATEGIES FOR THE PRODUCTION OF HUMAN-MOUSE MAbs

With the advances in genetic engineering, it is now possible to add certain human segments to a mouse antibody. This is truely a hybridized antibody and is referred to as ***humanized antibody*** or ***chimeric antibody***.

Substitution of Fv region of human Ig by mouse Fv

The DNA coding sequences for Fv regions of both L and H chains of human immunoglobulin are replaced by Fv DNA sequence (for L and H chains) from a mouse monoclonal antibody (***Fig. 17.4A***). The newly developed humanized MAb has Fc region of Ig being human. This stimulates proper immunological response.

The chimeric antibodies produced in this manner were found to be effective for the destruction of tumor cells *in vitro*.

Substitution of human Ig by mouse CDRs

Genetic engineers have been successful in developing human MAbs containing mouse complementary determining regions (CDRs). This is made possible by replacing CDRs genes (CDR_1, CDR_2, CDR_3) of humans by that of mouse. These chimeric antibodies (***Fig. 17.4B***) possess the antigen binding affinities of the mouse and they can serve as effective therapeutic agents. So far, about 50 monoclonal antibodies have been produced by this approach. However, this technique is costly and time consuming.

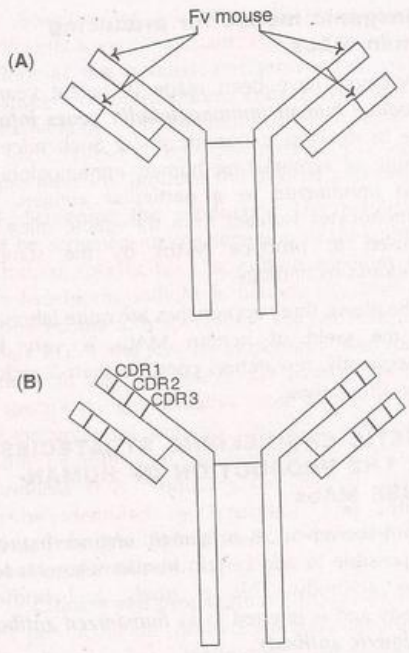


*Fig. 17.4 : Genetically engineered human-mouse antibodies (**A**) Substitution of Fv region of human Ig by mouse Fv (**B**) Substitution of human Ig by mouse CDRs (CDR-Complementary determining regions).*

Bispecific monoclonal antibodies

The MAbs in which the two arms of Fab (antigen-binding) have two different specificities for two different epitopes are referred to as bispecific MAbs. They may be produced by fusing two different hybridoma cell lines (***Fig. 17.5***) or by genetic engineering. Bispecific Fab MAbs theoretically, are useful for a simultaneous and combined treatment of two different diseases.

PRODUCTION OF MAbs IN *E. COLI*

The hybridoma technology is very laborious, expensive and time consuming. To overcome these limitation, researchers have been trying to genetically engineer bacteria, plants and animals. The objective is to develop ***bioreactors*** for the large scale production of monoclonal antibodies. It may be noted that the antigen binding regions of antibody (Fv or Fab fragments) are very crucial, while the Fc portion is dispensable.

A schematic representation of the procedure adopted for the production of functional antibody fragments is shown in ***Fig. 17.6***, and is briefly described.

The mRNA from isolated B-lymphocytes of either human or mouse is converted to cDNA. The H and L chain sequences of this cDNA are amplified by PCR. The so produced cDNAs are then cut by restriction endonucleases. H and L chain sequences are separately cloned in bacteriophage vectors. These sequences are put together and cloned in another bacteriophage vector. The combined H and L chains (forming Fv fragment) are screened for antigen binding activity. The specific H and L chains forming a part of the plasmid are transformed in *E. coli*. These *E. coli*, in turn, can be harvested to produce Fv fragments to bind to specific antigens.

SECOND GENERATION MONOCLONAL ANTIBODIES

In the recent years, a number of improvements have been made to produce more specific, sensitive and desired MAbs. This has been possible due to the rapid advances made in genetic engineering techniques. For instances, by employing ***site-directed mutagenesis***, it is possible to introduce cysteine residues at the predetermined positions on the MAb. These cysteine residues which facilitate the isotope labeling may be more useful in diagnostic imaging and radioimmunotherapy.

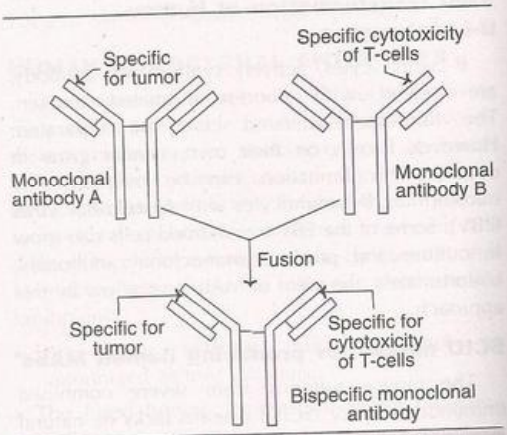


Fig. 17.5 : Production of a bispecific monoclonal antibody.

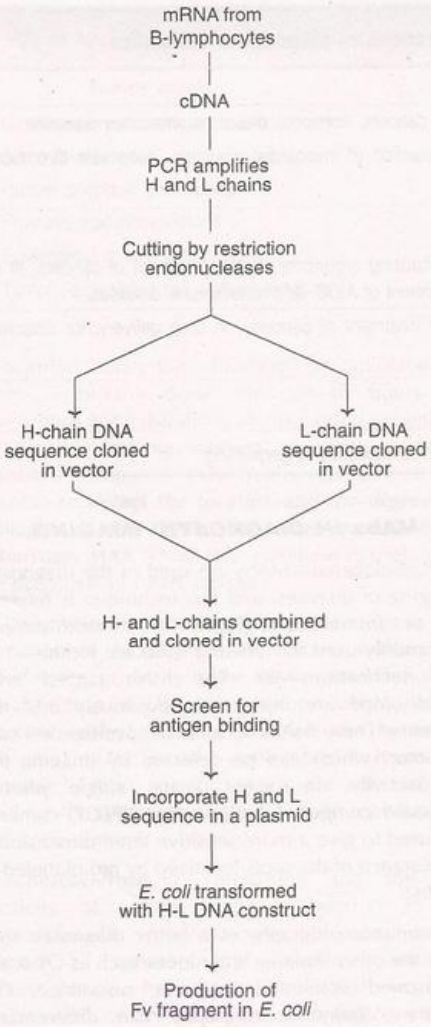


Fig. 17.6 : Production of monoclonal antibodies in E. coli.

ADVANTAGES OF MONOCLONAL ANTIBODIES

Monoclonal antibodies truely ***represent a homogeneous state of a single molecular species.*** Each MAb is specific to a given antigenic determinant. This is in contrast to the conventional antiserum that contains polyclonal antibodies. The wide range of applications of MAbs are described later.

LIMITATIONS OF MONOCLONAL ANTIBODIES

Hybridoma technology is laborious and time consuming. MAbs are produced against a single antigenic determinant, therefore, they cannot differentiate the molecule as a whole. Sometimes, they may be incapable of distinguishing groups of different molecules also.

The presence of retroviruses as a part of the mammalian chromosomes is a common occurrence. Mice used in MAb production carry several viruses (adenovirus, hepatic virus, retrovirus, reovirus, cytomegalovirus, thymic virus). The presence of some of these viruses has been detected in the hybridomas. This poses a great danger, since there is no guarantee that MAb produced is totally virus-free, despite the purification. For this reason, US Food and Drug Administration insists that ***MAb for human use should be totally free from all pathogenic organisms, including viruses.***

APPLICATIONS OF MONOCLONAL ANTIBODIES

Monoclonal antibodies with specificity and high purity have a wide range of applications, which can be broadly categorized as follows.

1. Diagnostic applications
2. Therapeutic uses
3. Protein purification
4. Miscellaneous applications.

A summary of the important applications of MAbs is given in ***Table 17.1***, and briefly discussed below.

1. DIAGNOSTIC APPLICATIONS

Monoclonal antibodies have revolutionized the laboratory diagnosis of various diseases. For this purpose, MAbs may be employed as diagnostic reagents for ***biochemical analysis*** or as tools for ***diagnostic imaging of diseases.***

(A) *MAbs IN BIOCHEMICAL ANALYSIS*

Diagnostic tests based on the use of MAbs as reagents are routinely used in radioimmunoassays

TABLE 17.1 A summary of important applications of monoclonal antibodies

1. **Diagnostic applications**
   (A) ***Biochemical analysis*** for the diagnosis of pregnancy, cancers, hormonal disorders, infectious diseases.
   (B) ***Diagnositic imaging (immunoscintigraphy)*** for the detection of myocardial infarction, deep vein thrombosis, atherosclerosis, cancers, bacterial infections.
2. **Therapeutic applications**
   (A) ***Direct use as therapeutic agents*** to destroy disease-causing organisms, in the treatment of cancers, in the immunosuppression of organ transplantation, in the treatment of AIDS, and autoimmune diseases.
   (B) ***As targeting agents in therapy*** as immunotoxins (for treatment of cancers), in drug delivery, for dissolving blood clots, in radioimmunotherapy (of tumors).
3. **Protein purification** by immunoaffinity techniques.
4. **Miscellaneous applications** as catalytic agents (abzymes), in autoantibody fingerprinting.

(RIA) and enzyme-linked immunosorbent assays (ELISA) in the laboratory. These assays measure the circulating concentrations of hormones (insulin, human chorionic gonadotropin, growth hormone, progesterone, thyroxine, triiodothyronine, thyroid stimulating hormone, gastrin, renin), and several other tissue and cell products (blood group antigens, blood clotting factors, interferons, interleukins, histocompatibility antigens, tumor markers).

In recent years, a number of diagnostic kits using MAbs have become commercially available. For instance, it is now possible to do the early diagnosis of the following conditions/diseases.

Pregnancy—by detecting the urinary levels of human chorionic gonadotropin.

Cancers—estimation of plasma carcino-embryonic antigen in colorectal cancer, and prostate specific antigen for prostate cancer. Besides diagnosis, estimation of tumor markers is also useful for the prognosis of cancers. That is a gradual fall in a specific tumor marker is observed with a reduction in tumor size, following treatment.

Hormonal disorders—analysis of thyroxine, triiodothyronine and thyroid stimulating hormone for thyroid disorders.

Infectious diseases—by detecting the circulatory levels of antigens specific to the infectious agent e.g., antigens of *Neisseria gonorrhoeae* and herpes simplex virus for the diagnosis of sexually transmitted diseases.

(B) *MAbs IN DIAGNOSTIC IMAGING*

Radiolabeled—MAbs are used in the diagnostic imaging of diseases, and this technique is referred to as ***immunoscintigraphy***. The radioisotopes commonly used for labeling MAb are iodine—131 and technetium—99. The MAb tagged with radioisotope are injected intravenously into the patients. These MAbs localise at specific sites (say a tumor) which can be detected by imaging the radioactivity. In recent years, ***single photon emission computed tomography (SPECT)*** cameras are used to give a more sensitive three dimensional appearance of the spots localized by radiolabeled—MAbs.

Immunoscintigraphy is a better diagnostic tool than the other imaging techniques such as CT scan, ultrasound scan and magnetic resonance. For instance, ***immunoscintigraphy can differentiate between cancerous and non-cancerous growth***, since radiolabeled—MAbs are tumor specific. This is not possible with other imaging techniques.

Monoclonal antibodies are successfully used in the diagnostic imaging of cardiovascular diseases, cancers and sites of bacterial infections.

Cardiovascular diseases

Myocardial infarction : The cardiac protein myosin gets exposed wherever myocardial necrosis (death of cardiac cells) occurs. Antimyosin MAb labeled with radioisotope indium chloride (^{111}In) is used for detecting myosin and thus the site of

Table 17.2 Selected tumor markers along with the associated cancers used in MAb imaging

Tumor marker	*Associated cancer(s)*
Carcinoembryonic antigen (CEA)	Cancers of colon, stomach, pancreas
Alpha fetoprotein	Cancers of liver, and germ cells of testes
Human chorionic gonadotropin	Choriocarcinoma
Prostatic acid phosphatase	Prostate cancer
Epiderminal growth factor receptor	Melanoma
Tumor–-associated cell surface antigens	Various cancers

myocardial infarction. Imaging of radiolabeled MAb, is usually done after 24–48 hours of intravenous administration. This is carried out either by planner gamma camera or single photon emission computed tomography (SPECT). It is possible to ***detect the location and the degree of damage to the heart by using radiolabeled antimyosin MAb***. Thus, this technique is useful for the diagnosis of heart attacks.

Deep vein thrombosis (DVT) : DVT refers to the formation of blood clots (thrombus) within the blood veins, primarily in the lower extremities. For the detection of DVT, radioisotope labeled ***MAb directed against fibrin or platelets can be used.*** The imaging is usually done after 4 hours of injection. Fibrin specific MAbs are successfully used for the detection of clots in thigh, pelvis, calf and knee regions.

Atherosclerosis : Thickening and loss of elasticity of arterial walls is referred to as atherosclerosis. Atherosclerotic plaques cause diseases of coronary and peripheral arteries. Atherosclerosis has been implicated in the development of heart diseases. ***MAb tagged with a radiolabel directed against activated platelets can be used to localize the atherosclerotic lesions by imaging technique***.

Cancers

Monoclonal antibodies against many types of human cancers are now available. A selected list of tumor markers (along with the associated cancers) that can be used for MAb imaging is given in ***Table 17.2***. Tumors can be located in patients using radioisotope labeled MAbs specific to the protein(s), particularly of membrane origin. It has been possible to detect certain cancers at early stages (lung cancer, breast cancer, ovariran cancer, malanoma, colorectal cancer) by employing MAbs. About 80 per cent specificity has been achieved for detecting cancers by this approach.

An iodine (^{131}I) labeled monoclonal antibody specific to breast cancer cells when administered to the patients detects (by imaging) the spread of cancer (metastasis) to other regions of the body. This is not possible by scanning techniques.

The imaging technique by using MAb can also be used to monitor therapeutic responses of a cancer. There are certain limitations in using MAb in cancer diagnosis and prognosis. These include the difficulty in the selection of a specific MAb and the access of MAb to the target site of the tumor which may be less vascularized.

MAbs in immunohistopathology of cancers : The pathological changes of the cancerous tissue can be detected by immunohistochemical techniques. This can be done by using MAb against a specific antigen.

MAbs in hematopoietic malignancies : Hematopoietic stem cells in bone marrow are the precursors for different blood cells, B- and T-lymphocytes which are produced in a stepwise transformation. During malignancy, transformation of lymphocytes stops at a particular stage of maturation. This can be detected by using stage-specific MAbs.

Bacterial infections

In recent years, attempts are made to detect the sites of infections by using MAbs. This is made possible by directing MAb against bacterial antigens. Further, monoclonal antibodies against inflammatory leucocytes which accumulate at

infection site are also useful to specifically detect localized infections.

2. THERAPEUTIC APPLICATIONS

Monoclonal antibodies have a wide range of therapeutic applications. MAbs are used in the treatment of cancer, transplantation of bone marrow and organs, autoimmune diseases, cardiovascular diseases and infectious diseases. The therapeutic applications of MAbs are broadly grouped into 2 types.

(A) Direct use of MAbs as therapeutic agents

(B) MAbs as targeting agents.

(A) *MAbs AS DIRECT THERAPEUTIC AGENTS*

Monoclonal antibodies can be directly used for enchancing the immune function of the host. Direct use of MAbs causes minimal toxicity to the target tissues or the host.

In destroying disease-causing organisms

MAbs promote efficient opsonization of pathogenic organisms (by coating with antibody) and enhance phagocytosis. In fact, MAbs were found to protect chimpanzes against certain ***viral*** (hepatitis B-virus) and ***bacterial*** (*E. coli Haemophilus influenza, Streptococcus* sp and *Pseudomonas* sp) infections.

In the treatment of cancer

MAbs, against the antigens on the surface of cancer cells, are useful for the treatment of cancer. The antibodies bind to the cancer cells and destroy them. This is brought out by antibody—dependent cell-mediated cytotoxicity, complement-mediated cytotoxicity and phagocytosis of cancer cells (coated with MAbs) by reticuloendothelial system. The patients suffering from leukemia, colorectal cancer, lymphoma and melanoma have been treated with MAbs. However, there was a wide variation in the success rate. A monoclonal antibody specific to the cells of leukemia is used to destroy the residual leukemia cells without affecting other cells.

MAbs are used *in vitro* to remove the residual tumor cells prior to autologous bone marrow transplantation (transplantation of the patient's own bone marrow cells, due to non-availability of a suitable donor).

Limitations for direct use of MAbs in cancer

1. The MAbs produced in mice and directly used for therapeutic purposes may lead to the development of antimouse antibodies and hypersensitivity reactions.

2. All the cancer cells may not carry the same antigen for which MAb has been produced. Thus, MAbs may not be attached to some cancer cells at all.

3. The free antigens (of target cells) present in the circulation may bind to MAbs and prevent them from their action on the target cells.

In the immunosuppression of organ transplantation

In the normal medical practice, immunosuppressive drugs such as cyclosporin and prednisone are administered to overcome the rejection of organ transplantation. In recent years, MAbs specific to T-lymphocyte surface antigens are being used for this purpose. The monoclonal antibody namely ***OKT_3***, was the ***first MAb to be licensed*** by U.S. Food and Drug Adminstration ***for use as immunosuppressive agent*** after organ transplantation in humans. OKT_3 specificially directed against CD_3 antigen of T-lymphocytes is successfully used in renal and bone marrow transplantations. In the normal course, CD_3 antigen activates T-lymphocytes and plays a key role in organ transplant rejection (destroys the foreign cells in the host). This is prevented by use of MAb against CD_3 antigen.

In the treatment of AIDS

Immunosuppression is the hall mark of AIDS. This is caused by reduction in CD_4 (cluster determinant antigen 4) cells of T-lymphocytes. The human immunodeficiency virus (HIV) binds to specific receptors on CD_4 cells by using surface membrane glycoprotein (gp120). Genetic engineers have been successful to ***attach Fc portion of mouse monoclonal antibody to human CD_4 molecule.*** This complex has high affinity to bind to membrane glycoprotein gp120 of virus infected cells. The ***Fc fragment induces cell-mediated destruction of HIV infected cells (Fig. 17.7).***

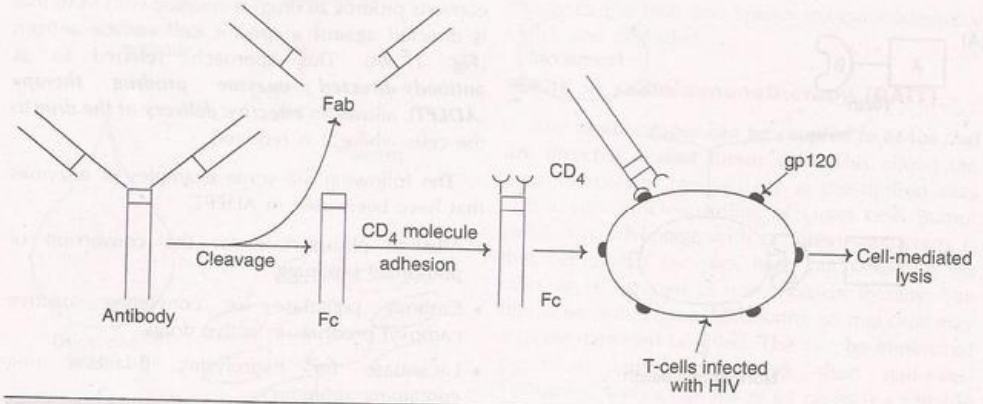


Fig. 17.7 : Modified monoclonal antibody in the treatment of AIDS.

In the treatment of autoimmune diseases

Autoimmune diseases like rheumatoid arthritis and multiple sclerosis are of great concern. Some success has been reported in the clinical trials of rheumatoid arthritis patients by using MAbs directed against T-lymphocytes and B-lymphocytes.

(B) *MAbs AS TARGETING AGENTS IN THERAPY*

Toxins, drugs, radioisotopes etc., can be attached or conjugated to the tissue-specific monoclonal antibodies and carried to target tissues for efficient action. This allows higher concentration of drugs to reach the desired site with minimal toxicity. In this way, MAbs are used for the appropriate delivery of drugs or isotopes.

MAbs in use as immunotoxins

The toxins can be coupled with MAbs to form immunotoxins and used in therapy e.g., diphtheria toxin, *Pseudomonas* exotoxin, toxins used for cancer treatment.

Anti-Tac MAb raised against IL2–R (T-cell growth factor receptor) can be conjugated with exotoxin of *Pseudomonas* sp. This immunotoxin can be used to destory the malignant T-cells in the patients suffering from T-cell leukemia (**Note :** IL2–R is expressed in abnormal T-cells with lymphoid malignancies).

Ricin is a cytotoxic protein derived from castor oil plant. It is composed of two polypeptide chains (A and B) held together by a disulfide linkage. The B-chain of ricin binds to the cell surface. This binding facilitates the A-chain of ricin to enter the cell and inhibit the function of ribosomes (i.e. biosynthesis of all proteins is blocked). This results in the death of cells (***Fig. 17.8A***). Ricin can be subjected to oxidation to separate to A and B chains. The toxic A-chain can be conjugated to MAb that is specific to cancer cells. The tumor-specific MAb bound to A-chain of ricin binds to cancer cells and not to normal cells. Once the ***A-chain*** enters the cells, it ***blocks ribosomal function, leading to the death of cancer cells (Fig. 17.8B)***.

MAbs in drug delivery

In general, the drugs are less effective *in vivo* (in the living body) when compared to *in vitro* (in laboratory when tested with cultured cells). This is mainly due to the fact that sufficient quantity of the drug does not reach the target tissue. This problem can be solved by using tissue-specific MAbs. The drugs can be coupled with MAb (directed against a cell surface antigen of the cells, say a tumor) and specifically targeted to reach the site of action (***Fig. 17.9A***).

In the treatment of certain diseases, a prodrug (an inactive form of the drug) can be used. This can be enzymatically converted to active drug in the target tissues. For this purpose, the enzyme (that

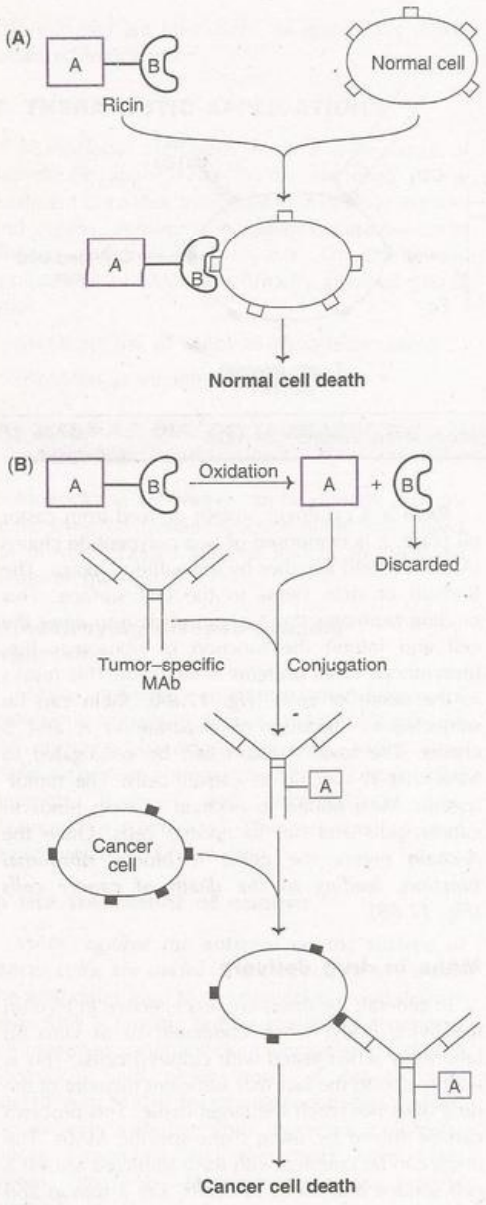


Fig. 17.8 : Use of ricin as a cytotoxic agent (A) Action of ricin on a normal cell; (B) A-chain of ricin conjugated to MAb causing cancer cell death (Note : A-chain of ricin cannot enter normal cell).

converts prodrug to drug) is coupled with MAb that is directed against a specific cell surface antigen (***Fig. 17.9A***). This approach, referred to as ***antibody-directed enzyme prodrug therapy*** (***ADEPT***), allows an ***effective delivery of the drug*** to the cells where it is required.

The following are some examples of enzymes that have been used in ADEPT.

- Alkaline phosphatase for the conversion of phosphate prodrugs.
- Carboxy peptidase for converting inactive carboxyl prodrugs to active drugs.
- Lactamase for hydrolysing β-lactam ring containing antibiotics.

MAbs in the dissolution of blood clots

A great majority of natural deaths are due to a blockage in cornary or cerebral artery by a blood

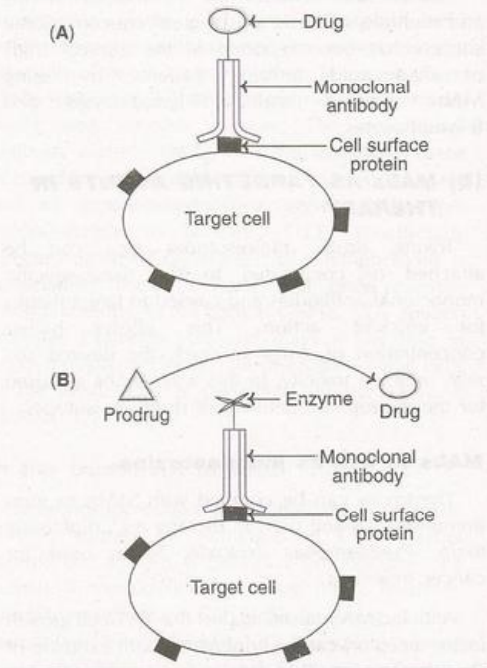


Fig. 17.9 : Monoclonal antibody based drug delivery to the target cells. (A) The drug is bound to MAb (B) The enzyme that converts prodrug to drug is bound to MAb.

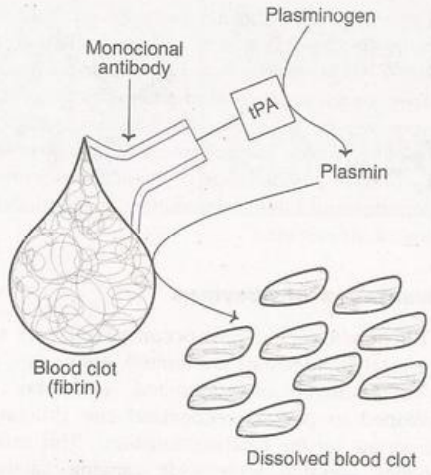


Fig. 17.10 : Monoclonal antibody in the dissolution of blood clot (tPA-Tissue plasminogen activator).

clot (thrombus). Fibrin is the major constituent of blood clot which gets dissolved by plasmin. Plasmin in turn is formed by the activation of plasminogen by plasminogen activator. The blockage of arteries occurs due to inadequate dissolution of blood clots. Tissue plasminogen activator (tPA) can be used as a therapeutic agent to remove the blood clots.

A ***monoclonal antibody directed against fibrin can be coupled to tPA and used for degradation of blood clots. MAb-tPA complex due to a high affinity gets attached to fibrin (Fig. 17.10).*** Due to the concentration of tPA at the target spots, there is more efficient conversion of plasminogen to plasmin which in turn dissolves blood clot (fibrin). Good success of clot lysis has been reported by using MAb-tPA complex in experimental animals.

Drug delivery through liposomes coupled to tissue-specific MAbs

Liposomes are sacs or vesicles formed spontaneously when certain lipid molecules are exposed to aqueous environment. Drug entrapped in liposomes that are coated with MAbs directed against tissue-specific antigens are being tried for drug delivery. Unfortunately, the progress in this approach has been limited, since such liposomes do not reach the target cells. They are retained mostly in the liver and spleen (reticuloendothelial cells), and degraded.

MAbs in radioimmunotherapy (RAIT)

The ***radioisotopes can be coupled to MAbs that are directed against tumor cells.*** This allows the concentration of radioactivity at the desired sites and a very efficient killing of target cells (tumor cells). The advantage with radioimmunotherapy is that conjugated complex need not penetrate the cells, as is required in immunotoxin therapy. The limitation is that the neighbouring normal cells may also get damaged or killed. This can be minimized by using radioisotopes with short half-lives. Yttrium–90 with a half-life of 64 hours is a suitable isotope to be employed in RAIT. Due to shortage in the supply of yttrium–90, indium–111 is more commonly used.

3. PROTEIN PURIFICATION

Monoclonal antibodies can be produced for any protein. And the so produced MAb can be conveniently used for the purification of the protein against which it was raised. MAbs columns can be prepared by coupling them to cyanogen bromide activated Sepharose (chromatographic matrix). The ***immobilized MAbs*** in this manner are very ***useful for the purification of proteins by immunoaffinity method.***

Advantages

There are certain advantages of using MAbs for protein purification. These include the specificity of the MAb to bind to the desired protein, very efficient elution from the chromatographic column and high degree of purification. Immunoaffinity chromatography is routinely used for the purification of recombinant interferons. The efficiency of this technique will be obvious from the fact that by a single step, it is possible to achieve more than 5,000 fold purification of interferon-α_2.

Disadvantages

It is not possible to achieve 100% puriy of the target protein by immunoaffinity. This is due to the fact that a small quantity of MAb leaks into the elution. Further, MAbs cannot distinguish between the intact target protein and a fragment of it with the antigenic site.

4. MISCELLANEOUS APPLICATIONS

CATALYTIC MAbs (ABZYMES)

Catalysis is the domain of enzymes. The most important common character between enzymes and antibodies is that both are proteins. Further, the binding of an antibody to its antigen is comparable to the binding of an enzyme to its substrate. In both instances, the binding is specific with high affinity and involves weak and non-covalent interactions (electrostatic, hydrogen and van der Waals forces). The striking difference is that the enzyme alters the substrate (to a product) while the antigen bound to antibody remains unaltered.

Certain similarities between enzyme-substrate interaction and antibody-antigen interaction have tempted researchers to explore the possibility of using antibodies in catalysis. The ***antibody enzymes***, appropriately regarded as ***abzymes***, are the ***catalytic antibodies***.

There is a difference in the antibody recognition of an antigen and enzyme recognition of a substrate. While the antibodies recognize in ***ground state***, the enzymes recognize in a ***transition state*** (associated with a conformational change of protein). In fact, it is in the transition condition the catalysis occurs. If a molecule resembling the transition state and conformation (between substrate and product) could be used as a hapten the antibodies so produced should bring about catalysis. This is what precisely is done to create abzymes.

Researchers have produced a ***hapten-carrier*** complex which resembles the transition state of an ester undergoing hydrolysis. This hapten conjugate is used to generate ***anti-hapten monoclonal antibodies***. These MAbs could bring about hydrolysis of esters with great degree of specificity (to the transition state to which MAbs were raised).

Besides ester hydrolysis, there are several other types of reactions wherein antibodies can be used. These include hydrolysis of amides and carbonates, cyclization reactions, elimination reactions and biomolecular chemical reactions. Certain enzymes require cofactors for their catalytic function. MAbs incorporating metal ions have been developed to carry out catalysis.

Lerner and his associates carried out pioneering work in the development of abzymes. They could create a large number of immunoglobulin-gene libraries for the production of antibodies that could be screened for their catalytic function. Abzymes represent a major biotechnological advancement that will have a wide range of applications (cutting of peptides and DNAs, dissolution of blood clots, killing of viruses etc.)

Advantages of abzymes

The number of naturally occurring enzymes and their catalytic functions are limited. Antibodies, on the other hand, are unlimited, and may be developed to possess recognized site structures, appropriate for the catalytic functions. This makes abzymes versatile with wide ranging catalytic applications. Thus, the area of abzyme technology is very promising, although the studies are at preliminary stages.

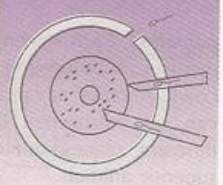
Limitations of abzymes

Despite the progress made in the production and utility of abzymes, it is doubtful whether they will ever match the natural enzymes in their catalytic function. However, the abzymes will be certainly useful for a variety of reactions where the natural enzymes do not have the desired specificities.

AUTOANTIBODY FINGERPRINTING

The occurrence of autoantibodies and their involvement in certain diseases is well known (e.g. rheumatic arthritis). A new category of ***individual specific (IS) autoantibodies*** have been discovered in recent years. These IS-autoantibodies are produced after birth and reach maximum in number by 2 years, and then remain constant for the later part of life. Monoclonal antibodies produced against IS-autoantibodies can be used for their detection, and ***identification of individuals***. This technique referred to as ***autoantibody fingerprinting***, is particularly useful for the detection of criminals, rapists etc. The autoantibodies collected from samples such as blood, saliva, semen and tears can be used.

18 Assisted Reproductive Technology



Traditional methods of breeding animals were only means of improving the genetic qualities. Despite the several advances made in biotechnology, genetic manipulations in relation to reproduction of animal and humans are very limited for obvious reasons, the most important being the unknown risks involved in the outcome.

The ***manipulations of reproduction in animals and humans*** are collectively considered under assisted reproductive technology (***ART***), and dealt with in this chapter. The objective of ART is to increase the number of progeny of the desired species.

Some of the important aspects of manipulation of reproduction in animals are first dealt with, and then the salient features of manipulations in human reproduction are described.

MANIPULATIONS OF REPRODUCTION IN ANIMALS

All the male mammals produce millions of sperms (male gametes) daily, and thus theoretically, can inseminate many females to produce a large number of offsprings. On the other hand, the female mammals produce a limited number of eggs (female gamets), usually one at a time (exception-pig), approximately at monthly intervals. The female also has the responsibility of developing the fertilized egg, besides providing nourishment to the newborn. Thus, the ruminant female mammals cannot produce more than one offspring in a year, in the normal circumstances.

The current approaches for the ***artificial breeding of animals*** with particular reference to the following are briefly described.

- Artificial insemination.
- Embryo transfer.
- *In vitro* fertilization.
- Embryo cloning.

ARTIFICIAL INSEMINATION

As the male produces millions of sperms daily, the semen can be used to produce several offsprings. This is made possible by artificial insemination (AI) of females. For an effective AI to produce the desired results, the following aspects must be considered.

Semen collection and storage

The semen ejaculate is collected, appropriately diluted and examined under microscope for the number of motile sperms. Normally, 0.2 ml of bull semen contains about 10 million motile sperms. The diluted semen can be used fresh within few days, or cryopreserved at –196°C in liquid nitrogen for long-term storage and transport.

Artificial insemination is done on standing animals through a technique, known a ***rectal***

palpation. Each semen ejaculate of a male bull, in theory, can be used to inseminate as many as 500 cows. Another advantage of AI is that the semen can be transported (in cryopreserved form) to different places/countries and used to develop superior animals.

Synchronization of ovulation

The female animals are inseminated after ovulation which can be detected by behavioural estrous. It is rather difficult to detect estrous, since it mostly occurs during night and lasts for only a few hours.

The animal breeders would like to inseminate a large number of females simultaneously, so that the management becomes easy. For this purpose, the females are induced (synchronized) to ovulate at a set time. This is achieved by administration of progesterone and/or prostaglandins, which regulate ovulation cycle. Although total synchrony of ovulation is not possible, about 80% of the females could be made to respond by this approach.

Sperm sexing

The livestock industries prefer to have animals belonging to one sex. For instance, the dairy and beef industries demand more females than males. It is possible ***to produce the animals of desired sex*** through sperm sexing.

Sperms and ova contain half of the chromosomes of a somatic cell. Thus, an ovum contains autosomes and one X chromosome while the sperm contains autosomes and one Y chromosome. Sex is determined genetically by the sex chromosomes (X and Y). X chromosome is present in all the ova, whereas half of the sperm (of a semen ejaculate) possess X and the other half Y chromosomes. The sex of the embryo is determined by the sperm (X or Y containing) that is successful in fertilization. Thus, if the embryo contains X chromosome, it is a female; while it is a male if it possesses Y chromosome. In the natural breeding, the sex ratio of progeny is close to 1:1.

It is possible to separate the sperms containing either X or Y chromosome, and use them selectively for the desired sex of the progeny.

By employing a fluorescent dye (Hoechst 33342) and an instrument namely ***fluorescent activated cell sorter (FACS),*** two populations of sperms (X or Y chromosome containing) can be separated. Using this approach and *in vitro* fertilization technique has produced pre- sexed calves.

FACS separation of sperms is expensive and time (about 24 hours)-consuming. Many sperms may die before they are actually separated. Attempts are being made to develop better separation techniques of sperms.

EMBRYO TRANSFER

As the ruminant animal produces one egg at time, it can carry one pregnancy at a time. It is possible to increase the production of female animals by increasing the number of mature eggs from a given female, fertilize them and transfer (implant) the embryos (fertilized eggs) into a ***foster mother (recipient)***. The foster mother ***serves as an incubator, and does not make any genetic contribution to the offspring***.

Embryo transfer is a costly technique, and is selectively used for the production of animals of high genetic or economic valve.

Superovulation (multiple ovulation)

In the normal reproductive cycle of a non-pregnant female, one ovarian follicle (out of the 20 that develop) matures and ruptures, releasing one fertile egg at a time. The time of ovulation varies in different animals—21 days for cow and horse; 16 days for sheep and goat.

The circulating ***gonadotrophic hormone*** is closely associated with ovulation and release of egg. By increasing the concentration of this hormone, ***more ovarian follicles can be induced to ripen and produce more eggs.*** This process, known as superovulation or multiple ovulation may yield not less than 8-10 eggs at a time. Some animal breeders were successful in superovulating animals to yield as many as 60 eggs at a time. This largely depends on the breed, nutrition and health of the animal, besides the environmental factors.

By administering prostaglandin F_{2a} (PGF_{2a}) and follicle stimulating hormone (FSH), estrous can be induced. Artificial insemination is carried out in the superovulated females. AI is preferred to natural mating, since the purpose of superovulation is to genetically improve the progeny.

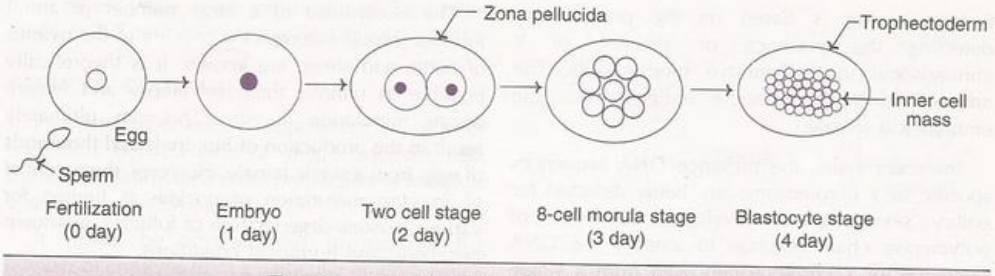


Fig. 18.1 : The early stages of embryo development.

As the eggs are fertilized, they undergo development to form embryos (***Fig. 18.1***).

Multiple ovulation with embryo transfer (MOET)

Sometimes, the process of multiple ovulation (superovulation described above) and embryo transfer are considered together which is referred to as MOET.

The embryos developed in the superovulated animals (described above) are recovered after 6-8 days of insemination. In case of cattle, the recovery process is easy and can be done by using a catheter. For certain animals with smaller reproductive tract, surgical procedures may be needed to expose the oviduct and recover the embryos. These embryos are examined microscopically and identified.

The embryos are then transferred into a synchronized recipient (i.e. the foster mother) by using a procedure, which is comparable to artificial insemination. Alternately, the embryos can be frozen and stored for use at an appropriate time and place later. About 50-60% of pregnancy could be achieved in cattle with transferred embryos. Thus, a superovulated female may result in 5-6 pregnancies.

The embryos may also be subjected to manipulations, as described below.

Embryo splitting

It is possible to increase the number of progenies (almost to double) by splitting the embryos. In addition, embryo splitting results in identical twins for various purposes (particularly genetic research).

Embryo splitting techniques have been refined, and are routinely used these days. The embryos are suspended in a hypertonic (high osmotic) sucrose and bovine serum albumin (BSA) containing culture medium. This results in shrinking of embryo cells and settling of the embryos to the bottom of the container (usually a petridish) due to increased density. Further, the embryos stick to the bottom due to electrostatic interaction of the negative charges on it (due to attachment of albumin) and the positive charges of the petridish. The inner cell mass (ICM) of the blastocyte can be bisected by using a surgical blade and a micromanipulation technique (with the help of an inverted microscope). Splitting of each embryo results in two equal halves.

As the embryos are split, each hemispherical cell mass reforms spheres. These split embryos can be transferred into the oviducts of synchronized recipients. The pregnancy rate with split embryos is about 5-10% less than the theoretical calculations. Thus, about 9 progencies (instead of only 5 in the original embryo transfer) can be developed through MOET.

By MOET, it is possible to produce at least four identical offsprings in animals at a time.

Embryo biopsy

Embryo biopsy involves the removal of a few cells from the embryo (mostly from the trophoblast cells of the trophectoderm) for analysis to determine the sex. In recent years, this technique is becoming popular for the detection of genetic diseases. By using embryo biopsy, it is possible to stop the transfer of embryos with genetic abnormalities and undesirable traits.

Embryo sexing

Before the transfer or implantation of the embryo, the progeny sex can be determined.

Embryo sexing is based on the principle of detecting the presence or absence of Y chromosome in the embryo biopsy cells. The absence of Y chromosome indicates that the embryo is a female.

In recent years, the presence DNA sequences specific to Y chromosome are being detected for embryo sexing. Another development is the use of polymerase chain reaction to amplify the DNA sequence (of Y chromosome) even from a single cell and determination the sex.

Limitations of embryo transfer

- The supply of embryo from superovulated donors is limited.
- Freezing and thawing of embryos requires a lot of care to keep them functionally intact.
- Embryo transfer requires technical skill, besides high cost factor.

IN VITRO FERTILIZATION

The limitations of embryo transfer could be successfully overcome by using *in vitro* fertilization (IVF). IVF basically involves fertilization of oocytes of a female animal in the laboratory conditions under artificial conditions. This is in contrast to the natural fertilization, which occurs in the uterus. IVF is reasonably successful as it results in 70–80% of the fertilized eggs.

Through IVF technology, a large number of offsprings can be produced from a single animal. Thus, a female animal, which normally produces 4-5 offsprings in her lifetime, can produce as many as 50–20 offsprings by employing IVF.

In vitro fertilization involves the following stages.

Oocyte recovery

During the course of estrous cycle, the ovarian follicles grow, get filled with fluid and become Graafian (antral) follicles. The oocytes from these follicls can be recovered by using laparoscopic surgery. It is possible to recover more number of oocytes from superovulated donors.

In vitro maturation (IVM) of oocytes

The immature oocytes recovered in the above step, when incubated *in vitro* will result in mature oocytes (eggs).

The occurrence of a large number of antral follicles throughout reproductive life in the ovaries of cattle and sheep are known. It is theoretically possible to remove them by biopsy and induce oocyte maturation *in vitro*. This may ultimately result in the production of hundreds and thousands of eggs from a single female. However, the potential of *in vitro* maturation of oocytes is limited for various reasons-degeneration of follicles, unknown metabolic and hormonal conditions.

Fertilization of eggs

In vitro fertilization of the eggs is carried out by using semen obtained from a superior male animal. For IVF, the eggs are carried in small droplets (microdroplets) of culture medium. Each microdroplet usually carries about 10 eggs. A dose of sperm approximately one million cells per ml is adequate for IVF. The penetration of the sperm into the egg is facilitated by supplementing the medium with certain compounds (penicillamine, epinephrine etc).

Embryo culture

The IVF embryos must be maintained in the *in vitro* conditions for a few days (about 7 days for sheep and goat, 8 days for cattle). This allows the development of embryos to blastocyte stage.

Approximately, 60% of the IVF embryos *in vitro* culture can form blastocytes.

Implantation of embryos

The 7 or 8 day old embryos from the *in vitro* culture are implanted in the reproductive tract of the recipient female which acts as a foster mother or surrogate mother.

Limitations of IVF

The pregnancy loss of IVF embryos, particularly during the first two months, is very high, although the reasons are not clearly known. The following parameters, which may be considered as limitations of IVF, may contribute to fetal losses of IVF embryos.

- Genetic defects in oocytes.
- Genetic defect in fertilizing sperms.
- Environmental mutagenesis.
- Inadequate supply of nutrients and hormones.
- Exposure to toxic agents and free radicals.

In recent years, some improvements have been made in IVF to minimize the fetal losses.

EMBRYO CLONING

A clone represents a population of cells or organisms derived from a single ancestor cell. Cloning basically means the production of identical copies of an individual.

It would be advantageous to increase the number of embryos from a particular embryo which possesses the desired characters. Two approaches are in use for embryo cloning.

1. Nuclear transfer.
2. Use of embryonic cells.

For more information on nuclear transfer and embryonic cells, the reader must refer the chapter on transgenic animals (Chapter 41).

MANIPULATIONS OF REPRODUCTION IN HUMANS

Assisted reproductive technology (ART) in humans is one of the greatest advances in reproductive medicine. ART has now become an accepted method for the treatment of infertile couple for whom there is no alternative therapy.

CAUSES OF INFERTILITY AND APPLICATION OF ART

Infertility in men and women may be due to a variety of reasons.

Male infertility

Oligospermia **:** Reduced concentration of sperms in semen (normal 15–20 million/ml).

Azoospermia : Total lack or very low concentration of motile sperm.

It is now possible to carry out ART even with a low concentration of motile sperms.

Female infertility

An outline of the female reproductive system with the essential organs is depicted in ***Fig 18.2.***

Tubal infertility : This occurs due to non-functional or damaged fallopian tubes. This can be corrected surgically or by tubal transplantation.

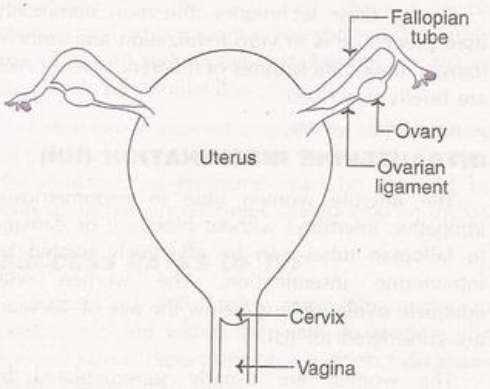


Fig. 18.2 : *An outline of the female reproduction organs.*

Non-functional ovaries : Some women may possess ovaries that are non-functional, or in some cases the ovaries may be totally absent. These women may serve as surrogate or foster mothers. The oocyte has to be obtained from a donar and ferlilized (*in vitro*) with the husbands' sperm.

Non-functional uterus : In some women, the uterus may be absent or non-functional. In such a case, the oocytes of these women can be fertilized by the husband's sperms, and then transferred into the uterus of surrogate mothers.

Idiopathic infertility : Some women may be infertile for unknown reasons which is regarded as idiopathic infertility. ART is useful for such women also.

The important ***techniques employed in assisted reproductive technology are listed*** below.

- Intrauterine insemination (IUI).
- *In vitro* fertilization and embryo transfer (IVF and ET).
- Gamete intrafallopian transfer (GIFT).
- Zygote intrafallopian transfer (ZIPT).
- Intravaginal culture (IVC).
- Cytoplasmic transfer (CT).
- Micromanipulation (Intracytoplasmic sperm injection (ICSI), subzonal insertion (SUZI).
- Cryopreservation.
- Assisted hatching (AH).

Among these techniques, the most commonly used procedure is *in vitro* fertilization and embryo transfer. Important features of different types of ART are briefly described.

INTRAUTERINE INSEMINATION (IUI)

The infertile women (due to endometriosis, idiopathic infertility) without blockage or damage to fallopian tubes can be effectively treated by intrauterine insemination. The women with adequate ovulation and below the age of 40 years are considered for IUI.

The women are usually superovulated by administering gonadotrophins. This results in multiple egg development. The IUI is timed to coincide with ovulation.

The semen is washed and the highly motile sperms are separated. By using a thin and soft catheter, the sperms are placed either in the cervix or in uterine cavity. The women subjects are advised to remain lying down for about 15-30 minutes following IUI.

Insemination should be carefully timed for good success. If it is done, a little before the expected time of ovulation, the chances for fertilization are much higher. IUI is usually successful in the first 3-4 attempts. In any case, this approach is not recommended for more than a maximum of 6 ovulation cycles.

The success rates of IUI vary considerably and are in the range of 15–30%.

IN VITRO FERTILIZATION AND EMBRYO TRANSFER (IVF and ET)

In vitro fertilization broadly deals with ***the removal of eggs from a women, fertilizing them in the laboratory, and then transferring the fertilized eggs (zygotes) into the uterus a few days later.***

Indications for IVF

Infertility due to the following causes may be considered for IVF.

- Failed ovulation induction
- Tubal diseases
- Cervical hostility
- Endometriosis
- Idiopathic infertility (in men and women).

Ideal subjects for IVF

Although it is not always possible to have a choice in the selection of subjects, the following criteria are preferred.

- Woman below 35 years.
- Presence of at least one functional ovary.
- Husband with normal motile sperm count (i.e. normal ***seminogram***).
- The couple must be negative for HIV and hepatitis.

METHODOLOGY OF IVF

The *in vitro* fertilization boardly involves the following steps.

1. Induction of superovulation.
2. Monitoring of ovarian response.
3. Oocyte retrieval.
4. Fertilization *in vitro*.
5. Embryo transfer.

Induction of superovulation

It is well known that the success rate IVF is much higher when more embryos (3-5) are transferred. This is possible only with ***controlled ovarian hyperstimulation (COH)***. The other advantages of COH include improvement in the quality of oocyte, control of ovulation timing, besides overcoming the ovulatory dysfunction. The following drug regimes are in use to induce superovulation.

- Clomiphene citrate (CC).
- CC + human menopausal gonadotrophin (hMG).
- CC + follicle stimulating hormone (FSH).
- Human menopausal gonadotrophin.
- Follicle stimulating hormone.
- Gonadotrophin releasing hormone agonists (GnRHa) + hMG (or FSH).

It is now common to use GnRH agonists to induce ovulation.

These compounds act through a process called downregulation of the physiologic hypothalamic-pituitary-ovarian feedback mechanism to effectively suppress spontaneous ovulation.

Monitoring of ovarian response

The follicular growth or ovarian response can be monitored by increase in serum estradiol level, increase in follicular diameter and thickening of endometrial bed.

Oocyte retrieval

The most common method for oocyte retrieval is carried out through vaginal route under ultrasound guidance. This method is simple and less invasive, and can be performed with analgesics only. It is easy to recognize the oocyte as a single cell surrounded by a mass of cumulus cells. The recovered oocytes are maintained *in vitro* culture for 4-6 hours.

Fertilization *in vitro*

The semen specimens are collected (just prior to oocyte retrival) via masturbation, processed, and incubated in protein-supplemented media for 3-4 hours prior to fertilization. The incubation results in sperm capacitation.

The retrieved oocytes are also cultured in protein-supplemented media for about 6-8 hours.

For the purpose of IVF, 50,000–1,00,000 capacitated sperms are placed in culture with a single oocyte. The signs of fertilization may be demonstrated 16–20 hours later by the presence of two pronuclei within the developing embryo.

There is no need to change the regime for a single failure of IVF. Many a times, success occurs in the subsequent cycles.

The two most important ***criteria for the success*** of IVF are ***sperm density and motility***.

Embryo transfer

Embryo at a stage between pronuclei and blastocyst stage are transferred. Conventionally, 4-8 cell stage embryos are transferred between 48–60 hours following insemination. The transfer procedure is carried out by use of a catheter. Not more than three embryos are transferred (per cycle) to minimize multiple pregnancies. However, in the women above the age of 40 years, higher number of embryo may be transferred. (**Note :** Excess oocytes and embryos are cryopreserved for further use. This will reduce the cost, besides the risk of ovarian hyperstimulation).

Luteal phase support is given by administration of progesterone for about two weeks. By this time, the diagnosis of pregnancy can be assessed by estimating human chorionic gonadotrophin (hCG).

SUCCESS RATES OF IVF

Success of IVF varies from programme to programme and within the same programme, the success rate is dependent on the correct diagnosis of the patient, and age.

The overall pregnancy rate in IVF is in the range of 25-35% per oocyte retrival. The ***take home baby rate is about 15-20%*** per procedure. The success rate of IVF is rather low due to the following reasons.

- Increased risk of abortion
- Multiple preghancy
- Ectopic pregnancy
- Low birth weigh baby
- Premature delivery.

THE WORLD'S PICTURE OF TEST TUBE BABIES

By employing *in vitro* fertilization and embryo transfer, the world's ***first test tube baby*** (Louise Brown) was born in UK on 28th ***July 1978***. The world's second test tube baby (Kanupriya alias Durga) was born in Kolkata on 3rd October 1978. A team led by Subhash Mukherjee carried IVF and ET in India. Scientists responsible for the birth of test tube babies were severely criticized then.

In fact, IVF turned out to be one of the major achievements of medical sciences in the last century. It has become a novel way of treating infertility. Today, there are more than a million test tube babies born all over the world.

In 2003, the world celebrated the silver jubilee of IVF with much fanfare.

GAMETE INTRAFALLOPIAN TRANSFER (GIFT)

Gamete intrafallopian transfer involves the transfer of both sperm and unfertilized oocyte into

the fallopian tube. This allows the fertilization to naturally occur *in vivo*. The prerequisite for GIFT procedure is that the woman should have at least one normal fallopian tube.

The induction of ovulation and the monitoring procedures for GIFT are almost the same as described for IVF. A couple of hours prior to oocyte retrieval, semen specimens are collected. Two oocytes along with 2-5 lakhs motile sperms for each fallopian tube are placed in a plastic tube container. It is then inserted (by laparoscopy) 4 cm into the distal end of the fallopian tube, and the oocyte sperm combination is injected.

The overall pregnancy rate is as high as 30-40%. The take home baby rate is about 25%. This is much higher when compared to IVF. But the ***major limitation*** is the requirement of ***laparoscopy*** (a major surgical procedure) to transfer oocytes and sperms into the fallopian tubes.

ZYGOTE INTRAFALLOPIAN TRANSFER (ZIFT)

ZIFT is suitable when the infertility lies in men, or in case of failure of GIFT.

The wife's oocytes are exposed to her husband's sperms in the laboratory. The fertilized eggs (zygotes) within 24 hours are transferred to the fallopian tube by using laparoscopy.

ZIPT has an advantage over GIFT with male factor infertility. Further, it can be known whether the wife's oocytes have been fertilized by her husbands' sperms.

INTRAVAGINAL CULTURE (IVC)

The ***body's own environment*** is appropriately ***utilized*** in intravaginal culture. The retrieved oocytes and sperms are placed in a culture medium inside a sealed container. This is inserted into the vagina. The container is held by a vaginal diaphragm. Thus, the oocytes and sperms are maintained at the normal body temperature (in contrast to any incubator in the laboratory). Two to 3 three days later, the container is opened, and the fertilized and dividing zygotes are transferred into the uterus.

This procedure appears simple, but the success rate is very low. Only a few centers practice this.

CYTOPLASMIC TRANSFER (CT)

Cytoplasm includes many things, the most important being mitochondria which provide energy to the cell. It is possible that deficiency in the mitochondria may leave the oocyte without the necessary power for cell division, after fertilization. This may result in abnormal cell division and poor development of embryo.

It is therefore logical to think of the transfer of cytoplasm from a donor (with active mitochondria) into the oocyte of a woman. The advantage with cytoplasmic transfer is that the ***mother's own genetic material is passed on to the offspring***.

Two methods of cytoplasmic transfer have been developed.

1. Transfer of a small amount of cytoplasm by a tiny needle from a donor to a recipient oocyte.

2. Transfer of a large amount of cytoplasm which is fused with the recipient's cytoplasm by applying electricity.

The procedure of cytoplasmic transfer is tedious and technically difficult, besides the cost factor. At least two viable pregnancies have been so far reported in literature by this approach.

MICROMANIPULATION

Micromanipulation involves *in vitro* microsurgically assisted fertilization procedures. This is required when the sperms are unable to penetrate the zona pellucida of oocyte and fertilize. Micromanipulations are usually done in severe cases of male factor infertility.

A diagrammatic representation of micro-manipulation is depicted in ***Fig 18.3.***

Intracytoplasmic sperm injection (ICSI)

Intracytoplasmic sperm injection is a new and novel infertility treatment utilizing the micromanipulation technology. Many of the previous treatment processes for male infertility have been abandoned in favour of ICSI. The male factor infertility could be due to low sperm counts, poor sperm motility, and poor quality of sperm to penetrate oocyte.

By ***partial zona dissection (PZD)***, the zona pellucida is opened using either chemical dissolution or a sharp instrument. A single

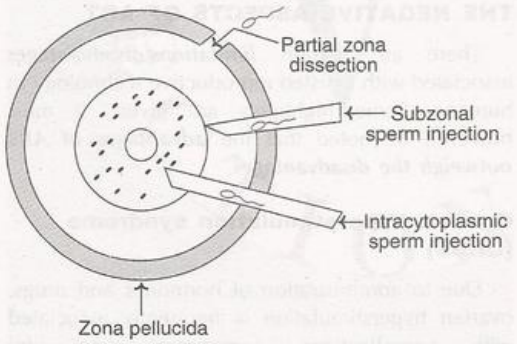


Fig. 18.3 : Micromanipulation for fertilization of an egg (microsurgically assisted fertilization).

spermatozoon can be directly injected into the cytoplasm of the oocyte through the micropuncture of zona pellucida. A micropipette is used to hold the oocyte while the spermatozoon is deposited inside the ooplasm of the oocyte.

Besides using normal sperms, round-headed sperms, sperms collected directly form the epididymis and previously cryopreserved sperms can be used in ICSI.

Among the ***micromanipulation techniques ICSI*** is the ***most successful one*** with a fertilization rate of about 65%. Attempts are on to improve this further. In fact, ICSI has revolutionized assistant reproductive technology by utilizing the sperms of husbands who were once considered to be unsuitable for fertilization process.

Subzonal insertion (SUZI)

In subzonal insertion, the zona pellucida is punctured and sperms (1-30 in number) are injected into an area between the zona and the egg. It is expected that one of the sperms will fertilize the egg.

The major limitation of SUZI is polyspermy since it is not possible to control the number of sperms that enter the egg.

Round spermid nucleus injection (ROSNI)

There are a few men who cannot manufacture sperms, and therefore they have a zero sperm count. For these men, it is possible to take out the round spermatids (immature cells) directly from the testicle, isolate the nucleus (containing the genetic material) and inject it into the partner's eggs.

ROSNI is a recent exciting breakthrough to solve the problem of male infertility through micromanipulation.

CRYOPRESERVATION

Preservation in a frozen state is regarded as cryopreservation. Cryopreservation is very useful in assisted reproductive technology.

- Semen can be cryopreserved. This may be from the donors, cancer patients (before the commencement of treatment).
- Fertilized eggs after IVP or ICSI can be preserved.
- Embryos can also be preserved for transfer at a later stage.

Human embryos have been successfully preserved in the presence of cryoprotectants (1, 2-propanediol/dimethyl sulfoxide/glycerol) and stored at –196°C under liquid nitrogen. At appropriate time, the embryos are thawed, cryoprotectants removed and then transferred. Many test tube babies in fact have been born as a result of application of ***freezing technology***.

ASSISTED HATCHING (AH)

Improper implantation of the embryo in the uterus is one of the limiting factors in the success of ART in humans. Assisted hatching is a novel approach for the proper implantation of the embryo in the endrometrium.

The embryos in the uterus possess an outer coating namely zona pellucida (the shell). These embryos must be hatched to remove the shell, a step necessary for implantation. In certain women, particularly above 40 years age, natural hatching does not occur, and requires outside assistance.

Assisted hatching is carried out by using a Laser to make a small hole in the shell of the embryo. These embryos when transferred into the uterus, hatch and get implanted.

During the course of AH for 3-4 days, the women are kept on steroids (to suppress mother's immunity) and antibiotics (to counter infections). Better results are reported with this approach.

PREIMPLANTATION GENETIC DIAGNOSIS (PGD)

The genetic defects in ovum before fertilization or in the embryo before implantation can be identified by a ***new medical tool*** namely preimplantation genetic diagnosis. It is estimated that about 60% of the ART driven pregnancies are lost due to chromosomal abnormalities. This can be minimized or prevented by using PGD.

A direct determination of chromosomal abnormalities prior to implantion ensures a successful pregnancy and ultimate delivery of a healthy baby. One group of workers has reported an increase in the pregnancy rate from 15 to 30% by employing preimplantation genetic diagnosis.

DNA amplification and analysis

The latest in PGD is the direct DNA analysis. This can be carried out by removing a single cell from 6-8-cell embryo. The DNA is removed and amplified by employing polymerase chain reaction.

Direct DNA analysis is useful for the diagnosis of several genetic diseases e.g. cystic fibrosis, sickle-cell anemia, hemophilia, Duchene's muscular dystrophy, Tay-Sachs disease.

Ethical advantages of PGD

PGD is highly advantageous from the ethical point of view, since the embryos with genetic disorders can be discarded in the very stages without the formation of offsprings with undesirable characteristics.

THE NEGATIVE ASPECTS OF ART

There are certain ***limitations/disadvantages*** associated with assisted reproductive technology in humans. Some highlights are given. It must however, be noted that the ***advantages of ART outweigh the disadvantages.***

Ovarian hyperstimulation syndrome (OHSS)

Due to administration of hormones and drugs, ovarian hyperstimulation is frequently associated with complications, sometimes even life-threatening. OHSS is more severe in women who conceived in the same cycle, and received hCG as luteal support (following embryo transfer).

Risks associated with pregnancy

ART is associated with multiple pregnancy, increased risk for anemia, gestational diabetes and premature labour. Low birth weight and prematurity are closed linked with mortality and morbidity.

Premature menopause

Controlled ovarian hyperstimulation (COH) causes multiple follicular utility. There is a risk of premature menopause as COH may reduce the ovarian follicles, besides faster aging. Sometimes, a single COH may use ovarian follicles, which in the normal course are equivalent to two years of ovulation during the natural menstrual cycle.

Ovarian cancer

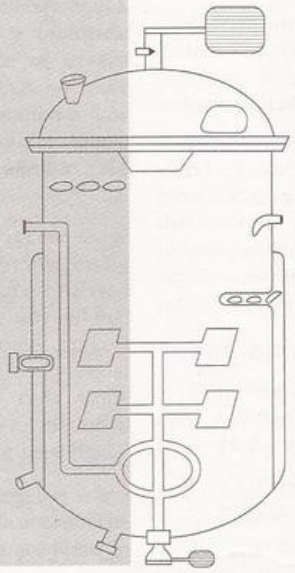
The use of fertility drugs and injuries to epithelium increase the risk of ovarian cancer at least by three times when compared to normal women.

SECTION V

MICROBIAL/ INDUSTRIAL BIOTECHNOLOGY

CONTENTS

19 Bioprocess/Fermentation Technology 239
20 Downstream Processing 270
21 Enzyme Technology 281
22 Biotransformation 306
23 Microbial Production of Organic Solvents 311
24 Microbial Production of Organic Acids 318
25 Microbial Production of Antibiotics 329
26 Microbial Production of Amino Acids 344
27 Microbial Production of Vitamins 355
28 Microbial Production of Foods and Beverages 362
29 Single-Cell Protein, and Mushrooms 373
30 Polysaccharides, Polyhydroxyalkanoates, and Lipids 382
31 Biomass and Bioenergy 393
32 Microbial Mining and Metal Biotechnology 400



Bioprocess/Fermentation Technology

For many centuries, man has been exploiting microorganisms for the production of foods (bread, cheese, yoghurts, pickles) and beverages (beer, wine). However, the organized use of microorganism for industrial purposes is about one and a half century old.

The word ***fermentation*** originates from a *Latin* verb ***fervere*** which literally means ***to boil***. During the production of alcohol (the first truely industrialised process), the gas bubbles (of CO_2) appear at the surface of the boiling liquid. ***Fermentation in a strict sense is a biological process that occurs in the absence of oxygen (anaerobic)***. This definition however, is no more valid, since the term ***industrial fermentation*** is now used for large-scale cultivation of microorganisms, even though ***most of them are aerobic*** (use oxygen).

Bioprocess technology is a more recent usage to replace fermentation technology. Bioprocessing broadly involves a multitude of enzyme-catalysed reactions carried out by living cells (or cell-free systems) for industrial purposes. Some workers prefer to use ***bioprocess technology*** for industrial use of ***higher plant and animal cells*** while ***fermentation technology*** is confined to ***microbial use***. This demarcation is however, not very rigid.

Bioprocess/fermentation technology is very widely exploited for industrial applications. The broad range of fermentation products are listed in ***Table 19.1***.

BIOREACTORS/FERMENTERS

The heart of fermentation (or bioprocessing) technology is the fermenter (or bioreactor). *A* ***bioreactor is basically a device in which the organisms (cells) are cultivated and motivated to form the desired product(s)***. It is a ***containment system*** designed to give right environment for optimal growth and metabolic activity of the organism.

A ***fermenter*** usually refers to the containment system for the ***cultivation of prokaryotic cells*** (bacteria, fungi), while a ***bioreactor*** grows the ***eukaryotic cells*** (mammalian, insect).

Traditional fermenters are open vats made up of wood or slate. In recent years, stainless steel bioreactors are in use. A high quality stainless steel that does not corrode or leak toxic metals into the growth medium is used. The size of a bioreactor is highly variable, ranging from 20 litres to 250 million litres or even more.

TYPES OF BIOREACTORS

Based on the designs of the bioreactors, they can be grouped into the following types (***Figs. 19.1–19.4***)

1. Continuous stirred tank bioreactors
2. Bubble column bioreactors
3. Airlift bioreactors
4. Fluidized bed bioreactors

TABLE 19.1 Industrial products of fermentation (bioprocess) technology

Group	*Products*
Foods	Dairy products (cheese, yogurt)
	Vitamins (B_1, B_{12})
	Amino acids (glutamic acid, lysine)
	Glucose and high fructose syrup
	Mushroom products
	Baker's yeast
	Food additives (antioxidants, colours, flavours)
	Beverages (beer, wine, whisky)
Chemicals	
Organic (bulk)	Ethanol, butanol, acetone, organic acids (citric acid, gluconic acid, lactic acid)
Organic (fine)	Enzymes, polymers (xanthan, dextran)
Inorganic	Bioaccumulation and leaching (Cu, U)
Pharmaceuticals (healthcare)	Antibiotics
	Vaccines
	Steroids
	Diagnostic enzymes
	Monoclonal antibodies
	Enzyme inhibitors
Agriculture	Single–cell protein
	Microbial pesticides
	Composting processes
	Plant cell and tissue culture

5. Packed bed bioreactors
6. Photobioreactors.

In all types of bioreactors, the ultimate aim is to ensure that all parts of the system are subjected to the same conditions.

Continuous stirred tank bioreactors

A continuous stirred tank bioreactor consists of a cylindrical vessel with motor driven central shaft that supports one or more ***agitators (impellers)***. The shaft is fitted at the bottom of the bioreactor (***Fig. 19.1A***). The number of impellers is variable and depends on the size of the bioreactor i.e., height to diameter ratio, referred to as ***aspect ratio***. The aspect ratio of a stirred tank bioreactor is usually between 3–5. However, for animal cell culture applications, the aspect ratio is less than 2. The diameter of the impeller is usually $\frac{1}{3}$rd of the vessel diameter. The distance between two impellers is approximately 1.2 impeller diameter. Different types of impellers (Rustom disc, concave bladded, marine propeller etc.) are in use.

In stirred tank bioreactors or in short ***stirred tank reactors (STRs)***, the air is added to the culture medium under pressure through a device called ***sparger***. The sparger may be a ring with many holes or a tube with a single orifice. The sparger along with impellers (agitators) enables better gas distribution system throughout the vessel. The bubbles generated by sparger are broken down to smaller ones by impellers and dispersed throughout the medium. This enables the creation of a uniform and homogeneous environment throughout the bioreactor.

Advantages of STRs : There are many advantages of STRs over other types. These include the efficient gas transfer to growing cells, good mixing of the contents and flexible operating conditions, besides the commercial availability of the bioreactors.

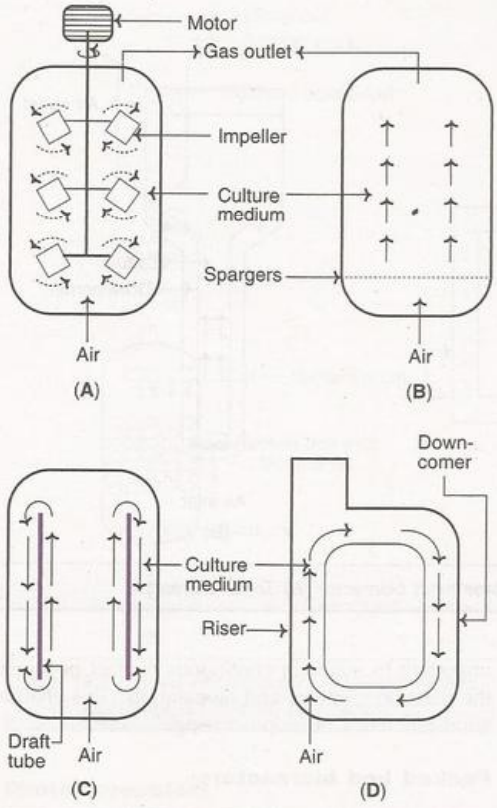


Fig. 19.1 : Types of bioreactors (A) Continuous stirred tank bioreactor (B) Bubble column bioreactor (C) Internal-loop airlift bioreactor (D) External-loop airlift bioreactor.

Bubble column bioreactors

In the bubble column bioreactor, the air or gas is introduced at the base of the column through perforated pipes or plates, or metal microporous ***spargers*** (***Fig. 19.1B***). The flow rate of the air/gas influences the performance factors — O_2 transfer, mixing. The bubble column bioreactors may be fitted with perforated plates to improve performance. The vessel used for bubble column bioreactors is usually cylindrical with an aspect ratio of 4-6 (i.e., height to diameter ratio).

Airlift bioreactors

In the airlift bioreactors, the medium of the vessel is divided into two interconnected zones by means of a baffle or draft tube. In one of the two zones referred to a ***riser***, the air/gas is pumped. The other zone that receives no gas is the ***downcomer***. The dispersion flows up the riser zone while the down flow occurs in the downcomer. There are two types of airlift bioreactors.

Internal-loop airlift bioreactor (***Fig. 11.1C***) has a single container with a central draft tube that creates interior liquid circulation channels. These bioreactors are simple in design, with volume and circulation at a fixed rate for fermentation.

External loop airlift bioreactor (***Fig. 19.1D***) possesses an external loop so that the liquid circulates through separate independent channels. These reactors can be suitably modified to suit the requirements of different fermentations. In general, the airlift bioreactors are more efficient than bubble columns, particularly for more denser suspensions of microorganisms. This is mainly because in these bioreactors, the mixing of the contents is better compared to bubble columns.

Airlift bioreactors are commonly ***employed for aerobic bioprocessing technology***. They ensure a controlled liquid flow in a recycle system by pumping. Due to high efficiency, airlift bioreactors are sometimes preferred e.g., methanol production, waste water treatment, single-cell protein production. In general, the performance of the airlift bioreactors is dependent on the pumping (injection) of air and the liquid circulation.

Two-stage airlift bioreactors

Two-stage airlift bioreactors are ***used for the temperature dependent formation of products***. Growing cells from one bioreactor (maintained at temperature 30°C) are pumped into another bioreactor (at temperature 42°C). There is a necessity for the two-stage airlift bioreactor, since it is very difficult to raise the temperature quickly from 30°C to 42°C in the same vessel. Each one of the bioreactors is fitted with valves and they are connected by a transfer tube and pump (***Fig. 19.2A***). The cells are grown in the first bioreactor and the bioprocess proper takes place in the second reactor.

Tower bioreactors

A ***pressure-cycle fermenter*** with large dimensions constitutes a tower bioreactor (***Fig. 19.2B***). A high hydrostatic pressure generated

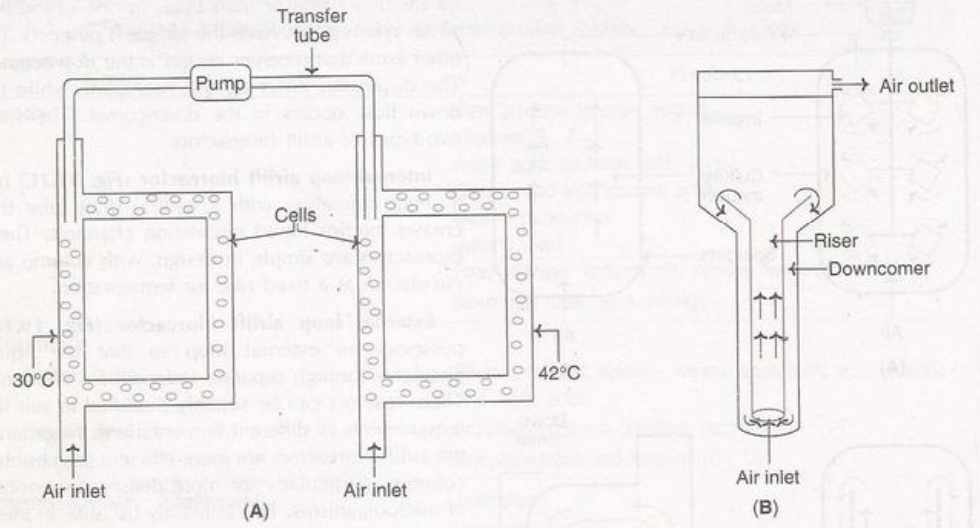


Fig. 19.2 : Types of bioreactors (A) Two–stage airlift bioreactor (B) Tower bioreactor.

at the bottom of the reactor increases the solubility of O_2 in the medium. At the top of the riser, (with expanded top) reduces pressure and facilitates expulsion of CO_2. The medium flows back in the downcomer and completes the cycle. The advantage with tower bioreactor is that it has high aeration capacities without having moving parts.

Fluidized bed bioreactors

Fluidized bed bioreactor is comparable to bubble column bioreactor except the top position is expanded to reduce the velocity of the fluid. The design of the fluidized bioreactors (expanded top and narrow reaction column) is such that the solids are retained in the reactor while the liquid flows out (***Fig. 19.3A***). These bioreactors are suitable for use ***to carry out reactions involving fluid suspended biocatalysts such as immobilized enzymes***, immobilized cells, microbial flocs.

For an efficient operation of fludized beds, gas is sparged to create a suitable gas-liquid-solid fluid bed. It is also necessary to ensure that the suspended solid particles are not too light or too dense (too light ones may float whereas to dense ones may settle at the bottom), and they are in a good suspended state. Recycling of the liquid is important to maintain continuous contact between the reaction contents and biocatalysts. This enable good efficiency of bioprocessing.

Packed bed bioreactors

A ***bed of solid particles, with biocatalysts*** on or within the matrix of solids, packed in a column constitutes a packed bed bioreactor (***Fig. 19.3B***). The solids used may be porous or non-porous gels, and they may be compressible or rigid in nature. A nutrient broth flows continuously over the immobilised biocatalyst. The products obtained in the packed bed bioreactor are released into the fluid and removed. While the flow of the fluid can be upward or downward, downflow under gravity is preferred.

The concentration of the nutrients (and therefore the products formed) can be increased by increasing the flow rate of the nutrient broth. Because of poor mixing, it is rather difficult to control the pH of packed bed bioreactors by the addition of acid or alkali. However, these bioreactors are preferred for bioprocessing technology involving product-inhibited reactions. The packed bed bioreactors do not allow accumulation of the products to any significant extent.

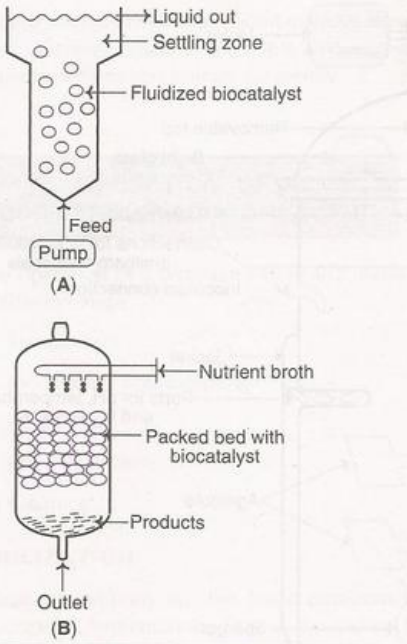


*Fig. 19.3 : Types of bioreactors (**A**) Fluidized bed bioreactor (**B**) Packed bed bioreactor.*

Photobioreactors

These are the bioreactors ***specialised for fermentation*** that can be carried out either ***by exposing to sunlight or artificial illumination***. Since artificial illumination is expensive, only the out door photobioreactors are preferred. Certain important compounds are produced by employing photobioreactors e.g., β-carotene, asthaxanthin.

The different types of photobioreactors are depicted in ***Fig. 19.4***. They are made up of glass or more commonly transparent plastic. The array of tubes or flat panels constitute light receiving systems (solar receivers). The culture can be circulated through the solar receivers by methods such as using centrifugal pumps or airlift pumps. It is essential that the cells are in continuous circulation without forming sediments. Further adequate penetration of sunlight should be maintained. The tubes should also be cooled to prevent rise in temperature.

Photobioreactors are usually operated in a continuous mode at a temperature in the range of

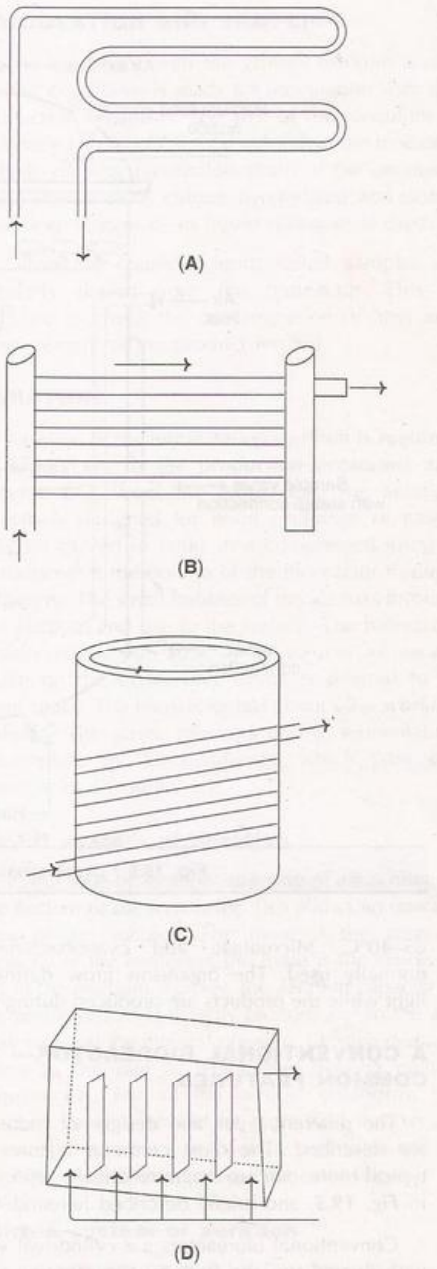


*Fig. 19.4 : Types of photobioreactors (**A**) Continuous run tubular loop (**B**) Multiple parallel tube (**C**) Helical wound tubular loop (**D**) Flat panel configuration.*

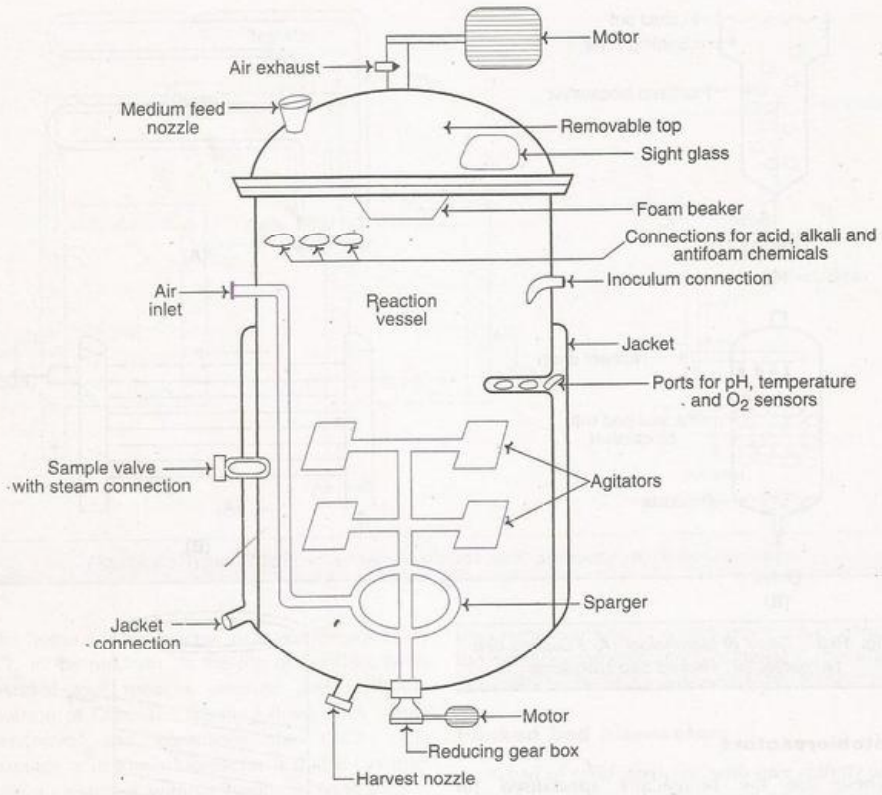


Fig. 19.5 : Diagrammatic representation of a typical bioreactor.

25–40°C. Microalgae and cyanobacteria are normally used. The organisms grow during day light while the products are produced during night.

A CONVENTIONAL BIOREACTOR — COMMON FEATURES

The different types and designs of bioreactors are described. The most common features of a typical bioreactor are diagrammatically represented in ***Fig. 19.5***, and briefly described hereunder.

Conventional bioreactors are cylindrical vessels with domed top and bottom. The reaction vessel, surrounded by a jacket, is provided with a sparger at the bottom through which air (or other gases such as CO_2 and NH_3 for pH maintenance) can be introduced. The agitator shaft is connected to a motor at the bottom. The reaction vessel has side ports for pH, temperature and dissolved O_2 sensors. Above the liquid level of the reaction vessel, connections for acid, alkali, antifoam chemicals and inoculum are located.

The bioreactor is usually designed to work at higher temperature (150–180°C), higher pressure (377-412 kPa). The reaction vessel is also designed to withstand vaccum, or else it may collapse while cooling. The materials used for the construction of bioreactor must be non-toxic and must withstand the repeated sterilization with high pressure steam.

The bioreactor vessel is usually made up of stainless steel. It should be free from crevices and stagnant areas so that no solids/liquids accumulate. Easy to clean channels and welded joints (instead of

couplings) are preferred. Transparent material should be used wherever possible, since it is advantageous to inspect medium and culture frequently.

OPERATION OF A CONVENTIONAL BIOREACTOR

The operation of a bioreactor basically involves the following steps.

1. Sterilization
2. Inoculation and sampling
3. Aeration
4. Control systems
5. Cleaning.

STERILIZATION

Aseptic conditions are the basic requirements for successful fermentation. That is the bioreactor and its accessories, the growth medium and the air supplied during fermentation must be sterile.

In situ sterilization

The bioreactor filled with the required medium is injected with pressurized steam into the jacket or coil surrounding the reaction vessel. The ***whole system is heated to about 120°C*** and held at this temperature ***for about 20 minutes***. *In situ* sterilization has certain limitations. It is not energy-efficient (i.e., energy is wasted) since the bioreactor has to be heated for a long period to rise the temperature of the whole system to 120°C. Prolonged heating may destroy vitamins, besides precipitating the medium components.

Continuous heat sterilization

In this technique, empty bioreactor is first sterilized by injecting pressurised steam. The ***medium is rapidly heated to 140°C for a short period***, by injecting the pressurised steam. Alternately, the medium can be sterilized by passing through a heat exchanger heated by pressurised steam. Subjecting the medium to high temperature for a short period does not precipitate medium components. Further, there is no energy wastage in continuous heat sterilization method.

INOCULATION AND SAMPLING

The bioreactor with the growth medium under aseptic conditions is ready for inoculation with the production organism. The size of the inoculum is generally 1–10% of the total volume of the medium. A high yielding production strain of the organism taken from a stock culture (lyophilized and stored in a deep freezer or in liquid nitrogen) is used.

During the course of fermentation, samples are regularly drawn from the bioreactor. This is required to check the contamination (if any) and measurement of the product formed.

AERATION

Aeration of the fermentation medium is ***required to supply O_2 to the production organisms and remove CO_2 from the bioreactor***. The aeration system is designed for good exchange of gases. Oxygen (stored in tanks in a compressed form) is introduced at the bottom of the bioreactor through a ***sparger***. The small bubbles of the air pass through the medium and rise to the surface. The bioreactor usually has about 20% of its volume as vacant space on the upper part which is referred to as ***head space***. The bioreactor has about 80% ***working volume***. The gases released during fermentation accumulate in the headspace which pass out through an air outlet.

Air-lift system of aeration

In this type of aeration, ***sparging of air is done at the bottom of the fermenter***. This allows an upward flow of air bubbles. The more is the aeration capacity of the fermenter, the more is the dissolved O_2 in the medium. Further, the aeration capacity of the air-lift system is directly proportional to the air-flow rate and the internal pressure. Oxygen demand refers to the rate at which the growing culture requires O_2. For all the aerobic organisms, the aeration capacity should be more than the oxygen demand or else the growth of the organisms will be inhibited due to oxygen depletion (starvation).

Stirred system of aeration

The ***aeration capacity*** of the medium can be ***enhanced by stirring***. This can be done by using impellers driven by a motor. The aeration capacity of the stirred fermenter is proportional to the stirring speed, rate of air flow and the internal pressure.

Stirred fermenters are better suited than air-lift fermenters to produce better aeration capacities.

CONTROL SYSTEMS

It is essential to maintain optimal growth environment in the reaction vessel for maximum product formation. ***Maximal efficiency of the fermentation can be achieved by continuously monitoring the variables*** such as the pH, temperature, dissolved oxygen, adequate mixing, nutrient concentration and foam formation. Improved sensors are now available for continous and automated monitoring of these variables (i.e., on line measurement of pH).

Most of the microorganisms employed in fermentation grow optimally between pH 5.5 and 8.5. In the bioreactor, as the microorganisms grow, they release metabolites into the medium which change pH. Therefore, the pH of the medium should be continuously monitored and maintained at the optimal level. This can be done by the addition of acid or alkali base (as needed) and a thorough mixing of the fermentation contents. Sometimes, an acid or alkaline medium component can be used to correct pH, besides providing nutrients to the growing microorganisms.

Temperature

Temperature control is absolutely essential for a good fermentation process. Lower temperature causes reduced product formation while higher temperature adversely affects the growth of microorganisms. The bioreactors are normally equipped with heating and cooling systems that can be used as per the requirement, to maintain the reaction vessel at optimal temperature.

Dissolved oxygen

Oxygen is sparingly soluble in water (0.0084 g/1 at 25°C). Continuous supply of oxygen in the form of sterilized air is done to the culture medium. This is carried out by introducing air into the bioreactor in the form of bubbles. Continuous monitoring of dissolved oxygen concentration is done in the bioreactor for optimal product formation.

Adequate mixing

Continuous and adequate mixing of the microbial culture ensures optimal supply of nutrients and O_2, besides preventing the accumulation of toxic metabolic byproducts (if any). Good mixing (by agitation) also creates favourable environment for optimal and homogeneous growth environment, and good product formation. However, excessive agitation may damage microbial cells and increase the temperature of the medium, besides increased foam formation.

Nutrient concentration

The nutrient concentration in a bioreactor is limited so that its wastage is prevented. In addition, limiting concentrations of nutrients may be advantageous for optimal product formation, since high nutrient concentrations are often associated with inhibitory effect on microbial growth. It is now possible to do on-line monitoring of the nutrient concentration, and suitably modify as per the requirements.

Foam formation

The media used in industrial fermentation is generally rich in proteins. When agitated during aeration, it invariably results in froth or foam formation that builds in head space of the bioreactor. Antifoam chemicals are used to lower surface tension of the medium, besides causing foam bubbles to collapse. Mineral oils based on silicone or vegetable oils are commonly used as antifoam agents.

Mechanical foam control devices, referred to as mechanical foam breakers, can also be used. Such devices, fitted at the top of the bioreactor break the foam bubbles and the throw back into the fermentation medium.

CLEANING

As the fermentation is complete, the bioreactor is ***harvested*** i.e. the contents are removed for processing. The bioreactor is then prepared for the next round of fermentation after cleaning (technically called ***turn round***). The ***time taken for turn round***, referred to as ***down time***, should be as short as possible (since it is non-productive). Due to large size of the bioreactors, it is not possible to clean manually. The cleaning of the bioreactors is carried out by using high-pressure water jets from the nozzles fitted into the reaction vessel.

Table 19.2 A selected list of solid state fermentations

Product	*Substrate*	*Microorganism(s) involved*
Edible mushrooms	Straw manure	*Agaricus bisporus*
		Lentinula edodes
Cheeses	Milk, curd	*Penicillium roquefortii*
Soy sauce	Soy beans, wheat	*Aspergillus oryzae*
Sauekraut	Cabbage	Lactic acid bacteria
Enzymes	Wheat bran	*Aspergillus niger*
Organic acids	Cane sugar, molasses	*Aspergillus niger*
Leaching of metals	Low grade ores	*Thiobacillus sp.*
Composting	Mixed organic material	Fungi, bacteria, actinomycetes
Sewage treatment	Sewage components	Bacteria, fungi, protozoa

SOLID SUBSTRATE (SOLID STATE) FERMENTATION

There are certain fermentation processes that do not involve liquid medium. For these bio-technological processes, the ***growth of the microorganisms is carried out on solid substrates*** in the complete absence or almost complete absence of free water. The presence of some moisture (about 15%) is necessary for solid substrate (or solid state) fermentation (***SSF***). The most commonly used solid substrates for SSF are cereal grains, wheat bran, sawdust, wood shavings and several other plant and animal materials. These solid substrates are polymeric in nature, insoluble or sparingly soluble in water, and contain concentrated source of nutrients for the growth of microorganisms.

SSF is a very old traditional technique carried out in many countries. It is used for the production of ***edible mushrooms***, cheese, soy sauce and many other fermented products (including enzymes and organic acids). A selected list of solid state fermentations is given ***Table 19.2***. ***Composting is a good example of SSF***.

Solid substrate fermentation has been very popular ***for the production of fermented foods*** (idli, dosa, dhokla, bread, beverages, fermented fish, meat, yogurt, cheese, pickles). Fermentation often makes the food more nutritious, easily digestable and better in flavour.

For solid substrate fermentation, single pure cultures, mixed cultures or mixed organisms may be used. Pretreatment of substrate raw materials is sometimes done to facilitate the availability of nutrients.

Solid substrate fermentation is normally carried out as a non-aseptic process. This ***saves sterilization costs***. It is important that the substrates used in SSF have adequate spaces in between to allow good air circulation. This facilitates adequate exchange of gases, besides promoting heat elimination. Forced air circulation may be done to maintain optimal conditions in SSF.

Bioreactors for SSF

Bioreactors designed for solid state fermentation are much simpler compared to liquid-state fermentation. In the ***Fig 19.6, tower reactor, drum reactor*** and ***forced aeration reactor***, used in SSF are depicted.

Advantages of SSF

- Solid substrate fermentation employs simple natural solids as the media.
- Low technology, low energy expenditure and requires less capital investment.
- No need for sterilization, less microbial contamination and easy downstream processing.
- Yield of the products is reasonably high.
- Bioreactor design, aeration process, and effluent treatment are quite simple.

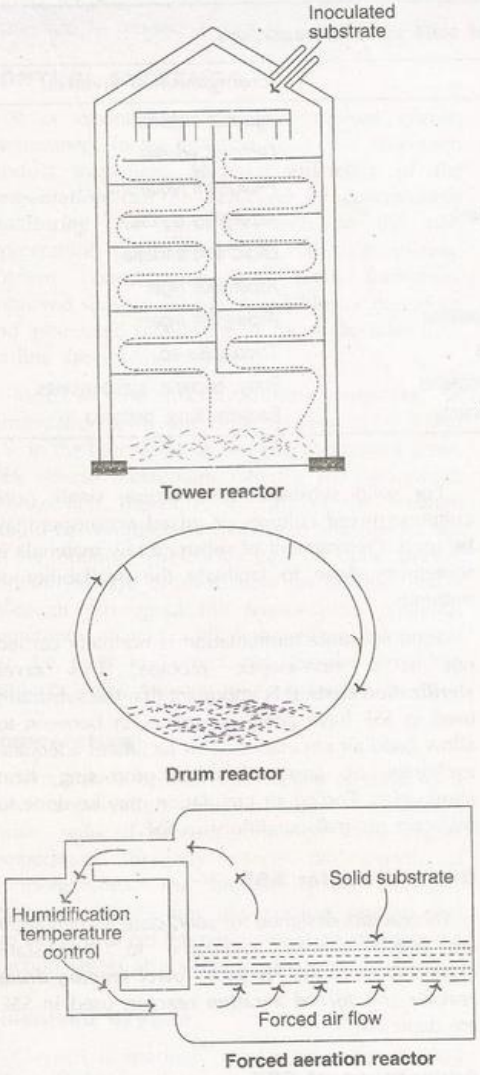


Fig. 19.6 : Bioreactors for solid–substrate fermentation.

- Many domestic, industrial and agricultural wastes can be fruitfully used in SSF.

Limitations of SSF

- The microorganisms that tolerate only low moisture content can be used.
- Precise monitoring of SSF (e.g., O_2 and CO_2 levels, moisture content) is not possible.
- The organisms grow slowly and consequently there is a limitation in product formation.
- Heat production creates problems, and it is very difficult to regulate the growth environment.

MEDIA (SUBSTRATES) FOR INDUSTRIAL FERMENTATION

The media used for the growth of microorganisms in industrial fermentation ***must contain all the elements in a suitable form for the synthesis of cellular substances as well as the metabolic products***. While designing a medium, several factors must be taken into consideration. The most important among them is the ultimate product desired in the fermentation. For growth-linked products (primary metabolites e.g. ethanol, citric acid), the product formations is directly dependent on the growth of the organisms, hence the medium should be such that it supports good growth. On the other hand, for products which are not directly linked to the growth (secondary metabolites e.g. antibiotics, alkaloids, gibberellins), the substrate requirements for product formation must also be considered.

In the laboratory, pure defined chemicals may be used for culturing microorganisms. However, for industrial fermentations, undefined and complex substrates are frequently used for economic reasons. Cheaper substrates are advantageous since they minimize the production cost of the fermented products. Wastes from agriculture, and byproducts of other industries are generally preferred, although they are highly variable in composition. Raw materials used in fermentation largely depend on their cost at a particular time, since there are seasonal variations.

The choice of the medium is very critical for successful product formation. For industrial fermentation, the microorganisms, in general, utilize a ***luxury metabolism***. Therefore, good production yields are expected with an abundant supply of carbon and nitrogen sources, besides requisite growth factors. The media used in fermentation processes may be ***synthetic*** or ***crude***.

Synthetic media

Media with all the requisite constituents in a pure form in the desired proportion represents synthetic media. Use of this type of media in fermentations is not practicable.

Crude media

The non-synthetic media with naturally available sources are better suited for fermentation.

In practice, crude media with an addition of requisite synthetic constituents is ideal for good product yield in fermentation.

The most frequently used substrates for industrial fermentation with special reference to the supply of carbon and nitrogen sources and growth factors are briefly described below.

SUBSTRATES USED AS CARBON SOURCES

Carbohydrates constitute the most predominant source of energy in fermentation industry. Refined and pure carbohydrates such as glucose or sucrose are rarely used for economic reasons.

Molasses

Molasses is a byproduct of sugar industry and is one of the ***cheapest sources of carbohydrates***. Sugar cane molasses (sucrose around 48%) and sugar beet molasses (sucrose around 33%) are commonly used. Besides being rich in sugar, molasses also contain nitrogenous substances, vitamins and trace elements. There occurs variation in the composition of the molasses which mostly depends on the climatic conditions and production process. ***Hydrol molasses***, a byproduct in glucose production from corn, is also used as a fermentation substrate.

Malt extract

Malt extract, an aqueous extract of malted barely, contains about 80% carbohydrates (glucose, fructose, sucrose, maltose). Nitrogen compounds constitute around 4.5% (proteins, peptides, amino acids, purines, pyrimidines).

Starch, dextrin and cellulose

The polysaccharides-starch, dextrin and cellulose can be metabolised by microorganisms. They are frequently used for the industrial production of alcohol. Due to its wide availability and low cost, the use of ***cellulose for alcohol production*** is extensively studied.

Whey

Whey is a ***byproduct of dairy industry*** and is produced worldwide. Most of it is consumed by humans and animals. Whey is a reasonably good source of carbon for the production of alcohol, single-cell protein, vitamin B_{12}, lactic acid and gibberellic acid. Storage of whey is a limiting factor for its widespread use in fermentation industry.

Methanol and ethanol

Some of the microorganisms are capable of utilizing methanol and/or ethanol as carbon source. ***Methanol is the cheapest substrate for fermentation***. However, it ***can be utilized by only a few bacteria and yeasts***. Methanol is commonly used for the production of single-cell protein. Ethanol is rather expensive. However, at present it is used for the production of acetic acid.

SUBSTRATES USED AS NITROGEN SOURCES

The nitrogen supply to the fermentation microorganisms may come from inorganic or organic sources.

Inorganic nitrogen sources

Ammonium salts and free ***ammonia*** are cheap inorganic nitrogen sources, particularly in industrialised countries. However, not all the microorganisms are capable of utilising them, hence their use is limited.

Organic nitrogen sources

Urea is fairly a good source of nitrogen. However, other cheaper organic forms of nitrogen sources are preferred.

Corn steep liquor : This is formed during starch production from corn. Corn steep liquor is rich in nitrogen (about 4%) and is very efficiently utilized by microorganisms. It is rich in several amino acids (alanine, valine, methionine, arginine, threonine, glutamate).

Yeast extracts : They contain about 8% nitrogen and are rich in amino acids, peptides and vitamins.

Glucose formed from glycogen and trehalose during yeast extraction is a good carbon source. Yeast extracts are produced from baker's yeast through autolysis (at 50–55°C) or through plasmolysis (high concentration of NaCl). Yeast extracts are very good sources for many industrially important microorganisms.

Soy meal : After extracting the soy bean oil from the soy bean seeds, the left out residue is soy meal. It is rich in proteins (about 50%) as well as carbohydrates (about 30%) contents. Soy meal is often used in antibiotic production.

Peptones : The protein hydrolysates are collectively referred to as peptones, and they are good sources for many microorganisms. The sources of peptones include meat, soy meal, peanut seeds, cotton seeds and sunflower seeds. The proteins namely ***casein, gelatin and keratin can also be hydrolysed to yield peptones.*** In general, peptones derived from animal sources have more nitrogen content while those from plant sources have more carbohydrate content. Peptones are relatively more expensive, hence not widely used in industries.

SOURCES OF GROWTH FACTORS

Some of the microorganisms are not capable of synthesising one or more growth factors such as vitamins. These growth factors are very expensive in pure form, hence crude sources are preferred. Yeast extract is a rich source of almost all growth factors.

Generally, the substrates derived from plant or animal sources in a crude form are reasonably rich in mineral content. Sometimes, however mineral (phosphate, sulfate) supplementation may be required.

STERILIZATION OF CULTURE MEDIA AND GASES

For successful fermentation, it is absolutely essential to ensure.

- Sterility of the media containing the nutrients.
- Sterility of incoming and outgoing air.
- Sterility of the bioreactor.
- Prevention of contamination during fermentation.

A brief account on the sterilization of the bioreactor has already been described (See p...). A bioreactor can be sterilized by destroying the organisms by heat/chemicals/radiation or sometimes by physical procedures such as filtration. Sterilization of media and air are discussed below.

STERILIZATION OF CULTURE MEDIA

The constituents of culture media, water and containers contribute to the contamination by vegetative cells and spores. The media must be free from contamination before use in fermentation. Sterilization of the media is most commonly achieved by applying heat, and to a lesser extent by other means (physical methods, chemical treatment, radiation).

Heat sterilization

Heat is the ***most widely used*** sterilization technique. The quality and quantity of contamination (i.e., the type and load of microorganisms), composition of the media and its pH and size of the suspended particles are the important factors that influence the success of heat sterilization. In general, ***vegetative cells are destroyed at lower temperature*** in a short time (around 60°C in 5–10 minutes). However, destruction of ***spores requires higher temperature*** and relatively ***longer time*** (around 80°C for 15–20 minutes). Spores of *Bacillus stearothermophilus* are the most heat resistant. In fact, this organism is exploited for testing the sterility of fermentation equipment.

Physical methods

The physical methods such as ***filtration, centrifugation,*** and ***adsorption*** (to ion-exchangers or activated carbon) are in use. Among these, filtration is most widely used. Certain constituents (vitamins, blood components, antibiotics) of culture media are heat labile and therefore, are destroyed by heat sterilization. Such components of the medium are completely dissolved (absolutely essential or else they will be removed along with microorganisms) and then subjected to filter sterilization. There are a couple of limitations of filtration technique.

1. Application of high pressure in filtration is unsuitable for industries.

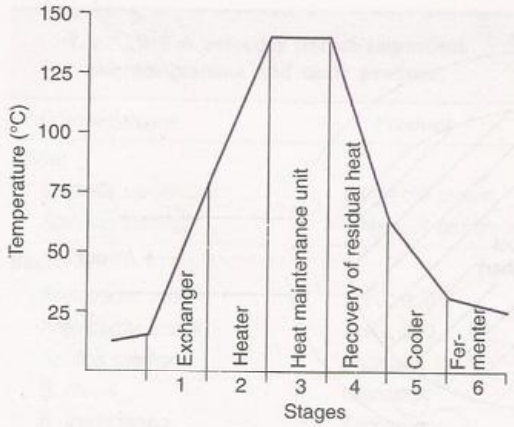


Fig. 19.7 : Different stages in continuous sterilization process in relation to temperature.

2. Some of the media components may be lost form the media during filtration.

Sometimes, a combination of filtration and heat sterilization are applied. For instance, the water used for media preparation is filtered while concentrated nutrient solution is subjected to heat sterilization. The filtered water is now added for appropriate dilution of the media.

The chemical methods (by using disinfectants) and ***radiation procedures*** (by using UV rays, γ rays, X-rays) are not commonly used for media sterilization.

Batch sterilization

The culture media are subjected to ***sterilization at 121°C in batch volumes, in the bioreactor.*** Batch sterilization can be done by injecting the steam into the medium (direct method) or injecting the steam into interior coils (indirect method). For the direct batch sterilization, the steam should be pure, and free from all chemical additives (that usually come from steam manufacturing process). There are two disadvantages of batch sterilization.

1. **Damage to culture media :** Alteration in nutrients, change in pH and discolouration of the culture media are common.

2. **High energy consumption :** It takes a few hours (2-4 hrs) for the entire contents of the bioreactor to attain the requisite temperature (i.e. 120°C). Another 20–60 minutes for the actual process of sterilization, followed by cooling for 1-2 hours. All this process ***involves wastage of energy***, and therefore batch sterilization is quite costly.

Continuous sterilization

Continuous sterilization is ***carried out at 140°C for a very short period of time ranging from 30 to 120 seconds.*** (This is in contrast to the batch fermentation done at 121°C for 20–60 minutes). This is based on the principle that the time required for killing microorganisms is much shorter at higher temperature. Continuous sterilization is carried out by directly injecting the steam or by means of heat exchangers. In either case, the temperature is very quickly raised to 140°C, and maintained for 30–120 seconds. The stages of continuous sterilization process and the corresponding temperatures are depicted in ***Fig. 19.7***. The different stages are— ***exchanger, heater, heat maintenance unit, recovery of residual heat, cooling and fermenter.***

In the continuous sterilization process, 3 types of heat exchangers are used. The first heat exchanger raises temperature to 90–120°C within 20–30 seconds. The second exchanger further raises temperature to 140°C and maintains for 30–120 seconds. The third heat exchanger brings down the temperature by cooling in the next 20–30 seconds. The actual time required for sterilization depends on the size of the suspended particles. The bigger is the size, the more is the time required.

The ***main advantage*** with continuous sterilization is that ***about 80–90% of the energy is conserved.*** The limitation however, is that certain compounds in the medium precipitate (e.g., calcium phosphate, calcium oxalate) due to very high temperature differences that occur in a very short time between sterilization and cooling. The starch-containing culture media becomes viscous in continuous sterilization and therefore is not used.

STERILIZATION OF AIR

In general, the industrial fermentations are carried out under vigorous and continuous aeration. ***For an effective fermentation, the air should be completely sterile, and free from all micro-organisms*** and suspended particles. There is a wide

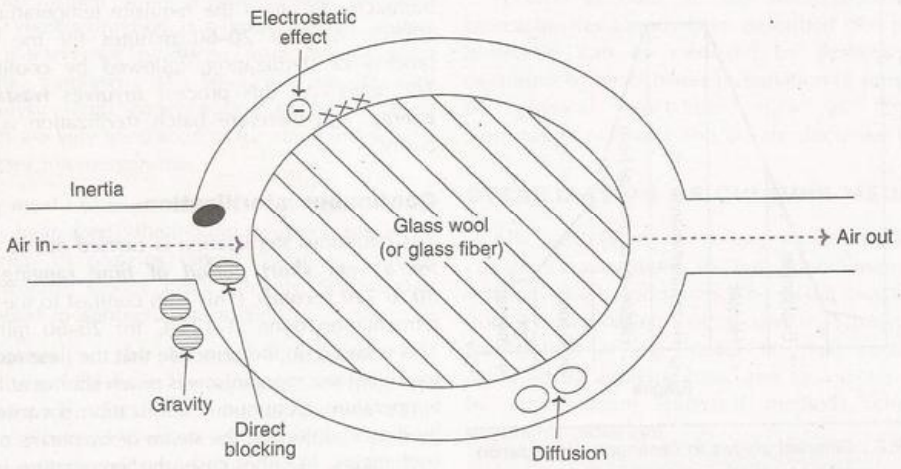


Fig. 19.8 : Use of depth filter in air sterilization.

variation in the quantity of suspended particles and microbes in the atmospheric outdoor air. The microorganisms may range from 10–2,000/m^3 while the suspended particles may be 20–100,00/m^3. Among the microorganisms present in the air, the fungal spores (50%) and Gram-negative bacteria (40%) dominate.

Air or other gases can be sterilized by filtration, heat, UV radiation and gas scrubbing. Among these, heat and filtration are most commonly used.

Air sterilization by heat

In the early years, air was passed over electrically heated elements and sterilized. But this is quite expense, hence not in use these days.

Air sterilization by filtration

Filtration of air is the most commonly used sterilization in fermentation industries.

Depth filters : When the air is passed through a glass wool containing depth filters the particles are trapped and removed (***Fig. 19.8***). This filtration technique primarily involves physical effects such as inertia, blocking, gravity, electrostatic attraction and diffusion. Glass wool filters can be subjected to steam sterilization and reused. But there is a limitation in their reuse since glass wool shrinks and solidifies on steam sterilization. In recent years, glass fiber filter cartridges (that do not have the limitations of glass wool filter) are being used.

Membrane cartridge filters : These are removable pleated membrane filters made up of cellulose ester, nylon or polysulfone. Membrane cartridge filters are smaller in size, simpler for operation and replacement.

The most important limitation of air sterilization is that there is no ***filter that can remove bacteriophages***. Bacteriophages are capable of crippling the industrial fermentation. e.g., bacteriophages interfere in the production of glutamic acid by *Corynebacterium glutamicum*.

ISOLATION OF MICROORGANISMS

There are over a million species of microorganisms widely distributed in nature. Less than 1% of the world's microorganisms have been studied. In fact, only ***a few hundred species are important for industrial use***. A selected list of organisms along with their products is given in ***Table 19.3***.

The good sources for the isolation of microorganisms are soils, lakes and river muds. It is estimated that a gram of soil contains 10^6–10^8

TABLE 19.3 A selected list of important microorganisms and their products

Microorganism	*Product*
Algae	
Chlorella sorokiniana	Single-cell protein
Spirulina maxima	Single-cell protein
Bacteria	
Acetobacter aceti	Acetic acid
Acetobacter woodii	Acetic acid
Bacillus subtilis	Bacitracin
B. brevis	Gramicidin
B. thuringiensis	Endotoxin
Clostridium aceticum	Acetic acid
Methylophilus methylotrophus	Glutamic acid
Pseudomonas denitrificans	Vitamin B_{12}
Actinomycetes	
Streptomyces aureofaciens	Tetracycline
S. griseus	Streptomycin
S. tradiae	Neomycin
Nocardia mediterranei	Rifamycin
Micromonospora purpurea	Gentamycin
Fungi	
Aspergillus niger	Citric acid
A. oryzae	Amylase, cellulase, single-cell protein
Candida lipolytica	Lipase
C. utilis	Single–cell protein
Penicillium chrysogenum	Penicillin
Saccharomyces cerevisiae	Ethanol, wine, single-cell protein
S. lipolytica	Citric acid, single-cell protein
Rhizopus nigricans	Steroids
Gibberella fujikuroi	Gibberellin
Trichoderma viride	Cellulase

bacteria, 10^4–10^6 actinomycete spores and 10^2–10^4 fungal spores. The common techniques employed for the isolation of microorganisms are given below.

1. Direct sponge of the soil
2. Soil dilution
3. Gradient plate method (pour plate and streak plate technique)
4. Aerosol dilution
5. Flotation
6. Centrifugation.

For full details on the isolation procedures, the reader must refer books on microbiology. The actual technique for the isolation of microorganisms depends on the source and the physiological properties of microorganisms. The general scheme adopted for isolating microorganisms from soil or water source is given below.

- The sample (soil or water) is diluted with sterile water to which an emulsifying agent (Tween) is added.
- Sample is throughly mixed and allowed to stand at room temperature.
- Supernatant is diluted, 10^{-1} to 10^{-10}.
- Various culture media are inoculated with diluted samples and incubated.
- Colonies from the plates are isolated and identified.
- The required pure strains are maintained and preserved.

Enrichment methods for isolation of microorganisms

The culture conditions can be appropriately modified to isolate certain types of microorganisms. The types of organisms that can be isolated by use of enrichment methods is given in ***Table 19.4***. For instance, thermophiles can be isolated by using high temperature while acidophiles can be isolated in acidic pH. Enrichment methods are certainly useful for quick isolation of specific types of organisms.

Strains of microorganisms from unusual environments

Biotechnologists often prefer to isolate microorganisms from very extreme and unusual environments. This is done with a hope that such strains may be capable of producing new products of industrial importance. The unusual environments such as ***cold habitats, high altitudes, deserts, deep sea and petroleum fields*** are constantly being tried for this purpose. The enrichment methods described above (***Table 19.4***) will be very useful for isolating unusual strains.

Table 19.4 Types of microorganisms that can be isolated by enrichment methods

Type of organisms	*Enrichment method*
Thermophiles	High temperature (42–100°C)
Psychrotrophs	Low temperature (5–15°C)
Acidophiles	Low pH (2-4)
Halophiles	High NaCl concentration
Anaerobes	N_2 atmosphere
Actinoplanes	Pollen grains
Myxobacteria	Wood bark

Screening of metabolites for isolation of microorganisms

The microorganisms can be tested directly for the product formation, and isolated. In fact, the water or soil samples can be directly used or suitably diluted for metabolite screening. Agar plates can be used for screening metabolites formed from the microorganisms. For instance, if the required product is an antibiotic, then the test system consists of the strains of organisms which inhibit the zones, on the agar plates. The inhibitory activity indicates the possible presence of some antibiotic being produced by the microorganisms. Another example is the isolation of microorganisms producing amylases. When grown on agar plates containing starch, and then stained with iodine, amylase-producing organisms can be identified and isolated.

Screening for new metabolites, and isolation of microorganisms

Industrial microbiologists continue their search for newer metabolites produced by microorganisms. Research work is particularly directed for identifying chemotherapeutically important products for the treatment of tumors, bacterial diseases (newer antibiotics against resistant strains) and viral diseases, besides several other substances (e.g. hormones, enzyme inhibitors). In addition, isolation of microorganisms for improvement of food industry, and for efficient degradation of the environmental pollutants and hazardous chemicals also assumes significance.

PRESERVATION OF MICROORGANISMS

There are distinct methods for preservation of microorganisms. The most important being storage by ***refrigeration***, ***freezing*** and ***lyophilization***.

MICROBIAL METABOLIC PRODUCTS — LOW MOLECULAR WEIGHT COMPOUNDS

The microorganisms possess tremendous capacity to produce a wide range of products that have commercial value. The primary and secondary metabolisms and bioconversions of microorganisms with special reference to their importance for the formation of biotechnologically important products are discussed hereunder (***Fig. 19.9***). Microbial growth in relation to primary and secondary metabolisms is depicted in ***Fig. 19.10***.

PRIMARY METABOLITES

Primary metabolism, also referred to as ***trophophase***, is characterized by balanced growth of microorganisms. It occurs when all the nutrients needed by the organisms are provided in the medium. Primary metabolism is essential for the very existence and reproduction of cells. In the trophophase, the cells possess optimal concentrations of almost all the macromolecules (proteins, DNA, RNA etc.).

It is during the period of trophophase, an exponential growth of microorganisms occurs. Several metabolic products, collectively referred to as ***primary metabolites***, are produced in trophophase (i.e., during the period of growth). The primary metabolites are divided into two groups.

1. **Primary essential metabolites :** These are the ***compounds*** produced in adequate quanties to ***sustain cell growth e.g. vitamins, amino acids, nucleosides***. The native microorganisms usually do not overproduce essential primary metabolites, since it is a wasteful exercise. However, for industrial overproduction, the regulatory mechanisms are suitably manipulated.

2. **Primary metabolic end products :** These are the normal and traditional ***end products of fermentation process of primary metabolism***. The end products may or may not have any significant function to perform in the microorganisms,

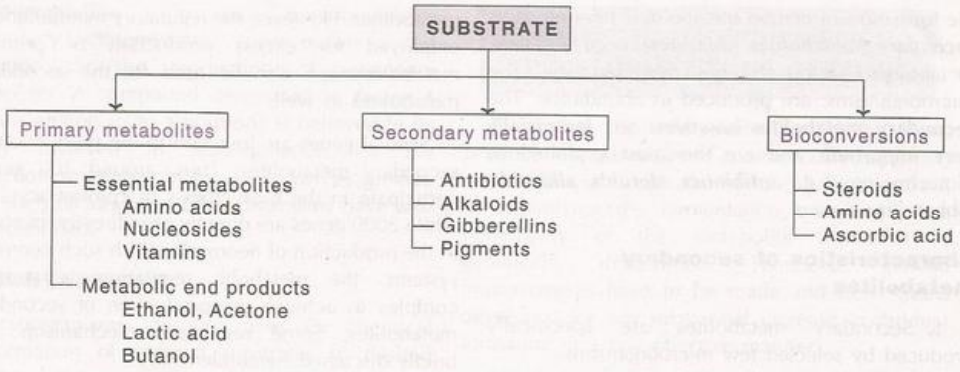


Fig. 19.9 : Different types of low–molecular weight compounds produced by microorganisms.

although they have many other industrial applications e.g. ethanol, acetone, lactic acid.

Carbon dioxide is a metabolic end product of *Saccharomyces cerevisiae*. This CO_2 is essential for leavening of dough in baking industry.

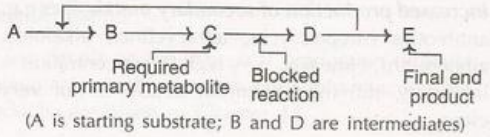
Limitations in growth : Due to insufficient/ limited supply of any nutrient (substrate or even O_2), the growth rate of microorganisms slows down. However, the metabolism does not stop. It continues as long as the cell lives, but the formation of products differs.

Overproduction of primary metabolites

Excessive production of primary metabolites is very important for their large scale use for a variety of purposes. Overproduction of several metabolites has been successfully accomplished by eliminating the feedback inhibition as briefly described below.

1. By using auxotrophic mutants with a block in one of the steps in the biosynthetic pathway concerned with the formation of primary metabolite (this should be an intermediate and not the final end product). In this manner, the end product (E) formation is blocked, hence no feedback inhibition. But overproduction of the required metabolite (C) occurs as illustrated below.

Feedback regulation :



(A is starting substrate; B and D are intermediates)

In the above example, an unbranched pathway is shown. This type of manipulation for overproduction of metabolites can be done for branched metabolic pathways also.

2. Mutant microorganisms with antimetabolite resistance which exhibit a defective metabolic regulation can also overproduce primary metabolites.

SECONDARY METABOLITES

As the exponential growth of the microorganisms ceases (i.e. as the trophophase ends), they enter ***idiophase***. Idiophase is characterized by ***secondary metabolism*** wherein

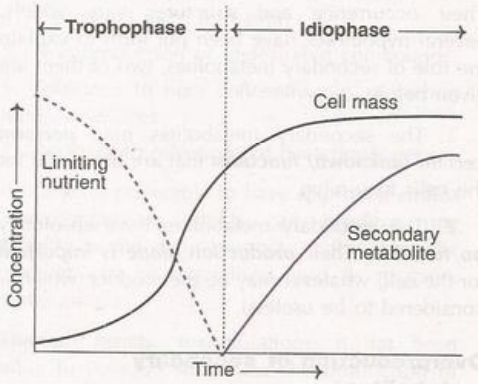


Fig. 19.10 : Microbial growth in relation to primary (trophophase) and secondary (idiophase) metabolism.

the formation of certain metabolites, referred to as ***secondary metabolites*** (***idiolites***) occurs. These metabolites, although not required by the microorganisms, are produced in abundance. The ***secondary metabolites*** however, are ***industrially very important***, and are the most exploited in biotechnology e.g., ***antibiotics, steroids, alkaloids,*** gibberellins, toxins.

Characteristics of secondary metabolites

1. Secondary metabolites are specifically produced by selected few microorganisms.

2. They are not essential for the growth and reproduction of organisms from which they are produced.

3. Environmental factors influence the production of secondary metabolites.

4. Some microorganisms produce secondary metabolites as a group of compounds (usually structurally related) instead of a single one e.g. about 35 anthracyclines are produced by a single strain of *Streptomyces*.

5. The biosynthetic pathways for most secondary metabolites are not clearly established.

6. The regulation of the formation of secondary metabolites is more complex and differs from that of primary metabolites.

Functions of secondary metabolites

As already stated, secondary metabolites are not essential for growth and multiplication of cells. Their occurrence and structures vary widely. Several hypotheses have been put forth to explain the role of secondary metabolites, two of them are given below.

1. The secondary metabolites may ***perform certain (unknown) functions*** that are beneficial for the cells to survive.

2. The secondary metabolites have absolutely ***no function***. Their ***production alone is important*** for the cell, whatever may be the product (which is considered to be useless).

Overproduction of secondary metabolites

As already stated, the production of secondary metabolites is more complex than primary metabolites. However, the regulatory manipulations employed for excess production of primary metabolites can also be used for the secondary metabolites as well.

Several genes are involved in the production of secondary metabolites. Thus, around 300 genes participate in the biosynthesis of chlortetracycline while 2000 genes are directly or indirectly involved in the production of neomycin. With such complex systems, the metabolic regulation is equally complex to achieve overproduction of secondary metabolites. Some regulatory mechanisms are briefly discussed hereunder.

Induction : Addition of methionine induces certain enzymes and ***enhances*** the production of ***cephalosporin***. Tryptophan regulates ergot alkaloid biosynthesis.

End product regulation : Some of the secondary metabolites inhibit their own biosynthesis, a phenomenon referred to as end product regulation e.g. penicillin, streptomycin, puromycin, chloramphenicol. It is possible to isolate ***mutants*** that are ***less sensitive to end product inhibition***, and in this manner the ***secondary metabolite production can be increased***.

Catabolite regulation : In this regulation process, a key enzyme involved in a catabolic pathway is inactivated, inhibited or repressed by adding a commonly used substrate. Catabolic repression can be achieved by using carbon or nitrogen sources. The mechanism of action of catabolite regulation is not very clearly understood.

The most commonly used ***carbon*** source is ***glucose***. It is found to ***inhibit the production of several antibiotics*** e.g. penicillin, streptomycin, bacitracin, chloramphenicol, puromycin.

The ***nitrogen sources*** such as ***ammonia*** also act as catabolite regulators (i.e. inhibitors) for the ***overproduction of certain antibiotics***.

Phosphate regulation : Inorganic phosphate (Pi) is required for the growth and multiplication of prokaryotes and eukaryotes. ***Increasing Pi concentration*** (up to 1mM) is associated with an ***increased production of secondary metabolites*** e.g. antibiotics (streptomycin, tetracycline), alkaloids, gibberellins. However, very high Pi concentration is inhibitory, the mechanism of action is not very clear.

Autoregulation : In some microorganisms (particularly actinomycetes), there occurs a self regulation for the production of secondary metabolites. A compound designated as ***factor A*** which is analogous to a hormone is believed to be closely involved in autoregulation for the production streptomycin by *Streptomyces griseus*. More such factors from other organisms have also been identified.

BIOCONVERSIONS

Microorganisms are also used for chemical transformation of unusual substrates to desired products. This process, also referred to as ***biotransformation***, is very important in producing several compounds e.g. conversion of ethanol to acetic acid (in vinegar), sorbitol to sorbose, synthesis of steroid hormones and certain amino acids.

In bioconversion, microorganisms convert a compound to a structurally related product in one or a few enzymatic reactions. The bioconversions can be carried out with resting cells, spores or even killed cells. Non-growing cells are preferred for bioconversions, since high substrate concentration can be used, besides washing the cells easily (to make them free from contamination).

Sometimes, mixed cultures are used for bioconversions to carry out different reactions. In recent years, the yield of bioconversion is increased by using immobilized cells at a lower cost. For more details on biotransformation, the reader must refer Chapter 22.

MICROBIAL METABOLIC PRODUCTS — HIGH MOLECULAR WEIGHT COMPOUNDS

Microorganisms are also used for the production of macromolecules i.e. high molecular weight compounds e.g. polysaccharides, proteins (including enzymes). Recombinant DNA technology has drastically improved the industrial production of various macromolecules in a cost-effective manner.

The details on the microbial production of high molecular weight compounds (pharmaceutical products) are given in Chapter 15.

GENETIC IMPROVEMENT (DEVELOPMENT) OF STRAINS

In general, the ***wild strains*** of microorganisms ***produce low quantities of commercially important metabolites***, although the yield can be increased by optimizing the fermentation conditions. The potentiality of the metabolite formation is genetically determined. Therefore, genetic improvements have to be made and new strains developed for any substantial increase in product formation in a cost-effective manner.

There are strain development programmes (mutation and recombination) to increase the product yield by 100 times or even more. The nature of the desired product determines the success associated with strain improvement. For example, if alterations in one or two genes (i.e. one or 2 key enzymes) can improve the product yield, it is simpler to achieve the target. This type of approach is sometimes possible with primary metabolites. As regards the secondary metabolites, the product formation and its regulation are quite complex. Hence, several genetic modifications have to be done to finally produce high-yielding strains. Ideally speaking, the ***improved strains*** should possess the following ***characteristics*** (as many as possible) to finally result in high product formation.

1. Shorter time of fermentation
2. Capable of metabolising low-cost substrates
3. Reduced O_2 demand
4. Decreased foam formation
5. Non-production of undesirable compounds
6. Tolerance to high concentrations of carbon or nitrogen sources
7. Resistant to infections of bacteriophages.

It is always preferable to have ***improved strains*** of microorganisms which ***can produce one metabolite*** as the main product. In this way, the production can be maximised, and its recovery becomes simpler.

Through genetic manipulations, it has been possible to develop strains for the production of modified or new metabolites which are of commercial value e.g. modified or newer antibiotics.

The ***major limitation*** of strain improvement is that for most of the industrially important microorganisms, there is lack of detailed information on the genetics, and molecular biology. This hinders the new strain development.

METHODS OF STRAIN DEVELOPMENT

There are three distinct approaches for improvement of strains-mutation, recombination and recombinant DNA technology.

MUTATION

Any change that occurs in the DNA of a gene is referred to as mutation. Thus, mutations result in a structural change in the genome. Mutations may be ***spontaneous*** (that occur naturally) or ***induced*** by mutagenic agents.

The spontaneous mutations occur at a very low frequency, and usually are not suitable for industrial purposes. Mutations may be induced by mutagenic agents such as ultraviolet light, various chemicals (nitrous oxide, nitrosoguanidine, hydroxylamine). Site-directed mutagenesis (For details Refer Chapter 10) is also important for strain improvement.

Selection of mutants

Selection and isolation of the appropriate mutant strains developed is very important for their industrial use. Two techniques commonly employed for this purpose are briefly described.

Random screening

The mutated strains are randomly selected and checked for their ability to produce the desired industrial product. This can be done with model fermentation units. The ***strains with maximum yield can be selected***. Random screening is costly and tedious procedure. But many a times, this is the only way to find the right strain of mutants developed.

Selective isolation of mutants

There are many methods for selective isolation of improved strains.

1. **Isolation of antibiotic resistant strains :** The mutated strains are grown on a selective medium containing an antibiotic. The wild strains are killed while the mutant strains with antibiotic resistance can grow. Such strains may be useful in industries.

Table 19.5 Antimetabolites used for screening of some common natural metabolites

Natural metabolite	*Antimetabolite(s)*
Amino acids	
Arginine	Canavanine
Histidine	2-Thiazolalanine
Valine	α-Aminobutyric acid, isoleucine
Leucine	4-Azaleucine
Methionine	α-Methylmethionine, ethionine
Phenylalanine	p-Fluorophenylalanine,
Tyrosine	p-Fluorophenylalanine
Tryptophan	5 (or 6)-Methyltryptophan
Threonine	β-Hydroxynorleucine
Proline	3, 4-Dehydroproline
Vitamins	
Thiamine	Pyrithiamine
Pyridoxine	Isoniazid
Niacin	3–Acetylpyridine
p–Aminobenzoic acid	Sulfonamide
Nitrogenous bases	
Adenine	2, 6–Diaminopurine
Guanine	8–Azaxanthine
Uracil	5–Fluorouracil

2. **Isolation of antimetabolite resistant strains :** Antimetabolites which have structural similarities with metabolites can block the normal metabolic pathways and kill the cells. The mutant strains resistant to antimetabolites can be selected for industrial purposes. In the ***Table 19.5***, a selected list antimetabolites used for screening the metabolites is given.

3. **Isolation of auxotrophic mutants :** An auxotrophic mutant is characterized by a defect in one of the biosynthetic pathways. As a result, it requires a specific compound for its normal growth. For instance, tyr mutants of *Corynebacterium glutamicus* require tyrosine for their growth while they can accumulate phenylalanine. The isolation of such mutants can be done by growing them on a complete agar medium that can specifically support the biochemically defective mutant.

GENETIC RECOMBINATION

The strain improvement can be made by combining genetic information from two genotypes, by a process called genetic recombination. The recombination can be brought out by ***transformation, transduction, conjugation*** (Chapter 6) and ***protoplast fusion*** (Chapter 44). There are many advantages of genetic recombination.

1. By crossing high product yielding mutant strains with wild-type strains, the fermentation process can be further increased.

2. Different mutant strains with high-yielding properties can be combined by recombination.

3. There is gradual decline in the product yield after each stage of mutation, due to undesirable mutations. This can be prevented by using recombination.

PRINCIPLES OF MICROBIAL GROWTH AND CULTURE SYSTEMS

The growth of microorganisms is a highly complex and coordinated process, ultimately expressed by increase in cell number or cell mass. The process of growth depends on the availability of requisite nutrients and their transport into the cells, and the environmental factors such as aeration, O_2 supply, temperature and pH.

Doubling time refers to the time period required for ***doubling*** the weight of the ***biomass*** while ***generation time*** represents the period for ***doubling the cell numbers***. Doubling times normally increase with increasing cell size and complexicity as given below.

Bacteria	0.30 – 1 hour
Yeasts	1 – 2 hours
Animal cells	25 – 48 hours
Plant cells	20 – 70 hours

In general, when all other conditions are kept ideal, growth of the microorganisms is dependent on the substrate (nutrient) supply. The microorganisms can be grown in ***batch, fed-batch, semicontinuous*** or ***continuous culture*** systems in a bioreactor.

A diagrammatic representation of microbial cell growth in relation to substrate is depicted in ***Fig. 19.11***. In batch fermentation, the growth medium containing the substrates is inoculated with microorganisms, and the fermentation proceeds without the addition of fresh growth medium. In fed-batch fermentation, substrates are added at short time intervals during fermentation. In batch and fed-batch fermentation, the growth of the cells is quite comparable. And in both cases, growth medium is not removed until the end of fermentation process.

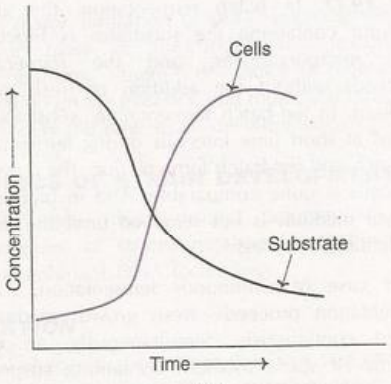
In case of continuous fermentation, as the fermentation proceeds, fresh growth medium is added continuously. Simultaneously, an equal volume of spent medium containing suspended microoganisms is removed. This enables the cells to grow optimally and continuously (Fig. ***19.11C***).

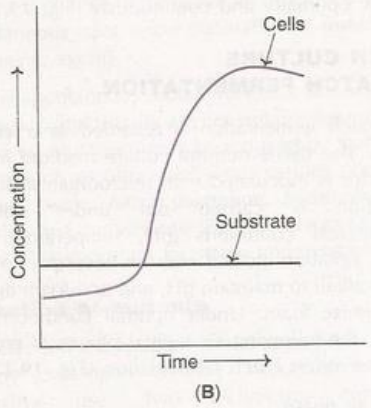
BATCH CULTURE OR BATCH FERMENTATION

A batch fermentation is regarded as a ***closed system***. The sterile nutrient culture medium in the bioreactor is inoculated with microorganisms. The incubation is carried out under optimal physiological conditions (pH, temperature, O_2 supply, agitation etc.). It may be necessary to add acid or alkali to maintain pH, and anti-foam agents to minimise foam. Under optimal conditions for growth, the following six typical phases of growth are observed in batch fermentation (***Fig. 19.12***).

1. Lag phase
2. Acceleration phase
3. Logarithmic (log) phase (exponential phase)
4. Deceleration phase
5. Stationary phase
6. Death phase.

1. **Lag phase :** The ***initial brief period of culturing*** after inoculation is referred to as lag phase. During the lag phase, the microorganisms adapt to the new environment—available nutrients, pH etc. There is no increase in the cell number, although the cellular weight may slightly increase. The length of the lag phase is variable and is mostly determined by the new set of physiological conditions, and the phase at which the micro-organisms were existing when inoculated. For instance, lag phase may not occur if the culture inoculated is at exponential phase (i.e., log phase), and growth may start immediately.





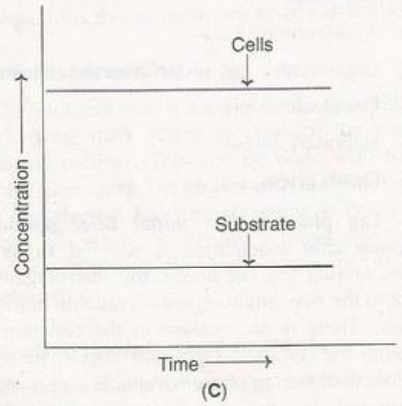


Fig. 19.11 : Diagrammatic representation of microbial cell growth in relation to substrate (A) Batch fermentation (B) Fed–batch fermentation (C) Continuous fermentation.

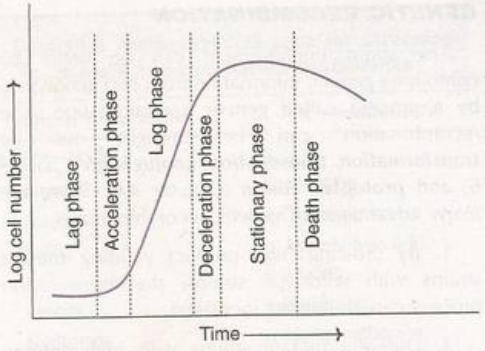


Fig. 19.12 : Pattern of microbial cell growth in batch culture or batch fermentation.

2. **Acceleration phase :** This is a brief transient period during which ***cells start growing slowly***. In fact, acceleration phase connects the lag phase and log phase.

3. **Log phase :** The most ***active growth of microorganisms and multiplication occur*** during log phase. The cells undergo several doublings and the cell mass increases. When the number of cells or biomass is plotted against time on a ***semilogarithmic graph, a straight line*** is obtained, hence the term log phase. Growth rate of microbes in log phase is independent of substrate (nutrient supply) concentration as long as excess substrate is present, and there are no growth inhibitors in the medium. In general, the specific growth rate of microorganisms for simpler substrates is greater than for long chain molecules. This is explained on the basis of extra energy needed to split long chain substrates.

Two log phases are observed when a complex nutrient medium with two substrates is used in fermentation, and this phenomenon is referred to as ***diauxy***. This happens since one of the substrates is preferentially metabolised first which represses the breakdown of second substrate. After the first substrate is completely degraded second lag phase occurs, during which period, the enzymes for the breakdown of second substrate the synthesized. Now a second log phase occurs.

More details on log phase with special reference to growth kinetics of microorganisms are discussed later.

4. **Deceleration phase :** As the growth rate of microorganisms during log phase decreases, they enter the deceleration phase. This phase is usually very short-lived and may not be observable.

5. **Stationary phase :** As the substrate in the growth medium gets depleted, and the metabolic end products that are formed inhibit the growth, the cells enter the stationary phase. The ***microbial growth may either slow down or completely stop***. The biomass may remain almost constant during stationary phase. This phase, however, is frequently associated with dramatic changes in the metabolism of the cells which may produce compounds (secondary metabolites) of biotechnological importance e.g. production of antibiotics.

6. **Death phase :** This phase is associated with ceasation of metabolic activity and depletion of energy reserves. The ***cells die at an exponential rate*** (a straight line may be obtained when the number of surviving cells are plotted against time on a semilogarithmic plot). In the commercial and industrial fermentations, the growth of the microorganisms is halted at the end of the log phase or just before the death phase begins, and the cells are harvested.

FED-BATCH CULTURE OR FED-BATCH FERMENTATION

Fed-batch fermentation (See ***Fig. 19.11***) is an improvement of batch fermentation wherein the ***substrate is added in increments at different times throughout the course of fermentation*** (**Note :** In batch culture method, substrate is added only at the beginning of the fermentation). Periodical substrate addition ***prolongs log and stationary phases*** which results in an increased biomass. Consequently, production of metabolites (e.g. antibiotics) during stationary phase is very much increased.

As it is difficult to directly measure substrate concentration in fed-batch fermentation, other indicators that correlate with substrate consumption are used. The formation of organic acids, production of CO_2 and changes in pH may be measured, and accordingly substrate addition carried out. In general, fed-batch fermentation requires more careful monitoring than batch fermentation, and is therefore not a preferred method by industrial biotechnologists.

Fed-batch fermentation for the production of recombinant proteins

In recent years, fed-batch fermentation has become popular, due to very high yield, for the production of recombinant proteins. Depending on the microorganism and the nature of recombinant protein, the fed-batch fermentation can increase the product yield from 25% to 1000% compared to batch fermentation. Careful monitoring of the fermentation reaction and appropriate addition of substrates (carbon and nitrogen sources, and trace metals) substantially increases the product yield.

Limitations : The major limitation of fed batch fermentation is that the microorganisms in the stationary phase produce proteolytic enzymes or proteases. These enzyme attack the recombinant proteins that are being produced. By carefully monitoring the fermentation, the log phase can be prolonged and the onset of stationary phase is delayed. By this way, the formation of proteases can be minimised.

Fed-batch cultures for higher organisms

Fed-batch cultures are successfully employed for mammalian and insect cells. This is very advantageous for the production of human therapeutic proteins with good yield.

SEMI-CONTINUOUS CULTURE OR SEMI-CONTINUOUS FERMENTATION

Some of the products of fermentation are growth-linked, and such products are formed at the end of the log phase e.g. ethanol production. In semi-continuous fermentation, a ***portion of the culture medium is removed from the bioreactor and replaced by fresh medium*** (identical in nutrients, pH, temperature etc.). This process of culture medium change can be repeated at appropriate intervals. In the semi-continuous fermentation, the lag phase and other non-productive phases are very much shortened. The product output is much higher compared to batch culture systems. Semi-continuous fermentation technique has been successfully used in the industrial production of alcohol.

There are however, certain disadvantages of semi-continuous fermentation. These include the technical difficulties of handling bioreactors, long

culture periods that may lead to contamination, mutation and mechanical breakdown.

CONTINUOUS CULTURE OR CONTINUOUS FERMENTATION

Continuous fermentation is an ***open system***. It involves the ***removal of culture medium continuously and replacement of this with a fresh sterile medium*** in a bioreactor. Both addition and removal are done at the same rate so that the working volume remains constant. Further, to maintain a steady state condition in continuous process, it is advisable that the cell loss as a result of outflow is balanced by growth of the organisms. The two common types of continuous fermentation and bioreactors are described below (***Fig. 19.13***).

Homogeneously mixed bioreactors

In this type, the culture solution is homogeneously mixed, and the bioreactors are of two types

Chemostat bioreactors : The concentration of any one of the substrates (carbohydrate, nitrogen source, salts, O_2) is adjusted to control the cell growth and maintain a steady state.

Turbidostat bioreactors : In this case, turbidity measurement is used to monitor the biomass concentration. The rate of addition of nutrient solution can be appropriately adjusted to maintain a constant cell growth.

Plug flow bioreactors

In plug flow bioreactors, the culture solution flows through a tubular reaction vessel without back mixing. The composition of the medium, the quantity of cells, O_2 supply and product formation vary at different locations in the bioreactor. Microorganisms along with nutrient medium are continuously added at the entrance of the bioreactor.

Industrial applications of continuous fermentation

Continuous fermentation processes have been used for the ***production of antibiotics, organic solvents, single-cell protein, beer and ethanol, besides waste-water treatment.***

Advantages of continuous fermentation

1. The size of the bioreactor and other equipment used in continuous fermentation are relatively smaller compared to batch fermentation for the production of the same quantity of product.

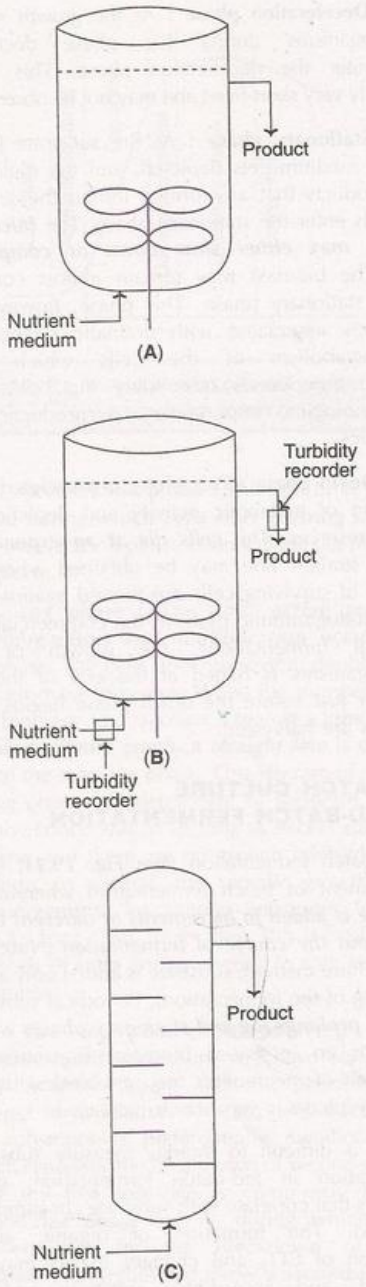


Fig. 19.13 : Continuous fermentation bioreactors (A) Chemostat bioreactor (B) Turbidostat bioreactor (C) Plug flow bioreactor.

2. The ***yield of the product is more consistent*** since the physiological state of the cells is uniform.

3. The 'down time' between two successive fermentations for cleaning and preparing the bioreactor for reuse is avoided in continuous fermentation.

4. Continuous fermentation can be run in a ***cost-effective*** manner.

Disadvantages of continuous fermentation

Despite many advantages of continuous fermentation (described above), it is not very widely used in industries. Some of the drawbacks are listed.

1. Continuous fermentation may run continuously for a period of 500 to 1,000 hours. ***Maintenance of sterile conditions for such a long period is difficult.***

2. The recombinant cells with plasmid constructs cannot function continuously and therefore the product yield decreases.

3. It is not easy to maintain the same quality of the culture medium for all the additions. Nutrient variations will alter the growth and physiology of the cells, and consequently the product yield.

In addition to the disadvantages listed above, industrial biotechnologists are rather reluctant to switch over to continuous fermentation from the batch fermentation. However, it is expected that continuous fermentation will also become, popular in due course.

GROWTH KINETICS OF MICROORGANISMS

The different types of fermentation processes-batch, fed-batch, semi-continuous and continuous are described above. The kinetics of microbial growth with special reference to log phase of batch fermentation are briefly discussed here.

After completion of lag phase, the cell enters log phase which is characterized by exponential growth (See ***Fig. 19.12***). If the initial number of cells is N_0, then

After 1st generation, the cell number will be $N_0 \times 2^1$.

After 2nd generation, the cell number will be $N_0 \times 2^2$.

After 3rd generation, $N_0 \times 2^3$, and so on. Thus, the number of cells after a given time (Nt) will be as follows :

$$Nt = N_0 \times 2^n$$

where n is the number of generations.

The term ***doubling time*** (td) or ***mean generation time*** (MGT) refers to the ***time taken for doubling the cell number or biomass***. The specific growth rate constant expressed by μ, is the direct measure of rate of growth of the organism. If N is the number of cells at a given time, then the increase in the number of cells (growth rate) with time is given by the formula.

$$\frac{dN}{dt} = \mu N \qquad (1)$$

If X is the biomass concentration at a given time, then the increase in the biomass (growth rate) with time is given by.

$$\frac{dX}{dt} = \mu X \qquad (2)$$

In general, the specific growth rate (μ) is a function of the concentration of limiting substrate (S), the maximum specific growth rate (μ_{max}) and a substrate specific constant (K_s). Their relationship was expressed by ***Monond*** by the following ***equation***

$$\mu = \mu_{max} \frac{S}{K_s + S} \qquad (3)$$

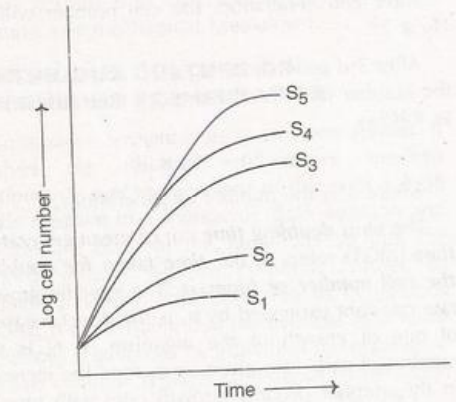
Both S and K_s are expressed as concentrations e.g., in moles or grams per liter.

The growth rate (μ) of an organism is not fixed but it is variable depending on the environmental conditions such as concentration of substrate and temperature. At a low concentration, the substrate is the limiting factor for growth (***Fig. 19.14A***). The ***Fig. 19.14B*** represents the growth rate for a given substrate concentration (by plotting μ against S).

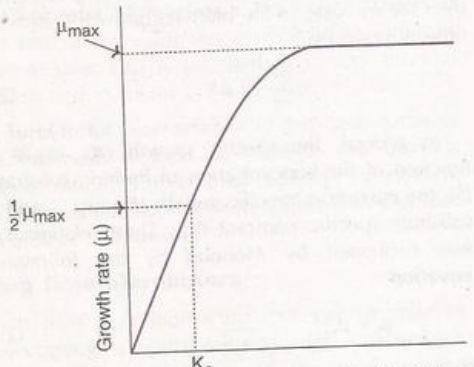
In batch culture, the substrate is initially present at a higher concentration i.e. (S) > K_s, hence the equation (3) is approximately 1.

$$\frac{S}{K_s + S} = 1$$

Thus, $\mu = \mu_{max}$.



(A)



(B)

Fig. 19.14 : Growth curves for unicellular organism in batch culture (A) With increasing concentrations of a substrate ($S_1<S_2<S_3<S_4<S_5$) (B) The effect of a given substrate concentration on growth rate (K_s = substrate concentration to produce half-maximal growth rate)

When the substrate concentration is low, as usually occurs at the end of growth phase, then,

$$\frac{S}{K_s + S} < 1$$

Hence $\mu < \mu_{max}$.

CLASSIFICATION OF FERMENTATION PROCESSES

There are different ways of classifying the fermentation processes. The major classification as batch, fed-batch, semi-continuous and continuous fermentation processes are already described in some detail (See p. 259). Another classification, based on the product formation in relation to energy metabolism is briefly discussed below (***Fig. 19.15***).

Type I fermentation

When the ***product*** is formed directly ***from the primary metabolism*** used for energy production, it is referred to as type I and may be represented as.

Substrate A ⟶ Product

Substrate A ⟶ B ⟶ C ⟶ D ⟶ Product

Growth, energy metabolism and product formation almost run in a parallel manner (***Fig. 19.15A***). In this type, trophophase and iodophase are not separated from each other e.g. production of ethanol, gluconic acid and single-cell protein.

Type II fermentation

In type II category, the product is also formed from the substrate used for primary energy metabolism. However, the ***product is produced in the secondary pathway***, as illustrated below.

Substrate A → B → C → D···Primary metabolism

↳ E → F → G → Product

At the beginning, the growth of the microorganisms is accompanied by high substrate utilization with little or no product formation. Now the growth is slowed down but the substrate consumption is high, and this is coupled with product formation. As is evident from ***Fig. 19.15B***, in type II fermentation, the trophophase and idiophase are separate. Production of some amino acids, citric acid and itaconic acid are good examples of type II fermentation.

Type III fermentation

There is a clear distinction between the primary metabolism and product formation in type III fermentation (***Fig. 19.15C***) as they occur at separate times. Substrate consumption and ***rapid growth occur in the first phase and the product formation***

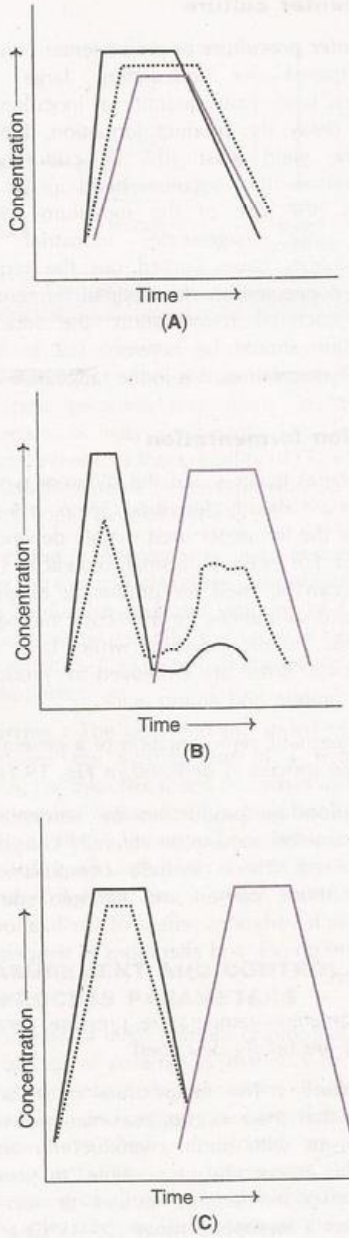


Fig. 19.15 : Types of fermentations in relation to product formation (A) Type I fermentation (B) Type II fermentation (C) Type III fermentation (— represents growth rate; ····· correspond to substrate consumption; — indicates product formation).

occurs in the second phase. The product is formed from amphibolic metabolic pathways and not from primary metabolism e.g. production of vitamins and antibiotics.

Overlap of different types of fermentations

Types I, II and III fermentations, originally categorized by Garden (in 1959) are not very rigid. There are intermediate forms based on the composition of the nutrient culture medium, strain of the microorganism used and product formation. For instance, industrial production of lactic acid falls between type I and II, while production of the antibiotic amyloglycoside is intermediate between types II and III.

It is sometimes difficult to categorize the industrial fermentations under any one of these types (I, II, III) due to complex nature of the process e.g. mycelium producing microorganisms in relation to antibiotic production.

THE FERMENTATION PROCESS

The fermentation process basically ***consists of inoculum preservation, inoculum build-up, prefermenter culture*** and finally ***production fermentation***. A brief account of the four stages of fermentation is given below.

Inoculum preservation (culture maintenance)

The preservation of high-yielding strains of microorganisms for fermentation is very important for product formation in substantial amounts. The ultimate purpose of preservation is to maintain the strains, as long as possible, without cell division. There are different methods of preservation.

Storage at low (2-6°C) temperature : In this method, the microorganisms can be stored in a refrigerator in liquid culture or as stab culture. Although this is the easiest method of preservation, there is a high risk of contamination.

Storage by freezing : The microbial cultures can be frozen and preserved for several years. In the freezers, the preservation can be done at –18°C or, at –80°C. For preservation at –196°C, liquid nitrogen must be used. It is very important that the freezing (and later thawing when required) is done slowly (usually with a change of 1°C/min) to

prevent damage and killing of the microorganisms. If proper care is not taken, as many as 95% of the cells may be killed by freezing and thawing.

Storage by lyophilization : Preservation of microorganisms by lyophilization (i.e., freeze drying) is the best method, although, it requires special equipment. In fact, ***lyophilization is the method of choice*** by many fermentation biotechnologists.

The storage of microorganisms can be done by any one of the three techniques described above. However, for each method, optimal conditions for preservation must be worked out for each strain separately. In general, the preserved ***master strains*** are cultivated once in two years for checking of their activity. When needed for use, the ***working strains*** can be obtained from the master strains.

Inoculum build up

The preserved cultures have to be revived for their industrial use. This can be done by growing the cultures in liquid or on solid media. The actual process and the conditions used for inoculum build-up largely depend on the preservation technique used. There are wide variations in the growth times which depend on the type of preservation and the organisms used as given below.

Refrigerated cultures (2-6°C) :

Bacteria 6–24 hours

Actinomycetes 1-3 days

Fungi 1-5 days

Frozen cultures (18°C, –80°C, –196°C) :

Bacteria 6–48 hours

Actinomycetes 1-5 days

Fungi 1-7 days

Lyophilized cultures :

For all organisms 4–10 days

For proper growth, and to obtain sufficient quantity of inoculum, a series of cultures are prepared. For good fermentation yield, the number of cells and spores, nutrient medium, temperature and age of the inoculum are important.

The inoculum build-up is suspended in a surface-active agent such as Tween 80 and transferred to the bioreactor for fermentation.

Prefermenter culture

Fermenter preculture or prefermenter culture is often required for inoculating large sized bioreactors. Inadequate quantity of inoculum will not only delay the product formation, but also reduce the yield drastically. By culturing the microorganisms (the inoculum build-up) in small fermenters, the size of the inoculum can be increased for large-scale industrial use. Biotechnologists have worked out the requisite inoculum concentrations for optimal fermentation e.g., for bacterial fermentation, the inoculum concentration should be between 0.2 to 3.0%; for fungal fermentation, it is in the range of 5–10%.

Production fermentation

The general features and the different types of bioreactors are already described (See p. 239-244). The size of the fermenter used mainly depends on the product. For example, a small bioreactor (1–20 litre size) can be used for producing diagnostic enzymes and substances for molecular biology by recombinant microorganisms, while large bioreactors (≥ 450 litres) are employed for producing single-cell protein and amino acids.

A diagrammatic representation of a generalized fermentation process is depicted in ***Fig. 19.16***.

For appropriate production by fermentation, several parameters need to be carefully considered and optimized. These include composition of nutrient medium, carbon and nitrogen sources, batch to batch variations, effect of sterilization on nutrients and on pH, and alterations in temperature and aeration.

The parameters—temperature, pressure, aeration and stirring are briefly described.

Temperature : The temperature must be so maintained that there occurs maximal growth of microorganisms with optimal product formation, although this is not always possible. In general, there are two temperature ranges to run the fermentations a ***mesophile range*** (20–45°C) and a ***thermophile range*** (> 45°C). Sometimes, two different temperatures are used for the same fermentation process—a higher temperature is employed for good growth (in trophophase), and then the temperature is decreased for optimizing product formation (in idiophase).

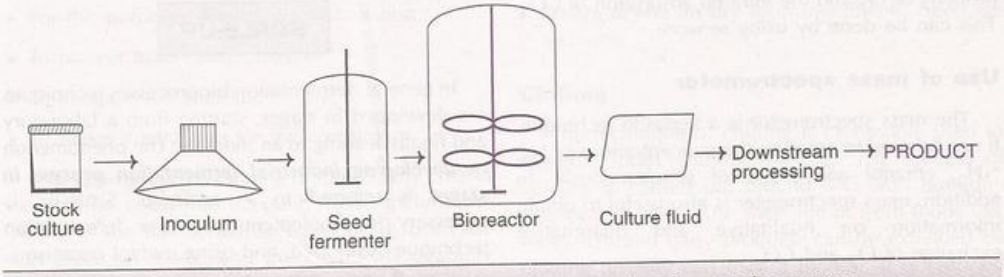


Fig. 19.16 : Diagrammatic representation of a generalized fermentation process.

Pressure : Appropriate maintenance of hydrostatic pressure, particularly in large sized bioreactors is very important. This is because pressure influences the solubility of O_2 and CO_2 in the culture medium. An overpressure in the range 0.2–0.5 bar is generally used.

Aeration : A bioreactor gets aerated by the supply of O_2 and therefore, adjustment must be made to furnish required amount of O_2 to the microorganisms. Usually, the aeration rate is in the range of 0.25–1.25 *vvm* (volume of air/volume of liquid/minute).

Stirring : The type and the speed of impellers determines the stirring rate in a fermenter. In general, the impeller speed decreases as the size of the fermenter increases. Thus, for a small bioreactor (size 1–20 litres), the impeller speed is in the range of 250–350 rpm, while for a large bioreactor (size around 450 litres, the impeller speed is 60–120 rpm.

MEASUREMENT AND CONTROL OF BIOPROCESS PARAMETERS

There are a large number of physical, chemical and biological parameters that can be measured during fermentation/bioprocessing (***Table 19.6***) for data analysis and appropriate control. Some ***special sensors*** have been developed to carry out measurements in the bioreactors. The basic requirement of all the sensors is that they must be sterilizable. The measurements of the parameters (listed in Table) can be done either directly in the bioreactor or in the laboratory.

pH measurement : There are pH electrodes that can withstand high temperature (sterilization) pressure and mechanical stresses, and yet measure the pH accurately. Combination electrodes (reference electrode, glass electrode) are being used. In fact, electrodes are also available for measuring several other inorganic ions.

O_2 and CO_2 measurement : Oxygen electrodes and CO_2 electrodes can be used to measure O_2 and CO_2 concentrations respectively. The electrodes are amperometric in nature. They are however, susceptible for damage on sterilization.

In a commonly used technique, O_2 and CO_2 respectively can be measured by the magnetic

TABLE 19.6 Important parameters that can be measured during bioprocessing

Physical parameters
Temperature
Pressure
Flow rates
Viscosity
Turbidity
Power consumption
Chemical parameters
pH
Substrate concentration
Product concentration
O_2 concentration (dissolved)
Waste gases concentration (e.g. CO_2)
Ionic strength
Biological parameters
Activities of specific enzymes
Protein concentration
Energetics (ATP concentration)
DNA/RNA content

property of O_2 and the infrared absorption of CO_2. This can be done by using sensors.

Use of mass spectrometer

The mass spectrometer is a versatile technique. It can be used to measure the concentrations of N_2, NH_3, ethanol and methanol simultaneously. In addition, mass spectrometer is also useful to obtain information on qualitative and quantitative exchange of O_2 and CO_2.

Use of gas-permeable membranes

The measurement of dissolved gases, up to 8 simultaneously, can be done almost accurately by using gas-permeable membranes. The advantage is that such measurement is possible to carry out in the nutrient medium.

Use of computers

Computers are used in industrial biotechnology for data acquisition, data analysis and developing fermentation models.

Data acquisition : By employing ***on-line sensors*** and computers in fermentation system, data can be obtained with regard to the concentration of O_2 and CO_2, pH, temperature, pressure, viscosity, turbidity, aeration rate etc. Certain other parameters (e.g. nutrient concentration, product formation, biomass concentration) can be measured in the laboratory i.e. ***off-line measurements***. The information collected from on-line and off-line measurements can be entered into a computer. In this fashion, the entire data regarding a fermentation can be processed, stored and retrieved.

Data analysis : The data collected on a computer can be used for various calculations e.g. rate of substrate utilization, rate of product formation, rates of O_2 uptake and CO_2 formation, heat balance, respiratory quotient. Through computer data analysis, it is possible to arrive at the optimal productivity for a given fermentation system.

Development of fermentation models : The computer can be used to develop mathematical models of fermentation processes. These models in turn will be useful to have a better control over fermentation systems with high productivity in a cost-effective manner.

SCALE-UP

In general, fermentation/bioprocesses techniques are developed in stages, starting from a laboratory and finally leading to an industry. The phenomenon of ***developing industrial fermentation process in stages*** is referred to as scale-up. Scale-up is necessary for implementing a new fermentation technique developed, and using mutant organisms.

The very purpose of scale-up is ***to develop optimal environmental and operating conditions at different levels for a successful fermentation industry***. The conditions that need to be studied include substrate concentration, agitation and mixing, aeration, power consumption and rate of O_2 transfer.

In a conventional scale-up, a fermentation technique is developed in 3-4 stages (i.e., scales). The initial stage involves a screening process using petri dishes or Erlenmeyer flasks. The next stage is a pilot project to determine the optimal operating conditions for a fermentation process with a capacity of 5–200 litres. The final stage involves the ***transfer of technology developed in the laboratory to industry***.

It has to be continuously noted that a fermentation process that works well at the ***laboratory scale*** may work poorly or may not work at all on ***industrial scale***. Therefore, it is not always possible to blindly apply the laboratory conditions of a fermentation technique developed to industry. At the laboratory scale, one is interested in the maximum yield of the product for unit time. At the industry level, besides the product yield, minimal operating cost is another important factor for consideration. An industrial biotechnologist has to apply his/her intellect and technical skill for establishing a successful fermentation set up in a cost-effective manner.

FOOD TECHNOLOGY

Food technology is a vast field. The salient features of food processing which has some relevance to bioprocess engineering and technology are given here. In general, the food processing is carried out to acheive the following objectives.

- For the purpose of storage and transport.
- To protect from contamination
- To increase the shelf-life
- To make it attractive for the consumers.

For appropriate processing of food, many criteria need to be taken into account. These include the ability of the microorganisms and pests to invade and grow on foods, and the chemical instability and biological activity of foods.

FOOD PRESERVATION

Among the many food processing techniques, food preservation is the most important one. Selected food preservation processes are briefly described.

Drying

Exposure of foods to sunlight and drying is a natural, and an earliest method of food preservation. Drying involves the removal of water. Consequently, the moisture content of the food falls, and the invading microorganisms cannot grow.

Hot air is most frequently used to remove moisture. In actual practice, foods are dried by using different types of equipment.

- ***Shallow-bed dryers*** (drying is carried out in perforated conveyor beds).
- ***Deep-bed dryers*** (drying occurs in bins).
- ***Spray dryers*** (to dry liquid foods and food slurries).
- ***Freeze dryers*** (to dry soup ingradients, beverage extracts).

Chilling

The technique of chilling is commonly used to preserve foods without freezing. The process of chilling is carried out around 0°C with humidity between 85–95%. The shelf–life of fresh foods like meat, fish and dairy products can be extended by chilling (usually less than a month). It is reported that the if chilling is done under low oxygen conditions, the shelf-life of foods increases.

Chilling is frequently used for long distance shipping of meat, apples, vegetables etc.

Freezing

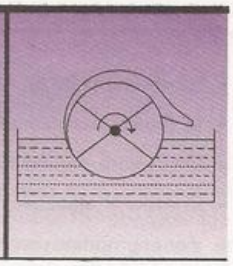
Foods can be preserved in a frozen state (–30° to –10°C). At a very low temperatures, the water activity and reactant mobility are low, hence the chemical reactions are reduced.

Foods can be frozen by using a stream of refrigerated air, or by passing inert refrigerant or cryogenic gas. Rapid freezing is advocated to minimize the adverse effects on the texture of foods.

Thermal processing

By applying heat (comparable to sterilization), majority of the microorganisms and spores present in the foods can be killed. Thermal processing is carried out in sealed containers devoid of oxygen, and thus aerobic organisms cannot grow. Further, higher temperature also leads to inactivation of enzymes, and consequently microbial growth will be retarded.

20 Downstream Processing



As the fermentation is complete (described in Chapter 19), it is necessary to recover the desired end product. The end products include antibiotics, amino acids, vitamins, organic acids, industrial enzymes, and vaccines. The ***extraction and purification of a biotechnological product from fermentation is*** referred to as ***downstream processing (DSP) or product recovery***. DSP is as complex and important as fermentation process. It often requires the expertise and technical skills of chemists, process engineers, besides bioscientists.

In the present day biotechnology, the fermentation and downstream processing are considered as an integrated system. The methodology adopted for downstream processing depends on the nature of the end product, its concentration, stability and the degree of purification required, besides the presence of other products. The product recovery yield in general is expected to be higher, if the number of steps in DSP is lower.

The desired products for isolation by DSP are most frequently metabolites which may be present as follows.

1. **Intracellular metabolites :** These products are located within the cells e.g. vitamins, enzymes.

2. **Extracellular metabolites :** They are present outside the cells (culture fluids) e.g. most antibiotics (penicillin, streptomycin), amino acids, alcohol, citric acid, some enzymes (amylases, proteases).

3. **Both intracellular and extracellular** e.g. vitamin B_{12}, flavomycin.

Sometimes, the microorganism may itself be the desired end product for isolation e.g. single-cell protein. The product recovery as a biomass is simpler. When subjected to heat, the cells aggregate and form clumps which can be separated by sedimentation.

STAGES IN DOWNSTREAM PROCESSING

Downstream processing of metabolites is a ***multistage operation***, and may be broadly divided into the following stages.

1. Solid-liquid separation
2. Release of intracellular products
3. Concentration
4. Purification
5. Formulation.

In ***Fig. 20.1***, an outline of the major steps in downstream processing is given, and they are described in some detail in the following pages.

SOLID-LIQUID SEPARATION

The first step in product recovery is the separation of whole cells (cell biomass) and other insoluble ingradients from the culture broth (**Note :** If the desired product is an intracellular metabolite, it must be released from the cells before

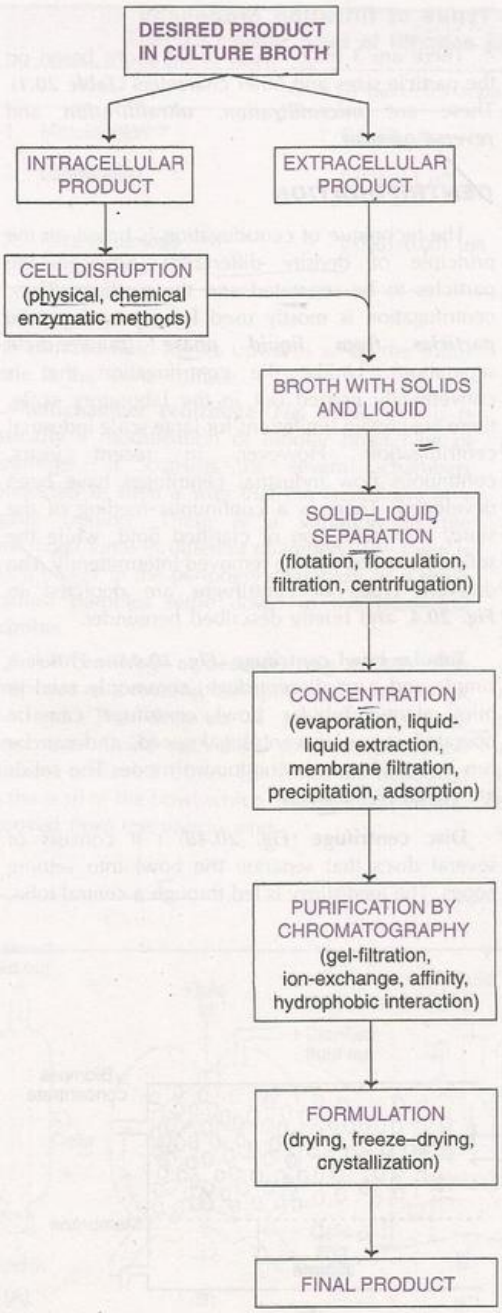


Fig. 20.1 : A summary of the major steps in downstream processing.

subjecting to solid-liquid separation. Details of cell disruption are described later). Some authors use the term ***harvesting of microbial cells*** for the separation of cells from the culture medium.

Several methods are in use for solid-liquid separation. These include flotation, flocculation, filtration and centrifugation.

FLOTATION

When a gas is introduced into the liquid broth, it forms bubbles. The cells and other solid particles get adsorbed on gas bubbles. These bubbles rise to the foam layer which can be collected and removed. The presence of certain substances, referred to as ***collector substances***, facilitates stable foam formation e.g., long chain fatty acids, amines.

FLOCCULATION

In flocculation, the cells (or cell debris) form large aggregates to settle down for easy removal. The process of flocculation depends on the nature of cells and the ionic constituents of the medium. Addition of flocculating agents (inorganic salt, organic polyelectrolyte, mineral hydrocolloid) is often necessary to achieve appropriate flocculation.

FILTRATION

Filtration is the most commonly used technique for separating the biomass and culture filtrate. The efficiency of filtration depends on many factors—the size of the organism, presence of other organisms, viscosity of the medium, and temperature. Several filters such as depth filters, absolute filters, rotary drum vacuum filters and membrane filters are in use.

Depth filters : They are composed of a filamentous matrix such as glass wool, asbestos or filter paper. The particles are trapped within the matrix and the fluid passes out. Filamentous fungi can be removed by using depth filters.

Absolute filters : These filters are with specific pore sizes that are smaller than the particles to be removed. Bacteria from culture medium can be removed by absolute filters.

Rotary drum vacuum filters : These filters are frequently used for separation of broth containing 10–40% solids (by volume) and particles in the size of 0.5–10μm. Rotary drum vacuum filters have been successfully used for filtration of yeast cells

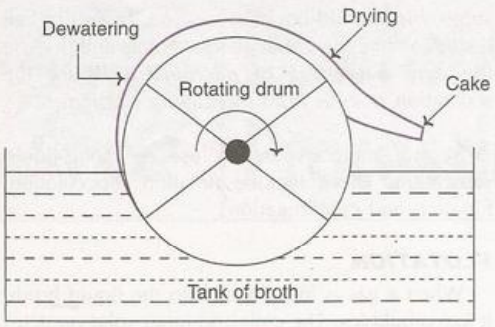


Fig. 20.2 : Diagrammatic representation of a rotary drum vacuum filter.

and filamentous fungi. The equipment is simple with low power consumption and is easy to operate. The filtration unit consists of a rotating drum partially immersed in a tank of broth (***Fig. 20.2***). As the drum rotates, it picks up the biomass which gets deposited as a cake on the drum surface. This filter cake can be easily removed.

Membrane filters : In this type of filtration, membranes with specific pore sizes can be used. However, clogging of filters is a major limitation. There are two types of membrane filtrations—***static filtration*** and ***cross-flow filtration*** (***Fig. 20.3***). In cross-flow filtration, the culture broth is pumped in a crosswise fashion across the membrane. This reduces the clogging process and hence better than the static filtration.

Types of filtration processes

There are 3 major types of filtrations based on the particle sizes and other characters (***Table 20.1***). These are ***microfiltration***, ***ultrafiltration*** and ***reverse osmosis***.

CENTRIFUGATION

The technique of centrifugation is based on the principle of density differences between the particles to be separated and the medium. Thus, centrifugation is mostly used for ***separating solid particles from liquid phase*** (fluid/particle separation). Unlike the centrifugation that is conveniently carried out in the laboratory scale, there are certain limitations for large scale industrial centrifugation. However, in recent years, continuous flow industrial centrifuges have been developed. There is a continuous feeding of the slurry and collection of clarified fluid, while the solids deposited can be removed intermittently. The different types of centrifuges are depicted in ***Fig. 20.4***, and briefly described hereunder.

Tubular bowl centrifuge (***Fig. 20.4A***) **:** This is a simple and a small centrifuge, commonly used in pilot plants. Tubular bowl centrifuge can be operated at a high centrifugal speed, and can be run in both batch or continuous mode. The solids are removed manually.

Disc centrifuge (***Fig. 20.4B***) **:** It consists of several discs that separate the bowl into settling zones. The feed/slurry is fed through a central tube.

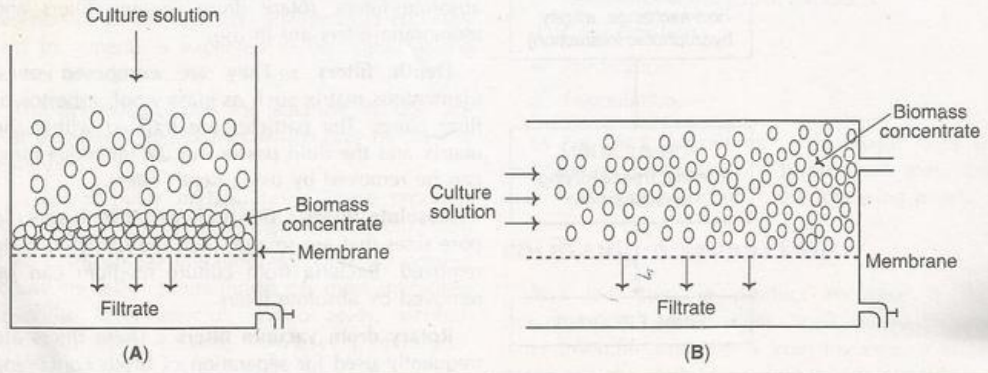


Fig. 20.3 : Filter systems for separation of biomass and culture filtrate (A) Static–flow filtration (B) Cross–flow filtration.

Table 20.1 Major types of filtration processes with characteristic features

	Type	Sizes of particles separated	Compound or particle separated
1.	Microfiltration	0.1–10 μm	Cells or cell fractions, viruses.
2.	Ultrafiltration	0.001–0.1 μm	Compounds with molecular weights greater than 1000 (e.g. enzymes).
3.	Reverse osmosis (hyperfiltration)	0.0001–0.001 μm	Compounds with molecular weights less than 1000 (e.g. lactose).

The clarified fluid moves upwards while the solids settle at the lower surface.

Multichamber centrifuge (*Fig. 20.4C*) : This is basically a modification of tubular bowl type of centrifuge. It consists of several chambers connected in such a way that the feed flows in a zigzag fashion. There is a variation in the centrifugal force in different chambers. The force is much higher in the periphery chambers, as a result smallest particles settle down in the outermost chamber.

Scroll centrifuge or decanter (*Fig. 20.4D*) : It is composed of a rotating horizontal bowl tapered at one end. The decanter is generally used to concentrate fluids with high solid concentration (biomass content 5–80%). The solids are deposited on the wall of the bowl which can be scrapped and removed from the narrow end.

RELEASE OF INTRACELLULAR PRODUCTS

As already stated, there are several biotechnological products (vitamins, enzymes) which are located within the cells. Such compounds have to be first released (maximally and in an active form) for their further processing and final isolation. The microorganisms or other cells can be disintegrated or disrupted by ***physical, chemical*** or ***enzymatic methods***. The outline of different techniques used for breakage of cells is given in ***Fig. 20.5***. The selection of a particular method depends on the nature of the cells, since there is a wide variation in the property of cell disruption or breakage. For instance, Gram-negative bacteria and filamentous fungi can be more easily broken compared to Gram-positive bacteria and yeasts.

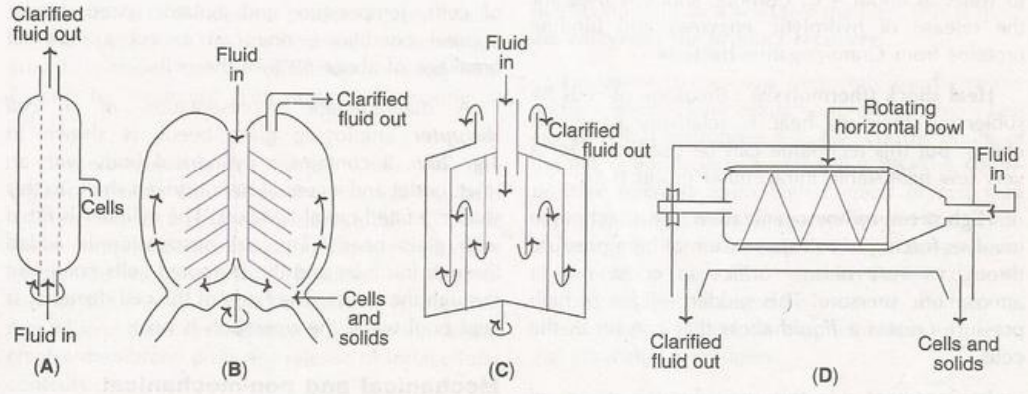


Fig. 20.4 : Centrifuges commonly used in downstream processing (A) Tubular bowl centrifuge (B) Disc centrifuge (C) Multichamber centrifuge (D) Scroll centrifuge (decanter).

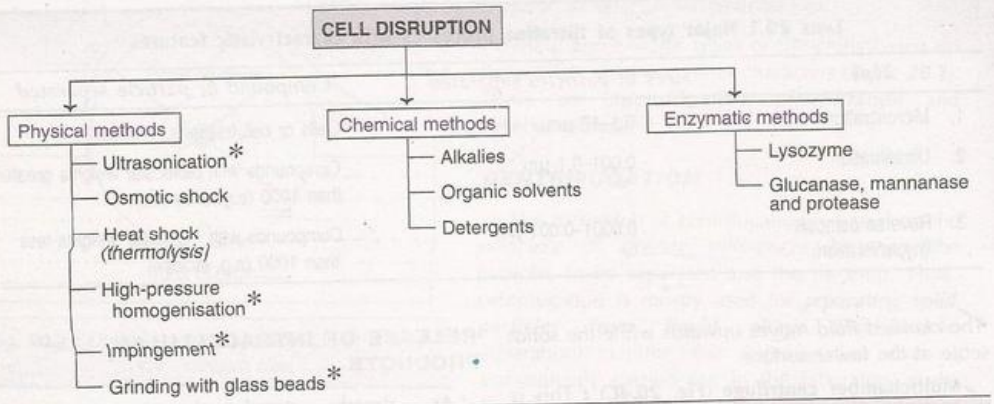


Fig. 20.5 : Major methods for cell disruption to release the intracellular products (indicate mechanical methods while all the remaining are non-mechanical).*

CELL DISRUPTION

Physical methods of cell disruption

The microorganisms or cells can be disrupted by certain physical methods to release the intracellular products.

Ultrasonication : Ultrasonic disintegration is widely employed in the laboratory. However, due to high cost, it is not suitable for large-scale use in industries.

Osmotic shock : This method involves the suspension of cells (free from growth medium) in 20% buffered sucrose. The cells are then transferred to water at about 4°C. Osmotic shock is used for the release of hydrolytic enzymes and binding proteins from Gram-negative bacteria.

Heat shock (thermolysis) : Breakage of cells by subjecting them to heat is relatively easy and cheap. But this technique can be used only for a very few heat-stable intracellular products.

High pressure homogenization : This technique involves forcing of cell suspension at high pressure through a very narrow orifice to come out to atmospheric pressure. This sudden release of high pressure creates a ***liquid shear*** that can break the cells.

Impingement : In this procedure, a stream of suspended cells at high velocity and pressure are forced to hit either a stationary surface or a second stream of suspended cells (impinge literally means to strike or hit). The cells are disrupted by the forces created at the point of contact. ***Microfluidizer*** is a device developed based on the principle of impingement. It has been successfully used for breaking *E. coli* cells. The advantage with impingement technique is that it can be effectively used for disrupting cells even at a low concentration.

Grinding with glass beads : The cells mixed with glass beads are subjected to a very high speed in a reaction vessel. The cells break as they are forced against the wall of the vessel by the beads. Several factors influence the cell breakage-size and quantity of the glass beads, concentration and age of cells, temperature and agitator speed. Under optimal conditions, one can expect a maximal breakage of about 80% of the cells.

A diagrammatic representation of a ***cell disrupter*** employing glass beeds is shown in ***Fig. 20.6***. It contains a cylindrical body with an inlet, outlet and a central motor-driven shaft. To this shaft are fitted radial agitators. The cylinder is fitted with glass beads. The cell suspension is added through the inlet and the disrupted cells come out through the outlet. The body of the cell disrupter is kept cool while the operation is on.

Mechanical and non-mechanical methods

Among the ***physical methods of cell disruption*** described above, ultrasonication, high-pressure

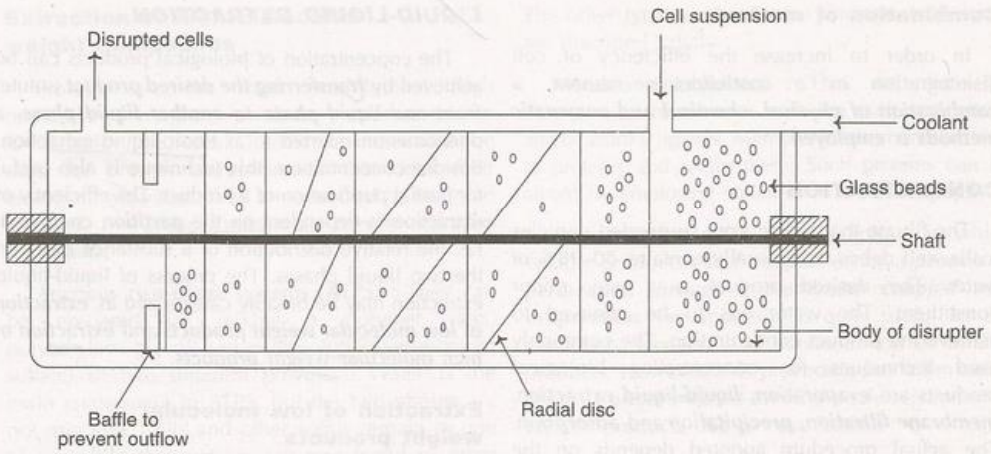


Fig. 20.6 : Diagrammatic representation of a cell disrupter.

homogenisation, impigement and grinding with glass beads are ***mechanical*** while osmotic shock and heat shock are non-mechanical. The ***chemical and enzymatic methods*** (described below) ***are non-mechanical in nature.***

Chemical methods of cell disruption

Treatment with alkalies, organic solvents and detergents can lyse the cells to release the contents.

Alkalies : Alkali treatment has been used for the extraction of some bacterial proteins. However, the alkali stability of the desired product is very crucial for the success of this method e.g., recombinant growth hormone can be efficiently released from *E. coli* by treatment with sodium hydroxide at pH 11.

Organic solvents : Several water miscible organic solvents can be used to disrupt the cells e.g., methanol, ethanol, isopropanol, butanol. These compounds are inflammable, hence require specialised equipment for fire safety. The organic solvent toluene is frequently used. It is believed that toluene dissolves membrane phospholipids and creates membrane pores for release of intracellular contents.

Detergents : Detergents that are ionic in nature, ***cationic***-cetyl trimethyl ammonium bromide or ***anionic***-sodium lauryl sulfate can denature membrane proteins and lyse the cells. Non-ionic detergents (although less reactive than ionic ones) are also used to some extent e.g., Triton X-100 or Tween. The problem with the use of detergents is that they affect purification steps, particularly the salt precipitation. This limitation can be overcome by using ultrafiltration or ion-exchange chromatography for purification.

Enzymatic methods of cell disruption

Cell disruption by enzymatic methods has certain advantages i.e., lysis of cells occurs under mild conditions in a selective manner. This is quite ***advantageous for product recovery.***

Lysozyme is the most frequently used enzyme and is commercially available (produced from hen egg white). It hydrolyses β-1, 4-glycosidic bonds of the mucopeptide in bacterial cell walls. The ***Gram-positive bacteria*** (with high content of cell wall mucopeptides) are more susceptible for the action of lysozyme. For Gram-negative bacteria, lysozyme in association with EDTA can break the cells. As the cell wall gets digested by lysozyme, the osmotic effects break the periplasmic membrane to release the intracellular contents.

Certain other enzymes are also used, although less frequently, for cell disruption. For the lysis of yeast cell walls, glucanase and mannanase in combination with proteases are used.

Combination of methods

In order to increase the efficiency of cell disintegration in a cost-effective manner, a ***combination of physical, chemical and enzymatic methods is employed.***

CONCENTRATION

The filtrate that is free from suspended particles (cells, cell debris etc.) usually contains 80–98% of water. The desired product is a very minor constituent. The water has to be removed to achieve the product concentration. The commonly used techniques for concentrating biological products are ***evaporation, liquid-liquid extraction, membrane filtration, precipitation*** and ***adsorption***. The actual procedure adopted depends on the nature of the desired product (quality and quantity to be retained as far as possible) and the cost factor.

EVAPORATION

Water in the broth filtrate can be removed by a simple evaporation process. The evaporators, in general, have a heating device for supply of steam, and unit for the separation of concentrated product and vapour, a condenser for condensing vapour, accessories and control equipment. The capacity of the equipment is variable that may range from small laboratory scale to industrial scale. Some of the important types of evaporators in common use are briefly described.

Plate evaporators : The liquid to be concentrated flows over plates. As the steam is supplied, the liquid gets concentrated and becomes viscous.

Falling film evaporators : In this case, the liquid flows down long tubes which gets distributed as a thin film over the heating surface. Falling film evaporators are suitable for removing water from viscous products of fermentation.

Forced film evaporators : The liquid films are mechanically driven and these devices are suitable for producing dry product concentrates.

Centrifugal forced film evaporators : These equipment evaporate the liquid very quickly (in seconds), hence suitable for concentrating even heat-labile substances. In these evaporators, a centrifugal force is used to pass on the liquid over heated plates or conical surfaces for instantaneous evaporation.

LIQUID-LIQUID EXTRACTION

The concentration of biological products can be achieved by ***transferring the desired product*** (solute) ***from one liquid phase to another liquid phase***, a phenomenon referred to as liquid-liquid extraction. Besides concentration, this technique is also useful for partial purification of a product. The efficiency of extraction is dependent on the ***partition coefficient*** i.e. the relative distribution of a substance between the two liquid phases. The process of liquid-liquid extraction may be broadly categorized as ***extraction of low molecular weight products*** and ***extraction of high molecular weight products***.

Extraction of low molecular weight products

By using organic solvents, the lipophilic compounds can be conveniently extracted. However, it is quite difficult to extract hydrophilic compounds. Extraction of lipophilic products can be done by the following techniques.

Physical extraction : The compound gets itself distributed between two liquid phases based on the physical properties. This technique is used for extraction of non-ionising compounds.

Dissociation extraction : This technique is suitable for the extraction of ionisable compounds. Certain antibiotics can be extracted by this procedure.

Reactive extraction : In this case, the desired product is made to react with a carrier molecule (e.g., phosphorus compound, aliphatic amine) and extracted into organic solvent. Reactive extraction procedure is quite useful for the extraction of certain compounds that are highly soluble in water (aqueous phase) e.g., organic acids.

Supercritical fluid (SCF) extraction : This technique differs from the above procedures, since the materials used for extraction are supercritical fluids (SCFs). ***SCFs are intermediates between gases and liquids*** and exist as fluids above their critical temperature and pressure. Supercritical CO_2, with a low critical temperature and pressure is commonly used in the extraction. Supercritical fluid extraction is rather expensive, hence not widely used (SCF has been used for the extraction of caffeine from coffee beans, and pigments and flavor ingradients from biological materials).

Extraction of high molecular weight compounds

Proteins are the most predominant high molecular weight products produced in fermentation industries. Organic solvents cannot be used for protein extraction, as they lose their biological activities. They are extracted by using an aqueous ***two-phase systems*** or ***reverse micelles formation***.

Aqueous two-phase systems (ATPS) : They can be prepared by mixing a polymer (e.g., polyethylene glycol) and a salt solution (ammonium sulfate) or two different polymers. Water is the main component in ATPS, but the two phases are not miscible. Cells and other solids remain in one phase while the proteins are transferred to other phase. The distribution of the desired product is based on its surface and ionic character and the nature of phases. Th separation takes much longer time by ATPS.

Reverse miceller systems : Reverse micelles are stable aggregates of surfactant molecules and water in organic solvents. The proteins can be extracted from the aqueous medium by forming reverse micelles. In fact, the enzymes can be extracted by this procedure without loss of biological activity.

MEMBRANE FILTRATION

Membrane filtration has become a common separation technique in industrial biotechnology. It can be conveniently used for the separation of biomolecules and particles, and for the concentration of fluids. The membrane filtration technique basically involves the use of a semi-permeable membrane that selectively retains the particles/molecules that are bigger than the pore size while the smaller molecules pass through the membrane pores.

Membranes used in filtration are made up of polymeric materials such as polyethersulfone and polyvinyl difluoride. It is rather difficult to sterilize membrane filters. In recent years, microfilters and ultrafilters composed of ceramics and steel are available. Cleaning and sterilization of such filters are easy.

The filtration techniques namely ***microfiltration, ultrafiltration,*** and ***hyperfiltration*** (reverse osmosis) have already been described (Refer ***Table 20.1***). The other types of membrane filtration techniques are discribed briefly.

Membrane adsorbers : They are micro- or macroporous membranes with ion exchange groups and/or affinity ligands. Membrane adsorbers can bind to proteins and retain them. Such proteins can be eluted by employing solutions in chromatography.

Pervaporation : This is a technique in which volatile products can be separated by a ***process of permeation through a membrane coupled with evaporation***. Pervaporation is quite useful for the extraction, recovery and concentration of volatile products. However, this procedure has a limitation since it cannot be used for large scale separation of volatile products due to cost factor.

Perstraction : This is an advanced technique working on the principle of membrane filtration coupled with solvent extraction. The hydrophobic compounds can be recovered/concentrated by this method.

PRECIPITATION

Precipitation is the most commonly used technique in industry for the concentration of macromolecules such as proteins and polysaccharides. Further, precipitation technique can also be employed for the removal of certain unwanted byproducts e.g. nucleic acids, pigments. Neutral salts, organic solvents, high molecular weight polymers (ionic or non-ionic), besides alteration in temperature and pH are used in precipitation. In addition to these non-specific protein precipitation reactions (i.e. the nature of the protein is unimportant), there are some protein specific precipitations e.g., affinity precipitation, ligand precipitation.

Neutral salts : The most commonly used salt is ***ammonium sulfate***, since it is highly soluble, non-toxic to proteins and low-priced. Ammonium sulfate ***increases hydrophobic interactions*** between protein molecules that result in their precipitation. The precipitation of proteins is dependent on several factors such as protein concentration, pH and temperature.

Organic solvents : ***Ethanol, acetone*** and ***propanol*** are the commonly used organic solvents for protein precipitation. They ***reduce the dielectric constant*** of the medium and enhance electrostatic

interaction between protein molecules that lead to precipitation. Since proteins are denatured by organic solvents, the precipitation process has to be carried out below 0°C.

Non-ionic polymers : ***Polyethylene glycol*** **(*PEG*)** is a high molecular weight non-ionic polymer that can precipitate proteins. It ***reduces*** the quantity of water available for ***protein solvation*** and precipitates protein. PEG does not denature proteins, besides being non-toxic.

Ionic polymers : The charged polymers such as ***polyacrylic acid*** and ***polyethyleneimine*** are used. They form complexes with oppositely charged protein molecules that causes charge neutralisation and precepitation.

Increase in temperature : The heat sensitive proteins can be precipitated by increasing the temperature.

Change in pH : Alterations in pH can also lead to protein precipitation.

Affinity precipitation : The affinity interaction (e.g., between antigen and antibody) is exploited for precipitation of proteins.

Precipitation by ligands : Ligands with specific binding sites for proteins have been successfully used for selective precipitation.

ADSORPTION

The biological products of fermentation can be concentrated by using solid adsorbent particles. In the early days, activated charcoal was used as the adsorbent material. In recent years, cellulose-based adsorbents are employed for protein concentration. And for concentration of low molecular weight compounds (vitamins, antibiotics, peptides) polystyrene, methacrylate and acrylate based matrices are used. The process of adsorption can be carried out by making a bed of adsorbent column and passing the culture broth through it. The desired product, held by the adsorbent, can be eluted.

PURIFICATION BY CHROMATOGRAPHY

The biological products of fermentation (proteins, pharmaceuticals, diagnostic compounds and research materials) are very effectively purified by chromatography. It is basically an analytical technique dealing with the separation of closely related compounds from a mixture. Chromatography usually consists of a ***stationary phase*** and ***mobile phase***. The stationary phase is the porous solid matrix packed in a column (equilibrated with a suitable solvent) on to which the mixture of compounds to be separated is loaded. The compounds are eluted by a mobile phase. A single mobile phase may be used continuously or it may be changed appropriately to facilitate the release of desired compounds. The eluate from the column can be monitored continuously (e.g. protein elution can be monitored by ultraviolet adsorption at 280 nm), and collected in fractions of definite volumes.

TABLE 20.2 Chromatographic techniques along with the principles for separation of proteins

Chromatography	*Principle*
Gel-filtration (size exclusion)	Size and shape
Ion-exchange	Net charge
Chromatofocussing	Net charge
Affinity	Biological affinity and molecular recognition
Hydrophobic interaction	Polarity (hydrophobicity of molecules)
Immobilized metal-ion affinity	Metal ion binding

The different types of chromatography techniques used for separation (mainly proteins) along with the principles are given in ***Table 20.2***. A large number of matrices are commercially available for purification of proteins e.g., agarose, cellulose, polyacrylamide, porous silica, cross-linked dextran, polystyrene. Some of the important features of selected chromatographic techniques are briefly described.

Gel-filtration chromatography : This is also referred to as ***size-exclusion chromatography***. In this technique, the separation of molecules is based on the size, shape and molecular weight. The sponge-like gel beads with pores serve as molecular sieves for separation of smaller and bigger molecules. A solution mixture containing molecules of different sizes (e.g. different proteins) is applied to the column and eluted. The smaller molecules enter the gel beads through their pores and get

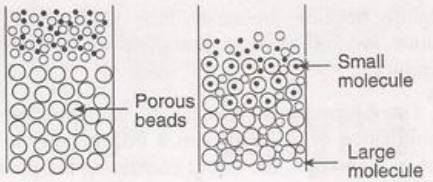


Fig. 20.7 : The principle of gel-filtration chromatography.

trapped. On the other hand, the larger molecules cannot pass through the pores and therefore come out first with the mobile liquid (***Fig. 20.7***). At the industrial scale, gel-filtration is particularly useful to remove salts and low molecular weight compounds from high molecular weight products.

Ion-exchange chromatography : It involves the separation of molecules based on their surface charges. Ion-exchangers are of two types (***cation-exchangers*** which have negatively charged groups like carboxymethyl and sulfonate, and ***anion-exchangers*** with positively charged groups like diethylaminoethyl (DEAE). The most commonly used cation-exchangers are Dowex HCR and Amberlite IR, the anion-exchangers are Dowex SAR and Amberlite IRA.

In ion-exchange chromatography, the pH of the medium is very crucial, since the net charge varies with pH. In other words, the pH determines the effective charge on both the target molecule and the ion-exchanger. The ionic bound molecules can be eluted from the matrix by changing the pH of the eluant or by increasing the concentration of salt solution. Ion-exchange chromatography is useful for the purification of antibiotics, besides the purification of proteins.

Affinity chromatography : This is an elegant method for the purification of proteins from a complex mixture. Affinity chromatography is based on an interaction of a protein with an immobilized ligand. The ligand can be a specific antibody, substrate, substrate analogue or an inhibitor. The immobilized ligand on a solid matrix can be effectively used to fish out complementary structures. In ***Table 20.3***, some examples of ligands used for the purification of proteins are given. The protein bound to the ligand can be eluted by reducing their interaction. This can be achieved by changing the pH of the buffer, altering the ionic strength or by using another free ligand molecule. The fresh ligand used has to be removed in the subsequent steps.

Hydrophobic interaction chromatography (HIC) : This is based on the principle of weak hydrophobic interactions between the hydrophobic ligands (alkyl, aryl side chains on matrix) and hydrophobic amino acids of proteins. The differences in the composition of hydrophobic amino acids in proteins can be used for their separation. The elution of proteins can be done by lowering the salt concentration, decreasing the polarity of the medium or reducing the temperature.

FORMULATION

Formulation broadly refers to the maintenance of activity and stability of a biotechnological products during storage and distribution. The formulation of low molecular weight products (solvents, organic acids) can be achieved by concentrating them with removal of most of the water. For certain small molecules, (antibiotics, citric acid), formulation can be done by crystallization by adding salts.

Proteins are highly susceptible for loss of biological activity, hence their formulation requires special care. Certain stabilizing additives are added to prolong the shelf life of protein. The ***stabilizers of protein formulation*** include ***sugars*** (sucrose, lactose), ***salts*** (sodium chloride, ammonium sulfate), ***polymers*** (polyethylene glycol) and ***polyhydric alcohols*** (glycerol). Proteins may be formulated in the form of solutions, suspensions or dry powders.

TABLE 20.3 Some examples of ligands used for separation of proteins by affinity chromatography

Ligand	*Type of protein*
Antibody	Antigen
Cofactor	Enzyme
Receptor	Hormone
Hapten	Antibody
Inhibitor	Enzyme
Lectins	Glycoproteins
Heparin	Coagulation factors
Metal ions	Metal ion binding proteins

Drying

Drying is an essential component of product formulation. It basically involves the transfer of heat to a wet product for removal of moisture. Most of the biological products of fermentation are sensitive to heat, and therefore require gentle drying methods.

Based on the method of heat transfer, drying devices may be categorized as ***contact-***, ***convection-***, ***radiation dryers***. These three types of dryers are commercially available.

Spray drying

Spray drying is used for drying large volumes of liquids. In spray drying, small droplets of liquid containing the product are passed through a nozzle directing it over a stream of hot gas. The water evaporates and the solid particles are left behind.

Freeze-drying

Freeze-drying or ***lyophilization*** is the most preferred method for drying and formulation of a wide-range of products—pharmaceuticals, foodstuffs, diagnostics, bacteria, viruses. This is mainly because freeze-drying usually does not cause loss of biological activity of the desired product.

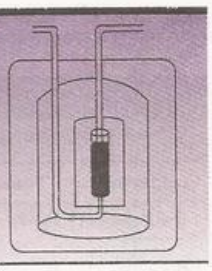
Lyophilization is based on the principle of sublimation of a liquid from a frozen state. In the actual technique, the liquid containing the product is frozen and then dried in a freeze-dryer under vacuum. The vacuum can now be released and the product containing vials can be sealed e.g., penicillin can be freeze dried directly in ampules.

INTEGRATION OF DIFFERENT PROCESSES

It is ideal to ***integrate the fermentation and downstream processing to finally get the desired product***. However, this has not been practicable for various reasons.

Integration of certain stages in downstream processing for purification of product has met with some success. For instance, protein concentration by extraction into two phase systems combined with clarification and purification can be done together.

Enzyme Technology



Enzymes are the biocatalysts synthesized by living cells. They are complex protein molecules that bring about chemical reactions concerned with life. The general features of enzymes are given in Chapter 66.

It is fortunate that enzymes continue to function (bring out catalysis) when they are separated from the cells i.e. *in vitro*. Basically, enzymes are non-toxic and biodegradable. They can be produced in large amounts by microorganisms for industrial applications.

Enzyme technology broadly involves production, isolation, purification and use of enzymes (in soluble or immobilized form) for the ultimate benefit of humankind. In addition, recombinant DNA technology and protein engineering involved in the production of more efficient and useful enzymes are also a part of enzyme technology. The commercial production and use of enzymes is a major part of biotechnology industry. The specialities like microbiology, chemistry and process engineering, besides biochemistry have largely contributed for the growth of enzyme technology.

APPLICATIONS OF ENZYMES

Enzymes have wide range of applications. These include their use in food production, food processing and preservation, washing powders, textile manufacture, leather industry, paper industry, medical applications, improvement of environment and in scientific research. As per recent estimates, a great majority of ***industrially produced enzymes are useful in processes related to foods*** (45%), ***detergents*** (35%), ***textiles*** (10%) and ***leather*** (3%). For details on the applications of individual enzymes, ***Tables 21.1–21.3*** (given later) must be referred.

COMMERCIAL PRODUCTION OF ENZYMES

Microbial enzymes have been utilized for many centuries without knowing them fully. The ***first enzyme produced industrially was takadiastase*** (a fungal amylase) in 1896, in United States. It was used as a pharmaceutical agent to cure digestive disorders.

In Europe, there existed a centuries old practice of softening the hides by using feces of dogs and pigeons before tanning. A German scientist (Otto Rohm) demonstrated in 1905 that extracts from animal organs (pancreases from pig and cow) could be used as the source of enzymes-proteases, for leather softening.

The utilization of enzymes (chiefly proteases) for laundry purposes started in 1915. However, it was not continued due to allergic reactions of impurities in enzymes. Now special techniques are available for manufacture, and use of enzymes in washing powders (without allergic reactions).

Table 21.1 Commercially produced enzymes from plant sources and their applications

Enzyme	*Source(s)*	*Application(s)*
β–Amylase	Barley, soy bean	Baking, preparation of maltose syrup
Bromelain	Pineapple	Baking
Esterase	Wheat	Ester hydrolysis
Ficin	Fig	Meat tenderiser
Papain	Papaya	Meat tenderiser, tanning, baking
Peroxidase	Horse radish	Diagnostic
Urease	Jack bean	Diagnostic

Commercial enzymes can be produced from a wide range of biological sources. At present, a great ***majority*** (80%) of them are ***from microbial sources.***

The different organisms and their relative contribution for the production of commercial enzymes are given below.

Fungi – 60%
Bacteria – 24%
Yeast – 4%
Streptomyces – 2%
Higher animals – 6%
Higher plants – 4%

A real breakthrough for large scale industrial production of enzymes from microorganisms occurred after 1950s.

Enzymes from animal and plant sources

In the early days, animal and plant sources largely contributed to enzymes. Even now, for certain enzymes they are the major sources. A selected list of plant (***Table 21.1***) and animal (***Table 21.2***) enzymes with their sources and applications are given.

Animal organs and tissues are very good sources for enzymes such as lipases, esterases and proteases. The enzyme lysozyme is mostly obtained from hen eggs. Some plants are excellent sources for certain enzymes-papain (papaya), bromelain (pineapple).

Limitations : There are several drawbacks associated with the manufacture of enzymes from animal and plant sources. The quantities are limited and there is a wide variation in their distribution. The most important limitations are the difficulties in isolating, purifying the enzymes, and the cost factor. As regards extraction of industrial enzymes from bovine sources, there is a heavy risk of contamination with bovine spongiform encephalopathy (BSE is prion disease caused by ingestion of abnormal proteins). For these reasons, ***microbial production of enzymes is preferred.***

Enzymes from mammalian cell cultures

There exists a possibility of producing commercial enzymes directly by mammalian cell cultures. But the main constraint will the cost factor which will be extremely high. However, certain therapeutic enzymes such as ***tissue plasminogen activator are produced by cell cultures.***

Enzymes from microbial sources

Microorganisms are the most significant and convenient sources of commercial enzymes. They can be made to produce abundant quantities of enzymes under suitable growth conditions. Microorganisms can be cultivated by using inexpensive media and production can take place in a short period. In addition, it is easy to manipulate microorganisms in genetic engineering

Table 21.2 Commercially produced enzymes from animal sources and their applications

Enzyme(s)	*Source(s)*	*Application(s)*
Amylase, esterase pepsin, trypsin, lipase, rennin (chymosin), phospholipase, phytase	Lamb, calf, bovine, porcine	Digestive aids, preparation of cheese
Lysozyme	Hen eggs	Cell wall breakage in bacteria
Human urine	Urokinase	For dissolution of blood clots

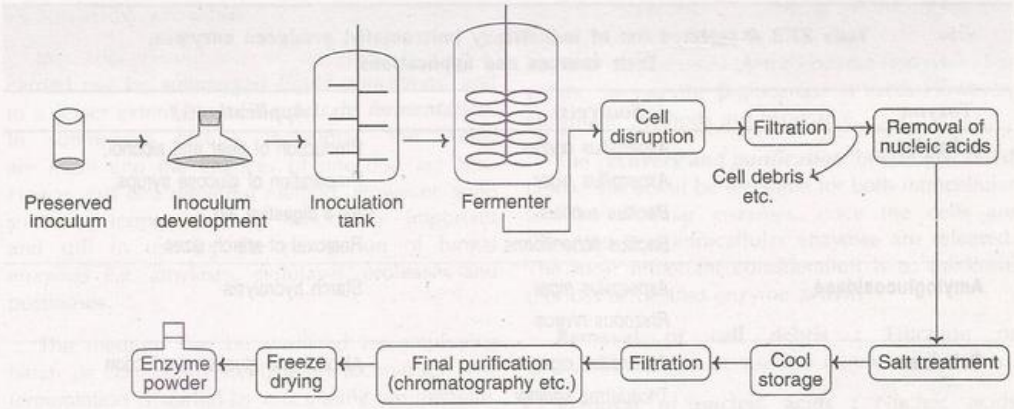


Fig. 21.1 : An outline of the flow chart for the production of enzymes by microorganisms.

techniques to increase the production of desired enzymes. Recovery, isolation and purification processes are easy with microbial enzymes than that with animal or plant sources.

In fact, most enzymes of industrial applications have been successfully produced by micro-organisms. Various fungi, bacteria and yeasts are employed for this purpose. A selected list of enzymes, microbial sources and the applications are given in ***Table 21.3***.

Aspergillus niger—A unique organism for production of bulk enzymes

Among the microorganisms, *A. niger* (a fungus) occupies a special position for the manufacture of a large number of enzymes in good quantities. There are well over 40 commercial enzymes that are conveniently produced by *A. niger*. These include α-amylase, cellulase, protease, lipase, pectinase, phytase, catalase and insulinase.

THE TECHNOLOGY OF ENZYME PRODUCTION—GENERAL CONSIDERATIONS

In general, the techniques employed for microbial production of enzymes are comparable to the methods used for manufacture of other industrial products (Refer Chapter 15). The salient features are briefly described.

1. Selection of organisms
2. Formulation of medium
3. Production process
4. Recovery and purification of enzymes.

An outline of the flow chart for enzyme production by microorganisms is depicted in ***Fig. 21.1***.

Selection of organism

The most important criteria for selecting the microorganism are that the organism should produce the maximum quantities of desired enzyme in a short time while the amounts of other metabolite produced are minimal. Once the organism is selected, strain improvement for optimising the enzyme production can be done by appropriate methods (mutagens, UV rays). From the organism chosen, inoculum can be prepared in a liquid medium.

Formulation of medium

The culture medium chosen should contain all the nutrients to support adequate growth of microorganisms, that will ultimately result in good quantities of enzyme production. The ingradients of the medium should be readily available at low cost and are nutritionally safe. Some of the commonly used substrates for the medium are starch hydrolysate, molasses, corn steep liquor, yeast extract, whey, and soy bean meal. Some cereals (wheat) and pulses (peanut) have also been used. The pH of the medium should be kept optimal for good microbial growth and enzyme production.

TABLE 21.3 A selected list of industrially (microbially) produced enzymes, their sources and applications

Enzyme	*Source(s)*	*Application(s)*
α-Amylase	*Aspergillus oryzae* *Aspergillus niger* *Bacillus subtilus* *Bacillus licheniforms*	Production of beer and alcohol, Preparation of glucose syrups, As a digestive aid Removal of starch sizes
Amyloglucosidase	*Aspergillus niger* *Rhizopus niveus*	Starch hydrolysis
Cellulase	*Aspergillus niger* *Tricoderma koningi*	Alcohol and glucose production
Glucoamylase	*Aspergillus niger* *Bacillus amyloliquefaciens*	Production of beer and alcohol Starch hydrolysis
Glucose isomerase	*Arthrobacter* sp *Bacillus* sp	Manufacture of high fructose syrups
Glucose oxidase	*Aspergillus niger*	Antioxidant in prepared foods
Invertase	*Saccharomyces cerevisiae*	Surcose inversion Preparation of aritficial honey, confectionaries
Keratinase	*Streptomyces fradiae*	Removal of hair from hides
Lactase	*Kluyveromyus* sp *Saccharomyces fragilis*	Lactose hydrolysis Removal of lactose from whey
Lipase	*Candida lipolytica* *Asperigillus niger*	Preparation of cheese Flavour production
Pectinase	*Aspergillus* sp *Sclerotina libertina*	Clarification of fruit juices and wines Alcohol production, coffee concentration
Penicillin acylase	*Escherichia coli*	Production of 6-aminopenicillanic acid
Penicillanase	*Bacillus subtilis*	Removal of penicillin
Protease, acid	*Aspergillus niger*	Digestive aid Substitute for calf rennet
Protease, neutral	*Bacillus amyloliquefaciens*	Fish and meat tenderiser
Protease, alkaline	*Aspergillus oryzae* *Streptomyces griseus* *Bacillus* sp	Meat tenderiser Detergent additive Beer stabilizer
Pollulanase	*Klebsiella aerogens*	Hydrolysis of starch
Takadiastase	*Aspergillus oryzae*	Supplement to bread, Digestive aid

Production process

Industrial production of enzymes is mostly carried out by ***submerged liquid conditions***, and to a lesser extent by ***solid-substrate fermentation***. In submerged culture technique, the yields are more and the chances of infection are less. Hence, this is a preferred method. However, solid substrate fermentation is historically important and still in use for the production of fungal enzymes e.g. amylases, cellulases, proteases and pectinases.

The medium can be sterilized by employing batch or continuous sterilization techniques. The fermentation is started by inoculating the medium. The growth conditions (pH, temperature, O_2 supply, nutrient addition) are maintained at optimal levels. The froth formation can be minimised by adding antifoam agents.

The ***production of enzymes*** is mostly carried out ***by batch fermentation***, and to a lesser extent by continuous process. The bioreactor system must be maintained sterile throughout the fermentation process. The duration of fermentation is variable around 2-7 days, in most production processes. Besides the desired enzyme(s), several other metabolites are also produced. The enzyme(s) have to be recovered and purified.

Recovery and purification of enzymes

The desired enzyme produced may be excreted into the culture medium (extracellular enzymes) or may be present within the cells (intracellular enzymes). Depending on the requirement, the commercial enzyme may be crude or highly purified. Further, it may be in the solid or liquid form. The steps involved in ***downstream processing*** i.e. recovery and purification steps employed will depend on the nature of the enzyme and the degree of purity desired.

In general, recovery of an extracellular enzyme which is present in the broth is relatively simpler compared to an intracellular enzyme. For the release of intracellular enzymes, special techniques are needed for cell disruption. The ***physical, chemical and enzymatic methods adopted to break the cells and release their contents are described elsewhere*** (Chapter 20). The reader must invariably refer them now and learn all the details, as they form part of enzyme technology. Microbial cells can be broken down by physical means (sonication, high pressure, glass beads). The cell walls of bacteria can be lysed by the enzyme lysozyme. For yeasts, the enzyme β-glucanase is used. However, enzymatic methods are expensive.

The ***recovery and purification*** (briefly described below) steps will be the same for both intracellular and extracellular enzymes, once the cells are disrupted and intracellular enzymes are released. The most important consideration is to minimise the loss of desired enzyme activity.

Removal of cell debris : Filtration or centrifugation can be used to remove cell debris.

Removal of nucleic acids : Nucleic acids interfere with the recovery and purification of enzymes. They can be precipitated and removed by adding polycations such as polyamines, streptomycin and polyethyleneimine.

Enzyme precipitation : Enzymes can be precipitated by using salts (ammonium sulfate) organic solvents (isopropanol, ethanol, acetone). Precipitation is advantageous since the precipitated enzyme can be dissolved in a minimal volume to concentrate the enzyme.

Liquid-liquid partition : Further concentration of desired enzymes can be achieved by liquid-liquid extraction using polyethylene glycol or polyamines.

Separation by chromatography : There are several chromatographic techniques for separation and purification of enzymes. These include ***ion-exchange***, ***size exclusion***, ***affinity***, ***hydrophobic interaction*** and ***dye ligand chromatography*** (Refer Chapters 20 and 64). Among these, ion-exchange chromatography is the most commonly used for enzyme purification.

Drying and packing : The concentrated form of the enzyme can be obtained by drying. This can be done by film evaporators or freeze dryers (lyophilizers). The dried enzyme can be packed and marketed. For certain enzymes, stability can be achieved by keeping them in ammonium sulfate suspensions.

All the ***enzymes used in foods or medical treatments must be of high grade purity***, and must meet the required specifications by the regulatory bodies. These enzymes should be totally ***free from toxic materials, harmful microorganisms and should not cause allergic reactions***.

REGULATION OF MICROBIAL ENZYME PRODUCTION —GENERAL CONSIDERATIONS

A maximal production of microbial enzymes can be achieved by optimising the fermentation conditions (nutrients, pH, O_2, temperature etc.). For this purpose, a clear understanding of the genetic regulation of enzyme synthesis is required. Some of the general aspects of microbial enzyme regulation are briefly described.

Induction

Several enzymes are inducible i.e. they are synthesized only in the presence of inducers. The inducer may be the substrate (sucrose, starch, galactosides) or product or intermediate (fatty acid, phenylacetate, xylobiose). A selected list of inducible enzymes and the respective inducers is given in ***Table 21.4***.

The inducer compounds are expensive and their handling (sterilization, addition at specific time) also is quite difficult. In recent years, attempts are being made to develop mutants of microorganisms in which inducer dependence is eliminated.

Feedback repression

Feedback regulation by the end product (usually a small molecule) significantly influences the enzyme synthesis. This occurs when the end product accumulates in large quantities. Large scale production of feedback regulated enzymes is rather difficult. However, mutants that lack feedback repression have been developed to overcome this problem.

Nutrient repression

The native metabolism of microorganism is so devised that there occurs no production of unnecessary enzymes. In other words, the microorgnaisms do not synthesize enzymes that are not required by them, since this is a wasteful exercise. The inhibition of unwanted enzyme production is done by nutrient repression. The nutrients may be carbon, nitrogen, phosphate or sulfate suppliers in the growth medium. For large scale production of enzymes, nutrient repression must be overcome.

Glucose repression is a classical example of nutrient (more appropriately ***catabolite*) *repression***.

Table 21.4 Selected examples of inducible enzymes along with the inducers

Enzyme	*Inducer*
Invertase	Sucrose
Amylase	Starch
Lipase	Fatty acids
β-Galactosidase	Galactosides
Penicillin G amidase	Phenylacetate
Xylanase	Xylobiose

That is in the presence of glucose, the enzymes needed for the metabolism of rest of the compounds are not synthesized. Glucose repression can be overcome by feeding of carbohydrate to the fermentation medium in such a way that the concentration of glucose is almost zero at any given time. In recent years, attempts are being made to select mutants that are resistant to catabolite repression by glucose.

For certain microorganisms, other carbon sources such as pyruvate, lactate, citrate and succinate also act as catabolite repressors.

Nitrogen source repression is also observed in microorganisms. This may be due to ammonium ions or amino acids. Most commonly inexpensive ammonium salts are used as nitrogen sources. The repression by ammonium salts can be overcome by developing mutants resistant to this nitrogen source.

GENETIC ENGINEERING FOR MICROBIAL ENZYME PRODUCTION

Enzymes are the functional products of genes. Therefore, theoretically, enzymes are good candidates for improved production through genetic engineering.

During the past 15 years, the advances in the recombinant DNA technology have certainly helped for increasing the microbial production of commercial enzymes. It is now possible to transfer the desired enzyme genes from one organism to the other. Once an enzyme with a potential use in industry is identified, the relevant gene can be cloned and inserted into a suitable production host.

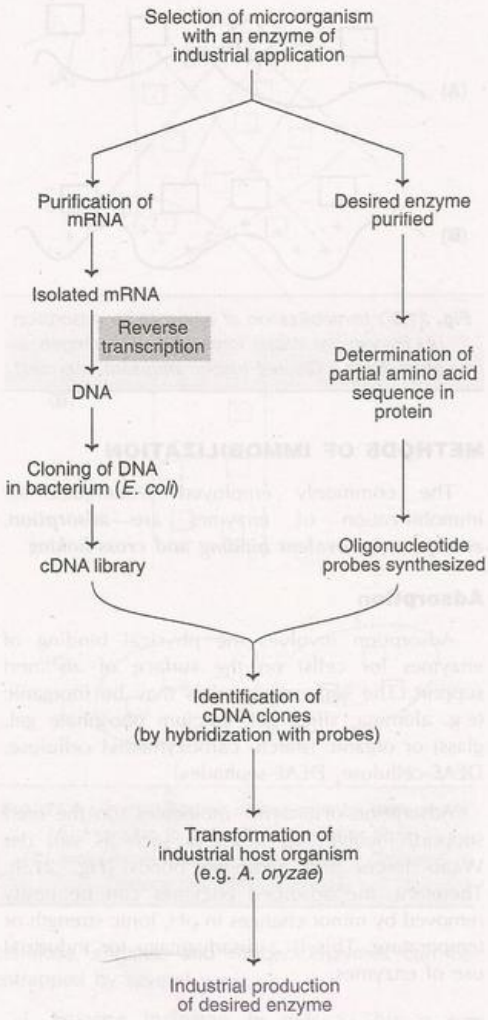


Fig. 21.2 : Schematic representation of a cloning strategy for industrial production of enzymes.

Cloning strategies

A diagrammatic representation of a cloning strategy for industrial production of enzymes is given in ***Fig. 21.2***. This involves the development of cDNA library for the mRNA, and creation of oligonucleotide probes for the desired enzyme. On hybridization with oligonucleotide probes, the specific cDNA clones can be identified. The next step is the transformation of industrially important host organism (e.g. *Aspergillus oryzae*) for the production of the desired enzyme. By this approach, it is possible to manufacture high quality industrial enzymes. A couple of enzymes produced by employing cloning strategies are described below.

1. The enzyme ***lipolase***, found in the fungus *Humicola languinosa* is very effective to remove fat stains in fabrics. However, industrial production of lipolase by this organism is not possible due to a very low level of synthesis. The gene responsible for lipolase was isolated, cloned and inserted into *Aspergillus oryzae*. Thus, large scale production of this enzyme was successfully achieved. Lipolase is very stable and resistant to degradation by proteases that are commonly used in detergents. All these properties make lipolase a strong candidate for its use fabric washing.

2. ***Rennet (chymosin)*** is an enzyme widely used in making cheese. It is mainly obtained from the stomachs of young calves. Consequently, there is a shortage in its supply. The gene for the synthesis of chymosin has been cloned for its large scale production.

Protein engineering for modification of industrial enzymes

It is now possible to alter the structure of a protein/enzyme by ***protein engineering and site-directed mutagenesis*** (For full details Refer Chapter 10). The changes in the enzymes are carried out with the objectives of increased enzyme stability and its catalytic function, resistance to oxidation, changed substrate preference and increased tolerance to alkali and organic solvents. By site-directed mutagenesis, selected amino acids at specific positions (in enzyme) can be changed to produce an enzyme with desired properties. For instance, protein engineering has been used to structurally modify ***phospholipase*** A_2 that can resist high concentration of acid. The modified enzyme is more efficiently used as a food emulsifier.

Genetic engineering has tremendous impact on the industrial production of enzymes with desired properties in a cost-effective manner.

IMMOBILIZATION OF ENZYMES AND CELLS

Traditionally, enzymes in free solutions (i.e. in soluble or free form) react with substrates to result in products. Such use of enzymes is wasteful, particularly for industrial purposes, since enzymes are not stable, and they cannot be recovered for reuse.

Immobilization of enzymes (or cells) refers to the ***technique of confining/anchoring the enzymes (or cells) in or on an inert support for their stability and functional reuse***. By employing this technique, enzymes are made more efficient and cost-effective for their industrial use. Some workers regard immobilization as a goose with a golden egg in enzyme technology.

Immobilized enzymes retain their structural conformation necessary for catalysis. There are several ***advantages of immobilized enzymes***.

- Stable and more efficient in function.
- Can be reused again and again.
- Products are enzyme-free.
- Ideal for multi-enzyme reaction systems.
- Control of enzyme function is easy.
- Suitable for industrial and medical use.
- Minimize effluent disposal problems.

There are however, certain ***disadvantages*** also associated with immobilization.

- The possibility of loss of biological activity of an enzyme during immobilization or while it is in use.
- Immobilization is an expensive affair often requiring sophisticated equipment.

Immobilized enzymes are generally preferred over immobilized cells due to specificity to yield the products in pure form. However, there are several advantages of using immobilized multienzyme systems such as organelles and whole cells over immobilized enzymes. The immobilized cells possess the natural environment with cofactor availability (and also its regeneration capability) and are particularly suitable for multiple enzymatic reactions.

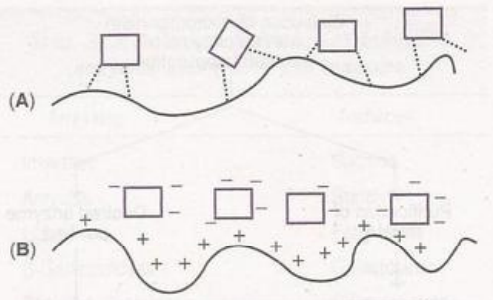


*Fig. 21.3 : Immobilization of enzymes by adsorption (**A**) By van der Waals forces (**B**) By hydrogen bonding (**Note** : Cloured blocks represent enzymes)*

METHODS OF IMMOBILIZATION

The commonly employed techniques for immobilization of enzymes are—***adsorption, entrapment, covalent binding and cross-linking***.

Adsorption

Adsorption involves the physical binding of enzymes (or cells) on the surface of an inert support. The support materials may be inorganic (e.g. alumina, silica gel, calcium phosphate gel, glass) or organic (starch, carboxymethyl cellulose, DEAE-cellulose, DEAE-sephadex).

Adsorption of enzyme molecules (on the inert support) involves weak forces such as van der Waals forces and hydrogen bonds (***Fig. 21.3***). Therefore, the adsorbed enzymes can be easily removed by minor changes in pH, ionic strength or temperature. This is a disadvantage for industrial use of enzymes.

Entrapment

Enzymes can be immobilized by physical entrapment inside a polymer or a gel matrix. The size of the matrix pores is such that the enzyme is retained while the substrate and product molecules pass through. In this technique, commonly referred to as ***lattice entrapment***, the enzyme (or cell) is not subjected to strong binding forces and structural distortions. Some deactivation may however, occur during immobilization process due to changes in pH or temperature or addition of solvents. The matrices used for entrapping of enzymes include polyacrylamide gel, collagen, gelatin, starch,

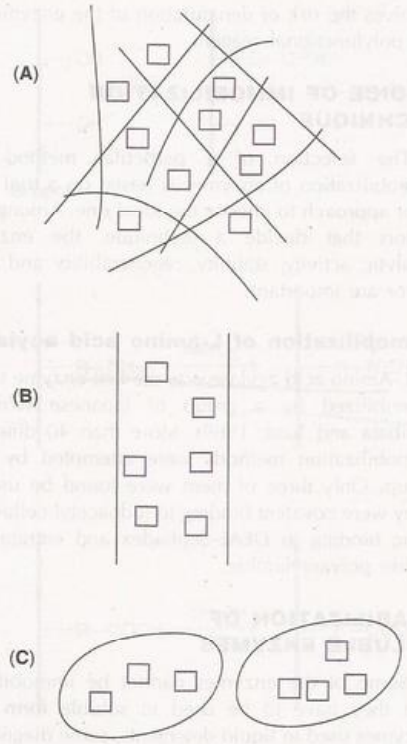


*Fig. 21.4 : Immobilization of enzymes by entrapment (A) Inclusion in gels (B) Inclusion in fibres (C) Inclusion in microcapsules (**Note** : Coloured blocks represent enzymes)*

cellulose, silicone and rubber. Enzymes can be entrapped by several ways.

1. **Enzyme inclusion in gels :** This is an entrapment of enzymes inside the gels (***Fig. 21.4A***).

2. **Enzyme inclusion in fibres :** The enzymes are trapped in a fibre format of the matrix (***Fig. 21.4B***).

3. **Enzyme inclusion in microcapsules :** In this case, the enzymes are trapped inside a microcapsule matrix (***Fig. 21.4C***). The hydrophobic and hydrophilic forms of the matrix polymerise to form a microcapsule containing enzyme molecules inside.

The major limitation for entrapment of enzymes is their leakage from the matrix. Most workers prefer to use the technique of entrapment for immobilization of whole cells. Entrapped cells are in use for industrial production of amino acids (L-isoleucine, L-aspartic acid), L-malic acid and hydroquinone.

Microencapsulation

Microencapsulation is a type of entrapment. It refers to the process of spherical particle formation wherein a liquid or suspension is enclosed in a semipermeable membrane. The membrane may be polymeric, lipoidal, lipoprotein-based or non-ionic in nature. There are three distinct ways of microencapsulation.

1. Building of special membrane reactors.
2. Formation of emulsions.
3. Stabilization of emulsions to form microcapsules.

Microencapsulation is recently being used for immobilization of enzymes and mammalian cells. For instance, pancreatic cells grown in cultures can be immobilized by microencapsulation. Hybridoma cells have also been immobilized successfully by this technique.

Covalent binding

Immobilization of the enzymes can be achieved by creation of covalent bonds between the chemical groups of enzymes and the chemical groups of the support (***Fig. 21.5***). This technique is widely used. However, covalent binding is often associated with loss of some enzyme activity. The inert support usually requires pretreatment (to form pre-activated support) before it binds to enzyme. The following are the common methods of covalent binding.

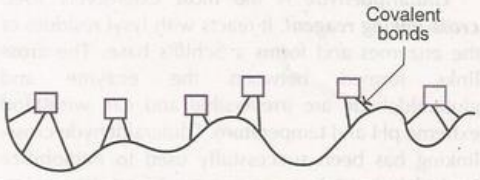


*Fig. 21.5 : A general representation of immobilization of enzymes by covalent binding (**Note** : coloured blocks represent enzymes).*

1. **Cyanogen bromide activation :** The inert support materials (cellulose, sepharose, sephadex) containing glycol groups are activated by CNBr, which then bind to enzymes and immobilize them (***Fig. 21.6A***).

2. **Diazotation :** Some of the support materials (aminobenzyl cellulose, amino derivatives of polystyrene, aminosilanized porous glass) are subjected to diazotation on treatment with $NaNO_2$ and HCl. They, in turn, bind covalently to tyrosyl or histidyl groups of enzymes (***Fig. 21.6B***).

3. **Peptide bond formation :** Enzyme immobilization can also be achieved by the formation of peptide bonds between the amino (or carboxyl) groups of the support and the carboxyl (or amino) groups of enzymes (***Fig. 21.6C***). The support material is first chemically treated to form active functional groups.

4. **Activation by bi- or polyfunctional reagents :** Some of the reagents such as ***glutaraldehyde*** can be used to create bonds between amino groups of enzymes and amino groups of support (e.g. aminoethylcellulose, albumin, amino alkylated porous glass). This is depicted in ***Fig. 21.6D***.

Cross-linking

The absence of a solid support is a characteristic feature of immobilization of enzymes by cross-linking. The enzyme molecules are immobilized by creating cross-links between them, through the involvement of polyfunctional reagents. These reagents in fact react with the enzyme molecules and create bridges which form the backbone to hold enzyme molecules (***Fig. 21.7***). There are several reagents in use for cross-linking. These include glutaraldehyde, diazobenzidine, hexamethylene diisocyanate and toluene di-isothiocyanate.

Glutaraldehyde is the most extensively used ***cross-linking reagent***. It reacts with lysyl residues of the enzymes and forms a Schiff's base. The cross links formed between the enzyme and glutaraldehyde are irreversible and can withstand extreme pH and temperature. Glutaraldehyde cross-linking has been successfully used to immobilize several industrial enzymes e.g. glucose isomerase, penicillin amidase.

The technique of cross-linking is quite simple and cost-effective. But the disadvantage is that it involves the risk of denaturation of the enzyme by the polyfunctional reagent.

CHOICE OF IMMOBILIZATION TECHNIQUE

The selection of a particular method for immobilization of enzymes is based on a trial and error approach to choose the ideal one. Among the factors that decide a technique, the enzyme catalytic activity, stability, regenerability and cost factor are important.

Immobilization of L-amino acid acylase

L-Amino acid acylase was the first enzyme to be immobilized by a group of Japanese workers (Chibata and Tosa, 1969). More than 40 different immobilization methods were attempted by this group. Only three of them were found be useful. They were covalent binding to iodoacetyl cellulose, ionic binding to DEAE-Sephadex and entrapment within polyacrylamide.

STABILIZATION OF SOLUBLE ENZYMES

Some of the enzymes cannot be immobilized and they have to be used in soluble form e.g. enzymes used in liquid detergents, some diagnostic reagents and food additives. Such enzymes can be stabilized by using certain additives or by chemical modifications. The stabilized enzymes have longer half-lives, although they cannot be recycled. Some important methods of enzyme stabilization are briefly described.

Solvent stabilization

Certain solvents at low concentrations, stabilize the enzymes, while at high concentrations the enzymes get denatured e.g. acetone (5%) and ethanol (5%) can stabilize benzyl alcohol dehydrogenase.

Substrate stabilization

The active site of an enzyme can be stabilized by adding substrates e.g. starch stabilizes α-amylase; glucose stabilizes glucose isomerase.

Stabilization by polymers

Enzymes can be stabilized, particularly against increased temperature, by addition of polymers such as gelatin, albumin and polyethylene glycol.

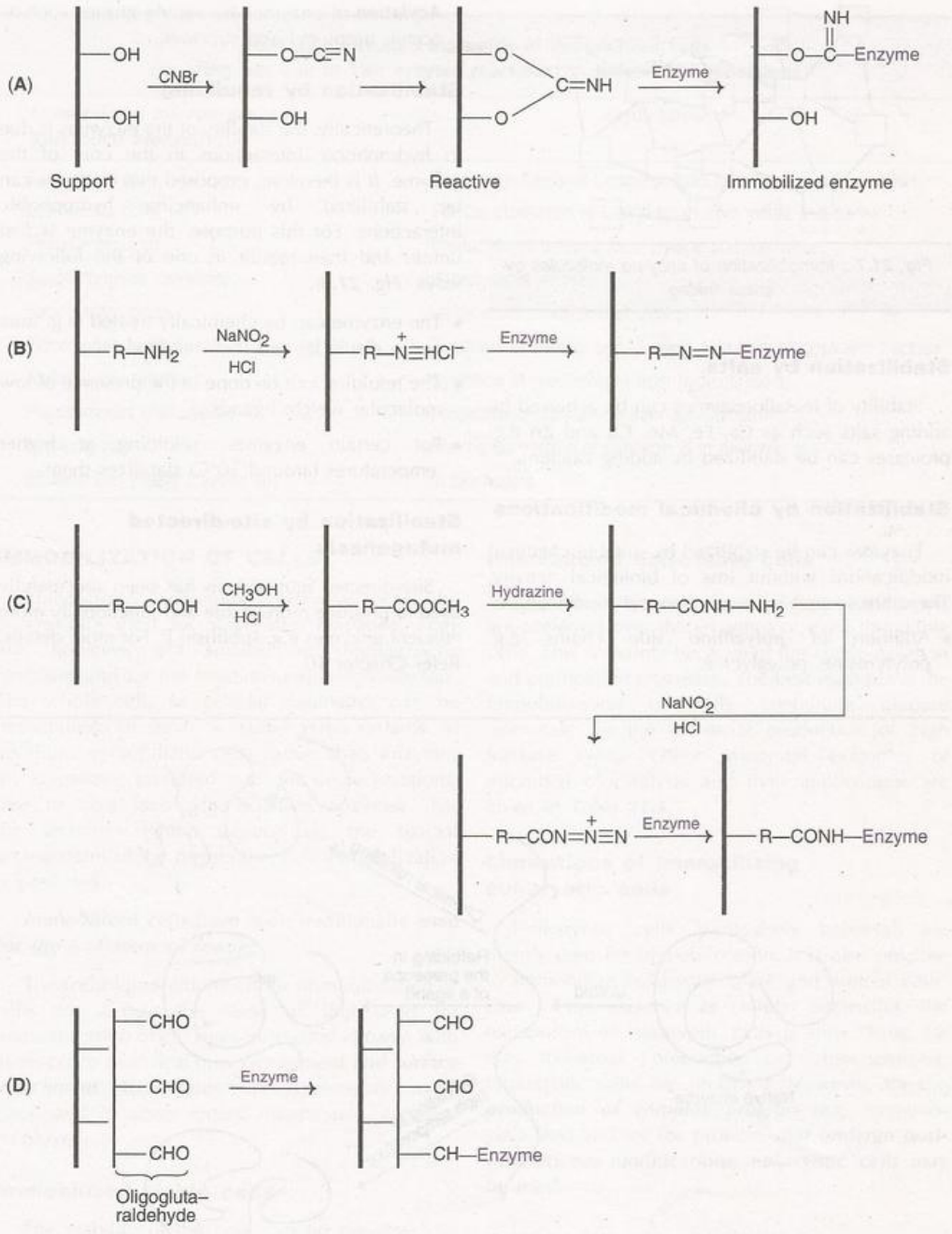


Fig. 21.6 : *Immobilization of enzymes by covalent binding (**A**) Cyanogen bromide activation, (**B**) Diazotation, (**C**) Peptide bond formation, (**D**) Activation by bifunctional agent.*

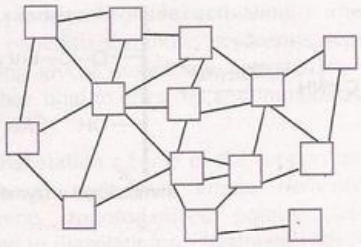


Fig. 21.7 : Immobilization of enzyme molecules by cross linking.

Stabilization by salts

Stability of metalloenzymes can be achieved by adding salts such as Ca, Fe, Mn, Cu and Zn e.g. proteases can be stabilized by adding calcium.

Stabilization by chemical modifications

Enzymes can be stabilized by suitable chemical modifications without loss of biological activity. There are several types of chemical modifications.

- Addition of ***polyamino*** side chains e.g. polytyrosine, polyglycine.
- ***Acylation*** of enzymes by adding groups such as acetyl, propionyl and succinyl.

Stabilization by rebuilding

Theoretically, the stability of the enzymes is due to hydrophobic interactions in the core of the enzyme. It is therefore, proposed that enzymes can be stabilized by enhancing hydrophobic interactions. For this purpose, the enzyme is first unfold and then rebuilt in one of the following ways (***Fig. 21.8***).

- The enzyme can be chemically treated (e.g. urea and a disulfide) and then refolded.
- The refolding can be done in the presence of low molecular weight ligands.
- For certain enzymes, refolding at higher temperatures (around 50°C) stabilizes them.

Stabilization by site-directed mutagenesis

Site-directed mutagenesis has been successfully used to produce more stable and functionally more efficient enzymes e.g. subtilisin E. For more details, Refer Chapter 10.

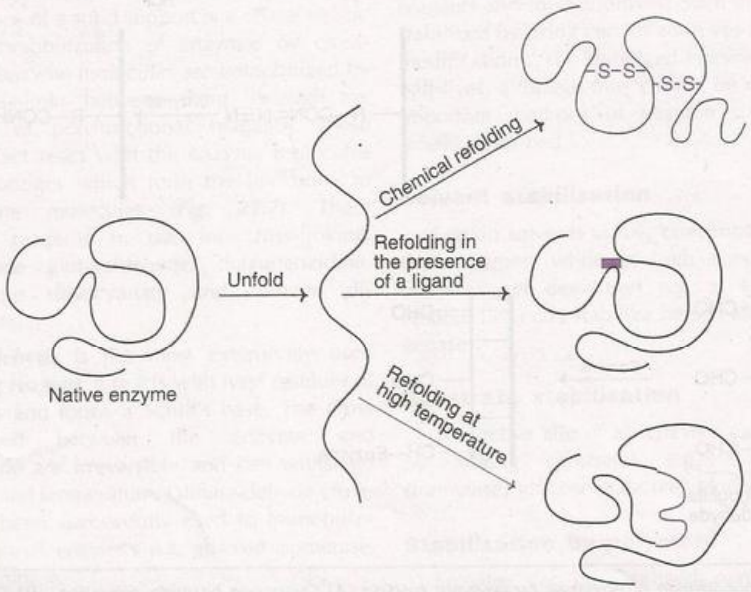


Fig. 21.8 : Stabilization of an enzyme by refolding.

TABLE 21.5 Selected examples of immobilized cells (to bring out one or two enzyme reactions) in industrial applications

Immobilized microorganism (microbial biocatalyst)	*Application(s)*
Escherichia coli	For the synthesis of L-aspartic acid from fumaric acid and NH_3
Escherichia coli	For the production of L-tryptophan from indole and serine
Pseudomonas sp	Production of L-serine from glycine and methanol
Saccharomyces cerevisiae	Hydrolysis of sucrose
Saccharomyces sp	Large scale production of alcohol
Zymomonas mobilis	Synthesis of sorbitol and gluconic acid from glucose and fructose
Anthrobacter simplex	Synthesis of prednisolone from hydrocortisone
Pseudomonas chlororaphis	Production of acrylamide from acrylonitrile
Humicola sp	For the conversion of rifamycin B to rifamycin S
Bacteria and yeasts (several sp)	In biosensors

IMMOBILIZATION OF CELLS

Immobilized individual enzymes can be successfully used for single-step reactions. They are, however, not suitable for multienzyme reactions and for the reactions requiring cofactors. The whole cells or cellular organelles can be immobilized to serve as multienzyme systems. In addition, immobilized cells rather than enzymes are sometimes preferred even for single reactions, due to cost factor in isolating enzymes. For the enzymes which depend on the spacial arrangement of the membrane, cell immobilization is preferred.

Immobilized cells have been traditionally ***used for the treatment of sewage***.

The techniques employed for immobilization of cells are almost the same as that used for immobilization of enzymes (discussed already) with appropriate modifications. ***Entrapment and surface attachment techniques are commonly used.*** Gels, and to some extent membranes, are also employed.

Immobilized viable cells

The viability of the cells can be preserved by mild immobilization. Such immobilized cells are particularly useful for fermentations. Sometimes mammalian cell cultures are made to function as immobilized viable cells.

Immobilized non-viable cells

In many instances, immobilized non-viable cells are preferred over the enzymes or even the viable cells. This is mainly because of the costly isolation and purification processes. The best example is the immobilization of cells containing glucose isomerase for the industrial production of high fructose syrup. Other important examples of microbial biocatalysts and their applications are given in ***Table 21.5***.

Limitations of immobilizing eukaryotic cells

Prokaryotic cells (particularly bacterial) are mainly used for immobilization. It is also possible to immobilize eukaryotic plant and animal cells. Due to the presence of cellular organelles, the metabolism of eukaryotic cells is slow. Thus, for the industrial production of biochemicals, prokaryotic cells are preferred. However, for the ***production of complex proteins*** (e.g. immunoglobulins) and for the proteins ***that undergo post-translational modifications, eukaryotic cells may be used***.

EFFECT OF IMMOBILIZATION ON ENZYME PROPERTIES

Enzyme immobilization is frequently associated with alterations in enzyme properties, particularly

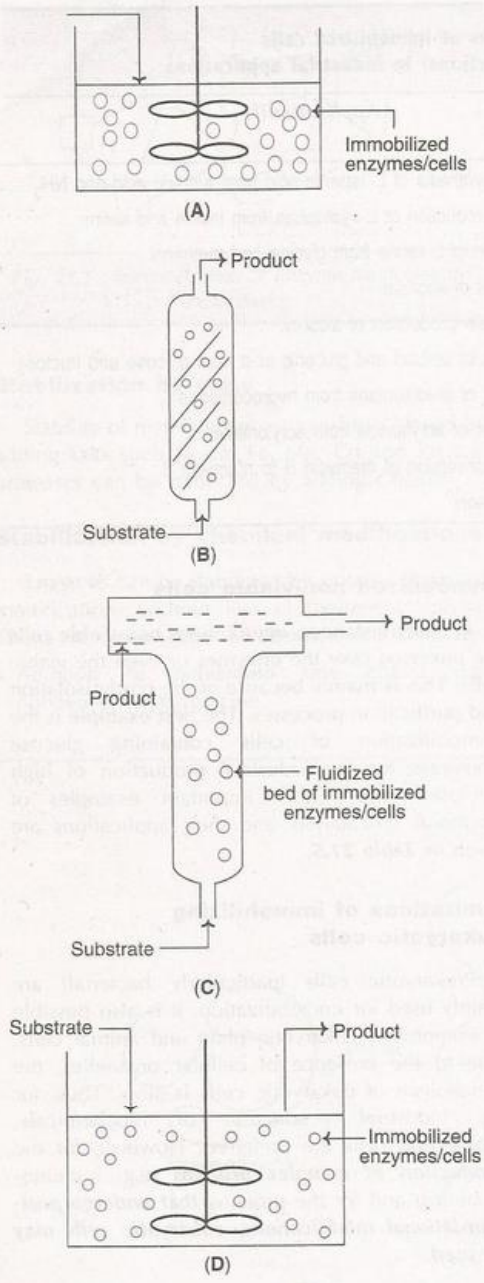


Fig. 21.9 : Immobilized enzyme (cell) reactors (A) Batch stirred tank reactor, (B) Packed bed reactor, (C) Fluidized bed reactor, (D) Continuous stirred tank reactor.

the kinetic properties of enzymes. Some of them are listed below.

1. There is a substantial decrease in the enzyme specificity. This may be due to conformational changes that occur when the enzyme gets immobilized.

2. The kinetic constants K_m and V_{max} of an immobilized enzyme differ from that of the native enzyme. This is because the conformational change of the enzyme will affect the affinity between enzyme and substrate.

IMMOBILIZED ENZYME REACTORS

The immobilized enzymes cells are utilized in the industrial processes in the form of enzyme reactors. They are broadly of two types — batch reactors and continuous reactors. The frequently used enzyme reactors are shown in ***Fig. 21.9***.

Batch reactors

In batch reactors, the immobilized enzymes and substrates are placed, and the reaction is allowed to take place under constant stirring. As the reaction is completed, the product is separated from the enzyme (usually by denaturation).

Soluble enzymes are commonly used in batch reactors. It is rather difficult to separate the soluble enzymes from the products, hence there is a limitation of their reuse. However, special techniques have been developed for recovery of soluble enzymes, although this may result in loss of enzyme activity.

Stirred tank reactors : The simplest form of batch reactor is the ***stirred tank reactor*** (***Fig. 21.9A***). It is composed of a reactor fitted with a stirrer that allows good mixing, and appropriate temperature and pH control. However, there may occur loss of some enzyme activity. A modification of stirred tank reactor is ***basket reactor***. In this system, the enzyme is retained over the impeller blades. Both stirred tank reactor and basket reactor have a ***well mixed flow pattern***.

Plug flow type reactors : These reactors are alternatives to flow pattern type of reactors. The flow rate of fluids controlled by a plug system. The plug flow type reactors may be in the form of ***packed bed or fluidized bed*** (***Fig. 21.9B*** and ***21.9C***). These reactors are particularly useful when

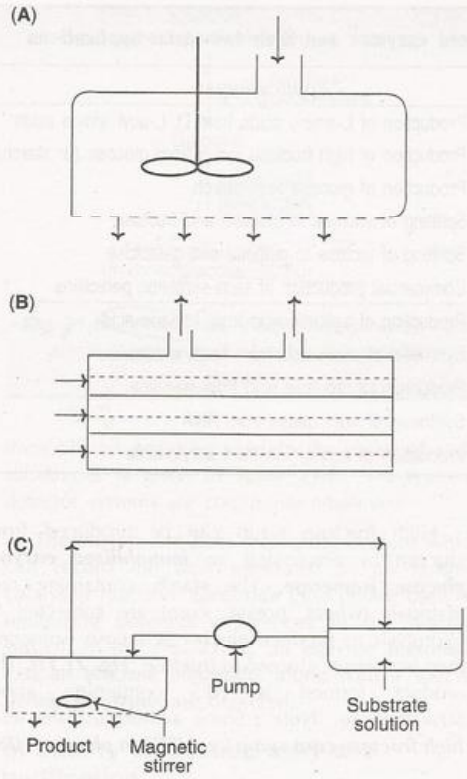


Fig. 21.10 : Membrane reactors (A) Batch membrane reactor, (B) Continuous membrane reactor, (C) Recycle membrane reactor (Coloured lines indicate membranes).

there occurs inadequate product formation in flow type reactors. Further, plug flow reactors are also useful for obtaining kinetic data on the reaction systems.

Continuous reactors

In continuous enzyme reactors, the substrate is added continuously while the product is removed simultaneously. Immobilized enzymes can also be used for continuous operation. Continuous reactors have certain advantages over batch reactors. These include control over the product formation, convenient operation of the system and easy automation of the entire process. There are mainly two types of continuous reactors-***continuous stirred tank reactor*** (***CSTR***) and ***plug reactor*** (***PR***). A diagrammatic representation of CSTR is depicted in ***Fig. 21.9D***. CSTR is ideal for good product formation.

Membrane reactors

Several membranes with a variety of chemical compositions can be used. The commonly used membrane materials include polysulfone, polyamide and cellulose acetate. The biocatalysts (enzymes or cells) are normally retained on the membranes of the reactor. The substrate is introduced into reactor while the product passes out. Good mixing in the reactor can be achieved by using ***stirrer*** (***Fig. 21.10A***). In a ***continuous membrane reactor***, the biocatalysts are held over membrane layers on to which substrate molecules are passed (***Fig. 21.10B***).

In a ***recycle model membrane reactor***, the contents (i.e. the solution containing enzymes, cofactors, and substrates along with freshly released product are recycled by using a pump (***Fig. 21.10C***). The product passes out which can be recovered.

APPLICATIONS OF IMMOBILIZED ENZYMES AND CELLS

Immobilized enzymes and cells are very widely used for ***industrial***, ***analytical*** and ***therapeutic purpose***, besides their involvement in ***food production*** and exploring the knowledge of biochemistry, microbiology and other allied specialities. A brief account of the industrial applications of immobilized cells is given in ***Table 21.5***.

MANUFACTURE OF COMMERCIAL PRODUCTS

A selected list of important immobilized enzymes and their industrial applications is given in ***Table 21.6***. Some details on the manufacture of L-amino acids and high fructose syrup are given hereunder.

Production of L-amino acids

L-Amino acids (and not D-amino acids) are very important for use in food and feed supplements and medical purposes. The chemical methods employed for their production result in a racemic mixture of D- and L-amino acids. They can be

TABLE 21.6 A selected list of important immobilized enzymes and their industrial applications

Immobilized enzyme	*Application(s)*
Aminoacylase	Production of L-amino acids from D, L-acyl amino acids
Glucose isomerase	Production of high fructose syrup from glucose (or starch)
Amylase	Production of glucose from starch
Invertase	Splitting of sucrose to glucose and fructose
β-Galactosidase	Splitting of lactose to glucose and galactose
Penicillin acylase	Commercial production of semi-synthetic penicillins
Aspartase	Production of aspartic acid from fumaric acid
Fumarase	Synthesis of malic acid from fumaric acid
Histidine ammonia lyase	Production of urocanic acid from histidine
Ribonuclease	Synthesis of nucleotides from RNA
Nitrilase	Production of acrylamide from acrylonitrile

acylated to form D, L-acyl amino acids. The immobilized enzyme ***aminoacylase*** (frequently immobilized on DEAE sephadex) can selectively hydrolyse D, L-acyl amino acids to produce L-amino acids.

$$\text{D, L-Acyl amino acids} \xrightarrow{\text{Aminoacylase}} \text{L-Amino acids} + \text{D, L-Acyl amino acids}$$

The free L-amino acids can separated from the unhydrolysed D-acyl amino acids. The latter can be recemized to D, L-acyl amino acids and recycled through the enzyme reactor containing immobilized aminoacylase. Huge quantities of L-methionine, L-phenylalanine L-tryptophan and L-valine are produced worldwide by this approach.

Production of high fructose syrup

Fructose is the sweetest among the monosaccharides, and has twice the sweetening strength of sucrose. Glucose is about 75% as sweet as sucrose. Therefore, glucose (the most abundant monosaccharide) cannot be a good substitute for sucrose for sweetening. Thus, there is a great demand for fructose which is very sweet, but has the same calorific value as that of glucose or sucrose.

High fructose syrup (HFS) contains approximately equivalent amounts of glucose and fructose. HFS is almost similar to sucrose from nutritional point of view. HFS is a good substitute for sugar in the preparation of soft drinks, processed foods and baking.

High fructose syrup can be produced from glucose by employing an ***immobilized enzyme glucose isomerase***. The starch containing raw materials (wheat, potato, corn) are subjected to hydrolysis to produce glucose. Glucose isomerase then isomerises glucose to fructose (***Fig. 21.11***). The product formed is HFS containing about 50% fructose. (**Note :** Some authors use the term ***high fructose corn syrup*** i.e. ***HFCS in place of HFS***).

Glucose isomerase : This is an intracellular enzyme produced by a number of microorganisms. The species of *Arthrobacter, Bacillus* and *Streptomyces* are the preferred sources. Being an intracellular enzyme, the isolation of glucose isomerase without loss of biological activity requires special and costly techniques. Many a times, whole cells or partly broken cells are immobilized and used.

IMMOBILIZED ENZYMES AND CELLS-ANALYTICAL APPLICATIONS

In biochemical analysis

Immobilized enzymes (or cells) can be used for the development of precise and specific analytical techniques for the estimation of several biochemical compounds. The principle of analytical assay primarily involves the action of the immobilized enzyme on the substrate. A decrease in the substrate concentration or an increase in the product level or an alteration in the cofactor concentration can be used for the assay. A selected list of examples of

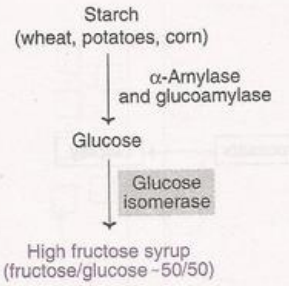


Fig. 21.11 : Production of high fructose syrup from starch (glucose isomerase is the immobilized enzyme).

immobilized enzymes used in the assay of some substances is given in ***Table 21.7***. Two types of detector systems are commonly employed.

Thermistors are heat measuring devices which can record the heat generated in an enzyme catalysed reaction. ***Electrode*** devices are used for measuring potential differences in the reaction system. In the ***Fig. 21.12***, an ***enzyme thermistor*** and an ***enzyme electrode,*** along with a specific ***urease electrode*** are depicted.

In affinity chromatography and purification

Immobilized enzymes can be used in affinity chromatography. Based on the property of affinity, it is possible to purify several compounds e.g. antigens, antibodies, cofactors.

TABLE 21.7 Selected examples of immobilized enzymes used in analytical biochemistry

Immobilized enzyme	*Substance assayed*
Glucose oxidase	Glucose
Urease	Urea
Cholesterol oxidase	Cholesterol
Lactate dehydrogenase	Lactate
Alcohol oxidase	Alcohol
Hexokinase	ATP
Galactose oxidase	Galactose
Penicillinase	Penicillin
Ascorbic acid oxidase	Ascorbic acid
L-Amino acid oxidase	L-Amino acids
Cephalosporinase	Cephalosporin
Monoamine oxidase	Monoamine

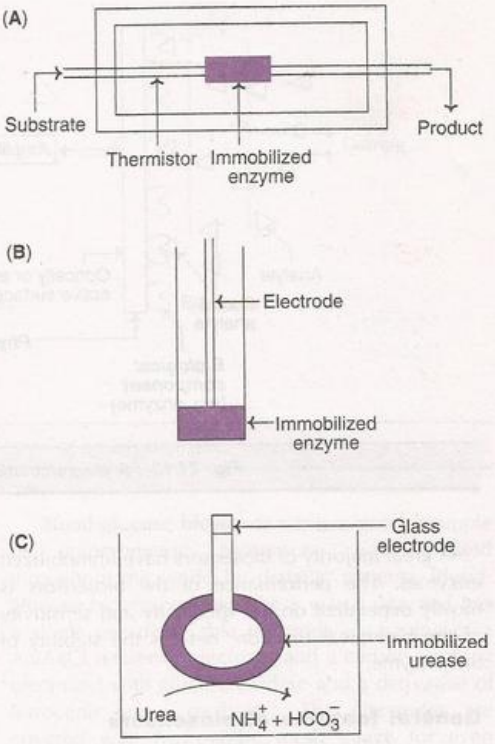


Fig. 21.12 : Immobilized enzymes or cells in analytical biochemistry (A) Enzyme thermistor, (B) Enzyme electrode, (C) Urease electrode.

BIOSENSORS

A biosensor is an analytical device containing an ***immobilized biological material*** (enzyme, antibody, nucleic acid, hormone, organelle or whole cell) which can specifically ***interact with an analyte and produce physical, chemical or electrical signals that can be measured***. An analyte is a compound (e.g. glucose, urea, drug, pesticide) whose concentration has to be measured. Biosensors basically involve the quantitative analysis of various substances by converting their biological actions into measurable signals.

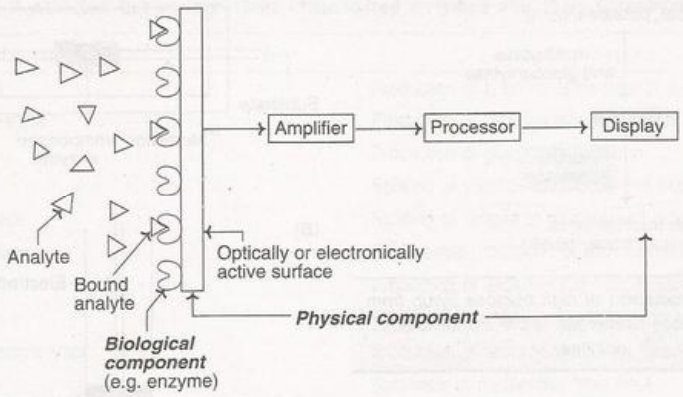


Fig. 21.13 : A diagrammatic representation of a biosensor.

A great majority of biosensors have immobilized enzymes. The performance of the biosensors is mostly dependent on the specificity and sensitivity of the biological reaction, besides the stability of the enzyme.

General features of biosensors

A biosensor has two distinct components (*Fig. 21.13*).

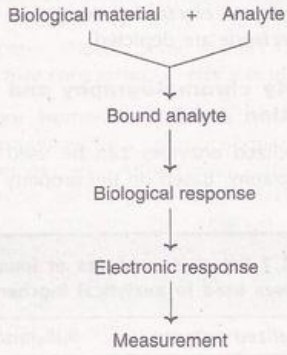
1. Biological component—enzyme, cell etc.

2. Physical component—transducer, amplifier etc.

The biological component recognises and interacts with the analyte to produce a physical change (a signal) that can be detected by the transducer. In practice, the biological material is appropriately immobilized on to the transducer and the so prepared biosensors can be repeatedly used several times (may be around 10,000 times) for a long period (many months).

Principle of a biosensor

The desired biological material (usually a specific enzyme) is immobilized by conventional methods (physical or membrane entrapment, non-covalent or covalent binding). This immobilized biological material is in intimate contact with the transducer. The analyte binds to the biological material to form a bound analyte which in turn produces the electronic response that can be measured.



In some instances, the analyte is converted to a product which may be associated with the release of heat, gas (oxygen), electrons or hydrogen ions. The transducer can convert the product linked changes into electrical signals which can be amplified and measured.

TYPES OF BIOSENSORS

There are several types of biosensors based on the sensor devices and the type of biological materials used. A selected few of them are discussed below.

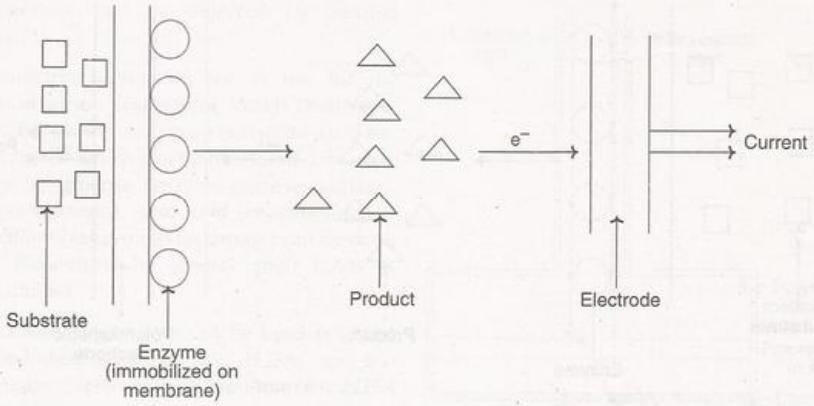


Fig. 21.14 : A diagrammatic representation of an amperometric biosensor.

ELECTROCHEMICAL BIOSENSORS

Electrochemical biosensors are simple devices based on the ***measurements of electric current***, ionic or conductance changes carried out by ***bioelectrodes***.

Amperometric biosensors

These biosensors are ***based on the movement of electrons*** (i.e. determination of electric current) as a result of enzyme-catalysed redox reactions. Normally, a constant voltage passes between the electrodes which can be determined. In an enzymatic reaction that occurs, the substrate or product can transfer an electron with the electrode surface to be oxidised or reduced (***Fig. 21.14***). This results in an altered current flow that can be measured. The magnitude of the current is proportional to the substrate concentration. Clark oxygen electrode which determines reduction of O_2, is the simplest form of amperometric biosensor. ***Determination of glucose by glucose oxidase*** is a good example.

In the ***first generation amperometric biosensors*** (described above), there is a direct transfer of the electrons released to the electrode which may pose some practical difficulties. A ***second generation amperometric biosensors*** have been developed wherein a mediator (e.g. ferrocenes) takes up the electrons and then transfers them to electrode. These biosensors however, are yet to become popular.

Blood-glucose biosensor : It is a good example of amperometric biosensors, widely used throughout the world by diabetic patients. Blood-glucose biosensor looks like a watch pen and has a single use disposable electrode (consisting of a Ag/AgCl reference electrode and a carbon working electrode) with glucose oxidase and a derivative of ferrocene (as a mediator). The electrodes are covered with hydrophilic mesh guaze for even spreading of a blood drop. The disposable test strips, sealed in aluminium foil have a shelf-life of around six months.

An ***amperometric biosensor for assessing the freshness of fish has been developed.*** The accumulation of ionosine and hypoxanthine in relation to the other nucleotides indicates freshness of fish-how long dead and stored. A biosensor utilizing immobilized nucleoside phosphorylase and xanthine oxidase over an electrode has been developed for this purpose.

Potentiometric biosensors

In these biosensors, changes in ionic concentrations are determined by use of ion-selective electrodes (***Fig. 21.15***). pH electrode is the most commonly used ion-selective electrode, since many enzymatic reactions involve the release or absorption of hydrogen ions. The other important electrodes are ammonia-selective and CO_2 selective electrodes. The potential difference obtained between the potentiometric electrode and

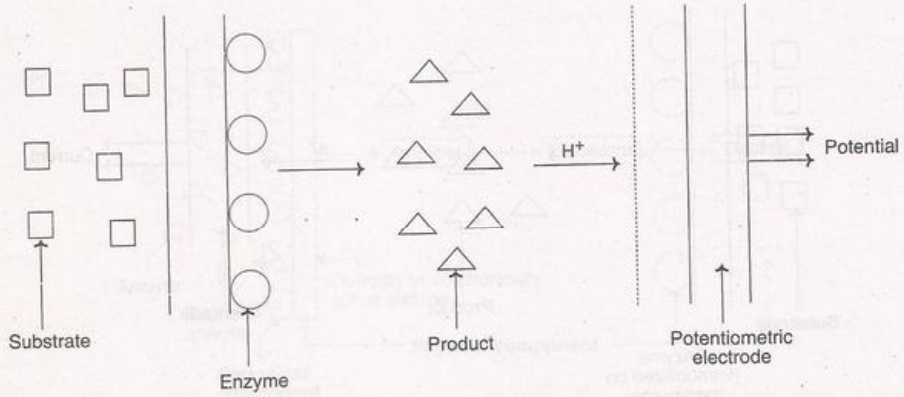


Fig. 21.15 : A diagrammatic representation of a potentiometric biosensor.

the reference electrode can be measured. It is proportional to the concentration of the substrate. The major limitation of potentiometric biosensors is the sensitivity of enzymes to ionic concentrations such as H^+ and NH_4^+.

Ion-selective field effect transistors (ISFET) are the low cost devices that can be used for miniaturization of potentiometric biosensors. A good example is an ISFET biosensor used to monitor intramyocardial pH during open-heart surgery.

Conductimetric biosensors

There are several reactions in the biological systems that bring about changes in the ionic species. These ionic species alter the electrical conductivity which can be measured. A good example of conductimetric biosensor is the ***urea biosensor*** utilizing immobilized urease. Urease catalyses the following reaction.

$$\begin{matrix} H_2N \\ \diagdown \\ \quad C{=}O \\ \diagup \\ H_2N \end{matrix} + 3H_2O \xrightarrow{\text{Urease}} 2NH_4^+ + HCO_3^- + OH^-$$

The above reaction is associated with drastic alteration in ionic concentration which can be used for monitoring urea concentration. In fact, urea biosensors are very successfully used during dialysis and renal surgery.

THERMOMETRIC BIOSENSORS

Several biological reactions are associated with the ***production of heat*** and this forms the basis of thermometric biosensors. They are more commonly referred to as ***thermal biosensors*** or ***calorimetric biosensors***. A diagrammatic representation of a thermal biosensor is depicted in ***Fig. 21.16***. It consists of a heat insulated box fitted with heat exchanger (aluminium cylinder). The reaction takes place in a small enzyme packed bed reactor. As the substrate enters the bed, it gets converted to a product and heat is generated. The difference in the temperature between the substrate and product is measured by thermistors. Even a small change in

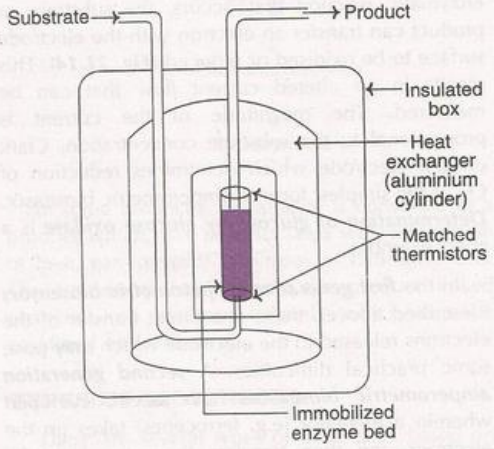


Fig. 21.16 : A diagrammatic representation of thermometric biosensor.

the temperature can be detected by thermal biosensors.

Thermometric biosensors are in use for the estimation of ***serum cholesterol***. When cholesterol gets oxidized by the enzyme cholesterol oxidase, heat is generated which can be measured. Likewise, estimations of ***glucose*** (enzyme-glucose oxidase), ***urea*** (enzyme-urease), ***uric acid*** (enzyme-uricase) and ***penicillin G*** (enzyme-β lactamase) can be done by these biosensors. In general, their utility is however, limited.

Thermometric biosensors can be used as a part of enzyme-linked immunoassay (ELISA) and the new technique is referred to as ***thermometric ELISA*** (***TELISA***).

OPTICAL BIOSENSORS

Optical biosensors are the devices that utilize the principle of ***optical measurements*** (absorbance, fluorescence, chemiluminescence etc.). They employ the use of fibre optics and optoelectronic transducers. The word ***optrode***, representing a condensation of the words ***op***tical and elec***trode***, is commonly used. Optical biosensors primarily involve enzymes and antibodies as the transducing elements.

Optical biosensors allow a safe non-electrical remote sensing of materials. Another advantage is that these biosensors usually do not require reference sensors, as the comparative signal can be generated using the same source of light as the sampling sensor. Some of the important optical biosensors are briefly described hereunder.

Fibre optic lactate biosensor

Fig. 21.17 represents the fibre optic lactate biosensor. Its working is based on the measurement of changes in molecular O_2 concentration by determining the quenching effect of O_2 on a fluorescent dye. The following reaction is catalysed by the enzyme lactate monooxygenase.

$$O_2 + \text{Lactate} \xrightarrow{\text{Lactate monooxygenase}} CO_2 + \text{Acetate} + H_2O$$

The amount of fluorescence generated by the dyed film is dependent on the O_2. This is because O_2 has a quenching (reducing) effect on the fluorescence. As the concentration of lactate in the reaction mixture increases, O_2 is utilized, and consequently there is a proportionate decrease in the quenching effect. The result is that there is an increase in the fluorescent output which can be measured.

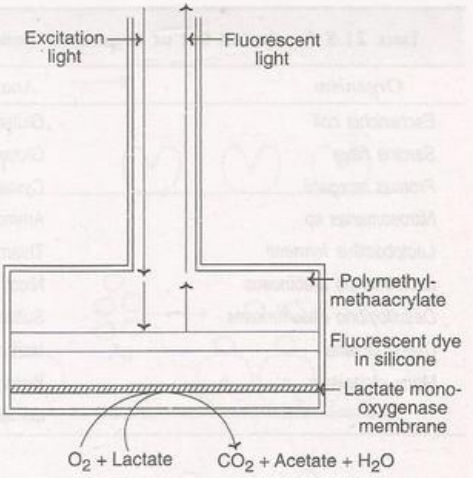


Fig. 21.17 : Diagrammatic representation of a fibre optic lactate biosensor.

Optical biosensors for blood glucose

Estimation of blood glucose is very important for monitoring of diabetes. A simple technique involving paper strips impregnated with reagents is used for this purpose. The strips contain glucose oxidase, horse radish peroxidase and a chromogen (e.g. toluidine). The following reactions occur.

$$\text{Glucose} \xrightarrow[\text{oxidase}]{\text{Glucose}} \text{Gluconic acid} + H_2O_2$$

$$\text{Chromogen} + 2H_2O_2 \xrightarrow{\text{Peroxidase}} \text{Colour dye} + 2H_2O$$

The intensity of the colour of the dye can be measured by using a portable reflectance meter. Glucose strip production is a very big industry worldwide.

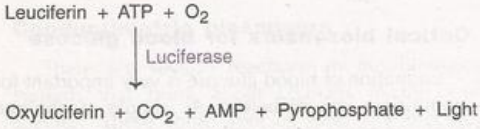
Colorimetric test strips of cellulose coated with appropriate enzymes and reagents are in use for the estimation of several blood and urine parameters.

Table 21.8 A selected list of organisms along with the analytes and the types of biosensors

Organism	*Analyte*	*Type of biosensor*
Escherichia coli	Glutamate	Potentiometric (CO_2)
Sarcina flava	Glutamine	Potentiometric (NH_3)
Proteus morganii	Cysteine	Potentiometric (H_2S)
Nitrosomanas sp	Ammonia	Amperometric (O_2)
Lactobacillus fermenti	Thiamine	Amperometric (mediated)
Lactobacillus arabinosus	Nicotinic acid	Potentiometric (H^+)
Desulfovibrio desulfuricans	Sulfate	Potentiometric (SO_3^-)
Cyanobacteria	Herbicides	Amperometric (mediated)
Many organisms	Biological oxygen demand (BOD)	Amperometric (O_2)

Luminescent biosensors to detect urinary infections

The microorganisms in the urine, causing urinary tract infections, can be detected by employing luminescent biosensors. For this purpose, the immobilized (or even free) enzyme namely ***luciferase*** is used. The microorganisms, on lysis release ATP which can be detected by the following reaction. The quantity of light output can be measured by electronic devices.



Other optical biosensors

Optical fibre sensing devices are in use for measuring pH, pCO_2 and pO_2 in critical care, and surgical monitoring.

PIEZOELECTRIC BIOSENSORS

Piezoelectric biosensors are based on the ***principle of acoustics*** (sound vibrations), hence they are also called as ***acoustic biosensors***. Piezoelectric crystals form the basis of these biosensors. The crystals with positive and negative charges vibrate with characteristic frequencies. Adsorption of certain molecules on the crystal surface alters the resonance frequencies which can be measured by electronic devices. Enzymes with gaseous substrates or inhibitors can also be attached to these crystals.

A piezoelectric biosensor for organophosphorus insecticide has been developed incorporating acetylcholine esterase. Likewise, a biosensor for formaldehyde has been developed by incorporating formaldehyde dehydrogenase. A biosensor for cocaine in gas phase has been created by attaching cocaine antibodies to the surface of piezoelectric crystal.

Limitations of piezoelectric biosensors

It is very difficult to use these biosensors to determine substances in solution. This is because the crystals may cease to oscillate completely in viscous liquids.

WHOLE CELL BIOSENSORS

Whole cell biosensors are particularly useful for multi-step or cofactor requiring reactions. These biosensors may ***employ live or dead microbial cells***. A selected list of some organisms along with the analytes and the types of biosensors used is given in ***Table 21.8***.

Advantages of microbial cell biosensors

The microbial cells are cheaper with longer half-lives. Further, they are less sensitive to variations in pH and temperature compared to isolated enzymes.

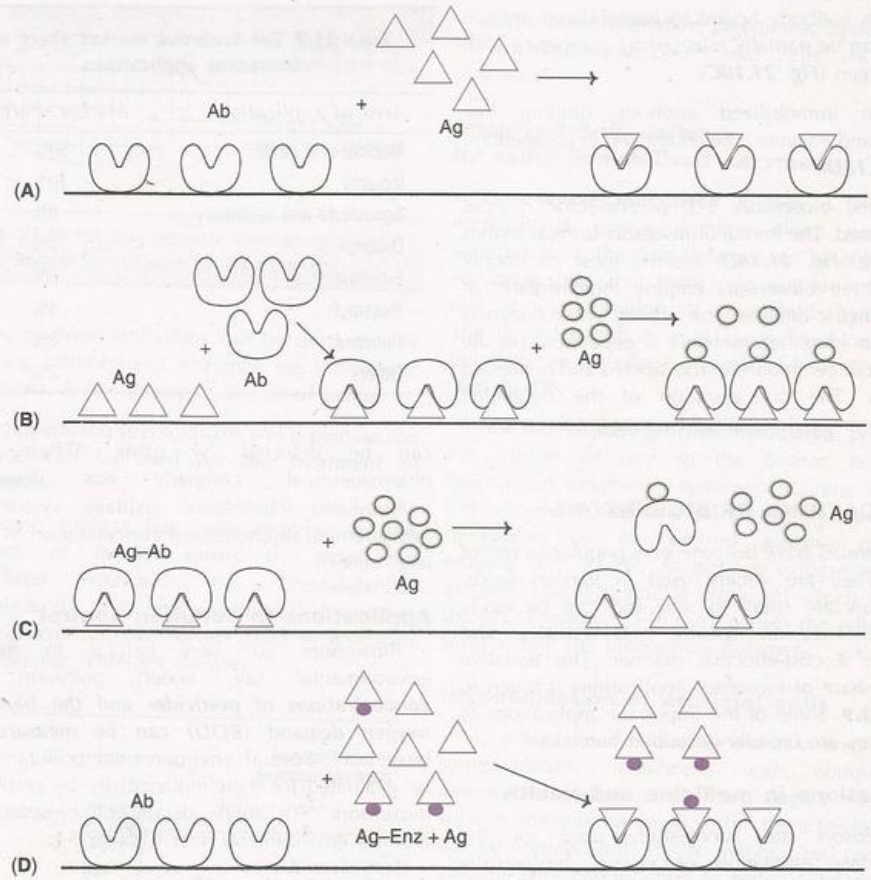


Fig. 21.18 : Diagrammatic representation of selected immunobiosensors (A) Direct binding of antigen to immobilized antibody, (B) Antigen–antibody sandwiches (immobilized antigen binds to antibody and then to a second antigen), (C) Antibody binds to immobilized antigen which gets partially released by a competitive free antigen, (D) Immobilized antibody binds to free antigen and enzyme labeled antigen (in competition).

Limitations of microbial cell biosensors

The whole cells, in general, require longer periods for catalysis. In addition, the specificity and sensitivity of whole cell biosensors may be lower compared to that of enzymes.

IMMUNOBIOSENSORS

Immunobiosensors or ***immunochemical biosensors*** work on the principle of immunological specificity, coupled with measurement (mostly) based on amperometric or potentiometric biosensors. There are several possible configurations for immunobiosensors and some of them are depicted in ***Fig. 21.18***, and briefly described hereunder.

1. An immobilized antibody to which antigen can directly bind (***Fig. 21.18A***).

2. An immobilized antigen that binds to antibody which in turn can bind to a free second antigen (***Fig. 21.18B***).

3. An antibody bound to immobilized antigen which can be partially released by competing with free antigen (***Fig. 21.18C***).

4. An immobilized antibody binding free antigen and enzyme labeled antigen in competition (***Fig. 21.18D***).

For the biosensors 1-3, piezoelectric devices can be used. The immunobiosensors using enzymes (4 above, ***Fig. 21.18D***) are the most commonly used. These biosensors employ thermometric or amperometric devices. The activity of the enzymes bound to immunobiosensors is dependent on the relative concentrations of the labeled and unlabeled antigens. The concentration of the unlabeled antigen can be determined by assaying the enzyme activity.

APPLICATIONS OF BIOSENSORS

Biosensors have become very popular in recent years. They are widely used in various fields. Biosensors are small in size and can be easily handled. They are specific and sensitive, and work in a cost-effective manner. The tentative market share of biosensor applications is given in ***Table 21.9***. Some of the important applications of biosensors are broadly described hereunder.

Applications in medicine and health

Biosensors are successfully used for the quantitative estimation of several biologically important substances in body fluids e.g. glucose, cholesterol, urea. ***Glucose biosensor*** is a boon for diabetic patients for regular monitoring of blood glucose. ***Blood gas monitoring*** for pH, pCO_2 and pO_2 is carried out during critical care and surgical monitoring of patients. Mutagenicity of several chemicals can be determined by using biosensors. Several toxic compounds produced in the body can also be detected.

Applications in industry

Biosensors can be used for ***monitoring of fermentation products*** and estimation of various ions. Thus, biosensors help for improving the fermentation conditions for a better yield.

Now a days, biosensors are employed to ***measure the odour and freshness of foods***. For instance, freshness of stored fish can be detected by ATPase. ATP is not found in spoiled fish and this can be detected by using ATPase. One pharmaceutical company has developed immobilized cholesterol oxidase system for measurement of cholesterol concentration in foods (e.g. butter).

Table 21.9 The tentative market share of biosensor applications

Area of application	*Market share (%)*
Medical and health	60%
Industry	10%
Agriculture and veterinary	8%
Defence	7%
Environmental	6%
Research	4%
Robotics	3%
Others	2%

Applications in pollution control

Biosensors are very helpful to monitor environmental (air, water) pollution. The ***concentrations of pesticides and the biological oxygen demand (BOD) can be measured*** by biosensors. Several environmental pollutants can be evaluated for their mutagenicity by employing biosensors. For more details or biosensors to monitor environment, refer Chapter 54.

Applications in military

Biosensors have been developed to detect the toxic gases and other chemical agents used during war.

IMMOBILIZED ENZYMES AND CELLS-THERAPEUTIC APPLICATIONS

The industrial and analytical applications of immobilized enzymes have been described in the preceeding pages. The therapeutic applications are dealt with here.

There are several limitations for the direct use of enzymes for therapeutic purposes. These include the poor availability of the enzyme at the site of action, sensitivity to natural inhibitors, degradation by endogenous proteases and immunogenicity of certain enzymes.

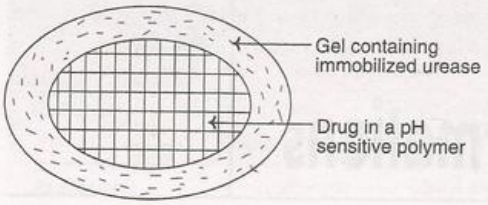


Fig. 21.19 : A diagrammatic representation of immobilized urease for drug delivery.

Some of these limitations can be overcome by employing immobilized enzymes for therapeutic applications. A few examples are listed below.

1. Immobilized streptokinase and urokinase (on Sephadex) can be used ***for the treatment of thromboses***.

2. Some success has been reported in the ***treatment of inborn errors*** by employing immobilized enzymes e.g. phenylalanine hydroxylase to treat phenylketonuria, lysosomal α 1, 4-glucosidase to correct type II glycogen storage disease (Pompe's disease).

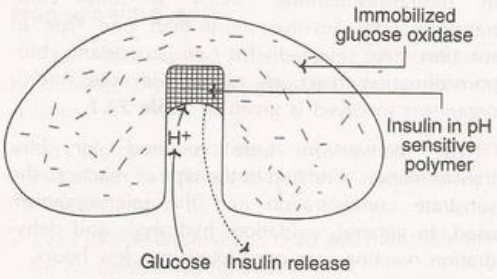


Fig. 21.20 : A diagrammatic representation of immobilized glucose oxidase for insulin delivery.

Some other important therapeutic applications of immobilized enzymes and cells are briefly described.

Improved drug delivery by using immobilized enzymes

Urea-urease modulated system : By using immobilized urease enzyme along with the substrate urea, the drug delivery can be enhanced. As urease splits urea, there is an increase in the pH due to the formation of ammonium hydroxide. The drug located in a pH sensitive bioerodible polymer can be effectively released for its action (***Fig. 21.19***).

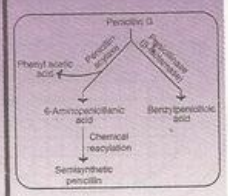
Glucose oxidase-glucose modulated system : For insulin delivery to the human body, a bioerodible polymeric system containing insulin has been developed. The delivery of insulin can be modulated by immobilized glucose oxidase enzyme (***Fig. 21.20***). As glucose oxidase acts on glucose, gluconic acid is produced which lowers the pH. The low pH in turn, causes the release of insulin from the bioerodible polymeric system.

Immobilization of artificial cells

An artificial cell primarily consists of a spherical semipermeable membrane with comparable dimensions of a living cell. The biological materials such as enzymes enclosed within the artificial cells can be immobilized. The so immobilized compact artificial cells can function as artificial organs. The important artifical organs constructed include artificial kidney, artificial liver, blood detoxifiers and immunosorbents. Their functioning is however, very limited.

Multienzyme system can be immobilized in the form of artificial cells for the conversion of a substrate to a product, through a series of reactions.

Biotransformations



Microorganisms possess the capability to enzymatically modify a wide range of organic compounds. Biotransformations (***biconversions*** or ***microbial transformations***) broadly refer to the ***processes in which microorganisms convert organic compounds into structurally related products***. In other words, biotransformation deals with microbial (enzymatic) conversion of a substrate into a product with a limited number (one or a few) enzymatic reactions. This is in contrast to ***fermentation*** which ***involves a large number reactions*** (often complex in nature).

Although there are hundreds of biotransformations known, only a selected few of them are useful for the synthesis of commercially important products. The significance of bioconversion reactions becomes obvious when the production of a particular compound is either difficult or costly by chemical methods. Further, biotransformations are generally preferred to chemical reactions because of substrate specificity, stereospecificity and mixed reaction conditions (pH, temperature, pressure). The environmental pollution due to biotransformation is almost insignificant or negligible. In addition, it is easy to apply recombinant DNA technology to make desired improvements in biotransformations. Another practical advantage of biotransformations is that it is easy to scale-up the processes due to limited number of reactions.

TYPES OF BIOTRANSFORMATION REACTIONS

Many types of chemical reactions occur in biotransformations. These include ***oxidation, reduction, hydrolysis, condensation, isomerization, formation of new C—C bonds, synthesis of chiral compounds*** and ***reversal of hydrolytic reactions***. Among these, oxidation, isomerization and hydrolysis reactions are more commonly observed in biotransformations. Many a times biotransformations involve more than one type of reaction. A selected list of important biotransformation reactions along with the microorganisms involved is given in ***Table 22.1***.

The conversion time required for biotransformation is related to the type of reaction, the substrate concentration and the microorganism used. In general, oxidation, hydrolysis and dehydration reactions are completed in a few hours.

SOURCES OF BIOCATALYSTS AND TECHNIQUES FOR BIOTRANSFORMATION

A wide variety of biological catalysts can be used for biotransformation reactions. These include ***growing cells***, ***resting cells***, ***killed cells***, ***immobilized cells***, cell-free extracts, enzymes and ***immobilized enzymes***. The most important sources of biocatalysts and the procedures employed for biotransformation are briefly described.

TABLE 22.1 A selected list of important biotransformation reactions

Type of reaction	*Example*	*Commonly used microorganism*
Oxidation	Tryptophan ⟶ 5–Hydroxytryptophan	*Bacillus subtitis*
	Naphthalene ⟶ Salicylic acid	*Corynebacterium* sp
Reduction	Benzaldehyde ⟶ Benzyl alcohol	*Saccharomyces cerevisiae*
	Nitropentachlorobenzol ⟶ Pentachloroaniline	*Streptomyces aureofaciens*
Hydrolysis	Anhydrotetracycline ⟶ Tetracycline	*Streptomyces aureofaciens*
	Menthyl laureate ⟶ Menthol	*Mycobacterium phlei*
Condensation	Streptomycin ⟶ Streptomycin-phosphate	*Streptomyces griseus*

Growing cells

The desired cells are cultivated in a suitable medium. As the growth of the cells occurs (6–24 hours), a concentrated substrate is added to the culture. Sometimes, addition of emulsifiers (Tween, organic solvents) is required to solubilize substrates and/or products e.g. steroid biotransformation. The substrate conversion to product can be monitored by spectroscopic or chromatographic techniques. Biotransformation can be terminated when the product formation is optimum.

Non-growing cells

The non-growing cells are preferred for biotransformation reactions for the following reasons.

- Very high concentration of substrate can be used (with high substrate concentration, growing cells stop their growth).
- Cells can be washed and used and thus there will be no contaminating substances.
- ***Conversion efficiency of substrate to product is high.***
- Biotransformation can be optimized by creating suitable environmental conditions (pH, temperature etc.).
- Product isolation and its recovery are easy.

Immobilized cells

Biotransformations can be carried out continuously by employing immobilized cells. Further, the same cells can be used again and again. Several bioconversions with single or multistage reactions are in fact carried out by using immobilized cells e.g. commercial production of L-alanine and malic acid (for details Refer Chapter 21).

Immobilized enzymes

Cell-free enzyme systems in the form of immobilized enzymes are ***most commonly used in biotransformations***, due to the following advantages.

- ***No occurrence of undesirable side reactions.***
- The desired products are not degraded.
- There is no transport barrier across the cell membrane for the substrate or product.
- The isolation and recovery of the product is simpler and easier.

Several immobilized enzyme systems have been developed for biotransformations e.g. glucose isomerase, penicillin acylase (For details, Refer Chapter 21).

PRODUCT RECOVERY IN BIOTRANSFORMATIONS

In most biotransformation reactions, the desired end products are extracellular. The product may be either in a soluble or suspended state. When whole cells are used, they have to be separated and repeatedly washed (with water or organic solvent) as required. The extracted product can be recovered by employing the commonly used techniques-precipitation by salts, extraction with solvents, adsorption to ion-exchangers etc. The volatile poducts can be recovered by direct distillation from the medium.

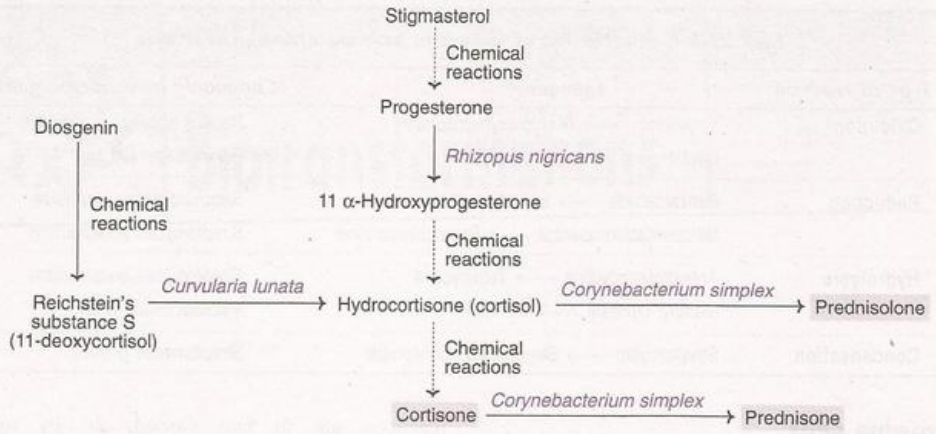


Fig. 22.1 : Biotransformation of commercially important steroids.

EXAMPLES OF BIOTRANSFORMATIONS

A large number of biotransformations are described in literature. Of these, only a selected few are important for industrial and commercial purposes. The major limitations with some of the biotransformations are that either yields are low or the process is expensive or the market itself is very limited.

BIOTRANSFORMATION TO PRODUCE COMMERCIAL PRODUCTS

As already stated, among the large number and a wide range of biotransformations, only a selected few are commercially important. Some of these are briefly described hereunder.

BIOTRANSFORMATION OF STEROIDS

All the ***steroids possess the basic structure*** namely ***cyclopentanoperhydrophenanthrene***. Steroids as hormones (glucocorticoids, mineralocorticoids, androgens, estrogens) perform a wide range of functions. They are very useful therapeutically. For instance, cortisone, due to its anti-inflammatory action is used in the treatment of rheumatoid arthritis and skin diseases; derivatives of progesterone and estrogens are employed as contraceptives. Certain derivatives of cortisone (e.g. prednisolone) are more effective in their therapeutic action.

Commercial production of steroids is very important. Earlier, cortisone was chemically synthesized, and this process involved as many as 37 reactions. The cost of the so obtained product was around $200/g (in 1950). With the introduction of biotransformation reactions, the number of steps (microbial and chemical put together) was reduced to II, and cost of the product was reduced to just $1/g in 1980! The credit obviously goes to the developments in biotransformation.

Types of reactions in biotransformation of steroids

The microbial transformation of steroids broadly involves ***oxidation*** (introduction of hydroxyl groups, splitting of side chains, production of epoxides etc.) ***reduction*** (conversion of aldehydes or ketones to alcohols, hydration of double bonds), ***hydrolysis*** and ***ester formation***.

Production process of steroids

The production of steroids, entirely by bio-transformation reactions is not practicable. Therefore, microbial transformation along with chemical reactions is carried out. The major steps involved in the biotransformation of steroids are depicted in ***Fig. 22.1***. ***Stigmasterol*** extracted from soybeans or ***diosgenin*** isolated from the roots of the Mexican barbasco plant can serve as the starting material. Stigmasterol can be chemically converted

to progesterone which is subjected to biotransformation to form 11α-hydroxyprogesterone by the microorganism, *Rhizopus nigricans*. Cortisol (hydrocortisone), produced from 11α-hydroxyprogesterone by chemical reactions, undergoes microbial transformation (organism-*Corynebacterium simplex*) to form prednisolone. Further, cortisone formed from cortisol can be subjected to biotransformation by *Corynebacterium simplex* to produce prednisone. When diosgenin is used as the starting compound, substance S can be produced by chemical reactions which can be converted to cortisol by biotransformation with the help of the microorganism *Curvularia lunata*.

Biotransformation of steroids is usually carried out by ***batch fermentation***. Use of immobilized cells or immobilized enzymes is gaining importance in recent years. This is advantageous since the biotransformation is more efficient with high substrate concentration, short conversion time and good product recovery. Since the steroids are not water soluble, the microbial transformation reactions have to be carried out in organic solvent (water-immiscible) system. However, the organic solvents are toxic to micro-organisms or enzymes. It is ideal to use an aqueous two phase system for biotransformation of steroids.

Biotransformation of cholesterol

Certain commercially important steroids (e.g. androstendione, androstadiendione) can be produced directly from cholesterol by biotransformation (***Fig. 22.2***).

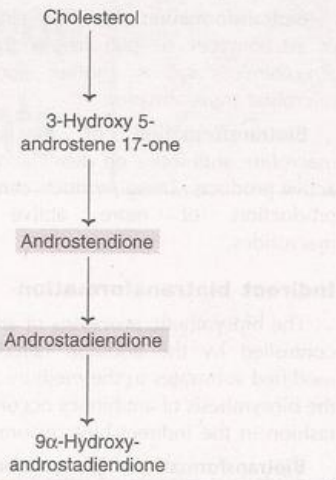


Fig. 22.2 : *Biotransformation of cholesterol by mycobacteria to commercial products.*

BIOTRANSFORMATION OF ANTIBIOTICS

Production of new antibiotics or modifications in the existing ones for more effective treatment of the diseases is always on the priority of the pharmaceutical industry. Further, antibiotics with wider antimicrobial spectrum, reduced toxicity, low allergic reactions and decreased resistance are highly advantageous. Biotransformation reactions significantly contribute for improving the pharmaceutical products.

Direct biotransformation

Acylation and deacylation, phosphorylation, adenylation and hydrolysis are some of the reactions involved in the microbial transformation of antibiotics.

Biotransformation of penicillin G : Microbial transformation, in association with chemical synthesis, is routinely used for the commercial production of semisynthetic penicillins and cephalosporins. The enzymatic cleavage of penicillin by ***penicillin acylase*** into 6-aminopenicillanic acid is a very important reaction (***Fig. 22.3***). Penicillin G gets inactivated by its conversion to benzylpenicilloic acid by the enzyme ***penicillinase*** (β-lactamase).

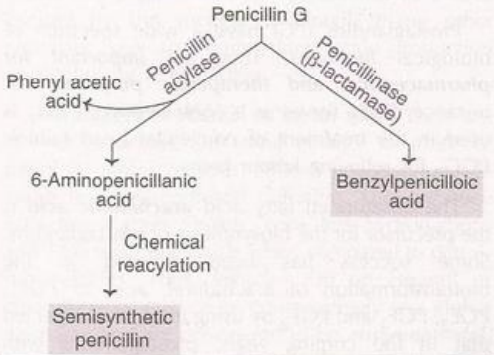


Fig. 22.3 : *Biotransformation of penicillin G.*

Biotransformation of narbomycin : Hydroxylation of narbomycin to picromycin (brought out by *Streptomyces* sp) is another good example of microbial transformation.

Biotransformation of macrolides : The macrolide antibiotics on deacylation will give less active products. These products can be used for the production of more active semisynthetic macrolides.

Indirect biotransformation

The biosynthetic processes of antibiotics can be controlled by the addition certain inhibitors or modified substrates to the medium. In other words, the biosynthesis of antibiotics occurs in a controlled fashion in the indirect biotransformation.

Biotransformation of actinomycins : The microorganism *Streptomyces parvulus* produces new actinomycins in the presence of 4-methyl-proline (proline analog) in the medium. The new antibiotics will have 4-methylproline in place of proline and these actinomycins are more efficient in their function.

Biotransformation of ribostamycin : In the biosynthesis of neomycin, ribostamycin is an intermediate. By employing mutant strains of *Streptomyces fradiae,* ribostamycin can be produced in large quantities.

Several other mutant strains of microorganisms have been created by recombinant DNA technology for the production of modified antibiotics of aminoglycosides and rifamycins.

BIOTRANSFORMATION OF ARACHIDONIC ACID TO PROSTAGLANDINS

Prostaglandins (PG) have a wide spectrum of biological functions. They are ***important for pharmaceutical and therapeutic purposes***. For instance, PGE_1 serves as a contraceptive; PGG_1 is used in the treatment of congenital heart failure; PGG_2 for relieving labour pains.

The unsaturated fatty acid ***arachidonic acid*** is the precursor for the biosynthesis of prostaglandins. Some success has been reported in the biotransformation of arachidonic acid to PGE_1, PGE_2, PGF_1 and PGF_2 by using fungi. It is expected that in the coming years, prostaglandins with improved efficiency will be produced by biotransformations.

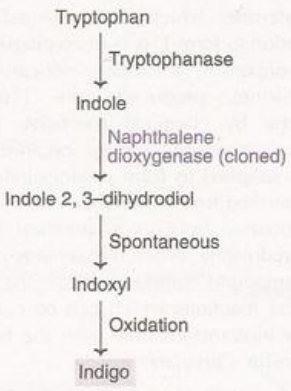


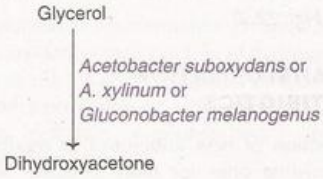
Fig. 22.4 : Microbial production of indigo.

BIOTRANSFORMATION FOR THE PRODUCTION OF ASCORBIC ACID

Ascorbic acid (vitamin C) can be commercially produced by a combination of chemical and microbial transformation processes. For full details, the reader must refer Chapter 24.

BIOTRANSFORMATION OF GLYCEROL TO DIHYDROXYACETONE

Dihydroxyacetone is used in cosmotics and suntan lotions. Certain acetic acid bacteria can convert glycerol to dihydroxyacetone through the process of biotransformation.

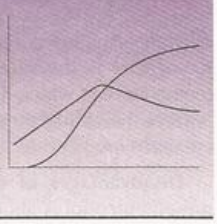


Good oxygen supply, temperature 26–28°C and pH 6.0 are ideal for the optimal biotransformation.

BIOTRANSFORMATION FOR THE PRODUCTION OF INDIGO

Indigo can be synthesized by microbial transformation. This has been made possible by cloning a single *Pseudomonas* gene that encodes naphthalene dioxygenase in the creation of *E. coli*. The relevant reactions of biotransformation for the production of indigo are depicted in ***Fig. 22.4***.

Microbial Production of Organic Solvents



The commercially important organic solvents are ***ethanol, acetone, butanol and glycerol***. These compounds were mostly being produced in petrochemical industries and little attention was paid for their microbial production. It was in mid 1970, due to economic and political reasons, petroleum and natural gas became scarce. Since then, scientists have started looking for suitable alternatives for commercial production of organic solvents. Microbial production of organic solvents utilizing low cost raw materials (e.g. wood, cellulose, starch), is considered to be an ideal alternative.

ALCOHOL

Alcohol, chemically ethanol (C_2H_5OH) has been produced by fermentation for thousands of years. However, this was mostly associated with brewer and distiller industries. At present, most developed countries produce industrial alcohol by chemical means. The petrochemical ethanol is manufactured by hydration of ethylene (C_2H_4).

In the developing countries, microbial fermentation processes are preferred for the production of alcohol. This is mainly because of the cheap raw materials available. At the current rate of oil price, it is cheaper to produce alcohol by petrochemical means rather than fermentation. However, large scale production of alcohol in a given country depends on the political and economic considerations, besides the availability of raw materials. With increasing oil prices, many countries now realise the potential of alcohol production by fermentation.

Ethanol as a motor fuel

Prior to Second World War, ethanol was extensively used as a motor fuel. In fact, some motor companies (e.g. Henry Ford) designed some vehicles to run on alcohol or petrol or a mixture of both.

Brazil was the first country to produce ethanol in large scale by yeast fermentation, utilizing sugarcane and cassava. This ***alcohol used as motor fuel is referred to as green petrol***. Brazil started growing ***alcohol-producing plants*** in a vast area, and installed several fermentation and distillation plants. By 1988, about 90% of the new cars in Brazil had engines that run on alcohol as a fuel. Excited by the success of Brazil, many other countries have also started large scale production of alcohol by fermentation.

Advantages of ethanol as motor fuel : There are distinct advantages of using ethanol in place of petrol for motor vehicles. Ethanol powered engines cause less environmental pollution. They produce 60% less carbon dioxide, 65% less hydrocarbons and 15% less nitric oxide when compared to petrol-run vehicles. Thus, vehicles run on ethanol or ethanol-petrol mix cause comparatively less global warming, and are considered to be ***environmental-***

friendly. Another advantage of ethanol use as a motor fuel is the ***safety***. The flash point (i.e. the temperature at which a substance ignites) of ethanol is three times higher than that of petrol (45°C for ethanol compared to 13°C for petrol).

Disadvantages of ethanol as motor fuel : At present, alcohol is about twice ***costilier than petrol*** in most countries (This may not however, be a limitation in the long run with the availability of cheap substrates and large scale industrial production of alcohol). Since the flash point for ethanol is higher, starting of engines in cold will be difficult. Ethanol may react with alloys containing aluminium and magnesium and may ***damage the containers***. The alcohol used in motor vehicles should be of high purity, and should not be allowed to pick up water from the air. A mixture of ethanol with water does not burn readily, and further it also causes corrosion of engines and tanks.

Gasohol as a fuel : ***A mixture of about 20% ethanol with 80% petrol***, known as gasohol is now being used as a motor fuel some countries like USA. The ethanol used for this purpose should of 100% purity, otherwise petrol and ethanol do not form a uniform mixture.

PRODUCTION OF ETHANOL BY FERMENTATION

As already stated, in recent years many countries have started production of ethanol by fermentation process. The organisms and the raw materials used, along with the biosynthetic pathway, the production and recovery processes for alcohol are briefly described.

Microorganisms

Certain yeasts and bacteria are employed for alcohol fermentation. The type of the organism chosen mostly depends on the nature of the substrate used (***Table 23.1***). Among the yeasts, *Saccharomyces cerevisiae* is the most commonly used, while among the bacteria, *Zymomonas mobilis* is the most frequently employed for alcohol production.

Raw materials

There are a large number of raw materials that can serve as substrates for alcohol fermentation. They may be broadly categorized as ***sugary materials*** (e.g. molasses, whey, glucose, sucrose), ***starchy materials*** (wheat, rice, maize, potato) and ***cellulosic materials*** (wood, agricultural wastes). Some more examples of potential raw materials are given in ***Table 23.2***. The biomass (starchy and cellulosic materials) significantly contributes for the production of alcohol. For more details on biomass, refer Chapter 31.

Table 23.1 Microorganisms used in alcohol fermentation

Microorganism	*Source of carbohydrate*
Yeasts	
Saccharomyces cerevisiae	Starch, sugar
S. ellipsoideus	Starch, sugar
Kluyveromyces fragilis	Starch, sugar
Bacteria	
Zymomonas mobilis	Starch
Candida pseudotropicalis	Lactose, whey
C. utilis	Sulfite waste liquor

Pretreatment of raw materials : Most of the raw materials of alcohol fermentation require some degree of pretreatment. The actual process depends on the chemical composition of the raw material. In general, the sugary raw materials require mild or no pretreatment while the ***cellulosic materials need***

Table 23.2 Potential raw materials for alcohol production

Sugary materials	*Starchy materials*	*Cellulosic materials*
Molasses	**Cereals**	Wood
Sugar cane	Wheat	Saw dust
Sugar beet	Maize	Agricultural wastes
Sweet potato	Barley	Paper wastes
Sweet sorghum	Sorghum	Municipal solid wastes
Whey	Corn	
Sulfite waste	Rice	
Sucrose	**Starchy roots**	
Lactose	Potato	
Glucose	Tapioca	
	Milled products	
	Wheat flour	
	Corn feed	

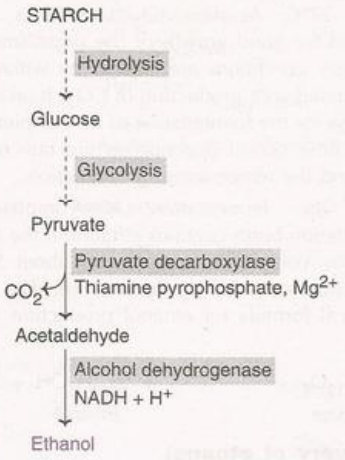


Fig. 23.1 : Biosynthesis of ethanol.

extensive pretreatment. This is because the cellulosic substances have to be subjected to acidic or enzyme hydrolysis to release monosaccharide units that are needed for alcohol production.

Biosynthesis of ethanol

The pathway for the biosynthesis of ethanol is depicted in ***Fig. 23.1***. Glucose gets broken down to pyruvate by glycolysis. Under anaerobic conditions (i.e. in the absence of O_2), pyruvate is converted to acetaldehyde by the enzyme pyruvate decarboxylase. Acetaldehyde is then reduced by alcohol dehydrogenase to form ethanol.

It is observed that under aerobic conditions (adequate O_2 supply) with excess glucose content in the medium, the microorganisms (e.g. *S. cerevisiae*) grow well without producing alcohol. However, ***under anaerobic conditions, growth slows down and alcohol production occurs***.

Regulation of synthesis : ***Ethanol at high*** concentration in the medium ***inhibits its own biosynthesis***. This is particularly observed when yeasts are the fermentation organisms. It is generally seen that growth of yeasts ceases at 5% ethanol concentration (volume/volume in water). It is striking to note that yeasts are sensitive to inhibition by endogenously synthesized ethanol and not to the ethanol added to the medium. Some biotechnologists prefer to use the bacterium *Zymononas mobilis*, since it can tolerate a high concentration of alcohol (up to 13% by some strains against 5% by yeasts).

Glucose to alcohol-conversion profile : Theoretically, one gram of glucose can be converted to 0.511 grams of ethanol. In fact, a conversion yield of 95% was observed when pure substrates (glucose, lactose, sucrose) are used. For industrial grade raw materials (corn starch) the yield is around 90%. It is estimated that ***for 100 g pure glucose, 48.5 g of ethanol is produced***, along with 46.5 g of CO_2, 3.3 g glycerol and 1.3 g of biomass (yeast cells).

Production process of ethanol

Ethanol production can be carried out in three stages-preparation of nutrient solution and inoculum, fermentation proper and recovery. A flow diagram for alcohol production is depicted in ***Fig. 23.2***.

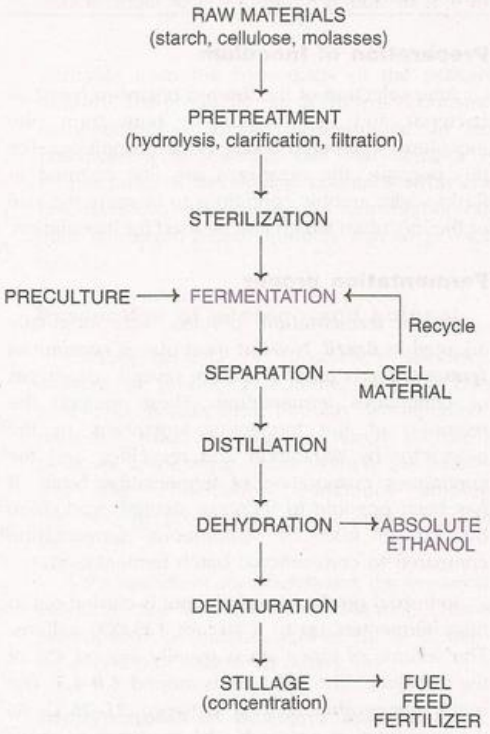


Fig. 23.2 : A schematic representation of ethanol production by fermentation.

Preparation of nutrient solution (media)

The various raw materials for alcohol production have already been described (Refer ***Table 23.1***). The most commonly used raw materials are molasses, whey, grains, potatoes and wood wastes.

When molasses is used for fermentation, it is diluted with water so that the sugar concentration is in the range of 10–18%. A concentration higher than this is detrimental to the yeast. When starchy materials (corn, barely) are used, they have to be first hydrolysed by pretreatment for use as nutrients. This may be done by barley malt, dilute acids or fungal amylases (e.g. *Aspergillus* sp, *Rhizopus* sp). Most frequently mashing with barely malt is used.

Molasses contains most of the nutrients required for alcohol fermentation. However, ammonium salts such as ammonium sulfate or phosphate are added to the nutrient solution to supply nitrogen and phosphorus. The pH of the medium is adjusted to 4-5, by adding sulfuric acid or lactic acid.

Preparation of inoculum

After selection of the desired organism (yeast or bacteria) and its isolation in pure form, the inoculum is prepared under aseptic conditions. For this purpose, the organisms are first cultured in flasks under aerobic conditions to increase the size of the inoculum which can be used for inoculation.

Fermentation proper

Batch fermentation process was originally adopted in ***Brazil***. Now, at most places ***continuous fermentation*** is used. There are several advantages of continuous fermentation. These include the retention of the fermenting organisms in the bioreactor by separation and recycling, and the continuous evaporation of fermentation broth. It has been possible to increase alcohol production by 10–12 fold by continuous fermentation compared to conventional batch fermentation.

Industrial production of alcohol is carried out in huge fermenters up to a size of 125,000 gallons. The volume of inoculum is usually around 4% of the fermenter. The ***ideal pH*** is around ***4.0-4.5***. The initial ***temperature*** is kept between ***21–26°C***. As the fermentation proceeds, the temperature raises to around 30°C. It is necessary to use cooling devices and bring down the temperature to less than 27°C. Ethanol gets evaporated at temperature above 27°C. Aeration (O_2 supply) is initially required for good growth of the organisms. Later, anaerobic conditions are created by withdrawl of O_2 coupled with production of CO_2. It takes about 2-3 days for the fermentation to be completed. The actual time period depends on the raw materials used and the temperature of incubation.

As the fermentation is complete, the fermentation broth contains ethanol in the range of 6-9% by volume. This represents about 90–95% conversion of substrate to ethanol. The overall empirical formula for ethanol production is given below.

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Fermentation}} \underset{\text{Ethanol}}{2C_2H_5OH} + 2CO_2$$

Recovery of ethanol

The cell mass is separated by centrifugation or sedimentation. Ethanol from the fermentation broth can be recovered by successive distillations. By this processes, it is easy to obtain ethanol of around 95%. For a concentration above 95%, special techniques of distillation have to be adopted. For preparation of absolute (100%) alcohol, an ***azeotropic mixture*** of benzene, water and alcohol is first prepared. This mixture is then distilled by gradually increasing the temperature. By this technique, it is possible to first remove benzene-ethanol-water mixture, and then ethanol-benzene mixture. Thus, absolute alcohol is left out.

Stillages in alcohol production

Large volumes of ***wastes*** which are technically referred to stillages are formed during the course of alcohol fermentation. Attempts are made to fruitfully utilize stillages for various purposes.

- To use as feed or fertilizers.
- For converting to single-cell protein.
- To use a fuel.
- For production of methanol.

Genetic engineering for improved alcohol production

With the advent of recombinant DNA technology, attempts are being made worldwide to develop new strains of organisms for increasing alcohol production. Some success has been achieved in creating more efficient microorganisms with the desired characteristics.

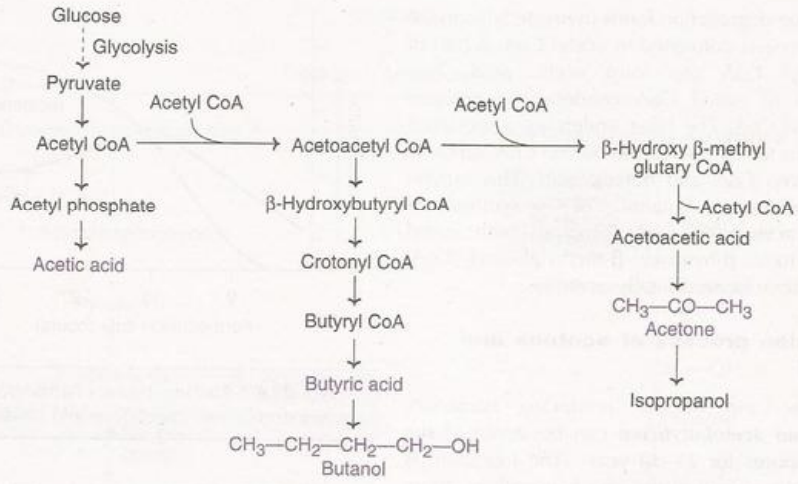


*Fig. 23.3 : Biosynthesis of acetone and butanol (**Note** : Acetic acid and butyric acid can also be produced by this pathway)*

- Improved production of alcohol.
- Resistance to high alcohol concentration.
- Resistance to high temperature.
- Rapid conversion of substrate to alcohol.

It has also been possible to improve alcohol production by using immobilized enzymes in reactor systems.

ACETONE AND BUTANOL

Both acetone and butanol are good ***organic solvents***. Decades ago, acetone was used as a gelatinizing agent for nitrocellulose in the manufacture of explosives. It is also used in dyestuff industry. Butanol is required for the synthesis of butadiene to produce synthetic rubber. At present, butanol is extensively used in break fluids, for antibiotic recovery, as amines for gasoline additives and as ester in protective-coating industry.

The production of acetone and butanol by ***anaerobic fermentation*** is almost a century old. This process was carried out by using starch or sugar-rich raw materials (potato, molasses) and the organism *Clostridium acetobutylicum*. After World War II, acetone and butanol became readily available from the byproducts of the petroleum industry. The production of acetone and butanol by fermentation is ***almost discontinued in many countries***. It is however, said that being a very simple process, developing countries (with cheap raw materials) may soon opt for fermentation rather than petroleum-based products due to economic reasons.

Production of acetone and butanol

Certain bacteria of *Clostridium* species are capable of producing acetone, butanol, butyric acid and isopropanol through fermentation of molasses, wood hydrolysates, starch, sucrose and pentoses. Depending on the strain of the bacterium and the fermentation conditions employed, the relative production of the four compounds varies.

For *Clostridium acetobutylicum*, the fermentation products are acetone and butanol. Butanol-isopropanol or butyric acid-acetic acid can be produced by *Clostridium butyricum*.

Biosynthesis of acetone and butanol

The sequence of reactions leading to the formation of acetone and butanol, along with acetic acid and butyric acid is depicted in ***Fig. 23.3***.

Glucose, on degradation forms pyruvate (glycolysis) which then gets converted to acetyl CoA. A part of this acetyl CoA can form acetic acid. Two molecules of acetyl CoA condense to produce acetoacetyl CoA. The latter undergoes a sequence of reactions to form β-hydroxybutyryl CoA, crotonyl CoA, butyryl CoA and butyric acid. This butyric acid is reduced to butanol. For the synthesis of acetone, acetoacetyl CoA combines with acetyl CoA to form β-hydroxy β-methylglutaryl CoA, acetoacetic acid, and finally acetone.

Production process of acetone and butanol

Acetone and butanol producing bacterium ***Clostridium acetobutylicum*** can be stored in the form of spores for 25–30 years. The inoculum is build up in several stages. Such inoculum, more efficiently ferments sugars to give a high yield. These organisms are also resistant to contamination.

The fermentation is usually carried out in corn or molasses based medium. The molasses medium contains molasses, ammonium sulfate, calcium carbonate and sometimes corn steep liquor.

The fermenters after sterilization are gased with CO_2. After the addition of the medium and inoculum (approximately 2-4% by volume), the fermenter contents are again stirred with CO_2. The fermentation is carried out with a starting pH 5.8-6.0, and temperature 34°C. The actual process of fermentation, spread over 36 hours, occurs in three phases.

Phase I is characterized by rapid growth of the organism and production acetic acid and butyric acid. There is an increase in titratable acidity. The pH of the medium drops to around 5.2.

In phase 2, an increased production of acetone and butanol (solvents) coupled with a decrease in titratable acidity (the latter phenomenon is referred to as ***acid break***).

During the ***third phase***, there is a decrease in the solvent production with a slight or no increase in pH.

In ***Fig. 23.4***, the fermentation processes with reference to solvent production and titratable acidity are depicted.

Contamination : Absolute sterile conditions are required for good production of acetone and butanol. Contamination due to bacteriophages and lactobacilli (particularly *Lactobacillus leichmannii*) is a major problem that hinders the yield.

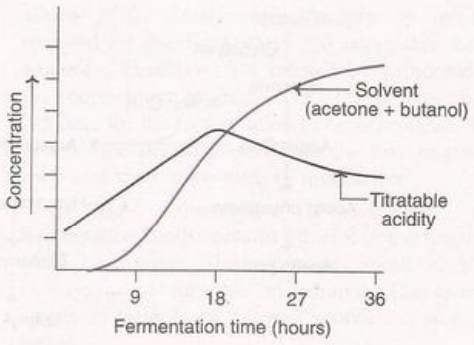


Fig. 23.4 : Acetone–butanol fermentation with reference to solvent production and titratable acidity.

Product yield : Approximately, 30% of the carbohydrate gets converted to neutral solvents in the fermentation. In molasses medium, the ratio of butanol and acetone is 7 : 3, while in corn medium it is 6 : 3. The actual production of butanol is influenced by its toxicity to the organism. Butanol is toxic at a concentration higher than 13.5% in the medium.

Product recovery : Acetone and butanol are recovered through continuous distillation and fractionation. The left over residue after distillation can be dried and used as animal feed.

GLYCEROL

Glycerol is widely used in industry and commerce. It is particularly important as a ***starting material for the manufacture of explosives***. Glycerol is commercially ***produced by the saponification of fats and oils***. It can also be chemically synthesized from propylene or propane. As such, glycerol is not usually produced by fermentation.

During the First World War, the Germans could not import vegetable oils to produce glycerol, due to British naval blockade. And for this reason, Germans were forced to produce glycerol by microbial fermentation. At present, glycerol

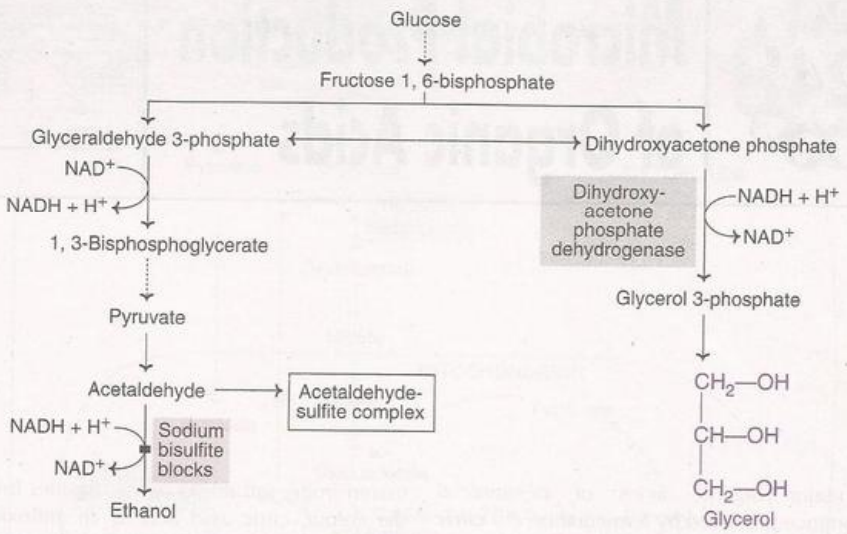


Fig. 23.5 : Biosynthesis of glycerol.

fermentation is of no significance except theoretical and historical importance. In addition, this fermentation process also demonstrates how a modification in the fermentation condition leads to a modification in the product formed.

Production process of glycerol

Basically, glycerol is produced by yeast during the course of alcohol fermentation. However, in the normal process, the quantity of glycerol formed is very low. Addition of sodium sulfite blocks alcohol production and diverts the pathway for the large scale production of glycerol (***Fig. 23.5***). Sodium sulfite reacts with CO_2 in the medium and gets converted to sodium bisulfite. The latter combines with acetaldehyde (an intermediate in ethanol formation) to form acetaldehyde-sulfite complex and thus blocks alcohol synthesis. NADH, generated in one of the early reactions of glycolysis is utilized for glycerol production (instead of ethanol formation due to inhibition).

The fermentation is run for 2-3 days. The yield of glycerol is in the range of 20–30% of the sugar used. This yield is about half of the theoretical value (~ 50%) due to the fact that ethanol fermentation cannot be completely inhibited. Further, the recovery of glycerol from the fermentation broth also is incomplete.

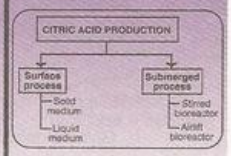
Glycerol production by alga

A novel fermentation process for glycerol production by the alga *Dunaliella salina* has been developed in Israel. This organism, commonly found in salty (hypersaline) lakes synthesizes glycerol within the cells to balance the high osmotic pressure in the surroundings (due to salts). Thus, the higher is the salt concentration in the surroundings, the more is the intracellular glycerol production. However, when the surrounding salt concentration is suddenly reduced, glycerol is excreted into the medium. This principle is successfully exploited for the production glycerol by *D. salina*.

Glycerol production by *Bacillus subtilis*

Bacillus subtilis is normally an aerobic microorganism. It is capable of converting glucose to glycerol, ethanol, lactic acid, butanediol etc. But the production of glycerol under aerobic conditions is very low. However, under anaerobic conditions, the production of glycerol by *B. subtilis* is quite high. And the yield of glycerol comparable to that obtained by yeast fermentation.

Microbial Production of Organic Acids



The major organic acids of commercial importance produced by fermentation are ***citric acid***, ***gluconic acid***, ***lactic acid***, ***acetic acid***, ***ascorbic acid*** and ***itaconic acid***. These compounds may be produced directly from glucose (e.g., gluconic acid) or formed as end products from pyruvate or ethanol (e.g., lactic acid and acetic acid). The production of most of the organic acids is directly or indirectly linked with Krebs cycle.

Besides microbiological production by fermentation, some organic acids are also produced by chemical methods e.g. acetic acid and lactic acid. Some details on the production of important organic acids by fermentation are given in the following pages.

CITRIC ACID

Citric acid was first discovered as a constituent of lemon. Today, we know citric acid as an intermediate of ubiquitous Krebs cycle (citric acid cycle), and therefore, it is present in every living organism. In the early days, citric acid was isolated from lemons (that contain 7-9% citric acid), and today about 99% of the world's citric acid comes from microbial fermentation.

Applications of citric acid

1. Citric acid, due to its pleasant taste and palatability, is used as a ***flavouring agent*** in foods and beverages e.g., jams, jellies, candies, desserts, frozen fruits, soft drinks, wine. Besides brightening the colour, citric acid acts as an antioxidant and preserves the flavors of foods.

2. It is used in the ***chemical industry*** as an antifoam agent, and for the treatment of textiles. In metal industry, pure metals are complexed with citrate and produced as metal citrates.

3. In ***pharmaceutical industry***, as trisodium citrate, it is used as a blood preservative. Citric acid is also used for preservation of ointments and cosmotic preparations. As iron citrate, it serve as a good source of iron.

4. Citric acid can be utilized as an ***agent for stabilization*** of fats, oils or ascorbic acid. It forms a complex with metal ions (iron, copper) and prevents metal catalysed reactions. Citric acid is also used as a stabiliser of emulsions in the preparation of cheese.

5. In ***detergent/cleaning*** industry, citric acid has slowly replaced polyphosphates.

Microbial strains for citric acid production

Many microorganisms can produce citric acid. The ***fungus Aspergillus niger*** is most ***commonly used*** for industrial production of citric acid. The other organisms (although less important) include *A. clavatus, A. wentii, Penicillium luteum, Candida catenula, C. guilliermondii* and *Corynebacterium* sp.

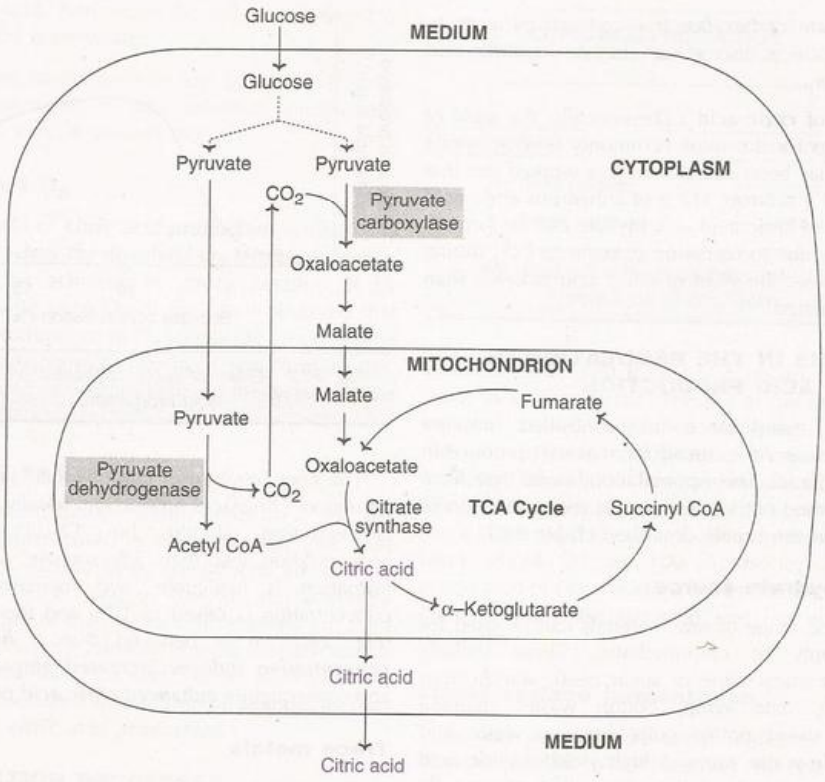


Fig. 24.1 : An outline of metabolic pathway for the biosynthesis of citric acid (TCA cycle-Tricarboxylic acid cycle).

For improved industrial production of citric acid, mutant strains of *A. niger* have been developed. The strains that can tolerate high sugar concentration and low pH with reduced synthesis of undersirable byproducts (oxalic acid, isocitric acid and gluconic acid) are industrially important.

Microbial biosynthesis of citric acid

Citric acid is a ***primary metabolic product*** (of primary metabolism) formed in the tricarboxylic acid (Krebs) cycle. Glucose is the predominant carbon source for citric acid production. The biosynthetic pathway for citric acid production involves ***glycolysis*** wherein ***glucose*** is converted ***to two molecules of pyruvate***. Pyruvate in turn forms acetyl CoA and oxaloacetate which condense to finally give citrate. The major steps in the biosynthesis of citric acid are depicted in ***Fig. 24.1***.

Enzymatic regulation of citric acid production : During the synthesis of citric acid, there is a tenfold increase in the activity of the enzyme ***citrate synthase*** while the activities of other enzymes (aconitase, isocitrate dehydrogenase) that degrade citric acid are reduced. However, recent evidence does not support the theory that reduction in the operation of tricarboxylic acid (i.e. degradation of citric acid) contributes to accumulation of citric acid. Increased citric acid is more likely due to enhanced biosynthesis rather than inhibited degradation. Further, there are anaplerotic reactions that replenish the TCA cycle intermediates to keep the cycle continuously in operation.

Pyruvate carboxylase that converts pyruvate to oxaloacetate is also a key enzyme in citric acid production.

Yield of citric acid : Theoretically, the yield of citric acid for the most commonly used substrate sucrose has been calculated. It is worked out that from 100 g sucrose, 112 g of anhydrous citric acid or 123 g of citric acid — 1 hydrate can be formed. However, due to oxidation of sugar to CO_2 during trophophase, the yield of citric acid is lower than the calculated.

FACTORS IN THE REGULATION OF CITRIC ACID PRODUCTION

Strict maintenance of controlled nutrient conditions is very crucial for maximal production of citric acid. The optimal conditions that have been worked out for *A. niger* for the production of citric acid are briefly described (***Table 24.1***).

Carbohydrate source

A wide range of raw materials can be used for the supply of carbohydrates. These include ***molasses*** (sugar cane or sugar beet), ***starch*** (from potatoes), date syrup, cotton wastes, banana extract, sweet potato pulp, brewery waste and pineapple waste water. A high yield of citric acid production occurs if the sugars that are rapidly metabolised are used e.g. sucrose, glucose, maltose. At present, cane molasses and beet molasses are commonly used. The variations in the composition of molasses (seasonal and production level), have to be carefully considered for optimising citric acid production.

Table 24.1 Optimal parameters/conditions for citric acid production

Condition/parameter	*Optimum*
Sugar concentration	10–25%
Trace metal concentration	
Manganese	$<10^{-8}$ M
Zinc	$<10^{-7}$ M
Iron	$<10^{-4}$ M
pH	1.5–2.5
Dissolved O_2 tension	150 mbr
Ammonium salts concentration	>0.2%
Time	150–250 hours

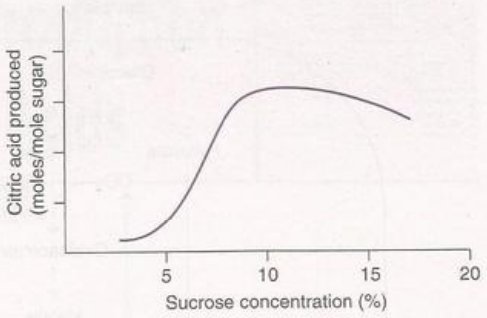


Fig. 24.2 : Effect of sugar concentration on citric acid production.

The concentration of carbohydrate significantly influences citric acid production. Ideally, the sugar concentration should be 12–25%. At a concentration less than 5% sucrose, citric acid formation is negligible, and increases as the concentration is raised to 10% and then stablises (***Fig. 24.2***). It is believed that a ***high sugar concentration induces increased glucose uptake*** and consequently ***enhanced citric acid production***.

Trace metals

Certain trace elements (Fe, Cu, Zn, Mn, Mg, Co) are essential for the growth of *A. niger*. Some of the trace metals particularly ***Mn^{2+}, Fe^{3+} and Zn^{2+} increase the yield of citric acid***.

The effect of manganese ions has been investigated to some extent. These ions promote glycolysis and reduce respiration, both these processes promote citric acid production.

As regards iron, it is a cofactor for the enzyme aconitase (of TCA cycle). It is estimated that an Fe concentration of 0.05-0.5 ppm is ideal for optimal citric acid production. At higher Fe concentration, the yield is lower which can be reversed to some extent by adding copper.

pH

The pH of the medium influences the yield of citric acid, and it is ***maximal when pH is below 2.5***. At this pH, the production of oxalic acid and gluconic acid is suppressed. Further, at low pH, transport of citric acid is much higher. If the pH is above 4, gluconic acid accumulates at the expense

of citric acid. And when the pH goes beyond 6, oxalic acid accumulates.

Another advantage with low pH is that the risk of contamination is very minimal, since many organisms cannot grow at this pH.

Dissolved O_2

The yield of ***citric acid production*** substantially ***increases when the dissolved O_2 tension is higher.*** This can be achieved by strong aeration or by sparging with pure O_2. It has been observed that sudden interruptions in O_2 supply (as occurs during power breakdowns) cause drastic reduction in citric acid production without harming the growth of the organism.

Nitrogen source

Ammonium salts, nitrates and urea are the nitrogen sources used in the media for citric acid production. All the three compounds are equally good sources, as long as they do not adversely effect the pH of the medium. If molasses are used for nutrient supply, addition of extra nitrogen source is not required. However, some workers have shown that exogenous ***addition of ammonium ions stimulates citric acid production.***

PRODUCTION PROCESSES FOR CITRIC ACID

There are two processes by which citric acid can be industrially produced — the surface process and submerged process (***Fig. 24.3***).

The surface process : This is characterized by growing the ***microorganisms*** as a layer or a film ***on a surface*** in contact with the nutrient medium, which may be ***solid or liquid*** in nature. Thus, the surface process has supported-growth systems.

The submerged process : In this case, the ***organisms*** are immersed in or ***dispersed throughout*** the nutrient ***medium.*** There are two types of submerged fermenters (bioreactors) ***stirred bioreactors*** and ***airlift bioreactors.***

SURFACE PROCESSES

Solid surface fermentation

Surface processes using solid substrates are particularly carried out in less developed areas of some Asian countries. The solid substrates such as wheat bran or pulp from sweet potato starch are used, as culture media. The pH of the medium is adjusted to 4-5, and then sterilized. Now the inoculum in the form of spores of *A. niger* is spread as layers (3-6 cm thickness) and incubated at 28°C. The growth of the organisms can be accelerated by the addition of α-amylase. Solid-state fermentation takes about 80 to 100 hours for maximal production of citric acid. At the end of the process, citric acid can be extracted into hot water and isolated.

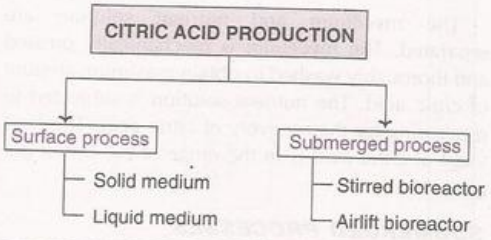


Fig. 24.3 : The industrial processes for the production of citric acid.

Liquid surface fermentation

Surface fermentation using liquid as nutrient medium is the oldest method for citric acid production. It is still in use due to a simple technology, low energy costs and higher reproducibility. Further, the interference of trace metals and dissolved O_2 tension are minimal. The labour costs are however, higher since the manpower requirements are more for cleaning the systems. About 20% of the citric acid in the world is produced by surface processes.

The nutrient supply for surface fermentation normally comes from beet molasses. The fermentation is usually carried out in aluminium trays filled with sterile nutrient medium. The inoculum in the form of spores is sprayed over the medium. A sterile air is passed for supplying O_2 as well as cooling. The temperature is maintained around 30°C during fermentation. As the spores germinate (that occurs within 24 hours of inoculation), a layer of mycelium is formed over the medium. The pH of the nutrient medium falls to less than 2, as the mycelium grows in size and forms a thick layer on the surface of the nutrient solution. The fermentation is stopped after 7–15 days.

The mycelium and nutrient solution are separated. The mycelium is mechanically pressed and thoroughly washed to obtain maximum amount of citric acid. The nutrient solution is subjected to processing for the recovery of citric acid. The final yield of citric acid is in the range of 0.7-0.9 of per gram of sugar.

SUBMERGED PROCESSES

Around 80% of the world's supply of citric acid is produced by submerged processes. This is the most preferred method due to its high efficiency and easy automation. The disadvantages of submerged fermentation are — adverse influence of trace metals and other impurities, variations in O_2 tension, and advanced control technology that requires highly trained personnel.

Two types of bioreactors are in use — ***stirred tanks*** and ***aerated towers***. The vessels of the bioreactors are made up of high-quality stainless steel. The sparging of air occurs from the base of the fermenter.

The success and yield of citric acid production mainly depend on the structure of mycelium. The mycelium with forked and bulbous hyphae and branches which aggregate into pellets is ideal for citric acid formation. On the other hand, no citric acid production occurs if the mycelium is loose and filamentous with limited branches. An adequate supply of O_2 (20–25% of saturation value) is required for good production of citric acid. The ideal aeration rate is in the range of 0.2-1 vvm (volume/volume/minute).

The submerged fermenters have the problem of foam formation which may occupy about 1/3rd of the bioreactor. Antifoam agents (e.g. lard oil) and mechanical antifoam devices are used to prevent foaming.

Nutrient concentration is very important in the industrial production of citric acid. A diagrammatic representation of sucrose, citric acid and biomass concentration with respect to cultivation time is shown in ***Fig. 24.4***. It is estimated that under optimal conditions, in about 250–280 hours, 100–110 g/l of citric acid is obtained from 140 g/l of sucrose with a biomass (dry weight) of 8–12 g/l.

PRODUCTION OF CITRIC ACID FROM ALKANES

Both yeasts and bacteria can be used for citric acid production from n-alkanes (C_9–C_{23} hydrocarbons). The citric acid yield is better from hydrocarbons compared to sugars i.e. 145% of citric acid from paraffin. The most commonly used organism is *Candida lipolytica*. The fermentation can be carried out in batch, semi-continuous or continuous modes. The pH should be kept above 5.

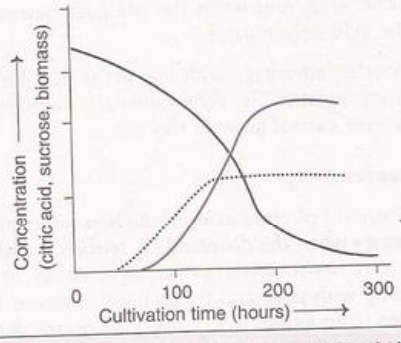


Fig. 24.4 : A diagrammatic representation of citric acid (—) production along with sucrose (—) and biomass (- - -) concentration in relation to time.

The major limitations of citric acid production from alkanes are—very low solubility of alkanes and increased production of unwanted isocitric acid.

RECOVERY OF CITRIC ACID

The steps for the recovery of citric acid either from surface process or submerged process are comparable (***Fig. 24.5***). The recovery starts with the filtration of the culture broth and washing of mycelium (which may contain about 10% of citric acid produced). Oxalic acid is an unwanted byproduct and it can be removed by precipitation by adding lime at pH < 3. The culture broth is then subjected to pH 7.2 and temperature 70–90°C for precipitating citric acid. For further purification, citric acid is dissolved in sulfuric acid (calcium sulfate precipitate separates). The final steps for ***citric acid recovery*** are — ***treatment with activated charcoal, cation and anion-exchangers and crystallization***. Citric acid monohydrate formed below 36°C is the main commercial product. Above 40°C, citric acid crystallizes in an anhydrous form. The degree of purity of citric acid produced depends on the purpose for which it is required. For instance, pure forms of citric acid are needed for use in food preparations, while for industrial use it can be crude form.

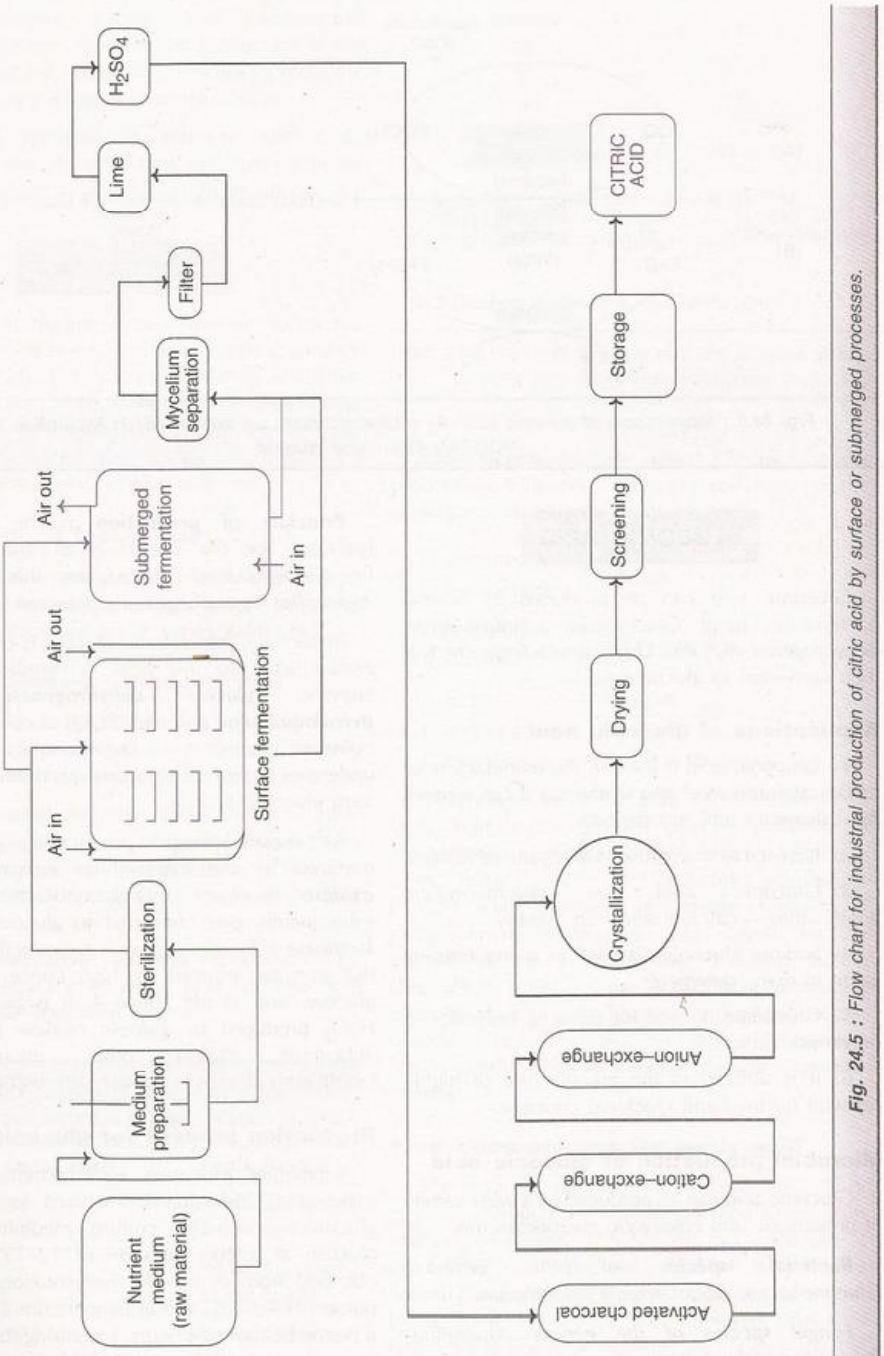


Fig. 24.5 : Flow chart for industrial production of citric acid by surface or submerged processes.

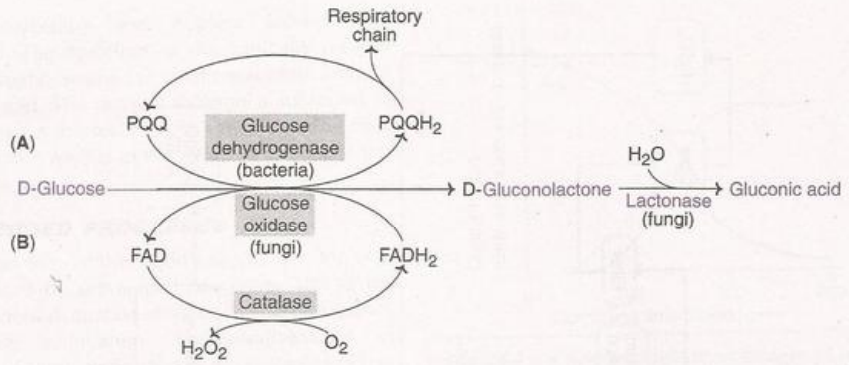


Fig. 24.6 : Biosynthesis of gluconic acid (A) in Gluconobacter suboxidans *(B) in* Aspergillus niger. *(PQQ-Pyrroloquinoline quinone)*

GLUCONIC ACID

Gluconic acid can be produced by several bacteria and fungi. Glucose, on a simple direct dehydrogenation, forms D-gluconolactone which is then converted to gluconic acid.

Applications of gluconic acid

1. Gluconic acid is used in the manufacture of metals, stainless steel and leather, as it can remove the calcareous and rust deposits.

2. It is used as an additive to foods and beverages.

3. Gluconic acid has pharmaceutical applications — calcium and iron therapy.

4. Sodium gluconate is used as a sequestering agent in many detergents.

5. Gluconate is used for desizing polyester or polyamide fabrics.

6. It is utilized in the manufacture of highly resistant (to frost and cracking) concrete.

Microbial production of gluconic acid

Gluconic acid can be produced by a wide variety of prokaryotic and eukaryotic microorganisms.

Bacterial species of the genera—*Gluconobacter, Acetobacter, Pseudomonas, Vibrio.*

Fungal species of the genera—*Aspergillus, Penicillium, Gliocladium.*

Principle of production : The enzymatic reactions for the formation of gluconic acid in *Gluconobacter suboxidans* (bacteria) and *Aspergillus niger* (fungus) are depicted in ***Fig. 24.6***.

In bacteria, intracellular glucose is converted to extracellular gluconic acid. A membrane bound enzyme, ***glucose dehydrogenase*** utilizes ***pyrroloquinoline quinone (PQQ)*** as coenzyme and converts glucose to δ-D-gluconolactone which undergoes hydrolysis (spontaneous or enzymatic) to form gluconic acid.

As regards fungal production, glucose is oxidized by the extracellular enzyme ***glucose oxidase*** to form δ-D-gluconolactone, which subsequently gets converted to gluconic acid by ***lactonase***. Glucose oxidase is an inducible enzyme that can be induced by high concentrations of glucose, and at pH above 4. It is believed that H_2O_2 produced by glucose oxidase acts as an antagonist against other microorganisms (antimicrobial activity) in the surroundings.

Production process for gluconic acid

Submerged processes, by employing either *A. niger* or *G. suboxidans,* are used for producing gluconic acid. The culture medium contains glucose at a concentration of 12–15% (usually obtained from corn). The fermentation is carried out at pH 4.5-6.5, and at temperature 28–30°C for a period of about 24 hours. Increasing the supply of O_2 enhances gluconic acid yield.

Biotechnologists exploit the fermentation process of gluconic acid for the production of the enzyme glucose oxidase, besides producing calcium gluconate and sodium gluconate.

Chemical synthesis of gluconic acid : By employing the immobilized enzyme glucose oxidase, gluconic acid can also be produced.

LACTIC ACID

Lactic acid occurs in two isomeric forms i.e. L(+) and D(–) isomers, and as a racemic mixture (DL-lactic acid). The isolation of lactic acid from milk was done in 1798. It was the first organic acid produced by microorganisms in 1880. Today, lactic acid is competitively produced both by microbiological and chemical methods.

Applications of lactic acid

There are different grades of lactic acid mainly based on the percentage of lactic acid. The grades and their applications are given in ***Table 24.2***.

Microorganisms for production of lactic acid

Lactic acid producing bacteria are broadly categorized into two types.

Heterofermentative bacteria—produce other byproducts, besides lactic acid, and therefore are not useful for industrial production of lactic acid. These bacteria are employed in food or feed preservation.

Homofermentative bacteria—specialised for exclusive production of lactic acid and therefore are suitable for industrial purpose.

TABLE 24.2 Commercial grades of lactic acid along with their applications

Grade (% lactic acid)	*Application(s)*
Technical grade (20–50%)	Ester manufacture, textile industry
Food grade (>80%)	Food additive (sour flour and dough)
Pharmaceutical grade (>90%)	Intestinal treatment (metal ion lactates)

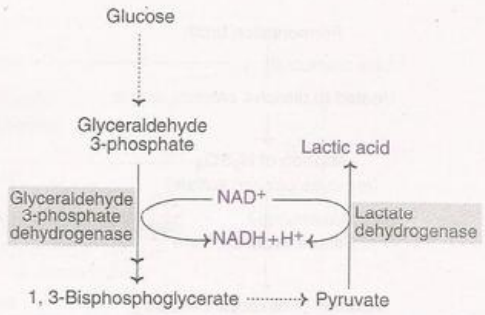


Fig. 24.7 : An outline of the pathway (glycolysis) for the biosynthesis of lactic acid.

Lactobacillus sp are used for lactic acid production. However, there are variations in the substrates utilised as indicated below.

L. delbrueckii, *L. leichamanni* } Glucose
L. bulgaricus, *L. helvetii* } Whey (lactose)
L. lactis — Maltose
L. amylophilus — Starch
L. pentosus — Sulfite waste liquor

Biosynthesis of lactic acid : The synthesis of lactic acid occurs through glucose oxidation by glycolysis to produce pyruvate which on reduction gives lactic acid. The reducing equivalents ($NADH^+ + H^+$) produced during the oxidation of glyceraldelyde 3-phosphate are utilised by the enzyme lactate dehydrogenase to form lactate (***Fig. 24.7***). Most of the lactic acid producing microorganisms normally produce only one isomer of lactic acid L(+) or D(–). However, some bacteria which usually occur as infection can form racemic mixture.

Production process for lactic acid

The fermentation medium contains 12–15% of glucose, nitrogen and phosphate containing salts and micronutrients. The process is carried out at pH 5.5-6.5 and temperature 45–50°C for about 75 hours. Generally, the strains operating at higher temperature (45–60°C) are preferred, since it reduces the need for medium sterilization. As the lactic acid is produced, it has to be removed since

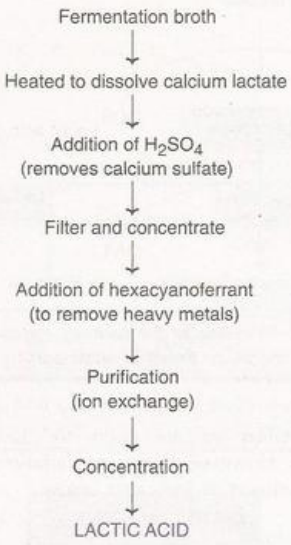


Fig. 24.8 : Flow chart for recovery of lactic acid from fermentation broth.

it is toxic to the organisms. This can achieved either by a continuous culture technique or by removal of lactic acid by electrodialysis. Theoretically, every molecule of glucose forms two molecules of lactic acid. About 90% of theoretical yield is possible in fermentation industry. L(+) Lactic acid is predominantly produced. The outline of the steps involved in the recovery lactic acid is depicted in ***Fig. 24.8***.

ACETIC ACID

The production of acetic acid, in the form of vinegar (used as a refreshing drink), from alcoholic liquids has been known for centuries.

Microorganisms used for production of acetic acid

The commercial production of acetic acid is carried out by a special group of ***acetic acid bacteria***, which are divided into two genera.

Gluconobacter that oxidizes ethanol exclusively to acetic acid.

Acetobacter that oxidizes ethanol first to acetic acid, and then to CO_2 and H_2O. These ***over-oxidizers*** are Gram-negative and acid tolerant e.g. *A. aceti, A. peroxidans, A. pasteurianus.*

Biosynthesis of acetic acid : Acetic acid is a product of incomplete oxidation of ethanol. Ethanol is first oxidized by alcohol dehydrogenase to acetaldehyde which then gets hydrated to form acetaldehyde hydrate. The latter is then acted upon by acetaldehyde dehydrogenase to form acetic acid (***Fig. 24.9***).

Production process for acetic acid

For every molecule of ethanol oxidised, one molecule of acetic acid is produced. Thus, high-yielding strains can produce 11–12% acetic acid from 12% alcohol. For optimal production, adequate supply of oxygen is very essential. Insufficient O_2, coupled with high concentration of alcohol and acetic acid result in the death of microorganisms.

Surface fermentation or submerged fermentation processes can be carried out to produce acetic acid. ***Trickling generation*** process, a type of surface fermentation, is very commonly used.

Recovery : The acetic acid produced is clarified by filtration and then subjected to decolourization by $K_4(FeCN)_6$.

Production of vinegar

Vinegar is an aqueous solution containing ***about 4%*** by volume ***acetic acid*** and small quantities of alcohol, salts, sugars and esters. It is widely used as a ***flavouring agent*** for processed liquid foods such as sauces and ketchups.

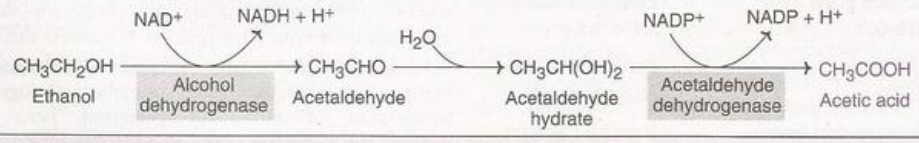


Fig. 24.9 : Biosynthesis of acetic acid from ethanol.

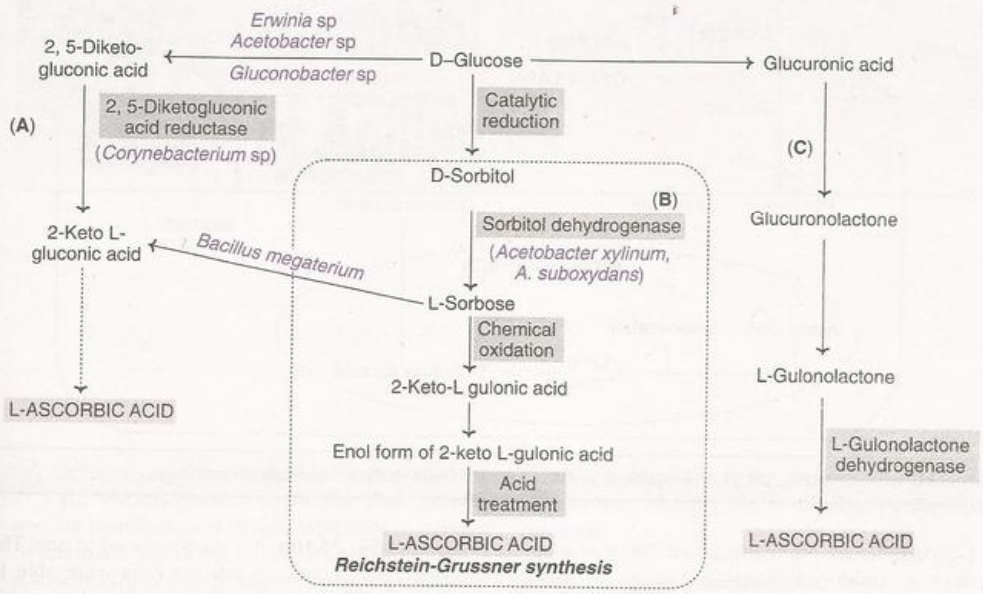


Fig. 24.10 : *Pathways for the commercial production of ascorbic acid. (**A**) Two-step fermentation process (**B**) Reichstein-Grussner synthesis (**C**) Production via L-gulonolactone.*

The starting materials for vinegar production are wine, whey, malt (with low alcohol content). Vinegar production can be carried out either by ***surface process*** (trickling generator) or by ***submerged process***.

Surface process : The fermentation material is sprayed over the surface which trickles through the shavings that contain the acetic acid producing bacteria. The temperature is around 30°C on the upper part while it is around 35°C on the lower part. Vinegar is produced in about 3 days.

Submerged process : The fermentation bioreactors are made up of stainless steel. Aeration is done by a suction pump from the top. The production rate in the submerged process is about 10 times higher than the surface process.

L-ASCORBIC ACID

L-Ascorbic acid is the commonly used chemical name for the water soluble ***vitamin C***. This vitamin forms a redox system and participates in several biological processes. It is intimately involved in the biosynthesis of collagen, the most abundant protein in the human body. Vitamin C also protects the body against carcinogenic nitrosamines and free radicals. The deficiency of ascorbic acid causes ***scurvy***.

Applications of ascorbic acid

Because of the wide range of physiological and beneficial functions of ascorbic acid, its commercial production assumes significance. Vitamin C is mainly used in ***food and pharmaceutical industries***.

Industrial production of ascorbic acid

Ascorbic acid is commercially produced by a combination of several chemical steps, and one reaction of ***biotransformation*** brought out by microorganisms. This process is referred to as ***Reichstein-Grussner synthesis*** (***Fig. 24.10B***). D-Glucose is first converted to D-sorbitol. Oxidation of D-sorbitol to L-sorbose is carried out by *Acetobacter xylinum* or *A. suboxydans* (The enzyme being sorbitol dehydrogenase). A submerged bioreactor fermentation process is ideal for this reaction. It takes about 24 hours at temperature 30–35°C. Sorbose by a couple of chemical reactions can be finally converted to

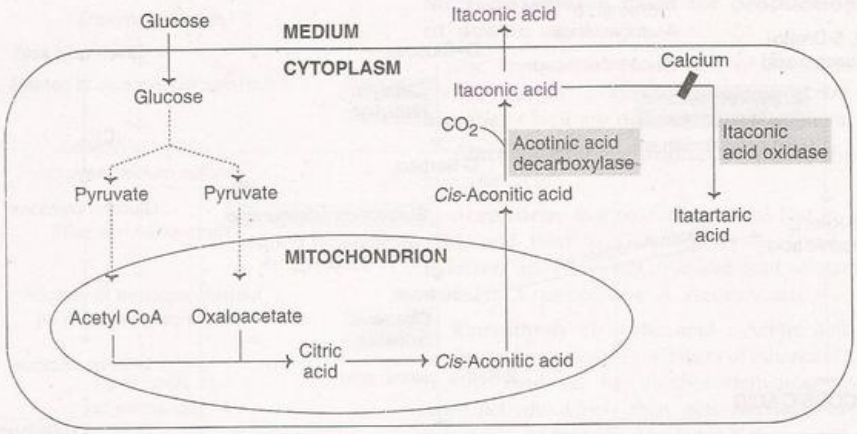


Fig. 24.11 : Biosynthesis of itaconic acid—an outline of metabolic pathway.

L-ascorbic acid. Normally, about 100 g of ascorbic acid is produced from 200 g of glucose in Reichstein-Grussner synthesis.

Two-step fermentation process : In this, D-glucose is converted to 2, 5-diketogluconic acid by *Erwinia, Acetobacter* or *Gluconobacter* sp. In the second step, *Corynebacterium* sp converts 2, 5-diketogluconic acid to 2-keto-L-gluconic acid, (***Fig. 24.10A***). It is also possible to involve ***Bacillus megaterium*** for converting L-sorbose to 2-keto-L-gluconic acid. The latter, by chemical reactions, can be converted to ascorbic acid.

Production via L-gulonolactone

Ascorbic acid can also be synthesized via-gulonolactone which can be directly converted to L-ascorbic acid by the enzyme L-gulonolactone dehydrogenase (***Fig. 24.10C***).

Direct production of ascorbic acid by fermentation

Several workers are trying to produce ascorbic acid directly from glucose. Microalgae of *Chlorella* have shown some promising results, although the yield is very low.

Genetic engineering for ascorbic acid production

Biotechnologists have been successful in ***cloning*** and expressing the gene for ***2, 5-diketogluconic acid reductase*** of *Corynebacterium* sp into *Erwinia herbicola*. By cloing this, the two step fermentation process (***Fig. 24.10A***) has been reduced to one. The genetically engineered *Erwinia* cells were able to convert D-glucose directly to 2-keto-L-gluconic acid. This is certainly advantageous since the metabolic capabilities of two different micro-organisms could be combined into one organism. However, the yield of ascorbic acid by the hybrid strain was very low. Scientists are now trying to alter certain amino acids in 2-5 diketogluconic acid reductase and increase the catalytic activity of this enzyme.

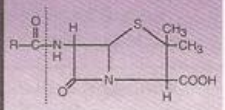
ITACONIC ACID

Itaconic acid is used in plastic industry, paper industry and in the manufacture of adhesives.

Itaconic acid can be commercially produced by *Aspergillus itoconicus* and *A. terreus*. The biosynthesis of itaconic acid occurs by way of Krebs cycle. The metabolite *cis*-aconitic acid (formed from citric acid) undergoes decarboxylation catalysed by the enzyme *cis*-aconitic decarboxylase (***Fig. 24.11***). Itaconic acid is oxidised to itatartaric acid by itaconic acid oxidase. This enzyme has to be inhibited for a maximum yield of itaconic acid. This can be achieved by adding calcium.

Batch submerged fermentation process is commonly used for itaconic acid production. The yield is around 75% of the theoretical calculation when the medium contains 15% sucrose.

Microbial Production of Antibiotics



Antibiotics are the ***chemical substances that can kill microorganisms or inhibit their growth***, and are therefore used to fight infections in humans or animals. Most of the antibiotics are produced by microorganisms (i.e. product of one organism that can kill other organism). Certain semi-synthetic antibiotics are the chemically modified natural antibiotics.

Antibiotics have undoubtedly changed the world we live in, and have certainly contributed to the increase in the human life-span. This is mainly due to the fact that several life-threatening infectious diseases could be conveniently cured by administration of antibiotics.

ANTIBIOTICS — GENERAL

A brief history of antibiotics along with the microorganisms producing them, and their applications are given hereunder.

History of antibiotic discovery

It was in 1928, Alexander Fleming made an accidental discovery that the fungus *Penicillium notatum* produced a compound (penicillin) that selectively killed a wide range of bacteria without adversely affecting the host cells. There are records that in some parts of Europe (in 1908) extracts of moldy bread were applied to wounds or abrasions to prevent infections, although the biochemical basis was not known. The ***penicillin discovery of Fleming has revolutionised antibiotic research***.

Wide range of antibiotics

Antibiotics are the most important class of pharmaceuticals produced by microbial biotechnological processes. They are the ***products of secondary metabolism***.

Around 10,000 different antibiotics are known, and 200–300 new ones are being added each year. Most of these antibiotics are not of commercial importance due to various reasons—toxicity, ineffectiveness or high cost of production. There are around 50 antibiotics which are most widely used.

In ***Table 25.1***, a selected list of important antibiotics, their properties and the producing organisms is given.

Broad spectrum antibiotics : They can control the growth of several unrelated organisms e.g. tetracyclines, chloramphenicol.

Narrow spectrum antibiotics : They are effective against selected species of bacteria e.g. penicillin, streptomycin.

Microorganisms producing antibiotics

A great majority of antibiotics are produced by ***actinomycetes*** particularly of the genus ***Streptomyces*** e.g. tetracyclines, actinomycin D.

TABLE 25.1 A selected list of important antibiotics along with producing organisms

Antibiotic activity specturm and antibiotic	*Producing microorganism*
Antibacterial	
Penicillin G	*Penicillium* sp
Cephalosporin	*Acremonium* sp
Streptomycin	*Streptomyces* sp
Tetracycline	*Streptomyces* sp
Chloramphenicol	*Cephalosporium* sp
Bacitracin	*Bacillus* sp
Antitumor	
Actinomycin D	*Streptomyces* sp
Mitomycin C	*Streptomyces* sp
Bleomycin	*Streptomyces* sp
Adriamycin	*Streptomyces* sp
Daunomycin	*Streptomyces* sp
Antifungal	
Griseofulvin	*Penicillium* sp
Food preservative	
Natamycin	*Streptomyces* sp
Nisin	*Streptomyces* sp
Antiprotozoal	
Daunorubicin	*Steptomyces* sp
Antituberculosis	
Refamycin	*Nocardia* sp
Antiamoebic	
Tetracycline	*Streptomyces* sp
Fumagillin	*Aspergillus* sp

The bacteria other than actinomyces also produce certain antibiotics e.g. bacitracin.

Among the ***fungi***, the two groups ***Aspergillaceae*** and ***Moniliales*** are important for antibiotic production e.g. penicillin, cephalosporin, griseofulvin.

APPLICATIONS OF ANTIBIOTICS

Antibiotics are particularly important ***as antimicrobial agents for chemotherapy***. A large number of bacterial diseases have been brought under control by use of antibiotics. These include pneumonia, cholera, tuberculosis and leprosy. The antifungal antibiotic griseofulvin has controlled the debilitating fungal skin diseases such as ring worm. Besides serving as antimicrobial agents, there are several other applications of antibiotics.

Antitumor antibiotics : There are a selected few antibiotics that are in use for control of cancer growth, although with a limited success e.g. actinomycin D, mitomycin C.

Food preservative antibiotics : Certain antibiotics are used in canning industry (e.g. chlortetracycline), and for preservation of fish, meat and poultry (e.g. pimaricin, nisin). The use of antibiotics in food preservation is usually under the control of the Governments.

Antibiotics used in animal feed and veterinary medicine : Till some time ago, antibiotics (penicillins, tetracyclines, erythromycins) were very widely used in processing of animal feeds. Such an indiscriminate use resulted in the development of antibiotic resistance in animals and humans. A new class of antibiotics have been developed for specific use in animal feed e.g. enduracidin, tylosin. Likewise, specific antibiotics have been developed for exclusive use in veterinary medicine e.g. hygromycin B, theostrepton, salinomycin.

Antibiotics for control of plant diseases : In recent years, several antibiotics have been developed for exclusive use to control plant diseases e.g. blasticidin, tetranactin, polyoxin.

Antibiotics as tools in molecular biology : Some of the antibiotics can selectively inhibit certain biological reactions at the molecular level. These antibiotics do in fact serve as tools for exploring the knowledge of life sciences. Thus, certain antibiotics have been used to obtain some important information on DNA replication, transcription and translation.

PRODUCTION OF ANTIBIOTICS — A MAJOR PHARMACEUTICAL INDUSTRY

The commercial production of antibiotics is a highly profitable industry worldover. Annual sales of antibiotics will run into several billions of dollars with an annual growth potential of about 10%.

Antibiotics may be ***produced by microbial fermentation***, or ***chemical synthesis***, or a ***combination of both***. For certain antibiotics, the

basic molecule is produced by fermentation and its therapeutic value can be increased by chemical modifications. The cost involved in production and chemical modifications, besides the efficacy of the antibiotic is very important in its manufacture.

Biotechnologists continue their efforts to increase the fermentation yield and recovery processes to produce pure antibiotics.

The industrial production of selected antibiotics is briefly described in the following pages.

PENICILLINS

Penicillins are a group of β-lactam containing bactericidal antibiotics. Being the first among the antibiotics to be discovered, penicillins are historically important. The structures of important synthetic and semi-synthetic penicillins are depicted in ***Fig. 25.1***. The basic structure of all the penicillins consists of a lactam ring and a thizolidine ring fused together to form 6-aminopenicillanic acid.

Action of penicillins

Natural penicillins (penicillins V and G) are ***effective against several Gram-positive bacteria***. They inhibit the bacterial cell wall (i.e. peptoglycan) synthesis and cause cell death. Some persons (approximately 0.5-2% of population) are allergic to penicillin.

Natural penicillins are ***ineffective against microorganisms that produce β-lactamase***, since this enzyme can hydrolyse penicillins e.g. *Staphylococcus aureus*. Several semi-synthetic penicillins that are resistant to β-lactamase have been developed and successfully used against a large number of Gram-negative bacteria. Cloxacillin, ampicillin, floxacillin and azlocillin are some examples of semi-synthetic penicillins. These are quite comparable in action to cephalosporins.

From the huge quantities of penicillins produced by fermentation, about 40% are used for human healthcare, 15% for animal healthcare and 45% for the preparation of semi-synthetic penIcillins.

Organisms for penicillin production

In the early days, ***Penicillium notatum*** was used for the large-scale production of penicillins.

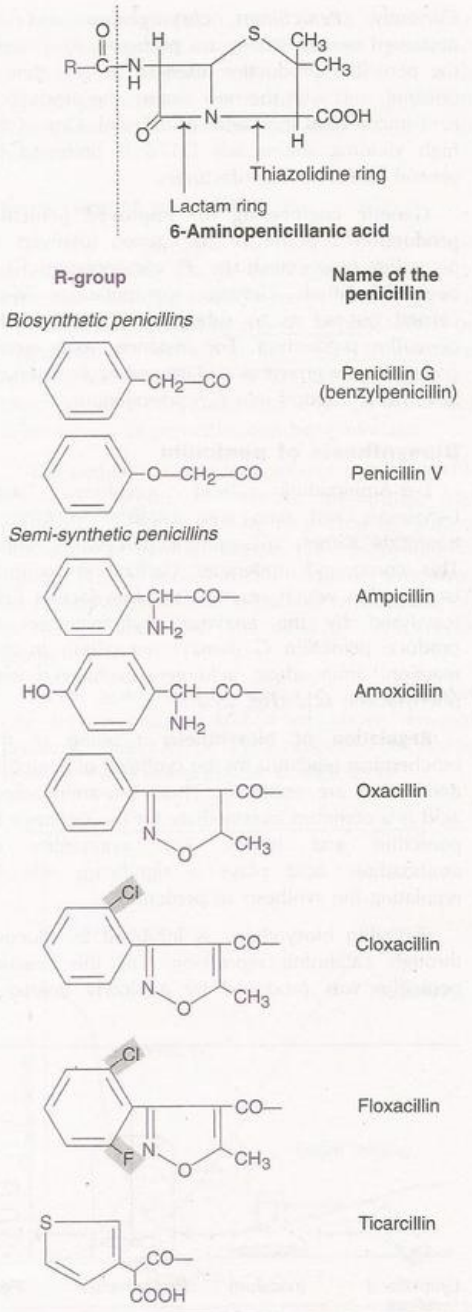


Fig. 25.1 : Structures of important penicillins.

Currently, ***Penicillium chrysogenum*** and its improved mutant strains are preferred. Previously, the penicillin production used to be less than 2 units/ml, and with the new strains, the production runs into several thousands of units/ml. One of the high yielding strains wis Q176 is preferred by several penicillin manufacturers.

Genetic engineering for improved penicillin production : Some of the genes involved in penicillin biosynthesis by *P. chrysogenum* have been identified. Genetic manipulations were carried out so as to substantially increase the penicillin production. For instance, extra genes coding for the enzymes cyclase and acyltransferase have been inserted into *C. chrysogenum.*

Biosynthesis of penicillin

L-α-Aminoadipic acid combines with L-cysteine, and then with L-valine to form a tripeptide namely α-L-aminoadipylcysteinylvaline. This compound undergoes cyclization to form isopenicillin which reacts with phenylacetyl CoA (catalysed by the enzyme acyltransferase) to produce penicillin G (benzyl penicillin). In this reaction, aminoadipic acid gets exchanged with phenylacetic acid (***Fig. 25.2***).

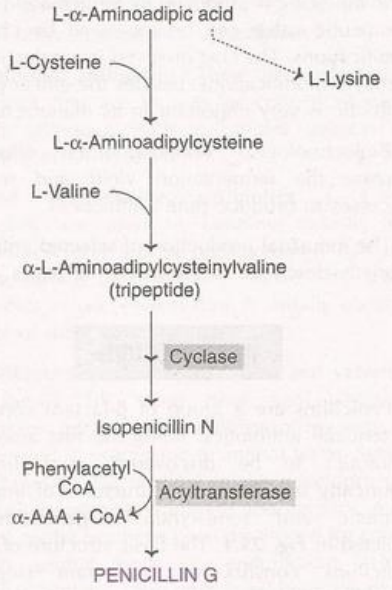


Fig. 25.2 : *Biosynthesis of penicillin by* Penicillium chrysogenum *(α-AAA — α-Amino adipic acid; CoA–Coenzyme A.*

Regulation of biosynthesis : Some of the biochemical reactions for the synthesis of penicillin and lysine are common. Thus, L-α-aminoadipic acid is a common intermediate for the synthesis of penicillin and lysine. The availability of aminoadipic acid plays a significant role in regulating the synthesis of penicillin.

Penicillin biosynthesis is inhibited by glucose through catabolite repression. For this reason, penicillin was produced by a slowly degraded sugar like lactose. The concentrations of phosphate and ammonia also influence penicillin synthesis.

PRODUCTION PROCESS OF PENICILLIN

An outline of the flow chart for the industrial production of penicillin is depicted in ***Fig. 25.3***. The lyophilized culture of spores is cultivated for inoculum development which is transferred to prefermenter, and then to fermenter.

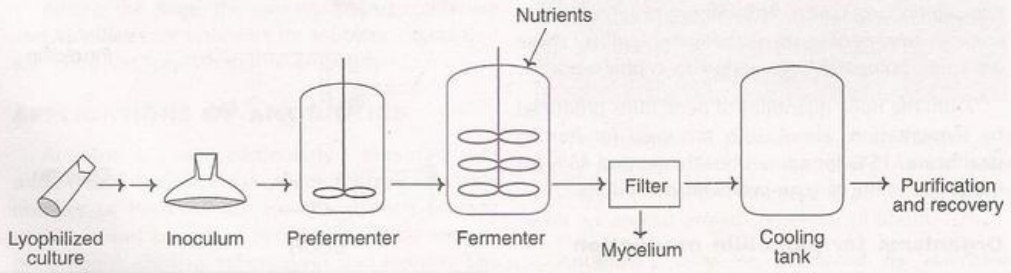


Fig. 25.3 : *An outline of the flow chart for penicillin fermentation.*

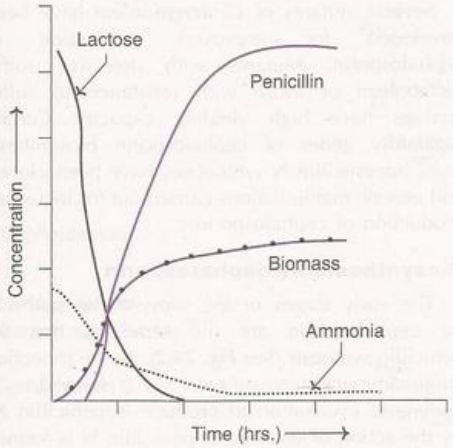


Fig. 25.4 : Penicillin production in relation to substrates utilization and biomass formation.

Penicillin production is an ***aerobic process*** and therefore, a continuous supply of O_2 to the growing culture is very essential. The required aeration rate is 0.5-1.0 vvm. The pH is maintained around 6.5, and the optimal temperature is in the range of 25–27°C. Pencillin production is usually carried out by ***submerged processes***.

The medium used for fermentation consists of corn steep liquor (4-5% dry weight) and carbon source (usually lactose). An addition of yeast extract, soy meal or whey is done for a good supply of nitrogen. Sometimes, ammonium sulfate is added for the supply of nitrogen. Phenylacetic acid (or phenoxyacetic acid) which serves as a precursor for penicillin biosynthesis is continuously fed. Further, continuous feeding of sugar is advantageous for a good yield of penicillin. The penicillin production profiles are depicted in ***Figs. 25.4*** and ***Fig. 25.5***.

It is estimated that approximately 10% of the metabolised carbon contributes to penicillin production, while 65% is utilised towards energy supply and 25% for growth of the organisms. The efficiency of penicillin production can be optimized by adequate supply of carbon source. Thus, by adding glucose and acetic acid, the yield can be increased by about 25%.

For efficient synthesis of penicillin, the growth of the organism from spores must be in a loose form and not as pellets. The growth phase is around 40 hours with a doubling time of 6-8 hours. After the growth phase is stabilized, the penicillin production exponentially increases with appropriate culture conditions. The penicillin production phase can be extended to 150–180 hours.

Recovery of penicillin

As the fermentation is complete, the broth containing about 1% penicillin is processed for extraction. The mycelium is removed by filtration. Penicillin is recovered by solvent (n-butylacetate or methylketone) extraction at low temperature (<10°C) and acidic pH (<3.0). By this way, the chemical and enzymatic (bacterial penicillinase) degradations of penicillin can be minimized.

The penicillin containing solvent is treated with activated carbon to remove impurities and pigments. Penicillin can be recovered by adding potassium or sodium acetate. The potassium or sodium salts of penicillin can be further processed (in dry solvents such as n-butanol or isopropanol) to remove impurities. The yield of penicillin is around 90%.

As the water is totally removed, penicillin salts can be crystallized and dried under required pressure. This can be then processed to finally produce the pharmaceutical dosage forms.

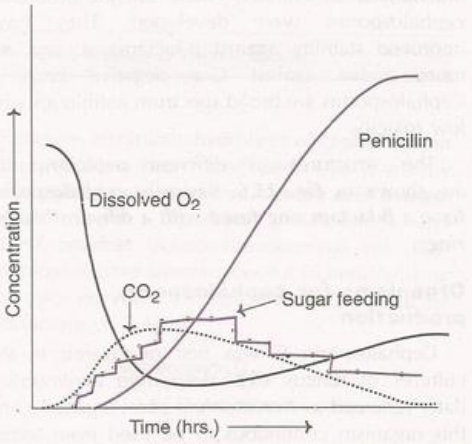


Fig. 25.5 : Penicillin production in relation to continuous feeding of sugar, O_2 utilization, and CO_2 formation.

Penicillins G and H are the fermented products obtained from the fungus *Penicillium chrysogenum*.

PRODUCTION OF 6-AMINO PENICILLANIC ACID

The penicillins G and H are mostly used as the starting materials for the production of several ***synthetic penicillins containing the basic nucleus namely 6-amino penicillanic acid*** (***6-APA***). About 10 years ago, only chemical methods were available for hydrolysis of penicillins to produce 6-APA. Now a days, enzymatic methods are preferred.

Immobilized penicillin amidases enzymes have been developed for specific hydrolysis of penicillin G and penicillin V. Penicillin salt of either G or V can be used for hydrolysis by immobilized enzyme system. The pH during hydrolysis is kept around 7-8, and the product 6-APA can be recovered by bringing down the pH to 4. At pH 4, 6-amino penicillanic acid gets precipitated almost completely in the presence of a water immiscible solvent.

In general, the enzymatic hydrolysis is more efficient for penicillin V than for penicillin G. However, penicillin G is a more versatile compound, as it is required for ring expansions.

CEPHALOSPORINS

The pharmaceutical uses of penicillins are associated with allergic reactions in some individuals. To overcome these allergic problems, cephalosporins were developed. They have improved stability against β-lactamases, and are more active against Gram-negative bacteria. Cephalosporins are broad spectrum antibiotics with low toxicity.

The structures of different cephalosporins are shown in ***Fig. 25.6***. Basically ***cephalosporins have a β-lactam ring fused with a dihydrothiazine ring***.

Organisms for cephalosporin production

Cephalosporin C was first discovered in the cultures of fungus *Cephalosporium acremonium* (later renamed as *Acremonium chreysogenum*) and this organism continuous to be used even today. The other organisms employed for cephalosporin production are *Emericellopsis* sp, *Paecilomyces* sp and *Streptomyces* sp.

Several mutants of ***C. acremonium*** have been developed for improved production of cephalosporin. Mutants with defective sulfur metabolism or those with resistance to sulfur analogs have high yielding capacity. Certain regulatory genes of cephalosporin biosynthesis (e.g., isopenicillin N synthetase) have been cloned and genetic manipulations carried out for increased production of cephalosporins.

Biosynthesis of cephalosporin

The early stages of the biosynthetic pathway for cephalosporin are the same as that for penicillin synthesis (See ***Fig. 25.2***). As the tripeptide (aminoadipylcysteinylvaline) is formed, it undergoes cyclization to produce isopenicillin N. By the action of epimerase, penicillin N is formed from isopenicillin N. Then, penicillin N gets converted to cephalosporin C by a three stage reaction catalysed by three distinct enzymes namely expandase, hydroxylase and acetyl transferase (***Fig. 25.7***).

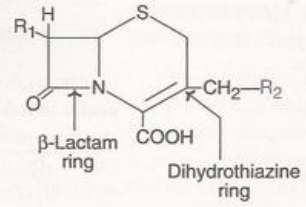
Regulation of biosynthesis : A low concentration of lysine promotes cephalosporin synthesis. The inhibitory effect of lysine at a higher concentration can be overcome by adding L-aminoadipic acid. The carbon sources that get rapidly degraded (e.g. glucose, glycerol) reduce cephalosporin production. Methionine promotes cephalosporin synthesis in *C. acremonium*, but has no influence on *Streptomycetes*.

PRODUCTION PROCESS OF CEPHALOSPORIN

The fermentation process concerned with the production of cephalosporin is similar to that of penicillin. The culture media consists of corn steep liquor and soy flour-based media in a continuous feeding system. The other ingradients of the medium include sucrose, glucose and ammonium salts. Methionine is added as a source of sulfur.

The fermentation is carried out at temperature 25–28°C and pH 6-7. The growth of micro-organisms substantially increases with good O_2 supply, although during production phase, O_2 consumption declines.

Cephalosporin C from the culture broth can be recovered by ion-exchange resins, and by using column chromatography. Cephalosporin C can be precipitated as zinc, sodium or potassium salt, and isolated.



Cephalosporin	R_1	R_2
7-Aminocephalosporanic acid (7-ACA)*	$-NH_2$	$-O-CO-CH_3$
Cephalosporin C	H_2N, CO, COOH	$-O-CO-CH_3$
Cephalexin	CO—, NH_2	$-CH_3$
Cefadroxil	HO, CO—, NH_2	$-CH_3$
Cefaclor	CO—, NH_2	$-Cl$
Cefazolin	N=N, N, N, CO—	S, S, N—N

Fig. 25.6 : A selected list of cephalosporins (–Active nucleus of cephalosporins).*

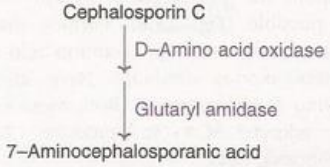
Chemical synthesis of cephalosporin

In recent years, by using penicillin V as the starting material, chemical synthesis of cephalosporin has become possible. This is being done due to low cost of production of penicillin.

PRODUCTION OF 7-AMINOCEPHALOSPORANIC ACID

7-Aminocephalosporonic acid (7-ACA) is the nucleus structure present in all the cephalosporins. Cephalosporin C, produced by fermentation, can be subjected to chemical hydrolysis to form 7-ACA. This is tedious, and is associated with several drawbacks.

Recently, enzymatic hydrolysis of cephalosporin C to 7-ACA has been developed. This is mainly carried out by two enzymes-*D-**amino acid oxidase*** (isolated from *Trigonopsis variabilis*) and ***glutaryl amidase*** (source-*Pseudomonas* sp). Biotechnologists have been successful in immobilizing these two enzymes for efficient and large scale manufacture of 7-ACA.



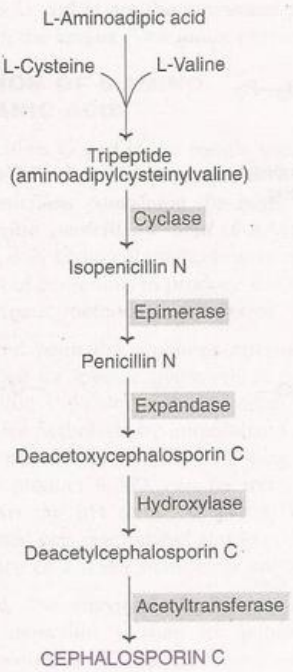


Fig. 25.7 : Biosynthesis of cephalosporin C by A. chrysogenum.

NEW β-LACTAM TECHNOLOGY FOR PRODUCTION OF 7-ACA

Scientists have been successful in producing 7-aminocephalosporanic acid by *P. chrysogenum* fermentation. This is possible through genetic manipulations. As already described in cephalosporin biosynthesis (***Fig. 25.7***), penicillin N is the substrate for the enzyme expandase. Adipyl-6-aminopenicillanic acid (produced by *P. chrysogenum* on adding adipic acid) which resembles in structure with penicillin N, can also serve as a substrate for expandase. By inserting expandase and hydroxylase gene (*cefEF*), and acetyl transferase gene (*cefG*) from *S. clavuligerus* into *P. chrysogenum*, the production of adipyl-7-ACA has become possible (***Fig. 25.8***). Further, the genes responsible for the enzymes D-amino acid oxidase (from *Pseudomonas diminuta*) have also been inserted into *P. chrysogenum*. Both these enzymes act on adipyl-7-ACA to produce 7-amino-cephalosporanic acid.

AMINOGLYCOSIDES

Aminoglycosides are ***oligosaccharide*** (***carbohydrate***) ***antibiotics***. They contain an aminocyclohexanol moiety which is bound to other amino sugars by glycosidic linkages. More than 100 aminoglycosides are known e.g. streptomycin, neomycin, kanamycin, gentamicin, hygromycin, sisomicin.

Aminoglycosides are very potent antibiotics and act against Gram-positive and Gram-negative bacteria, besides mycobacteria. At the molecular level, aminoglycosides bind to 30S ribosome and block protein biosynthesis. Prolonged use of aminoglycosides causes damage to kidneys, and hearing impairment.

For the treatment of severe and chronic infections, aminoglycosides are the antibiotics of choice. Streptomycin was the first aminoglycoside that was successfully used to treat tuberculosis (i.e. against *Mycobacterium tuberculosis*). Usually, aminoglycosides are regarded as reserve antibiotics, since resistance may develop easily.

Organisms for aminoglycoside production

Aminoglycoside antibiotics are produced by *Actinomyces* sp. Some examples are given in ***Table 25.2***. Recombinant DNA techniques have

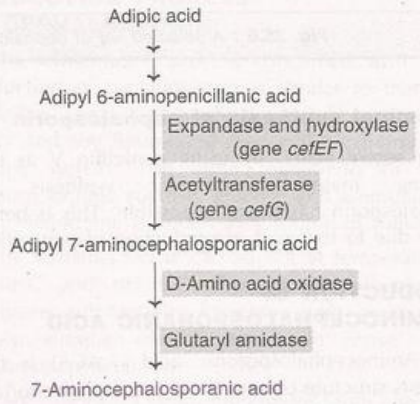


Fig. 25.8 : Biosynthesis of 7-aminocephalosporanic acid by P. chrysogenum *transformants (after suitable gene insertions).*

Table 25.2 Selected examples of aminoglycosides with the organisms responsible for their production

Aminoglycoside	*Organism*
Streptomycin	*Streptomyces griseus*
Neomycin B and C	*S. fradiae*
Kanamycin A, B and C	*S. kanamyceticus*
Hygromycin B	*S. hygroscopicus*
Gentamicin	*Micromonospora purpurea*
Sisomicin	*M. inyoensis*

been used to produce hybrid aminoglycosides, and for increasing the fermentation yield.

Biosynthesis of aminoglycosides

All the ring structures in the molecules of aminoglycosides are ultimately derived from glucose. Most of the biosynthetic pathways concerned with the formation of at least some aminoglycosides have been elucidated.

Biosynthesis of streptomycin

The outline of the pathway for the synthesis of streptomycin is depicted in ***Fig. 25.9***. More than 30 enzymatic steps have been identified. ***Glucose 6-phosphate*** obtained from glucose takes three independent routes to respectively produce streptidine 6-phosphate, L-dehydrostreptose and N-methylglucosamine. The former two compounds condense to form an intermediate which later combines with methylglucosamine to produce dihydrostreptomycin-6-phosphate. This compound, in the next of couple of reactions, gets converted to streptomycin.

Regulation of biosynthesis : Very little is known about the regulation of streptomycin synthesis. A compound named as ***factor A*** (chemically isocapryloyl-hydroxymethyl-γ-butyrate) has been isolated from streptomycin-producing strains of *S. griseus*. Factor A promotes streptomycin production. In fact, factor A^- mutants that cannot synthesize streptomycin have been isolated. They can synthesize streptomycin on adding factor A. The nutrient sources-carbohydrates (glucose), ammonia and phosphate also regulate (by feedback mechanism) streptomycin production.

PRODUCTION PROCESS OF STREPTOMYCIN

The medium used for streptomycin usually consists of soy meal or soy flour or corn syrup that can supply glucose at a slow rate (amylase activity is poor in *Streptomyces* sp). The initial supply of nitrogen (NH_3) and phosphate is also obtained from soy meal. This is required since glucose, ammonia and phosphate in high quantities inhibit streptomycin synthesis.

The fermentation conditions for optimal production of streptomycin are — temperature 27–30°C, pH 6.5-7.5, aeration rate 0.5-1.0 vvm. The duration of fermentation process depends on the strain used, and is between 6 to 8 days.

Recovery of streptomycin

Streptomycin or other aminoglycosides are basic in nature. They can be recovered

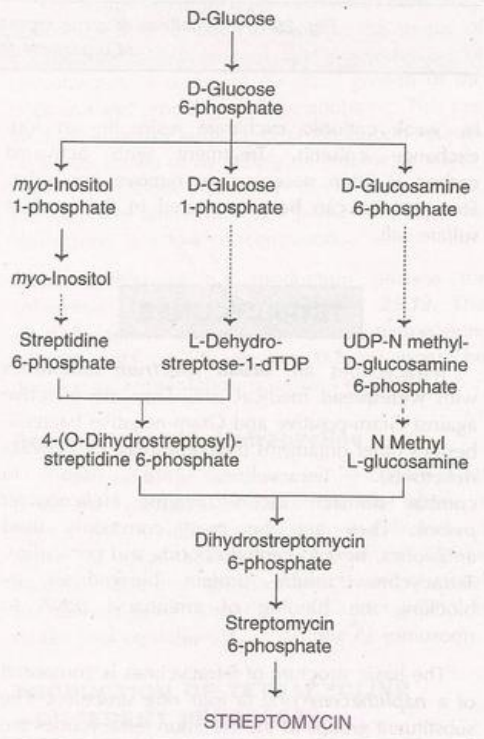
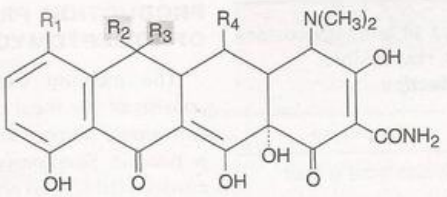


Fig. 25.9 : An outline of streptomycin biosynthesis.



Tetracycline	R_1	R_2	R_3	R_4	Examples of producing organisms
Tetracycline	H	CH_3	OH	H	*Streptomyces aureus, S. flavus, S. antibioticus*
Chlortetracycline	Cl	CH_3	OH	H	*S. aureofaciciens, S. viridifaciens, S. flavus*
Oxytetracycline	H	CH_3	OH	OH	*S. antibioticus, S. cellulosae S. parvus, S. rimosus*
Minocycline	$N(CH_3)_2$	H	H	H	Semisynthetic
Doxycycline	H	CH_3	H	OH	Semisynthetic

***Fig. 25.10** : Structures of some important tetracyclines along with the examples of organisms for their production.*

by weak cationic exchange resins in an ion-exchange column. Treatment with activated carbon is often necessary to remove impurities. Streptomycin can be precipitated in the form of sulfate salt.

TETRACYCLINES

Tetracyclines are ***broad spectrum antibiotics*** with widespread medical use. They are effective against Gram-positive and Gram-negative bacteria, besides other organisms (mycoplasmas, chlamydias rickettsias). Tetracyclines are used to combat stomach ulcers (against *Helicobacter pylori*). They are the most commonly used antibiotics, next to cephalosporins and penicillins. Tetracyclines inhibit protein biosynthesis by blocking the binding of aminoacyl tRNA to ribosomes (A site).

The basic structure of tetracyclines is composed of a ***naphthacene*** ring (a four ring structure). The substituent groups of the common tetracyclines are given in ***Fig. 25.10***. Among these, chlortetracycline and oxytetracycline are most commonly used in the treatment of human and veterinary deseases, besides in the preservation of fish, meat and poultry (in some countries).

Organisms for tetracycline production

The first tetracycline antibiotic that was isolated was chlortetracycline from the cultures of *Streptomyces aureofaciens* (in 1945). There are at least 20 streptomycetes identified now that usually produce a mixture of tetracyclines. In the ***Fig. 25.10***, a selected list of these organisms for producing tetracyclines is also given.

High-yielding strains of *S. aureofaciens* and *S. rimosus* have been developed by using ultraviolet radiation and/or other mutagens (nitrosoguanidine). Such strains are very efficient for the production of chlortetracycline. Further, genetically engineered strains of *S. rimosus* have been developed for increased synthesis of oxytetracycline.

Biosynthesis of tetracyclines

The pathway for the biosynthesis of tetracyclines is very complex. An outline of the synthesis of chlortetracycline by *S. aureofaciens* is given in

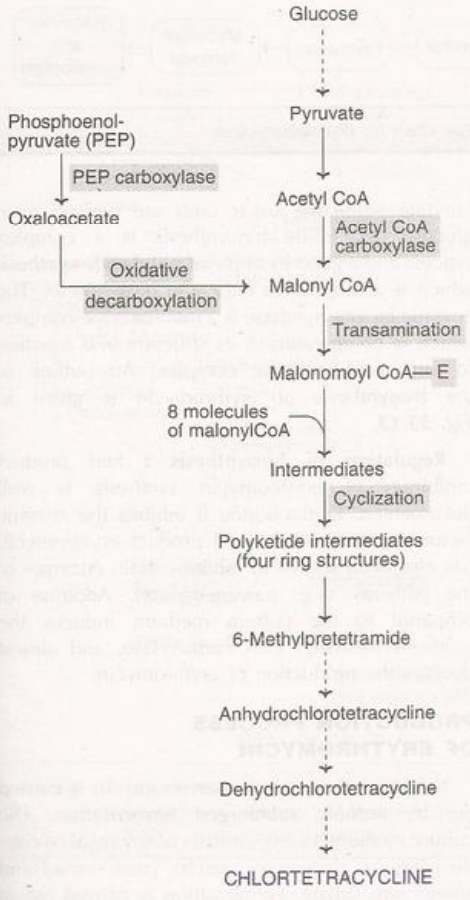


Fig. 25.11 : An outline of the biosynthesis of chlortetracycline by S. aureofaciens
(E —Anthracene synthase enzyme complex)

Fig. 25.11. There are at least 72 intermediates formed during the course of chlortetracycline biosynthesis, some of them have not been fully characterized.

Polyketide antibiotic synthesis : The term polyketide refers to a group of ***antibiotics*** that are ***synthesized*** by ***successive condensation of small carboxylic acids*** such as acetate, butyrate, propionate and malonate. The synthesis of polyketide antibiotics is comparable to that of long chain fatty acids. That is the carbon chain grows by cyclic condensation process. The synthesis of tetracyclines is a good example of polyketide antibiotic synthesis.

As glucose gets oxidised, it forms acetyl CoA and then malonyl CoA. On transamination, the later gives malonomoyl CoA. The enzyme anthracene synthase complex binds to malonomoyl CoA and brings out the condensation of 8 molecules of malonyl CoA to form a polyketide intermediates (four ring structures). These intermediates undergo a series of reactions to finally produce chlortetracycline.

Regulation of biosynthesis : Carbohydrate metabolism (particularly glycolysis) controls chlortetracycline synthesis. For more efficient synthesis of the antibiotic, glycolysis has to be substantially low. The addition of phosphate reduces chlortetracycline production.

PRODUCTION PROCESS OF CHLORTETRACYCLINE

The fermentation medium consists of corn steep liquor, soy flour or peanut meal for the supply of nitrogen and carbon sources. Continuous feeding of carbohydrate is desirable for good growth of the organism and production of the antibiotic. This can be done either by addition of crude carbon sources (as given above) or by supplying glucose or starch. For more efficient production of chlortetracycline, the supply of ammonium and phosphate has to be maintained at a low concentration.

An outline of the production process for chlortetracycline is depicted in ***Fig. 25.12***. The ideal fermentation conditions are — temperature 27–30°C, pH-6.5-7.5, aeration 0.8-1.0 vvm. The duration of fermentation is around 4 days.

Recovery of chlortetracycline

At the end of the fermentation, the culture broth is filtered to remove the mycelium. The filtrate is treated with n-butanol or methylisobutylketone in acidic or alkaline condition for extracting the antibiotic. It is then absorbed to activated charcoal to remove other impurities. Chlortetraycline is eluted and crystallized.

PRODUCTION OF TETRACYCLINE —DIFFERENT PROCESSES

The production of tetracycline can be achieved by one or more of the following ways.

Fig. 25.12 : An outline of production chart for chlortetracycline.

- By chemical treatment of chlortetracycline.
- By carrying out fermentation in a chloride-free culture medium.
- By employing mutants in which chlorination reaction does not occur.
- By blocking chlorination reaction by the addition of inhibitors e.g. thiourea, 2-thiouracil.

MACROLIDES

Macrolides are a group of ***antibiotics with large lactone rings*** (i.e. macrocylic lactone rings). They consist of 12-, 14-, or 16-membered lactone rings with 1-3 sugars linked by glycosidic bonds. The sugars may be 6-deoxyhexoses or aminosugars. ***Erythromycin*** and ***oleandomycin*** are 14-membered (lactone ring containing) macrolides while leucomycin and tylosin are examples for 16-membered microlides.

Erythromycin and its derivative clarithromycin are the most commonly prescribed microlides. They are effective against Gram-positive bacteria, and are frequently used to kill penicillin-resistant organisms. Clarithromycin is currently used to combat stomach ulcers caused by *H. pylori*. The macrolides inhibit the protein biosynthesis by binding to 50S ribosome.

Polyene macrolides is the term applied for very large ring macrolides that many contain lactone rings in the range of 26–28. e.g. nystatin, amphotericin. These polyene macrolides are ***antifungal***.

Production of macrolides

Macrolides are produced ***by actinomycetes***. The major macrolide antibiotics and the corresponding organisms synthesizing them are given in ***Table 25.3***.

Biosynthesis of erythromycin

In the biosynthesis of erythromycin, the lactone rings are contributed by acetate, propionate or butyrate while the sugar units are derived from glucose. Macrolide biosynthesis is a complex process and a good example of ***polyketide synthesis*** which is analogous to fatty acid biosynthesis. The enzyme lactone synthase is a multienzyme complex which is comparable in its structure and function to fatty acid synthase complex. An outline of the biosynthesis of erythromycin is given in ***Fig. 25.13***.

Regulation of biosynthesis : End product inhibition of erythromycin synthesis is well documented. Erythronolide B inhibits the enzyme lactone synthase. The final product erythromycin has also been shown to inhibit certain enzymes of the pathway (e.g. transmethylase). Addition of propanol to the culture medium induces the synthesis of acetyl CoA carboxylase, and almost doubles the production of erythromycin.

PRODUCTION PROCESS OF ERYTHROMYCIN

Industrial production of erythromycin is carried out by ***aerobic submerged fermentation***. The culture medium mainly consists of soy meal or corn steep liquor, glucose (or starch), yeast extract and ammonium sulfate. Fermentation is carried out at 30–34°C for about 3-7 days. Conventional methods are used for the recovery and purification of erythromycin.

TABLE 25.3 Important macrolide antibiotics with the organisms producing them

Macrolid antibiotic	*Producing organism*
Erythromycin	*Streptomyces erythreus*
Oleandomycin	*S. antibiotics*
Pikromycin	*S. felleus*
Megalomicin	*Micromonospora inositola*
Tylosin	*S. fradiae*
Carbomycin A	*S. halstedii*
Leucomycins	*Streptoverticillium kitasatoensis*

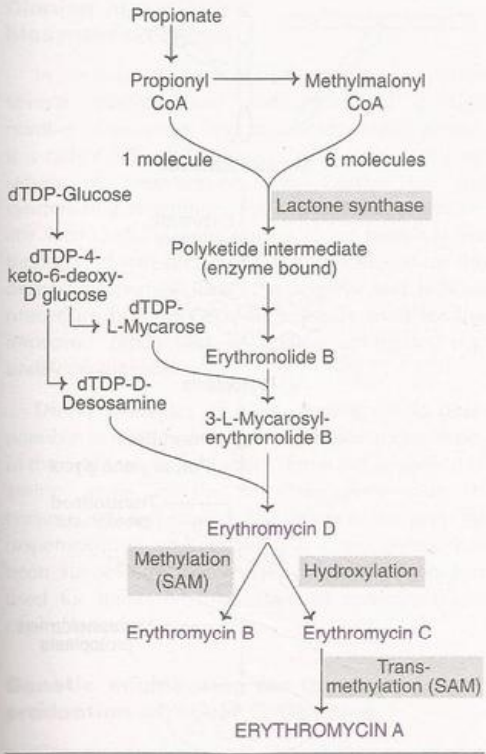


Fig. 25.13 : An outline of the biosynthesis of erythromycin by Streptomyces erythreus *(SAM — S-adenosylmethionine)*

AROMATIC ANTIBIOTICS

The antibiotics with aromatic rings in their structure are regarded as aromatic antibiotics. In a strict since, all the antibiotics containing aromatic nuclei should be considered in this group. However, most authors prefer to treat the three important antibiotics namely chloramphenicol, griseofulvin (***Fig. 25.14***) and novobiocin in the category of aromatic antibiotics, and the same is done in this book also.

CHLORAMPHENICOL

Chloramphenicol is a ***broad spectrum antibiotic*** that can act against Gram-positive and Gram-negative bacteria, besides rickettsias, actinomycetes and chlamydias. However, administration of chloramphenicol is associated with side effects, the most significant being damage to bone-marrow. As such, chloramphenicol is treated as a reserve antibiotic and selectively used.

Chloramphenicol binds to 50S ribosmal subunit and blocks (peptidyltransferase reaction) protein biosynthesis.

Production of chloramphenicol

Chloramphenicol can be produced by ***Streptomyces venezuelae*** and *S. omiyanesis.* However, chemical synthesis is mostly preferred for the commercial production of chloramphenicol.

GRISEOFULVIN

Griseofulvin is an antibiotic that acts specifically on fungi with chitinous cell walls. It is used in the ***treatment*** of various ***fungal skin infections***. Further, griseofulvin is also employed in the treatment of plant diseases caused by *Biotrytis* and *Alternaria solani.*

Although the exact mechanism of action of griseofulvin is not known, it is believed that chitin biosynthesis is adversely affected.

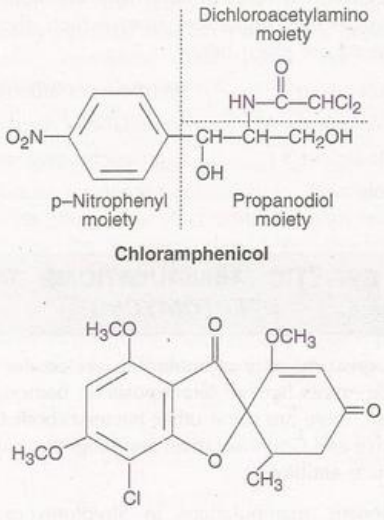


Fig. 25.14 : Structures of chloramphenicol and griseofulvin.

Production of griseofulvin

Commercial production of griseofulvin is carried out by employing ***Penicillium patulum***. The chemical synthesis is less frequently used due to high cost.

The fermentation is carried out by an ***aerobic submerged process*** with a glucose rich medium. Nitrogen is supplied by sodium nitrate. The optimal conditions for fermentation are—temperature 23–26°C, pH 6.8-7.3, aeration 0.8-1 vvm, and the period is 7–10 days.

NUCLEOSIDE ANTIBIOTICS

There are several antibiotics (more than 200 or so) which have nucleoside like structures e.g. puromycin, blasticidin S. Nucleoside antibiotics have diverse structures and biological activities.

Puromycin is used to understand the ribosomal function in protein biosynthesis. Neplanosin possesses antiviral activity. Blasticidin S is a fungicide antibiotic used in plant pathology.

Production of nucleoside antibiotics

Selected examples of nucleoside antibiotics and the respective organisms from which they are produced are given below.

Puromycin	—	*Streptomyces alboniger*
Neplamosin A	—	*Ampullariella regularis*
Blasticidin S	—	*S. griseochromogenes*
Polyoxins	—	*S. cacaoi*

GENETIC MANIPULATIONS OF *STEPTOMYCES*

A great majority of antibiotics are produced by *Streptomyces* sp, a Gram-positive bacteria. Of course, there are some other bacteria (both Gram-positive and Gram-negative) and fungi that can also produce antibiotics.

Genetic manipulations in *Streptomyces* have been extensively carried out ***to enhance the yield of antibiotics and reduce cost of production,*** besides developing newer and more effective antibiotics.

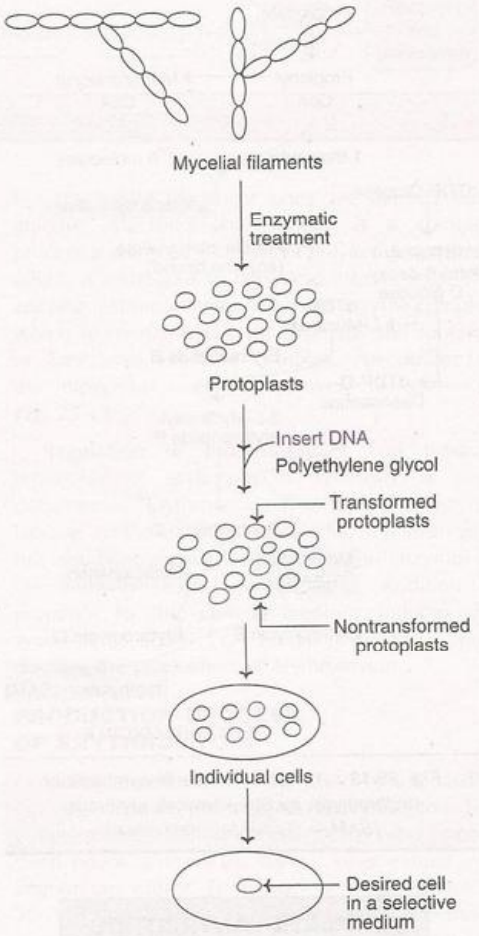


Fig. 25.15 : A diagrammatic representation of transformation in Streptomyces *sp.*

Transformation of streptomyces

Streptomyces strains exist as aggregates of mycelial filaments and not as individual cells. This is in contrast to *E. coli*. It is therefore essential that the cell walls of *Streptomyces* are broken to release protoplasts for transformation (***Fig. 25.15***). By adding the desired DNA (in plasmids) and polyethylene glycol, the cells can be transformed. These protoplasts are grown on a solid medium to regenerate the cell walls. The transformed cells with desired properties can be isolated for further use.

Cloning of antibiotic biosynthesis genes

In general, biosynthesis of antibiotics involves several reactions and participation of a large number of enzymes (and of course, several genes). It is rather difficult to clone so many genes. Mutant strains of *Streptomyces* that totally lack the synthesizing machinery for a particular antibiotic are very useful. Genes from a clone bank can be incorported into such mutants and screened for the desired properties. This is a lengthy and tedious procedure but has been successfully used for the improved production of certain antibiotics e.g. undeylprodigiosin.

Direct strategies of gene cloning : It is often possible to identify one or a few important enzymes in the synthesis of antibiotics. From the sequence of amino acids in the enzyme, gene can be constructed, and cloned. For instance, the gene for isopenicillin N synthase from *P. chrysogenum* has been successfully constructed in this fashion and used for increased production of penicillins and cephalosporins.

Genetic engineering for the production of novel antibiotics

Genetic manipulations can be done in the antibiotic synthesizing organisms to ultimately produce totally new and novel antibiotics.

Wild-type *Streptomyces coelicolor* genes encode the enzymes to produce the antibiotic actinorhodine. *S. violaceoruber* produces a related antibiotic namely granaticin. Genetic manipulations can be done between these organisms to produce novel hybrid antibiotics such as medarrhodine A and dihydrogranatirhodine.

The newly synthesized antibiotics are in fact structural variants of the existing antibiotics. As the biosynthetic pathways for antibiotic production and their corresponding genes are better understood, it becomes possible to design newer antibiotics with more efficient action.

Genetic engineering for improving antibiotic production

For aerobic microorganisms (e.g. *Streptomyces* sp), there is often the limitation of oxygen supply that hinders antibiotic production. Some workers have isolated a hemoglobin-like protein produced by the bacterium *Vitreoscilla* sp. The gene synthesizing this protein was isolated and cloned in a plasmid vector. The hemoglobin gene of *Vitroscilla* sp was finally incorporated into *Streptomyces*. These newly transformed strains have better capacity to take O_2 from the medium even at a low concentration.

The new strain of *S. coelicolor* (with hemoglobin gene) was found to produce 10 times more antibiotic actinorhodine than the wild strain, even at a low concentration of oxygen.

GOOD ANTIBIOTIC MANUFACTURING PRACTICES

Manufacture of antibiotics is highly commercialised due to a heavy demand worldwide. It is mandatory that detailed clinical trials are carried out before considering the manufacture of any antibiotic. More than half of the antibiotics produced are for human use. If is therefore ***absolutely essential that each antibiotic produced is safe, consistent and poses no health complications***. Government authorities play a predominant role in regulating the production of antibiotics. These guidelines ensure that correct procedures are followed at each stage of manufacturing the antibiotics. The product produced should be of consistently high quality. Quality of the products should be checked at different stages of manufacture.

It is expected that the antibiotic manufactured should be in the purest form, although 100% purity is not practicable. It is mandatory that the impurities (if any), their quantities, and their ill effects should be made known.

26 Microbial Production of Amino Acids

The first commercial production of the amino acid, glutamic acids, was started in Japan in as early as 1908. It was Ikeda (of Japan) who first identified that glutamic acid (in the form of monosodium glutamate) possesses taste-enchancing properties. In the early days, monosodium glutamate (MSG) was extracted from the vegetable proteins (wheat and soy). It was only in 1957, the large scale industrial production of MSG by using microorganisms commenced. Today, commercial production of amino acids is one of the biggest industries worldover with an annual increase in the demand by about 10%.

Glutamic acid continues to be ***largest producer*** among the amino acids, followed by lysine, methionine, threonine and aspartic acid.

AMINO ACID PRODUCTION — GENERAL CONSIDERATIONS

Some general considerations on the commercial applications of amino acids, their production methods, and the development of strains of microorganisms for improved amino acid production are briefly described.

COMMERCIAL APPLICATIONS OF AMINO ACIDS

Amino acids have a wide range of applications. The proportionate use of amino acids is given in the next column.

Food industry — 65%

Feed additives — 30%

Pharmaceutical — 5%

The individual L-amino acids (except glycine) along with their production methods and uses are given in ***Table 26.1***. The applications are broadly discussed hereunder.

Food industry

Amino acids are used either alone or in combination, as flavour enhancers. ***Monosodium glutamate is the most frequently*** used in food industry. Glycine and alanine also enhance taste and flavour. Tryptophan, in association with histidine, acts as an antioxidant to preserve milk powder. For the preservation of fruit juices, cysteine serves as an antioxidant.

Aspartame, a dipeptide (aspartyl-phenylalanine methyl ester) produced by a ***combination of aspartic acid and phenylalanine***, is about 200 times sweeter than sucrose. It is used as a low-calorie artificial sweetener in soft-drink industry.

There are certain essential amino acids that are deficient or limiting in plant proteins. These include lysine, methionine, threonine and tryptophan. Addition of the deficient amino acid(s) improves the nutritional quality of human foods as well as animal feeds. Thus, bread enriched with lysine, soy products supplemented with methionine are of better nutritional value. Methionine added soy

bean meal is a better feed for pigs and other animals.

Pharmaceutical industry

The amino acids can be used as medicines. Essential amino acids are useful as ingradients of infusion fluids, for administration to patients in post-operative treatment.

Chemical industry

Amino acids serve as starting materials for producing several compounds. Glycine is used as a precursor for the synthesis of ***glyphosate*** (a herbicide), while threonine is the starting material for the production of azthreonam (another herbicide). Poly-methylglutamate is utilised for manufacturing synthetic leather. Some amino acids in the form of N-acyl derivatives are useful for the preparation of cosmotics.

METHODS FOR PRODUCTION OF AMINO ACIDS

The industrial production of amino acids is carried out by one or more of the following three processes.

1. **Extraction :** Amino acids are the building blocks in protein structure. The proteins can be subjected to hydrolysis, and the requisite amino acids can be isolated e.g. cysteine, tyrosine, leucine.

2. **Chemical synthesis :** Chemical synthesis results in a mixture of D- and L-amino acids. Most of the amino acids required for commercial

Table 26.1 Amino acids along with their production methods and applications

*Amino acid**	*Preferred production method*	*Application(s)*
Glutamic acid	Fermentation	Flavour enhancer
Lysine	Fermentation	Feed additive, infusion solution
Methionine	Chemical synthesis	Feed additive
Threonine	Fermentation	Feed additive
Phenylalanine	Fermentation	Aspartame production
Aspartate	Enzymatic synthesis, extraction	Aspartame production
Glycine	Chemical synthesis	Sweetener, food additive
Arginine	Fermentation, extraction	Infusions, therapy for liver diseases, cosmotics
Valine	Fermentation, extraction	Infusions, pesticides
Tryptophan	Extraction, immobilized cells	Infusions, antioxidant
Isoleucine	Extraction, fermentation	Infusions
Alanine	Extraction, enzymatic	Flavour enhancer
Leucine	Extraction, enzymatic	Infusions
Proline	Extraction	Infusions
Serine	Extraction, chemical synthesis	Cosmotics
Histidine	Extraction, fermentation	Therapy for ulcers
Asparagine	Extraction, chemical synthesis	Diuretic
Glutamine	Fermentation	Therapy for ulcers
Cysteine	Extraction	Infusion
Tyrosine	Extraction, enzymatic	Infusion
Ornithine	Extraction, enzymatic	Therapy for liver diseases

* *The order of amino acids is approximately in the decreasing order of their annual production; Extraction refers to the extraction of the specific amino acid from protein hydrolysates.*

applications are of L-category. However, for the synthesis of glycine (optically inactive) and some other amino acids which can be used in L- or D-form (D, L-alanine, D, L-methionine) for certain purposes, chemical methods are employed.

3. **Microbiological production : *For the large-scale production of amino acids***, microbiological methods are employed. There are three different approaches.

(a) **Direct fermentation methods :** Amino acids can be produced by microorganisms by utilising several carbon sources e.g. glucose, fructose, alkanes, ethanol, glycerol, propionate. Certain industrial byproducts like molasses and starch hydrolysate can also be used. Methanol, being a cheap carbon source, is tried for amino acid production, but with limited success.

(b) **Conversion of metabolic intermediates into amino acids :** In this approach, the microorganisms are used to carry out selected reactions for amino acid production e.g. conversion of glycine to serine.

(c) **Direct use of microbial enzymes or immobilized cells :** Sometimes resting cells, immobilized cells, crude cell extracts or enzyme-membrane reactors can be used for the production of amino acids. Some examples are given below.

Amino acid dehydrogenases from certain bacteria (e.g. *Bacillus megaterium*) can be used for the amination of α-keto acids to produce L-amino acids e.g. alanine (from pyruvate), leucine (from α-ketoisocaproic acid) and phenylalanine (from phenylpyruvate). Immobilized cells or enzyme-membrane reactors can be used.

Enzymes or immobilised cells are also employed for the production of several other amino acids e.g. tryptophan, tyrosine, lysine, valine.

STRAIN DEVELOPMENT FOR AMINO ACID PRODUCTION

The metabolic pathways, for the synthesis of amino acids by microorganisms, are tightly controlled and they operate in an economical way. Therefore, a natural overproduction of amino acids is a rare occurrence. Some strains that excrete certain amino acids have been isolated e.g. glutamic acid, alanine, valine.

In order to achieve an overproduction of any amino acid by a microorganism, methods have to be devised for the elimination of the metabolic regulatory/control processes. In fact, several amino acid-producing ***microorganisms have been developed by mutagenesis and screening programmes.***

The following are the major ways of strain development. In fact, several methods are combined to successfully develop a new strain for producing amino acids.

Auxotrophic mutation : These mutants are characterized by a lack of the formation of regulatory end product (i.e. repressor or regulatory effector). The intermediates of the metabolic pathways accumulate and get excreted.

Genetic recombination : Mutants can be developed by genetic recombination for over-production of amino acids. Protoplast fusion in certain bacteria is used for development of hybrids e.g. *Corynebacterium glutamicum* and *Bacillus flavum*.

Recombinant DNA technology : The classical techniques of genetic engineering can be used for strain development. Strains with increasing activities of rate-limiting enzymes have been developed. In one of the techniques, *E. coli* and cloning vector pBR322 were used to increase the genes for the production of amino acids e.g. glutamic acid, lysine, phenylalanine, valine.

Functional genomics : a new approach

Analysis of genomes from the wild and mutant strains of microorganisms will help in creating improved strains. Once the entire sequence of the chromosomes in the organisms (e.g. *C. glutamicum, E. coli*) is established, efforts can be made to carry out genetic manipulations for efficient overproduction of desired amino acids. ***Chip technology*** can be used to detect new mutations and consequently the fermentation processes.

L-GLUTAMIC ACID

L-Glutamic acid was the ***first amino acid to be produced by microorganisms.*** The original

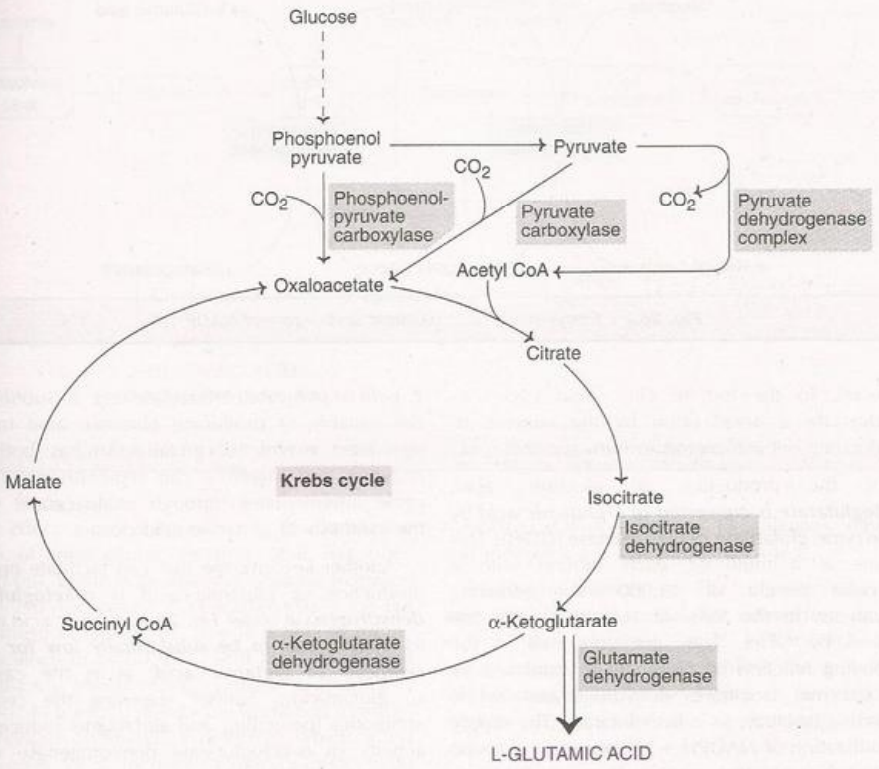


Fig. 26.1 : Biosynthesis of L-glutamic acid in Corynebacterium glutamicum.

bacterium, ***Corynebacterium glutamicum***, that was first used for large scale manufacture of glutamic acid continues to be successfully used even today. The other important organisms (although used to a lesser extent due to low yield) employed for glutamic acid production belong to genera *Microbacterium, Brevibacterium* and *Arthrobacter*. All these organisms have certain morphological and physiological characters comparable to *C. glutamicum*. Biochemically, ***glutamic acid-producing bacteria have a high activity of glutamate dehydrogenase and a low activity of α-ketoglutarate dehydrogenase***. They also require the vitamin biotin.

Improved production strains

Several improvements have been made, particularly in *C. glutamicum*, for improving the strains to produce and excrete more and more of glutamic acid. These include the strains that can tolerate high concentrations of biotin, and lysozyme-sensitive mutants with high yield.

Biosynthesis of L-glutamic acid

The pathway for the synthesis of glutamic acid with glucose as the carbon source is depicted in ***Fig. 26.1***. Glucose is broken down to phosphoenol pyruvate and then to pyruvate. Pyruvate is converted to acetyl CoA. Phosphoenol pyruvate (by the enzyme phosphoenol pyruvate carboxylase) can be independently converted to oxaloacetate. Both these carboxylation reactions are quite critical, and require biotin as the cofactor.

The next series of reactions that follow are the familiar citric acid (Krebs) cycle reactions wherein the key metabolite namely α-ketoglutarate is

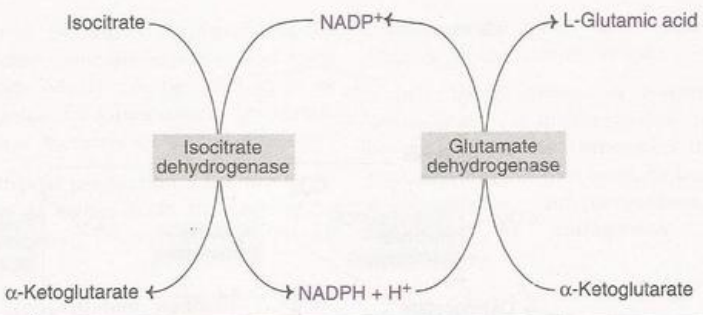


Fig. 26.2 : Biosynthesis of L. glutamic acid — role of NADP+.

produced. In the routine citric acid cycle, α-ketoglutarate is acted upon by the enzyme α-ketoglutarate dehydrogenase to form succinyl CoA.

For the production of glutamic acid, ***α-ketoglutarate is converted to L-glutamic acid*** by the enzyme ***glutamate dehydrogenase*** (GDH). This enzyme is a multimer, each subunit with a molecular weight of 49,000. The reducing equivalents, in the form of NADPH + H^+, are required by GDH. They are generated in the preceeding reaction of Krebs cycle (catalysed by the enzyme isocitrate dehydrogenase) while converting isocitrate to α-ketoglutarate. The supply and utilization of NADPH + H^+ occurs in a cyclic fashion through the participation of the two enzymes, namely isocitrate dehydrogenase and glutamate dehydrogenase (***Fig. 26.2***).

Theoretically, one molecule of glutamic acid can be formed from one molecule of glucose. In practice, the conversion efficiency of glucose to glutamic acid was found to be around 70%.

Regulation of glutamic acid biosynthesis

The essential requirement for glutamic acid production is the high capability for the supply of the citric acid cycle metabolites. This is made possible by an efficient conversion of phosphoenol pyruvate as well as pyruvate to oxaloacetate (Refer ***Fig. 26.1***). Thus, there are two enzymes (phosphoenol pyruvate carboxylase and pyruvate carboxylase) to efficiently produce oxaloacetate, while there is only one enzyme (pyruvate dehydrogenase) for the formation of acetyl CoA. Certain microorganisms which have either phosphoenol pyruvate carboxylase (e.g., *E. coli*) or pyruvate carboxylase (e.g. *B. subtilis*) are not capable of producing glutamic acid to any significant extent. *C. glutamicum* has both the enzymes and therefore can replenish citric acid cycle intermediates (through oxaloacetate) while the synthesis of glutamic acid occurs.

Another key enzyme that can facilitate optimal production of glutamic acid is ***α-ketoglutarate dehydrogenase*** (Refer ***Fig. 26.1***) of citric acid cycle. Its ***activity has to be substantially low for good synthesis of glutamic acid***, as is the case in *C. glutamicum*. Further, exposing the cells to antibiotics (penicillin) and surfactants reduces the activity of α-ketoglutarate dehydrogenase while glutamate dehydrogenase activity remains unaltered. By this way, oxidation of α-ketoglutarate via citric acid cycle can be minimised, while the formation of glutamic acid is made maximum possible.

Release of glutamic acid

Glutamic acid is ***synthesized intracellularly***, and therefore ***its release or export is*** equally important. It now appears that there is a ***carrier-mediated energy-dependent active process*** involved for the export of glutamic acid. There are several ways of increasing the membrane permeability for exporting glutamic acid.

- Biotin limitation
- Addition of saturated fatty acids
- Addition of penicillin
- Use of oleic acid auxotrophs
- Use of glycerol auxotrophs
- Addition of local anesthetics
- Addition of surfactants (Tween 40).

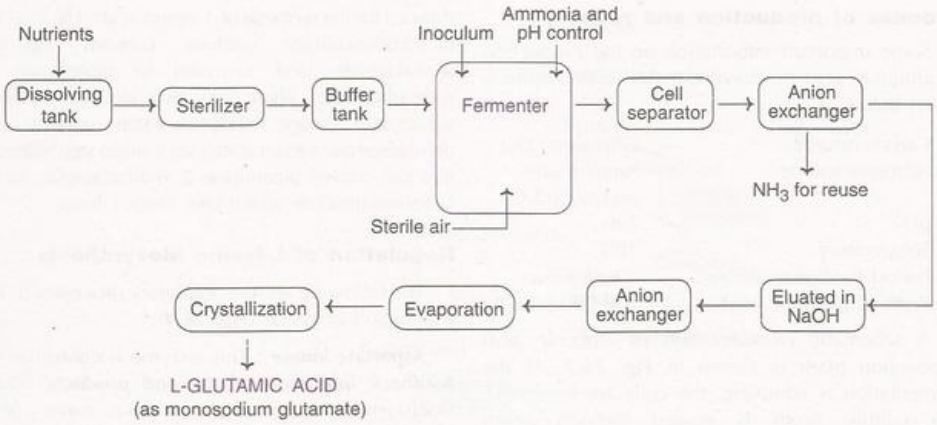


Fig. 26.3 : Diagrammatic representation of glutamic acid production plant.

The effect of biotin deficiency in facilitating the release of intracellular glutamic acid has been worked out. Biotin is an essential cofactor (required by the enzyme acetyl CoA carboxylase) for the biosynthesis of fatty acids. Due to a limited supply or deficiency of biotin, fatty acid biosynthesis and consequently phospholipid synthesis is drastically reduced. As a result, membrane formation (protein-phospholipid complex) is defective which alters permeability for an increased export of intracellular glutamic acid.

It is found that there is an alteration in the membrane composition of phospholipids in oleic acid and glycerol auxotroph mutants. This facilitates release of intracellular glutamic acid.

The knowledge on the membrane permeability of glutamic acid is successfully exploited for increased industrial production of glutamic acid.

Production of glutamic acid-requirements and influencing factors

The industrial production of glutamic acid is influenced by carbon sources, nitrogen sources, growth factors, pH and O_2 supply. The relevant aspects are briefly described.

Carbon sources : Either ***refined*** (glucose, sucrose, fructose, maltose) ***or unrefined*** (sugar beet molasses, sugar cane molasses) ***carbon sources*** are used. In countries like Japan, acetate (inexpensive) is utilized. Other substrates like alkanes, ethanol and methanol are less frequently used.

Nitrogen sources : The concentration of ***ammonia is very crucial for converting carbon source to glutamic acid.*** However, high concentration of ammonia inhibits the growth of the organisms. In the beginning of fermentation, ammonium salts and a low concentration of ammonia are added. During the course of fermentation, ammonia in aqueous solution is continuously fed. In this way, pH can be controlled, besides continuous supply of nitrogen source. Sometimes, urea is also used as a nitrogen source, since glutamic acid-producing bacteria possess urease that can split urea and release ammonia.

Growth factors : ***Biotin*** is an important growth factor and its concentration in the medium is influenced by the carbon source. For instance, a supply 5 µg of biotin per liter medium is recommended if the carbon source is 10% glucose, while for acetate as the carbon source, the biotin requirement is much lower (0.1-1.0 µg/l). Addition of L-cysteine in the medium is recommended for certain strains.

Supply of O_2 : O_2 supply should be adequately and continuously maintained. It is observed that a high O_2 concentration inhibits growth of the organisms while a low O_2 supply leads to the production of lactic acid and succinic acid. In both instances, glutamic acid formation is low.

Process of production and recovery

Some important information on the production of glutamic acid by *Brevibacterium divaricatum* is given below.

Carbon source	—	Glucose (12%)
Nitrogen source	—	Ammonium acetate (0.5%)
pH	—	7.8
Temperature	—	38°C
Period for fermentation	—	30–35 hours
Yield of glutamic acid	—	100 g/l medium.

A schematic representation of glutamic acid production plant is shown in ***Fig. 26.3***. As the fermentation is complete, the cells are separated, the culture broth is passed through anion exchanger. The glutamic acid bound to the resins is eluted in NaOH, while the ammonia released can be reused. With NaOH, glutamic acid forms monosodium glutamate (MSG) which can be purified by passing through anion exchanger. MSG can be subjected to evaporation and crystallization.

L-LYSINE

Lysine is present at a low concentration in most of the plant proteins. Being an essential amino acid, supplementation of plant foods with lysine increases their nutritional quality.

L-Lysine is predominantly produced by ***Corynebacterium glutamicum*** and to some extent by *Brevibacterium flavum* or *B. lactofermentum*.

Biosynthesis of L-lysine

The pathway for the synthesis of L-lysine is complex, and an outline of it is depicted in ***Fig. 26.4***. This metabolic pathway is also involved in the formation of 3 other amino acids, namely methionine, threonine and isoleucine.

As the glucose gets oxidised by glycolysis, phosphoenol pyruvate and pyruvate are formed. Both these metabolites can be converted to oxaloacetate, a key component of citric acid cycle. On transamination, oxaloacetate forms aspartate. The enzyme aspartate kinase converts aspartate to aspartyl phosphate which later forms aspartate semialdehyde. Aspartate semialdehyde has two fates—the biosynthesis of lysine and formation of 3 other amino acids (methionine, threonine and isoleucine). When homoserine dehydrogenase acts on aspartate semialdehyde, it is diverted for the synthesis of 3 amino acids. The enzyme dihydrodipicolinate synthase converts aspartate semialdehyde (and pyruvate) to piperideine 2, 6-dicarboxylate. There are two distinct enzymes succinylase variant (catalyses 4-step reaction) and dehydrogenase variant (catalyses a single step reaction) that can convert piperideine 2, 6-dicarboxylate to D, L-diaminopimelate which later forms L-lysine.

Regulation of L-lysine biosynthesis

The following are the regulatory processes in the production of lysine (***Fig. 26.4***).

Aspartate kinase : This enzyme is controlled by ***feedback inhibition of the end products***. Three isoenzymes of aspartate kinase have been identified-one repressed by L-methionine, the second one repressed by L-threonine and L-isoleucine, and the third one being inhibited and repressed by L-lysine. The amino acid sequence and structure of aspartate kinase have been elucidated. And by genetic manipulations, it has been possible to create mutants (of aspartate kinase) that are insensitive to feedback regulation by L-lysine.

Dihydrodipicolinate synthase : This enzyme competes with homoserine dehydrogenase to act on aspartate semialdehyde. Overexpression of dihydrodipicolinate synthase has been shown to increase the production of L-lysine.

Succinylase and dehydrogenase variants : The conversion of piperideine 2, 6-dicarboxylate to D, L-diaminopimelate is carried out by these two enzymes. At the start of the fermentation, dehydrogenase variant predominantly acts, and later succinylase variant comes into picture for the biosynthesis of L-lysine.

Role of D, L-diaminopimelate : This amino acid, an immediate precursor for the synthesis of L-lysine, is also required for the synthesis of a tripeptide (L-Ala-γ-D-Glu-D, L-Dap) which is part of the peptidoglycan of cell wall. The activities of both the enzymes (succinylase and dehydrogenase) that form diaminopimelate (Dap) are important for the production of L-lysine and for the proper formation of cell wall structure.

Improved production strains

Based on the biosynthetic pathway and the regulatory steps (discussed above), certain improvements have been made in the strains of

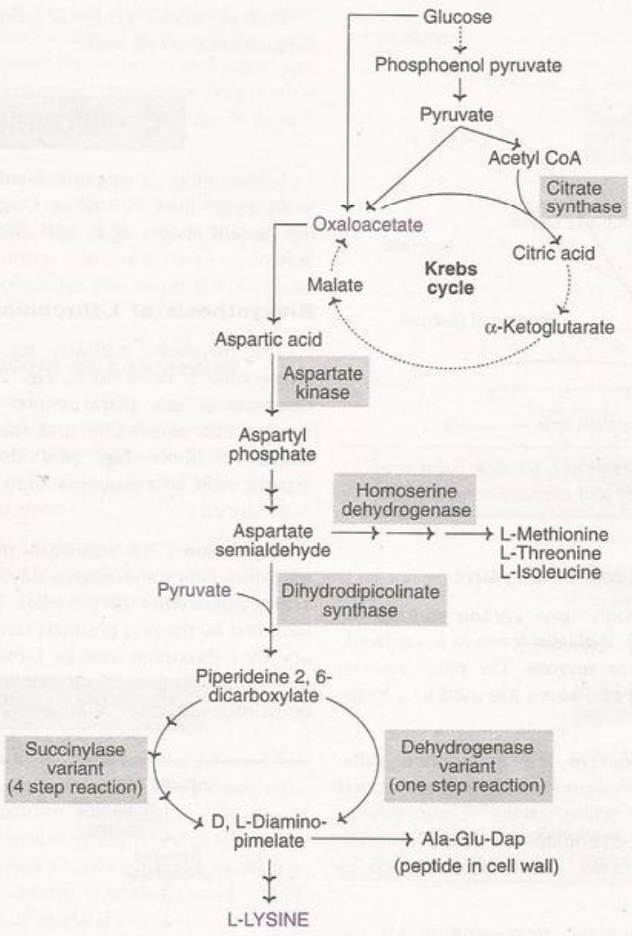


Fig. 26.4 : *Biosynthesis of L-lysine in* C. glutamicum
(Ala-Glu-Dap — Alanyl-glutamyl-diaminopimelate, a tripeptide)

C. glutamicum and *B. flavum* for overproduction of lysine.

- Mutant organisms resistant to lysine antimetabolites (e.g. b-aminoethyl-L-lysine).
- A mutant strain with an altered enzyme aspartokinase, so that it is not regulated by end product inhibition.
- A strain with a decreased homoserine dehydrogenase activity (so that diversion for the synthesis of methionine, threonine and isoleucine is minimised).
- A strain with reduced citrate synthase activity (to lower the occurrence of citric acid cycle).

Release of L-lysine

The export or release of L-lysine from the cells into the surrounding medium occurs through a ***lysine-export (LysE)*** carrier protein. It is a trans-membrane protein (mol. wt-25,400) with six segments that participate in lysine transport. The exporter system is very efficient active process to export large quantities of intracellular lysine.

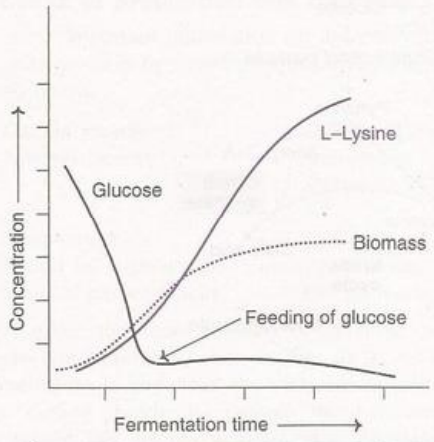


Fig. 26.5 : Production of L-lysine in relation to substrate (glucose) and biomass concentration.

Production process of L-lysine

The most commonly used ***carbon sources*** for lysine manufacture is ***molasses*** (cane or sugar beet), starch hydrolysates or ***sucrose***. The other sources like acetate, ethanol or alkanes are used to a lesser extent.

The ***nitrogen sources*** are ***ammonium salts***, gaseous ***ammonia***. Protein hydrolysates are added to supply certain amino acids (L-methionine, L-homoserine, L-threonine). The protein hydrolysates also supply growth factors such as biotin.

A time-course graphic representation for the formation of lysine is depicted in ***Fig. 26.5***. As is evident, a continuous supply of glucose (or other sugar) is required for sustained production of lysine. Under optimal fermentation conditions, the yield of lysine (in the form of L-lysine HCl) is 40–50 g per 100 g carbon source.

There are different recovery processes for lysine depending on its application.

- An alkaline solution containing about 50% L-lysine can be obtained after biomass separation, evaporation and filtration.
- A crystalline preparation with 98–99% L-lysine (as L-lysine HCl) can be obtained by subjecting the culture broth to ion-exchange chromatography, evaporation and crystallization.

Both the above grades of lysine are suitable for supplementation of feeds.

L-THREONINE

L-Threonine is manufactured industrially by employing either ***E. coli*** or ***C. glutamicum***. With the mutant strains of *E. coli*, the product yield is better.

Biosynthesis of L-threonine

The metabolic pathway for the synthesis of L-threonine is depicted in ***Fig. 26.6***. Some of the reactions of this pathway are common for the biosynthesis of L-lysine and methionine, besides isoleucine (Refer ***Fig. 26.4*** also). Starting with aspartic acid, in a sequence of five steps, threonine is produced.

Regulation : The regulatory reactions in *E. coli* for L-threonine biosynthesis have been elucidated. Three isoenzymes of aspartate kinase, separately inhibited by the end products have been identified-one by L-threonine, one by L-methionine and one by L-lysine. Further, two isoenzymes of homoserine dehydrogenase-one inhibited by L-threonine and

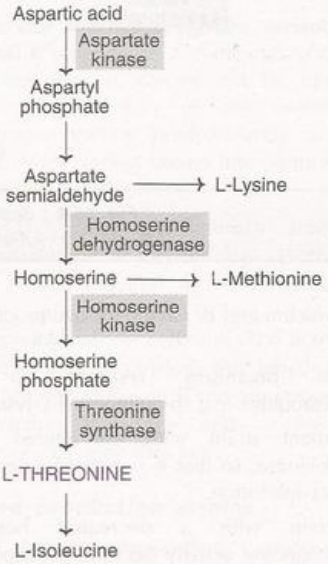


Fig. 26.6 : Biosynthesis of L-threonine in E. coli.

other by L-methionine are also known. A gene *thrABC* that encodes three polypeptides (one polypeptide possesses the activity of kinase and homoserine dehydrogenase, the second homoserine kinase and the third threonine synthase) in *E. coli* has been identified.

Improved production strains : The efficiency of the producer strains can be increased by creating *E. coli* mutants with high-level expression of the gene *thrABC*. Further, mutants with minimal production of L-isoleucine also result is high yield of L-threonine.

Production process of L-threonine

The culture medium containing glucose or sucrose, yeast extract and ammonium salts is adequate for L-threonine production. The sugar feeding has to be continued for good yield (about 60% of the carbon source).

The downstream processing for the isolation of L-threonine consists of coagulation of the cell mass (by heat), filtration, concentration by evaporation, and crystallization.

L-PHENYLALANINE

Both ***E. coli*** and ***C. glutamicum*** can be used for the production of L-phenylalanine. The biosynthetic pathway is quite complex and an outline is shown in ***Fig. 26.7***. An interesting feature is that the same pathway is responsible for the synthesis of all the three aromatic amino acids-tyrosine and tryptophan, besides phenylalanine.

The synthetic pathway commences with the condensation of erythrose 4-phosphate with phosphoenol pyruvate to form deoxy-arabinoheptulosonate phosphate (DAHP). DAHP in the next series of reactions is converted to chorismate which can form L-tryptophan. Chorismate mutase converts chorismate to prephenate which forms L-phenylalanine through the participation of prephenate dehydrogenase. Prephenate also serves as a precursor for the synthesis of tyrosine.

The genes responsible for the formation of the regulatory enzymes of L-phenylalanine have been identified. By employing genetic manipulations, strains for improved production of L-phenylalanine have been developed.

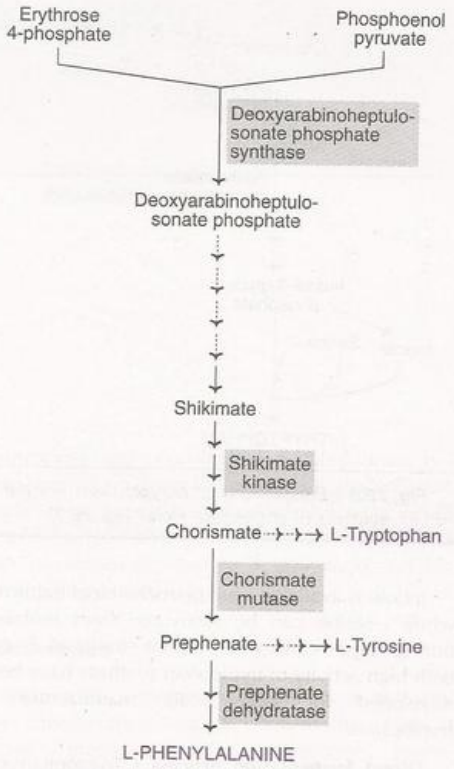
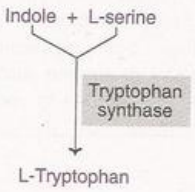


Fig. 26.7 : An outline of the pathway for the synthesis of L-phenylalanine, L-tyrosine and L-tryptophan.

L-TRYPTOPHAN

There are different ways of synthesizing L-tryptophan-chemical, enzymatic and fermentation methods. At present, large scale manufacture of tryptophan is carried out by using the enzyme ***tryptophan synthase*** of ***E. coli***. Tryptophan synthase combines indole with L-serine to form tryptophan.



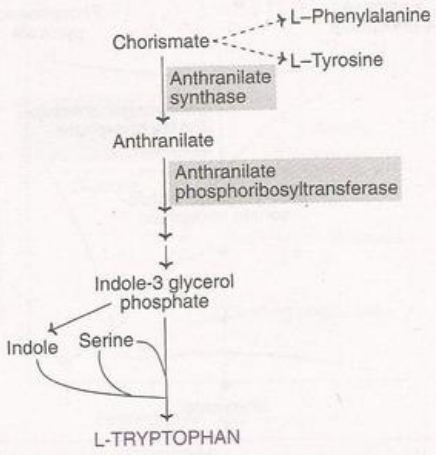


Fig. 26.8 : Biosynthesis of L-tryptophan (For the syntheis of chorismate Refer Fig. 26.7).

Indole is available from petrochemical industries while L-serine can be recovered from molasses during sugar refinement. Mutant strains of *E. coli* with high activity of tryptophan synthase have been developed for large scale manufacture of tryptophan.

Direct fermentation process : Tryptophan can also be produced by fermentation employing *C. glutamicum*, or *E. coli*. For the biosynthetic pathway, refer ***Fig. 26.7***. Mutant strains of both these organisms have been developed for increased yield of tryptophan.

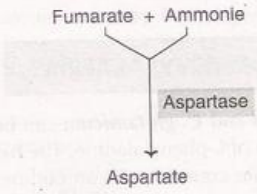
Mutant strains for overproduction L-tryptophan

The production of tryptophan by *C. glutamicum* was increased by introducing a second gene encoding ***anthranilate synthase***, a key enzyme in its biosynthesis (***Fig. 26.8***). Further, genes encoding other important enzymes (deoxyarabino-heptulosonate phosphate synthase, anthranilate phosphoribosyltransferase) were also be modified. The result is that the pathway becomes insensitive to feedback inhibition by end products, leading to an overproduction of L-tryptophan.

L-ASPARTIC ACID

There is a growing demand for aspartate, as it is a ***component of aspartame*** (an artificial sweetener), besides its use as a food additive, and in pharmaceutical preparations.

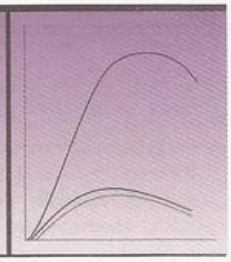
The preferred method for aspartate production is enzymatic in nature. The enzyme aspartase converts fumarate and ammonia to aspartate. Although this reaction is reversible, aspartate formation is favoured.



The aspartase of *E. coli* is used. It is a tetramer with a molecular weight 196,000. This enzyme is quite unstable. Immobilization of aspartase in polyacrylamide or carrageenan that enhances the stability of the enzyme is commonly used.

Immobilized *E. coli* cells with good activity of aspartase are also used for aspartate production.

27 Microbial Production of Vitamins



Vitamins are organic compounds that perform specific biological functions for normal maintenance and optimal growth of an organism. These ***vitamins cannot be synthesized by the higher organisms***, including man, and therefore they have to be supplied in small amounts in the diet.

Microorganisms are capable of synthesizing the vitamins. In fact, the bacteria in the gut of humans can produce some of the vitamins, which if appropriately absorbed can partially meet the body's requirements. It is an accepted fact that after administration of strong antibiotics to humans (which kill bacteria in gut), additional consumption of vitamins is recommended.

Microorganisms can be successfully used for the commercial production of many of the vitamins e.g. thiamine, riboflavin, pyridoxine, folic acid, pantothenic acid, biotin, vitamin B_{12}, ascorbic acid, β-carotene (provitamin A), ergosterol (provitamin D). However, from economic point of view, it is feasible to produce vitamin B_{12}, riboflavin, ascorbic acid and β-carotene by microorganisms. ***For the production of ascorbic acid (vitamin C), the reader must refer Chapter 24.***

VITAMIN B_{12}

The disease, ***pernicious anemia***, characterized by low levels of hemoglobin, decreased number of erythrocytes and neurological manifestations, has been known for several decades. It was in 1926 some workers reported the liver extracts could cure pernicious anemia. The active principle was later identified as vitamin B_{12}, a water soluble B-complex vitamin.

Occurrence

Vitamin B_{12} is present in animal tissue at a very low concentration (e.g. 1 ppm in the liver). It occurs mostly in the coenzyme forms-methylcobalamin and deoxyadenosylcobalamin. Isolation of vitamin B_{12} from animal tissues is very expensive and tedious.

Chemistry

Vitamin B_{12} (cyanocobalamin) is a water soluble vitamin with complex structure. The empirical formula of cyanocobalamin is $C_{63}H_{90}N_{14}O_{14}PCO$. The structure of vitamin B_{12} consists of a corrin ring with a central cobalt atom. The corrin ring is almost similar to the tetrapyrrole ring structure found in other porphyrin compounds e.g. heme (with Fe) and chlorophyll (with Mg).

The corrin ring has four pyrrole units. Cobalt present at the centre of the corrin ring is bonded to the four pyrrole nitrogens. Cobalt also binds to dimethylbenzimidazole and aminoisopropanol. Thus, cobalt atom present in vitamin B_{12} is in a coordination state of six.

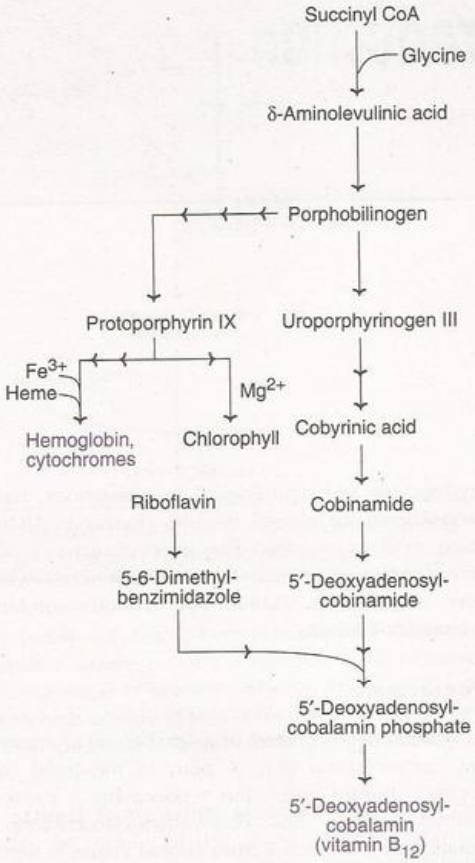


Fig. 27.1 : An outline of the biosynthesis of vitamin B_{12}

Biosynthesis

Vitamin B_{12} is ***exclusively synthesized*** in nature ***by microorganisms***. An outline of the pathway is depicted in ***Fig. 27.1***. The biosynthesis of B_{12} is comparable with that of chlorophyll and hemoglobin. Many of the reactions in the synthesis of vitamin B_{12} are not yet fully understood.

COMMERCIAL PRODUCTION OF VITAMIN B_{12}

Vitamin B_{12} is commercially produced by fermentation. It was first obtained as a byproduct of *Streptomyces* fermentation in the production of certain antibiotics (streptomycin, chloramphenicol, or neomycin). But the yield was very low. Later, high-yielding strains were developed. And at present, vitamin B_{12} is entirely produced by fermentation. It is estimated that the world's annual production of vitamin B_{12} is around 15,000 kg.

High concentrations of vitamin B_{12} are detected in sewage-sludge solids. This is produced by microorganisms. Recovery of vitamin B_{12} from sewage-sludge was carried out in some parts of United States.

Unlike most other vitamins, the chemical synthesis of vitamin B_{12} is not practicable, since about 20 complicated reaction steps need to be carried out. ***Fermentation of vitamin B_{12} is the only choice.***

Microorganisms and yields of vitamin B_{12}

Several microorganisms can be employed for the production of vitamin B_{12}, with varying yields. Glucose is the most commonly used carbon source. Some examples of microbes and their corresponding yields are given in ***Table 27.1***. The most commonly used microorganisms are — *Propionibacterium freudenreichii, Pseudomonas denitrificans, Bacillus megaterium* and *Streptomyces olivaceus.*

Genetically engineered strains for vitamin B_{12} production : By employing modern techniques of genetic engineering, vitamin B_{12} production can be enhanced. A protoplast fusion technique between *Protaminobacter rubber* and *Rhodopseudomonas spheroides* resulted in a hybrid strain called *Rhodopseudomonas protamicus.* This new strain can produce as high as 135 mg/l of vitamin B_{12} utilizing carbon source.

TABLE 27.1 Microorganisms with corresponding yields of vitamin B_{12}

Microorganism	*Yield (mg/l)*
Bacillus megaterium	0.51
Streptomyces olivaceus	3.31
Butyribacterium rettgeri	5.0
Micromonospora sp	11.5
Propionibacterium freudenreichii	19.0
Propionibacterium shermanii	35.0
Pseudomonas denitrificans	60.0
Hybrid strain	
Rhodopseudomonas protamicus	135.0

Production of vitamin B_{12} using *Propionibacterium* sp

Propionibacterium freudenreichii and *P. shermanii*, and their mutant strains are commonly used for vitamin B_{12} production. The process is carried out by adding cobalt in two phases.

Anaerobic phase : This is a preliminary phase that may take 2-4 days. In the anaerobic phase 5'-deoxyadenosylcobinamide is predominantly produced.

Aerobic phase : In this phase, 5, 6-dimethyl-benzimidazole is produced from riboflavin which gets incorporated to finally form coenzyme of vitamin B_{12} namely 5'-deoxyadenosylcobalamin.

In recent years, some fermentation technologists have successfully clubbed both an anaerobic and aerobic phases to carry out the operation continuously in two reaction tanks.

The ***bulk production of vitamin B_{12}*** is mostly done ***by submerged bacterial fermentation with beet molasses medium supplemented with cobalt chloride***. The specific details of the process are kept as a guarded secret by the companies.

Recovery of vitamin B_{12} : The cobalamins produced by fermentation are mostly bound to the cells. They can be solubilized by heat treatment at 80–120°C for about 30 minutes at pH 6.5-8.5. The solids and mycelium are filtered or centrifiged and the fermentation broth collected. The cobalamins can be converted to more stable cyanocobalamins. This vitamin B_{12} is around 80% purity and can be directly used as a feed additive. However, for medical use (particularly for treatment of pernicious anemia), vitamin B_{12} should be further purified (95–98% purity).

Production of vitamin B_{12} using *Pseudomonas* sp

Pseudomonas denitrificans is also used for large scale production of vitamin B_{12} in a cost-effective manner. Starting with a low yield (0.6 mg/l) two decades ago, several improvements have been made in the strains of *P. denitrificans* for a tremendous improvement in the yield (60 mg/l).

Addition of cobalt and 5, 6-dimethyl benzimidazole to the medium is essential. The yield of vitamin B_{12} increases when the medium is supplemented with betaine (usual source being sugar beet molasses).

Carbon sources for vitamin B_{12} production

Glucose is the most commonly used carbon source for large scale manufacture of vitamin B_{12}. Other carbon sources like ***alcohols*** (methanol, ethanol, isopropanol) and ***hydrocarbons*** (alkanes, decane, hexadecane) with varying yields can also be used.

A yield of 42 mg/l of vitamin B_{12} was reported using methanol as the carbon source by the microorganism *Methanosarcina barkeri*, in fed-batch culture system.

RIBOFLAVIN

Riboflavin (vitamin B_{12}) is a water soluble vitamin, essential for growth and reproduction in man and animals. Deficiency of riboflavin in rats causes growth retardation, dermatitis and eye lesions. In humans, ***vitamin B_2 deficiency*** results in ***cheilosis*** (fissures at the corner of mouth), ***glossitis*** (purplish tounge) and ***dermatitis***. Riboflavin exerts its biochemical functions through the coenzymes namely ***flavin adenine dinucleotide (FAD)*** and ***flavin mononucleotide (FMN)***.

Occurrence

Riboflavin occurs in milk and milk products, meat, eggs, liver and kidney. While in milk and eggs, it is present in free form, in other foods it is found in the form of flavoproteins (i.e. coenzymes of riboflavin bound to proteins).

Chemistry

Riboflavin contains ***6, 7-dimethyl isoalloxazine*** (a heterocyclic 3 ring structure) attached to ***D-ribitol*** by a nitrogen atom. The isoalloxazine ring participates in the oxidation-reduction reactions brought out by the coenzymes (FAD and FMN).

Biosynthesis

The biosynthetic pathway of riboflavin, elucidated for the microorganisms *Ashbya gossypii* and *Eremothecium ashbyii* is depicted in ***Fig. 27.2***. The overproduction of riboflavin in these organisms takes place mainly due to the constitutive nature of the riboflavin synthesizing enzymes. Iron which inhibits the production of vitamin B_{12} in clostridia and yeasts, has no effect on *A. gossypii* and *E. ashbyii*.

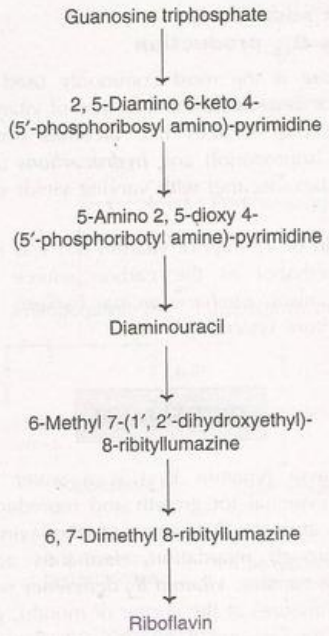


Fig. 27.2 : Biosynthesis of riboflavin.

COMMERCIAL PRODUCTION OF RIBOFLAVIN

There are three processes employed for the large scale production of riboflavin. The worldwide requirement of riboflavin is estimated to be around 2,500 tones per year.

1. **Biotransformation :** About 50% of the world's requirement of riboflavin is produced by biotransformation, followed by chemical synthesis. For this purpose, glucose is first converted to D-ribose by mutant strains of *Bacillus pumilus*. The D-ribose so produced is converted to riboflavin by chemical reactions.

2. **Chemical synthesis :** Approximately 20% of the world's riboflavin is produced by direct chemical synthesis.

3. **Fermentation :** At least one third of world's riboflavin requirements are met by direct fermentation processes.

Microorganisms and yields of riboflavin

Several microorganisms (bacteria, yeasts and fungi) can be employed for the production of riboflavin. In the acetone-butanol fermentation, employing the organisms *Clostridium acetobutylicum* and *Clostridium butylicum*, riboflavin is formed as a byproduct.

Commercial production of riboflavin is predominantly carried out by direct fermentation using the ascomycetes. The different organisms used and the corresponding yields of riboflavin are given in ***Table 27.2***. The two plant pathogens namely *Ashbya gossypii* and *Eremothecium ashbyii* are most commonly employed due to high yield. Among these two organisms, *A. gossypii* is preferred as it is more stable with a high producing capacity of riboflavin.

Genetically engineered strains for riboflavin production : High yielding strains of *Ashbya gossypii* have been developed by genetic manipulations. Such strains can yield as high as 15 g/1 riboflavin.

Production process of riboflavin

Industrial production of riboflavin is mostly carried out with the organism, ***Ashbya gossypii*** by using simple sugars such as glucose and corn steep liquor. Glucose can be replaced by sucrose or maltose for the supply of carbon source. In recent years, lipids such as corn oil, when added to the medium for energy purpose, have a profound influence on riboflavin production. Further, supplementation of the medium with yeast extract, peptones, glycine, inositol, purines (not pyrimidines) also increase the yield of riboflavin.

It is essential to carefully sterilize the medium for good yield of riboflavin. The initial pH of the culture medium is adjusted to around 6-7.5. The

TABLE 27.2 Microorganisms with corresponding yields of riboflavin

Microorganism	*Yield (mg/l)*
Clostridium acetobutylicum	0.097
Clostridium butylicum	0.120
Mycobacterium smegmatis	0.060
Mycocandida riboflavina	0.200
Candida flareri	0.575
Eremothecium ashbyii	2.500
Ashbya gossypii	7.500

fermentation is conducted at temperature 26–28°C with an aeration rate 0.3 vvm. The process is carried out for about 5-7 days by submerged aerated fermentation.

Riboflavin fermentation by *Eremothecium ashbyii* is comparable to that described above for *Ashbya gossypii*. *Candida* sp can also produce riboflavin, but this fermentation process is extremely sensitive to the presence of iron. Consequently, iron or steel equipment cannot be used. Such equipment have to be lined with plastic material.

Fermentation through phases

Some studies have been carried out to understand the process of fermentation of riboflavin particularly by ascomycetes. It is now accepted that the fermentation occurs through ***three phases***.

Phase I : This phase is characterized by rapid growth of the organism utilizing glucose. As pyruvic acid accumulates, pH becomes acidic. The growth of the organism stops as glucose gets exhausted. In phase I, there is no production of riboflavin.

Phase II : Sporulation occurs in this phase, and pyruvate concentration decreases. Simultaneously, there is an accumulation of ammonia (due to enhanced deaminase activity) which makes the medium alkaline. Phase II is characterized by a maximal production of riboflavin. But this is mostly in the form of FAD and a small portion of it as FMN.

Phase III : In this last phase, cells get disrupted by a process of autolysis. This allows release of FAD, FMN and free riboflavin into the medium.

Recovery : Riboflavin is found in fermentation broth and in a bound form to the cells. The latter can be released by heat treatment i.e. 120°C for about 1 hour. The cells can be discarded after filtration or centrifugation. The filtrate can be further purified and dried, as per the requirements.

Other carbon sources for riboflavin production

Besides ***sugars***, other carbon sources have also been used for riboflavin production. A pure grade of riboflavin can be prepared by using *Saccharomyces* sp, utilizing ***acetate*** as sole carbon source. Methanol-utilizing organism *Hansenula polymorpha* was found to produce riboflavin. The other carbon sources used with limited success for riboflavin production are aliphatic hydrocarbons (organism *Pichia guilliermoudii*) and n-hexadecane (organisms — *Pichia miso*).

β-CAROTENE

β-Carotene is the ***provitamin A***. When ingested, it gets converted to vitamin A in the intestine. Vitamin A is a fat soluble vitamin required for vision, proper growth and reproduction. The ***deficiency of vitamin A causes night blindness***, changes in the skin and mucosal membranes.

Occurrence and chemistry

β-Carotene is found in many animal and plant tissues. However, it originates exclusively from plants or microorganisms. Yellow and dark green vegetables and fruits are rich in β-carotene e.g. carrots, spinach, amaranthus, mango, papaya.

Carotenoids are isoprene derivatives. Chemically, they are tetraterpenoids with eight isoprene residues. There are around 400 naturally occurring carotenoids. The most important carotenoids are β-carotene, α-carotene, δ-carotene, lycopene and zeaxanthin.

Carotenoids are mainly used as colouring agents e.g., β-carotene, lycopene, xanthophylls. Several foods (cheese, meat, egg products) can be made attractive by coloration. It may be noted that the demand for β-carotene as the provitamin A is comparatively less.

Biosynthesis

The pathway for the biosynthesis of β-carotene and some other important carotenoids, elucidated in plants and fungi, is shown in ***Fig. 27.3***.

COMMERCIAL PRODUCTION OF β-CAROTENE

β-Carotene can be produced by microbial fermentation. However, for economic reasons, direct chemical synthesis of vitamin A is preferred rather than using its provitamin (β-carotene).

Microorganisms

The organisms *Blakeslea trispora*, *Phycomyces blakesleeanus* and *Choanephora cucurbitarum* are most frequently used for the production of

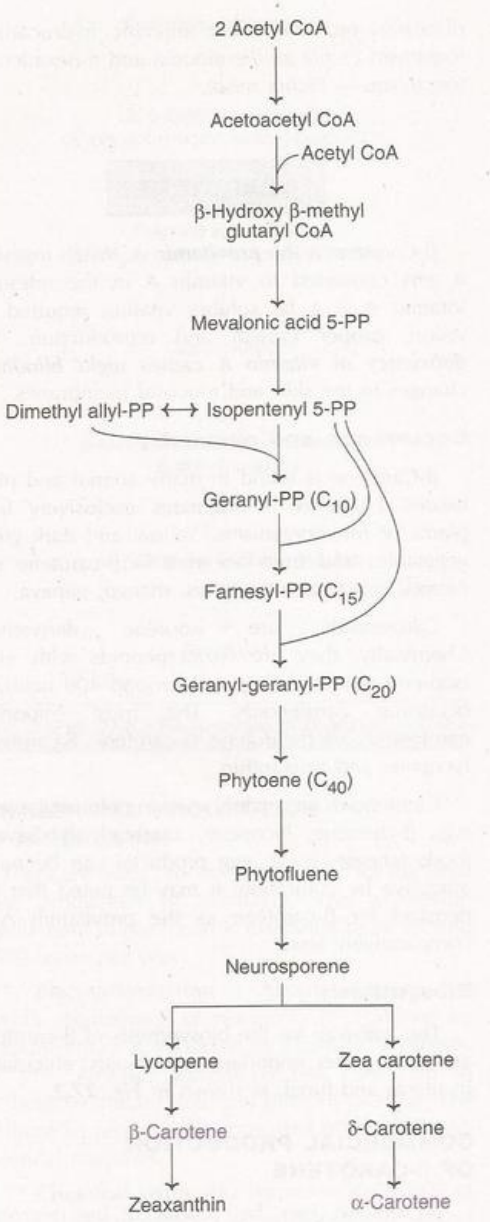


Fig. 27.3 : Biosynthesis of carotenes.

β-carotene. Among these, *Blakeslea trispora* is preferred due to high yield. In the ***Table 27.3***, some important carotenoids, the organisms and the production yields are given.

TABLE 27.3 Microbial production of important carotenoids

Carotenoid	*Organism*	*Production yield (g/l)*
β-Carotene	*Blakeslea trispora* (mixed cultures of + and – sexual forms)	3.0
Lycopene	*Blakeshlea trispora* (mixed culture)	0.4
	Streptomyces chrestomyceticus	0.5
Zeaxanthin	*Flavobacterium* sp	0.4

Production process of β-carotene

As already stated, the industrial production of β-carotene is mostly carried out by *Blakeslea trispora*. The fermentation medium contains corn starch, soybean meal, β-ionone, antioxidants etc. Addition of antioxidants improves the stability of β-carotene with in the cells. The ***fermentation is carried out by submerged process.***

The fermentation is usually started by mixing the cultures of both sexual forms, (+) and (–) strains of *B. trispora*. The yield of β-carotene is significantly higher with mixed cultures, compared to + or – strains (***Fig. 27.4***). This is due to the fact that

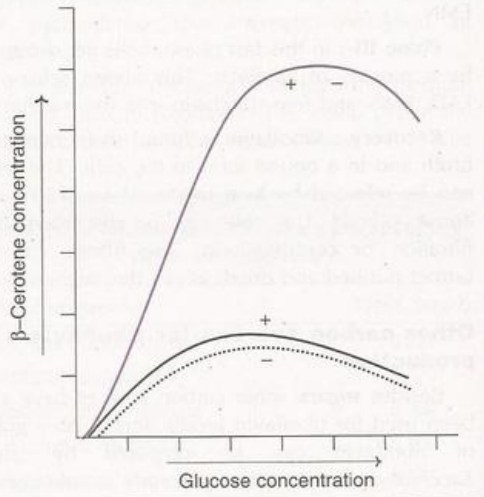


Fig. 27.4 : Yield of β–carotene by +, – and mixed (+ –) cultures of Blakesba trispora.

β-carotene production predominantly occurs during the process of zygospore formation. It may be stated here that the use of mixed strains does not improve the yield for other microorganisms (as observed in case of *Blakeslea trispora*).

Factors affecting production : Trisporic acid which can act as a microbial sexual hormone improves production yield of β-carotene. β-Ionones enhance β-carotene synthesis by increasing the activity of enzymes, and not by their direct incorporation into β-carotene. When the fermentation medium is supplemented with purified kerosene, β-carotene production is almost doubled. Kerosene increases the solubility of hydrophobic substrates.

Recovery : The mycelium rich in β-carotene can be directly used as a feed additive. For purification, mycelium is removed, subjected to dehydration (by methanol) and extracted in methylene chloride. This product is of 70–85% purity which can be further purified as per the requirements.

GIBBERELLINS — PLANT GROWTH STIMULANTS

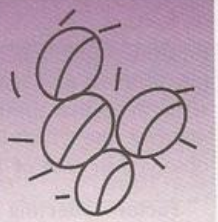
Gibberellins are plant hormones that stimulate plant growth. They promote growth by cell enlargement and cell division. The observable effects of gibberellins include stimulus to seed germination, flowering and lengthening of stems.

MICROBIAL PRODUCTION OF GIBBERELLINS

So far only one microorganism, ***the fungus*** namely ***Gibberella fujikuroi has been found to produce gibberellins***. This is actually a pathogenic fungus of rice seedlings.

Gibberellin production can be carried out by using a glucose-salt medium at pH 7.5 and temperature 25°C for 2-3 days. The fermentation process is conducted in aerated submerged process. After the growth of the fungus is maximum, the production of gibberellins commences.

Microbial Production of Foods and Beverages



It is estimated that about 20–30% of the household budget is spent towards foods in the developed countries. This may be a little less in the developing nations. Therefore, food and beverage biotechnology occupies a prominent place world-over.

There are records that man was making bread, wine, curd etc., as early as 4000 BC. These processes, collectively referred to as ***traditional*** or ***old biotechnology***, mostly employed for the preparation of foods and beverages, were ***based on the natural capabilities of the microorganisms*** (although their existence was unknown at that time).

With the advances made in microbiology and recently biotechnology, food and beverage production is a major industry. Food biotechnology is also concerned with the improved quality, nutrition, consistency, colour, safety and preservation of foods, besides making them available round the year (**Note :** Most foods are seasonal in nature and therefore as such are not available throughout the year). In addition, modern biotechnological processes also take into account the health aspects of the people.

FERMENTED FOODS

The production of fermented foods is variable. This depends on geographical region, availability of raw materials, traditions and food habits of the people. A selected list of fermented foods along with raw materials and fermenting microorganisms is given in ***Table 28.1***.

Advantages of fermented foods

- Enhanced nutritive value.
- Increased digestibility.
- Improved flavour and texture.
- Serve as supplements in preparing several dishes.

Besides the list given in ***Table 28.1***, there are many other compounds that are directly or indirectly used as food supplements which must also be referred by the reader. These include vitamins (Chapter 27), amino acids (Chapter 26), organic acids (Chapter 23), enzymes (Chapter 21), biopolymers (Chapter 30), single-cell proteins and mushrooms (Chapter 29).

A selected few of the fermented foods with special reference to their production are described briefly in this chapter.

CHEESE

Cheese production is the largest dairy industry in the world. There are around 1,000 types of different cheeses. They are broadly of two types — ***unripened cheeses*** (cottage cheese with low fat, cream cheese with high fat) and ***ripened cheeses*** (hard cheese e.g. chedder, blue cheese; soft cheese e.g. limburger, camembert). Irrespective of the type of the cheese, all of them are invariably ***made from the casein of milk***, that is produced after separating

Table 28.1 A selected list of fermented foods along with the raw materials and fermenting organisms

Fermented food/food product (country)	*Raw material (substrate)*	*Fermenting organism(s)*
Dairy products		
Cheese (worldwide)	Milk	*Streptococcus* sp
		Penicillium roquefortii,
		P. camembertii
Yogurt (worldwide)	Milk	*Streptococcus thermophilus*
		Lactobacillus bulgaricus
Kefir (Russia)	Milk	*Lactobacillus* sp, *Candida* sp
Vegetarian products		
Cocoa beans (worldwide)	Cocoa fruit	*Candida krusei*, *Geotrichum* sp
Coffee beans (worldwide)	Coffee cherries	*Edwinia dissolvens*, *Saccharomyces* sp
Tempeh (Indonesia)	Soy beans	*Rhizopus oryzae*, *Lactobacillus delbrueckiia*
Soy sauce (worldwide)	Soy beans	*Aspergillus orzyae*, *A. soyae*
Sauekraut (Europe)	Cabbage	*Leuconostoc mesenteroides*, *L. plantarum*
Breads (worldwide)	Wheat flour	*Saccharomyces cerevisiae*
Rolls, cakes (worldwide)	Wheat flour	*Saccharomyces cerevisiae*
Idli (India)	Rice and black gram	*Leuconotoc mesenteroides*
Non-vegetarian products		
Dry sausages (worldwide)	Beef, pork	*Pedicoccus cerevisiae*
Fish sauces (worldwide)	Small fish	*Halophilic* sp, *Bacillus* sp
Country-cured hams (worldwide)	Pork, hams	*Aspergillus* sp, *Penicillium* sp

the whey (liquid portion of milk). Milk from different animals can be used e.g. sheep, cow, goat, buffalo.

Historical perspective

The use of animal stomachs for carrying liquids is centuries old. When milk was transported in this fashion, the formation of solids (that were tasty) was observed. The solids were concentrated after draining liquids. These solids were salted and consumed later. A good example of food preservation, long long ago! We now know that this solid portion is the cheese. It is produced by the combined action of enzymes (rennet) of the stomach living and the bacterial contamination.

Production process

As already stated, cheese is produced from milk. This is carried out by a process of dehydration wherein casein (milk protein) and fats are concentrated 5–15 fold. Cheese production is very complicated, and broadly involves four stages-acidification of milk, coagulum formation, separation of curd from whey and ripening of cheese.

1. **Acidification of milk :** By employing lactic acid bacteria (*Streptococcus lactis, Lactobacillus lactis*) the sugar of milk (lactose) can be converted to lactic acid. This lowers the pH to around 4.6, and thus acidifies milk.

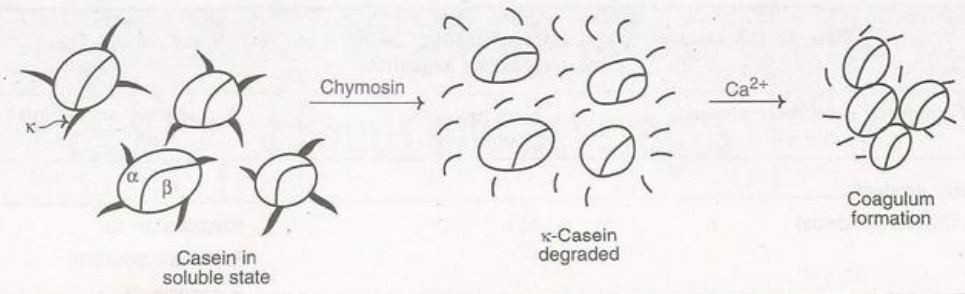


Fig. 28.1 : Formation of coagulum (curd) in the production of cheese.

2. **Coagulum formation :** When the acidified milk is treated with **rennet** (i.e. the enzyme chymosin of animal or fungal origin), casein gets coagulated. Casein mainly consists of three components-insoluble α and β caseins and a κ-casein that keeps them in soluble state. By the action of chymosin, κ-casein is degraded. Consequently, α and β caseins and the degraded products of κ casein combine to form a coagulum (curd) (***Fig. 28.1***). This process of coagulation is dependent on calcium ions.

3. **Separation of curd from whey :** When the temperature of the coagulum is raised to around 40°C; the coagulum (curd) and whey (fluid portion) get separated. The separated curd is cut into blocks, drained and pressed into different shapes.

4. **Ripening of cheese :** The flavour of raw cheese (with rubber texture) such as ***cheddar*** is bland. Ripening imparts flavours, besides making changes in its texture. The procedures adopted for ripening (or maturation) are highly variable depending on the type of cheese to be prepared. The blocks of curd separated are subjected to the action of proteases and/or lipases. Alternatively, they may be inoculated with certain fungi (e.g. *Penicillium roquefortii*). The hydrolysis of proteins and fats (either by enzymes or microorganisms) results in certain compounds which imparts flavour to the cheese. Mild hydrolysis of fats (or cheese), usually carried out by lipases or *Aspergillus niger* or *Mucor maihai* results in butyric acid formation with characteristic flavour.

A diagrammatic representation of cheese production is depicted in ***Fig. 28.2***.

Sources of chymosin for cheese production

There are several sources of rennet (chymosin enzyme) for cheese production. These include calves, adult cows, pigs and fungal sources. Fortunately, the fungal (e.g. *Mucor meihei*) sources of chymosin are almost comparable to the animal sources and are widely used in some countries. However, some people always prefer animal rennet used cheese due to its slight superior flavour.

Genetically engineered microorganisms for chymosin : Some workers could successfully clone the genes of animal chymosin and transfer the same into microorganisms. The chymosin so produced by genetic manipulation is very widely used these

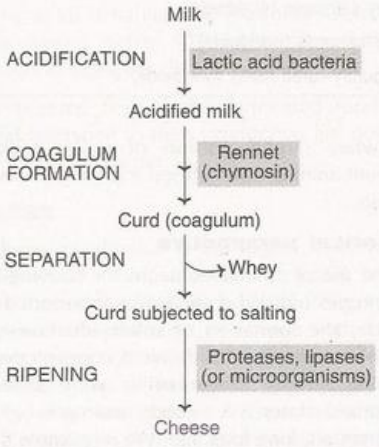


Fig. 28.2 : A diagrammatic representation of cheese production.

days in some countries. The taste of cheese manufactured by microbial chymosin (i.e. genetically engineered) and the animal chymosin are identical. Public as well as the vegetarian societies in some countries have accepted the cheese produced by the genetic manipulation of the enzyme, chymosin.

YOGHURT

Yoghurt is produced ***by fermenting whole milk*** by employing a mixed culture of *Lactobacillus bulgaricus* and *Streptococcus thermophilus*. While *L. bulgaricus* produces acetaldehyde that imparts a characteristic taste, *S. thermophilus* results in the formation of lactic acid to give acid flavour. In addition, both these bacteria produce extracellular polymers that increase the viscosity of the fermented milk. Yoghurt is very delicious and in fact frozen yoghurt is becoming popular as an alternative to ice cream.

SAUEKRAUT

Sauekraut is prepared in most western countries. It is a ***fermented and preserved form of cabbage***. The shredded cabbage is mixed with salt (approximately at 2.5% concentration) and packed anaerobically. High salt concentration promotes leakage of sugars from the cabbage while reducing the water activity. As the growth of the lactic acid bacteria occurs, the pH is lowered. At this low pH, the putrifying bacteria cannot grow. In this way, sauekraut can be preserved for long. Sauekraut is nutritious as well as delicious.

The traditional pickles (mango, lemon, etc.) in India are also good examples of fermentation and preservation, based on the principles of biotechnology.

BREAD

Bread making from the dough by employing microorganisms is one of the oldest examples of fermentation processes known to mankind. There is evidence that bread was prepared in Egypt in 3000 BC.

Primarily, bread is a ***fermented product of cereal flours such as wheat and rye***. The cereal flour mixed with water, salt, sugar, fat and other ingradients (as desired for enrichment of bread) is subjected to fermentation by yeast, *Saccharomyces cerevisiae* (top fermenting strain). The main reaction that occurs during bread formation is the fermentation of hexoses to CO_2 and ethanol.

$$C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2$$

The ethanol produced either gets evaporated or forms esters. The CO_2 gets entrapped in the dough resulting in its expansion. The expansion and stretching of the dough, particularly with wheat is due to the unique ***elastic protein*** namely ***gluten***. Gluten is mainly responsible for retaining the shape of bread.

Besides yeast enzymes, the enzymes (e.g. amylases) of other microorganisms also help in fermentation and baking of bread. The texture of bread is influenced by fats and emulsifiers added to the dough. The bread making is carried out with three objectives :

- Good leavening due to CO_2 formation.
- Flavour development.
- Good texture.

The yeast fermented bread has the above characteristics. This is in contrast to the bread produced with baking powder which also produces CO_2. But this does not have the same flavour and texture as that produced by yeast. Thus the yeast, which is appropriately referred to as ***baker's yeast*** is a package of enzymes to give a desired product.

In recent years, some workers have reported the development of ***genetically engineered strains*** of ***Saccharomyces cerevisiae*** with improved fermentation properties. Such organisms, when used in baking industry, are believed to further enhance the quality of bread with regard to flavour, texture etc.

Sour-dough breads : In some parts of the world, sour breads are prepared by using the yeast, *Candida milleri*. and bacterium, *Lactobacillus sanfrancisco*.

BAKER'S YEAST

The living cells of aerobically grown Saccharomyces cerevisiae are collectively referred to as baker's yeast. Baker's yeast is commercially available either as a dried powder i.e. dry yeast with about 95% dry weight or in the form of cakes (about 25–30% dry weight). These commercially available yeast preparations can be used in bread making.

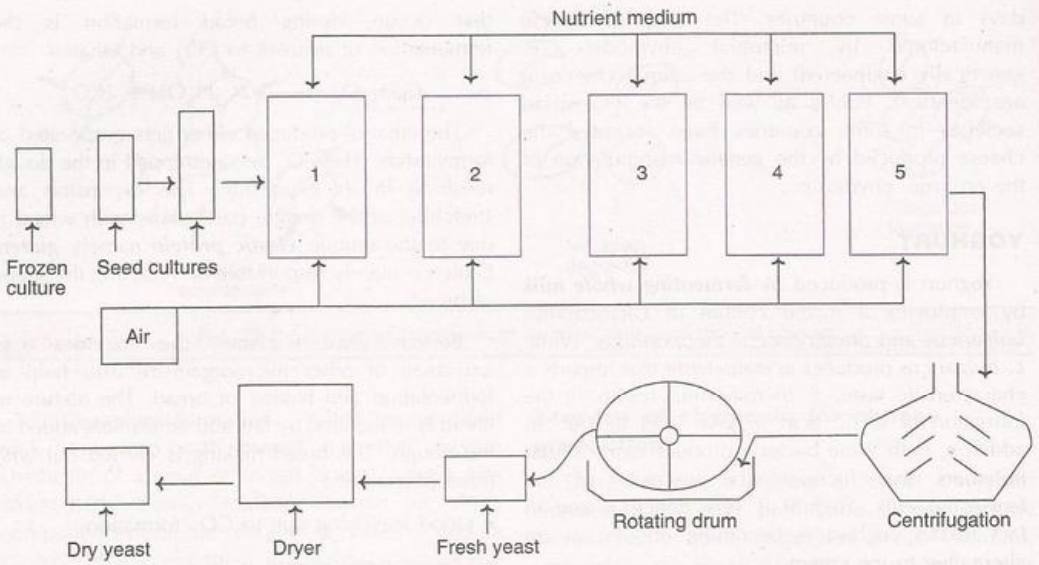


Fig. 28.3 : Flow chart for the production of baker's yeast (1 is the inoculation reactor, 2–5 are production reactors).

Production of baker's yeast : The medium for baker's yeast production contains molasses, ammonium salts (or ammonia), vitamins, phosphates and antifoam agents. Sugar cane or sugar beet molasses can be used. A commercially available molasses with a sugar concentration of 45–50% is usually preferred. Baker's yeast production is carried out by an ***aerobic fed-batch process***.

The flow chart for the production of baker's yeast is depicted in ***Fig. 28.3***. The actual production process and the strain of *Saccharomyces cerevisiae* used depends on the company. The desired strain is maintained in a frozen state. The inoculum is prepared in stages so that large volumes are finally obtained. From the inoculation reactor (fermenter), the culture is transferred to production reactors. The process is carried out in air-lift or bubble column type reactors at pH 4-5, temperature 28–30°C for about 12–18 hours. The yeast cells are washed, centrifuged and then dewatered on a rotating drum. The fresh yeast obtained can be either directly used or dried and stored.

The yeast cells may be mixed with plasticizer (e.g. vegetable oil) and prepared in a block form. This block can be cut into small pieces, wraped and stored (at –4°C) until used. The samples of yeast are usually tested for baking properties before they are put in use.

OTHER VEGETARIAN PRODUCTS

Besides the bread which is consumed worldwide, there are several other vegetarian fermented foods. These include ***rolls*** and ***cakes*** (worldwide) ***soy sauce*** (worldwide) ***idli*** (India) and ***tempeh*** (Indonesia).

Coffee, tea and cocoa

Coffee, tea and cocoa are very popular non-alcoholic fermented beverages, and extensively consumed throughout the world.

When the tea leaves are crushed, the chemical components of tea are released by enzymatic activity. For coffee and cocoa, a natural fermentation process (by bacteria, yeast and fungi) on the pulp surrounding beans results in the development of flavour and aroma. The exact nature of the fermentation processes involved in tea, coffee and cocoa are not fully known.

The dried products namely tea leaves, and coffee and cocoa beans are commercially available for beverage preparation.

SWEETENERS

Sweeteners occupy a prominent place in food consumption, by all the people. The most predominantly used ***natural sweetener*** is ***sucrose*** (obtained from sugar cane or beet). Sweeteners are invariably required for the preparation of soft drinks, ice creams, jams, jellies, sauces, bread, confectionery, pickles etc. Hence there is a continuous search for low calorie sweeteners.

Saccharin, a chemically synthesized sweetener (300 times sweeter than sucrose) was used for several years, and its use is now being restricted due to certain adverse health effects. ***High fructose corn syrup*** (***HFCS***), produced from sucrose is sweeter (1.5 times) than sucrose. ***Aspartame*** (aspartyl phenylalanine) is a product of innovative biotechnology. It is about ***200 times sweeter than sucrose*** and is truely a ***low calorie sweetener***. Aspartame is approved for human consumption and is widely used in soft drink and other industries.

THAUMATIN-A SWEET PROTEIN

Thaumatin is a protein extracted from the berries of the *Thaumatococcus daniellii*, a native plant in Africa. Thaumatin is about ***3000 times sweeter than sucrose.*** Thaumatin certainly appears to be a good sugar substitute and a low-calorie sweetener. It is in fact being used as a sweetener and flavour enhancer of various foods. At least five different forms of thaumatins with a molecular weights in the range of 20,000–22,000 have been isolated and characterized.

Genetic engineering to produce thaumatin

Natural production of thaumatin is very limited and therefore can never meet the demands of consumption. Biotechnological production of this sweetener is the method of choice for its large scale manufacture.

Thaumatin-encoding mRNA species have been isolated, identified and converted to cDNA and finally to double-stranded DNA. These DNAs were cloned and transferred to *E. coli*. But the yield of thaumatin production was very low. Attempts were also made to express thaumatin genes in yeasts such as *S. cerevisiae* and *Kluyveromyces lactis*. Some success has been reported in this direction.

MONELLIN-A CANDIDATE FOR SUGAR SUBSTITUTE

Monellin is a ***protein*** found in the fruit of an African plant *Discorephyllum cumminsii*. It is about ***100,000 times sweeter than sucrose*** (on molar basis). Monellin is dimer composed of an A-chain (45 amino acids) and a B-chain (50 amino acids). This protein is optimally sweet when both the chains are held together and not when they are separated.

It has been possible to chemically synthesize a gene that encodes both A and B chains as a single polypeptide. This gene was in fact introduced and expressed in tomatoes for the production of monellin. However, the success has been very limited.

FLAVOUR ENHANCERS

Taste and flavour are very important for the acceptability of foods. ***Monosodium glutamate is the most commonly used flavour enhancer.*** Its production and other aspects are described elsewhere (Chapter 26). The degraded products of proteins and RNAs also impart flavour. Thaumatin and monellin (described above) are also flavour enhancing agents, besides sweetening.

The other flavouring agents are citric acid (produced by *Aspergillus niger*), acetone and butanol (produced by *Clostridium acetobutylicum*). The lactones formed by some microbial fermentations also enhance flavour e.g. γ-butyrolactone produced in yeast fermentation of wine and beer. Mention must be made about alcohol also which can increase flavour and taste when added at appropriate concentrations e.g. certain medicinal syrups.

ALCOHOLIC BEVERAGES

Alcoholic beverages are produced throughout the world. The type of the beverage and the substrate used for its production mostly depends on the crops grown in the region. For instance, in

Table 28.2 Alcoholic beverages along with the common substrates used for their production

Beverage	*Alcohol (%)*	*Substrate(s)*
Beer (ale, lager)	4–8	Barley and other cereals
Wine (red, white)	10–16	Grapes
Sake	10–14	Rice
Champagne	10–12	Grapes
Cidar	7–12	Apple juice
Brandy	35–45	Grapes or other fruits
Whisky	40–55	Cereals (barley, rye, corn)
Rum	50–60	Molasses
Gin, vodka	40–50	Potato, wheat, rye
Chinese brandy	30–40	Rice
Toddy	5–15	Palmyra juice
Arrack	60–75	Rice, jaggery, waste carbohydrates

Poland, Russia and Scandinavia, beers manufactured from barely are consumed while in France, Italy and Greece, wines obtained from palmyra juice are mostly consumed by rural people.

The common types of alcoholic beverages with their alcoholic contents and substrates used for their production are given in ***Table 28.2***.

Historical perspective

Archaelogical evidence indicates that there existed fermentation of grains as early as 4000 B.C. Production of alcoholic beverages is considered to be a part of human civilization. The biological and chemical principles underlying alcoholic beverage production were elucidated by Louis Pasteur in the second half of nineteenth century.

Saccharomyces cerevisiae—the key organism for the production of alcoholic beverages

The most commonly used organism for the production of beverages is the yeast, *Saccharomyces cerevisiae*. This organism is capable of utilizing simple sugars (glucose, fructose) and converting them to ethanol. It is fortunate that *S. cerevisiae* is a unique organism that can ***tolerate*** and grow even at a ***high concentration of alcohol***. This is advantageous for good production yield of alcoholic beverages.

There are two types of *S. cerevisiae* strains—***bottom yeast*** and ***top yeast***. The bottom yeast settles to the bottom whereas the top yeast raises to the top while carrying out fermentation. Top yeast are frequently used for the production of beer while bottom yeast are employed for wine production. However, both strains can be used for beverage production.

Genetically engineered strains of *S. cerevisiae* : By employing recombinant DNA technology, new strains of *Saccharomyces cerevisiae* are constantly being developed. Some of these strains with ***improved production yields of alcoholic beverages***, are in fact approved for use in beverage industries.

ALCOHOLIC BEVERAGE PRODUCTION —GENERAL ASPECTS

The starting materials for beverage production are sugary or starchy materials.

Sugary materials : Fruit juice (or fruits), plant sap and honey that can be directly used.

Starchy materials—various grains and roots : They have to be subjected to hydrolysis to yield simple fermentable sugars. This process, referred to as ***saccharification***, is carried out either by plant materials (barley malt, rye malt or millet malt) or microorganisms (*Aspergillus* sp, *Rhizopus* sp. *Mucor sp*) rich in starch hydrolytic enzymes.

Fermentation process : When subjected to fermentation under ideal conditions, alcohol

production ranging from 2 to 16% is usually observed. These alcoholic beverages may be appropriately processed and drunk. e.g. beers. However, for the production of beverages with higher concentration of alcohol (brandy, whisky, rum, gin etc.), distillation has to be carried out to increase the alcohol content.

Brewing is the technical term used for the ***production of malt beverages***. Besides *S. cerevisiae* (described already), *S. carlsbergensis* is also used in brewing industry. The finished products in brewing namely the alcoholic beverages differ from the other industrial fermented products. This is because besides the alcohol content, several other factors are important in finished beverages. These include flavour, aroma, colour, clarity, foam production and stability, and satiety. These factors decide the consumer acceptance and the commercial value of alcoholic beverages. All these beverages can be consumed when freshly produced. However, they are normally kept for storing or aging before consumption. Aging certainly improves flavour, aroma etc.

The commercial production processes employed for the manufacture of beer and wine are briefly described hereunder.

BEER

Beer is an ***undistilled alcoholic beverage*** with alcohol content in the range of 4-8%. It is produced by fermentation of barley or other cereals, by employing the yeasts (most frequently) *S. cerevisiae* or (sometimes) *S. carlsbergensis*.

It is often necessary to add starch — rich materials (wheat, maize, rice) known as ***adjuncts***, to enhance fermentation, besides reducing the cost of raw materials. There are four major steps involved in the production of beer (***Fig. 28.4***).

1. **Malting :** Dried barely are cleaned and soaked in water for a period of two days. The excess water is drained and the soaked barley are incubated for 4-6 days to germinate. Germination process is associated with the formation of enzymes-***amylases*** (starch hydrolysing) and ***proteases*** (protein degrading). The germinated seeds are slowly subjected to increased temperature (up to 80°C) so that the germination process is halted, but the enzymes retain their activities. Malt is prepared by powdering the seeds.

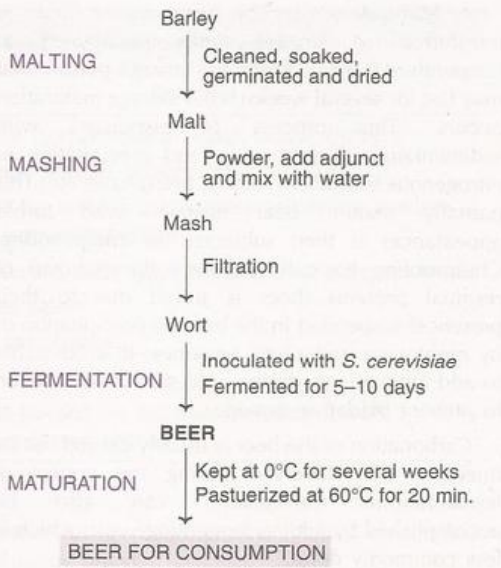


Fig. 28.4 : *Flow chart representing an outline of beer production.*

2. **Mashing :** The powdered malt is mixed with hot water (55–65°C). During the course of mashing, the soluble materials from malt and malt adjuncts are extracted. Mashing is also associated with degradation of starches (by amylases) to produce dextrins, maltose and glucose and hydrolysis of protein (by proteases) to peptones, peptides and amino acids. The temperature and pH influence the activities of enzymes. At the end of mashing, the medium for fermentation which is referred to as ***beer wort*** is developed. This is rich in sugars, amino acids, minerals and vitamins.

Hops are the dried female flowers obtained from hop plant. Addition of hops to wort is often done to provide characteristic flavour, aroma and stabilizing effect to the beer. Besides providing a mild antibacterial activity.

3. **Fermentation :** The beer wort, kept in open or closed bioreactors, is inoculated with pure strains of yeast. *S. cerevisiae* (usually top fermenting strains). The fermentation is carried out at temperature 20–28°C for 5–10 days. In some countries, the bottom fermenting yeast *S. uvarum* is used. For this organism, the ideal temperature for optimal fermentation is 10–15°C.

4. **Maturation :** The fermented fluid is transferred to storage tanks maintained at temperature 0-3°C. During the storage period (that may last for several weeks), cold storage maturation occurs. This process is associated with sedimentation of yeast cells and precipitation of nitrogenous substances, resins, phosphates etc. This partially mature beer (usually with turbid appearance) is then subjected to ***chillproofing***. Chillproofing basically involves the removal of residual proteins (beer is turbid due to their presence) suspended in the beer by precipitation or by employing proteolytic enzymes. It is advisable to add antioxidants during cold-storage maturation to prevent oxidative damage.

Carbonation of the beer is usually carried out by injecting CO_2 (evolved during the course of fermentation). Carbonation can also be accomplished by adding fermenting yeast, which is less commonly done.

The matured and carbonated beer is bottled or canned, and then pasteurized at 60°C for about 20 minutes.

As already stated, there is variation in the raw materials used for beer production. For instance, in Africa it is ***sorghum beer*** that are commonly produced. This is in contrast to the ***barley beer*** produced in most Western countries.

WINES

Wines are originally Middle East and European drinks, although almost every country now produces them. Large scale production of wines is carried out by using ***grapes*** of species *Vitis vinifera*. Grape juice is a good source for wine production because of its high concentration of sugar and other nutrients, natural acidity (that can inhibit unwanted growth of microorganisms) and the capability to produce pleasant aroma and flavour.

Louis Pasteur often used to state ***'wine is the most healthy and most hygienic of beverages'***. There is some recent scientific evidence also in support of this view, since moderate wine consumption reduces the risk for coronary heart disease (CHD).

Production of wines

Quality of the grapes is very important for the production of wines. The grapes are crushed (mechanically or by treading of feet) and the juice extracted. This ***grape juice ready for fermentation*** process is technically referred to as ***must***. It is a practice to add sulfur dioxide to must to inhibit the growth of non-wine yeast and contaminating bacteria. Sulfur dioxide which can kill other organisms can be tolerated by wine-fermenting yeasts. Sometimes, the must may also be subjected to partial or complete sterilization. The must in suitable bioreactors is inoculated with desired strains of the yeast *Saccharomyces cerevisiae*. Initially, oxygen is bubbled through the fermentation medium to promote good growth of yeast cells and gradually anaerobic conditions are established. The wine production normally takes a few days (2-5 days). The fermentation conditions (temperature, time etc.) are actually dependent on the type of wine produced. At the end of fermentation, wines are transferred to storage tanks (or vats) and allowed to age, which may take some months or years. ***Ageing*** of wine is very important ***for the development of characteristic flavour and aroma***. The alcohol content of wines is in the range of 10–16%.

Types of wines

There are hundreds of different types of wines produced in different parts of the world.

Red wine : The red colour of this wine is due the colour extracted from the black grape skins (when crushed totally). Red wines are commonly drunk, by many people in the west, along with lunch and dinner.

White wine : Black grapes after removal of skins or white grapes are used for the production of white wine.

Rose wine : This can be produced with a limited contact to the skins of grapes during fermentation.

Dry wine : It contain relatively higher alcohol content and produced when sugars are completely fermented.

Sweet wine : This is sweet to taste since it contains some residual sugars after fermentation.

Fortified wines : The normal concentration of alcohol in wines is less than 16%. The wines can be fortified with addition of alcohol (usually done by adding brandy or other distilled spirits) to the desired concentration. Some examples of fortified wines are sherry, port and vermouth.

Table 28.3 Enzymes used in various food and beverage industries

Industry	*Enzymes*
Dairy	Chymosins (animal/microbial), lipase, lactase, lysozyme
Baking	α–Amylase, protease, phospholipases, xylanase
Starch and sugar	Amylases (α– and β–), glucoamylase, pollulanase, invertase, glucose isomerase, xylanase
Fruit and vegetable processing	Pectin esterase, pectin lyase, hemicellulases, polygalacturanase
Meat processing	Proteases, papain
Animal and fish bone processing	Alkaline phosphatase
Egg white processing	Glucose oxidase, catalase
Brewing production of beer and alcoholic beverage	Amylases (α– and β–), proteases, cellulases, papain, aminoglucosidases, xylanase

OTHER MICROBIALLY-DERIVED FOOD PRODUCTS

There are a large number food products or their ingradients, produced by microorganisms. For full details on them, the reader must refer the corresponding Chapters — single-cell protein and mushrooms (Chapter 29), vitamins (Chapter 27), amino acids (Chapter 26), organic acids (Chapter 24), polysaccharides (Chapter 30), and vinegar (Chapter 24).

ENZYMES IN FOOD AND BEVERAGE INDUSTRIES

Food and beverage biotechnology is very closely linked with the use of enzymes. While a majority of enzymes are microbial in nature, some other sources of enzymes can also be used. In ***Table 28.3***, a selected list of enzymes employed in food processing and beverage industries is given. Some of the aspects of enzymes in dairy, baking and brewing industries are described in the foregoing pages. A few more important ones are described hereunder.

Proteases in food industry

Proteases (proteolytic enzymes) are used ***for tenderisation of meat*** which flavours the meat, besides easy digestion. Proteases are employed for removal of meal attached to bones in the form of a slurry. This can be used for the preparation of soups and canned meats. Partial hydrolysis of certain vegetable proteins also increases their flavour e.g. proteins of soy bean; protein (gluten) of dough in preparing bread.

Lactase in dairy industry

Milk and whey contain about 4-5% lactose. ***Lactose intolerance***, due to a defect in the intestinal enzyme lactase, is ***very common in humans***. It is estimated that half of the world's population suffers from lactose intolerance. These individuals when consume milk and milk products containing lactose (particularly in large quantities) suffer from diarrhea and flatulance (abdominal cramps and increased intestinal motility). Obviously, lactose intolerant persons require milk products with a low lactose concentration.

The enzyme lactase cleaves lactose to glucose and galactose which can be easily assimilated by the body. The lactase enzyme may be obtained from microbial sources e.g. *Kluyveromyces fragilis, Aspergillus niger*. Sometimes, immobilized lactase can also be used to degrade lactose.

For the preparation of flavoured and condensed milks and ice creams, lactase is widely used.

ENZYMES IN THE PREPARATION OF FRUIT JUICES

The turbidity and cloudiness, commonly seen in fruit juices, is due to the presence of pectins and cell debris. Enzyme preparation containing ***pectin esterase, polygalacturonase, pectin lyase*** and hemicellulase is added to the fruit pulp ***for the removal of pectins and other turbid-causing materials***.

Removal of glucose and oxygen from foodstuffs

The presence of glucose and oxygen in the foods is often associated with certain amount of damage. Their removal enhances the storability of foods. Glucose from the foodstuffs can be removed by the combined action of glucose oxidase and catalase, as illustrated below.

$$\text{Glucose} + O_2 \xrightarrow{\text{Glucose oxidase}} \text{Gluconic acid} + H_2O_2$$

$$2H_2O_2 \xrightarrow{\text{Catalase}} 2H_2O + O_2$$

The egg white used in baking industry is made free from glucose by the enzymatic treatment, given above. This pair of enzymes (glucose oxidase + catalase) is also useful for the removal of O_2 present in the head space of bottled and canned foods and drinks.

DIAGNOSTICS IN FOOD BIOTECHNOLOGY

It is essential to ensure that the foods produced by biotechnological processes are free from disease-causing organisms and toxins. Then only the foods are suitable for human consumption. The most common pathogens present in foods are *Salmonella* sp, *Campylobacter* sp and *Listeria* sp. The food toxins are derived from bacterial (endototoxins) and fungal (mycotoxins) sources.

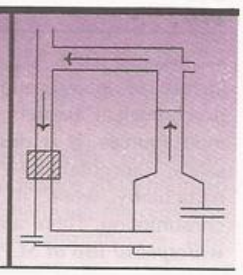
Several novel and innovative techniques have been developed in recent years for testing the safety of foods. For instance, the presence of *Salmonella* in foods can be detected by an immunoassy on the same day (This is in contrast to the traditional methods that take 5-7 days).

In fact, ***kits are available for the detection of concentrations of several undesirable compounds in the foods*** e.g. endotoxins, mycotoxins, antibiotics, hormones.

FOOD BIOTECHNOLOGY-PUBLIC ACCEPTANCE

In general, the public have a ***negative attitude towards*** the biotechnology based foods, particularly involving ***genetic manipulations***. For this reason, the application of genetic engineering techniques for food biotechnology is rather slow. Several factors influence the public acceptance. These include the type of biotechnology adopted, the regulatory procedures, economics of the people and consumer acceptance. For more details on this aspect, refer Chapter 61.

Single-Cell Protein, and Mushrooms



Single-cell protein (SCP) refers to the microbial cells or total protein extracted from pure microbial cell culture (monoculture) which can be used as protein supplement for humans or animals. The word SCP is considered to be appropriate, since most of the microorganisms grow as single or filamentious individuals. This is in contrast to complete multicellular plants and animals. If the ***SCP*** is suitable for human consumption, it is considered as ***food grade***. SCP is regarded as ***feed grade***, when it is used as animal feed supplement, but not suitable for human consumption.

Single-cell protein broadly refers to the ***microbial biomass*** or ***protein extract used as food or feed additive***. Besides high protein content (about 60–80% of dry cell weight), SCP also contains fats, carbohydrates, nucleic acids, vitamins and minerals. Another advantage with SCP is that it is rich in certain essential amino acids (lysine, methionine) which are usually limiting in most plant and animal foods. Thus, SCP is of high nutritional value for human or animal consumption.

It is estimated that about 25% of the world's population currently suffers from hunger and malnutrition. Most of these people live in developing countries. Therefore, SCP deserves a serious consideration for its use as food or feed supplement. In addition to its utility as a nutritional supplement, SCP can also be used for the isolation of several compounds e.g. carbohydrates, fats, vitamins, minerals.

Advantages of using microorganisms for SCP production

The protein-producing capabilities of a 250 kg cow and 250 g of microorganisms are often compared. The cow can produce about 200 g protein per day. On the other hand, microorganisms, theoretically, when grown under ideal conditions, could produce about 20–25 tonnes of protein. There are many advantages of using microorganisms for SCP production.

1. Microorganisms grow at a very rapid rate under optimal culture conditions. Some microbes double their mass in less than 30 minutes.

2. The ***quality and quantity of protein content in microorganisms is better compared to higher plants and animals.***

3. A ***wide range of raw materials***, which are otherwise wasted, ***can be fruitfully used*** for SCP production.

4. The culture conditions and the fermentation processes are very simple.

5. Microorganisms can be easily handled, and subjected to genetic manipulations.

Safety, acceptability and toxicology of SCP

There are many non-technological factors that influence the production of SCP. These include the geographical, social, political and psychological

factors. In many countries, there are social and psychological barriers to use microorganisms as food sources. It is desirable to first consider the safety, acceptability and toxicology of SCP, particularly when it is considered for human consumption. There are several ***limitations for the widespread use of SCP***.

1. The ***nucleic acid content*** of microbial biomass is ***very high*** (4-6% in algae; 10–15% in bacteria; 5–10% in yeast). This is highly hazardous, since humans have a limited capacity to degrade nucleic acids.

2. The presence of ***carcinogenic and other toxic substances*** is often observed in association with SCP. These include the hydrocarbons, heavy metals, mycotoxins and some contaminants. The nature and production of these compounds depends on the raw materials, and the type of organism used.

3. There is a possibility of contamination of pathogenic microorganisms in the SCP.

4. The digestion of microbial cells is rather slow. This is frequently ***associated with indigestion and allergic reactions*** in individuals.

5. Food grade production of SCP is more expensive than some other sources of proteins e.g. soy meal. Of course, this mainly depends on the cost of raw materials. In general, SCP for human consumption is 10 times more expensive than SCP for animal feed.

For the above said reasons, many countries give low priority for the use of SCP for human consumption. In fact, mass production of SCP using costly raw materials has been discontinued in some countries e.g. Japan, Britain, Italy. However, these countries continue their efforts to produce SCP from cheap raw materials such as organic wastes.

MICROORGANISMS AND SUBSTRATES USED FOR PRODUCTION OF SCP

Several microorganisms that include bacteria, yeasts, fungi, algae and actiomycetes utilizing a wide range of substrates are used for the production of SCP. A selected list is given in ***Table 29.1***.

The selection of microorganisms for SCP production is based on several criteria. These include their nutritive value, non-pathogenic nature, production cost, raw materials used and growth pattern.

Substrates

The nature of the raw materials supplying substrates is very crucial for SCP production. The cost of raw material significantly influences the

TABLE 29.1 A selected list of microorganisms and substrates used for single-cell protein production

Microorganism	*Substrate(s)*
Bacteria	
Methylophilus methylotrophus	Methane, methanol
Methylomonas sp	Methanol
Pseudomonas sp	Alkanes
Brevibacterium sp	C_1–C_4 hydrocarbons
Yeasts	
Saccharomycopsis lipolytica (previous name—*Candida lipolytica*)	Alkanes
Candida utilis	Sulfite liquor
Kluyveromyces fragilis	Whey
Saccharomyces cerevisiae (baker's yeast)	Molasses
Lactobacillus bulgaricus	Whey
Tosulopsis sp	Methanol
Fungi	
Chaetomium cellulolyticum	Cellulosic wastes
Paecilomyces varioti	Sulfite liquor
Aspergillus niger	Molasses
Trichoderma viride	Straw, starch
Algae	
Spirulina maxima	Carbon dioxide
Chlorella pyrenoidosa	Carbon dioxide
Scenedesmus acutus	Carbon dioxide
Actinomycetes	
Nocardia sp	Alkanes
Thermomonospora fusca	Cellulose
Mushrooms (a type of fungi)	
Agaricus biosporus	Compost, rice straw
Morchella crassipes	Whey, sulfite liquor
Auricularia sp	Saw dust, rice bran
Lentinus edodes	Saw dust, rice bran
Volvariella volvaceae	Cotton, straw

final cost of SCP. The most commonly used raw materials may be grouped in the following categories.

1. **High-energy sources** e.g. alkanes, methane, methanol, ethanol, gas oil.

2. **Waste products** e.g. molasses, whey, sewage, animal manures, straw, begasse.

3. **Agricultural and foresty sources** e.g. cellulose, lignin.

4. **Carbon dioxide**, the simplest carbon source.

Some details on the production of SCP from most important raw materials are briefly described.

PRODUCTION OF SCP FROM HIGH ENERGY SOURCES

There are a large number of energy-rich carbon compounds or their derivatives which serve as raw materials for SCP production. These include ***alkanes, methane, methanol, ethanol and gas oil.*** Bacteria and yeasts are mostly employed for SCP production from high energy sources. Some scientists question the wisdom of using (rather misusing) high-energy compounds for the production of food, since they regard it as a wasteful exercise.

Production of SCP from alkanes

Alkanes can be degraded by many yeasts, certain bacteria and fungi. The major limitation of alkanes is that they are not easily soluble, hence they cannot enter the cells rapidly. It is believed that the cells produce emulsifying substances which convert insoluble alkanes into small droplets (0.01-0.5 μm) that can enter the cells by passive diffusion. It is observed that when cells are grown on a medium of alkanes enriched with lipids, the diffusion of alkanes into the cells is enhanced.

Certain yeasts have been successfully used for producing SCP from alkanes e.g. *Saccharomycopsis lipolytica, Candida tropicalis, Candida oleophila.*

Petroleum products for SCP production : Several oil companies have developed fermentation systems, employing petroleum products for large scale manufacture of SCP by yeasts. Two types of petroleum products are mainly used for this purpose.

1. ***Gas oil*** or ***diesel oil*** containing 10-25% of alkanes with carbon length C_{15}–C_{30} (i.e. long chain alkanes).

2. ***Short chain alkanes*** with carbon length in the range of C_{10}–C_{17}, isolated from gas oil by use of molecular sieves.

Airlift bioreactor system with continuous operation was once used (in France and Britain) to produce SCP from gas oil employing the organism *Saccharomycopsis lipolytica.* But this is now discontinued for political reasons.

Degradation of alkanes : Alkanes have to be first broken down to appropriate metabolites for their utilization to form SCP. The most important step in this direction is the introduction of oxygen into alkanes which can be brought out by two pathways-terminal oxidation and subterminal oxidation (***Fig. 29.1***).

In ***terminal oxidation***, the terminal carbon gets oxidized to the corresponding monocarboxylic acid. The latter then undergoes β-oxidation to form acetic acid. In some microorganisms, the oxidation may occur at both the terminal carbon atoms (by a process referred to as ω-oxidation) to form a dicarboxylic acid. This can be further broken down to acetate and succinate by β-oxidation. Terminal oxidation is the predominant pathway occurring in majority of yeasts and bacteria.

Subterminal oxidation involves the oxidation of interminal carbon atoms (any carbon other than terminal i.e. C_2, C_3, C_4, and so on). The corresponding ketone produced undergoes α-oxidation, decarboxylation, and finally β-oxidation to form acetate and propionate.

The individual enzymes responsible for terminal oxidation or subterminal oxidation have not been fully identified.

Limitations of SCP production from alkanes

The production of SCP from alkanes is a very complex biotechnological process and has been extensively studied. The major drawback of alkanes as substrates is the formation of carcinogens, along with SCP which are highly harmful. For this reason, many countries have discontinued alkane-based production of SCP.

Production of SCP from methane

Methane is the chief constituent of natural gas in many regions. Although methane can be isolated in pure gas form, it cannot be liquified. The handling

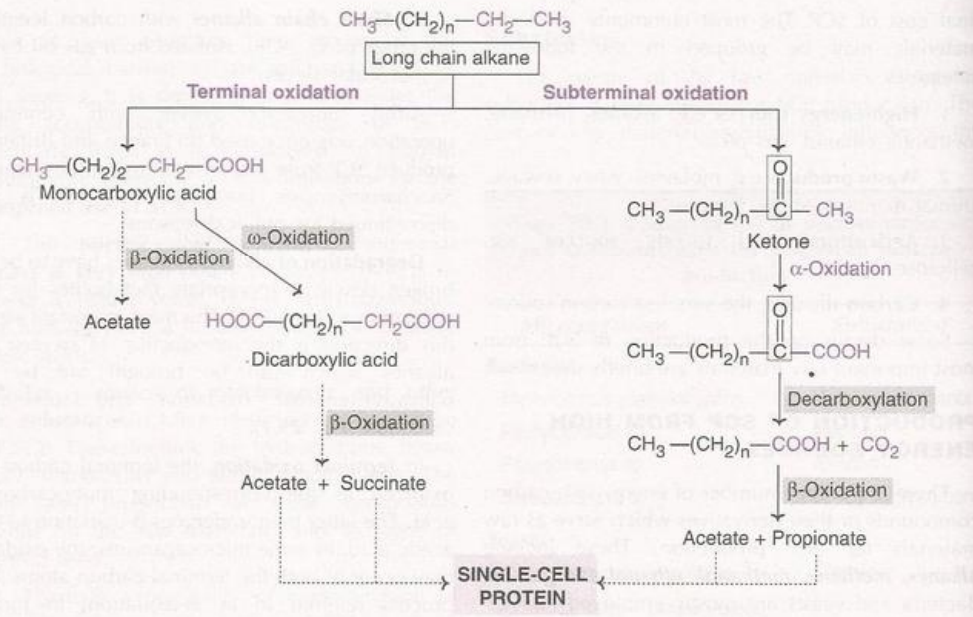


***Fig. 29.1** : Oxidation of alkanes by yeasts (e.g.* Saccharomycopsis lipolytica*) to produce single-cell protein.*

and transportation of methane (an explosive gas) are very difficult and expensive.

Certain bacteria that can utilize methane for SCP production have been identified e.g. *Methylococcus capsulatus, Methylomonas methanica, Methylovibrio soehngenii.* So far, yeasts that can utilize methane have not been identified.

The bacterial enzyme ***methane oxygenase*** oxidizes methane to methanol, which can be converted to formaldehyde and then to formic acid.

Although methane was extensively researched for its use as a source of SCP, it is not widely used due to technical difficulties.

Production of SCP from methanol

Methanol is a good substrate for producing SCP. Methanol as a carbon source for SCP has several advantages over alkanes and methane. Methanol is easily soluble in aqueous phase at all concentrations, and no residue of it remains in the harvested biomass. Technically, methanol can be easily handled. The sources for methanol are natural gas, coal, oil and methane. Many species of ***bacteria*** *(Methylobacter, Arthrobacter, Bacillus, Pseudomonas, Vibrio)* ***yeasts*** *(Candida biodinii, Hansenula* sp, *Torulopsis* sp) and ***fungi*** *(Trichoderma lignorum, Gliocladium delinquescens)* are capable of producing SCP from methanol. Bacteria are mostly preferred because they require simple fermentation conditions, grow rapidly and possess high content of protein.

Oxidation of methanol : Methanol gets oxidized to formaldehyde, then to formic acid and finally to carbon dioxide, as depicted in ***Fig. 29.2***.

The products obtained from methanol have to form C_3 compounds (such as pyruvate) for final production of SCP. Carbon dioxide formed from methanol can be utilized by photosynthetic organisms for the formation of ribulose diphosphate. Alternately, formaldehyde may condense with ribulose 5-phosphate to form 3-keto 6-phosphohexulose which then gives fructose 6-phosphate and finally pyruvate. This pathway is referred to as ***ribulose monophosphate*** (or ***Quayle***) ***cycle***.

Formaldehyde can condense with glycine to form serine which in a series of reactions forms phosphoenol pyruvate. This is referred to as ***serine pathway***.

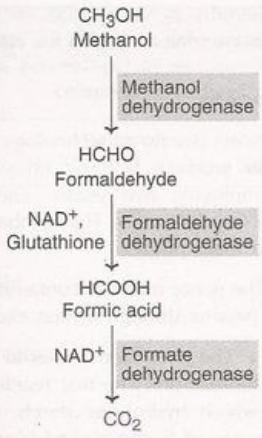


Fig. 29.2 : Oxidation of methanol.

Production process : Imperial Chemical Industries (ICI), U.K. was the first company to develop a process for continuous methanol fermentation for large scale production of SCP. Later, Hoechst (Germany) and Mistubishi (Japan) also developed similar fermentation systems.

ICI employed *Methylophilus methylotrophus* (formerly called *Pseudomonas methylotrophus*) for producing SCP from methanol. A bioreactor, referred to as ***ICI pressure cycle fermenter*** was used for this purpose (***Fig. 29.3***). This fermenter has three components-airlift column, down-flow tube and gas release space. The operation was carried out at temperature 35–37°C and pH 6.5-7.0. The cells were subjected to disruption by heat or acid treatment. The nutrient solution can be clarified by decanting.

ICI pruteen

The ***single-cell protein*** produced by ICI ***from methanol and ammonia*** using *M. methylotrophus* was referred to as ICI pruteen. This SCP was exclusively used for animal feeding. ICI invested a huge amount (around £40 million) in 1979 and installed a continuous culture system for SCP production. This was the world's largest continuous airlift fermenter. Unfortunately, the plant could not be operated for long due to economic reasons. For instance, in 1984 the cost of soy meal was around $125–200 per ton while ICI pruteen was sold at $600 per ton! This is mainly because of the high cost of methanol which represents approximately half of the production cost expenses. In the Middle East, due to high availability and low cost of methanol, the production of SCP appeared to be attractive. In the erstwhile Russia, there were several plants producing SCP from methanol which were later closed.

Genetic engineering for improved SCP production from methanol

The efficiency of SCP production has been improved by using genetic engineering. The assimilation of ammonia by *M. methylotrophus* is an essential step for cellular growth. This organism lacks glutamate dehydrogenase. It possesses glutamine synthase and glutamine ketoglutarate transaminase to utilize ammonia for the formation of glutamate (***Fig. 29.4A***). This is an energy (ATP) dependent reaction. By employing recombinant DNA technology, the gene for the enzyme glutamate dehydrogenase from *E. coli* was cloned and expressed in *M. methylotrophus*. These genetically transformed organisms were more efficient in assimilating ammonia. They could grow rapidly and convert more methanol to SCP. However, the overall increase in the production of SCP did not exceed 10%.

Production of SCP from ethanol

Ethanol is a good substrate for the production of SCP for human consumption (feed grade SCP).

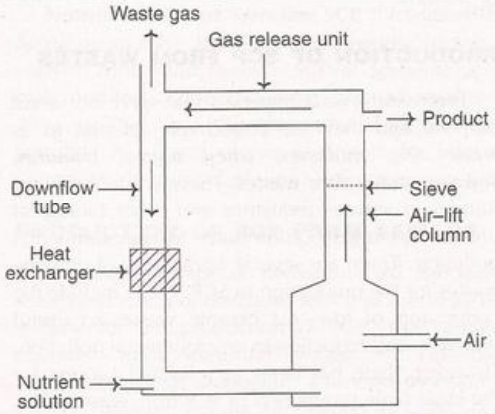


Fig. 29.3 : A diagrammatic representation of ICI pressure cycle fermenter to produce SCP from methanol.

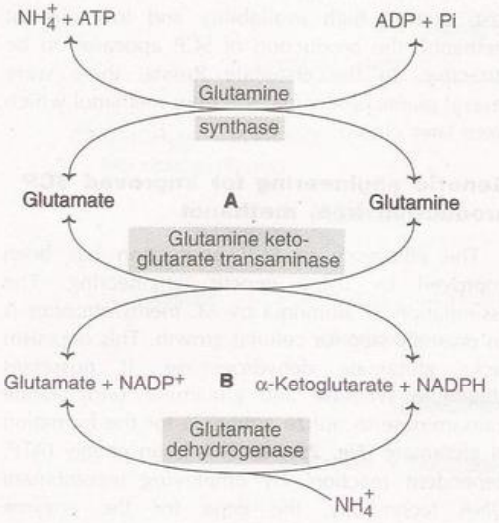


Fig. 29.4 : Assimilation of ammonia by Methylophilus methylotrophus *(A) Normal pathway, (B) Genetically engineered pathway.*

However, this process, as such, is not economically feasible. However, several factors-local raw materials, innovative fermentation technology, political decisions and foreign trade balances influence production of SCP. It may not be surprising if large scale production of SCP commences, on one day, from ethanol for a variety of reasons.

PRODUCTION OF SCP FROM WASTES

There are several materials that serve no useful purpose and they are collectively referred to as wastes e.g. ***molasses, whey, animal manures, sewage, straw, date wastes***. These waste products, formed in various industries and other biological processes, largely contribute to environmental pollution. There are several advantages of utilizing wastes for the production of SCP. These include the conversion of low-cost organic wastes to useful products, and reduction in environmental pollution. However, there has been very limited success for the large scale production of SCP from wastes. This is mainly because of transportation cost and technical difficulties. The technology adopted and the organism employed for SCP production depends on the waste being used as the substrate. Thus, *Saccharomyces cerevisiae* is used for molasses, *Kluyveromyces fragilis* for cheese whey.

Symba process

Symba process is a novel technology developed in Sweden to produce SCP by utilising starchy wastes by employing two yeasts, *Endomycopsis fibuligira* and *Candida utilis*. The Symba process is carried out in three phases.

Phase I : The waste material containing starch is sterilized by passing through a heat exchanger.

Phase II : The sterilized material is passed through two bioreactors. The first reactor contains *E. fibuligira* which hydrolyses starch. When this hydrolysate is passed to the second bioreactor, the organism, *C. utilis* grows to form biomass.

Phase III : The microbial biomass can be separated by centrifugation. The samples of SCP can be dried, packaged and stored.

Applications of Symba product : The yeast biomass produced in Symba process is of good nutritive value. It is widely used as an animal feed for pigs, calves and chicken. The animals grow quite well and no adverse effects have been reported.

Pekilo — a fungal protein rich product

A filamentous fungus, *Paecilomyces variotii*, with good fibrous structure was used for the production of Pekilo. This protein, rich in fungal biomass, was produced by fermentation of wastes such as molasses, whey, sulfite liquor and agricultural wastes. It can be produced by a continuous fermentation process. Pekilo is ***rich in proteins*** (containing essential amino acids), ***vitamins and minerals***. It was used as an animal feed in supplementing the diets of calves, pigs, chickens and hens without any adverse effects. It is unfortunate that the production of Pekilo has been discontinued at most places due to economic and commercial considerations.

Quorn-the mycoprotein for humans

The protein Quorn is the mycoprotein produced by the fungus *Fusarium graminearum*. Many companies in the developed countries are engaged in the production of fungal proteins for human consumption. Quorn is the trade name for ***Fusarium***

Table 29.2 Nutritional composition (in percentage) of mycoprotein in comparison to beef

Nutrient	*Mycoprotein*	*Beef (raw lean beefstreak)*
Protein	4	68
Lipid	15	30
Carbohydrate	10	0
Fiber	25	Traces
Ash (minerals etc.)	3	2
RNA	1	Traces

mycoprotein produced in Britain by Marlow Foods (ICI in association with Bank-Hovis-McDougall).

The fungus *Fusarium* can be grown continuously on simple carbohydrate sources (like glucose). Ammonium ions supply nitrogen. Mineral salts and vitamins are also added. The fermentation is carried out at pH 6.0 and temperature 30°C. At the end of fermentation, the culture is heated to 65°C to activate RNases. This is necessary to degrade RNA and reduce the content from 10% to around 1%. The breakdown products of RNA namely the nucleotides diffuse out from the cells and can be easily removed. (Reduction in RNA content is desirable to make the product acceptable for human consumption. This is because humans have a very limited capacity to digest nucleic acids). It is possible to produce 1 kg of fungal biomass with a protein content of about 135 g from 1 kg glucose utilized in the culture medium.

The dried *Fusarium* product is artificially flavoured and marketed in pieces that resemble beef, pork and chicken. The nutritional composition of mycoprotein when compared to beef is given in ***Table 29.2***. Besides being ***rich in essential nutrients***, mycoprotein has a ***good content of dietary fiber***. There are several advantages of fiber consumption-prevents constipation, decreases intestinal cancers, improves glucose tolerance and reduces serum cholesterol.

PRODUCTION OF SCP FROM WOOD

The natural waste wood sources containing ***cellulose, hemicellulose*** and ***lignin*** are attractive natural sources for the production of SCP. It is however, essential to breakdown these cellulosic compounds into fermentable sugars. For this purpose, extracellular cellulases can be used. Certain bacteria (*Cellulomonas* sp) and fungi (*Trichoderma* sp, *Penicillium* sp) are good sources for cellulases.

Techniques for the production of cellulases have been well standardized from several organisms. The cost of production of cellulases is a critical factor in determining the ultimate production cost of SCP.

In some instances, the cellulosic materials can be directly used for biomass production. The resultant SCP is used as animal feed.

PRODUCTION OF SCP FROM CO_2

Certain algae grown in open ponds require only CO_2 as the carbon source. In the presence of sunlight, they can effectively carry out photosynthesis, and produce SCP. The examples of these algae are *Chlorella* sp, *Senedesmus* sp and *Spirulina* sp. *Chlorella* is used as a protein and vitamin supplement for enriching ice-creams, breads and yoghurts in some countries.

In some parts of the world, the algae in ponds are used for the removal of organic pollutants. The resultant algae biomass can be harvested, dried and powdered. Algae SCP are very useful as animal supplements.

Nutritive value of *Spirulina* SCP : Traditionally *Spirulina* sp have been eaten by people in some parts of Africa and Mexico. SCP of *Spirulina* is of high nutritive value (protein-65%, carbohydrate-20%, fat-4%, fibre-3%, chlorophyll-5%, ash-3%). Spirulina is a good source of protein for human consumption, particularly in developing countries.

PRODUCTION OF SCP FROM SEWAGE

Domestic sewage is normally used for large scale production of methane, which in turn may be utilized for the production of SCP. The sewage obtained from industrial wastes in cellulose processing, starch production and food processing can be utilized for the production of SCP. The organism *Candida utilis* is used to produce SCP by using effluent formed during the course of paper manufacture. Other microorganisms namely *Candida tropicalis, Paecilomyces varioti* are

Table 29.3 A selected list of the edible mushrooms cultivated on commercial scale

Mushroom species	*Common name*	*Substrate(s)*
Agaricus bisporus	Button mushroom	Straw, horse manure, compost
Leutinule edodes	Oak or shiitake mushroom	Saw dust, wooden logs, rice bran
Pleurotus ostreatus	Oyster mushroom	Straw, saw dust, paper
Volvariella volvacea	Chinese mushroom or padi–straw mushroom	Straw, cotton
Auricularia sp	Wood–ear mushroom	Saw dust, rice bran
Coprinus sp	— — —	Straw

employed to use sulfite waste liquor for the production of SCP.

GENETICALLY ENGINEERED ARTIFICIAL PROTEIN AS ANIMAL FEED

Rumen bacteria can synthesize amino acids. Some workers have developed genetically engineered strains of rumen bacteria that can produce a protein rich in methionine, threonine, lysine and leucine. This artificial protein has a total of 100 amino acids, of which 57 are essential.

The gene for artificial protein was synthesized by 14 overlapping oligonucleotides held to maltose binding protein gene. This gene was expressed in *E. coli* under the transcriptional control of *tac* promoter. The production of this artificial protein accounts to around 12% of the intracellular proteins.

However, the large scale production of artificial protein by rumen bacteria is yet to be clearly established and commercialised.

MUSHROOMS

Mushrooms are fungi belonging to the classes basidiomycetes (*Agaricus* sp, *Auricularia* sp, *Tremella* sp) and ascomycetes (*Morchella* sp, *Tuber* sp). Majority of edible mushrooms are the species of basidomyces. It is estimated that there are around 4,000 species of basidiomyces. Of these, around 200 are edible, and a dozen of them are cultivated on large scale. Some of the most important edible mushrooms, their common names and the substrates used are given in ***Table 29.3***.

The cultivation of edible mushrooms is one of the rare examples of a microbial culture wherein the cultivated macroscopic organism itself is directly used as ***human food***. Mushroom growing is one of the fastest developing biotechnological industries worldover. Further growth of mushroom industry is expected ***for the production of enzymes, and pharmaceutical compounds, including antitumor agents and antibiotics***.

Poisonous mushrooms

There are certain poisonous mushrooms also. They usually possess unpleasant taste and odour. These mushrooms produce some poisonous substances like ***phallin*** and ***muscarine***. The examples of poisonous mushrooms are *Amanita phalloides, A. muscaria, A. viraosa, Lepiota morgani* and *Boletus satanas*.

Nutritive value of edible mushrooms

Some people regard ***edible mushrooms as vegetable meat***. Mushrooms contain 80–90% water, depending on the growth conditions (temperature, humidity). Edible mushrooms are rich sources of protein (35–45% of dry weight). However, all these proteins are not easily digestible by humans. Mushrooms also contain fats and free fatty acids (7–10%), carbohydrates (5–15%) and minerals in good concentration. Certain undesirable substances may also be present in edible mushrooms e.g. cadmium, chromium.

Many delicious recipes of edible mushrooms can be prepared. This actually depends on the dietary habits of the people. Some of the common recipes are mushroom soup, mushroom paneer, mushroom pulao, and mushroom omelette.

Advantages of edible mushroom biotechnology

1. Mushrooms can be produced by utilizing cheap and often waste substrates (industrial and wood wastes).

2. They are of high nutritive value being rich in proteins, vitamins and minerals.

3. Many delicious recipes can be prepared from mushrooms.

4. Due to low carbohydrate content, consumption of mushrooms is advocated to diabetic patients.

PRODUCTION OF EDIBLE MUSHROOMS

Mushroom production is basically a fermentation process. This is mostly carried out by ***solid-substrate fermentation***. A wide range of substrates (straw, saw dust, compost, wooden logs) depending the organism can be used (Refer ***Table 29.3***). Mushroom production is a good example of a low technology utilization in an otherwise sophisticated modern biotechnology.

The most common edible mushroom cultivated worldover (that may constitute about 20% world mushroom produce) is the ***white button mushroom, Agaricus bisporus. Lentinula edodes*** is the second most cultivated mushroom in the world. The substrates straw, compost or horse manure can be used. The substrate selection depends on the local factors.

A schematic representation of mushroom production is depicted in ***Fig. 29.5***. The compost with desired formulation is prepared and sterilized. It is spread into the trays which are then transferred to production room and inoculated with ***spawn***. Spawn is the term used for the ***mushroom inoculum containing spores*** and/or small pieces of fruiting body. After inoculation (spawning), the culture is maintained at optimal growth conditions. The trays are regularly watered to maintain 70–80% humidity. The ideal temperature is about 15°C, and pH about 7.0. It takes about 7–10 days for each crop of mushroom production. It is possible to have 3-4 crops, before terminating the production process. The mushrooms can be harvested and marketed.

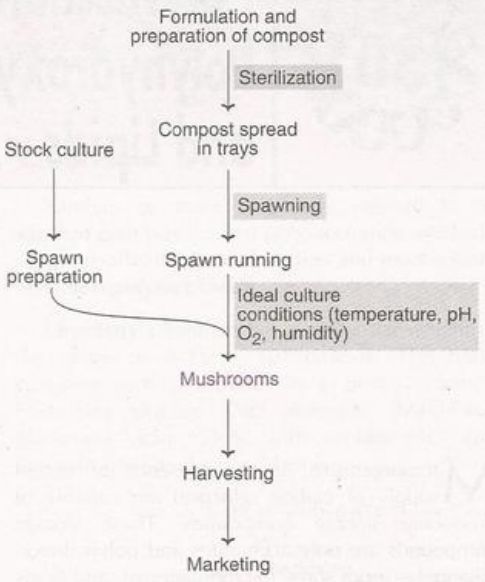


Fig. 29.5 : An overview of edible mushroom production.

Mushrooms have a very short life 8–12 hours, unless stored at low temperature (refrigerator 2-5°C). Therefore, they should be immediately consumed, stored or canned.

Variations in culturing mushrooms : The production of mushrooms is highly variable and mostly depends on the organism and the substrate used, besides several other local factors. There are distinct differences in the mushroom cultivation methods between different countries. For instance, ***garden and field cultivation*** methods are used in Europe, while in USA, ***cave and house cultivation*** techniques are employed.

Some mushrooms (e.g. *Volirariella* sp) are suitable for cultivation in summer and rainy reason while others grow well in winter (*Agaricus bisporus, Pleurotus* sp). It is however, possible to grow these mushrooms any time in a year with appropriate temperature and humidity control arrangements.